Literature search report – run 20260828T160426

scitech-librarian 3.2

August 28, 2026

Generated by scitech-librarian 3.2 (report level: full). Every number below is reproducible from the archived run directory `20260828T160426`.

Item	Value
Run started	2026-08-28 16:04:26
Duration	42 s
Query file	queries.example.json
Blocks	PSC, MLP, NOV, QC
Backends / sources	openalex, arxiv, inspire
Mode	full fetch, up to 300 records per block and backend
Non-curated venue filter	on
Open-access lookup	not run
Journal metrics	on file for 103 journals
Filters	none
Interrupted	no

Search strategy

Each block is one structural query – a conjunction of synonym groups, (a OR b) AND (c OR d) – rendered into every backend's native grammar. The strings below are exactly what was sent (PRISMA-S item 8).

Block PSC: Perovskite solar cell degradation under humidity

Purpose: grounding block – expect thousands of hits; use it to sanity-check that every backend is reachable and the counts are plausible

```
(perovskite solar cell OR halide perovskite photovoltaic) AND (degradation OR stability OR encapsulation)
 AND (humidity OR moisture OR water ingress)
```

Backend	Query string sent
openalex	("perovskite solar cell" OR "halide perovskite photovoltaic") AND (degradation OR stability OR encapsulation) AND (humidity OR moisture OR "water ingress")
arxiv	(all:"perovskite solar cell" OR all:"halide perovskite photovoltaic") AND (all:"degradation" OR all:"stability" OR all:"encapsulation")
inspire	("perovskite solar cell" or "halide perovskite photovoltaic") and ("degradation" or "stability" or "encapsulation") and ("humidity" or "moisture" or "water ingress")

Block MLP: Machine-learned interatomic potentials for amorphous materials

Purpose: a narrower intersection – expect hundreds; good for testing record fetch and RIS export

```
(machine learning potential OR machine-learned interatomic potential OR neural network potential) AND
(amorphous OR glass OR disordered solid) AND (molecular dynamics OR structure prediction)
```

arXiv receives groups [0, 1] only (nested-boolean limitation).

Backend	Query string sent
openalex	("machine learning potential" OR "machine-learned interatomic potential" OR "neural network potential") AND (amorphous OR glass OR "disordered solid") AND ("molecular dynamics" OR "structure prediction")
arxiv	(all:"machine learning potential" OR all:"machine-learned interatomic potential" OR all:"neural network potential") AND (all:"amorphous" OR all:"glass" OR all:"disordered solid")
inspire	("machine learning potential" or "machine-learned interatomic potential" or "neural network potential") and ("amorphous" or "glass" or "disordered solid") and ("molecular dynamics" or "structure prediction")

Block NOV: EXAMPLE NOVELTY CHECK: origami metamaterials as topological acoustic pumps

Purpose: novelty-check pattern – a SMALL number is the good outcome; read every hit by hand before claiming a gap

```
(origami OR kirigami) AND (acoustic metamaterial OR phononic crystal) AND (topological pumping OR
Thouless pump OR edge state)
```

Backend	Query string sent
openalex	(origami OR kirigami) AND (”acoustic metamaterial” OR ”phononic crystal”) AND (”topological pumping” OR ”Thouless pump” OR ”edge state”)
arxiv	(all:”origami” OR all:”kirigami”) AND (all:”acoustic metamaterial” OR all:”phononic crystal”)
inspire	(”origami” or ”kirigami”) and (”acoustic metamaterial” or ”phononic crystal”) and (”topological pumping” or ”Thouless pump” or ”edge state”)

Block QC: Quasicrystal photonics (arXiv-tuned example)

Purpose: demonstrates arxiv_groups: arXiv chokes on deeply nested booleans, so name the two groups it should get

```
(quasicrystal OR quasiperiodic lattice) AND (photonic OR waveguide OR optical lattice) AND (localization
OR critical states OR fractal spectrum)
```

arXiv receives groups [0, 2] only (nested-boolean limitation).

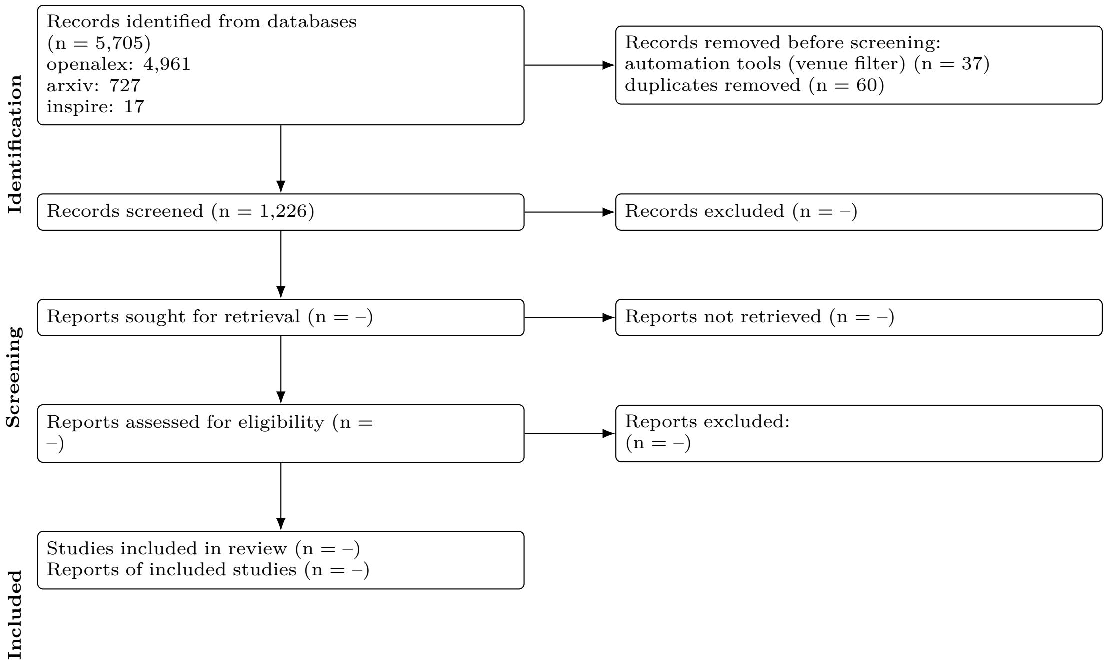
Backend	Query string sent
openalex	(quasicrystal OR ”quasiperiodic lattice”) AND (photonic OR waveguide OR ”optical lattice”) AND (localization OR ”critical states” OR ”fractal spectrum”)
arxiv	(all:”quasicrystal” OR all:”quasiperiodic lattice”) AND (all:”localization” OR all:”critical states” OR all:”fractal spectrum”)
inspire	(”quasicrystal” or ”quasiperiodic lattice”) and (”photonic” or ”waveguide” or ”optical lattice”) and (”localization” or ”critical states” or ”fractal spectrum”)

Results summary

Block	openalex	arxiv	inspire	Identified	Retrieved	Unique
PSC	4569	175	0	4,744	475	474
MLP	238	138	0	376	344	313
NOV	0	0	0	0	0	0
QC	154	414	17	585	467	439
Total	4,961	727	17	5,705	1286	1226

Identified = database hit counts (not comparable across backends: proximity operators are dropped and stemming differs). Retrieved = records actually downloaded after the venue filter, capped by `--limit`. Unique = after DOI/title deduplication across all sources.

PRISMA 2020 flow



Stage	n
Records identified from databases	5,705
openalex	4,961
arxiv	727
inspire	17
Records retrieved (downloaded / ingested)	1,323
Removed before screening: automation (non-curated venues)	37
Removed before screening: duplicates	60
Records to screen (unique)	1,226
Records screened	1,226 (assumed = unique)
Records excluded at screening	–
Reports sought for retrieval	–
Reports not retrieved	–
Reports assessed for eligibility	–
Studies included	–
Reports of included studies	–

Automation stages are computed from the data; '–' marks manual stages not yet recorded in prisma.json. Note that 'identified' counts hits reported by each database while 'retrieved' is what was downloaded within `--limit`, so the two differ on large blocks.

PRISMA-S search-reporting checklist

Item	Requirement	This search
1	Database name	openalex, arxiv, inspire (documented public APIs)
2	Multi-database searching	3 databases, one structural query per block rendered into each native grammar; see Search strategy
3	Study registries	not applicable
4	Online resources and browsing	none recorded
5	Citation searching	not performed
6	Contacts	none recorded
7	Other methods	none
8	Full search strategies	reported verbatim per backend under Search strategy; archived in queries.json
9	Limits and restrictions	record download capped at 300 per block and backend, most-cited first; no date, language or document-type limits applied
10	Search filters	venue filter on: records from non-curated repositories (Zenodo, Figshare, SSRN...) removed
11	Prior work	none
12	Updates	3 earlier run(s) archived; counts tracked in counts_history.csv
13	Dates of searches	searched on 2026-08-28 16:04:26
14	Peer review	none
15	Total records	5,705 identified from databases; 1,323 retrieved; 1,226 unique
16	Deduplication	exact DOI match, else first 90 characters of the lower-cased title; 60 duplicates removed

Records

Deduplicated across sources, sorted by cited.

Block PSC (474 unique)

Incorporation of rubidium cations into perovskite solar cells improves photovoltaic performance

Michael Saliba; Taisuke Matsui; Konrad Domanski; Ji-Youn Seo; Amita Ummadisingu; Shaik M. Zakeeruddin; Juan-Pablo Correa-Baena; Wolfgang Tress; Antonio Abate; Anders Hagfeldt; Michaël Grätzel. Science (2016). Cited by 3691. Found by: openalex. DOI: 10.1126/science.aah5557

Abstract: Improving the stability of perovskite solar cells Inorganic-organic perovskite solar cells have poor long-term stability because ultraviolet light and humidity degrade these materials. Bella et al. show that coating the cells with a water-proof fluorinated polymer that contains pigments to absorb ultraviolet light and re-emit it in the visible range can boost cell efficiency and limit photodegradation. The performance and stability of inorganic-organic perovskite solar cells are also limited by the size of the cations required for forming a correct lattice. Saliba et al. show that the rubidium cation, which is too small to form a perovskite by itself, can form a lattice with cesium and organic cations. Solar cells based on these materials have efficiencies exceeding 20% for over 500 hours if given environmental protection by a polymer coating. Science , this issue pp. 203 and 206

High-efficiency two-dimensional Ruddlesden–Popper perovskite solar cells

Hsinhan Tsai; Wanyi Nie; Jean-Christophe Blancon; Constantinos C. Stoumpos; Reza Asadpour; Boris Harutyunyan; Amanda J. Neukirch; Rafael Verduzco; Jared Crochet; Sergei Tretiak; Laurent Pédesseau; Jacky Even; Muhammad A. Alam; Gautam Gupta; Jun Lou; Pulickel M. Ajayan; Michael J. Bedzyk; Mercouri G. Kanatzidis; Aditya D. Mohite. Nature (2016). Cited by 3361. Found by: openalex. DOI: 10.1038/nature18306

Efficient, stable and scalable perovskite solar cells using poly(3-hexylthiophene)

Eui Hyuk Jung; Nam Joong Jeon; Eun Young Park; Chan Su Moon; Tae Joo Shin; Tae-Youl Yang; Jun Hong Noh; Jangwon Seo. Nature (2019). Cited by 2318. Found by: openalex. DOI: 10.1038/s41586-019-1036-3

A Layered Hybrid Perovskite Solar-Cell Absorber with Enhanced Moisture Stability

Ian C. P. Smith; Eric T. Hoke; Diego Solís-Ibarra; Michael D. McGehee; Hemamala I. Karunadasa. Angewandte Chemie International Edition (2014). Cited by 2006. Found by: openalex. openalex_2yr: 15.006 (2026). DOI: 10.1002/anie.201406466

Abstract: Two-dimensional hybrid perovskites are used as absorbers in solar cells. Our first-generation devices containing (PEA)2(MA)2[Pb3I10] (1; PEA=C6H5(CH2)2NH3(+), MA=CH3NH3(+)) show an open-circuit voltage of 1.18 V and a power conversion efficiency of 4.73%. The layered structure allows for high-quality films to be deposited through spin coating and high-temperature annealing is not required for device fabrication. The 3D perovskite (MA)[PbI3] (2) has recently been identified as a promising absorber for solar cells. However, its instability to moisture requires anhydrous processing and operating conditions. Films of 1 are more moisture resistant than films of 2 and devices containing 1 can be fabricated under ambient humidity levels. The larger bandgap of the 2D structure is also suitable as the higher bandgap absorber in a dual-absorber tandem device. Compared to 2, the layered perovskite structure may offer greater tunability at the molecular level for material optimization.

Review of recent progress in chemical stability of perovskite solar cells

Guangda Niu; Xudong Guo; Liduo Wang. Journal of Materials Chemistry A (2014). Cited by 1945. Found by: openalex. DOI: 10.1039/c4ta04994b

Abstract: The understanding of how the chemical stability of PSCs is affected by oxygen and moisture, UV light, the solution process, and temperature was reviewed.

Understanding Degradation Mechanisms and Improving Stability of Perovskite Photovoltaics

Caleb C. Boyd; Rongrong Cheacharoen; Tomas Leijtens; Michael D. McGehee. Chemical Reviews (2018). Cited by 1767. Found by: openalex. openalex_2yr: 53.886 (2026). DOI: 10.1021/acs.chemrev.8b00336

Abstract: This review article examines the current state of understanding in how metal halide perovskite solar cells can degrade when exposed to moisture, oxygen, heat, light, mechanical stress, and reverse bias. It also highlights strategies for improving stability, such as tuning the composition of the perovskite, introducing hydrophobic coatings, replacing metal electrodes with carbon or transparent conducting oxides, and packaging. The article concludes with recommendations on how accelerated testing should be performed to rapidly develop solar cells that are both extraordinarily efficient and stable.

Formamidinium and Cesium Hybridization for Photo- and Moisture-Stable Perovskite Solar Cell

Jin-Wook Lee; Deok-Hwan Kim; Hui-Seon Kim; Seung-Woo Seo; Sung-Min Cho; Nam-Gyu Park. Advanced Energy Materials (2015). Cited by 1661. Found by: openalex. openalex_2yr: 18.63 (2026). DOI: 10.1002/aenm.201501310

Abstract: Although power conversion efficiency (PCE) of state-of-the-art perovskite solar cells has already exceeded 20%, photo- and/or moisture instability of organolead halide perovskite have prevented further commercialization. In particular, the underlying weak interaction of organic cations with surrounding iodides due to eight equivalent orientations of the organic cation along the body diagonals in unit cell and chemically non-inertness of organic cation result in photo- and moisture instability of organometal halide perovskite. Here, a perovskite light absorber incorporating organic–inorganic hybrid cation in the A-site of 3D APbI 3 structure with enhanced photo- and moisture stability is reported. A partial substitution of Cs + for HC(NH 2) 2 + in HC(NH 2) 2 PbI 3 perovskite is found to substantially improve photo- and moisture stability along with photovoltaic performance. When 10% of HC(NH 2) 2 + is replaced by Cs + , photo- and moisture stability of perovskite film are significantly improved, which is attributed to the enhanced interaction between HC(NH 2) 2 + and iodide due to contraction of cubo-octahedral volume. Moreover, trap density is reduced by one order of magnitude upon incorporation of Cs + , which is responsible for the increased open-circuit voltage and fill factor, eventually leading to enhancement of average PCE from 14.9% to 16.5%.

23.6%-efficient monolithic perovskite/silicon tandem solar cells with improved stability

Kevin A. Bush; Axel F. Palmstrom; Zhengshan J. Yu; Mathieu Boccard; Rongrong Cheacharoen; Jonathan P. Mailoa; David P. McMeekin; Robert L. Z. Hoye; Colin D. Bailie; Tomas Leijtens; Ian Marius Peters; Maxmillian C. Minichetti; Nicholas Rolston; Rohit Prasanna; Sarah E. Sofia; Duncan Harwood; Wen Ma; Farhad Moghadam; Henry J. Snaith; Tonio Buonassisi; Zachary C. Holman; Stacey F. Bent; Michael D. McGehee. Nature Energy (2017). Cited by 1533. Found by: openalex. DOI: 10.1038/nenergy.2017.9

Investigation of CH 3 NH 3 PbI 3 Degradation Rates and Mechanisms in Controlled Humidity Environments Using in Situ Techniques

Jinli Yang; Braden D. Siempelkamp; Dianyi Liu; Timothy L. Kelly. ACS Nano (2015). Cited by 1342. Found by: openalex. openalex_2yr: 16.894 (2026). DOI: 10.1021/nn506864k

Abstract: Perovskite solar cells have rapidly advanced to the forefront of solution-processable photovoltaic devices, but the CH3NH3PbI3 semiconductor decomposes rapidly in moist air, limiting their commercial utility. In this work, we report a quantitative and systematic investigation of perovskite degradation processes. By carefully controlling the relative humidity of an environmental chamber and using in situ absorption spectroscopy and in situ grazing incidence X-ray diffraction to monitor phase changes in perovskite degradation process, we demonstrate the formation of a hydrated intermediate containing isolated PbI6(4-) octahedra as the first step of the degradation mechanism. We also show that the identity of the hole transport layer can have a dramatic impact on the stability of the underlying perovskite film, suggesting a route toward perovskite solar cells with long device lifetimes and a resistance to humidity.

Carbon Nanotube/Polymer Composites as a Highly Stable Hole Collection Layer in Perovskite Solar Cells

Severin N. Habisreutinger; Tomas Leijtens; Giles E. Eperon; Samuel D. Stranks; R. J. Nicholas; Henry J. Snaith. Nano Letters (2014). Cited by 1198. Found by: openalex. DOI: 10.1021/nl501982b

Abstract: Organic-inorganic perovskite solar cells have recently emerged at the forefront of photovoltaics research. Power conversion efficiencies have experienced an unprecedented increase to reported values exceeding 19% within just four years. With the focus mainly on efficiency, the aspect of stability has so far not been thoroughly addressed. In this paper, we identify thermal stability as a fundamental weak point of perovskite solar cells, and demonstrate an elegant approach to mitigating thermal degradation by replacing the organic hole transport material with polymer-functionalized single-walled carbon nanotubes (SWNTs) embedded in an insulating polymer matrix. With this composite structure, we achieve JV scanned power-conversion efficiencies of up to 15.3% with an average efficiency of 10 ± 2%. Moreover, we observe strong retardation in thermal degradation as compared to cells employing state-of-the-art organic hole-transporting materials. In addition, the resistance to water ingress is remarkably enhanced. These are critical developments for achieving long-term stability of high-efficiency perovskite solar cells.

Improved performance and stability of perovskite solar cells by crystal crosslinking with alkylphosphonic acid -ammonium chlorides

Xiong Li; M. Ibrahim Dar; Chenyi Yi; Jingshan Luo; Manuel Tschumi; Shaik M. Zakeeruddin; Mohammad Khaja Nazeeruddin; Hongwei Han; Michaël Grätzel. Nature Chemistry (2015). Cited by 1176. Found by: openalex. DOI: 10.1038/nchem.2324

All-Inorganic Perovskite Solar Cells

Jia Liang; Caixing Wang; Yanrong Wang; Zhaoran Xu; Zhipeng Lü; Yue Ma; Hongfei Zhu; Yi Hu; Chengcan Xiao; Xu Yi; Guoyin Zhu; Hongling Lv; Lianbo Ma; Tao Chen; Zuoxiu Tie; Zhong Jin; Jie Liu. Journal of the American Chemical Society (2016). Cited by 1092. Found by: openalex. DOI: 10.1021/jacs.6b10227

Abstract: The research field on perovskite solar cells (PSCs) is seeing frequent record breaking in the power conversion efficiency (PCE). However, organic-inorganic hybrid halide perovskites and organic additives in common hole-transport materials (HTMs) exhibit poor stability against moisture and heat. Here we report the successful fabrication of all-inorganic PSCs without any labile or expensive organic components. The entire fabrication process can be operated in ambient environment without humidity control (e.g., a glovebox). Even without encapsulation, the all-inorganic PSCs present no performance degradation in humid air (90-95% relative humidity, 25 °C) for over 3 months (2640 h) and can endure extreme temperatures (100 and -22 °C). Moreover, by elimination of expensive HTMs and noble-metal electrodes, the cost was significantly reduced. The highest PCE of the first-generation all-inorganic PSCs reached 6.7%. This study opens the door for next-generation PSCs with long-term stability under harsh conditions, making practical application of PSCs a real possibility.

Study on the stability of CH 3 NH 3 PbI 3 films and the effect of post-modification by aluminum oxide in all-solid-state hybrid solar cells

Guangda Niu; Wenzhe Li; Fanqi Meng; Liduo Wang; Haopeng Dong; Yong Qiu. Journal of Materials Chemistry A (2013). Cited by 1067. Found by: openalex. DOI: 10.1039/c3ta13606j

Abstract: Degradation of perovskite has been a big problem in all-solid-state perovskite solar cells, although many researchers mainly focus on the high efficiency of these solar cells. This paper studies the stability of CH3NH3PbI3 films and finds that CH3NH3PbI3 is sensitive to moisture. The degradation reaction is proposed according to UV-Vis spectra and XRD results. In order to improve the degradation of CH3NH3PbI3, we introduce aluminum oxide as a post-modification material into all-solid-state perovskite solar cells for the first time. UV-Vis spectra show that Al2O3 modification could maintain the absorption of CH3NH3PbI3 after degradation. XRD results reveal that Al2O3 could protect perovskite from degradation. Moreover, the device post-modified by Al2O3 has shown more brilliant stability than that without modification when exposed to moisture. EIS results and dark current illustrate that the modification increased interface resistance in the dark, indicating the restrained electron recombination process.

Organometallic-functionalized interfaces for highly efficient inverted perovskite solar cells

Zhen Li; Bo Li; Xin Wu; Stephanie A. Sheppard; Shoufeng Zhang; Danpeng Gao; Nicholas J. Long; Zonglong Zhu. Science (2022). Cited by 1031. Found by: openalex. DOI: 10.1126/science.abm8566

Abstract: Further enhancing the performance and stability of inverted perovskite solar cells (PSCs) is crucial for their commercialization. We report that the functionalization of multication and halide perovskite interfaces with an organometallic compound, ferrocenyl-bis-thiophene-2-carboxylate (FcTc 2), simultaneously enhanced

the efficiency and stability of inverted PSCs. The resultant devices achieved a power conversion efficiency of 25.0% and maintained >98% of their initial efficiency after continuously operating at the maximum power point for 1500 hours under simulated AM1.5 illumination. Moreover, the FcTc 2 -functionalized devices passed the international standards for mature photovoltaics (IEC61215:2016) and have exhibited high stability under the damp heat test (85°C and 85% relative humidity).

Degradation observations of encapsulated planar CH 3 NH 3 PbI 3 perovskite solar cells at high temperatures and humidity

Yu Han; S. Meyer; Yasmina Dkhissi; Karl Weber; Jennifer M. Pringle; Udo Bach; Leone Spiccia; Yi-Bing Cheng. Journal of Materials Chemistry A (2015). Cited by 997. Found by: openalex. DOI: 10.1039/c5ta00358j

Abstract: The stability of encapsulated planar-structured CH 3 NH 3 PbI 3 (MAPbI 3) perovskite solar cells (PSCs) was investigated under various simulated environmental conditions.

Stability of perovskite solar cells

Dian Wang; Matthew Wright; Naveen Kumar Elumalai; Ashraf Uddin. Solar Energy Materials and Solar Cells (2016). Cited by 993. Found by: openalex. DOI: 10.1016/j.solmat.2015.12.025

Strained hybrid perovskite thin films and their impact on the intrinsic stability of perovskite solar cells

Jingjing Zhao; Yehao Deng; Haotong Wei; Xiaopeng Zheng; Zhenhua Yu; Yuchuan Shao; Jeffrey E. Shield; Jinsong Huang. Science Advances (2017). Cited by 965. Found by: openalex. DOI: 10.1126/sciadv.aao5616

Abstract: Organic-inorganic hybrid perovskite (OIHP) solar cells have achieved comparable efficiencies to those of commercial solar cells, although their instability hinders their commercialization. Although encapsulation techniques have been developed to protect OIHP solar cells from external stimuli such as moisture, oxygen, and ultraviolet light, understanding of the origin of the intrinsic instability of perovskite films is needed to improve their stability. We show that the OIHP films fabricated by existing methods are strained and that strain is caused by mismatched thermal expansion of perovskite films and substrates during the thermal annealing process. The polycrystalline films have compressive strain in the out-of-plane direction and in-plane tensile strain. The strain accelerates degradation of perovskite films under illumination, which can be explained by increased ion migration in strained OIHP films. This study points out an avenue to enhance the intrinsic stability of perovskite films and solar cells by reducing residual strain in perovskite films.

Scalable fabrication and coating methods for perovskite solar cells and solar modules

Nam-Gyu Park; Kai Zhu. Nature Reviews Materials (2020). Cited by 902. Found by: openalex. DOI: 10.1038/s41578-019-0176-2

Improving efficiency and stability of perovskite solar cells with photocurable fluoropolymers

Federico Bella; Gianmarco Griffini; Juan-Pablo Correa-Baena; Guido Saracco; Michaël Grätzel; Anders Hagfeldt; Stefano Turri; Claudio Gerbaldi. Science (2016). Cited by 866. Found by: openalex. DOI: 10.1126/science.aah4046

Abstract: Organometal halide perovskite solar cells have demonstrated high conversion efficiency but poor long-term stability against ultraviolet irradiation and water. We show that rapid light-induced free-radical polymerization at ambient temperature produces multifunctional fluorinated photopolymer coatings that confer luminescent and easy-cleaning features on the front side of the devices, while concurrently forming a strongly hydrophobic barrier toward environmental moisture on the back contact side. The luminescent photopolymers re-emit ultraviolet light in the visible range, boosting perovskite solar cells efficiency to nearly 19% under standard illumination. Coated devices reproducibly retain their full functional performance during prolonged operation, even after a series of severe aging tests carried out for more than 6 months.

Thermochromic halide perovskite solar cells

Jia Lin; Minliang Lai; Letian Dou; Christopher S. Kley; Hong Chen; Fei Peng; Junliang Sun; Dylan Lu; Steven A. Hawks; Chenlu Xie; Fan Cui; A. Paul Alivisatos; David T. Limmer; Peidong Yang. Nature Materials (2018).

Cited by 854. Found by: openalex. DOI: 10.1038/s41563-017-0006-0

Damp heat–stable perovskite solar cells with tailored-dimensionality 2D/3D heterojunctions

Randi Azmi; Esma Ugur; Akmaral Seitkhan; Faisal Aljamaan; Anand S. Subbiah; Jiang Liu; George T. Harrison; Mohamad Insan Nugraha; Mathan K. Eswaran; Maxime Babics; Yuan Chen; Fuzong Xu; Thomas G. Allen; Atteq ur Rehman; Chien-Lung Wang; Thomas D. Anthopoulos; Udo Schwingenschlögl; Michele De Bastiani; Erkan Aydın; Stefaan De Wolf. Science (2022). Cited by 843. Found by: openalex. DOI: 10.1126/science.abm5784

Abstract: If perovskite solar cells (PSCs) with high power conversion efficiencies (PCEs) are to be commercialized, they must achieve long-term stability, which is usually assessed with accelerated degradation tests. One of the persistent obstacles for PSCs has been successfully passing the damp-heat test (85°C and 85% relative humidity), which is the standard for verifying the stability of commercial photovoltaic (PV) modules. We fabricated damp heat-stable PSCs by tailoring the dimensional fragments of two-dimensional perovskite layers formed at room temperature with oleylammonium iodide molecules; these layers passivate the perovskite surface at the electron-selective contact. The resulting inverted PSCs deliver a 24.3% PCE and retain >95% of their initial value after >1000 hours at damp-heat test conditions, thereby meeting one of the critical industrial stability standards for PV modules.

Tailored interfaces of unencapsulated perovskite solar cells for >1,000 hour operational stability

Jeffrey A. Christians; Philip Schulz; Jonathan S. Tinkham; Tracy H. Schloemer; Steven P. Harvey; Bertrand J. Tremolet de Villers; Alan Sellinger; Joseph J. Berry; Joseph M. Luther. Nature Energy (2018). Cited by 826. Found by: openalex. DOI: 10.1038/s41560-017-0067-y

Stable High-Performance Perovskite Solar Cells via Grain Boundary Passivation

Tianqi Niu; Jing Lü; Rahim Munir; Jianbo Li; Dounya Barrit; Xu Zhang; Hanlin Hu; Zhou Yang; Aram Amassian; Kui Zhao; Shengzhong Liu. Advanced Materials (2018). Cited by 800. Found by: openalex. openalex_2yr: 22.086 (2026). DOI: 10.1002/adma.201706576

Abstract: The trap states at grain boundaries (GBs) within polycrystalline perovskite films deteriorate their optoelectronic properties, making GB engineering particularly important for stable high-performance optoelectronic devices. It is demonstrated that trap states within bulk films can be effectively passivated by semiconducting molecules with Lewis acid or base functional groups. The perovskite crystallization kinetics are studied using in situ synchrotron-based grazing-incidence X-ray scattering to explore the film formation mechanism. A model of the passivation mechanism is proposed to understand how the molecules simultaneously passivate the Pb-I antisite defects and vacancies created by under-coordinated Pb atoms. In addition, it also explains how the energy offset between the semiconducting molecules and the perovskite influences trap states and intergrain carrier transport. The superior optoelectronic properties are attained by optimizing the molecular passivation treatments. These benefits are translated into significant enhancements of the power conversion efficiencies to 19.3%, as well as improved environmental and thermal stability of solar cells. The passivated devices without encapsulation degrade only by 13% after 40 d of exposure in 50% relative humidity at room temperature, and only 10% after 24 h at 80 °C in controlled environment.

In Situ Growth of 2D Perovskite Capping Layer for Stable and Efficient Perovskite Solar Cells

Peng Chen; Yang Bai; Songcan Wang; Miaoqiang Lyu; Jung-Ho Yun; Lianzhou Wang. Advanced Functional Materials (2018). Cited by 772. Found by: openalex. openalex_2yr: 14.975 (2026). DOI: 10.1002/adfm.201706923

Abstract: Abstract 2D halide perovskites have recently been recognized as a promising avenue in perovskite solar cells (PSCs) in terms of encouraging stability and defect passivation effect. However, the efficiency (less than 15%) of ultrastable 2D Ruddlesden–Popper PSCs still lag far behind their traditional 3D perovskite counterparts. Here, a rationally designed 2D-3D perovskite stacking-layered architecture by in situ growing 2D PEA 2 PbI 4 capping layers on top of 3D perovskite film, which drastically improves the stability of PSCs without compromising their high performance, is reported. Such a 2D perovskite capping layer induces larger Fermi-level splitting in the 2D-3D perovskite film under light illumination, resulting in an enhanced open-circuit voltage (V oc) and thus a higher efficiency of 18.51% in the 2D-3D PSCs. Time-resolved photoluminescence decay measurements indicate the facilitated hole extraction in the 2D-3D stacking-layered perovskite films, which is ascribed to the optimized energy band alignment and reduced nonradiative recombination at the subgap states. Benefiting from the high moisture resistivity as well as suppressed ion migration of the 2D perovskite, the 2D-3D PSCs show significantly improved long-term stability, retaining nearly 90% of the initial power conversion efficiency after 1000 h exposure in the ambient conditions with a high relative humidity level of 60 ± 10%.

2D perovskite stabilized phase-pure formamidinium perovskite solar cells

Jin-Wook Lee; Zhenghong Dai; Tae-Hee Han; Chungseok Choi; Sheng-Yung Chang; Sung-Joon Lee; Nicholas De Marco; Hongxiang Zhao; Pengyu Sun; Yu Huang; Yang Yang. Nature Communications (2018). Cited by 747. Found by: openalex. DOI: 10.1038/s41467-018-05454-4

Abstract: Abstract Compositional engineering has been used to overcome difficulties in fabricating high-quality phase-pure formamidinium perovskite films together with its ambient instability. However, this comes alongside an undesirable increase in bandgap that sacrifices the device photocurrent. Here we report the fabrication of phase-pure formamidinium-lead tri-iodide perovskite films with excellent optoelectronic quality and stability. Incorporation of 1.67 mol% of 2D phenylethylammonium lead iodide into the precursor solution enables the formation of phase-pure formamidinium perovskite with an order of magnitude enhanced photoluminescence lifetime. The 2D perovskite spontaneously forms at grain boundaries to protect the formamidinium perovskite from moisture and suppress ion migration. A stabilized power conversion efficiency (PCE) of 20.64% (certified stabilized PCE of 19.77%) is achieved with a short-circuit current density exceeding 24 mA cm − 2 and an open-circuit voltage of 1.130 V, corresponding to a loss-in-potential of 0.35 V, and significantly enhanced operational stability.

Suppression of atomic vacancies via incorporation of isovalent small ions to increase the stability of halide perovskite solar cells in ambient air

Makhsud I. Saidaminov; Junghwan Kim; Ankit Jain; Rafael Quintero-Bermudez; Hairen Tan; Guankui Long; Furui Tan; Andrew Johnston; Yicheng Zhao; Oleksandr Voznyy; Edward H. Sargent. Nature Energy (2018). Cited by 745. Found by: openalex. DOI: 10.1038/s41560-018-0192-2

A Layered Hybrid Perovskite Solar-Cell Absorber with Enhanced Moisture Stability

Ian C. P. Smith; Eric T. Hoke; Diego Solís-Ibarra; Michael D. McGehee; Hemamala I. Karunadasa. Angewandte Chemie (2014). Cited by 692. Found by: openalex. openalex_2yr: 2.173 (2026). DOI: 10.1002/ange.201406466

Abstract: Abstract Two-dimensional hybrid perovskites are used as absorbers in solar cells. Our first-generation devices containing (PEA)2(MA)2[Pb3I10] (1; PEA=C6H5(CH2)2NH3+, MA=CH3NH3+) show an open-circuit voltage of 1.18 V and a power conversion efficiency of 4.73 %. The layered structure allows for high-quality films to be deposited through spin coating and high-temperature annealing is not required for device fabrication. The 3D perovskite (MA)[PbI3] (2) has recently been identified as a promising absorber for solar cells. However, its instability to moisture requires anhydrous processing and operating conditions. Films of 1 are more moisture resistant than films of 2 and devices containing 1 can be fabricated under ambient humidity levels. The larger bandgap of the 2D structure is also suitable as the higher bandgap absorber in a dual-absorber tandem device. Compared to 2, the layered perovskite structure may offer greater tunability at the molecular level for material optimization.

Stable high efficiency two-dimensional perovskite solar cells via cesium doping

Xu Zhang; Xiaodong Ren; Bin Liu; Rahim Munir; Xuejie Zhu; Dong Yang; Jianbo Li; Yucheng Liu; Detlef-M. Smilgies; Ruipeng Li; Zhou Yang; Tianqi Niu; Xiuli Wang; Aram Amassian; Kui Zhao; Shengzhong Liu. Energy & Environmental Science (2017). Cited by 675. Found by: openalex. openalex_2yr: 30.475 (2026). DOI: 10.1039/c7ee01145h

Abstract: Cs + doping into 2D (BA) 2 (MA) 3 Pb 4 I 13 perovskites boosts power conversion efficiency (PCE) to 13.7% and yields superior humidity and thermal stability.

A polymer scaffold for self-healing perovskite solar cells

Yicheng Zhao; Jing Wei; Heng Li; Yin Yan; Wenke Zhou; Dapeng Yu; Qing Zhao. Nature Communications (2016). Cited by 666. Found by: openalex. DOI: 10.1038/ncomms10228

Abstract: Advancing of the lead halide perovskite solar cells towards photovoltaic market demands large-scale devices of high-power conversion efficiency, high reproducibility and stability via low-cost fabrication technology, and in particular resistance to humid environment for long-time operation. Here we achieve uniform perovskite film based on a novel polymer-scaffold architecture via a mild-temperature process. These solar cells exhibit efficiency of up to 16% with small variation. The unencapsulated devices retain high output for up to 300 h in highly humid environment (70% relative humidity). Moreover, they show strong humidity resistant and self-healing behaviour, recovering rapidly after removing from water vapour. Not only the film can self-heal in this case, but the corresponding devices can present power conversion efficiency recovery after the water vapour is removed. Our work demonstrates the value of cheap, long chain and hygroscopic polymer scaffold in perovskite solar cells towards commercialization.

Accelerated degradation of methylammonium lead iodide perovskites induced by exposure to iodine vapour

Shenghao Wang; Yan Jiang; Emilio J. Juárez-Pérez; Luis K. Ono; Yabing Qi. Nature Energy (2016). Cited by 657. Found by: openalex. DOI: 10.1038/nenergy.2016.195

All-Inorganic CsPbX 3 Perovskite Solar Cells: Progress and Prospects

Jingru Zhang; Gary Hodes; Zhiwen Jin; Shengzhong Liu. Angewandte Chemie International Edition (2019). Cited by 611. Found by: openalex. openalex_2yr: 15.006 (2026). DOI: 10.1002/anie.201901081

Abstract: Abstract Recently, lead halide-based perovskites have become one of the hottest topics in photovoltaic research because of their excellent optoelectronic properties. Among them, organic-inorganic hybrid perovskite solar cells (PSCs) have made very rapid progress with their power conversion efficiency (PCE) now at 23.7 %. However, the intrinsically unstable nature of these materials, particularly to moisture and heat, may be a problem for their long-term stability. Replacing the fragile organic group with more robust inorganic Cs + cations forms the cesium lead halide system (CsPbX 3 , X is halide) as all-inorganic perovskites which are much more thermally stable and often more stable to other factors. From the first report in 2015 to now, the PCE of CsPbX 3 -based PSCs has abruptly increased from 2.9 % to 17.1 % with much enhanced stability. In this Review, we summarize the field up to now, propose solutions in terms of development bottlenecks, and attempt to boost further research in CsPbX 3 PSCs.

Intact 2D/3D halide junction perovskite solar cells via solid-phase in-plane growth

Yeoun-Woo Jang; Seungmin Lee; Kyung Mun Yeom; Kiwan Jeong; Kwang Choi; Mansoo Choi; Jun Hong Noh. Nature Energy (2021). Cited by 606. Found by: openalex. DOI: 10.1038/s41560-020-00749-7

Trapped charge-driven degradation of perovskite solar cells

Namyoung Ahn; Kwisung Kwak; Min Seok Jang; Heetae Yoon; Byung Yang Lee; Jong Kwon Lee; Peter V. Pikhitsa; Junseop Byun; Mansoo Choi. Nature Communications (2016). Cited by 601. Found by: openalex. DOI: 10.1038/ncomms13422

Abstract: Perovskite solar cells have shown unprecedent performance increase up to 22% efficiency. However, their photovoltaic performance has shown fast deterioration under light illumination in the presence of humid air even with encapulation. The stability of perovskite materials has been unsolved and its mechanism has been elusive. Here we uncover a mechanism for irreversible degradation of perovskite materials in which trapped charges, regardless of the polarity, play a decisive role. An experimental setup using different polarity ions revealed that the moisture-induced irreversible dissociation of perovskite materials is triggered by charges trapped along grain boundaries. We also identified the synergetic effect of oxygen on the process of moisture-induced degradation. The deprotonation of organic cations by trapped charge-induced local electric field would be attributed to the initiation of irreversible decomposition.

Stability of Perovskite Solar Cells: A Prospective on the Substitution of the A Cation and X Anion

Ze Wang; Zejiao Shi; Taotao Li; Yonghua Chen; Wei Huang. Angewandte Chemie International Edition (2016). Cited by 595. Found by: openalex. openalex_2yr: 15.006 (2026). DOI: 10.1002/anie.201603694

Abstract: In recent years, organometal trihalide perovskites have emerged as promising materials for low-cost, flexible, and highly efficient solar cells. Despite their processing advantages, before the technology can be commercialized the poor stability of the organic-inorganic hybrid perovskite materials with regard to humidity, heat, light, and oxygen has be to overcome. Herein, we distill the current state-of-the-art and highlight recent advances in improving the chemical stability of perovskite materials by substitution of the A-cation and X-anion. Our hope is to pave the way for the rational design of perovskite materials to realize perovskite solar cells with unprecedented improvement in stability.

Efficient and stable perovskite solar cells prepared in ambient air irrespective of the humidity

Qidong Tai; Peng You; Hongqian Sang; Zhike Liu; Chenglong Hu; H.L.W. Chan; Feng Yan. Nature Communications (2016). Cited by 582. Found by: openalex. DOI: 10.1038/ncomms11105

Abstract: Poor stability of organic-inorganic halide perovskite materials in humid condition has hindered the success of perovskite solar cells in real applications since controlled atmosphere is required for device fabrication and operation, and there is a lack of effective solutions to this problem until now. Here we report the use of lead (II) thiocyanate (Pb(SCN)2) precursor in preparing perovskite solar cells in ambient air. High-quality CH3NH3PbI3-x(SCN)x perovskite films can be readily prepared even when the relative humidity exceeds 70%. Under optimized processing conditions, we obtain devices with an average power conversion efficiency of 13.49% and the maximum efficiency over 15%. In comparison with typical CH3NH3PbI3-based devices, these solar cells without encapsulation show greatly improved stability in humid air, which is attributed to the incorporation of thiocyanate ions in the crystal lattice. The findings pave a way for realizing efficient and stable perovskite solar cells in ambient atmosphere.

Efficient and stable Ruddlesden–Popper perovskite solar cell with tailored interlayer molecular interaction

Hui Ren; Shidong Yu; Lingfeng Chao; Yingdong Xia; Yuanhui Sun; Shouwei Zuo; Fan Li; Tingting Niu; Yingguo Yang; Huanxin Ju; Bixin Li; Haiyan Du; Xingyu Gao; Jing Zhang; Jianpu Wang; Lijun Zhang; Yonghua Chen; Wei Huang. Nature Photonics (2020). Cited by 581. Found by: openalex. DOI: 10.1038/s41566-019-0572-6

Quantum-size-tuned heterostructures enable efficient and stable inverted perovskite solar cells

Hao Chen; Sam Teale; Bin Chen; Yi Hou; Luke Grater; Tong Zhu; Koen Bertens; So Min Park; Harindi R. Atapattu; Yajun Gao; Mingyang Wei; Andrew Johnston; Qi-Lin Zhou; Kaimin Xu; Danni Yu; Congcong Han; Teng Cui; Eui Hyuk Jung; Chun Zhou; Wenjia Zhou; Andrew H. Proppe; Sjoerd Hoogland; Frédéric Laquai; Tobin Filleter; Kenneth R. Graham; Zhijun Ning; Edward H. Sargent. Nature Photonics (2022). Cited by 581. Found by: openalex. DOI: 10.1038/s41566-022-00985-1

Perovskite-polymer composite cross-linker approach for highly-stable and efficient perovskite solar cells

Tae-Hee Han; Jin-Wook Lee; Chungseok Choi; Shaun Tan; Changsoo Lee; Yepin Zhao; Zhenghong Dai; Nicholas De Marco; Sung-Joon Lee; Sang-Hoon Bae; Yonghai Yuan; Hyuck Mo Lee; Yu Huang; Yang Yang. Nature Communications (2019). Cited by 581. Found by: openalex. DOI: 10.1038/s41467-019-08455-z

Abstract: Manipulation of grain boundaries in polycrystalline perovskite is an essential consideration for both the optoelectronic properties and environmental stability of solar cells as the solution-processing of perovskite films inevitably introduces many defects at grain boundaries. Though small molecule-based additives have proven to be effective defect passivating agents, their high volatility and diffusivity cannot render perovskite films robust enough against harsh environments. Here we suggest design rules for effective molecules by considering their molecular structure. From these, we introduce a strategy to form macromolecular intermediate phases using long chain polymers, which leads to the formation of a polymer-perovskite composite cross-linker. The cross-linker functions to bridge the perovskite grains, minimizing grain-to-grain electrical decoupling and yielding excellent environmental stability against moisture, light, and heat, which has not been attainable with small molecule defect passivating agents. Consequently, all photovoltaic parameters are significantly enhanced in the solar cells and the devices also show excellent stability.

Stabilizing the Efficiency Beyond 20% with a Mixed Cation Perovskite Solar Cell Fabricated in Ambient Air under Controlled Humidity

Trilok Singh; Tsutomu Miyasaka. Advanced Energy Materials (2017). Cited by 573. Found by: openalex. openalex_2yr: 18.63 (2026). DOI: 10.1002/aenm.201700677

Abstract: Abstract Perovskite solar cells have evolved to have compatible high efficiency and stability by employing mixed cation/halide type perovskite crystals as pinhole-free large grain absorbers. The cesium (Cs)–formamidium–methylammonium triple cation-based perovskite device fabricated in a glove box enables reproducible high-voltage performance. This study explores the method to reproduce stable and high power conversion efficiency (PCE) of a triple cation perovskite prepared using a one-step solution deposition and low-temperature annealing fully conducted in controlled ambient humidity conditions. Optimizing the perovskite grain size by Cs concentration and solution processes, a route is created to obtain highly uniform, pinhole-free large grain perovskite films that work with reproducible PCE up to 20.8% and high preservation stability without cell encapsulation for more than 18 weeks. This study further investigates the light intensity characteristics of open-circuit voltage (Voc) of small (5 × 5 mm2, PCE > 20%) and large (10 × 10 mm2, PCE of 18%) devices. Intensity dependence of Voc shows an ideality factor in the range of 1.7-1.9 for both devices, implying that the triple cation perovskite involves trap-assisted recombination loss at the hetero junction interfaces that influences Voc. Despite relatively high ideality factor, perovskite device is capable of supplying high power conversion efficiency under low light intensity (0.01 Sun) whereas maintaining Voc over 0.9 V.

Two-dimensional Ruddlesden–Popper layered perovskite solar cells based on phase-pure thin films

Chao Liang; Hao Gu; Yingdong Xia; Zhuo Wang; Xiaotao Liu; Junmin Xia; Shouwei Zuo; Yue Hu; Xingyu Gao; Wei Hui; Lingfeng Chao; Tingting Niu; Min Fang; Hui Lu; Hanshan Dong; Hui Yu; Shi Chen; Xueqin Ran; Lin Song; Bixin Li; Jing Zhang; Yong Peng; Guosheng Shao; Jianpu Wang; Yonghua Chen; Guichuan Xing; Wei Huang. Nature Energy (2020). Cited by 554. Found by: openalex. DOI: 10.1038/s41560-020-00721-5

Low-loss contacts on textured substrates for inverted perovskite solar cells

So Min Park; Mingyang Wei; Nikolaos Lempesis; Wenjin Yu; Tareq Hossain; Lorenzo Agosta; Virginia Carnevali; Harindi R. Atapattu; Peter Serles; Felix T. Eickemeyer; Heejong Shin; Maral Vafaie; Deokjae Choi; Kasra Darabi; Eui Dae Jung; Yi Yang; Da Bin Kim; Shaik M. Zakeeruddin; Bin Chen; Aram Amassian; Tobin Filleter; Mercouri G. Kanatzidis; Kenneth R. Graham; Lixin Xiao; Ursula Röthlisberger; Michaël Grätzel; Edward H. Sargent. Nature (2023). Cited by 548. Found by: openalex. DOI: 10.1038/s41586-023-06745-7

Multifunctional Chemical Linker Imidazoleacetic Acid Hydrochloride for 21% Efficient and Stable Planar Perovskite Solar Cells

Jiangzhao Chen; Xing Zhao; Seul-Gi Kim; Nam-Gyu Park. Advanced Materials (2019). Cited by 534. Found by: openalex. openalex_2yr: 22.086 (2026). DOI: 10.1002/adma.201902902

Abstract: Abstract Chemical interaction at a heterojunction interface induced by an appropriate chemical linker is of crucial importance for high efficiency, hysteresis-less, and stable perovskite solar cells (PSCs). Effective interface engineering in PSCs is reported via a multifunctional chemical linker of 4-imidazoleacetic acid hydrochloride (ImAcHCl) that can provide a chemical bridge between SnO 2 and perovskite through an ester bond with SnO 2 via esterification reaction and an electrostatic interaction with perovskite via imidazolium cation in ImAcHCl and iodide anion in perovskite. In addition, the chloride anion in ImAcHCl plays a role in the improvement of crystallinity of perovskite film crystallinity. The introduction of ImAcHCl onto SnO 2 realigns the positions of the conduction and valence bands upwards, reduces nonradiative recombination, and improves carrier life time. As a consequence, average power conversion efficiency (PCE) is increased from 18.60% ± 0.50% to 20.22% ± 0.34% before and after surface modification, respectively, which mainly results from an enhanced voltage from 1.084 ± 0.012 V to 1.143 ± 0.009 V. The best PCE of 21% is achieved by 0.1 mg mL −1 ImAcHCl treatment, along with negligible hysteresis. Moreover, an unencapsulated device with ImAcHCl-modified SnO 2 shows much better thermal and moisture stability than unmodified SnO 2 .

A review on perovskite solar cells: Evolution of architecture, fabrication techniques, commercialization issues and status

Priyanka Roy; Numeshwar Kumar Sinha; Sanjay Tiwari; Ayush Khare. Solar Energy (2020). Cited by 530. Found by: openalex. DOI: 10.1016/j.solener.2020.01.080

CsPb0.9Sn0.1IBr2 Based All-Inorganic Perovskite Solar Cells with Exceptional Efficiency and Stability

Jia Liang; Peiyang Zhao; Caixing Wang; Yanrong Wang; Yi Hu; Guoyin Zhu; Lianbo Ma; Jie Liu; Zhong Jin. Journal of the American Chemical Society (2017). Cited by 528. Found by: openalex. DOI: 10.1021/jacs.7b07949

Abstract: The emergence of perovskite solar cells (PSCs) has generated enormous interest in the photovoltaic research community. Recently, cesium metal halides (CsMX 3, M = Pb or Sn; X = I, Br, Cl or mixed halides) as a class of inorganic perovskites showed great promise for PSCs and other optoelectronic devices. However, CsMX 3 -based PSCs usually exhibit lower power conversion efficiencies (PCEs) than organic–inorganic hybrid PSCs, due to the unfavorable band gaps. Herein, a novel mixed-Pb/Sn mixed-halide inorganic perovskite, CsPb 0.9 Sn 0.1 IBr 2, with a suitable band gap of 1.79 eV and an appropriate level of valence band maximum, was prepared in ambient atmosphere without a glovebox. After thoroughly eliminating labile organic components and noble metals, the all-inorganic PSCs based on CsPb 0.9 Sn 0.1 IBr 2 and carbon counter electrodes exhibit a high open-circuit voltage of 1.26 V and a remarkable PCE up to 11.33%, which is record-breaking among the existing CsMX 3 -based PSCs. Moreover, the all-inorganic PSCs show good long-term stability and improved endurance against heat and moisture. This study indicates a feasible way to design inorganic halide perovskites through energy-band engineering for the construction of high-performance all-inorganic PSCs.

Functionalization of perovskite thin films with moisture-tolerant molecules

Shuang Yang; Yun Wang; Porun Liu; Yi-Bing Cheng; Huijun Zhao; Hua Gui Yang. Nature Energy (2016). Cited by 527. Found by: openalex. DOI: 10.1038/nenergy.2015.16

Surface Passivation Using 2D Perovskites toward Efficient and Stable Perovskite Solar Cells

Guangbao Wu; Rui Liang; Mingzheng Ge; G.C. Sun; Yuan Zhang; Guichuan Xing. Advanced Materials (2021). Cited by 524. Found by: openalex. openalex_2yr: 22.086 (2026). DOI: 10.1002/adma.202105635

Abstract: 3D perovskite solar cells (PSCs) have shown great promise for use in next-generation photovoltaic devices. However, some challenges need to be addressed before their commercial production, such as enormous defects formed on the surface, which result in severe SRH recombination, and inadequate material interplay between the composition, leading to thermal-, moisture-, and light-induced degradation. 2D perovskites, in which the organic layer functions as a protective barrier to block the erosion of moisture or ions, have recently emerged and attracted increasing attention because they exhibit significant robustness. Inspired by this, surface passivation by employing 2D perovskites deposited on the top of 3D counterparts has triggered a new wave of research to simultaneously achieve higher efficiency and stability. Herein, we exploited a vast amount of literature to comprehensively summarize the recent progress on 2D/3D heterostructure PSCs using surface passivation. The review begins with an introduction of the crystal structure, followed by the advantages of the combination of 2D and 3D perovskites. Then, the surface passivation strategies, optoelectronic properties, enhanced stability, and photovoltaic performance of 2D/3D PSCs are systematically discussed. Finally, the perspectives of passivation techniques using 2D perovskites to offer insight into further improved photovoltaic performance in the future are proposed.

Engineering Stress in Perovskite Solar Cells to Improve Stability

Nicholas Rolston; Kevin A. Bush; Adam D. Printz; Aryeh Gold-Parker; Yichuan Ding; Michael F. Toney; Michael D. McGehee; Reinhold H. Dauskardt. Advanced Energy Materials (2018). Cited by 510. Found by: openalex. openalex_2yr: 18.63 (2026). DOI: 10.1002/aenm.201802139

Abstract: Abstract An overlooked factor affecting stability: the residual stresses in perovskite films, which are tensile and can exceed 50 MPa in magnitude, a value high enough to deform copper, is reported. These stresses provide a significant driving force for fracture. Films are shown to be more unstable under tensile stress—and conversely more stable under compressive stress—when exposed to heat or humidity. Increasing the formation temperature of perovskite films directly correlates with larger residual stresses, a result of the high thermal expansion coefficient of perovskites. Specifically, this tensile stress forms upon cooling to room temperature, as the substrate constrains the perovskite from shrinking. No evidence of stress relaxation is observed, with the purely elastic film stress attributed to the thermal expansion mismatch between the perovskite and substrate. Additionally, the authors demonstrate that using a bath conversion method to form the perovskite film at room temperature leads to low stress values that are unaffected by further annealing, indicating complete perovskite formation prior to annealing. It is concluded that reducing the film stress is a novel method for improving

perovskite stability, which can be accomplished by lower formation temperatures, flexible substrates with high thermal expansion coefficients, and externally applied compressive stress after fabrication.

Interfacial Residual Stress Relaxation in Perovskite Solar Cells with Improved Stability

Hao Wang; Cheng Zhu; Lang Liu; Sai Ma; Pengfei Liu; Jiafeng Wu; Congbo Shi; Qin Du; Yanmin Hao; Sisi Xiang; Haining Chen; Pengwan Chen; Yang Bai; Huanping Zhou; Yujing Li; Qi Chen. Advanced Materials (2019). Cited by 496. Found by: openalex. openalex_2yr: 22.086 (2026). DOI: 10.1002/adma.201904408

Abstract: To improve the photovoltaic performance (both efficiency and stability) in hybrid organic-inorganic halide perovskite solar cells, perovskite lattice distortion is investigated with regards to residual stress (and strain) in the polycrystalline thin films. It is revealed that residual stress is concentrated at the surface of the as-prepared film, and an efficient method is further developed to release this interfacial stress by A site cation alloying. This results in lattice reconstruction at the surface of polycrystalline thin films, which in turn results in low elastic modulus. Thus, a "bone-joint" configuration is constructed within the interface between the absorber and the carrier transport layer, which improves device performance substantially. The resultant photovoltaic devices exhibit an efficiency of 21.48% with good humidity stability and improved resistance against thermal cycling.

Enhanced Efficiency and Stability of Inverted Perovskite Solar Cells Using Highly Crystalline SnO 2 Nanocrystals as the Robust Electron-Transporting Layer

Zonglong Zhu; Yang Bai; Xiao Liu; Chu-Chen Chueh; Shihe Yang; Alex K.-Y. Jen. Advanced Materials (2016). Cited by 496. Found by: openalex. openalex_2yr: 22.086 (2026). DOI: 10.1002/adma.201600619

Abstract: Highly crystalline SnO2 is demonstrated to serve as a stable and robust electron-transporting layer for high-performance perovskite solar cells. Benefiting from its high crystallinity, the relatively thick SnO2 electron-transporting layer (120 nm) provides a respectable electron-transporting property to yield a promising power conversion efficiency (PCE)(18.8%) Over 90% of the initial PCE can be retained after 30 d storage in ambient with 70% relative humidity.

Degradation mechanism of hybrid tin-based perovskite solar cells and the critical role of tin (IV) iodide

Luis Lanzetta; Thomas Webb; Nourdine Zibouche; Xinxing Liang; Dong Ding; Ganghong Min; Robert J. E. Westbrook; Benedetta Gaggio; Thomas J. Macdonald; M. Saïful Islam; Saif A. Haque. Nature Communications (2021). Cited by 493. Found by: openalex. DOI: 10.1038/s41467-021-22864-z

Abstract: Abstract Tin perovskites have emerged as promising alternatives to toxic lead perovskites in next-generation photovoltaics, but their poor environmental stability remains an obstacle towards more competitive performances. Therefore, a full understanding of their decomposition processes is needed to address these stability issues. Herein, we elucidate the degradation mechanism of 2D/3D tin perovskite films based on (PEA)0.2(FA)0.8SnI3(where PEA is phenylethylammonium and FA is formamidinium). We show that SnI4, a product of the oxygen-induced degradation of tin perovskite, quickly evolves into iodine via the combined action of moisture and oxygen. We identify iodine as a highly aggressive species that can further oxidise the perovskite to more SnI4, establishing a cyclic degradation mechanism. Perovskite stability is then observed to strongly depend on the hole transport layer chosen as the substrate, which is exploited to tackle film degradation. These key insights will enable the future design and optimisation of stable tin-based perovskite optoelectronics.

Precise Control of Crystal Growth for Highly Efficient CsPbI2Br Perovskite Solar Cells

Weijie Chen; Haiyang Chen; Guiying Xu; Rongming Xue; Shuhui Wang; Yaowen Li; Yongfang Li. Joule (2018). Cited by 491. Found by: openalex. DOI: 10.1016/j.joule.2018.10.011

Gas chromatography–mass spectrometry analyses of encapsulated stable perovskite solar cells

Lei Shi; Martin P. Bucknall; Trevor L. Young; Meng Zhang; Long Hu; Jueming Bing; Da Seul Lee; Jincheol Kim; Tom Wu; Noboru Takamure; David R. McKenzie; Shujuan Huang; Martin A. Green; Anita Ho-Baillie. Science (2020). Cited by 489. Found by: openalex. DOI: 10.1126/science.aba2412

Abstract: Although perovskite solar cells have produced remarkable energy conversion efficiencies, they cannot become commercially viable without improvements in durability. We used gas chromatography-mass spectrometry (GC-MS) to reveal signature volatile products of the decomposition of organic hybrid perovskites under thermal stress. In addition, we were able to use GC-MS to confirm that a low-cost polymer/glass stack encapsulation is effective in suppressing such outgassing. Using such an encapsulation scheme, we produced multi-cation, multi-halide perovskite solar cells containing methylammonium that exceed the requirements of the International Electrotechnical Commission 61215:2016 standard by surviving more than 1800 hours of the Damp Heat test and 75 cycles of the Humidity Freeze test.

Engineering ligand reactivity enables high-temperature operation of stable perovskite solar cells

So Min Park; Mingyang Wei; Jian Xu; Harindi R. Atapattu; Felix T. Eickemeyer; Kasra Darabi; Luke Grater; Yi Yang; Cheng Liu; Sam Teale; Bin Chen; Hao Chen; Tonghui Wang; Lewei Zeng; Aidan Maxwell; Zaiwei Wang; K.R.P.M. Rao; Zhuoyun Cai; Shaik M. Zakeeruddin; Jonathan T. Pham; Chad Risko; Aram Amassian; Mercouri G. Kanatzidis; Kenneth R. Graham; Michaël Grätzel; Edward H. Sargent. Science (2023). Cited by 482. Found by: openalex. DOI: 10.1126/science.adi4107

Abstract: Perovskite solar cells (PSCs) consisting of interfacial two- and three-dimensional heterostructures that incorporate ammonium ligand intercalation have enabled rapid progress toward the goal of uniting performance with stability. However, as the field continues to seek ever-higher durability, additional tools that avoid progressive ligand intercalation are needed to minimize degradation at high temperatures. We used ammonium ligands that are nonreactive with the bulk of perovskites and investigated a library that varies ligand molecular structure systematically. We found that fluorinated aniliniums offer interfacial passivation and simultaneously minimize reactivity with perovskites. Using this approach, we report a certified quasi–steady-state power-conversion efficiency of 24.09% for inverted-structure PSCs. In an encapsulated device operating at 85°C and 50% relative humidity, we document a 1560-hour T 85 at maximum power point under 1-sun illumination.

Advances in hole transport materials engineering for stable and efficient perovskite solar cells

Zinab H. Bakr; Qamar Wali; Azhar Fakharuddin; Lukas Schmidt-Mende; Thomas M. Brown; Rajan Jose. Nano Energy (2017). Cited by 448. Found by: openalex. DOI: 10.1016/j.nanoen.2017.02.025

Flexible Perovskite Solar Cells

Hyun Suk Jung; Gill Sang Han; Nam-Gyu Park; Min Jae Ko. Joule (2019). Cited by 447. Found by: openalex. DOI: 10.1016/j.joule.2019.07.023

Perovskite Solar Cell Stability in Humid Air: Partially Reversible Phase Transitions in the PbI 2 -CH 3 NH 3 I-H 2 O System

Zhaoning Song; Antonio Abate; Suneth C. Watthage; Geethika K. Liyanage; Adam B. Phillips; Ullrich Steiner; Michaël Grätzel; Michael J. Heben. Advanced Energy Materials (2016). Cited by 435. Found by: openalex. openalex_2yr: 18.63 (2026). DOI: 10.1002/aenm.201600846

Abstract: After rapid progress over the past five years, organic–inorganic perovskite solar cells (PSCs) currently exhibit photoconversion efficiencies comparable to the best commercially available photovoltaic technologies. However, instabilities in the materials and devices, primarily due to reactions with water, have kept PSCs from entering the marketplace. Here, laser beam induced current imaging is used to investigate the spatial and temporal evolution of the quantum efficiency of perovskite solar cells under controlled humidity conditions. Several interesting mechanistic aspects are revealed as the degradation proceeds along a four-stage process. Three of the four stages can be reversed, while the fourth stage leads to irreversible decomposition of the photoactive perovskite material. A series of reactions in the PbI 2 -CH 3 NH 3 I-H 2 O system explains the interplay between the interactions with water and the overall stability. Understanding of the degradation mechanisms of PSCs on a microscopic level gives insight into improving the long-term stability.

Is Excess PbI2 Beneficial for Perovskite Solar Cell Performance?

Fangzhou Liu; Qi Dong; Man Kwong Wong; Aleksandra B. Djurišić; Annie Ng; Zhiwei Ren; Qian Shen; C. Surya; Wai Kin Chan; Jian Wang; Alan Man Ching Ng; Changzhong Liao; Hangkong Li; Kaimin Shih; Chengrong Wei; Huimin Su; Junfeng Dai. Advanced Energy Materials (2016). Cited by 422. Found by: openalex. openalex_2yr: 18.63 (2026). DOI: 10.1002/aenm.201502206

Abstract: Unreacted lead iodide is commonly believed to be beneficial to the efficiency of methylammonium lead iodide perovskite based solar cells, since it has been proposed to passivate the defects in perovskite grain boundaries. However, it is shown here that the presence of unreacted PbI2 results in an intrinsic instability of the film under illumination, leading to the film degradation under inert atmosphere and faster degradation upon exposure to illumination and humidity. The perovskite films without lead iodide have improved stability, but lower efficiency due to inferior film morphology (smaller grain size, the presence of pinholes). Optimization of the deposition process resulted in PbI2-free perovskite films giving comparable efficiency to those with excess PbI2 (14.2 ± 1.3% compared to 15.1 ± 0.9%) Thus, optimization of the deposition process for PbI2-free films leads to dense, pinhole-free, large grain size perovskite films which result in cells with high efficiency without detrimental effects on the film photostability caused by excess PbI2. However, it should be noted that for encapsulated devices illuminated through the substrate (fluorine-doped tin oxide glass, TiO2 film), film photostability is not a key factor in the device degradation.

Device stability of perovskite solar cells – A review

Muhammad Imran Asghar; Jun Zhang; H. Wang; Peter D. Lund. Renewable and Sustainable Energy Reviews (2017). Cited by 418. Found by: openalex. DOI: 10.1016/j.rser.2017.04.003

Multifunctional Enhancement for Highly Stable and Efficient Perovskite Solar Cells

Yuan Cai; Jian Cui; Ming Chen; Miaomiao Zhang; Yu Han; Qian Fang; Huan Zhao; Shaomin Yang; Zhou Yang; Hongtao Bian; Tao Wang; Kunpeng Guo; Molang Cai; Songyuan Dai; Zhike Liu; Shengzhong Liu. Advanced Functional Materials (2020). Cited by 418. Found by: openalex. openalex_2yr: 14.975 (2026). DOI: 10.1002/adfm.202005776

Abstract: Abstract With a certified efficiency as high as 25.2%, perovskite has taken the crown as the highest efficiency thin film solar cell material. Unfortunately, serious instability issues must be resolved before perovskite solar cells (PSCs) are commercialized. Aided by theoretical calculation, an appropriate multifunctional molecule, 2,2-difluoropropanediamide (DFPDA), is selected to ameliorate all the instability issues. Specifically, the carbonyl groups in DFPDA form chemical bonds with Pb 2+ and passivate under-coordinated Pb 2+ defects. Consequently, the perovskite crystallization rate is reduced and high-quality films are produced with fewer defects. The amino groups not only bind with iodide to suppress ion migration but also increase the electron density on the carbonyl groups to further enhance their passivation effect. Furthermore, the fluorine groups in DFPDA form both an effective barrier on the perovskite to improve its moisture stability and a bridge between the perovskite and HTL for effective charge transport. In addition, they show an effective doping effect in the HTL to improve its carrier mobility. With the help of the combined effects of these groups in DFPDA, the PSCs with DFPDA additive achieve a champion efficiency of 22.21% and a substantially improved stability against moisture, heat, and light.

Enhancing stability and efficiency of perovskite solar cells with crosslinkable silane-functionalized and doped fullerene

Yang Bai; Qingfeng Dong; Yuchuan Shao; Yehao Deng; Qi Wang; Liang Shen; Dong Wang; Wei Wei; Jinsong Huang. Nature Communications (2016). Cited by 416. Found by: openalex. DOI: 10.1038/ncomms12806

Abstract: The instability of hybrid perovskite materials due to water and moisture arises as one major challenge to be addressed before any practical application of the demonstrated high efficiency perovskite solar cells. Here we report a facile strategy that can simultaneously enhance the stability and efficiency of p-i-n planar heterojunction-structure perovskite devices. Crosslinkable silane molecules with hydrophobic functional groups are bonded onto fullerene to make the fullerene layer highly water-resistant. Methylammonium iodide is introduced in the fullerene layer for n-doping via anion-induced electron transfer, resulting in dramatically increased conductivity over 100-fold. With crosslinkable silane-functionalized and doped fullerene electron transport layer, the perovskite devices deliver an efficiency of 19.5% with a high fill factor of 80.6%. A crosslinked silane-modified fullerene layer also enhances the water and moisture stability of the non-sealed perovskite devices by retaining nearly 90% of their original efficiencies after 30 days' exposure in an ambient environment.

Tailored Amphiphilic Molecular Mitigators for Stable Perovskite Solar Cells with 23.5% Efficiency

Hongwei Zhu; Yuhang Liu; Felix T. Eickemeyer; Linfeng Pan; Dan Ren; Marco A. Ruiz-Preciado; Brian Carlsen; Bowen Yang; Xiaofei Dong; Zaiwei Wang; Hongli Liu; Shirong Wang; Shaik M. Zakeeruddin; Anders Hagfeldt;

M. Ibrahim Dar; Xianggao Li; Michaël Grätzel. Advanced Materials (2020). Cited by 416. Found by: openalex. openalex_2yr: 22.086 (2026). DOI: 10.1002/adma.201907757

Abstract: Passivation of interfacial defects serves as an effective means to realize highly efficient and stable perovskite solar cells (PSCs). However, most molecular modulators currently used to mitigate such defects form poorly conductive aggregates at the perovskite interface with the charge collection layer, impeding the extraction of photogenerated charge carriers. Here, a judiciously engineered passivator, 4-tert-butyl-benzylammonium iodide (tBBAI), is introduced, whose bulky tert-butyl groups prevent the unwanted aggregation by steric repulsion. It is found that simple surface treatment with tBBAI significantly accelerates the charge extraction from the perovskite into the spiro-OMeTAD hole-transporter, while retarding the nonradiative charge carrier recombination. This boosts the power conversion efficiency (PCE) of the PSC from 20% to 23.5% reducing the hysteresis to barely detectable levels. Importantly, the tBBAI treatment raises the fill factor from 0.75 to the very high value of 0.82, which concurs with a decrease in the ideality factor from 1.72 to 1.34, confirming the suppression of radiation-less carrier recombination. The tert-butyl group also provides a hydrophobic umbrella protecting the perovskite film from attack by ambient moisture. As a result, the PSCs show excellent operational stability retaining over 95% of their initial PCE after 500 h full-sun illumination under maximum-power-point tracking under continuous simulated solar irradiation.

Unveiling facet-dependent degradation and facet engineering for stable perovskite solar cells

Chunqing Ma; Felix T. Eickemeyer; Sun-Ho Lee; Dong-Ho Kang; Seok Joon Kwon; Michaël Grätzel; Nam-Gyu Park. Science (2023). Cited by 415. Found by: openalex. DOI: 10.1126/science.adf3349

Abstract: A myriad of studies and strategies have already been devoted to improving the stability of perovskite films; however, the role of the different perovskite crystal facets in stability is still unknown. Here, we reveal the underlying mechanisms of facet-dependent degradation of formamidinium lead iodide (FAPbI 3) films. We show that the (100) facet is substantially more vulnerable to moisture-induced degradation than the (111) facet. With combined experimental and theoretical studies, the degradation mechanisms are revealed; a strong water adhesion following an elongated lead-iodine (Pb-I) bond distance is observed, which leads to a -phase transition on the (100) facet. Through engineering, a higher surface fraction of the (111) facet can be achieved, and the (111)-dominated crystalline FAPbI 3 films show exceptional stability against moisture. Our findings elucidate unknown facet-dependent degradation mechanisms and kinetics.

2D/3D heterojunction engineering at the buried interface towards high-performance inverted methylammonium-free perovskite solar cells

Jing Li; Cong Zhang; Cheng Gong; Daliang Zhang; Hong Zhang; Qixin Zhuang; Xuemeng Yu; Shaokuan Gong; Xihan Chen; Jiabao Yang; Xuanhua Li; Ru Li; Jingwei Li; Jinfei Zhou; Hua Gui Yang; Qianqian Lin; Junhao Chu; Michaël Grätzel; Jiangzhao Chen; Zhigang Zang. Nature Energy (2023). Cited by 395. Found by: openalex. DOI: 10.1038/s41560-023-01295-8

Orientation Regulation of Phenylethylammonium Cation Based 2D Perovskite Solar Cell with Efficiency Higher Than 11%

Xinqian Zhang; Gang Wu; Weifei Fu; Minchao Qin; Weitao Yang; Jielin Yan; Zhongqiang Zhang; Xinhui Lu; Hongzheng Chen. Advanced Energy Materials (2018). Cited by 389. Found by: openalex. openalex_2yr: 18.63 (2026). DOI: 10.1002/aenm.201702498

Abstract: Abstract Increasing the power conversion efficiency (PCE) of the two-dimensional (2D) perovskite-based solar cells (PVSCs) is really a challenge. Vertical orientation of the 2D perovskite film is an efficient strategy to elevate the PCE. In this work, vertically orientated highly crystalline 2D (PEA)2(MA)n–1PbnI3n+1 (PEA= phenylethylammonium, MA = methylammonium, n = 3, 4, 5) films are fabricated with the assistance of an ammonium thiocyanate (NH4SCN) additive by a one-step spin-coating method. Planar-structured PVSCs with the device structure of indium tin oxide (ITO)/poly(3,4-ethylenedioxythiophene):poly(styrenesulfonate)/(PEA)2(MA)n–1PbnI3n+1/[6,6]-phenyl-C61-butyric acid methyl ester/bahocuproine/Ag are fabricated. The PCE of the PVSCs is boosted from the original 0.56% (without NH4SCN) to 11.01% with the optimized NH4SCN addition at n = 5, which is among the highest PCE values for the low-n (n < 10) 2D perovskite-based PVSCs. The improved performance is attributed to the vertically orientated highly crystalline 2D perovskite thin films as well as the balanced electron/hole transportation. The humidity stability of this oriented 2D perovskite thin film is also confirmed by the almost unchanged X-ray diffraction patterns after 28 d exposed to the moisture in a humidity-controlled cabinet (Hr = 55 ± 5%). The unsealed device retains 78.5% of its original PCE after

160 h storage in air atmosphere with humidity of 55 ± 5%. The results provide an effective approach toward a highly efficient and stable PVSC for future commercialization.

Isomer-Pure Bis-PCBM-Assisted Crystal Engineering of Perovskite Solar Cells Showing Excellent Efficiency and Stability

Fei Zhang; Wenda Shi; Jingshan Luo; Norman Pellet; Chenyi Yi; Xiong Li; Xiaoming Zhao; T. John S. Dennis; Xianggao Li; Shirong Wang; Yin Xiao; Shaik M. Zakeeruddin; Dongqin Bi; Michaël Grätzel. Advanced Materials (2017). Cited by 388. Found by: openalex. openalex_2yr: 22.086 (2026). DOI: 10.1002/adma.201606806

Abstract: -butyric acid methyl ester (bis-[60]PCBM) isomer mixture by preparative peak-recycling, high-performance liquid chromatography, and is employed as a templating agent for solution processing of metal halide perovskite films via an antisolvent method. The resulting -bis-PCBM-containing perovskite solar cells achieve better stability, efficiency, and reproducibility when compared with analogous cells containing PCBM. -bis-PCBM fills the vacancies and grain boundaries of the perovskite film, enhancing the crystallization of perovskites and addressing the issue of slow electron extraction. In addition, -bis-PCBM resists the ingression of moisture and passivates voids or pinholes generated in the hole-transporting layer. As a result, a power conversion efficiency (PCE) of 20.8% is obtained, compared with 19.9% by PCBM, and is accompanied by excellent stability under heat and simulated sunlight. The PCE of unsealed devices dropped by less than 10% in ambient air (40% RH) after 44 d at 65 °C, and by 4% after 600 h under continuous full-sun illumination and maximum power point tracking, respectively.

UV Degradation and Recovery of Perovskite Solar Cells

Sang-Won Lee; Seongtak Kim; Soohyun Bae; Kyungjin Cho; Taewon Chung; Laura E. Mundt; Seunghun Lee; Sungeun Park; Hyomin Park; Martin C. Schubert; Stefan W. Glunz; Yohan Ko; Yongseok Jun; Yoonmook Kang; Hae-Seok Lee; Dong-Hwan Kim. Scientific Reports (2016). Cited by 383. Found by: openalex. DOI: 10.1038/srep38150

Abstract: Abstract Although the power conversion efficiency of perovskite solar cells has increased from 3.81% to 22.1% in just 7 years, they still suffer from stability issues, as they degrade upon exposure to moisture, UV light, heat, and bias voltage. We herein examined the degradation of perovskite solar cells in the presence of UV light alone. The cells were exposed to 365 nm UV light for over 1,000 h under inert gas at <0.5 ppm humidity without encapsulation. 1-sun illumination after UV degradation resulted in recovery of the fill factor and power conversion efficiency. Furthermore, during exposure to consecutive UV light, the diminished short circuit current density (J sc) and EQE continuously restored. 1-sun light soaking induced recovery is considered to be caused by resolving of stacked charges and defect state neutralization. The J sc and EQE bounce-back phenomenon is attributed to the beneficial effects of PbI 2 which is generated by the decomposition of perovskite material.

Polymer Doping for High-Efficiency Perovskite Solar Cells with Improved Moisture Stability

Jiexuan Jiang; Qian Wang; Zhiwen Jin; Xisheng Zhang; Jie Lei; Haijun Bin; Zhiguo Zhang; Yongfang Li; Shengzhong Liu. Advanced Energy Materials (2017). Cited by 378. Found by: openalex. openalex_2yr: 18.63 (2026). DOI: 10.1002/aenm.201701757

Abstract: Abstract Each component layer in a perovskite solar cell plays an important role in the cell performance. Here, a few types of polymers including representative p-type and n-type semiconductors, and a classical insulator, are chosen to dope into a perovskite film. The long-chain polymer helps to form a network among the perovskite crystalline grains, as witnessed by the improved film morphology and device stability. The dewetting process is greatly suppressed by the cross-linking effect of the polymer chains, thereby resulting in uniform perovskite films with large grain sizes. Moreover, it is found that the polymer-doped perovskite shows a reduced trap-state density, likely due to the polymer effectively passivating the perovskite grain surface. Meanwhile the doped polymer formed a bridge between grains for efficient charge transport. Using this approach, the solar cell efficiency is improved from 17.43% to as high as 19.19%, with a much improved stability. As it is not required for the polymer to have a strict energy level matching with the perovskite, in principle, one may use a variety of polymers for this type of device design.

A chemically inert bismuth interlayer enhances long-term stability of inverted perovskite solar cells

Shaohang Wu; Rui Chen; Shasha Zhang; B. Hari Babu; Youfeng Yue; Hongmei Zhu; Zhichun Yang; Chuanliang Chen; Weitao Chen; Weitao Chen; Yuqian Huang; Shaoying Fang; ; Liyuan Han; Wei Chen; Wei Chen. Nature Communications (2019). Cited by 378. Found by: openalex. DOI: 10.1038/s41467-019-09167-0

Abstract: Long-term stability remains a key issue impeding the commercialization of halide perovskite solar cells (HPVKSCs). The diffusion of molecules and ions causes irreversible degradation to photovoltaic device performance. Here, we demonstrate a facile strategy for producing highly stable HPVKSCs by using a thin but compact semimetal Bismuth interlayer. The Bismuth film acts as a robust permeation barrier that both insulates the perovskite from intrusion by undesirable external moisture and protects the metal electrode from iodine corrosion. The Bismuth-interlayer-based devices exhibit greatly improved stability when subjected to humidity, thermal and light stresses. The unencapsulated device retains 88% of its initial efficiency in ambient air in the dark for over 6000 h; the devices maintain 95% and 97% of their initial efficiencies after 85 °C thermal aging and light soaking in nitrogen atmosphere for 500 h, respectively. These sound stability parameters are among the best for planar structured HPVKSCs reported to date.

Humidity-Induced Degradation via Grain Boundaries of HC(NH2)2PbI3 Planar Perovskite Solar Cells

Jae Sung Yun; Jincheol Kim; Trevor L. Young; Robert Patterson; Dohyung Kim; Jan Seidel; Sean Lim; Martin A. Green; Shujuan Huang; Anita Ho-Baillie. Advanced Functional Materials (2018). Cited by 373. Found by: openalex. openalex_2yr: 14.975 (2026). DOI: 10.1002/adfm.201705363

Abstract: Abstract The sensitivity of organic–inorganic perovskites to environmental factors remains a major barrier for these materials to become commercially viable for photovoltaic applications. In this work, the degradation of formamidinium lead iodide (FAPbI3) perovskite in a moist environment is systematically investigated. It is shown that the level of relative humidity (RH) is important for the onset of degradation processes. Below 30% RH, the black phase of the FAPbI3 perovskite shows excellent phase stability over 90 d. Once the RH reaches 50%, degradation of the FAPbI3 perovskite occurs rapidly. Results from a Kelvin probe force microscopy study reveal that the formation of nonperovskite phases initiates at the grain boundaries and the phase transition proceeds toward the grain interiors. Also, ion migration along the grain boundaries is greatly enhanced upon degradation. A post-thermal treatment (PTT) that removes chemical residues at the grain boundaries which effectively slows the degradation process is developed. Finally, it is demonstrated that the PTT process improves the performance and stability of the final device.

Enhancing Stability of Perovskite Solar Cells to Moisture by the Facile Hydrophobic Passivation

Insung Hwang; Inyoung Jeong; Jinwoo Lee; Min Jae Ko; Kijung Yong. ACS Applied Materials & Interfaces (2015). Cited by 353. Found by: openalex. openalex_2yr: 8.382 (2026). DOI: 10.1021/acsami.5b04490

Abstract: In this study, a novel and facile passivation process for a perovskite solar cell is reported. Poor stability in ambient atmosphere, which is the most critical demerit of a perovskite solar cell, is overcome by a simple passivation process using a hydrophobic polymer layer. Teflon, the hydrophobic polymer, is deposited on the top of a perovskite solar cell by a spin-coating method. With the hydrophobic passivation, the perovskite solar cell shows negligible degradation after a 30 day storage in ambient atmosphere. Suppressed degradation of the perovskite film is proved in various ways: X-ray diffraction, light absorption spectrum, and quartz crystal microbalance. This simple but effective passivation process suggests new kind of approach to enhance stability of perovskite solar cells to moisture.

Optimal Interfacial Engineering with Different Length of Alkylammonium Halide for Efficient and Stable Perovskite Solar Cells

Hyeonwoo Kim; Seungun Lee; Do Yoon Lee; Min Jae Paik; Hyejin Na; Jaemin Lee; Sang Il Seok. Advanced Energy Materials (2019). Cited by 353. Found by: openalex. openalex_2yr: 18.63 (2026). DOI: 10.1002/aenm.201902740

Abstract: Abstract Recently, two-dimensional (2D) structure on three-dimensional (3D) perovskites (graded 2D/3D) has been reported to be effective in significantly improving both efficiency and stability. However, the electrical properties of the 2D structure as a passivation layer on the 3D perovskite thin film and resistance to the penetration of moisture may vary depending on the length of the alkyl chain. In addition, the surface defects

of the 2D itself on the 3D layer may also be affected by the correlation between the 2D structure and the hole conductive material. Therefore, systematic interfacial study with the alkyl chain length of long-chained alkylammonium iodide forming a 2D structure is necessary. Herein, the 2D interfacial layers formed are compared with butylammonium iodide (BAI), octylammonium iodide (OAI), and dodecylammonium iodide (DAI) iodide on a 3D (FAPbI 3) 0.95 (MAPbBr 3) 0.05 perovskite thin film in terms of the PCE and humidity stability. As the length of the alkyl chain increased from BA to OA to DA, the electron-blocking ability and humidity resistance increase significantly, but the difference between OA and DA is not large. The PSC post-treated with OAI has slightly higher PCE than those treated with BAI and DAI, achieving a certified stabilized efficiency of 22.9%.

Passivation of Grain Boundaries by Phenethylammonium in Formamidinium-Methylammonium Lead Halide Perovskite Solar Cells

Da Seul Lee; Jae Sung Yun; Jincheol Kim; Arman Mahboubi Soufiani; Sheng Chen; Yongyoon Cho; Xiaofan Deng; Jan Seidel; Sean Lim; Shujuan Huang; Anita Ho-Baillie. ACS Energy Letters (2018). Cited by 349. Found by: openalex. openalex_2yr: 16.712 (2026). DOI: 10.1021/acsenergylett.8b00121

Abstract: In this work, we report the benefits of incorporating phenethylammonium cation (PEA +) into (HC(NH 2) 2 PbI 3) 0.85 (CH 3 NH 3 PbBr 3) 0.15 perovskite for the first time. After adding small amounts of PEA cation (<10%), the perovskite film morphology is changed but, most importantly, grain boundaries are passivated. This is supported by Kelvin Probe Force Microscopy (KPFM). The passivation results in the increase in photoluminescence intensity and carrier lifetimes of test structures and open-circuit voltages (V OC) of the devices as long as the addition of PEA + is 4.5%. The presence of higher-band-gap quasi-2D PEA incorporated perovskite is responsible for the grain boundary passivation, and the quasi-2D perovskites are also found to be concentrated near the TiO 2 layer, revealed by PL spectroscopy. Results of moisture exposure tests show that PEA + incorporation is effective in slowing down the degradation of unencapsulated devices compared to the control devices without PEA + . These findings provide insights into the operation of perovskite solar cells when large cations are incorporated.

Material and Device Stability in Perovskite Solar Cells

Hui-Seon Kim; Ja-Young Seo; Nam-Gyu Park. ChemSusChem (2016). Cited by 347. Found by: openalex. openalex_2yr: 5.476 (2026). DOI: 10.1002/cssc.201600915

Abstract: Organic-inorganic halide perovskite solar cells have attracted great attention because of their superb efficiency reaching 22 % and low-cost, facile fabrication processing. Nevertheless, stability issues in perovskite solar cells seem to block further advancements toward commercialization. Thus, device stability is one of the important topics in perovskite solar cell research. In the beginning, the poor moisture resistivity of the perovskite layer was considered as a main problem that hindered further development of perovskite solar cells, which encouraged engineering of the perovskite or protection of the perovskite by a buffer layer. Soon after, other parameters affecting long-term stability were sequentially found and various attempts have been made to enhance intrinsic and extrinsic stability. Here we review the recent progresses addressing stability issues in perovskite solar cells. In this report, we investigated factors affecting stability from material and device points of view. To gain a better understanding of the stability of the bulk perovskite material, decomposition mechanisms were investigated in relation to moisture, photons, and heat. Stability of full device should also be carefully examined because its stability is dependent not only on bulk perovskite but also on the interfaces and selective contacts. In addition, ion migration and current-voltage hysteresis were found to be closely related to stability.

Mixed Dimensional 2D/3D Hybrid Perovskite Absorbers: The Future of Perovskite Solar Cells?

Anurag Krishna; Sébastien Gottis; Mohammad Khaja Nazeeruddin; Frédéric Sauvage. Advanced Functional Materials (2018). Cited by 347. Found by: openalex. openalex_2yr: 14.975 (2026). DOI: 10.1002/adfm.201806482

Abstract: Abstract The cost-effective processability and high efficiency of the organic–inorganic metal halide perovskite solar cells (PSCs) have shown tremendous potential to intervene positively in the generation of clean energy. However, prior to an industrial scale-up process, there are certain critical issues such as the lack of stability against over moisture, light, and heat, which have to be resolved. One of the several proposed strategies to improve the stability that has lately emerged is the development of lower-dimensional (2D) perovskite structures derived from the Ruddlesden–Popper (RP) phases. The excellent stability under ambient conditions shown by 2D RP phase perovskites has made the scalability expectations burgeon since it is one of the most credible paths toward stable PSCs. In this review, the 2D/3D mixed system for photovoltaics (PVs) is elaborately discussed with the focus on the crystal structure, optoelectronic properties, charge carrier dynamics, and their impact on the photovoltaic performances. Finally, some of the further challenges are highlighted while outlining the perspectives of 2D/3D perovskites for high-efficiency stable solar cells.

Mixed Cation FAxPEA1–xPbI3 with Enhanced Phase and Ambient Stability toward High-Performance Perovskite Solar Cells

Nan Li; Zonglong Zhu; Chu-Chen Chueh; Hongbin Liu; Bo Peng; Alessio Petrone; Xiaosong Li; Liduo Wang; Alex K.-Y. Jen. Advanced Energy Materials (2016). Cited by 345. Found by: openalex. openalex_2yr: 18.63 (2026). DOI: 10.1002/aenm.201601307

Abstract: In this work, different from the commonly explored strategy of incorporating a smaller cation, MA+ and Cs+ into FAPbI3 lattice to improve efficiency and stability, it is revealed that the introduction of phenylethylammonium iodide (PEAI) into FAPbI3 perovksite to form mixed cation FAxPEA1–xPbI3 can effectively enhance both phase and ambient stability of FAPbI3 as well as the resulting performance of the derived devices. From our experimental and theoretical calculation results, it is proposed that the larger PEA cation is capable of assembling on both the lattice surface and grain boundaries to form quais-3D perovskite structures. The surrounding of PEA+ ions at the crystal grain boundaries not only can serve as molecular locks to tighten FAPbI3 domains but also passivate the surface defects to improve both phase and moisture stablity. Consequently, a high-performance (PCE:17.7%) and ambient stable FAPbI3 solar cell could be developed.

Improvement of the humidity stability of organic–inorganic perovskite solar cells using ultrathin Al 2 O 3 layers prepared by atomic layer deposition

Xu Dong; Xiang Fang; Minghang Lv; Bencai Lin; Shuai Zhang; Jianning Ding; Ningyi Yuan. Journal of Materials Chemistry A (2015). Cited by 343. Found by: openalex. DOI: 10.1039/c4ta06128d

Abstract: The high polarity of water molecules inevitably causes the decomposition of perovskites. We retard the degradation by introducing an ultrathin ALD–Al 2 O 3 layer, which has almost no negative effect on performance.

Encapsulation for long-term stability enhancement of perovskite solar cells

Fabio Matteocci; Lucio Cinà; Enrico Lamanna; Stéfania Cacovich; Giorgio Divitini; Paul A. Midgley; Caterina Ducati; Aldo Di Carlo. Nano Energy (2016). Cited by 337. Found by: openalex. DOI: 10.1016/j.nanoen.2016.09.041

Phase Pure 2D Perovskite for High-Performance 2D–3D Heterostructured Perovskite Solar Cells

Pengwei Li; Yiqiang Zhang; Chao Liang; Guichuan Xing; Xiaolong Liu; Fengyu Li; Xiaotao Liu; Xiaotian Hu; Guosheng Shao; Yanlin Song. Advanced Materials (2018). Cited by 336. Found by: openalex. openalex_2yr: 22.086 (2026). DOI: 10.1002/adma.201805323

Abstract: Abstract Three-dimensional (3D) metal-halide perovskite solar cells (PSCs) have demonstrated exceptional high efficiency. However, instability of the 3D perovskite is the main challenge for industrialization. Incorporation of some long organic cations into perovskite crystal to terminate the lattice, and function as moisture and oxygen passivation layer and ion migration blocking layer, is proven to be an effective method to enhance the perovskite stability. Unfortunately, this method typically sacrifices charge-carrier extraction efficiency of the perovskites. Even in 2D–3D vertically aligned heterostructures, a spread of bandgaps in the 2D due to varying degrees of quantum confinement also results in charge-carrier localization and carrier mobility reduction. A trade-off between the power conversion efficiency and stability is made. Here, by introducing 2D C 6 H 18 N 2 O 2 PbI 4 (EDBEPbI 4) microcrystals into the precursor solution, the grain boundaries of the deposited 3D perovskite film are vertically passivated with phase pure 2D perovskite. The phases pure (inorganic layer number n = 1) 2D perovskite can minimize photogenerated charge-carrier localization in the low-dimensional perovskite. The dominant vertical alignment does not affect charge-carrier extraction. Therefore, high-efficiency (21.06%) and ultrastable (retain 90% of the initial efficiency after 3000 h in air) planar PSCs are demonstrated with these 2D–3D mixtures.

Boosting the efficiency of carbon-based planar CsPbBr3 perovskite solar cells by a modified multistep spin-coating technique and interface engineering

Xingyue Liu; Xianhua Tan; Zhiyong Liu; Haibo Ye; Bo Sun; Tielin Shi; Zirong Tang; Guanglan Liao. Nano Energy (2018). Cited by 336. Found by: openalex. DOI: 10.1016/j.nanoen.2018.11.053

High-Efficiency Perovskite Solar Cells with Imidazolium-Based Ionic Liquid for Surface Passivation and Charge Transport

Xuejie Zhu; Minyong Du; Jiangshan Feng; Hui Wang; Zhuo Xu; Likun Wang; Shengnan Zuo; Chenyu Wang; Ziyu Wang; Cong Zhang; Xiaodong Ren; Shashank Priya; Dong Yang; Shengzhong Liu. Angewandte Chemie International Edition (2020). Cited by 335. Found by: openalex. openalex_2yr: 15.006 (2026). DOI: 10.1002/anie.202010987

Abstract: Abstract Surface defects have been a key constraint for perovskite photovoltaics. Herein, 1,3-dimethyl-3-imidazolium hexafluorophosphate (DMIMPF6) ionic liquid (IL) is adopted to passivate the surface of a formamidinium-cesium lead iodide perovskite (Cs0.08FA0.92PbI3) and also reduce the energy barrier between the perovskite and hole transport layer. Theoretical simulations and experimental results demonstrate that Pb-cluster and Pb-I antisite defects can be effectively passivated by [DMIM]+ bonding with the Pb2+ ion on the perovskite surface, leading to significantly suppressed non-radiative recombination. As a result, the solar cell efficiency was increased to 23.25 % from 21.09 %. Meanwhile, the DMIMPF6-treated perovskite device demonstrated long-term stability because the hydrophobic DMIMPF6 layer blocked moisture permeation.

Mixed 3D–2D Passivation Treatment for Mixed-Cation Lead Mixed-Halide Perovskite Solar Cells for Higher Efficiency and Better Stability

Yongyoon Cho; Arman Mahboubi Soufiani; Jae Sung Yun; Jincheol Kim; Da Seul Lee; Jan Seidel; Xiaofan Deng; Martin A. Green; Shujuan Huang; Anita Ho-Baillie. Advanced Energy Materials (2018). Cited by 334. Found by: openalex. openalex_2yr: 18.63 (2026). DOI: 10.1002/aenm.201703392

Abstract: Abstract Layered low-dimensional perovskite structures employing bulky organic ammonium cations have shown significant improvement on stability but poorer performance generally compared to their 3D counterparts. Here, a mixed passivation (MP) treatment is reported that uses a mixture of bulky organic ammonium iodide (iso-butylammonium iodide, iBAI) and formammidinium iodide (FAI), enhancing both power conversion efficiency and stability. Through a combination of inactivation of the interfacial trap sites, characterized by photoluminescence measurement, and formation of an interfacial energetic barrier by which ionic transport is reduced, demonstrated by Kelvin probe force microscopy, MP treatment of the perovskite/hole transport layer interface significantly suppresses photocurrent hysteresis. Using this MP treatment, the champion mixed-halide perovskite cell achieves a reverse scan and stabilized power conversion efficiency of 21.7%. Without encapsulation, the devices show excellent moisture stability, sustaining over 87% of the original performance after 38 d storage in ambient environment under 75 ± 20% relative humidity. This work shows that FAI/iBAI, is a new and promising material combination for passivating perovskite/selective-contact interfaces.

Role of 4- tert -Butylpyridine as a Hole Transport Layer Morphological Controller in Perovskite Solar Cells

Shen Wang; Mahsa Sina; Pritesh Parikh; Taylor Uekert; Brian Shahbazian; Arun Devaraj; Ying Shirley Meng. Nano Letters (2016). Cited by 333. Found by: openalex. DOI: 10.1021/acs.nanolett.6b02158

Abstract: Hybrid organic-inorganic materials for high-efficiency, low-cost photovoltaic devices have seen rapid progress since the introduction of lead based perovskites and solid-state hole transport layers. Although majority of the materials used for perovskite solar cells (PSC) are introduced from dye-sensitized solar cells (DSSCs), the presence of a perovskite capping layer as opposed to a single dye molecule (in DSSCs) changes the interactions between the various layers in perovskite solar cells. 4-tert-Butylpyridine (tBP), commonly used in PSCs, is assumed to function as a charge recombination inhibitor, similar to DSSCs. However, the presence of a perovskite capping layer calls for a re-evaluation of its function in PSCs. Using TEM (transmission electron microscopy), we first confirm the role of tBP as a HTL morphology controller in PSCs. Our observations suggest that tBP significantly improves the uniformity of the HTL and avoids accumulation of Li salt. We also study degradation pathways by using FTIR (Fourier transform infrared spectroscopy) and APT (atom probe tomography) to investigate and visualize in 3-dimensions the moisture content associated with the Li salt. Long-term effects, over 1000 h, due to evaporation of tBP have also been studied. Based on our findings, a PSC failure mechanism associated with the morphological change of the HTL is proposed. tBP, the morphology

controller in HTL, plays a key role in this process, and thus this study highlights the need for additive materials with higher boiling points for consistent long-term performance of PSCs.

Room-Temperature Molten Salt for Facile Fabrication of Efficient and Stable Perovskite Solar Cells in Ambient Air

Lingfeng Chao; Yingdong Xia; Bixin Li; Guichuan Xing; Yonghua Chen; Wei Huang. Chem (2019). Cited by 324. Found by: openalex. openalex_2yr: 11.7 (2026). DOI: 10.1016/j.chempr.2019.02.025

Design of low bandgap tin–lead halide perovskite solar cells to achieve thermal, atmospheric and operational stability

Rohit Prasanna; Tomas Leijtens; Sean P. Dunfield; James A. Raiford; Eli J. Wolf; Simon A. Swifter; Jérémie Werner; Giles E. Eperon; Camila de Paula; Axel F. Palmstrom; Caleb C. Boyd; Maikel F. A. M. van Hest; Stacey F. Bent; Glenn Teeter; Joseph J. Berry; Michael D. McGehee. Nature Energy (2019). Cited by 323. Found by: openalex. DOI: 10.1038/s41560-019-0471-6

Fundamental Understanding of Photocurrent Hysteresis in Perovskite Solar Cells

Pengyun Liu; Wei Wang; Shaomin Liu; Hua Gui Yang; Zongping Shao. Advanced Energy Materials (2019). Cited by 321. Found by: openalex. openalex_2yr: 18.63 (2026). DOI: 10.1002/aenm.201803017

Abstract: Abstract Organic–inorganic hybrid perovskite solar cells (PSCs) have become a promising candidate in the photovoltaic field due to their high power conversion efficiency and low material cost. However, the development of PSCs is limited by their poor stability under practical conditions in the presence of oxygen, moisture, sunlight, heat, and the current–voltage (I – V) hysteresis. In particular, the hysteretic I – V issue casts doubt on the validity of the photovoltaic performance results that are achieved, making it difficult to evaluate the authentic performance of PSCs. This review article focuses on understanding the I – V hysteresis behavior in PSCs and on exploring the possible reasons leading to this hysteresis phenomenon. The various strategies attempted to suppress the I – V hysteresis in PSCs are summarized, and a brief future recommendation is provided.

Red-Carbon-Quantum-Dot-Doped SnO2 Composite with Enhanced Electron Mobility for Efficient and Stable Perovskite Solar Cells

Wei Hui; Yingguo Yang; Quan Xu; Hao Gu; Shanglei Feng; Zhenhuang Su; Miaoran Zhang; Jiaou Wang; Xiaodong Li; Junfeng Fang; Fei Xia; Yingdong Xia; Yonghua Chen; Xingyu Gao; Wei Huang. Advanced Materials (2019). Cited by 313. Found by: openalex. openalex_2yr: 22.086 (2026). DOI: 10.1002/adma.201906374

Abstract: Abstract An efficient electron transport layer (ETL) plays a key role in promoting carrier separation and electron extraction in planar perovskite solar cells (PSCs). An effective composite ETL is fabricated using carboxylic-acid- and hydroxyl-rich red-carbon quantum dots (RCQs) to dope low-temperature solution-processed SnO2, which dramatically increases its electron mobility by 20 times from 9.32 × 10−4 to 1.73 × 10−2 cm2 V−1 s−1. The mobility achieved is one of the highest reported electron mobilities for modified SnO2. Fabricated planar PSCs based on this novel SnO2 ETL demonstrate an outstanding improvement in efficiency from 19.15% for PSCs without RCQs up to 22.77% and have enhanced long-term stability against humidity, preserving over 95% of the initial efficiency after 1000 h under 40–60% humidity at 25 °C. These significant achievements are solely attributed to the excellent electron mobility of the novel ETL, which is also proven to help the passivation of traps/defects at the ETL/perovskite interface and to promote the formation of highly crystallized perovskite, with an enhanced phase purity and uniformity over a large area. These results demonstrate that inexpensive RCQs are simple but excellent additives for producing efficient ETLs in stable high-performance PSCs as well as other perovskite-based optoelectronics.

Reduction of bulk and surface defects in inverted methylammonium- and bromide-free formamidinium perovskite solar cells

Rui Chen; Jianan Wang; Zonghao Liu; Fumeng Ren; Sanwan Liu; Jing Zhou; Haixin Wang; Xin Meng; Zheng Zhang; Xinyu Guan; Wenxi Liang; Pavel A. Troshin; Yabing Qi; Liyuan Han; Wei Chen. Nature Energy (2023). Cited by 299. Found by: openalex. DOI: 10.1038/s41560-023-01288-7

High-Performance Perovskite Solar Cells with Enhanced Environmental Stability Based on a (p-FC6H4C2H4NH3)2[PbI4] Capping Layer

Qin Zhou; Lusheng Liang; Junjie Hu; Bingbing Cao; Longkai Yang; Tingjun Wu; Xin Li; Bao Zhang; Peng Gao. Advanced Energy Materials (2019). Cited by 295. Found by: openalex. openalex_2yr: 18.63 (2026). DOI: 10.1002/aenm.201802595

Abstract: Abstract Supported by the density functional theory (DFT) calculations, for the first time, a fluorinated aromatic cation, 2-(4-fluorophenyl)ethyl ammonium iodide (FPEAI), is introduced to grow in situ a low dimensional perovskite layer atop 3D perovskite film with excess PbI2. The resulted (p-FC6H4C2H4NH3)2[PbI4] perovskite functions as a protective capping layer to protect the 3D perovskite from moisture. In the meantime, the thin layer facilitates charge transfer at the interfaces, thereby reducing the nonradiative recombination pathways. Laser scanning confocal microscopy unveils visually the distribution of the 2D perovskite layer on top of the 3D perovskite. When employing the 3D–2D perovskite as the absorbing layer in the photovoltaic cells, a high power conversion efficiency of 20.54% is realized. Superior device performance and moisture stability are observed with the modified perovskite over the whole stability test period.

Stable and Efficient Organo-Metal Halide Hybrid Perovskite Solar Cells via -Conjugated Lewis Base Polymer Induced Trap Passivation and Charge Extraction

Pingli Qin; Guang Yang; Zhiwei Ren; Sin Hang Cheung; Shu Kong So; Li Chen; Jianhua Hao; Jianhui Hou; Gang Li. Advanced Materials (2018). Cited by 293. Found by: openalex. openalex_2yr: 22.086 (2026). DOI: 10.1002/adma.201706126

Abstract: High-quality pinhole-free perovskite film with optimal crystalline morphology is critical for achieving high-efficiency and high-stability perovskite solar cells (PSCs). In this study, a p-type -conjugated polymer poly[(2,6-(4,8-bis(5-(2-ethylhexyl) thiophen-2-yl)-benzo[1,2-b:4,5-b'] dithiophene))-alt-(5,5-(1',3'-di-2-thienyl-5',7'-bis(2-ethylhexyl) benzo[1',2'-c:4',5'-c'] dithiophene-4,8-dione))] (PBDB-T) is introduced into chlorobenzene to form a facile and effective template-agent during the anti-solvent process of perovskite film formation. The -conjugated polymer PBDB-T is found to trigger a heterogeneous nucleation over the perovskite precursor film and passivate the trap states of the mixed perovskite film through the formation of Lewis adducts between lead and oxygen atom in PBDB-T. The p-type semiconducting and hydrophobic PBDB-T polymer fills in the perovskite grain boundaries to improve charge transfer for better conductivity and prevent moisture invasion into the perovskite active layers. Consequently, the PSCs with PBDB-T modified anti-solvent processing leads to a high-efficiency close to 20%, and the devices show excellent stability, retaining about 90% of the initial power conversion efficiency after 150 d storage in dry air.

Unveiling the Role of tBP–LiTFSI Complexes in Perovskite Solar Cells

Shen Wang; Zihan Huang; Xuefeng Wang; Yingmin Li; Marcella Günther; Sophia Valenzuela; Pritesh Parikh; Amanda Cabreros; Wei Xiong; Ying Shirley Meng. Journal of the American Chemical Society (2018). Cited by 290. Found by: openalex. DOI: 10.1021/jacs.8b09809

Abstract: Lead halide-based perovskite materials have been applied as an intrinsic layer for next-generation photovoltaic devices. However, the stability and performance reproducibility of perovskite solar cells (PSCs) needs to be further improved to match that of silicon photovoltaic devices before they can be commercialized. One of the major bottlenecks that hinders the improvement of device stability/reproducibility is the additives in the hole-transport layer, lithium bis(trifluoromethanesulfonyl)imide (LiTFSI) and 4- tert-butylpyridine (tBP). Despite the positive effects of these hole-transport layer additives, LiTFSI is hygroscopic and can adsorb moisture to accelerate the perovskite decomposition. On the other hand, tBP, the only liquid component in PSCs, which evaporates easily, is corrosive to perovskite materials. Since 2012, the empirical molar ratio 6:1 tBP:LiTFSI has been wildly applied in PSCs without further concerns. In this study, the formation of tBP-LiTFSI complexes at various molar ratios has been discovered and investigated thoroughly. These complexes in PSCs can alleviate the negative effects (decomposition and corrosion) of individual components tBP and LiTFSI while maintaining their positive effects on perovskite materials. Consequently, a minor change in tBP:LiTFSI ratio results in huge influences on the stability of perovskite. Due to the existence of uncomplexed tBP in the 6:1 tBP:LiTFSI mixture, this empirical tBP-LiTFSI molar ratio has been demonstrated not as the ideal ratio in PSCs. Instead, the 4:1 tBP:LiTFSI mixture, in which all components are complexed, shows all positive effects of the hole-transport layer components with dramatically reduced negative effects. It minimizes the hygroscopicity of LiTFSI, while lowering the evaporation speed and corrosive effect of tBP. As a result, the PSCs fabricated with this tBP:LiTFSI ratio have the highest average device efficiency and obviously decreased efficiency variation with enhanced device stability, which is proposed as the golden ratio in PSCs. Our understanding of interactions

between hole-transport layer additives and perovskite on a molecular level shows the pathway to further improve the PSCs' stability and performance reproducibility to make them a step closer to large-scale manufacturing.

High-Performance Perovskite Solar Cells with Excellent Humidity and Thermo-Stability via Fluorinated Perylenediimide

Jia Yang; Cong Liu; C.S. Cai; Xiaotian Hu; Zengqi Huang; Xiaopeng Duan; Xiangchuan Meng; Zhongyi Yuan; Licheng Tan; Yiwang Chen. Advanced Energy Materials (2019). Cited by 289. Found by: openalex. openalex_2yr: 18.63 (2026). DOI: 10.1002/aenm.201900198

Abstract: Abstract The notoriously poor stability of perovskite solar cells is a crucial issue restricting commercial applications. Here, a fluorinated perylenediimide (F-PDI) is first introduced into perovskite film to enhance the device's photovoltaic performance, as well as thermal and moisture stability simultaneously. The conductive F-PDI molecules filling at grain boundaries (GBs) and surface of perovskite film can passivate defects and promote charge transport through GBs due to the chelation between carbonyl of F-PDI and noncoordinating lead. Furthermore, an effective multiple hydrophobic structure is formed to protect perovskite film from moisture erosion. As a result, the F-PDI-incorporated devices based on MAPbI3 and Cs0.05 (FA0.83MA0.17)0.95 Pb (Br0.17I0.83)3 absorber achieve champion efficiencies of 18.28% and 19.26%, respectively. Over 80% of the initial efficiency is maintained after exposure in air for 30 days with a relative humidity (RH) of 50%. In addition, the strong hydrogen bonding of F · · · H-N can immobilize methylamine ion (MA+) and thus enhances the thermal stability of device, remaining nearly 70% of the initial value after thermal treatment (100 °C) for 24 h at 50% RH condition.

Strain in perovskite solar cells: origins, impacts and regulation

Jinpeng Wu; Shunchang Liu; Zongbao Li; Shuo Wang; Ding-Jiang Xue; Yuan Lin; Jin-Song Hu. National Science Review (2021). Cited by 289. Found by: openalex. DOI: 10.1093/nsr/nwab047

Abstract: Metal halide perovskite solar cells (PSCs) have seen an extremely rapid rise in power conversion efficiencies in the past few years. However, the commercialization of this class of emerging materials still faces serious challenges, one of which is the instability against external stimuli such as moisture, heat and irradiation. Much focus has deservedly been placed on understanding the different origins of intrinsic instability and thereby enhancing their stability. Among these, tensile strain in perovskite films is an important source of instability that cannot be overcome using conventionally extrinsic stabilization approaches such as encapsulation. Here we review recent progress in the understanding of the origin of strain in perovskites as well as its corresponding characterization methods, and their impacts on the physical properties of perovskites and the performance of PSCs including efficiency and stability. We then summarize the latest advances in strain-regulation strategies that improve the intrinsic stability of perovskites and photovoltaic devices. Finally, we provide a perspective on how to make further progress in stable and high-efficiency PSCs via strain engineering.

Encapsulation of Organic and Perovskite Solar Cells: A Review

Ashraf Uddin; Mushfika Baishakhi Upama; Haimang Yi; Leiping Duan. Coatings (2019). Cited by 285. Found by: openalex. openalex_2yr: 3.91 (2026). DOI: 10.3390/coatings9020065

Abstract: Photovoltaic is one of the promising renewable sources of power to meet the future challenge of energy need. Organic and perovskite thin film solar cells are an emerging cost-effective photovoltaic technology because of low-cost manufacturing processing and their light weight. The main barrier of commercial use of organic and perovskite solar cells is the poor stability of devices. Encapsulation of these photovoltaic devices is one of the best ways to address this stability issue and enhance the device lifetime by employing materials and structures that possess high barrier performance for oxygen and moisture. The aim of this review paper is to find different encapsulation materials and techniques for perovskite and organic solar cells according to the present understanding of reliability issues. It discusses the available encapsulate materials and their utility in limiting chemicals, such as water vapour and oxygen penetration. It also covers the mechanisms of mechanical degradation within the individual layers and solar cell as a whole, and possible obstacles to their application in both organic and perovskite solar cells. The contemporary understanding of these degradation mechanisms, their interplay, and their initiating factors (both internal and external) are also discussed.

Metal Oxide Charge Transport Layers for Efficient and Stable Perovskite Solar Cells

Seong Sik Shin; Seon Joo Lee; Sang Il Seok. Advanced Functional Materials (2019). Cited by 285. Found by: openalex. openalex_2yr: 14.975 (2026). DOI: 10.1002/adfm.201900455

Abstract: Abstract Currently, the efficiency of perovskite solar cells (PSCs) is 24%. For the fabrication of such high efficiency PSCs, it is necessary to use both electron and hole transport layers to effectively separate the charges generated by light absorption of the perovskite layer and selectively transfer the separated electrons and holes. In addition to the efficiency, the materials used for transporting charges must be resilient to light, heat, and moisture to ensure long-term stability of PSCs; furthermore, low-cost fabrication is required to form a charge transport layer at low temperatures by a solution process. For this purpose, metal oxides are best suited as charge transport materials for PSCs because of their advantages such as low cost, long-term stability, and high efficiency. In this Review, the metal oxide electron and hole transport materials used in PSCs are reviewed and preparation of these materials is summarized. Finally, the challenges and future research direction for metal oxide-based charge transport materials are described.

Encapsulation for improving the lifetime of flexible perovskite solar cells

Hasitha C. Weerasinghe; Yasmina Dkhissi; Andrew D. Scully; Rachel A. Caruso; Yi-Bing Cheng. Nano Energy (2015). Cited by 284. Found by: openalex. DOI: 10.1016/j.nanoen.2015.10.006

Fluorinated Low-Dimensional Ruddlesden–Popper Perovskite Solar Cells with over 17% Power Conversion Efficiency and Improved Stability

Jishan Shi; Yerun Gao; Xiang Gao; Yun Zhang; Junjie Zhang; Xin Jing; Ming Shao. Advanced Materials (2019). Cited by 283. Found by: openalex. openalex_2yr: 22.086 (2026). DOI: 10.1002/adma.201901673

Abstract: Abstract Low-dimensional Ruddlesden–Popper perovskites (RPPs) exhibit excellent stability in comparison with 3D perovskites; however, the relatively low power conversion efficiency (PCE) limits their future application. In this work, a new fluorine-substituted phenylethlammonium (PEA) cation is developed as a spacer to fabricate quasi-2D (4FPEA)2(MA)4Pb5I16 (n = 5) perovskite solar cells. The champion device exhibits a remarkable PCE of 17.3% with a Jsc of 19.00 mA cm−2, a Voc of 1.16 V, and a fill factor (FF) of 79%, which are among the best results for low-dimensional RPP solar cells (n 5). The enhanced device performance can be attributed as follows: first, the strong dipole field induced by the 4-fluoro-phenethylammonium (4FPEA) organic spacer facilitates charge dissociation. Second, fluorinated RPP crystals preferentially grow along the vertical direction, and form a phase distribution with the increasing n number from bottom to the top surface, resulting in efficient charge transport. Third, 4FPEA-based RPP films exhibit higher film crystallinity, enlarged grain size, and reduced trap-state density. Lastly, the unsealed fluorinated RPP devices demonstrate superior humidity and thermal stability. Therefore, the fluorination of the long-chain organic cations provides a feasible approach for simultaneously improving the efficiency and stability of low-dimensional RPP solar cells.

Influence of Electrode Interfaces on the Stability of Perovskite Solar Cells: Reduced Degradation Using MoOx/Al for Hole Collection

Erin M. Sanehira; Bertrand J. Tremolet de Villers; Philip Schulz; Matthew O. Reese; Suzanne Ferrere; Kai Zhu; Lih Y. Lin; Joseph J. Berry; Joseph M. Luther. ACS Energy Letters (2016). Cited by 282. Found by: openalex. openalex_2yr: 16.712 (2026). DOI: 10.1021/acsenergylett.6b00013

Abstract: We investigated and characterized the stability of the power output from methylammonium lead iodide perovskite photovoltaic devices produced with various hole-collecting anode configurations consisting of Au, Ag, MoO x /Au, MoO x /Ag, and MoO x /Al. The unencapsulated devices were operated under constant illumination and constant load conditions in laboratory ambient with periodic current–voltage testing. Although the initial efficiencies of devices were comparable across these configurations, the stability of these devices varied significantly due to subtle differences in the electrode structure. Specifically, we found that devices with MoO x /Al electrodes are more stable than devices with more conventional, and more costly, Au and Ag electrodes. We demonstrate that a thin MoO x layer inhibits decomposition of the perovskite films under illumination in ambient laboratory conditions and greater improvements in device stability are achieved specifically with MoO x /Al electrodes. We investigated the role of the MoO x interlayer in the MoO x /Al electrodes by exploring the effect of relative humidity and the MoO x interlayer thickness on the perovskite solar cell stability.

Towards linking lab and field lifetimes of perovskite solar cells

Qi Jiang; Robert Tirawat; Ross A. Kerner; E. Ashley Gaulding; Yeming Xian; Xiaoming Wang; Jimmy M. Newkirk; Yanfa Yan; Joseph J. Berry; Kai Zhu. Nature (2023). Cited by 281. Found by: openalex. DOI: 10.1038/s41586-023-06610-7

Highly Air-Stable Carbon-Based -CsPbI3 Perovskite Solar Cells with a Broadened Optical Spectrum

Sisi Xiang; Zhongheng Fu; Weiping Li; Ya Wei; Jiaming Liu; Huicong Liu; Liqun Zhu; Ruifeng Zhang; Haining Chen. ACS Energy Letters (2018). Cited by 279. Found by: openalex. openalex_2yr: 16.712 (2026). DOI: 10.1021/acsenergylett.8b00820

Abstract: Inorganic cesium lead halide perovskites with superb thermal stability show promise to fabricate long-term operational photovoltaic devices. However, the cubic phase ( ) of CsPbI 3 with an appropriate band gap is unstable in air. We discover that highly stable -CsPbI 3 can be obtained in dry air (temperature: 20–30 °C; humidity: 10–20%) by replacing PbI 2 with HPbI 3 in a one-step deposition solution. Furthermore, the band gap of HPbI 3 -processed -CsPbI 3 is advantageously reduced from 1.72 to 1.68 eV due to the existence of tensile lattice strain. By employing such an -CsPbI 3 film in carbon-based perovskite solar cells (C-PSCs), a power conversion efficiency (PCE) of 9.5% is achieved, which is a record value for the -CsPbI 3 PSCs without hole transport material. Most importantly, over 90% of the initial PCE is retained for nonencapsulated devices after 3000 h of storage in dry air. Therefore, HPbI 3 -based one-step deposition presents a promising strategy to prepare high-performance and air-stable -CsPbI 3 PSCs.

Fabrication of perovskite solar cells in ambient air by blocking perovskite hydration with guanabenz acetate salt

Luyao Yan; Hao Huang; Peng Cui; Shuxian Du; Zhineng Lan; Yingying Yang; Shujie Qu; Xinxin Wang; Qiang Zhang; Benyu Liu; Xiaopeng Yue; Xing Zhao; Yingfeng Li; Haifang Li; Jun Ji; Meicheng Li. Nature Energy (2023). Cited by 278. Found by: openalex. DOI: 10.1038/s41560-023-01358-w

Interfacial Defect Passivation and Stress Release via Multi-Active-Site Ligand Anchoring Enables Efficient and Stable Methylammonium-Free Perovskite Solar Cells

Baibai Liu; Huan Bì; Dongmei He; Le Bai; Wenqi Wang; Hongkuan Yuan; Qunliang Song; Pengyu Su; Zhigang Zang; Tingwei Zhou; Jiangzhao Chen. ACS Energy Letters (2021). Cited by 276. Found by: openalex. openalex_2yr: 16.712 (2026). DOI: 10.1021/acsenergylett.1c00794

Abstract: Interfacial trap-assisted non-radiative recombination and residual stress impede the further increase of power conversion efficiency (PCE) and stability of the methylammonium-free (MA-free) perovskite solar cells (PSCs). Here, we report an interfacial defect passivation and stress release strategy through employing the multi-active-site Lewis base ligand (i.e., (5-mercapto-1,3,4-thiadiazol-2-ylthio)acetic acid (MTDAA)) to modify the surface and grain boundaries (GBs) of MA-free perovskite films. Both experimental and theoretical results confirm strong chemical interactions between multiple active sites in the MTDAA molecule and undercoordinated Pb 2+ at the surface or GBs of perovskite films. It is demonstrated theoretically that multi-active-site adsorption is more favorable thermodynamically as compared to single-active-site adsorption, regardless of PbI 2 termination and formamidinium iodide (FAI) termination types. MTDAA modification results in much reduced defect density, increased carrier lifetime, and almost thoroughly released interfacial residual stress. Upon MTDAA passivation, the PCE is boosted from 20.26% to 21.92%. The unencapsulated device modified by MTDAA maintains 99% of its initial PCE after aging under the relative humidity range of 10–20% for 1776 h, and 91% after aging at 60 °C for 1032 h.

Suppressing Vacancy Defects and Grain Boundaries via Ostwald Ripening for High-Performance and Stable Perovskite Solar Cells

Yuqian Yang; Jihuai Wu; Xiaobing Wang; Qiyao Guo; Xuping Liu; Weihai Sun; Yuelin Wei; Yunfang Huang; Zhang Lan; Miaoliang Huang; Jianming Lin; Hongwei Chen; Zhanhua Wei. Advanced Materials (2019). Cited by 275. Found by: openalex. openalex_2yr: 22.086 (2026). DOI: 10.1002/adma.201904347

Abstract: Abstract As one kind of promising next-generation photovoltaic devices, perovskite solar cells (PVSCs) have experienced unprecedented rapid growth in device performance over the past few years. However,

the practical applications of PVSCs require much improved device long-term stability and performance, and internal defects and external humidity sensitivity are two key limitation need to be overcome. Here, gadolinium fluoride (GdF 3) is added into perovskite precursor as a redox shuttle and growth-assist; meanwhile, aminobutanol vapor is used for Ostwald ripening in the formation of the perovskite layer. Consequently, a high-quality perovskite film with large grain size and few grain boundaries is obtained, resulting in the reduction of trap state density and carrier recombination. As a result, a power conversion efficiency of 21.21% is achieved with superior stability and negligible hysteresis.

Quantum Dots Supply Bulk- and Surface-Passivation Agents for Efficient and Stable Perovskite Solar Cells

Xiaopeng Zheng; Joel Troughton; Nicola Gasparini; Yuanbao Lin; Mingyang Wei; Yi Hou; Jiakai Liu; Kepeng Song; Zhaolai Chen; Chen Yang; Bekir Türedi; Abdullah Y. Alsalloum; Jun Pan; Jie Chen; Ayan A. Zhumekenov; Thomas D. Anthopoulos; Yu Han; Derya Baran; Omar F. Mohammed; Edward H. Sargent; Osman M. Bakr. Joule (2019). Cited by 272. Found by: openalex. DOI: 10.1016/j.joule.2019.05.005

Bifunctional Organic Spacers for Formamidinium-Based Hybrid Dion–Jacobson Two-Dimensional Perovskite Solar Cells

Yang Li; Jovana V. Milić; Amita Ummadisingu; Ji-Youn Seo; Jeong-Hyeok Im; Hui-Seo Kim; Yuhang Liu; M. Ibrahim Dar; Shaik M. Zakeeruddin; Peng Wang; Anders Hagfeldt; Michaël Grätzel. Nano Letters (2018). Cited by 271. Found by: openalex. DOI: 10.1021/acs.nanolett.8b03552

Abstract: Three-dimensional (3D) perovskite materials display remarkable potential in photovoltaics owing to their superior solar-to-electric power conversion efficiency, with current efforts focused on improving stability. Two-dimensional (2D) perovskite analogues feature greater stability toward environmental factors, such as moisture, owing to a hydrophobic organic cation that acts as a spacer between the inorganic layers, which offers a significant advantage over their comparatively less stable 3D analogues. Here, we demonstrate the first example of a formamidinium (FA) containing Dion–Jacobson 2D perovskite material characterized by the BFA n −1 Pb n I 3 n +1 formulation through employing a novel bifunctional organic spacer (B), namely 1,4-phenylenedimethanammonium (PDMA). We achieve remarkable efficiencies exceeding 7% for (PDMA)FA 2 Pb 3 I 10 based 2D perovskite solar cells resisting degradation when exposed to humid ambient air, which opens up new avenues in the design of stable perovskite materials.

Over 20% PCE perovskite solar cells with superior stability achieved by novel and low-cost hole-transporting materials

Fei Zhang; Zhi-Qiang Wang; Hongwei Zhu; Norman Pellet; Jingshan Luo; Chenyi Yi; Xicheng Liu; Hongli Liu; Shirong Wang; Xianggao Li; Yin Xiao; Shaik M. Zakeeruddin; Dongqin Bi; Michaël Grätzel. Nano Energy (2017). Cited by 269. Found by: openalex. DOI: 10.1016/j.nanoen.2017.09.035

Effect of Selective Contacts on the Thermal Stability of Perovskite Solar Cells

Xing Zhao; Hui-Seon Kim; Ja-Young Seo; Nam-Gyu Park. ACS Applied Materials & Interfaces (2017). Cited by 264. Found by: openalex. openalex_2yr: 8.382 (2026). DOI: 10.1021/acsami.6b15673

Abstract: Thermal stability of CH 3 NH 3 PbI 3 (MAPbI 3)-based perovskite solar cells was investigated for normal structure including the mesoporous TiO 2 layer and spiro-MeOTAD and the inverted structure with PCBM and NiO. MAPbI 3 was found to be intrinsically stable from 85 °C to 120 °C in the absence of moisture. However, fast degradation was observed for the encapsulated device including spiro-MeOTAD upon thermal stress at 85 °C. Photoluminescence (PL) intensity and the time constant for charge separation increased with thermal exposure time, which is indicative of inhibition of charge separation from MAPbI 3 into spiro-MeOTAD. A full recovery of photovoltaic performance was observed for the 85 °C-aged device after renewal with fresh spiro-MeOTAD, which clearly indicates that thermal instability of the normal structured device is mainly due to spiro-MeOTAD, and MAPbI 3 is proved to be thermally stable. Spiro-MeOTAD with additives was crystallized at 85 °C due to a low glass transition temperature, and hole mobility was significantly deteriorated, which was responsible for the thermal instability. Thermal stability was significantly improved for the inverted structure with the NiO hole transporting layer, where the power conversion efficiency (PCE) was maintained at 74% of its initial PCE of 14.71% after the 80th thermal cycle (one cycle: heating at 85 °C for 2 h and cooling at 25 °C for 2 h). This work implies that the thermal stability of perovskite solar cells depends on selective contacts.

Well-Defined Thiolated Nanographene as Hole-Transporting Material for Efficient and Stable Perovskite Solar Cells

Jing Cao; Yumin Liu; Xiaojing Jing; Jun Yin; Jing Li; Bin Xu; Yuan-Zhi Tan; Nanfeng Zheng. Journal of the American Chemical Society (2015). Cited by 261. Found by: openalex. DOI: 10.1021/jacs.5b06493

Abstract: Perovskite solar cells (PSCs) have been demonstrated as one of the most promising candidates for solar energy harvesting. Here, for the first time, a functionalized nanographene (perthiolated trisulfur-annulated hexa-peri-hexabenzocoronene, TSHBC) is employed as the hole transporting material (HTM) in PSCs to achieve efficient charge extraction from perovskite, yielding the best efficiency of 12.8% in pristine form. The efficiency is readily improved up to 14.0% by doping with graphene sheets into TSHBC to enhance the charge transfer. By the HOMO-LUMO level engineering of TSHBC homologues, we demonstrate that the HOMO levels are critical for the performance of PSCs. Moreover, beneficial from the hydrophobic nature of TSHBC, the devices show the improved stability under AM 1.5 illumination in the humidity about 45% without encapsulation. These findings open the opportunities for efficient HTMs based on the functionalized nanographenes utilizing the strong interactions of their functional groups with perovskite.

Stability Issues on Perovskite Solar Cells

Xing Zhao; Nam-Gyu Park. Photonics (2015). Cited by 259. Found by: openalex. DOI: 10.3390/photonics2041139

Abstract: Organo lead halide perovskite materials like methylammonium lead iodide (CH3NH3PbI3) and formamidinium lead iodide (HC(NH2)2PbI3) show superb opto-electronic properties. Based on these perovskite light absorbers, power conversion efficiencies of the perovskite solar cells employing hole transporting layers have increased from 9.7% to 20.1% within just three years. Thus, it is apparent that perovskite solar cell is a promising next generation photovoltaic technology. However, the unstable nature of perovskite was observed when exposing it to continuous illumination, moisture and high temperature, impeding the commercial development in the long run and thus becoming the main issue that needs to be solved urgently. Here, we discuss the factors affecting instability of perovskite and give some perspectives about further enhancement of stability of perovskite solar cell.

Interface Modification by Ionic Liquid: A Promising Candidate for Indoor Light Harvesting and Stability Improvement of Planar Perovskite Solar Cells

Meng Li; Chao Zhao; Zhao-Kui Wang; Cong-Cong Zhang; Harrison Ka Hin Lee; Adam Pockett; Jérémy Barbé; Wing Chung Tsoi; Yingguo Yang; Matthew J. Carnie; Xingyu Gao; Wenxing Yang; James R. Durrant; Liang-Sheng Liao; Sagar M. Jain. Advanced Energy Materials (2018). Cited by 257. Found by: openalex. openalex_2yr: 18.63 (2026). DOI: 10.1002/aenm.201801509

Abstract: Abstract Organic–inorganic hybrid perovskite solar cells (PSCs) are currently attracting significant interest owing to their promising outdoor performance. However, the ability of indoor light harvesting of the perovskites and corresponding device performance are rarely reported. Here, the potential of planar PSCs in harvesting indoor light for low-power consumption devices is investigated. Ionic liquid of 1-butyl-3-methylimidazolium tetrafluoroborate ([BMIM]BF4) is employed as a modification layer of [6,6]-phenyl-C61-butyric acid methyl ester) (PCBM) in the inverted PSCs. The incorporation of [BMIM]BF4 not only paves the interface contact between PCBM and electrode, but also facilitates the electron transport and extraction owing to the efficient passivation of the surface trap states. Moreover, [BMIM]BF4 with excellent thermal stability can act as a protective layer by preventing the erosion of moisture and oxygen into the perovskite layer. The resulting devices present a record indoor power conversion efficiency (PCE) of 35.20% under fluorescent lamps of 1000 lux, and an impressive PCE of 19.30% under 1 sun illumination. The finding in this work verifies the excellent indoor performance of PSCs to meet the requirements of eco-friendly economy.

Vertically Oriented 2D Layered Perovskite Solar Cells with Enhanced Efficiency and Good Stability

Xinqian Zhang; Gang Wu; Shida Yang; Weifei Fu; Zhongqiang Zhang; Chen Chen; Wenqing Liu; Jielin Yan; Weitao Yang; Hongzheng Chen. Small (2017). Cited by 255. Found by: openalex. DOI: 10.1002/smll.201700611

Abstract: Vertically oriented highly crystalline 2D layered (BA)2(MA)n−1PbnI3n+1 (BA = CH3(CH2)3NH3, MA = CH3NH3, n = 3, 4) perovskite thin-films are fabricated with the aid of ammonium thiocyanate (NH4SCN) additive through one-step spin-coating process. The humidity-stability of the film is certified by the almost unchanged X-ray diffraction patterns after exposed to humid atmosphere (Hr = 55 ± 5%) for 40 d. The photovoltaic devices with the structure of indium tin oxide(ITO)/poly(3,4-ethylenedioxythiophene):poly(styrene-sulfonate)/(BA)2(MA)n− = 3,4)/[6,6]-phenyl-C61-butyric acid methyl ester/Bathocuproine/Ag are fabricated. The devices based on (BA)2(MA)2Pb3I10 perovskite (n = 3) with the precursor composition of BAI:methylammonium iodide:PbI2:NH4SCN = 2:2:3:1 (by molar ratio) show an averaged power conversion efficiency (PCE) of 6.82%. In the case of (BA)2(MA)3Pb4I13 (n = 4), a higher PCE of 8.79% is achieved. Both of the unsealed devices perform unique stability with almost unchanged PCE during the period of storage in purified N2 glove box. This work provides a simple and effective method to enhance the efficiency of the 2D perovskite solar cell.

A Review: Thermal Stability of Methylammonium Lead Halide Based Perovskite Solar Cells

Tanzila Tasnim Ava; Abdullah Al Mamun; Sylvain Marsillac; Gon Namkoong. Applied Sciences (2019). Cited by 254. Found by: openalex. openalex_2yr: 3.811 (2026). DOI: 10.3390/app9010188

Abstract: Perovskite solar cells have achieved photo-conversion efficiencies greater than 20%, making them a promising candidate as an emerging solar cell technology. While perovskite solar cells are expected to eventually compete with existing silicon-based solar cells on the market, their long-term stability has become a major bottleneck. In particular, perovskite films are found to be very sensitive to external factors such as air, UV light, light soaking, thermal stress and others. Among these stressors, light, oxygen and moisture-induced degradation can be slowed by integrating barrier or interface layers within the device architecture. However, the most representative perovskite absorber material, CH3NH3PbI3 (MAPbI3), appears to be thermally unstable even in an inert environment. This poses a substantial challenge for solar cell applications because device temperatures can be over 45 °C higher than ambient temperatures when operating under direct sunlight. Herein, recent advances in resolving thermal stability problems are highlighted through literature review. Moreover, the most recent and promising strategies for overcoming thermal degradation are also summarized.

Multifunctional Phosphorus-Containing Lewis Acid and Base Passivation Enabling Efficient and Moisture-Stable Perovskite Solar Cells

Zhi Yang; Jinjuan Dou; Song Kou; Jialin Dang; Yongqiang Ji; Guan-Jun Yang; Wu-Qiang Wu; Dai-Bin Kuang; Minqiang Wang. Advanced Functional Materials (2020). Cited by 254. Found by: openalex. openalex_2yr: 14.975 (2026). DOI: 10.1002/adfm.201910710

Abstract: Abstract Multiple-cation lead mixed-halide perovskites (MLMPs) have been recognized as ideal candidates in perovskite solar cells in terms of high efficiency and stability due to decreased open-circuit voltage loss and suppressed yellow phase formation. However, they still suffer from an unsatisfactory long-term moisture stability. In this study, phosphorus-containing Lewis acid and base molecules are employed to improve device efficiency and stability based on their multifunction including recombination reduction, phase segregation suppression, and moisture resistance. The strong fluorine-containing Lewis acid treatment can achieve a champion PCE of 22.02%. Unencapsulated and encapsulated devices retain 63% and 80% of the initial efficiency after 14 days of aging under 75% and 85% relative humidity, respectively. The better passivation of Lewis acid implies more halide defects than Pb defects at the MLMP surface. This unbalanced defect type results from phase segregation that is the synergistic effect of Cs and halide ion migrations. Identifying defect type based on different passivation effects is beneficial to not only choose suitable passivators to boost the efficiency and slow down the moisture degradation of MLMP solar cells, but also to understand the mechanism of defect-assisted moisture degradation.

Highly Efficient and Stable Perovskite Solar Cells via Modification of Energy Levels at the Perovskite/Carbon Electrode Interface

Zhifang Wu; Zonghao Liu; Zhanhao Hu; Zafer Hawash; Longbin Qiu; Yan Jiang; Luis K. Ono; Yabing Qi. Advanced Materials (2019). Cited by 254. Found by: openalex. openalex_2yr: 22.086 (2026). DOI: 10.1002/adma.201804284

Abstract: Perovskite solar cells (PSCs) have attracted great attention in the past few years due to their rapid increase in efficiency and low-cost fabrication. However, instability against thermal stress and humidity is a big issue hindering their commercialization and practical applications. Here, by combining thermally stable formamidinium-cesium-based perovskite and a moisture-resistant carbon electrode, successful fabrication of stable PSCs is reported, which maintain on average 77% of the initial value after being aged for 192 h under conditions of 85 °C and 85% relative humidity (the "double 85" aging condition) without encapsulation. However, the mismatch of energy levels at the interface between the perovskite and the carbon electrode limits charge collection and leads to poor device performance. To address this issue, a thin-layer of poly(ethylene

oxide) (PEO) is introduced to achieve improved interfacial energy level alignment, which is verified by ultraviolet photoemission spectroscopy measurements. Indeed as a result, power conversion efficiency increases from 12.2% to 14.9% after suitable energy level modification by intentionally introducing a thin layer of PEO at the perovskite/carbon interface.

Intermolecular Exchange Boosts Efficiency of Air-Stable, Carbon-Based All-Inorganic Planar CsPbIBr2 Perovskite Solar Cells to Over 9%

Weidong Zhu; Qianni Zhang; Dazheng Chen; Zeyang Zhang; Zhenhua Lin; Jingjing Chang; Jincheng Zhang; Chunfu Zhang; Yue Hao. Advanced Energy Materials (2018). Cited by 251. Found by: openalex. openalex_2yr: 18.63 (2026). DOI: 10.1002/aenm.201802080

Abstract: Abstract Among all inorganic halide perovskite photovoltaic materials, CsPbIBr2 exhibits the most balanced features in terms of bandgap and stability. However, the poor quality of solution-processed CsPbIBr2 films impedes further optimization of cells performance. Herein, a facile intermolecular exchange strategy for CsPbIBr2 film is demonstrated, wherein an optimized methanol solution of CsI is spin-coated on CsPbIBr2 precursor film in conventional one-step solution route. It surprisingly produces full-coverage and pure-phase CsPbIBr2 films featured with average grain size of 0.65 µm, few grain boundaries, high crystallinity, preferable (100) orientation, stoichiometric composition along with favorable electronic structures for effective dissociation and transfer of carriers. Hence, the cost-effective, carbon-based all-inorganic planar perovskite solar cells based on them, yield an optimized efficiency of 9.16% with a stabilized value of 8.46% in ambient air conditions that highlight a particularly superb open-circuit voltage of 1.245 V, all of which represent the highest values reported in pure CsPbIBr2 based cells so far. Moreover, the optimized cell without encapsulation shows excellent long-term stability because it can retain 90% over 60 days and 97% over 7 days of its initial efficiency, when is stored controllably in 45% relative humidity at 25 or 85 °C at zero humidity, respectively.

Interfacial defect passivation and stress release by multifunctional KPF6 modification for planar perovskite solar cells with enhanced efficiency and stability

Huan Bì; Baibai Liu; Dongmei He; Le Bai; Wenqi Wang; Zhigang Zang; Jiangzhao Chen. Chemical Engineering Journal (2021). Cited by 250. Found by: openalex. openalex_2yr: 10.953 (2026). DOI: 10.1016/j.cej.2021.129375

Green-Solvent-Processable, Dopant-Free Hole-Transporting Materials for Robust and Efficient Perovskite Solar Cells

Junwoo Lee; Mahdi Malekshahi Byranvand; Gyeongho Kang; Sung Yun Son; Seulki Song; Guan-Woo Kim; Taiho Park. Journal of the American Chemical Society (2017). Cited by 250. Found by: openalex. DOI: 10.1021/jacs.7b04949

Abstract: In addition to having proper energy levels and high hole mobility (h) without the use of dopants, hole-transporting materials (HTMs) used in n-i-p-type perovskite solar cells (PSCs) should be processed using green solvents to enable environmentally friendly device fabrication. Although many HTMs have been assessed, due to the limited solubility of HTMs in green solvents, no green-solvent-processable HTM has been reported to date. Here, we report on a green-solvent-processable HTM, an asymmetric D–A polymer (asy-PBTBDT) that exhibits superior solubility even in the green solvent, 2-methylanisole, which is a known food additive. The new HTM is well matched with perovskites in terms of energy levels and attains a high h (1.13 × 10 −3 cm 2 /(V s)) even without the use of dopants. Using the HTM, we produced robust PSCs with 18.3% efficiency (91% retention after 30 days without encapsulation under 50%–75% relative humidity) without dopants; with dopants (bis(trifluoromethanesulfonyl) imide and tert -butylpyridine, a 20.0% efficiency was achieved. Therefore, it is a first report for a green-solvent-processable hole-transporting polymer, exhibiting the highest efficiencies reported so far for n-i-p devices with and without the dopants.

Investigation of Thermally Induced Degradation in CH3NH3PbI3 Perovskite Solar Cells using In-situ Synchrotron Radiation Analysis

Nam-Koo Kim; Young Hwan Min; Seokhwan Noh; Eunkyung Cho; Gitaeg Jeong; Minho Joo; Seh-Won Ahn; Jeong Soo Lee; Seongtak Kim; Kyuwook Ihm; Hyungju Ahn; Yoonmook Kang; Hae-Seok Lee; Dong-Hwan Kim. Scientific Reports (2017). Cited by 249. Found by: openalex. DOI: 10.1038/s41598-017-04690-w

Abstract: Abstract In this study, we employ a combination of various in-situ surface analysis techniques to investigate the thermally induced degradation processes in MAPbI 3 perovskite solar cells (PeSCs) as a function

of temperature under air-free conditions (no moisture and oxygen). Through a comprehensive approach that combines in-situ grazing-incidence wide-angle X-ray diffraction (GIWAXD) and high-resolution X-ray photoelectron spectroscopy (HR-XPS) measurements, we confirm that the surface structure of MAPbI 3 perovskite film changes to an intermediate phase and decomposes to CH 3 I, NH 3 , and PbI 2 after both a short (20 min) exposure to heat stress at 100 °C and a long exposure (>1 hour) at 80 °C. Moreover, we observe clearly the changes in the orientation of CH 3 NH 3 + organic cations with respect to the substrate in the intermediate phase, which might be linked directly to the thermal degradation processes in MAPbI 3 perovskites. These results provide important progress towards improved understanding of the thermal degradation mechanisms in perovskite materials and will facilitate improvements in the design and fabrication of perovskite solar cells with better thermal stability.

Inorganic Rubidium Cation as an Enhancer for Photovoltaic Performance and Moisture Stability of HC(NH2)2PbI3 Perovskite Solar Cells

Yun Hee Park; Inyoung Jeong; Seunghwan Bae; Hae Jung Son; Phillip Lee; Jinwoo Lee; Chul-Ho Lee; Min Jae Ko. Advanced Functional Materials (2017). Cited by 248. Found by: openalex. openalex_2yr: 14.975 (2026). DOI: 10.1002/adfm.201605988

Abstract: Perovskite solar cells (PSCs) based on organic monovalent cation (methylammonium or formamidinium) have shown excellent optoelectronic properties with high efficiencies above 22%, threatening the status of silicon solar cells. However, critical issues of long-term stability have to be solved for commercialization. The severe weakness of the state-of-the-art PSCs against moisture originates mainly from the hygroscopic organic cations. Here, rubidium (Rb) is suggested as a promising candidate for an inorganic–organic mixed cation system to enhance moisture-tolerance and photovoltaic performances of formamidinium lead iodide (FAPbI3). Partial incorporation of Rb in FAPbI3 tunes the tolerance factor and stabilizes the photoactive perovskite structure. Phase conversion from hexagonal yellow FAPbI3 to trigonal black FAPbI3 becomes favored when Rb is introduced. The authors find that the absorbance and fluorescence lifetime of 5% Rb-incorporated FAPbI3 (Rb0.05FA0.95PbI3) are enhanced than bare FAPbI3. Rb0.05FA0.95PbI3-based PSCs exhibit a best power conversion efficiency of 17.16%, which is much higher than that of the FAPbI3 device (13.56%). Moreover, it is demonstrated that the Rb0.05FA0.95PbI3 film shows superior stability against high humidity (85%) and the full device made with the mixed perovskite exhibits remarkable long-term stability under ambient condition without encapsulation, retaining the high performance for 1000 h.

A Review on Encapsulation Technology from Organic Light Emitting Diodes to Organic and Perovskite Solar Cells

Qian Lu; Zhichun Yang; Xin Meng; Youfeng Yue; Muhammad Ashfaq Ahmad; Wenjun Zhang; Shasha Zhang; Yiqiang Zhang; Zonghao Liu; Wei Chen. Advanced Functional Materials (2021). Cited by 246. Found by: openalex. openalex_2yr: 14.975 (2026). DOI: 10.1002/adfm.202100151

Abstract: Abstract Organic light emitting diodes (OLEDs) employing organic thin-film based emitters have attracted tremendous attention due to their widespread applications in lighting and as displays in mobile devices and televisions. The novel thin-film photovoltaic techniques using organic or organic–inorganic hybrid materials such as organic photovoltaics (OPVs) and perovskite solar cells (PSCs) have become emerging competitive candidates with regard to the traditional photovoltaic techniques on account of high-efficiency, low-cost, and simple manufacturing processing properties. However, OLEDs, OPVs, and PSCs are vulnerable to the undesired degradation induced by moisture and oxygen. To afford long-term stability, a robust encapsulation technique by employing materials and structures that possess high barrier performance against oxygen and moisture must be explored and employed to protect these devices. Herein, the recent progress on specific encapsulation materials and techniques for three types of devices on the basis of fundamental understanding of device stability is reviewed. First, their degradation mechanisms, as well as, influencing factors are discussed. Then, the encapsulation technologies and materials are classified and discussed. Moreover, the advantages and disadvantages of various encapsulation technologies and materials coupled with their encapsulation applications in different devices are compared. Finally, the ongoing challenges and future perspectives of encapsulation frontier are provided.

Accelerated Lifetime Testing of Organic–Inorganic Perovskite Solar Cells Encapsulated by Polyisobutylene

Lei Shi; Trevor L. Young; Jincheol Kim; Yun Sheng; Lei Wang; Yifeng Chen; Zhiqiang Feng; Mark Keevers; Xiaojing Hao; Pierre Verlinden; Martin A. Green; Anita Ho-Baillie. ACS Applied Materials & Interfaces (2017).

Cited by 244. Found by: openalex. openalex_2yr: 8.382 (2026). DOI: 10.1021/acsami.7b07625

Abstract: Metal halide perovskite solar cells (PSCs) have undergone rapid progress. However, unstable performance caused by sensitivity to environmental moisture and high temperature is a major impediment to commercialization of PSCs. In the present work, a low-temperature, glass–glass encapsulation technique using high performance polyisobutylene (PIB) as the moisture barrier is investigated on planar glass/FTO/TiO 2 /FAPbI 3 /PTAA/gold perovskite solar cells. PIB was applied as either an edge seal or blanket layer. Electrical connections to the encapsulated PSCs were provided by either the FTO or Au layers. Results of a "calcium test" demonstrated that a PIB edge-seal effectively prevents moisture ingress. A shelf life test was performed and the PIB-sealed PSC was stable for at least 200 days. Damp heat and thermal cycling tests, in compliance with IEC61215:2016, were used to evaluate different encapsulation methods. Current–voltage measurements were performed regularly under simulated AM1.5G sunlight to monitor changes in PCE. The best results we have achieved to date maintained the initial efficiency after 540 h of damp heat testing and 200 thermal cycles. To the best of the authors' knowledge, these are among the best damp heat and thermal cycle test results for perovskite solar cells published to date. Given the modest performance of the cells (8% averaged from forward and reverse scans) especially with the more challenging FAPbI 3 perovskite material tested in this work, it is envisaged that better stability results can be further achieved when higher performance perovskite solar cells are encapsulated using the PIB packaging techniques developed in this work. We propose that heat rather than moisture was the main cause of our PSC degradation. Furthermore, we propose that preventing the escape of volatile decomposition products from the perovskite solar cell materials is the key for stability. PIB encapsulation is a very promising packaging solution for perovskite solar cells, given its demonstrated effectiveness, ease of application, low application temperature, and low cost.

All-inorganic CsPbBr 3 perovskite solar cell with 10.26% efficiency by spectra engineering

Haiwen Yuan; Yuanyuan Zhao; Jialong Duan; Yudi Wang; Xiya Yang; Qunwei Tang. Journal of Materials Chemistry A (2018). Cited by 242. Found by: openalex. DOI: 10.1039/c8ta08900k

Abstract: Through fabricating a perovskite/photoactive layer mixed light-harvester, the all-inorganic CsPbBr 3 PSC achieved a champion PCE of 10.26% and excellent stability in high humidity or high temperature atmosphere.

Copper Salts Doped Spiro-OMeTAD for High-Performance Perovskite Solar Cells

Meng Li; Zhao-Kui Wang; Yingguo Yang; Yun Hu; Shanglei Feng; Jinmiao Wang; Xingyu Gao; Liang-Sheng Liao. Advanced Energy Materials (2016). Cited by 241. Found by: openalex. openalex_2yr: 18.63 (2026). DOI: 10.1002/aenm.201601156

Abstract: The development of effective and stable hole transporting materials (HTMs) is very important for achieving high-performance planar perovskite solar cells (PSCs). Herein, copper salts (cuprous thiocyanate (CuSCN) or cuprous iodide (CuI)) doped 2,2,7,7-tetrakis(N , N -di- p -methoxyphenylamine)-9,9-spirobifluorene (spiro-OMeTAD) based on a solution processing as the HTM in PSCs is demonstrated. The incorporation of CuSCN (or CuI) realizes a p-type doping with efficient charge transfer complex, which results in improved film conductivity and hole mobility in spiro-OMeTAD:CuSCN (or CuI) composite films. As a result, the PCE is largely improved from 14.82% to 18.02% due to obvious enhancements in the cell parameters of short-circuit current density and fill factor. Besides the HTM role, the composite film can suppress the film aggregation and crystallization of spiro-OMeTAD films with reduced pinholes and voids, which slows down the perovskite decomposition by avoiding the moisture infiltration to some extent. The finding in this work provides a simple method to improve the efficiency and stability of planar perovskite solar cells.

A comprehensive device modelling of perovskite solar cell with inorganic copper iodide as hole transport material

Syed Zulqarnain Haider; Hafeez Anwar; Mingqing Wang. Semiconductor Science and Technology (2018). Cited by 241. Found by: openalex. DOI: 10.1088/1361-6641/aaa596

Abstract: Abstract Hole transport material (HTM) plays an important role in the efficiency and stability of perovskite solar cells (PSCs). Spiro-MeOTAD, the commonly used HTM, is costly and can be easily degraded by heat and moisture, thus offering hindrance to commercialize PSCs. There is dire need to find an alternate inorganic and stable HTM to exploit PSCs with their maximum capability. In this paper, a comprehensive device simulation is used to study various possible parameters that can influence the performance of perovskite solar cell with CuI as HTM. These include the effect of doping density, defect density and thickness of absorber

layer, along with the influence of diffusion length of carriers as well as electron affinity of electron transport layer (ETM) and HTM on the performance of PSCs. In addition, hole mobility and doping density of HTM is also investigated. CuI is a p-type inorganic material with low cost and relatively high stability. It is found that concentration of dopant in absorber layer and HTM, the electron affinity of HTM and ETM affect the performance of solar cell minutely, while cell performance improves greatly with the reduction of defect density. Upon optimization of parameters, power conversion efficiency for this device is found to be 21.32%. The result shows that lead-based PSC with CuI as HTM is an efficient system. Enhancing the stability and reduction of defect density are critical factors for future research. These factors can be improved by better fabrication process and proper encapsulation of solar cell.

Self-assembled bilayer for perovskite solar cells with improved tolerance against thermal stresses

Bitao Dong; Mingyang Wei; Yuheng Li; Yingguo Yang; Wei Ma; Yueshuai Zhang; Yanbiao Ran; Meijie Cui; Zhongbo Su; Qunping Fan; Zhaozhao Bi; Tomas Edvinsson; Zhiqin Ding; Huanxin Ju; Shuai You; Shaik M. Zakeeruddin; Xiong Li; Anders Hagfeldt; Michaël Grätzel; Yuhang Liu. Nature Energy (2025). Cited by 240. Found by: openalex. DOI: 10.1038/s41560-024-01689-2

Abstract: The adoption of perovskite solar cells (PSCs) requires improved resistance to high temperatures and temperature variations. Hole-selective self-assembled monolayers (SAMs) have enabled progress in the performance of inverted PSCs, yet they may compromise temperature stability owing to desorption and weak interfacial contact. Here we developed a self-assembled bilayer by covalently interconnecting a phosphonic acid SAM with a triphenylamine upper layer. This polymerized network, formed through Friedel–Crafts alkylation, resisted thermal degradation up to 100 °C for 200 h. Meanwhile, the face-on-oriented upper layer exhibited adhesive contact with perovskites, leading to a 1.7-fold improvement in adhesion energy compared with the SAM–perovskite interface. We reported power conversion efficiencies exceeding 26% for inverted PSCs. The champion devices demonstrated less than 4% and 3% efficiency loss after 2,000 h damp heat exposure (85 °C and 85% relative humidity) and over 1,200 thermal cycles between −40 °C and 85 °C, respectively, meeting the temperature stability criteria outlined in the International Electrotechnical Commission 61215:2021 standards. To improve the tolerance of perovskite solar cells against high temperatures and temperature variations, Dong et al. covalently cross-link two molecules in the charge transport layer to strengthen adhesion with the perovskite layer.

Boron Doping of Multiwalled Carbon Nanotubes Significantly Enhances Hole Extraction in Carbon-Based Perovskite Solar Cells

Xiaoli Zheng; Haining Chen; Qiang Li; Yinglong Yang; Zhanhua Wei; Yang Bai; Yongcai Qiu; Dan Zhou; Kam Sing Wong; Shihe Yang. Nano Letters (2017). Cited by 239. Found by: openalex. DOI: 10.1021/acs.nanolett.7b00200

Abstract: Compared to the conventional perovskite solar cells (PSCs) containing hole-transport materials (HTM), carbon materials based HTM-free PSCs (C-PSCs) have often suffered from inferior power conversion efficiencies (PCEs) arising at least partially from the inefficient hole extraction at the perovskite–carbon interface. Here, we show that boron (B) doping of multiwalled carbon nanotubes (B-MWNTs) electrodes are superior in enabling enhanced hole extraction and transport by increasing work function, carrier concentration, and conductivity of MWNTs. The C-PSCs prepared using the B-MWNTs as the counter electrodes to extract and transport hole carriers have achieved remarkably higher performances than that with the undoped MWNTs, with the resulting PCE being considerably improved from 10.70% (average of 9.58%) to 14.60% (average of 13.70%). Significantly, these cells show negligible hysteretic behavior. Moreover, by coating a thin layer of insulating aluminum oxide (Al 2 O 3) on the mesoporous TiO 2 film as a physical barrier to substantially reduce the charge losses, the PCE has been further pushed to 15.23% (average 14.20%). Finally, the impressive durability and stability of the prepared C-PSCs were also testified under various conditions, including long-term air exposure, heat treatment, and high humidity.

FAPbI3-Based Perovskite Solar Cells Employing Hexyl-Based Ionic Liquid with an Efficiency Over 20% and Excellent Long-Term Stability

Seçkin Akın; Erdi Akman; Savaş Sönmezoğlu. Advanced Functional Materials (2020). Cited by 236. Found by: openalex. openalex_2yr: 14.975 (2026). DOI: 10.1002/adfm.202002964

Abstract: Abstract Formamidinium lead triiodide (FAPbI3)-based perovskite materials are of interest for photovoltaics in view of their close-to-ideal bandgap, allowing absorption of photons over a broad solar spectrum. However, FAPbI3-based materials suffer from a notorious phase transition from the photoactive black phase (-FAPbI3) to nonperovskite yellow phase (-FAPbI3) under ambient conditions. This transition dramatically

reduces light absorbtion, thus, degrading the photovoltaic performance and stability of ensuring solar cells. In this study, 1-hexyl-3-methylimidazolium iodide (HMII) ionic liquid (IL) is employed as an additive for the first time in FAPbI3 perovskite to overcome the above-mentioned issues. HMII incorporation facilitates the grain coarsening of FAPbI3 crystal owing to its high-polarity and high-boiling point, which yields liquid domains between neighboring grains to reduce the activation energy of the grain-boundary migration. As a result, the FAPbI3 active layer exhibits micron-sized grains with substantially suppressed parasitic traps with an Urbach energy reduced by 2 meV. Hence, the resulting perovskite solar cell achieves an efficiency of 20.6% with notable increase in open circuit voltage (VOC) of 80 mV compared with HMII-free cells (17.1%). More importantly, the HMII-doped FAPbI3-based cells show a striking enhancement in shelf-stability under high humidity and thermal stress, retaining >80% of their initial efficiencies at 60 ± 10% relative humidity and 95% at 65 °C.

Enhancing Stability of Perovskite Solar Cells to Moisture by the Facile Hydrophobic Passivation

; ; ; ; . (2015). Cited by 235. Found by: openalex. URL: https://openalex.org/W3029134055

Inorganic CsPb1−xSnxIBr2 for Efficient Wide-Bandgap Perovskite Solar Cells

Nan Li; Zonglong Zhu; Jiangwei Li; Alex K.-Y. Jen; Liduo Wang. Advanced Energy Materials (2018). Cited by 235. Found by: openalex. openalex_2yr: 18.63 (2026). DOI: 10.1002/aenm.201800525

Abstract: Abstract Recently, the stability of organic–inorganic perovskite thin films under thermal, photo, and moisture stresses has become a major concern for further commercialization due to the high volatility of the organic cations in the prototype perovskite composition (CH3NH3PbI3). All inorganic cesium (Cs) based perovskite is an alternative to avoid the release or decomposition of organic cations. Moreover, substituting Pb with Sn in the organic–inorganic lead halide perovskites has been demonstrated to narrow the bandgap to 1.2–1.4 eV for high-performance perovskite solar cells. In this work, a series of CsPb1−xSnxIBr2 perovskite alloys via one-step antisolvent method is demonstrated. These perovskite films present tunable bandgaps from 2.04 to 1.64 eV. Consequently, the CsPb0.75Sn0.25IBr2 with homogeneous and densely crystallized morphology shows a remarkable power conversion efficiency of 11.53% and a high Voc of 1.21 V with a much improved phase stability and illumination stability. This work provides a possibility for designing and synthesizing novel inorganic halide perovskites as the next generation of photovoltaic materials.

Review on Chemical Stability of Lead Halide Perovskite Solar Cells

Jing Zhuang; Jizheng Wang; Feng Yan. Nano-Micro Letters (2023). Cited by 232. Found by: openalex. DOI: 10.1007/s40820-023-01046-0

Abstract: Lead halide perovskite solar cells (PSCs) have become a promising next-generation photovoltaic technology due to their skyrocketed power conversion efficiency. However, the device stability issues may restrict their commercial applications, which are dominated by various chemical reactions of perovskite layers. Hence, a comprehensive illustration on the stability of perovskite films in PSCs is urgently needed. In this review article, chemical reactions of perovskite films under different environmental conditions (e.g., moisture, oxygen, light) and with charge transfer materials and metal electrodes are systematically elucidated. Effective strategies for suppressing the degradation reactions of perovskites, such as buffer layer introduction and additives engineering, are specified. Finally, conclusions and outlooks for this field are proposed. The comprehensive review will provide a guideline on the material engineering and device design for PSCs.

Enhancement of thermal stability for perovskite solar cells through cesium doping

Guangda Niu; Wenzhe Li; Jiangwei Li; Xingyao Liang; Liduo Wang. RSC Advances (2017). Cited by 231. Found by: openalex. DOI: 10.1039/c6ra28501e

Abstract: Organic–inorganic hybrid perovskite solar cells are found to be sensitive to moisture, oxygen, UV light, light soaking, heat, electric field, etc.

Inhibition of halide oxidation and deprotonation of organic cations with dimethylammonium formate for air-processed p–i–n perovskite solar cells

Hongguang Meng; Kaitian Mao; Fengchun Cai; Kai Zhang; Shaojie Yuan; Tieqiang Li; Fangfang Cao; Zhenhuang Su; Zhengjie Zhu; Xingyu Feng; Wei Peng; Jiahang Xu; Jiahang Xu; Yan Gao; W Q Chen; Chuanxiao

Xiao; Xiaojun Wu; Michael D. McGehee; Jixian Xu; Jixian Xu. Nature Energy (2024). Cited by 229. Found by: openalex. DOI: 10.1038/s41560-024-01471-4

Perovskite Solar Cells: A Review of the Recent Advances

Priyanka Roy; Aritra Ghosh; Fraser Barclay; Ayush Khare; Erdem Cüce. Coatings (2022). Cited by 228. Found by: openalex. openalex_2yr: 3.91 (2026). DOI: 10.3390/coatings12081089

Abstract: Perovskite solar cells (PSC) have been identified as a game-changer in the world of photovoltaics. This is owing to their rapid development in performance efficiency, increasing from 3.5% to 25.8% in a decade. Further advantages of PSCs include low fabrication costs and high tunability compared to conventional silicon-based solar cells. This paper reviews existing literature to discuss the structural and fundamental features of PSCs that have resulted in significant performance gains. Key electronic and optical properties include high electron mobility (800 cm2/Vs), long diffusion wavelength (>1 m), and high absorption coefficient (105 cm−1). Synthesis methods of PSCs are considered, with solution-based manufacturing being the most cost-effective and common industrial method. Furthermore, this review identifies the issues impeding PSCs from large-scale commercialisation and the actions needed to resolve them. The main issue is stability as PSCs are particularly vulnerable to moisture, caused by the inherently weak bonds in the perovskite structure. Scalability of manufacturing is also a big issue as the spin-coating technique used for most laboratory-scale tests is not appropriate for large-scale production. This highlights the need for a transition to manufacturing techniques that are compatible with roll-to-roll processing to achieve high throughput. Finally, this review discusses future innovations, with the development of more environmentally friendly lead-free PSCs and high-efficiency multi-junction cells. Overall, this review provides a critical evaluation of the advances, opportunities and challenges of PSCs.

Crown Ether Modulation Enables over 23% Efficient Formamidinium-Based Perovskite Solar Cells

Tzu-Sen Su; Felix T. Eickemeyer; Michael A. Hope; Farzaneh Jahanbakhshi; Marko Mladenović; Jun Li; Zhiwen Zhou; Aditya Mishra; Jun-Ho Yum; Dan Ren; Anurag Krishna; Olivier Ouellette; Tzu-Chien Wei; Hua Zhou; Hsin-Hsiang Huang; Mounir Mensi; Kevin Sivula; Shaik M. Zakeeruddin; Jovana V. Milić; Anders Hagfeldt; Ursula Röthlisberger; Lyndon Emsley; Hong Zhang; Michaël Grätzel. Journal of the American Chemical Society (2020). Cited by 227. Found by: openalex. DOI: 10.1021/jacs.0c08592

Abstract: The use of molecular modulators to reduce the defect density at the surface and grain boundaries of perovskite materials has been demonstrated to be an effective approach to enhance the photovoltaic performance and device stability of perovskite solar cells. Herein, we employ crown ethers to modulate perovskite films, affording passivation of undercoordinated surface defects. This interaction has been elucidated by solid-state nuclear magnetic resonance and density functional theory calculations. The crown ether hosts induce the formation of host-guest complexes on the surface of the perovskite films, which reduces the concentration of surface electronic defects and suppresses nonradiative recombination by 40%, while minimizing moisture permeation. As a result, we achieved substantially improved photovoltaic performance with power conversion efficiencies exceeding 23%, accompanied by enhanced stability under ambient and operational conditions. This work opens a new avenue to improve the performance and stability of perovskite-based optoelectronic devices through supramolecular chemistry.

Zwitterionic-Surfactant-Assisted Room-Temperature Coating of Efficient Perovskite Solar Cells

Kuan Liu; Qiong Liang; Minchao Qin; Dong Shen; Hang Yin; Zhiwei Ren; Yaokang Zhang; Hengkai Zhang; W.K. Fong; Zehan Wu; Jiaming Huang; Jianhua Hao; Zijian Zheng; Shu Kong So; Chun-Sing Lee; Xinhui Lu; Gang Li. Joule (2020). Cited by 226. Found by: openalex. DOI: 10.1016/j.joule.2020.09.011

Large-area perovskite solar cells employing spiro-Naph hole transport material

Mingyu Jeong; In Woo Choi; Kanghoon Yim; Seonghun Jeong; Minjin Kim; Seung Ju Choi; Yongjoon Cho; Jeong-Ho An; Hak-Beom Kim; Yimhyun Jo; So-Huei Kang; Jin-Hyuk Bae; Chan-Woo Lee; Dong Suk Kim; Changduk Yang. Nature Photonics (2022). Cited by 225. Found by: openalex. DOI: 10.1038/s41566-021-00931-7

Compositional Engineering for Thermally Stable, Highly Efficient Perovskite Solar Cells Exceeding 20% Power Conversion Efficiency with 85 °C/85% 1000 h Stability

Taisuke Matsui; Teruaki Yamamoto; Takashi Nishihara; Ryosuke Morisawa; Tomoyasu Yokoyama; Takashi Sekiguchi; Takayuki Negami. Advanced Materials (2019). Cited by 222. Found by: openalex. openalex_2yr: 22.086 (2026). DOI: 10.1002/adma.201806823

Abstract: Abstract Perovskite solar cells have received great attention because of their rapid progress in efficiency, with a present certified highest efficiency of 23.3%. Achieving both high efficiency and high thermal stability is one of the biggest challenges currently limiting perovskite solar cells because devices displaying stability at high temperature frequently suffer from a marked decrease of efficiency. In this report, the relationship between perovskite composition and device thermal stability is examined. It is revealed that Rb can suppress the growth of PbI2 even under PbI2-rich conditions and decreasing the Br ratio in the perovskite absorber layer can prevent the generation of unwanted RbBr-based aggregations. The optimized device achieved by engineering perovskite composition exhibits 92% power conversion efficiency retention in a stress test conducted at 85 °C/85% relative humidity (RH) according to an international standard (IEC 61215) while exceeding 20% power conversion efficiency (certified efficiency of 20.8% at 1 cm2). These results reveal the great potential for the practical use of perovskite solar cells in the near future.

High-Performance Flexible Perovskite Solar Cells via Precise Control of Electron Transport Layer

Keqing Huang; Yongyi Peng; Yaxin Gao; Jiao Shi; Hengyue Li; Xindi Mo; Han Huang; Yongli Gao; Liming Ding; Junliang Yang. Advanced Energy Materials (2019). Cited by 221. Found by: openalex. openalex_2yr: 18.63 (2026). DOI: 10.1002/aenm.201901419

Abstract: Abstract Flexible perovskite solar cells (f-PSCs) have attracted great attention due to their promising commercial prospects. However, the performance of f-PSCs is generally worse than that of their rigid counterparts. Herein, it is found that the unsatisfactory performance of planar heterojunction (PHJ) f-PSCs can be attributed to the undesirable morphology of electron transport layer (ETL), which results from the rough surface of the flexible substrate. Precise control over the thickness and morphology of ETL tin dioxide (SnO2) not only reduces the reflectance of the indium tin oxide (ITO) on polyethylene 2,6-naphthalate (PEN) substrate and enhances photon collection, but also decreases the trap-state densities of perovskite films and the charge transfer resistance, leading to a great enhancement of device performance. Consequently, the f-PSCs, with a structure of PEN/ITO/SnO2/perovskite/Spiro-OMeTAD/Ag, exhibit a power conversion efficiency (PCE) up to 19.51% and a steady output of 19.01%. Furthermore, the f-PSCs show a robust bending resistance and maintain about 95% of initial PCE after 6000 bending cycles at a bending radius of 8 mm, and they present an outstanding long-term stability and retain about 90% of the initial performance after >1000 h storage in air (10% relative humidity) without encapsulation.

Encapsulation of Perovskite Solar Cells for High Humidity Conditions

Qi Dong; Fangzhou Liu; Man Kwong Wong; Ho Won Tam; Aleksandra B. Djurišić; Annie Ng; C. Surya; Wai Kin Chan; Alan Man Ching Ng. ChemSusChem (2016). Cited by 219. Found by: openalex. openalex_2yr: 5.476 (2026). DOI: 10.1002/cssc.201600868

Abstract: Abstract We examined different encapsulation strategies for perovskite solar cells by testing the device stability under continuous illumination, elevated temperature (85 °C) and ambient humidity of 65 %. The effects of the use of different epoxies, protective layers and the presence of desiccant were investigated. The best stability (retention of 80 % of initial efficiency on average after 48 h) was obtained for devices protected by a SiO2 film and encapsulated with a UV-curable epoxy including a desiccant sheet. However, the stability of ZnO-based cells encapsulated by the same method was found to be inferior to that of TiO2-based cells. Finally, outdoor performance tests were performed for TiO2-based cells (30–90 % ambient humidity). All the stability tests were performed following the established international summit on organic photovoltaic stability (ISOS) protocols for organic solar cell testing (ISOS-L2 and ISOS-O1).

Perovskite Solar Cells: A Review of the Latest Advances in Materials, Fabrication Techniques, and Stability Enhancement Strategies

Rakesh A. Afre; Diego Pugliese. Micromachines (2024). Cited by 218. Found by: openalex. DOI: 10.3390/mi15020192

Abstract: Perovskite solar cells (PSCs) are gaining popularity due to their high efficiency and low-cost fabrication. In recent decades, noticeable research efforts have been devoted to improving the stability of these

cells under ambient conditions. Moreover, researchers are exploring new materials and fabrication techniques to enhance the performance of PSCs under various environmental conditions. The mechanical stability of flexible PSCs is another area of research that has gained significant attention. The latest research also focuses on developing tin-based PSCs that can overcome the challenges associated with lead-based perovskites. This review article provides a comprehensive overview of the latest advances in materials, fabrication techniques, and stability enhancement strategies for PSCs. It discusses the recent progress in perovskite crystal structure engineering, device construction, and fabrication procedures that has led to significant improvements in the photo conversion efficiency of these solar devices. The article also highlights the challenges associated with PSCs such as their poor stability under ambient conditions and discusses various strategies employed to enhance their stability. These strategies include the use of novel materials for charge transport layers and encapsulation techniques to protect PSCs from moisture and oxygen. Finally, this article provides a critical assessment of the current state of the art in PSC research and discusses future prospects for this technology. This review concludes that PSCs have great potential as a low-cost alternative to conventional silicon-based solar cells but require further research to improve their stability under ambient conditions in view of their definitive commercialization.

A Low-Temperature Thin-Film Encapsulation for Enhanced Stability of a Highly Efficient Perovskite Solar Cell

Young Il Lee; Nam Joong Jeon; Bong Jun Kim; Hyunjeong Shim; Tae-Youl Yang; Sang Il Seok; Jangwon Seo; Sung Gap Im. Advanced Energy Materials (2017). Cited by 217. Found by: openalex. openalex_2yr: 18.63 (2026). DOI: 10.1002/aenm.201701928

Abstract: Abstract The stability of a perovskite solar cell (PSC) is enhanced significantly by applying a customized thin-film encapsulation (TFE). The TFE is composed of a multilayer stack of organic/inorganic layers deposited by initiated chemical vapor deposition and atomic layer deposition, respectively, whose water vapor transmission rate is on the order of 10 −4 g m −2 d −1 at an accelerated condition of 38 °C and 90% relative humidity (RH). The TFE is optimized, taking into consideration various aspects of thermosensitive PSCs. Lowering the process temperature is one of the most effective methods for minimizing the thermal damage to the PSC during the monolithic integration of the TFE onto PSC. The direct deposition of TFE onto a PSC causes less than 0.3% degradation (from 18.5% to 18.2%) in the power conversion efficiency, while the long-term stability is substantially improved; the PSC retains 97% of its original efficiency after a 300 h exposure to an accelerated condition of 50 °C and 50% RH, confirming the enhanced stability of the PSC against moisture. This is the first demonstration of a TFE applied directly onto PSCs in a damage-free manner, which will be a powerful tool for the development of highly stable PSCs with high efficiency.

Electric-Field-Induced Degradation of Methylammonium Lead Iodide Perovskite Solar Cells

Soohyun Bae; Seongtak Kim; Sang-Won Lee; Kyung Jin Cho; Sungeun Park; Seunghun Lee; Yoonmook Kang; Hae-Seok Lee; Dong-Hwan Kim. The Journal of Physical Chemistry Letters (2016). Cited by 216. Found by: openalex. DOI: 10.1021/acs.jpclett.6b01176

Abstract: Perovskite solar cells have great potential for high efficiency generation but are subject to the impact of external environmental conditions such as humidity, UV and sun light, temperature, and electric fields. The long-term stability of perovskite solar cells is an important issue for their commercialization. Various studies on the stability of perovskite solar cells are currently being performed; however, the stability related to electric fields is rarely discussed. Here the electrical stability of perovskite solar cells is studied. Ion migration is confirmed using the temperature-dependent dark current decay. Changes in the power conversion efficiency according to the amount of the external bias are measured in the dark, and a significant drop is observed only at an applied voltage greater than 0.8 V. We demonstrate that perovskite solar cells are stable under an electric field up to the operating voltage.

Efficient and stable tin-based perovskite solar cells by introducing -conjugated Lewis base

Tianhao Wu; Xiao Liu; Xin He; Yanbo Wang; Xiangyue Meng; Takeshi Noda; Xudong Yang; Liyuan Han. Science China Chemistry (2019). Cited by 212. Found by: openalex. DOI: 10.1007/s11426-019-9653-8

Tailoring the Functionality of Organic Spacer Cations for Efficient and Stable Quasi-2D Perovskite Solar Cells

Weifei Fu; Hongbin Liu; Xueliang Shi; Lijian Zuo; Xiaosong Li; Alex K.-Y. Jen. Advanced Functional Materials (2019). Cited by 211. Found by: openalex. openalex_2yr: 14.975 (2026). DOI: 10.1002/adfm.201900221

Abstract: Abstract The recent development of quasi-2D perovskite solar cells have drawn significant attention due to the improved stability of these materials and devices against moisture compared to their 3D counterparts. However, the optoelectronic properties of 2D perovskites need to be optimized in order to achieve high efficiency. In this work, the effect of spacer cations, i.e., phenethylammonium (PEA), 4-fluorophenethylammonium (F-PEA), and 4-methoxyphenethylammonium (MeO-PEA) on the optoelectronic properties and device performance of quasi-2D perovskites is systematically studied. It is observed that both larger and more hydrophobic cations can improve perovskite stability against moisture, while larger size can adversely influence the device performance. Interestingly, with F-PEA or MeO-PEA, distribution of n value can be shifted toward high 3D content in quasi-2D perovskite layers, which enables lower bandgaps and possibly lower exciton binding energy. Due to the best charge transport and lowest bandgap, the F-PEAI-based quasi-2D perovskite (n = 5) solar cell shows a highest power conversion efficiency (PCE) of 14.5% with excellent stability in air with a humidity of 40–50%, keeping 90% of the original PCE after 40 d. It is believed that the approach may open a way for the design of new organic spacer cations for stable low-dimensional hybrid perovskites with high performance.

Importance of Functional Groups in Cross-Linking Methoxysilane Additives for High-Efficiency and Stable Perovskite Solar Cells

Lin Xie; Jiangzhao Chen; Parth Vashishtha; Xing Zhao; Gwang Su Shin; Subodh G. Mhaisalkar; Nam-Gyu Park. ACS Energy Letters (2019). Cited by 211. Found by: openalex. openalex_2yr: 16.712 (2026). DOI: 10.1021/acsenergylett.9b01356

Abstract: Here we report an efficient and reproducible multifunctional additive engineering strategy via methoxysilane cross-linking agents functionalized by the different terminal group, moderate electron-donating −SH, weak electron-donating −CH 3, or strong electron-withdrawing −CN, into a PbI 2 precursor solution. The power conversion efficiency (PCE) is increased from 18.4 to 20.8% after introduction of (3-mercaptopropyl)trimethoxysilane (MPTS) containing a −SH group as a consequence of improved voltage and current density, while 3-cyanopropyltriethoxysilane (CPTS) containing a −CN group deteriorates the overall photovoltaic performance. Moreover, −SH in MPTS is found to passivate defects effectively through a Lewis acid–base interaction with PbI 2, resulting in a larger grain size and a longer carrier lifetime. Owing to the formation of a cross-linking siloxane network as a protective layer on the grain boundary, the thermal and moisture stability of the device are improved remarkably. The present work provides a guideline for multifunctional additive engineering for the purpose of simultaneous achievement of a high PCE and long-term stability.

Perovskite Solar Cells with Inorganic Electron- and Hole-Transport Layers Exhibiting Long-Term (500 h) Stability at 85 °C under Continuous 1 Sun Illumination in Ambient Air

Seongrok Seo; Seonghwa Jeong; Changdeuck Bae; Nam-Gyu Park; Hyunjung Shin. Advanced Materials (2018). Cited by 209. Found by: openalex. openalex_2yr: 22.086 (2026). DOI: 10.1002/adma.201801010

Abstract: Despite the high power conversion efficiency (PCE) of perovskite solar cells (PSCs), poor long-term stability is one of the main obstacles preventing their commercialization. Several approaches to enhance the stability of PSCs have been proposed. However, an accelerating stability test of PSCs at high temperature under the operating conditions in ambient air remains still to be demonstrated. Herein, interface-engineered stable PSCs with inorganic charge-transport layers are shown. The highly conductive Al-doped ZnO films act as efficient electron-transporting layers as well as dense passivation layers. This layer prevents underneath perovskite from moisture contact, evaporation of components, and reaction with a metal electrode. Finally, inverted-type PSCs with inorganic charge-transport layers exhibit a PCE of 18.45% and retain 86.7% of the initial efficiency for 500 h under continuous 1 Sun illumination at 85 °C in ambient air with electrical biases (at maximum power point tracking).

Thermodynamically Self-Healing 1D–3D Hybrid Perovskite Solar Cells

Jiandong Fan; Yunping Ma; Cuiling Zhang; Chong Liu; Wenzhe Li; R.E.I. Schropp; Yaohua Mai. Advanced Energy Materials (2018). Cited by 208. Found by: openalex. openalex_2yr: 18.63 (2026). DOI: 10.1002/aenm.201703421

Abstract: Abstract Thermal degradation in perovskite solar cells is still an unsettled issue that limits its further development. In this study, 2-(1 H -pyrazol-1-yl)pyridine is introduced into lead halide 3D perovskites, which allows 1D–3D hybrid perovskite materials to be obtained. The heterostructural 1D–3D perovskites are proved to be capable of remarkably prolonging the photoluminescence decay lifetime and suppressing charge carrier recombination in comparison to conventional 3D perovskites. The intrinsic properties of thermodynamically

stable yet kinetically labile 1D materials allow the system to alleviate the lattice mismatch and passivate the interface traps of heterojunction region of 1D–3D hybrid perovskites that may occur during the crystal growth process. Importantly, the as-fabricated 1D–3D perovskite solar cells display a thermodynamic self-healing ability, which is induced through blocking the ion-migration channels of A-site ions by the flexible 1D perovskite with less densely close-packed structure. Particularly, the power conversion efficiency of as-fabricated unencapsulated 1D–3D perovskite solar cells is demonstrated to be reversible under temperature cycling (25–85 °C) at 55% relative humidity, which largely outperforms the pure 3D perovskite solar cell. The present study provides a facile approach to fabricate 1D–3D perovskite solar cells with high efficiency and long-term stability.

Hydrophobic Organic Hole Transporters for Improved Moisture Resistance in Metal Halide Perovskite Solar Cells

Tomas Leijtens; Tommaso Giovenzana; Severin N. Habisreutinger; Jonathan S. Tinkham; Nakita K. Noel; Brett A. Kamino; Golnaz Sadoughi; Alan Sellinger; Henry J. Snaith. ACS Applied Materials & Interfaces (2016). Cited by 207. Found by: openalex. openalex_2yr: 8.382 (2026). DOI: 10.1021/acsami.5b10093

Abstract: Solar cells based on organic-inorganic perovskite semiconductor materials have recently made rapid improvements in performance, with the best cells performing at over 20% efficiency. With such rapid progress, questions such as cost and solar cell stability are becoming increasingly important to address if this new technology is to reach commercial deployment. The moisture sensitivity of commonly used organic-inorganic metal halide perovskites has especially raised concerns. Here, we demonstrate that the hygroscopic lithium salt commonly used as a dopant for the hole transport material in perovskite solar cells makes the top layer of the devices hydrophilic and causes the solar cells to rapidly degrade in the presence of moisture. By using novel, low cost, and hydrophobic hole transporters in conjunction with a doping method incorporating a preoxidized salt of the respective hole transporters, we are able to prepare efficient perovskite solar cells with greatly enhanced water resistance.

Wafer-scale monolayer MoS 2 film integration for stable, efficient perovskite solar cells

Huachao Zai; Pengfei Yang; Jie Su; Ruiyang Yin; Rundong Fan; Yuetong Wu; Xiao Zhu; Yue Ma; Tong Zhou; Wentao Zhou; Yu Zhang; Yu Zhang; Zijian Huang; Yiting Jiang; Nengxu Li; Yang Bai; Cheng Zhu; Zhaohui Huang; Jingjing Chang; Qi Chen; Yanfeng Zhang; Yanfeng Zhang; Huanping Zhou. Science (2025). Cited by 207. Found by: openalex. DOI: 10.1126/science.ado2351

Abstract: One of the primary challenges in commercializing perovskite solar cells (PSCs) is achieving both high power conversion efficiency (PCE) and sufficient stability. We integrate wafer-scale continuous monolayer MoS 2 buffers at the top and bottom of a perovskite layer through a transfer process. These films physically block ion migration of perovskite into carrier transport layers and chemically stabilize the formamidinium lead iodide phase through strong coordination interaction. Effective chemical passivation results from the formation of Pb-S bonds, and minority carriers are blocked through a type-I band alignment. Planar p-i-n PSCs (0.074 square centimeters) and modules (9.6 square centimeters) with MoS 2 /perovskite/MoS 2 configuration achieve PCEs up to 26.2% (certified steady-state PCE of 25.9%) and 22.8%, respectively. Moreover, the devices show excellent damp heat (85°C and 85% relative humidity) stability with <5% PCE loss after 1200 hours and notable high temperature (85°C) operational stability with <4% PCE loss after 1200 hours.

Synergistic Effects of Eu-MOF on Perovskite Solar Cells with Improved Stability

Jie Dou; Cheng Zhu; Hao Wang; Ying Han; Sai Ma; Xiuxiu Niu; Nengxu Li; Congbo Shi; Zhiwen Qiu; Huanping Zhou; Yang Bai; Qi Chen. Advanced Materials (2021). Cited by 206. Found by: openalex. openalex_2yr: 22.086 (2026). DOI: 10.1002/adma.202102947

Abstract: Abstract Enhancing device lifetime is one of the essential challenges in perovskite solar cells. The ultrathin Eu-MOF layer is introduced at the interface between the electron-transport layer and the perovskite absorber to improve the device stability. Both Eu ions and organic ligands in the MOF can reduce the defect concentration and improve carrier transport. Moreover, due to the Förster resonance energy transfer effect, Eu-MOF in perovskite films can improve light utilization and reduce the decomposition under ultraviolet light. Meanwhile, Eu-MOF also turns tensile strain to compressive strain in the perovskite films. As a result, the corresponding devices achieve a champion power conversion efficiency (PCE) of 22.16%. In addition, the devices retain 96% of their original PCE after 2000 h under the relative humidity of 30% and 91% of their original PCE after 1200 h after continuous 85 °C aging condition in N 2 .

Molecular materials as interfacial layers and additives in perovskite solar cells

Maria Vasilopoulou; Azhar Fakharuddin; Athanassios G. Coutsolelos; Polycarpos Falaras; Panagiotis Argitis; Abd. Rashid bin Mohd Yusoff; Mohammad Khaja Nazeeruddin. Chemical Society Reviews (2020). Cited by 206. Found by: openalex. openalex_2yr: 42.455 (2026). DOI: 10.1039/c9cs00733d

Abstract: Solar cells based on organo-metal halide perovskites have gained unprecedented research interest over the last few years due to their low-cost solution processability, high power conversion efficiency, which has recently reached a certified value of 25.2%, and abundance of raw materials. Nevertheless, the best efficiencies remain below the Shockley-Queisser theoretical limit of 32.5% due to several losses arising from either defect traps present in the bulk of the perovskite absorber or at the device heterointerfaces. While bulk defects are detrimental for the device performance by mainly limiting the open circuit voltage, interfacial layers are also crucial. They dictate the charge transfer/transport from the perovskite layer to the collecting electrodes, hence influencing the device photocurrent, but also act as protective barriers against oxygen and moisture penetration. Molecular materials and additives are widely used to improve the bulk properties of perovskite absorbers through the formation of high-quality perovskite films with superior optoelectronic properties, and improved crystallinity, and also of electronically clean interfaces with minimum losses during charge transfer/transport. In this review, we analyze the predominant pathways that contribute to voltage and current losses due to poor interfaces and also due to non-radiative recombination losses arising from inferior perovskite morphology and its inherent polycrystalline and highly defective nature. We then discuss strategies for achieving interfacial organic and inorganic molecular materials for application as electron and hole transport layers in perovskite solar cells with ideal energy levels, high charge mobilities and improved thermal, photo, and structural stability. Moreover, the prerequisites for molecular additives to achieve dimensionality engineering, defect passivation, molecular cross-linking, interfacial energy alignment and electronic doping are thoroughly discussed. Finally, we examine prospects for future research directions and commercialization.

Continuous Grain-Boundary Functionalization for High-Efficiency Perovskite Solar Cells with Exceptional Stability

Yingxia Zong; Yuanyuan Zhou; Yi Zhang; Zhipeng Li; Lin Zhang; Ming-Gang Ju; Min Chen; Shuping Pang; Xiao Cheng Zeng; Nitin P. Padture. Chem (2018). Cited by 204. Found by: openalex. openalex_2yr: 11.7 (2026). DOI: 10.1016/j.chempr.2018.03.005

In situ studies of the degradation mechanisms of perovskite solar cells

Soumya Kundu; Timothy L. Kelly. EcoMat (2020). Cited by 201. Found by: openalex. openalex_2yr: 7.557 (2026). DOI: 10.1002/eom2.12025

Abstract: Abstract The last decade has seen an extraordinary rise in the performance of perovskite solar cells (PSCs). State-of-the-art devices now have efficiencies of over 25%, putting them on par with the best silicon solar cells. Yet despite their impressive performance, their longevity lags behind that of conventional silicon technology. Environmental factors like moisture, heat, and light can all adversely affect PSC performance and limit device lifetime. Systematically elucidating and eliminating PSC degradation pathways will be critical to the success of this technology. In situ techniques provide powerful tools to this end, as they allow structural, compositional, morphological, and optoelectronic changes to be tracked in real-time. Because they follow a single film or device over the course of the degradation process, they can help eliminate the statistical variation that negatively affects many studies. Here we provide an overview of perovskite degradation processes, with an emphasis on in situ studies. image

Inkjet-Printed Triple Cation Perovskite Solar Cells

Florian Mathies; Helge Eggers; Bryce S. Richards; Gerardo Hernandez-Sosa; Uli Lemmer; Ulrich W. Paetzold. ACS Applied Energy Materials (2018). Cited by 200. Found by: openalex. openalex_2yr: 5.576 (2026). DOI: 10.1021/acsaem.8b00222

Abstract: Noncontact inkjet printing offers rapid and digital deposition combined with excellent control over the layer formation for printed perovskite solar cells. In this work, inkjet printing is used to deposit triple cation perovskite layers with 10% cesium in a mixed formamidinium/methylammonium lead iodide/bromide composite for solar cells with high temperature and moisture stability. A reliable process control over a wide range of perovskite layer thickness from 175 to 780 nm and corresponding grain sizes is achieved by adjusting the drop spacing of the inkjet printer cartridge. A continuous power output at constant voltage, resulting in a power conversion efficiency of 12.9%, is demonstrated, representing a major improvement from previously

reported inkjet-printed methylammonium lead triiodide perovskite solar cells. Moreover, this work highlights the extended resistance of triple cation perovskite solar cells against heat and moisture for our ambient inkjet printing approach. The presented results are a proof of concept for the processability of high efficiency perovskite solar cells using digital inkjet printing for next generation photovoltaic applications.

Passivation of the grain boundaries of CH 3 NH 3 PbI 3 using carbon quantum dots for highly efficient perovskite solar cells with excellent environmental stability

Qiang Guo; Fanglong Yuan; Bing Zhang; Shijie Zhou; Jin Zhang; Yiming Bai; Louzhen Fan; Tasawar Hayat; Ahmed Alsaedi; Zhan'ao Tan. Nanoscale (2018). Cited by 200. Found by: openalex. DOI: 10.1039/c8nr08295b

Abstract: Organic-inorganic hybrid perovskites are prone to defect formation due to iodine and methylamine ion/defect migration, leading to the formation of lots of defects at the perovskite surface and grain boundaries. Passivation of the defects is an effective method to improve the power conversion efficiency (PCE) and stability of perovskite solar cells (PSCs). To achieve stable passivation, the interaction between the perovskite and additive materials should be taken into consideration. In this work, we for the first time introduced carbon quantum dots (CQDs) as an additive for the stabilization of MAPbI3via passivation of the grain boundaries of the perovskite. Because the carboxylic groups, hydroxyl groups and amino-groups on the edge of CQDs can bond with the uncoordinated Pb in MAPbI3, strong and stable interactions between the perovskite and CQDs can be generated, inducing a lower trap-state density and better optoelectronic properties. The typical PCE of the PSCs based on CQD modified MAPbI3 films increases from 17.59% to 18.81% and the PCE of the optimized champion PSCs reaches 19.38%. Furthermore, the hydrophobic CQD molecules can block the contact between water and MAPbI3, and even if the CQD modified perovskite is kept under ambient atmosphere without controlling the humidity for 4 months, the MAPbI3 film still retained its original black color.

Synergistic Effect of Fluorinated Passivator and Hole Transport Dopant Enables Stable Perovskite Solar Cells with an Efficiency Near 24%

Hongwei Zhu; Yameng Ren; Linfeng Pan; Olivier Ouellette; Felix T. Eickemeyer; Yinghui Wu; Xianggao Li; Shirong Wang; Hongli Liu; Xiaofei Dong; Shaik M. Zakeeruddin; Yuhang Liu; Anders Hagfeldt; Michaël Grätzel. Journal of the American Chemical Society (2021). Cited by 200. Found by: openalex. DOI: 10.1021/jacs.0c12802

Abstract: Long-term durability is critically important for the commercialization of perovskite solar cells (PSCs). The ionic character of the perovskite and the hydrophilicity of commonly used additives for the hole-transporting layer (HTL), such as lithium bis(trifluoromethanesulfonyl)imide (Li-TFSI) and tert -butylpyridine (t BP), render PSCs prone to moisture attack, compromising their long-term stability. Here we introduce a trifluoromethylation strategy to overcome this drawback and to boost the PSC's solar to electric power conversion efficiency (PCE). We employ 4-(trifluoromethyl)benzylammonium iodide (TFMBAI) as an amphiphilic modifier for interfacial defect mitigation and 4-(trifluoromethyl)pyridine (TFP) as an additive to enhance the HTL's hydrophobicity. Surface treatment of the triple-cation perovskite with TFMBAI largely suppressed the nonradiative charge carrier recombination, boosting the PCE from 20.9% to 23.9% and suppressing hysteresis, while adding TFP to the HTL enhanced the PCS's resistance to moisture while maintaining its high PCE. Taking advantage of the synergistic effects resulting from the combination of both fluoromethylated modifiers, we realize TFMBAI/TFP-based highly efficient PSCs with excellent operational stability and resistance to moisture, retaining over 96% of their initial efficiency after 500 h maximum power point tracking (MPPT) under simulated 1 sun irradiation and 97% of their initial efficiency after 1100 h of exposure under ambient conditions to a relative humidity of 60–70%.

Chiral-structured heterointerfaces enable durable perovskite solar cells

Tianwei Duan; Shuai You; Min Chen; Wenjian Yu; Yanyan Li; Peijun Guo; Joseph J. Berry; Joseph M. Luther; Kai Zhu; Yuanyuan Zhou. Science (2024). Cited by 199. Found by: openalex. DOI: 10.1126/science.ado5172

Abstract: Mechanical failure and chemical degradation of device heterointerfaces can strongly influence the long-term stability of perovskite solar cells (PSCs) under thermal cycling and damp heat conditions. We report chirality-mediated interfaces based on R -/ S -methylbenzyl-ammonium between the perovskite absorber and electron-transport layer to create an elastic yet strong heterointerface with increased mechanical reliability. This interface harnesses enantiomer-controlled entropy to enhance tolerance to thermal cycling–induced fatigue and material degradation, and a heterochiral arrangement of organic cations leads to closer packing of benzene rings, which enhances chemical stability and charge transfer. The encapsulated PSCs showed retentions of 92% of

power-conversion efficiency under a thermal cycling test (−40°C to 85°C; 200 cycles over 1200 hours) and 92% under a damp heat test (85% relative humidity; 85°C; 600 hours).

Compositional and Interface Engineering of Organic-Inorganic Lead Halide Perovskite Solar Cells

Haizhou Lu; Anurag Krishna; Shaik M. Zakeeruddin; Michaël Grätzel; Anders Hagfeldt. iScience (2020). Cited by 198. Found by: openalex. DOI: 10.1016/j.isci.2020.101359

Abstract: -based compositions. Owing to the enormous engineering works, perovskite absorber layers with monolithic grains could be achieved, in which the interior defects are negligible compared with the surface defects. Therefore, interface engineering, which can passivate the surface defects and/or isolate the perovskite from the environmental moistures, has been playing a more and more important role to further boost the PCE and stability of the PSCs. Herein, a compact review study of the compositional and interface engineering is presented and promising strategies and directions of the PSCs are discussed.

Efficient Passivation with Lead Pyridine-2-Carboxylic for High-Performance and Stable Perovskite Solar Cells

Sheng Fu; Xiaodong Li; Li Wan; Yulei Wu; Wenxiao Zhang; Yueming Wang; Qinye Bao; Junfeng Fang. Advanced Energy Materials (2019). Cited by 198. Found by: openalex. openalex_2yr: 18.63 (2026). DOI: 10.1002/aenm.201901852

Abstract: Abstract Stability has become the main obstacle for the commercialization of perovskite solar cells (PSCs) despite the impressive power conversion efficiency (PCE). Poor crystallization and ion migration of perovskite are the major origins of its degradation under working condition. Here, high-performance PSCs incorporated with pyridine-2-carboxylic lead salt (PbPyA2) are fabricated. The pyridine and carboxyl groups on PbPyA2 can not only control crystallization but also passivate grain boundaries (GBs), which result in the high-quality perovskite film with larger grains and fewer defects. In addition, the strong interaction among the hydrophobic PbPyA2 molecules and perovskite GBs acts as barriers to ion migration and component volatilization when exposed to external stresses. Consequently, superior optoelectronic perovskite films with improved thermal and moisture stability are obtained. The resulting device shows a champion efficiency of 19.96% with negligible hysteresis. Furthermore, thermal (90 °C) and moisture (RH 40–60%) stability are improved threefold, maintaining 80% of initial efficiency after aging for 480 h. More importantly, the doped device exhibits extraordinary improvement of operational stability and remains 93% of initial efficiency under maximum power point (MPP) tracking for 540 h.

Highly Air-Stable Tin-Based Perovskite Solar Cells through Grain-Surface Protection by Gallic Acid

Tianyue Wang; Qidong Tai; Xuyun Guo; Jiupeng Cao; Chun-Ki Liu; Naixiang Wang; Dong Shen; Ye Zhu; Chun-Sing Lee; Feng Yan. ACS Energy Letters (2020). Cited by 194. Found by: openalex. openalex_2yr: 16.712 (2026). DOI: 10.1021/acsenergylett.0c00526

Abstract: Maintaining the stability of tin halide perovskites is a major challenge in developing lead-free perovskite solar cells (PSCs). Adding extra SnX 2 (X = F, Cl, or Br) in the precursor solution to inhibit Sn 2+ oxidation is an essential strategy to improve device efficiency and stability. However, SnX 2 on the surface of perovskite grains tends to prohibit charge transfer across perovskite films. Here, we report a coadditive engineering approach by introducing antioxidant gallic acid (GA) together with SnCl 2 to improve the performance of tin-based PSCs. The SnCl 2 –GA complex can not only protect the perovskite grains but also more effectively conduct electrons across it, leading to highly stable and efficient PSCs. The unencapsulated devices can maintain 80% of their initial efficiency after 1000 h of storage in ambient air with a relative humidity of 20%, which is the best air stability achieved in tin-based PSCs to date.

Propylammonium Chloride Additive for Efficient and Stable FAPbI3 Perovskite Solar Cells

Yong Zhang; Yan Li; Lin Zhang; Hanlin Hu; Zikang Tang; Baomin Xu; Nam-Gyu Park. Advanced Energy Materials (2021). Cited by 192. Found by: openalex. openalex_2yr: 18.63 (2026). DOI: 10.1002/aenm.202102538

Abstract: Abstract To achieve high efficiency perovskite solar cells (PSCs) based on -phase formamidinium lead iodide (FAPbI3), addition of methylammonium chloride (MACl) in the precursor solution is commonly used, mainly because of phase stability and improvement of grain size and crystallinity. However, the instability of MA in the perovskite limits the device long-term stability. In this report, n-propylammonium chloride (PACl)

is proposed as an alternative to MACl for more stable and efficient FAPbI3-based PSCs. Perovskite grain size is increased after addition of PACl. Unlike the MA cation, the propylammonium cation passivates the grain boundary rather than being incorporated into the perovskite lattice due to larger ionic size, which minimizes the change in bandgap. Carrier lifetime is significantly increased by more than five times from 405 to 2110 ns with the PACl additive with negligible trap-mediated recombination, while only four times longer carrier lifetime is observed by MACl additive. As a result, a power conversion efficiency over 22.2% is achieved by 20 mol% PACl additive, which is one of the best efficiencies among the MA-free and Br-free PSCs. In addition, stability against moisture is much better for PACl than for MACl due to an in situ formed barrier at the bulk perovskite.

Highly Efficient and Stable Perovskite Solar Cells by Interfacial Engineering Using Solution-Processed Polymer Layer

Feijiu Wang; Ai Shimazaki; Fengjiu Yang; Kaito Kanahashi; Keiichiro Matsuki; Yuhei Miyauchi; Taishi Takenobu; Atsushi Wakamiya; Yasujiro Murata; Kazunari Matsuda. The Journal of Physical Chemistry C (2017). Cited by 191. Found by: openalex. DOI: 10.1021/acs.jpcc.6b12137

Abstract: Solution-processed organo-lead halide perovskite solar cells with deep pinholes in the perovskite layer lead to shunt-current leakage in devices. Herein, we report a facile method for improving the performance of perovskite solar cells by inserting a solution-processed polymer layer between the perovskite layer and the hole-transporting layer. The photovoltaic conversion efficiency of the perovskite solar cell increased to 18.1% and the stability decreased by only about 5% during 20 days of exposure in moisture ambient conditions through the incorporation of a poly(methyl methacrylate) (PMMA) polymer layer. The improved photovoltaic performance of devices with a PMMA layer is attributed to the reduction of carrier recombination loss from pinholes, boundaries, and surface states of perovskite layer. The significant gain generated by this simple procedure supports the use of this strategy in further applications of thin-film optoelectronic devices.

Poly(4-Vinylpyridine)-Based Interfacial Passivation to Enhance Voltage and Moisture Stability of Lead Halide Perovskite Solar Cells

Bhumika Chaudhary; Ashish Kulkarni; Ajay Kumar Jena; M. Ikegami; Yosuke Udagawa; Hideyuki Kunugita; Kazuhiro Ema; Tsutomu Miyasaka. ChemSusChem (2017). Cited by 190. Found by: openalex. openalex_2yr: 5.476 (2026). DOI: 10.1002/cssc.201700271

Abstract: Abstract It is well known that the surface trap states and electronic disorders in the solution-processed CH 3 NH 3 PbI 3 perovskite film affect the solar cell performance significantly and moisture sensitivity of photoactive perovskite material limits its practical applications. Herein, we show the surface modification of a perovskite film with a solution-processable hydrophobic polymer (poly(4-vinylpyridine), PVP), which passivates the undercoordinated lead (Pb) atoms (on the surface of perovskite) by its pyridine Lewis base side chains and thereby eliminates surface-trap states and non-radiative recombination. Moreover, it acts as an electron barrier between the perovskite and hole-transport layer (HTL) to reduce interfacial charge recombination, which led to improvement in open-circuit voltage (V oc) by 120 to 160 mV whereas the standard cell fabricated in same conditions showed V oc as low as 0.9 V owing to dominating interfacial recombination processes. Consequently, the power conversion efficiency (PCE) increased by 3 to 5 % in the polymer-modified devices (PCE=15 %) with V oc more than 1.05 V and hysteresis-less J – V curves. Advantageously, hydrophobicity of the polymer chain was found to protect the perovskite surface from moisture and improved stability of the non-encapsulated cells, which retained their device performance up to 30 days of exposure to open atmosphere (50 % humidity).

Monoammonium Porphyrin for Blade-Coating Stable Large-Area Perovskite Solar Cells with >18% Efficiency

Congping Li; Jun Yin; Ruihao Chen; Xudong Lv; Xiaoxia Feng; Yiying Wu; Jing Cao. Journal of the American Chemical Society (2019). Cited by 190. Found by: openalex. DOI: 10.1021/jacs.9b01305

Abstract: Efficient control of crystallization and defects of perovskite films are the key factors toward the performance and stability of perovskite solar cells (PSCs), especially for the preparation of large-area PSCs devices. Herein, we directly embedded surfactant-like monoammonium zinc porphyrin (ZnP) compound into the methylammonium (MA +) lead iodide perovskite film to blade-coat large-area uniform perovskite films as large as 16 cm 2 . Efficiency as high as 18.3% for blade-coating large-area (1.96 cm 2) PSCs with ZnP was unprecedentedly achieved, while the best efficiency of fabricated small-area (0.1 cm 2) device was up to 20.5%. The detailed analyses demonstrated the functions of ZnP in crystallization control and defects passivation of perovskite surfaces and grain boundaries. As a consequence, the ZnP-encapsulated devices retained over 90%

of its initial efficiency after 1000 h with a humidity of about 45% at 85 °C. This research presents a facile way to achieve the synergistic effect of large-area coating, morphology tailoring, and defect suppression based on the molecular encapsulation strategy for perovskite films, further improving the photovoltaic performance and stability of PSCs.

Perovskite solar cells based on spiro-OMeTAD stabilized with an alkylthiol additive

Xu Liu; Bolin Zheng; Lei Shi; Shujie Zhou; Jiangtao Xu; Ziheng Liu; Jae Sung Yun; Eun Young Choi; Meng Zhang; Yinhua Lv; Wen-Hua Zhang; Jialiang Huang; Caixia Li; Kaiwen Sun; Jan Seidel; Mingrui He; Jun Peng; Xiaojing Hao; Martin A. Green. Nature Photonics (2022). Cited by 189. Found by: openalex. DOI: 10.1038/s41566-022-01111-x

NbF 5 : A Novel -Phase Stabilizer for FA-Based Perovskite Solar Cells with High Efficiency

Shihao Yuan; Qian Fang; Shaomin Yang; Yuan Cai; Qiang Wang; Jie Sun; Zhike Liu; Shengzhong Liu. Advanced Functional Materials (2019). Cited by 189. Found by: openalex. openalex_2yr: 14.975 (2026). DOI: 10.1002/adfm.201807850

Abstract: Abstract The HC(NH 2) 2 + (FA +) is a well-known substitute to CH 3 NH 3 + (MA +) for its capability to extend light utilization for improved power conversion efficiency for perovskite solar cells; unfortunately, the dark cubic phase (-phase) can easily transition to the yellow orthorhombic phase (-phase) at room temperature, an issue that prevents its commercial application. In this report, an inorganic material (NbF 5) is developed to stabilize the desired -phase perovskite material by incorporating NbF 5 additive into the perovskite films. It is found that the NbF 5 additive effectively suppresses the formation of the yellow -phase in the perovskite synthesis and aging process, thus enhancing the humidity and light-soaking stability of the perovskite film. As a result, the perovskite solar cells with the NbF 5 additive exhibit improved air stability by tenfold, retaining nearly 80% of their initial efficiency after aging in air for 50 d. In addition, under full-sun AM 1.5 G illumination of a xenon lamp without any UV-reduction, the perovskite solar cells with the NbF 5 additive also show fivefold improved illumination stability than the control devices without NbF 5 .

Enhancing the Performance of Inverted Perovskite Solar Cells via Grain Boundary Passivation with Carbon Quantum Dots

Yuhui Ma; Heyi Zhang; Yewei Zhang; Ruiyuan Hu; Mao Fa Jiang; Rui Zhang; Hao Lv; Jingjing Tian; Liang Chu; Jian Zhang; Qifan Xue; Hin-Lap Yip; Ruidong Xia; Xing'ao Li; Wei Huang. ACS Applied Materials & Interfaces (2018). Cited by 187. Found by: openalex. openalex_2yr: 8.382 (2026). DOI: 10.1021/acsami.8b18867

Abstract:), is one of the crucial problems for achieving high power conversion efficiency (PCE) in inverted perovskite solar cells (PSCs). Usually, grain boundary passivation is considered as an effective way to reduce nonradiative recombination because the defects (uncoordinated ions) on grain boundaries are passivated. We added the hydroxyl and carbonyl functional groups containing carbon quantum dots (CQDs) into a perovskite precursor solution to passivate the uncoordinated lead ions on grain boundaries. Higher photoluminescence intensity and longer carrier lifetime were demonstrated in the perovskite film with the CQD additive. This confirmed that the addition of CQDs can reduce nonradiative recombination by grain boundary passivation. Additionally, the introduction of CQDs could increase the thickness of the perovskite film. Consequently, we achieved a champion device with a PCE of 18.24%. The device with CQDs retained 73.4% of its initial PCE after being aged for 48 h under 80% humidity in the dark at room temperature. Our findings reveal the mechanisms of how CQDs passivate the grain boundaries of perovskite, which can improve the efficiency and stability of PSCs.

Defect-Engineering-Enabled High-Efficiency All-Inorganic Perovskite Solar Cells

Jia Liang; Xiao Han; Ji-Hui Yang; Boyu Zhang; Qiyi Fang; Jing Zhang; Qing Ai; Meredith M. Ogle; Tanguy Terlier; Angel A. Martí; Jun Lou. Advanced Materials (2019). Cited by 186. Found by: openalex. openalex_2yr: 22.086 (2026). DOI: 10.1002/adma.201903448

Abstract: Abstract The emergence of cesium lead iodide (CsPbI 3) perovskite solar cells (PSCs) has generated enormous interest in the photovoltaic research community. However, in general they exhibit low power conversion efficiencies (PCEs) because of the existence of defects. A new all-inorganic perovskite material, CsPbI 3 :Br:InI 3 , is prepared by defect engineering of CsPbI 3 . This new perovskite retains the same bandgap as CsPbI 3 , while the intrinsic defect concentration is largely suppressed. Moreover, it can be prepared in an

extremely high humidity atmosphere and thus a glovebox is not required. By completely eliminating the labile and expensive components in traditional PSCs, the all-inorganic PSCs based on CsPbI 3 :Br:InI 3 and carbon electrode exhibit PCE and open-circuit voltage as high as 12.04% and 1.20 V, respectively. More importantly, they demonstrate excellent stability in air for more than two months, while those based on CsPbI 3 can survive only a few days in air. The progress reported represents a major leap for all-inorganic PSCs and paves the way for their further exploration in order to achieve higher performance.

Interface Engineering of Imidazolium Ionic Liquids toward Efficient and Stable CsPbBr 3 Perovskite Solar Cells

Wenyu Zhang; Xiaojie Liu; Benlin He; Zekun Gong; Jingwei Zhu; Yang Ding; Haiyan Chen; Qunwei Tang. ACS Applied Materials & Interfaces (2020). Cited by 186. Found by: openalex. openalex_2yr: 8.382 (2026). DOI: 10.1021/acsami.9b20831

Abstract: The defect passivation of perovskite films is an efficacious way to further boost the power conversion efficiency (PCE) and long-term stability of perovskite solar cells (PSCs). In this work, ionic liquids (ILs) of 1-butyl-2,3-dimethylimidazolium chloride ([BMMIm]Cl) are used as a modification layer in perovskite films in carbon-based CsPbBr 3 PSCs without a hole-transporting material (HTM) for passivating the surface defects. The preliminary results demonstrate that the [BMMIm]Cl modifier passivates the surface defects of the perovskite film and reduces the valence band of perovskite close to the work function of the carbon electrode, which causes a remarkably inhibited nonradiative and radiative charge recombination, improved energy-level matching, and decreased energy loss. After optimization, a champion efficiency of 9.92% with a V oc as high as 1.61 V is achieved for the [BMMIm]Cl tailored carbon-based CsPbBr 3 PSC without HTM, which is improved by 61.3% in comparison with 6.15% for the control device. Furthermore, the encapsulation-free PSC presents good long-term stability after storage in an air atmosphere with 70% RH at 20 °C or 0% RH at 80 °C as well as under continuous illumination conditions for 30 days. The significantly improved PCE and stability in high humidity or temperature suggest that the perovskite passivation by ILs is an effective strategy for fabricating high-PCE and stable PSCs.

Review of Novel Passivation Techniques for Efficient and Stable Perovskite Solar Cells

Jincheol Kim; Anita Ho-Baillie; Shujuan Huang. Solar RRL (2019). Cited by 185. Found by: openalex. DOI: 10.1002/solr.201800302

Abstract: Perovskite solar cells contain various defects within the perovskite absorber and the corresponding interfaces, affecting device performance and stability. Fortunately, there have been tremendous efforts in advancing passivation techniques contributing to high-efficiency perovskite solar cell with improved stability. Here, the state-of-the-art passivation approaches for each layer of the perovskite cell with the aim of improving carrier extraction, reducing carrier recombination, and/or improving cell stability are reviewed. Passivation of the electron transport layer can improve the stability of perovskite solar cells by reducing trap states or by physically separating the transport layer from contacting perovskite. Controlling the amount of PbI2 in the perovskite precursor has been found to be effective in passivating defect states at the grain boundaries and on the surface. Additives such as elemental iodine, organic surfactants, and Group 1 metal compounds incorporated in perovskite precursors have been reported to passivate recombination trap centers. These approaches have also contributed to improved energy band alignment between carrier transport layers and perovskite absorber improving device performance. An effective strategy to improve moisture stability is the use of 2D perovskites or hydrophobic large cation molecules forming 2D or quasi-2D phases at grain boundaries or film surfaces providing passivation and preventing moisture ingress.

Thermal Stability of CuSCN Hole Conductor-Based Perovskite Solar Cells

Minsu Jung; Young Chan Kim; Nam Joong Jeon; Woon Seok Yang; Jangwon Seo; Jun Hong Noh; Sang Il Seok. ChemSusChem (2016). Cited by 183. Found by: openalex. openalex_2yr: 5.476 (2026). DOI: 10.1002/cssc.201600957

Abstract: Although perovskite solar cells (PSCs) surpassing 20 % in certified power conversion efficiency (PCE) have been demonstrated with organic hole-transporting layers (HTLs), thermal degradation remains one of the key issues for practical applications. We fabricated PSCs using low temperature solution-processed CuSCN as the inorganic hole-transport layer (HTL), which possesses a highly stable crystalline structure and is robust even at high temperatures. The best-performing cell delivers a PCE of 18.0 %, with 15.9 % measured at the stabilized power output. Here we report the thermal stability of PSCs based on CuSCN in comparison with commonly used 2,2',7,7'-tetrakis-(N,N-di-4-methoxyphenylamino)-9,9'-spirobifluorene (spiro-OMeTAD). The PSC

fabricated with organic spiro-OMeTAD degrades to 25 % of initial PCE after annealing for 2 h at 125 °C in air under 40 % average relative humidity. However, CuSCN-based PSCs maintain approximately 60 % of the initial value, exhibiting superior thermal stability under identical conditions. This work demonstrates that high efficiency and improved thermal stability are simultaneously achieved when CuSCN is used as an HTL in PSCs.

Investigation of the influence of different hole-transporting materials on the performance of perovskite solar cells

Elham Karimi; Seyed Mohammad Bagher Ghorashi. Optik (2016). Cited by 183. Found by: openalex. DOI: 10.1016/j.ijleo.2016.10.122

Effect of Side-Group-Regulated Dipolar Passivating Molecules on CsPbBr 3 Perovskite Solar Cells

Jialong Duan; Meng Wang; Yingli Wang; Junshuai Zhang; Qiyao Guo; Qiaoyu Zhang; Yanyan Duan; Qunwei Tang. ACS Energy Letters (2021). Cited by 183. Found by: openalex. openalex_2yr: 16.712 (2026). DOI: 10.1021/acsenergylett.1c01060

Abstract: Interfacial passivation of defective perovskite films with a dipole molecule shows a great potential for high-efficiency perovskite solar cells (PSCs). Herein, by regulating the side group of an aniline-based molecule with an electron-withdrawing nitryl (−NO 2) or electron-donating methoxyl (−OCH 3), the binding energy between −NH 2 and under-coordinated Pb 2+ in the CsPbBr 3 lattice is reinforced, and the dipole-moment-induced surface energy level reconstruction allows for efficient hole extraction for the −OCH 3 -tailored molecule because of the changed intramolecular electron distribution, whereas −NO 2 displays the opposite effect. Finally, the all-inorganic CsPbBr 3 PSC achieves a power conversion efficiency (PCE) of 9.81% with an open-circuit voltage (V oc) of 1.632 V and an excellent stability in both high humidity and light irradiation. Upon doping Sm 3+ into CsPbBr 3, the PCE and V oc are further increased to 10.75% and 1.675 V. This work confirms the importance of a side group on an interface dipole molecule to maximize the charge scavenging, especially for V oc improvement.

Crystallization Kinetics in 2D Perovskite Solar Cells

Youkui Xu; Meng Wang; Yutian Lei; Zhipeng Ci; Zhiwen Jin. Advanced Energy Materials (2020). Cited by 181. Found by: openalex. openalex_2yr: 18.63 (2026). DOI: 10.1002/aenm.202002558

Abstract: Abstract 2D perovskites demonstrate higher moisture stability, oxygen content, thermal stability, and a significantly lower ion migration/phase transition occurrence in comparison to 3D perovskite. These advantages imply huge potential for 2D perovskite in commercial applications in the photovoltaic field. However, the horizontal arrangement of the organic layer severely hinders the transport of carriers, and thus, the power conversion efficiency of 2D perovskite solar cells (PSCs) is significantly lower than that of 3D. Controlling the crystallization orientation to achieve rapid carrier transport can effectively avoid or reduce such adverse effects. Hence, an in-depth understanding of the formation mechanism and crystallization kinetics of 2D perovskite films is crucial to the development of high-performing 2D PSCs. This review explores the studies conducted on crystallization kinetics, which is the key issue for 2D perovskite, and discusses their effects on the performance of various types of 2D PSCs to date. The crystal/natural quantum well structures and origin of the stability for 2D perovskite are also summarized. Finally, the remaining challenges in terms of development bottlenecks for 2D PSCs are discussed, alongside the proposal of possible solutions.

Efficiency and stability enhancement of perovskite solar cells by introducing CsPbI3 quantum dots as an interface engineering layer

Chang Liu; Manman Hu; Xianyong Zhou; Jianchang Wu; Luozheng Zhang; Weiguang Kong; Xiangnan Li; Xingzhong Zhao; Songyuan Dai; Baomin Xu; Chun Cheng. NPG Asia Materials (2018). Cited by 181. Found by: openalex. DOI: 10.1038/s41427-018-0055-0

Abstract: Although great efforts have been devoted to enhancing the efficiency and stability of perovskite solar cells (PSCs), the performance of PSCs has been far lower than anticipated. Interface engineering is helpful for obtaining high efficiency and stability through control of the interfacial charge transfer in PSCs. This paper demonstrates that the efficiency and stability of PSCs can be enhanced by introducing stable - CsPbI3 quantum dots (QDs) as an interface layer between the perovskite film and the hole transport material (HTM) layer. By synergistically controlling the valence band position (VBP) of the perovskite and the interface

layer, an interface engineering strategy was successfully used to increase the efficiency of hole transfer from the perovskite to the HTM layer, resulting in the power conversion efficiency increasing from 15.17 to 18.56%. In addition, the enhancement of the stability of PSCs can be attributed to coating inorganic CsPbI3 QDs onto the perovskite layer, which have a high moisture stability and result in long-term stability of the PSCs in ambient air. Researchers have used tiny inorganic crystals known as quantum dots to engineer improved moisture resistance and enhanced power conversion into low-cost solar cells. Perovskites, a family of crystals that includes methylammonium lead tri-iodide, can be easily assembled into photovoltaic devices that harvest sunlight nearly as well as silicon. Baomin Xu and Chun Cheng from China's Southern University of Science and Technology, Shenzhen, and co-workers report a solar cell containing cesium–lead–iodine quantum dots inserted between a perovskite film and an organic conducting material. The dots introduce new electronic states that improve extraction of photogenerated charge. They also migrate through pores in the perovskite film to form a water-repelling surface layer. This strategy enabled the team to operate the device with high efficiency for over a month in humid conditions that normally degrade perovskite films. Due to the matching energy level of CsPbI3 quantum dots and perovskite layer, the efficiency of perovskite solar cells was substantially enhanced from 15.17% to 18.56% by introducing CsPbI3 quantum dots as interface engineering layer. Moreover, perovskite solar cells with CsPbI3 QDs maintained 82% of its initial performance for 30 days at over 30% RH without any encapsulation.

Water-Soluble Triazolium Ionic-Liquid-Induced Surface Self-Assembly to Enhance the Stability and Efficiency of Perovskite Solar Cells

Shuangjie Wang; Zhen Li; Yuanyuan Zhang; Xingrui Liu; Jian Han; Xuanhua Li; Zhike Liu; Shengzhong Liu; Wallace C. H. Choy. Advanced Functional Materials (2019). Cited by 181. Found by: openalex. openalex_2yr: 14.975 (2026). DOI: 10.1002/adfm.201900417

Abstract: Abstract Despite being a promising candidate for next-generation photovoltaics, perovskite solar cells (PSCs) exhibit limited stability that hinders their practical application. In order to improve the humidity stability of PSCs, herein, a series of ionic liquids (ILs) "1-alkyl-4-amino-1,2,4-triazolium" (termed as RATZ; R represents alkyl chain, and ATZ represents 4-amino-1,2,4-triazolium) as cations are designed and used as additives in methylammonium lead iodide (MAPbI 3) perovskite precursor solution, obtaining triazolium ILs-modified PSCs for the first time (termed as MA/RATZ PSCs). As opposed to from traditional methods that seek to improve the stability of PSCs by functionalizing perovskite film with hydrophobic molecules, humidity-stable perovskite films are prepared by exploiting the self-assembled monolayer (SAM) formation of water-soluble triazolium ILs on a hydrophilic perovskite surface. The mechanism is validated by experimental and theoretical calculation. This strategy means that the MA/RATZ devices exhibit good humidity stability, maintaining around 80% initial efficiency for 3500 h under $40 \pm 5\%$ relative humidity. Meanwhile, the MA/RATZ PSCs exhibit enhanced thermal stability and photostability. Tuning the molecule structure of the ILs additives achieves a maximum power conversion efficiency (PCE) of 20.03%. This work demonstrates the potential of using triazolium ILs as additives and SAM and molecular design to achieve high performance PSCs.

Perovskite solar cells: Materials, configurations and stability

Isabel Mesquita; Luísa Andrade; Adélio Mendes. Renewable and Sustainable Energy Reviews (2017). Cited by 178. Found by: openalex. DOI: 10.1016/j.rser.2017.09.011

Enhancing Moisture and Water Resistance in Perovskite Solar Cells by Encapsulation with Ultrathin Plasma Polymers

Jesús Idígoras; Francisco J. Aparicio; Lidia Contreras-Bernal; Susana Ramos-Terrón; María Alcaire; Juan R. Sánchez-Valencia; Ana Borrás; Ángel Barranco; Juan A. Anta. ACS Applied Materials & Interfaces (2018). Cited by 177. Found by: openalex. openalex_2yr: 8.382 (2026). DOI: 10.1021/acsami.7b17824

Abstract: A compromise between high power conversion efficiency and long-term stability of hybrid organic-inorganic metal halide perovskite solar cells is necessary for their outdoor photovoltaic application and commercialization. Herein, a method to improve the stability of perovskite solar cells under water and moisture exposure consisting of the encapsulation of the cell with an ultrathin plasma polymer is reported. The deposition of the polymer is carried out at room temperature by the remote plasma vacuum deposition of adamantane powder. This encapsulation method does not affect the photovoltaic performance of the tested devices and is virtually compatible with any device configuration independent of the chemical composition. After 30 days under ambient conditions with a relative humidity (RH) in the range of 35-60%, the absorbance of encapsulated perovskite films remains practically unaltered. The deterioration in the photovoltaic performance of the

corresponding encapsulated devices also becomes significantly delayed with respect to devices without encapsulation when vented continuously with very humid air (RH > 85%). More impressively, when encapsulated solar devices were immersed in liquid water, the photovoltaic performance was not affected at least within the first 60 s. In fact, it has been possible to measure the power conversion efficiency of encapsulated devices under operation in water. The proposed method opens up a new promising strategy to develop stable photovoltaic and photocatalytic perovskite devices.

Enhanced Stability of Perovskite Solar Cells with Low-Temperature Hydrothermally Grown SnO2 Electron Transport Layers

Qin Liu; Minchao Qin; Weijun Ke; Xiaolu Zheng; Chen Zhao; Pingli Qin; Liangbin Xiong; Hongwei Lei; Jiawei Wan; Jian Wen; Guang Yang; Junjie Ma; Zhenyu Zhang; Guojia Fang. Advanced Functional Materials (2016). Cited by 176. Found by: openalex. openalex_2yr: 14.975 (2026). DOI: 10.1002/adfm.201600910

Abstract: Perovskite solar cells (PSCs) may offer huge potential in photovoltaic conversion, yet their practical applications face one major obstacle: their low stability, or quick degradation of their initial efficiencies. Here, a new design scheme is presented to enhance the PSC stability by using low-temperature hydrothermally grown hierarchical nano-SnO2 electron transport layers (ETLs). The ETL contains a thin compact SnO2 layer underneath a mesoporous layer of SnO2 nanosheets. The mesoporous layer plays multiple roles of enhancing photon collection, preventing moisture penetration and improving the long-term stability. Through such simple approaches, PSCs with power conversion efficiencies of 13% can be readily obtained, with the highest efficiency to be 16.17%. A prototypical PSC preserves 90% of its initial efficiency even after storage in air at room temperature for 130 d without encapsulation. This study demonstrates that hierarchical SnO2 is a potential ETL for fabricating low-cost and efficient PSCs with long-term stability.

Targeted Therapy for Interfacial Engineering Toward Stable and Efficient Perovskite Solar Cells

Shuhui Wang; Haiyang Chen; Jiandong Zhang; Guiying Xu; Weijie Chen; Rongming Xue; Moyao Zhang; Yaowen Li; Yongfang Li. Advanced Materials (2019). Cited by 176. Found by: openalex. openalex_2yr: 22.086 (2026). DOI: 10.1002/adma.201903691

Abstract: Abstract The poor long-term stability of organic–inorganic hybrid halide perovskite solar cells (pero-SCs) remains a big challenge for their commercialization. Although strategies such as encapsulation, doping, and passivation have been reported, there remains a lack of understanding of the water resistance and thermal stability of pero-SCs. A fullerene derivative, [6,6]-phenyl-C61-butyric acid-N,N-dimethyl-3-(2-thienyl)propanam ester (PCBB-S-N) containing a functional sulfur atom and C60, is synthesized and employed as electron transporting layer (ETL)/intermediary layer to targetedly heal the multitype defects in pero-SCs or assist the growth of ETL, such as [6,6]-phenyl-C61-butyric acid methyl ester (PCBM), in planar p-i-n pero-SCs. The repaired pero-SCs can not only dramatically improve their power conversion efficiencies, but also address stability issues under moisture and high temperature. The corresponding mechanism of PCBB-S-N with targeted therapy effect in a device is systematically investigated by both experiments and theoretical calculation. This work demonstrates that the proposed fullerene derivative with finely tuned chemical structure can be a promising ETL candidate or intermediary to approach stable and efficient planar p-i-n pero-SCs.

Sulfonate-Assisted Surface Iodide Management for High-Performance Perovskite Solar Cells and Modules

Ruihao Chen; Yongke Wang; Siqing Nie; Hui Shen; Hui Yong; Jian Peng; Binghui Wu; Jun Yin; Jing Li; Nanfeng Zheng. Journal of the American Chemical Society (2021). Cited by 176. Found by: openalex. DOI: 10.1021/jacs.1c03419

Abstract: Owing to the ionic nature of lead halide perovskites, their halide-terminated surface is unstable under light-, thermal-, moisture-, or electric-field-driven stresses, resulting in the formation of unfavorable surface defects. As a result, nonradiative recombination generally occurs on perovskite films and deteriorates the efficiency, stability, and hysteresis performances of perovskite solar cells (PSCs). Here, a surface iodide management strategy was developed through the use of cesium sulfonate to stabilize the perovskite surface. It was found that the pristine surface of common perovskite was terminated with extra iodide, that is, with an I – /Pb 2+ ratio larger than 3, explaining the origination of surface-related problems. Through post-treatment of perovskite films by cesium sulfonate, the extra iodide on the surface was facilely removed and the as-exposed Pb 2+ cations were chelated with sulfonate anions while maintaining the original 3D perovskite structure. Such iodide replacement and lead chelating coordination on perovskite could reduce the commonly existing surface

defects and nonradiative recombination, enabling assembled PSCs with an efficiency of 22.06% in 0.12 cm 2 cells and 18.1% in 36 cm 2 modules with high stability.

Recent efficient strategies for improving the moisture stability of perovskite solar cells

Faming Li; Mingzhen Liu. Journal of Materials Chemistry A (2017). Cited by 169. Found by: openalex. DOI: 10.1039/c7ta01325f

Abstract: Current popular and efficient strategies to improve the long-term stability regarding protection against moisture in the field of PSCs.

Advances in stability of perovskite solar cells

Qamar Wali; Faiza Jan Iftikhar; Muhammad Ejaz Khan; Abid Ullah; Yaseen Iqbal; Rajan Jose. Organic Electronics (2019). Cited by 169. Found by: openalex. DOI: 10.1016/j.orgel.2019.105590

Spontaneously Self-Assembly of a 2D/3D Heterostructure Enhances the Efficiency and Stability in Printed Perovskite Solar Cells

Jinlong Hu; Chuan Wang; Shudi Qiu; Yicheng Zhao; Ening Gu; Linxiang Zeng; Yuzhao Yang; Chaohui Li; Xianhu Liu; Karen Forberich; Christoph J. Brabec; Mohammad Khaja Nazeeruddin; Yaohua Mai; Fei Guo. Advanced Energy Materials (2020). Cited by 169. Found by: openalex. openalex_2yr: 18.63 (2026). DOI: 10.1002/aenm.202000173

Abstract: Abstract As perovskite solar cells (PSCs) are highly efficient, demonstration of high-performance printed devices becomes important. 2D/3D heterostructures have recently emerged as an attractive way to relieving the film inhomogeneity and instability in perovskite devices. In this work, a 2D/3D ensemble with 2D perovskites self-assembled atop 3D methylammonium lead triiodide (MAPbI3) via a one-step printing process is shown. A clean and flat interface is observed in the 2D/3D bilayer heterostructure for the first time. The 2D perovskite capping layer significantly suppresses nonradiative charge recombination, resulting in a marked increase in open-circuit voltage (VOC) of the devices by up to 100 mV. An ultrahigh VOC of 1.20 V is achieved for MAPbI3 PSCs, corresponding to 91% of the Shockley–Queisser limit. Moreover, notable enhancement in light, thermal, and moisture stability is obtained as a result of the protective barrier of the 2D perovskites. These results suggest a viable approach for scalable fabrication of highly efficient perovskite solar cells with enhanced environmental stability.

Constructing Efficient and Stable Perovskite Solar Cells via Interconnecting Perovskite Grains

Xian Hou; Sumei Huang; Wei Ou-Yang; Likun Pan; Zhuo Sun; Xiaohong Chen. ACS Applied Materials & Interfaces (2017). Cited by 165. Found by: openalex. openalex_2yr: 8.382 (2026). DOI: 10.1021/acsami.7b08488

Abstract: A high-quality perovskite film with interconnected perovskite grains was obtained by incorporating terephthalic acid (TPA) additive into the perovskite precursor solution. The presence of TPA changed the crystallization kinetics of the perovskite film and promoted lateral growth of grains in the vicinity of crystal boundaries. As a result, sheet-shaped perovskite was formed and covered onto the bottom grains, which made some adjacent grains partly merge together to form grains-interconnected perovskite film. Perovskite solar cells (PSCs) with TPA additive exhibited a power conversion efficiency (PCE) of 18.51% with less hysteresis, which is obviously higher than that of pristine cells (15.53%). PSCs without and with TPA additive retain 18 and 51% of the initial PCE value, respectively, aging for 35 days exposed to relative humidity 30% in air without encapsulation. Furthermore, MAPbI 3 film with TPA additive shows superior thermal stability to the pristine one under 100 °C baking. The results indicate that the presence of TPA in perovskite film can greatly improve the performance of PSCs as well as their moisture resistance and thermal stability.

Fluorine-Containing Passivation Layer via Surface Chelation for Inorganic Perovskite Solar Cells

Hao Zhang; Wanchun Xiang; Xuejiao Zuo; Xiaojing Gu; Shiang Zhang; Yachao Du; Zhiteng Wang; Yali Liu; Haifeng Wu; Peijun Wang; Qingyue Cui; Hang Su; Qingwen Tian; Shengzhong Liu. Angewandte Chemie International Edition (2022). Cited by 165. Found by: openalex. openalex_2yr: 15.006 (2026). DOI: 10.1002/anie.202216634

Abstract: Abstract Minimizing surface defect is vital to further improve power conversion efficiency (PCE) and stability of inorganic perovskite solar cells (PSCs). Herein, we designed a passivator trifluoroacetamidine (TFA)

to suppress CsPbI3−xBrxfilm defects. The amidine group of TFA can strongly chelate onto the perovskite surface to suppress the iodide vacancy, strengthened by additional hydrogen bonds. Moreover, three fluorine atoms allow strong intermolecular connection via intermolecular hydrogen bonds, thus constructing a robust shield against moisture. The TFA-treated PSCs exhibit remarkably suppressed recombination, yielding the record PCEs of 21.35 % and 17.21 % for 0.09 cm2and 1.0 cm2device areas, both of which are the highest for all-inorganic PSCs so far. The device also achieves a PCE of 39.78 % under indoor illumination, the highest for all-inorganic indoor photovoltaic devices. Furthermore, TFA greatly improves device ambient stability by preserving 93 % of the initial PCE after 960 h.

Reducing Energy Disorder in Perovskite Solar Cells by Chelation

Yiting Jiang; Jiabin Wang; Huachao Zai; Dongyuan Ni; Jiayu Wang; Peiyao Xue; Nengxu Li; Boyu Jia; Huanjun Lu; Yu Zhang; Feng Wang; Zhenyu Guo; Zhaozhao Bi; Haipeng Xie; Qian Wang; Wei Ma; Yingfeng Tu; Huanping Zhou; Xiaowei Zhan. Journal of the American Chemical Society (2022). Cited by 164. Found by: openalex. DOI: 10.1021/jacs.1c12732

Abstract: In inverted perovskite solar cells (PSCs), the fullerene derivative [6,6]-phenyl-C 61 -butyric acid methyl ester (PCBM) is a widely used electron transport material. However, a high degree of energy disorder and inadequate passivation of PCBM limit the efficiency of devices, and severe self-aggregation and unstable morphology limit the lifespan of devices. Here, we design a series of fullerene dyads FP-C n (n = 4, 8, 12) to replace PCBM as an electron transport layer, where [60]fullerene is linked with a terpyridine chelating group via a flexible alkyl chain of different lengths as a spacer. Among three fullerene dyads, FP-C8 shows the most enhanced molecule ordering and adhesion with the perovskite surface due to the balanced decoupling between the chelation effect from terpyridine and the self-assembly of fullerene, leading to lower energy disorder and higher morphological stability relative to PCBM. The FP-C8/C60-based devices using Cs 0.05 FA 0.90 MA 0.05 PbI 2.85 Br 0.15 as a light absorber show a power conversion efficiency of 21.69%, higher than that of PCBM/C60 (20.09%), benefiting from improved electron extraction and transport as well as reduced charge recombination loss. When employing FAPbI 3 as a light absorber, the FP-C8/C60-based devices exhibit an efficiency of 23.08%, which is the champion value of inverted PSCs with solution-processed fullerene derivatives. Moreover, the FP-C8/C60-based devices show better moisture and thermal stability than PCBM/C60-based devices and maintain 96% of their original efficiency after 1200 h of operation, while their counterpart PCBM/C60 maintains 60% after 670 h.

Poly(N,N -bis-4-butylphenyl-N,N -bisphenyl)benzidine-Based Interfacial Passivation Strategy Promoting Efficiency and Operational Stability of Perovskite Solar Cells in Regular Architecture

Erdi Akman; Seçkin Akın. Advanced Materials (2020). Cited by 163. Found by: openalex. openalex_2yr: 22.086 (2026). DOI: 10.1002/adma.202006087

Abstract: The failure of perovskite solar cells (PSCs) to maintain their maximum efficiency over a prolonged time is due to the deterioration of the light harvesting material under environmental factors such as humidity, heat, and light. Systematically elucidating and eliminating such degradation pathways are critical to imminent commercial use of this technology. Here, a straightforward approach is introduced to reduce the level of defect-states present at the perovskite and hole transporting layer interface by treating the various perovskite surfaces with poly(N,N'-bis-4-butylphenyl-N,N'-bisphenyl)benzidine (polyTPD) molecules. This strategy significantly suppresses the defect-mediated non-radiative recombination in the ensuing devices and prevents the penetration of degrading agents into the inner layers by passivating the perovskite surface and grain boundaries. Suppressed non-radiative recombination and improved interfacial hole extraction result in PSCs with stabilized efficiency exceeding 21% with negligible hysteresis (19.1% for control device). Moreover, ultra-hydrophobic polyTPD passivant considerably alleviates moisture penetration, showing 91% retention of initial efficiencies after 300 h storage at high relative humidity of 80%. Similarly, passivated device retains 94% of its initial efficiency after 800 h under operational conditions (maximum power point tracking under continuous illumination at 60 °C). In addition to interfacial passivation function, hole-selective role of dopant-free polyTPD is also evaluated and discussed in this study.

Enhanced Performance and Stability of 3D/2D Tin Perovskite Solar Cells Fabricated with a Sequential Solution Deposition

Efat Jokar; Po-Yuan Cheng; Chia-Yi Lin; Sudhakar Narra; Saeed Shahbazi; Eric Wei-Guang Diau. ACS Energy Letters (2021). Cited by 161. Found by: openalex. openalex_2yr: 16.712 (2026). DOI: 10.1021/acsenergylett.0c02305

Abstract: To solve the toxic issue for new-generation photovoltaic applications, tin-based perovskite solar cells are a promising alternative to their lead counterparts, but they suffer from poor stability because of their tendency to exhibit tin oxidation. Herein we report a new sequential method of deposition based on solution processing using hexafluoro-2-propanol as a solvent to deposit eight bulky ammonium cations on top of the 3D layer to form a 3D/quasi-2D layer to protect the tin perovskite grains from penetration by moisture. The formation of the 2D layer was confirmed with grazing-incidence wide-angle X-ray scattering, scanning electron microscopy, conducive atomic force microscopy, photoluminescence, and transient absorption spectroscopy measurements. The anilinium (AN) device showed a remarkable performance with an efficiency of 10.6% and with great stability in ambient air without encapsulation. The AN device also showed a self-healing effect of performance when it was subjected to a severe environment under continuous light soaking in one-sun illumination and thermal stress between 20 and 50 °C for 10 cycles.

The Role of Charge Selective Contacts in Perovskite Solar Cell Stability

Bart Roose; Qiong Wang; Antonio Abate. Advanced Energy Materials (2018). Cited by 155. Found by: openalex. openalex_2yr: 18.63 (2026). DOI: 10.1002/aenm.201803140

Abstract: Abstract Lead halide perovskite solar cells have rapidly achieved high efficiencies comparable to established commercial photovoltaic technologies. The main focus of the field is now shifting toward improving the device lifetime. Many efforts have been made to increase the stability of the perovskite compound and charge-selective contacts. The electron and hole selective contacts are responsible for the transport of photogenerated charges out of the solar cell and are in intimate contact with the perovskite absorber. Besides the intrinsic stability of the selective contacts themselves, the interfaces at perovskite/selective contact and metal/selective contact play an important role in determining the overall operational lifetime of perovskite solar cells. This review discusses the impact of external factors, i.e., heat, UV-light, oxygen, and moisture, and measured conditions, i.e., applied bias on the overall stability of perovskite solar cells (PSCs). The authors summarize and analyze the reported strategies, i.e., material engineering of selective contacts and interface engineering via the introduction of interlayers in the aim of enhancing the device stability of PSCs at elevated temperatures, high humidity, and UV irradiation. Finally, an outlook is provided with an emphasis on inorganic contacts that is believed to be the key to achieving highly stable PSCs.

Over 24% efficient MA-free CsxFA1−xPbX3 perovskite solar cells

Siyang Wang; Liguo Tan; Junjie Zhou; Minghao Li; Xing Zhao; Hang Li; Wolfgang Tress; Liming Ding; Michaël Grätzel; Chenyi Yi. Joule (2022). Cited by 155. Found by: openalex. DOI: 10.1016/j.joule.2022.05.002

Bifunctional alkyl chain barriers for efficient perovskite solar cells

Jing Zhang; Zhelu Hu; Like Huang; Guoqiang Yue; Jinwang Liu; Xingwei Lu; Ziyang Hu; Minghui Shang; Liyuan Han; Yuejin Zhu. Chemical Communications (2015). Cited by 154. Found by: openalex. openalex_2yr: 4.166 (2026). DOI: 10.1039/c5cc00128e

Abstract: Perovskite solar cells as a hot research topic show the necessity of controlling the interface. In this work, an insulating alkyl chain layer is self-assembled at the perovskite/hole transport material interface, which successfully exhibits a dual function: blocking electron recombination and resisting moisture at the same time. Improved solar energy conversion efficiency and stability of the device are both achieved.

Self-Encapsulating Thermostable and Air-Resilient Semitransparent Perovskite Solar Cells

Jie Zhao; Kai Oliver Brinkmann; Ting Hu; Neda Pourdavoud; Tim Becker; Tobias Gahlmann; R. Heiderhoff; Andreas Polywka; Patrick Görrn; Y. Chen; Baochang Cheng; Thomas Riedl. Advanced Energy Materials (2017). Cited by 151. Found by: openalex. openalex_2yr: 18.63 (2026). DOI: 10.1002/aenm.201602599

Abstract: Semitransparent perovskite solar cells (PSCs) are of interest for application in tandem solar cells and building-integrated photovoltaics. Unfortunately, several perovskites decompose when exposed to moisture or elevated temperatures. Concomitantly, metal electrodes can be degraded by the corrosive decomposition products of the perovskite. This is even the more problematic for semitransparent PSCs, in which the semitransparent top electrode is based on ultrathin metal films. Here, we demonstrate outstandingly robust PSCs with semitransparent top electrodes, where an ultrathin Ag layer is sandwiched between SnO x grown by low-temperature atomic layer deposition. The SnO x forms an electrically conductive permeation barrier, which protects both the perovskite and the ultrathin silver electrode against the detrimental impact of moisture. At the same time, the

SnO_x cladding layer underneath the ultra-thin Ag layer shields the metal against corrosive halide compounds leaking out of the perovskite. Our semitransparent PSCs show an efficiency higher than 11% along with about 70% average transmittance in the near-infrared region (> 800 nm) and an average transmittance of 29% for = 400–900 nm. The devices reveal an astonishing stability over more than 4500 hours regardless if they are exposed to ambient atmosphere or to elevated temperatures.

Paradoxical Approach with a Hydrophilic Passivation Layer for Moisture-Stable, 23% Efficient Perovskite Solar Cells

Chunqing Ma; Nam-Gyu Park. ACS Energy Letters (2020). Cited by 151. Found by: openalex. openalex_2yr: 16.712 (2026). DOI: 10.1021/acsenergylett.0c01848

Abstract: Since the first report on the 10% efficient solid-state perovskite solar cell (PSC) in 2012, perovskite photovoltaics surged swiftly. Although there has been a quantum leap in the power conversion efficiency (PCE), stability has still been an issue despite considerable progress in material and device stability. In this report, we propose a paradoxical approach for improving the stability of perovskite against moisture, which is a surface treatment of perovskite film with hydrophilic 2-aminoethanol hydroiodide (2AEI) or 4-amino-1-butanol hydroiodide (4ABI). The post-treated hydrophilic materials protect effectively the underneath perovskite layer from moisture because water molecules are absorbed by the hydrophilic layer. Over 90% of the initial PCE is retained after 1000 h stored under relative humidity of 20%, while a faster degradation is monitored from the untreated perovskite layer. The surface anchored hydrophilic materials are found to improve PCE simultaneously, leading to a value of 23.25%, due to lowered trap-state distribution by passivating surface defects.

Research Update: Strategies for improving the stability of perovskite solar cells

Severin N. Habisreutinger; David P. McMeekin; Henry J. Snaith; R. J. Nicholas. APL Materials (2016). Cited by 150. Found by: openalex. openalex_2yr: 5.589 (2026). DOI: 10.1063/1.4961210

Abstract: The power-conversion efficiency of perovskite solar cells has soared up to 22.1% earlier this year. Within merely five years, the perovskite solar cell can now compete on efficiency with inorganic thin-film technologies, making it the most promising of the new, emerging photovoltaic solar cell technologies. The next grand challenge is now the aspect of stability. The hydrophilicity and volatility of the organic methylammonium makes the work-horse material methylammonium lead iodide vulnerable to degradation through humidity and heat. Additionally, ultraviolet radiation and oxygen constitute stressors which can deteriorate the device performance. There are two fundamental strategies to increasing the device stability: developing protective layers around the vulnerable perovskite absorber and developing a more resilient perovskite absorber. The most important reports in literature are summarized and analyzed here, letting us conclude that any long-term stability, on par with that of inorganic thin-film technologies, is only possible with a more resilient perovskite incorporated in a highly protective device design.

Dion-Jacobson 2D-3D perovskite solar cells with improved efficiency and stability

Xiaoqing Jiang; Jiafeng Zhang; Sajjad Ahmad; Dandan Tu; Xuan Liu; Guoqing Jia; Xin Guo; Can Li. Nano Energy (2020). Cited by 150. Found by: openalex. DOI: 10.1016/j.nanoen.2020.104892

Perfluorinated Self-Assembled Monolayers Enhance the Stability and Efficiency of Inverted Perovskite Solar Cells

Christian M. Wolff; Laura Canil; Carolin Rehermann; Nguyen Ngoc Linh; Fengshuo Zu; Maryline Ralaiarisoa; Pietro Caprioglio; Lukas Fiedler; Martin Stolterfoht; Sergio Kogikoski; Ilko Bald; Norbert Koch; Eva Unger; Thomas Dittrich; Antonio Abate; Dieter Neher. ACS Nano (2020). Cited by 150. Found by: openalex. openalex_2yr: 16.894 (2026). DOI: 10.1021/acsnano.9b03268

Abstract: Perovskite solar cells are among the most exciting photovoltaic systems as they combine low recombination losses, ease of fabrication, and high spectral tunability. The Achilles heel of this technology is the device stability due to the ionic nature of the perovskite crystal, rendering it highly hygroscopic, and the extensive diffusion of ions especially at increased temperatures. Herein, we demonstrate the application of a simple solution-processed perfluorinated self-assembled monolayer (p-SAM) that not only enhances the solar cell efficiency, but also improves the stability of the perovskite absorber and, in turn, the solar cell under increased temperature or humid conditions. The p-i-n-type perovskite devices employing these SAMs exhibited power

conversion efficiencies surpassing 21%. Notably, the best performing devices are stable under standardized maximum power point operation at 85 °C in inert atmosphere (ISOS-L-2) for more than 250 h and exhibit superior humidity resilience, maintaining 95% device performance even if stored in humid air in ambient conditions over months (3000 h, ISOS-D-1). Our work, therefore, demonstrates a strategy towards efficient and stable perovskite solar cells with easily deposited functional interlayers.

Improving Efficiency and Stability of Perovskite Solar Cells Enabled by A Near-Infrared-Absorbing Moisture Barrier

Qin Hu; Wei Chen; Wenqiang Yang; Yu Li; Yecheng Zhou; Bryon W. Larson; Justin C. Johnson; Yi-Hsien Lu; Wenkai Zhong; Jinqiu Xu; Liana M. Klivansky; Cheng Wang; Miquel Salmerón; Aleksandra B. Djurišić; Feng Liu; Zhubing He; Rui Zhu; Thomas P. Russell. Joule (2020). Cited by 149. Found by: openalex. DOI: 10.1016/j.joule.2020.06.007

Review on persistent challenges of perovskite solar cells' stability

Maithili K. Rao; D. Sangeetha; M. Selvakumar; Y.N. Sudhakar; M.G. Mahesha. Solar Energy (2021). Cited by 149. Found by: openalex. DOI: 10.1016/j.solener.2021.03.005

High-Performance Thickness Insensitive Perovskite Solar Cells with Enhanced Moisture Stability

Jiehuan Chen; Lijian Zuo; Yingzhu Zhang; Xiaomei Lian; Weifei Fu; Jielin Yan; Jun Li; Gang Wu; Chang-Zhi Li; Hongzheng Chen. Advanced Energy Materials (2018). Cited by 148. Found by: openalex. openalex_2yr: 18.63 (2026). DOI: 10.1002/aenm.201800438

Abstract: Abstract High-performance perovskite solar cells (PVSCs) with absorber layer thickness insensitive features are important for practical fabrication, however these features are difficult to be realized. There are very few reports of the fabrication of polycrystalline PVSCs with power conversion efficienies (PCE) insensitive to film thickness beyond 600 nm. The main reason lies in more serious recombination of the thick perovskite layer compared to the thin layer. Herein, this challenge is addressed by a simple hot casting method to formulate high-quality perovskite film with enlarged grain size, high carrier mobility, and reduced defects. It is found that increasing the temperature to 70 °C can dramatically increase the film thickness and enlarge the perovskite crystal, therefore boost the efficiency from 16% to 19%. Notably, a record PCE of 19.54% is achieved with 850 nm thick perovskite film, which is among the highest efficiency for thick-film PVSCs. The PCE remains steady around 19% when modifying the perovskite layer from 700 to 1150 nm. Moreover, these thick-film PVSCs show good stability with 80% of its initial efficiency after 30 d in air with a humidity of 50%. Overall, this simple yet effective method has a great potential in the mass manufacture of PVSCs.

Moisture-Resistant FAPbI3 Perovskite Solar Cell with 22.25 % Power Conversion Efficiency through Pentafluorobenzyl Phosphonic Acid Passivation

Erdi Akman; Ahmed Esmail Shalan; Faranak Sadegh; Seçkin Akın. ChemSusChem (2020). Cited by 148. Found by: openalex. openalex_2yr: 5.476 (2026). DOI: 10.1002/cssc.202002707

Abstract: Abstract Perovskite solar cells (PSCs) have shown great promise for photovoltaic applications, owing to their low-cost assembly, exceptional performance, and low-temperature solution processing. However, the advancement of PSCs towards commercialization requires improvements in efficiency and long-term stability. The surface and grain boundaries of perovskite layer, as well as interfaces, are critical factors in determining the performance of the assembled cells. Defects, which are mainly located at perovskite surfaces, can trigger hysteresis, carrier recombination, and degradation, which diminish the power conversion efficiencies (PCEs) of the resultant cells. This study concerns the stabilization of the -FAPbI3 perovskite phase without negatively affecting the spectral features by using 2,3,4,5,6-pentafluorobenzyl phosphonic acid (PFBPA) as a passivation agent. Accordingly, high-quality PSCs are attained with an improved PCE of 22.25 % and respectable cell parameters compared to the pristine cells without the passivation layer. The thin PFBPA passivation layer effectively protects the perovskite layer from moisture, resulting in better long-term stability for unsealed PSCs, which maintain >90 % of the original efficiency under different humidity levels (40–75 %) after 600 h. PFBPA passivation is found to have a considerable impact in obtaining high-quality and stable FAPbI3 films to benefit both the efficiency and the stability of PSCs.

In situ growth of graphene on both sides of a Cu–Ni alloy electrode for perovskite solar cells with improved stability

Xuesong Lin; Hongzhen Su; Sifan He; Yenan Song; Yanbo Wang; Zhenzhen Qin; Yongzhen Wu; Xudong Yang; Qifeng Han; Junfeng Fang; Yiqiang Zhang; Hiroshi Segawa; Michaël Grätzel; Liyuan Han. Nature Energy (2022). Cited by 148. Found by: openalex. DOI: 10.1038/s41560-022-01038-1

Self-Seeding Growth for Perovskite Solar Cells with Enhanced Stability

Fei Zhang; Chuanxiao Xiao; Xihan Chen; Bryon W. Larson; Steven P. Harvey; Joseph J. Berry; Kai Zhu. Joule (2019). Cited by 147. Found by: openalex. DOI: 10.1016/j.joule.2019.03.023

Ionic liquids engineering for high-efficiency and stable perovskite solar cells

Xiaoyu Deng; Lisha Xie; Shurong Wang; Chengbo Li; Aili Wang; Yuan Yuan; Zhiyuan Cao; Tingshuai Li; Liming Ding; Feng Hao. Chemical Engineering Journal (2020). Cited by 146. Found by: openalex. openalex_2yr: 10.953 (2026). DOI: 10.1016/j.cej.2020.125594

Multifunctional Small Molecule as Buried Interface Passivator for Efficient Planar Perovskite Solar Cells

Meizi Wu; Yuwei Duan; Lu Yang; Peng You; Zhijun Li; Jungang Wang; Hui Zhou; Shaomin Yang; Dongfang Xu; Hong Zou; Zhike Liu. Advanced Functional Materials (2023). Cited by 146. Found by: openalex. openalex_2yr: 14.975 (2026). DOI: 10.1002/adfm.202300128

Abstract: Abstract The improvement of power conversion efficiency (PCE) and stability of the perovskite solar cell (PSC) is hindered by carrier recombination originating from the defects at the buried interface of the PSC. It is crucial to suppress the nonradiative recombination and facilitate carrier transfer in PSC via interface engineering. Herein, P-biguanylbenzoic acid hydrochloride (PBGH) is developed to modify the tin oxide (SnO 2)/perovskite interface. The effects of PBGH on carrier transportation, perovskite growth, defect passivation, and PSC performance are systematically investigated. On the one hand, the PBGH can effectively passivate the trap states of Sn dangling bonds and O vacancies on the SnO 2 surface via Lewis acid/base coordination, which is conducive to improving the conductivity of SnO 2 film and accelerating the electron extraction. On the other hand, PBGH modification assists the formation of high-quality perovskite film with low defect density due to its strong interaction with PbI 2 . Consequently, the PBGH-modified PSC exhibits a champion power conversion efficiency (PCE) of 24.79%, which is one of the highest PCEs among all the FACsPbI 3 -based PSCs reported to date. In addition, the stabilities of perovskite films and devices under high temperature/humidity and light illumination conditions are also systematically studied.

1000 h Operational Lifetime Perovskite Solar Cells by Ambient Melting Encapsulation

Sai Ma; Yang Bai; Hao Wang; Huachao Zai; Jiafeng Wu; Liang Li; Sisi Xiang; Na Liu; Lang Liu; Cheng Zhu; Guilin Liu; Xiuxiu Niu; Haining Chen; Huanping Zhou; Yujing Li; Qi Chen. Advanced Energy Materials (2020). Cited by 145. Found by: openalex. openalex_2yr: 18.63 (2026). DOI: 10.1002/aenm.201902472

Abstract: Abstract Improving device lifetime is one of the critical challenges for the practical use of metal halide perovskite solar cells (PSCs), wherein a reliable encapsulation is indispensable. Herein, based on an in-depth understanding of the degradation mechanism for the PSCs, a solvent-free and low-temperature melting encapsulation technique, by employing low-cost paraffin as the encapsulant that is compatible with perovskite absorbers, is demonstrated. The encapsulation strategy enables the full encapsulating operations to be undertaken under an ambient environment. It is found that the strategy not only removes residual oxygen and moisture to prevent the perovskite from phase segregation, but also suppresses the species volatilization to impede absorber decomposition, enabling a PSC devices with good thermal and moisture stability. As a result, the as-encapsulated PSCs achieve a 1000 h operational lifetime for the encapsulated device at continuous maximum power point output under an ambient environment. This work paves the way for scalable and robust encapsulation strategy feasible to hybrid perovskite optoelectronics in an economic manner.

The balance between efficiency, stability and environmental impacts in perovskite solar cells: a review

Antonio Urbina. Journal of Physics Energy (2019). Cited by 145. Found by: openalex. DOI: 10.1088/2515-7655/ab5eee

Abstract: Abstract Photovoltaic technology is progressing very fast, both in a new installed capacity, now reaching a total of more than 400 GW worldwide, and in a big research effort to develop more efficient and sustainable technologies. Organic and hybrid solar cells have been pointed out as a technological breakthrough due to their potential for low economical cost and low environmental impact; but despite impressive laboratory progress, the market is still beyond reach for these technologies, especially for perovskite-based technology. In this review, the historical evolution and relationship of efficiency and stability is addressed, including Life Cycle Assessment studies which provide a quantitative evaluation of environmental impacts in several categories, such as human health or freshwater ecotoxicity, with special focus on lead toxicity. The main conclusion is that there is no unsurmountable barrier for the massive deployment of photovoltaic systems with perovskite solar modules, if the stability is extended to lifetimes similar to technologies already in the market. The results of this review provide some recommendations mainly focused on the best options for improved stability (avoiding mainly moisture and oxygen degradation) by using metal oxides, ternary or quaternary cations, or the novel 2D/3D approach, and the encapsulation effort which should also take into account the recyclability of the materials and the low environmental impact processes for up-scaled industrial production. Research guidelines should take into account the end-of-life of the devices and cleaner routes for production avoiding toxic solvents.

Dual-Protection Strategy for High-Efficiency and Stable CsPbI2Br Inorganic Perovskite Solar Cells

Sheng Fu; Wenxiao Zhang; Xiaodong Li; Li Wan; Yulei Wu; Lijun Chen; Xiaohui Liu; Junfeng Fang. ACS Energy Letters (2020). Cited by 143. Found by: openalex. openalex_2yr: 16.712 (2026). DOI: 10.1021/acsenergylett.9b02716

Abstract: Large opening circuit voltage (V oc) loss and poor moisture stability have significantly hindered the progress of inorganic perovskite solar cells (IPSCs). Here, we report a dual-protection strategy via incorporating monomer trimethylolpropane triacrylate (TMTA) into CsPbI 2 Br perovskite bulk and capping the surface with 2-thiophenemethylammonium iodide (Th−NI) to ameliorate the above issues. Benefiting from growth control and effective suppression of both bulk and surface recombination, the resulting devices show a great improved efficiency from 12.17 to 15.58% with a champion V oc of 1.286 V. Also, the dual-protection strategy endows films with promoted moisture tolerance, and the nonencapsulated device retains 83.4% of its initial efficiency after aging at 25% relative humidity for 1540 h. More importantly, the target device shows good operational stability and reveals the unfallen efficiency after maximum power point tracking for 350 h at 45 °C. Our work offers a feasible method to fabricate efficient and stable CsPbI 2 Br IPSCs for future commercialization.

Quantifying the Interface Defect for the Stability Origin of Perovskite Solar Cells

Jionghua Wu; Jiangjian Shi; Yiming Li; Hongshi Li; Huijue Wu; Yanghong Luo; Dongmei Li; Qingbo Meng. Advanced Energy Materials (2019). Cited by 142. Found by: openalex. openalex_2yr: 18.63 (2026). DOI: 10.1002/aenm.201901352

Abstract: Abstract The stability issue that is obstructing commercialization of the perovskite solar cell is widely recognized, and tremendous effort has been dedicated to solving this issue. However, beyond the apparent thermal and moisture stability, more intrinsic semiconductor mechanisms regarding defect behavior have yet to be explored and understood. Herein, defects are quantified; especially interface defects, within the cell to reveal their impact on device performance and especially stability. Both the bulk and interface defects are distinguished and traced in situ using an expanded admittance model when the cell degrades in its efficiency under illumination or voltage. The electric field-induced interface, rather than bulk defects, is found to have a direct correlation to stability. Releasing the interface strain using a fullerene derivative is an effective way to suppress interface defect formation and improve stability. Overall, this work provides a quantitative approach to probing the semiconductor mechanism behind the stability issue, and the inherent correlation discovered here among the electric field, interface strain, interface defects, and cell stability has important implications for ongoing device stability engineering.

Critical Role of Water in Defect Aggregation and Chemical Degradation of Perovskite Solar Cells

Yun-Hyok Kye; Chol-Jun Yu; Un-Gi Jong; Yue Chen; Aron Walsh. The Journal of Physical Chemistry Letters (2018). Cited by 141. Found by: openalex. DOI: 10.1021/acs.jpclett.8b00406

Abstract: The chemical stability of methylammonium lead iodide (MAPbI 3) under humid conditions remains the primary challenge facing halide perovskite solar cells. We investigate defect processes in the water-intercalated iodide perovskite (MAPbI 3 _H 2 O) and monohydrated phase (MAPbI 3 ·H 2 O) within a first-principles thermodynamic framework. We consider the formation energies of isolated and aggregated vacancy defects with different charge states under I-rich and I-poor conditions. It is found that a PbI 2 (partial Schottky) vacancy complex can be formed readily, while the MAI vacancy complex is difficult to form in the hydrous compounds. Vacancies in the hydrous phases create deep charge transition levels, indicating the degradation of the lead halide perovskite upon exposure to moisture. Electronic structure analysis supports a mechanism of water-mediated vacancy pair formation.

Impact of Peripheral Groups on Phenothiazine-Based Hole-Transporting Materials for Perovskite Solar Cells

Fei Zhang; Shirong Wang; Hongwei Zhu; Xicheng Liu; Hongli Liu; Xianggao Li; Yin Xiao; Shaik M. Zakeeruddin; Michaël Grätzel. ACS Energy Letters (2018). Cited by 141. Found by: openalex. openalex_2yr: 16.712 (2026). DOI: 10.1021/acsenergylett.8b00395

Abstract: Three hole-transporting materials (HTMs) based on the phenothiazine core containing 4,4-dimethyltriphenylamine (Z28), N -ethylcarbazole (Z29), and 4,4-dimethoxytriphenylamine (Z30) as the peripheral groups connected by double bonds were designed and synthesized. The HTMs were tested in mixed cation/anion perovskite solar cells (PSCs) of the composition [(FAPbI 3) 0.85 (MAPbBr 3) 0.15]. A power conversion efficiency (PCE) of 19.17% under 100 Mw cm −2 standard AM 1.5G solar illumination was obtained using Z30. Importantly, the devices based on Z30 show better stability compared to those using Z28 and Z29 when aged under ambient air of 40% relative humidity in the dark for 1008 h and under continuous sunlight soaking without encapsulation for 600 h. These results indicate that the 4,4-dimethoxytriphenylamine is a promising peripheral group in combination with the phenothiazine core, providing an alternative to develop small molecular HTMs for efficient and stable PSCs.

Dually-Passivated Perovskite Solar Cells with Reduced Voltage Loss and Increased Super Oxide Resistance

Qin Zhou; Yifeng Gao; C.S. Cai; Zhuangzhuang Zhang; Jianbin Xu; Zhongyi Yuan; Peng Gao. Angewandte Chemie International Edition (2021). Cited by 140. Found by: openalex. openalex_2yr: 15.006 (2026). DOI: 10.1002/anie.202017148

Abstract: In recent years, the power conversion efficiency (PCE) of perovskite solar cells (PSCs) has witnessed rapid progress. Nevertheless, the pervasive defects prone to non-radiative recombination and decomposition exist at the surface and the grain boundaries (GBs) of the polycrystalline perovskite films. Herein, we report a comprehensive dual-passivation (DP) strategy to effectively passivate the defects at both surface and GBs to enhance device performance and stability further. Firstly, a fluorinated perylene-tetracarboxylic diimide derivative is permeated in the perovskite metaphase during antisolvent treatment, and then a fluorinated bulky aromatic ammonium salt is introduced over the annealed perovskite. The reduction of defect density can be unambiguously proved by the superoxide species generation/quenching reaction. As a result, optimized planar PSCs demonstrate a decreased open-circuit voltages deficit from 0.47 to 0.39 V and the best efficiency of 23.80 % from photocurrent scanning with a stabilized maximum power output efficiency of 22.99 %. Without encapsulation, one typical device can maintain over 85 % of the initial efficiency after heating on a hot plate at 100 °C for 30 h under relative humidity (RH) of 70 %. When the device is aged under 30±5 % RH, over 97 % of its initial PCE is retained after 1700 h.

Effect of Electron-Transport Material on Light-Induced Degradation of Inverted Planar Junction Perovskite Solar Cells

Azat F. Akbulatov; Lyubov A. Frolova; Monroe P. Griffin; Ioana R. Gearba; Andrei Dolocan; David A. Vanden Bout; Sergey Tsarev; Eugene A. Katz; Alexander F. Shestakov; Keith J. Stevenson; Pavel A. Troshin. Advanced Energy Materials (2017). Cited by 139. Found by: openalex. openalex_2yr: 18.63 (2026). DOI: 10.1002/aenm.201700476

Abstract: Abstract This paper presents a systematic study of the influence of electron-transport materials on the operation stability of the inverted perovskite solar cells under both laboratory indoor and the natural outdoor conditions in the Negev desert. It is shown that all devices incorporating a Phenyl C61 Butyric Acid Methyl ester ([60]PCBM) layer undergo rapid degradation under illumination without exposure to oxygen and moisture. Time-of-flight secondary ion mass spectrometry depth profiling reveals that volatile products from the decomposition of methylammonium lead iodide (MAPbI3) films diffuse through the [60]PCBM layer, go all the way toward the top metal electrode, and induce its severe corrosion with the formation of an interfacial AgI layer. On the contrary, alternative electron-transport material based on the perylendiimide derivative provides good isolation for the MAPbI3 films preventing their decomposition and resulting in significantly improved device operation stability. The obtained results strongly suggest that the current approach to design inverted perovskite solar cells should evolve with respect to the replacement of the commonly used fullerene-based electron-transport layers with other types of materials (e.g., functionalized perylene diimides). It is believed that these findings pave a way toward substantial improvements in the stability of the perovskite solar cells, which are essential for successful commercialization of this photovoltaic technology.

A review of stability and progress in tin halide perovskite solar cell

Asim Aftab; Md. Imteyaz Ahmad. Solar Energy (2021). Cited by 137. Found by: openalex. DOI: 10.1016/j.solener.2020.12.06

Moisture-Resilient Perovskite Solar Cells for Enhanced Stability

Randi Azmi; Shynggys Zhumagali; Helen Bristow; Shanshan Zhang; Aren Yazmaciyan; Anil Reddy Pininti; Drajad Satrio Utomo; Anand S. Subbiah; Stefaan De Wolf. Advanced Materials (2023). Cited by 134. Found by: openalex. openalex_2yr: 22.086 (2026). DOI: 10.1002/adma.202211317

Abstract: With the rapid rise in device performance of perovskite solar cells (PSCs), overcoming instabilities under outdoor operating conditions has become the most crucial obstacle toward their commercialization. Among stressors such as light, heat, voltage bias, and moisture, the latter is arguably the most critical, as it can decompose metal-halide perovskite (MHP) photoactive absorbers instantly through its hygroscopic components (organic cations and metal halides). In addition, most charge transport layers (CTLs) commonly employed in PSCs also degrade in the presence of water. Furthermore, photovoltaic module fabrication encompasses several steps, such as laser processing, subcell interconnection, and encapsulation, during which the device layers are exposed to the ambient atmosphere. Therefore, as a first step toward long-term stable perovskite photovoltaics, it is vital to engineer device materials toward maximizing moisture resilience, which can be accomplished by passivating the bulk of the MHP film, introducing passivation interlayers at the top contact, exploiting hydrophobic CTLs, and encapsulating finished devices with hydrophobic barrier layers, without jeopardizing device performance. Here, existing strategies for enhancing the performance stability of PSCs are reviewed and pathways toward moisture-resilient commercial perovskite devices are formulated.

Engineering fluorinated-cation containing inverted perovskite solar cells with an efficiency of >21% and improved stability towards humidity

Xiao Wang; Kasparas Rakštys; Kevin S. Jack; Hui Jin; Jonathan Lai; Hui Li; Chandana Sampath Kumara Ranasinghe; Jaber Saghaei; Guanran Zhang; Paul L. Burn; I. Gentle; Paul E. Shaw. Nature Communications (2021). Cited by 134. Found by: openalex. DOI: 10.1038/s41467-020-20272-3

Abstract: Efficient and stable perovskite solar cells with a simple active layer are desirable for manufacturing. Three-dimensional perovskite solar cells are most efficient but need to have improved environmental stability. Inclusion of larger ammonium salts has led to a trade-off between improved stability and efficiency, which is attributed to the perovskite films containing a two-dimensional component. Here, we show that addition of 0.3 mole percent of a fluorinated lead salt into the three-dimensional methylammonium lead iodide perovskite enables low temperature fabrication of simple inverted solar cells with a maximum power conversion efficiency of 21.1%. The perovskite layer has no detectable two-dimensional component at salt concentrations of up to 5 mole percent. The high concentration of fluorinated material found at the film-air interface provides greater hydrophobicity, increased size and orientation of the surface perovskite crystals, and unencapsulated devices with increased stability to high humidity.

Review of recent developments and persistent challenges in stability of perovskite solar cells

Billel Salhi; Y.S. Wudil; Mohammad Kamal Hossain; Amir Al-Ahmed; Fahad A. Al-Sulaiman. Renewable and Sustainable Energy Reviews (2018). Cited by 134. Found by: openalex. DOI: 10.1016/j.rser.2018.03.058

Impact of moisture on efficiency-determining electronic processes in perovskite solar cells

Manuel Salado; Lidia Contreras-Bernal; Laura Calió; Anna Todinova; Carmen López-Santos; Shahzada Ahmad; Ana Borrás; Jesús Idígoras; Juan A. Anta. Journal of Materials Chemistry A (2017). Cited by 133. Found by: openalex. DOI: 10.1039/c7ta02264f

Abstract: Moisture-induced degradation in perovskite solar cells was thoroughly investigated by structural (SEM, EDS, XRD and XPS) and device characterization (impedance and intensity modulated photocurrent spectroscopy) techniques.

Stability Improvement of Perovskite Solar Cells by Compositional and Interfacial Engineering

Weiguang Chi; Sanjay K. Banerjee. Chemistry of Materials (2021). Cited by 132. Found by: openalex. openalex_2yr: 6.387 (2026). DOI: 10.1021/acs.chemmater.0c04931

Abstract: Solar cells based on metal halide perovskites continue to approach their theoretical efficiency limits thanks to worldwide research efforts. The next challenge is to develop perovskite devices that can retain these efficiencies but exhibit acceptable degradation and decent stability for real-life practical applications. The degradation can be triggered and significantly affected by external environmental factors, such as moisture, oxygen, light, and heat. Although the encapsulation allows effective suppression of moisture- and oxygen-induced degradation, the reduction of light degradation and heat degradation is primarily dependent on the improvement of materials and interfaces of cells. Herein, the degradation mechanisms caused by light and heat are elucidated for each of the major layers in the device. The methodologies for the corresponding degradation reduction and stability enhancement are interpreted from compositional and interfacial engineering strategies with quantitative analysis including the site-based substitution in perovskite lattice, doping in charge transporting layers, passivation by using various materials (small molecules, polymers, ligands, perovskite quantum dots, and low-dimensional perovskites), and a protective layer for vulnerable layers. This Review will provide important insight into degradation suppression and stability enhancement of perovskite solar cells and give a clue to optimal design toward high-efficiency and stable devices.

Encapsulation and Outdoor Testing of Perovskite Solar Cells: Comparing Industrially Relevant Process with a Simplified Lab Procedure

Quiterie Emery; Marko Remec; Gopinath Paramasivam; S. Janke; Janardan Dagar; Carolin Ulbrich; Rutger Schlatmann; Bernd Stannowski; Eva Unger; Mark Khenkin. ACS Applied Materials & Interfaces (2022). Cited by 132. Found by: openalex. openalex_2yr: 8.382 (2026). DOI: 10.1021/acsami.1c14720

Abstract: Perovskite solar cells (PSCs) have shown great potential for next-generation photovoltaics. One of the main barriers to their commercial use is their poor long-term stability under ambient conditions and, in particular, their sensitivity to moisture and oxygen. Therefore, several encapsulation strategies are being developed in an attempt to improve the stability of PSCs in a humid environment. The lack of common testing procedures makes the comparison of encapsulation strategies challenging. In this paper, we optimized and investigated two common encapsulation strategies: lamination-based glass-glass encapsulation for outdoor operation and commercial use (COM) and a simple glue-based encapsulation mostly utilized for laboratory research purposes (LAB). We compare both approaches and evaluate their effectiveness to impede humidity ingress under three different testing conditions: on-shelf storage at 21 °C and 30% relative humidity (RH) (ISOS-D1), damp heat exposure at 85 °C and 85% RH (ISOS-D3), and outdoor operational stability continuously monitoring device performance for 10 months under maximum power point tracking on a roof-top test site in Berlin, Germany (ISOS-O3). LAB encapsulation of perovskite devices consists of glue and a cover glass and can be performed at ambient temperature, in an inert environment without the need for complex equipment. This glue-based encapsulation procedure allowed PSCs to retain more than 93% of their conversion efficiency after 1566 h of storage in ambient atmosphere and, therefore, is sufficient and suitable as an interim encapsulation for cell transport or short-term experiments outside an inert atmosphere. However, this simple encapsulation does not pass the IEC 61215 damp heat test and hence results in a high probability of fast degradation of the cells under outdoor conditions. The COM encapsulation procedure requires the use of a vacuum laminator and the cells to be able to withstand a short period of air exposure and at least 20 min at elevated temperatures (in our case, 150 °C). This encapsulation method enabled the cells to pass the IEC 61215 damp heat test and even to retain over 95% of their initial efficiency after 1566 h in a damp heat chamber. Above all, passing the damp heat test for COM-encapsulated devices translates to devices fully retaining their initial efficiency for the full duration of the outdoor test (>10 months). To the best of the authors' knowledge, this is one of the longest outdoor stability demonstrations for PSCs published to date. We stress that both encapsulation approaches described in this work are useful for the scientific community as they fulfill different purposes: the COM for the

realization of prototypes for long-term real-condition validation and, ultimately, commercialization of perovskite solar cells and the LAB procedure to enable testing and carrying out experiments on perovskite solar cells under noninert conditions.

The Effect of Stoichiometry on the Stability of Inorganic Cesium Lead Mixed-Halide Perovskites Solar Cells

Qingshan Ma; Shujuan Huang; Sheng Chen; Meng Zhang; Cho Fai Jonathan Lau; Mark Lockrey; Hemant Kumar Mulmudi; Yuchao Shan; Jizhong Yao; Jianghui Zheng; Xiaofan Deng; Kylie Catchpole; Martin A. Green; Anita Ho-Baillie. The Journal of Physical Chemistry C (2017). Cited by 131. Found by: openalex. DOI: 10.1021/acs.jpcc.7b06268

Abstract: Metal halide perovskite solar cells that use the inorganic cation Cs have been shown to have better thermal stability than the organic cation containing counterparts, and CsPbI 2 Br has a more suitable (lower) band gap than CsPbIBr 2 as a photovoltaic energy harvesting material. However, increase in iodine content reduces structural stability due to the preference toward the non-perovskite orthorhombic phase when the film is exposed to air. In this work, the effect of varying stoichiometry of CsPbI 2 Br perovskite on film quality such as the grain size, presence of impurities and nature of impurity grains, photoluminescence, morphology, and elemental distribution are studied. Details on how to vary the stoichiometry during the dual source thermal evaporation process are reported. It is found that the air stability of CsPbI 2 Br film correlates with the CsBr-to-PbI 2 deposition rate ratio, in which the CsBr-rich CsPbI 2 Br is the most stable upon air exposure, while the stoichiometrically balanced CsPbI 2 Br perovskite film gives the best photovoltaic performance. The encapsulated device maintains 90% of the initial performance after 240 h damp and heat test at 85 °C and 85% relative humidity.

Progress and Challenges Toward Effective Flexible Perovskite Solar Cells

Xiongjie Li; Haixuan Yu; Zhirong Liu; Junyi Huang; Xiaoting Ma; Yuping Liu; Qiang Sun; Letian Dai; Shahzada Ahmad; Yan Shen; Mingkui Wang. Nano-Micro Letters (2023). Cited by 129. Found by: openalex. DOI: 10.1007/s40820-023-01165-8

Abstract: The demand for building-integrated photovoltaics and portable energy systems based on flexible photovoltaic technology such as perovskite embedded with exceptional flexibility and a superior power-to-mass ratio is enormous. The photoactive layer, i.e., the perovskite thin film, as a critical component of flexible perovskite solar cells (F-PSCs), still faces long-term stability issues when deformation occurs due to encountering temperature changes that also affect intrinsic rigidity. This literature investigation summarizes the main factors responsible for the rapid destruction of F-PSCs. We focus on long-term mechanical stability of F-PSCs together with the recent research protocols for improving this performance. Furthermore, we specify the progress in F-PSCs concerning precise design strategies of the functional layer to enhance the flexural endurance of perovskite films, such as internal stress engineering, grain boundary modification, self-healing strategy, and crystallization regulation. The existing challenges of oxygen-moisture stability and advanced encapsulation technologies of F-PSCs are also discussed. As concluding remarks, we propose our viewpoints on the large-scale commercial application of F-PSCs.

A facile molecularly engineered copper (II) phthalocyanine as hole transport material for planar perovskite solar cells with enhanced performance and stability

Guang Yang; Yulong Wang; Jiaju Xu; Hongwei Lei; Cong Chen; Haiquan Shan; Xiao-Yuan Liu; Zong-Xiang Xu; Guojia Fang. Nano Energy (2016). Cited by 126. Found by: openalex. DOI: 10.1016/j.nanoen.2016.11.039

Hermetic seal for perovskite solar cells: An improved plasma enhanced atomic layer deposition encapsulation

Haoran Wang; Yepin Zhao; Zhenyu Wang; Yunfei Liu; Zipeng Zhao; Guangwei Xu; Tae-Hee Han; Jin-Wook Lee; Chen Chen; Daqian Bao; Yu Huang; Yu Duan; Yang Yang. Nano Energy (2019). Cited by 124. Found by: openalex. DOI: 10.1016/j.nanoen.2019.104375

Indigo: A Natural Molecular Passivator for Efficient Perovskite Solar Cells

Junjun Guo; Jianguo Sun; Long Hu; Shiwen Fang; Xufeng Ling; Xuliang Zhang; Yao Wang; Hehe Huang; Chenxu Han; Claudio Cazorla; Yingguo Yang; Dewei Chu; Tom Wu; Jianyu Yuan; Wanli Ma. Advanced Energy Materials (2022). Cited by 124. Found by: openalex. openalex_2yr: 18.63 (2026). DOI: 10.1002/aenm.202200537

Abstract: Abstract Organic–inorganic hybrid lead halide perovskite solar cells have made unprecedented progress in improving photovoltaic efficiency during the past decade, while still facing critical stability challenges. Herein, the natural organic dye Indigo is explored for the first time to be an efficient molecular passivator that assists in the preparation of high-quality hybrid perovskite film with reduced defects and enhanced stability. The Indigo molecule with both carbonyl and amino groups can provide bifunctional chemical passivation for defects. In-depth theoretical and experimental studies show that the Indigo molecules firmly binds to the perovskite surfaces, enhancing the crystallization of perovskite films with improved morphology. Consequently, the Indigo-passivated perovskite film exhibits increased grain size with better uniformity, reduced grain boundaries, lowered defect density, and retarded ion migration, boosting the device efficiency up to 23.22%, and 21% for large-area device (1 cm2). Furthermore, the Indigo passivation can enhance device stability in terms of both humidity and thermal stress. These results provide not only new insights into the multipassivation role of natural organic dyes but also a simple and low-cost strategy to prepare high-quality hybrid perovskite films for optoelectronic applications based on Indigo derivatives.

CuSCN as Hole Transport Material with 3D/2D Perovskite Solar Cells

Vinod E. Madhavan; Iwan Zimmermann; Ahmer A.B. Baloch; Afsal Manekkathodi; Abdelhak Belaidi; Nouar Tabet; Mohammad Khaja Nazeeruddin. ACS Applied Energy Materials (2019). Cited by 124. Found by: openalex. openalex_2yr: 5.576 (2026). DOI: 10.1021/acsaem.9b01692

Abstract: We report stable perovskite solar cells having 3D/2D perovskite absorber layers and CuSCN as an inorganic hole transporting material (HTM). (Phenylethyl)ammonium (PEA) and [(4-fluorophenyl)ethyl]ammonium (FPEA) have been chosen as 2D cations, creating thin layers of (PEA) 2 PbI 4 or (FPEA) 2 PbI 4 on top of the 3D perovskite. The 2D perovskite as an interfacial layer, neutralizes defects at the surface of the 3D perovskite absorber, and can protect from moisture-induced degradations. We demonstrate excellent charge extraction through the modified interfaces into the inorganic CuSCN HTM, with device efficiencies above 18%, compared to 19.3% with conventional spiro-OMeTAD. Furthermore, we show significantly enhanced ambient stability.

Stability of Halide Perovskite Solar Cell Devices: In Situ Observation of Oxygen Diffusion under Biasing

Hee Joon Jung; Daehan Kim; Sung-Kyu Kim; Joonsuk Park; Vinayak P. Dravid; Byungha Shin. Advanced Materials (2018). Cited by 123. Found by: openalex. openalex_2yr: 22.086 (2026). DOI: 10.1002/adma.201802769

Abstract: Abstract Using in situ electrical biasing transmission electron microscopy, structural and chemical modification to n–i–p-type MAPbI3 solar cells are examined with a TiO2 electron-transporting layer caused by bias in the absence of other stimuli known to affect the physical integrity of MAPbI3 such as moisture, oxygen, light, and thermal stress. Electron energy loss spectroscopy (EELS) measurements reveal that oxygen ions are released from the TiO2 and migrate into the MAPbI3 under a forward bias. The injection of oxygen is accompanied by significant structural transformation; a single-crystalline MAPbI3 grain becomes amorphous with the appearance of PbI2. Withdrawal of oxygen back to the TiO2, and some restoration of the crystallinity of the MAPbI3, is observed after the storage in dark under no bias. A subsequent application of a reverse bias further removes more oxygen ions from the MAPbI3. Light current–voltage measurements of perovskite solar cells exhibit poorer performance after elongated forward biasing; recovery of the performance, though not complete, is achieved by subsequently applying a negative bias. The results indicate negative impacts on the device performance caused by the oxygen migration to the MAPbI3 under a forward bias. This study identifies a new degradation mechanism intrinsic to n–i–p MAPbI3 devices with TiO2.

Machine learning analysis on stability of perovskite solar cells

Çağla Odabaşı; Ramazan Yıldırım. Solar Energy Materials and Solar Cells (2019). Cited by 122. Found by: openalex. DOI: 10.1016/j.solmat.2019.110284

Intrinsic and Extrinsic Stability of Formamidinium Lead Bromide Perovskite Solar Cells Yielding High Photovoltage

Neha Arora; M. Ibrahim Dar; Mojtaba Abdi-Jalebi; Fabrizio Giordano; Norman Pellet; Gwénolé Jacopin; Richard H. Friend; Shaik M. Zakeeruddin; Michaël Grätzel. Nano Letters (2016). Cited by 122. Found by: openalex. DOI: 10.1021/acs.nanolett.6b03455

Abstract: We report on both the intrinsic and the extrinsic stability of a formamidinium lead bromide [CH(NH 2) 2 PbBr 3 = FAPbBr 3] perovskite solar cell that yields a high photovoltage. The fabrication of FAPbBr 3 devices, displaying an outstanding photovoltage of 1.53 V and a power conversion efficiency of over 8%, was realized by modifying the mesoporous TiO 2 –FAPbBr 3 interface using lithium treatment. Reasons for improved photovoltaic performance were revealed by a combination of techniques, including photothermal deflection absorption spectroscopy (PDS), transient-photovoltage and charge-extraction analysis, and time-integrated and time-resolved photoluminescence. With lithium-treated TiO 2 films, PDS reveals that the TiO 2 –FAPbBr 3 interface exhibits low energetic disorder, and the emission dynamics showed that electron injection from the conduction band of FAPbBr 3 into that of mesoporous TiO 2 is faster than for the untreated scaffold. Moreover, compared to the device with pristine TiO 2, the charge carrier recombination rate within a device based on lithium-treated TiO 2 film is 1 order of magnitude lower. Importantly, the operational stability of perovskites solar cells examined at a maximum power point revealed that the FAPbBr 3 material is intrinsically (under nitrogen) as well as extrinsically (in ambient conditions) stable, as the unsealed devices retained over 95% of the initial efficiency under continuous full sun illumination for 150 h in nitrogen and dry air and 80% in 60% relative humidity (T = 60 °C). The demonstration of high photovoltage, a record for FAPbBr 3, together with robust stability renders our work of practical significance.

Versatility of Carbon Enables All Carbon Based Perovskite Solar Cells to Achieve High Efficiency and High Stability

Xiangyue Meng; Junshuai Zhou; Jie Hou; Xia Tao; Sin Hang Cheung; Shu Kong So; Shihe Yang. Advanced Materials (2018). Cited by 122. Found by: openalex. openalex_2yr: 22.086 (2026). DOI: 10.1002/adma.201706975

Abstract: Abstract Carbon-based perovskite solar cells (PVSCs) without hole transport materials are promising for their high stability and low cost, but the electron transporting layer (ETL) of TiO2 is notorious for inflicting hysteresis and instability. In view of its electron accepting ability, C60 is used to replace TiO2 for the ETL, forming a so-called all carbon based PVSC. With a device structure of fluorine-doped tin oxide (FTO)/C60/methylammonium lead iodide (MAPbI3)/carbon, a power conversion efficiency (PCE) is attained up to 15.38% without hysteresis, much higher than that of the TiO2 ones (12.06% with obvious hysteresis). The C60 ETL is found to effectively improve electron extraction, suppress charge recombination, and reduce the sub-bandgap states at the interface with MAPbI3. Moreover, the all carbon based PVSCs are shown to resist moisture and ion migration, leading to a much higher operational stability under ambient, humid, and light-soaking conditions. To make it an even more genuine all carbon based PVSC, it is further attempted to use graphene as the transparent conductive electrode, reaping a PCE of 13.93%. The high performance of all carbon based PVSCs stems from the bonding flexibility and electronic versatility of carbon, promising commercial developments on account of their favorable balance of cost, efficiency, and stability.

2-CF3-PEAI to eliminate Pb0 traps and form a 2D perovskite layer to enhance the performance and stability of perovskite solar cells

Junjie Zhou; Minghao Li; Siyang Wang; Liguo Tan; Yue Liu; Chaofan Jiang; Xing Zhao; Liming Ding; Chenyi Yi. Nano Energy (2022). Cited by 122. Found by: openalex. DOI: 10.1016/j.nanoen.2022.107036

Encapsulation: The path to commercialization of stable perovskite solar cells

Qianqian Chu; Zhijian Sun; Dong Wang; Bo Cheng; Haijiang Wang; Ching-Ping Wong; Baizeng Fang. Matter (2023). Cited by 120. Found by: openalex. DOI: 10.1016/j.matt.2023.08.016

Ambient Fabrication of Organic–Inorganic Hybrid Perovskite Solar Cells

Yuan Zhang; Ashleigh Kirs; Filip Ambrož; Chieh-Ting Lin; Abdulaziz S. R. Bati; Ivan P. Parkin; Joseph G. Shapter; Munkhbayar Batmunkh; Thomas J. Macdonald. Small Methods (2020). Cited by 118. Found by: openalex. DOI: 10.1002/smtd.202000744

Abstract: Organic-inorganic hybrid perovskite solar cells (PSCs) have attracted significant attention in recent years due to their high-power conversion efficiency, simple fabrication, and low material cost. However, due to their high sensitivity to moisture and oxygen, high efficiency PSCs are mainly constructed in an inert environment. This has led to significant concerns associated with the long-term stability and manufacturing costs, which are some of the major limitations for the commercialization of this cutting-edge technology. Over the past few years, excellent progress in fabricating PSCs in ambient conditions has been made. These advancements have drawn considerable research interest in the photovoltaic community and shown great promise for the successful commercialization of efficient and stable PSCs. In this review, after providing an overview to the influence of an ambient fabrication environment on perovskite films, recent advances in fabricating efficient and stable PSCs in ambient conditions are discussed. Along with discussing the underlying challenges and limitations, the most appropriate strategies to fabricate efficient PSCs under ambient conditions are summarized along with multiple roadmaps to assist in the future development of this technology.

Recent Progress of Carbon-Based Inorganic Perovskite Solar Cells: From Efficiency to Stability

Xiang Zhang; Zhenhua Yu; Dan Zhang; Qidong Tai; Xingzhong Zhao. Advanced Energy Materials (2022). Cited by 117. Found by: openalex. openalex_2yr: 18.63 (2026). DOI: 10.1002/aenm.202201320

Abstract: Abstract Organic–inorganic hybrid perovskite solar cells (PSCs) are regarded as a promising next-generation photovoltaic technology. However, poor device stability limits their commercialization. Carbon-based inorganic PSCs (C-IPSCs) can meet stability challenge owing to the absence of organic components. Meanwhile, the hydrophobic carbon materials as hole transport layers and back electrodes simplify fabrication process, decrease costs, and protect devices from moisture erosion. Since the first attempt for C-IPSCs in 2016, a series of strategies have been proposed to improve device performance, including the fabrication technique optimization, solvent engineering, composition engineering, interface engineering, charge transport layer optimization, and so on, and the power conversion efficiency of C-IPSCs are rapidly increased from initial 5% to exceeding 15%. In this review, the recent progress of C-IPSCs is summarized, the existing challenges in this field are discussed, followed by a prospect for future development.

Film Grain-Size Related Long-Term Stability of Inverted Perovskite Solar Cells

Chien-Hung Chiang; Chun-Guey Wu. ChemSusChem (2016). Cited by 117. Found by: openalex. openalex_2yr: 5.476 (2026). DOI: 10.1002/cssc.201600887

Abstract: Abstract The power conversion efficiency (PCE) of the perovskite solar cell is high enough to be commercially viable. The next important issue is the stability of the device. This article discusses the effect of the perovskite grain-size on the long-term stability of inverted perovskite solar cells. Perovskite films composed of various sizes of grains were prepared by controlling the solvent annealing time. The grain-size related stability of the inverted cells was investigated both in ambient atmosphere at relative humidity of approximately 30–40 % and in a nitrogen filled glove box (H2O<0.1 ppm, O2<10 ppm). The PCE of the solar cell based on a perovskite film having the grain size larger than 1 m (D-10) decreases less than 10 % with storage in a glove box and less than 15 % when it was stored under an ambient atmosphere for 30 days. However, the cell using the perovskite film composed of small (100 nm) perovskite grains (D-0) exhibits complete loss of PCE after storage under the ambient atmosphere for only 15 days and a PCE loss of up to 70 % with storage in the glove box for 30 days. These results suggest that, even under H2O-free conditions, the chemical- and thermal-induced production of pin holes at the grain boundaries of the perovskite film could be the reason for long-term instability of inverted perovskite solar cells.

Gradated Mixed Hole Transport Layer in a Perovskite Solar Cell: Improving Moisture Stability and Efficiency

Guan-Woo Kim; Gyeongho Kang; Mahdi Malekshahi Byranvand; Gang-Young Lee; Taiho Park. ACS Applied Materials & Interfaces (2017). Cited by 116. Found by: openalex. openalex_2yr: 8.382 (2026). DOI: 10.1021/acsami.7b07071

Abstract: We demonstrate a simple and facile way to improve the efficiency and moisture stability of perovskite solar cells using commercially available hole transport materials, 2,2',7,7'-tetrakis-(N,N-di-4-methoxyphenylamino)-9,9'-spirobifluorene (spiro-OMeTAD) and poly(3-hexylthiophene) (P3HT). The hole transport layer (HTL) composed of mixed spiro-OMeTAD and P3HT exhibited favorable vertical phase separation. The hydrophobic P3HT was more distributed near the surface (the air atmosphere), whereas the hydrophilic spiro-OMeTAD was more distributed near the perovskite layer. This vertical separation resulted in improved moisture stability by effectively blocking moisture in air. In addition, the optimized composition of spiro-OMeTAD and P3HT improved the efficiency of the solar cells by enabling fast intramolecular charge transport. In addition, a suitable energy level alignment facilitated charge transfer. A device fabricated using the mixed HTL exhibited enhanced performance, demonstrating 18.9% power conversion efficiency and improved moisture stability.

Printable WO3 electron transporting layer for perovskite solar cells: Influence on device performance and stability

Alexandre Gheno; Trang Thi Thu Pham; Catherine Di Bin; Johann Bouclé; Bernard Ratier; Sylvain Vedraine. Solar Energy Materials and Solar Cells (2016). Cited by 116. Found by: openalex. DOI: 10.1016/j.solmat.2016.10.002

Barrier reinforcement for enhanced perovskite solar cell stability under reverse bias

Nengxu Li; Zhifang Shi; Chengbin Fei; Haoyang Jiao; Mingze Li; Hangyu Gu; Steven P. Harvey; Yifan Dong; Matthew C. Beard; Jinsong Huang. Nature Energy (2024). Cited by 114. Found by: openalex. DOI: 10.1038/s41560-024-01579-7

Additive engineering for stable halide perovskite solar cells

Carlos Pereyra; Haibing Xie; Mónica Lira-Cantú. Journal of Energy Chemistry (2021). Cited by 113. Found by: openalex. DOI: 10.1016/j.jechem.2021.01.037

Mechanochemistry Advances High-Performance Perovskite Solar Cells

Yuzhuo Zhang; Yanju Wang; Xiaoyu Yang; Lichen Zhao; Rui Su; Jiang Wu; Deying Luo; Shunde Li; Peng Chen; Maotao Yu; Qihuang Gong; Rui Zhu. Advanced Materials (2021). Cited by 112. Found by: openalex. openalex_2yr: 22.086 (2026). DOI: 10.1002/adma.202107420

Abstract: Abstract A prerequisite for commercializing perovskite photovoltaics is to develop a swift and eco-friendly synthesis route, which guarantees the mass production of halide perovskites in the industry. Herein, a green-solvent-assisted mechanochemical strategy is developed for fast synthesizing a stoichiometric -phase formamidinium lead iodide (-FAPbI 3) powder, which serves as a high-purity precursor for perovskite film deposition with low defects. The presynthesized -FAPbI 3 precursor possesses high concentration of micrometer-sized colloids, which are in favor of preferable crystallization by spontaneous nucleation. The resultant perovskite films own preferred crystal orientations of cubic (100) plane, which is beneficial for superior carrier transport compared to that of the films with isotropic crystal orientations using “mixture of PbI 2 and FAI” as precursors. As a result, high-performance perovskite solar cells with a maximum power conversion efficiency of 24.2% are obtained. Moreover, the -FAPbI 3 powder shows superior storage stability for more than 10 months in ambient environment (40 ± 10% relative humidity), being conducive to a facile and practical storage for further commercialization.

High Efficiency and Stability of Inverted Perovskite Solar Cells Using Phenethyl Ammonium Iodide-Modified Interface of NiOx and Perovskite Layers

Yuning Liu; Jiaji Duan; Jiankai Zhang; Sumei Huang; Wei Ou-Yang; Qinye Bao; Zhuo Sun; Xiaohong Chen. ACS Applied Materials & Interfaces (2019). Cited by 112. Found by: openalex. openalex_2yr: 8.382 (2026). DOI: 10.1021/acsami.9b18217

Abstract: Hole transport layer NiO x -based inverted perovskite solar cells (PSCs) have advantages of simple fabrication, low temperature, and low cost. Furthermore, the p-type NiO x material compared to that of typical n-type SnO x for PSCs has better photostability potential due to its lower photocatalytic ability. However, the NiO x layer modified by some typical materials show relatively simple functions, which limit the synthesized performance of NiO x -based inverted PSCs. Phenethyl ammonium iodide (PEAI) was introduced to modify the NiO x /perovskite interface, which can synchronously contribute to better crystallinity and stability of the perovskite layer, passivating interface defects, formed quasi-two-dimensional PEA 2 PbI 4 perovskite layers, and superior interface contact properties. The PCEs of PSCs with the PEAI-modified NiO x /perovskite interface was obviously increased from 20.31 from 16.54% compared to that of the reference PSCs. The PSCs with PEAI modification remained 75 and 72% of the original PCE values aging for 10 h at 85 °C and 65 days in a relative humidity of 15%, which are superior to the original PCE values (47 and 51%, respectively) for the reference PSCs. Therefore, PSCs with the PEAI-modified NiO x /perovskite interface show higher PCEs and better thermal stability and moisture resistance.

Achieving Resistance against Moisture and Oxygen for Perovskite Solar Cells with High Efficiency and Stability

Weiguang Chi; S. Banerjee. Chemistry of Materials (2021). Cited by 107. Found by: openalex. openalex_2yr: 6.387 (2026). DOI: 10.1021/acs.chemmater.1c00773

Abstract: Realization of high efficiency along with long-term stability of perovskites solar cells has been a goal of researchers in this field. Breakthroughs in efficiency have been achieved in the past decade, exceeding those of most thin film solar cells. However, the challenge of degradation and instability of perovskite solar cells is currently the bottleneck for their real-life application. Extensive research has been conducted to achieve simultaneously high efficiency and high stability, leading to considerable progress in device performance. A significant source causing degradation is the ingression of external materials, such as moisture and oxygen. In this review, the mechanisms of moisture and oxygen degradation occurring in perovskite absorber and charge transporting layers are interpreted. Primary approaches to simultaneously achieve high efficiency and high stability focus on encapsulation, interfacial layer engineering, process engineering, ion engineering, and dopant and alternative engineering. In brief, we review the materials development and engineering strategies to improve performance and identify the obstacles that hinder the realization of high stability along with high efficiency.

Interface degradation of perovskite solar cells and its modification using an annealing-free TiO2 NPs layer

Jian Xiong; Bingchu Yang; Chenghao Cao; Runsheng Wu; Yulan Huang; Jia Sun; Jian Zhang; Chengbin Liu; Shaohua Tao; Yongli Gao; Junliang Yang. Organic Electronics (2015). Cited by 107. Found by: openalex. DOI: 10.1016/j.orgel.2015.12.010

Boosting mechanical durability under high humidity by bioinspired multisite polymer for high-efficiency flexible perovskite solar cells

Zhihao Li; Zhihao Li; Chunmei Jia; Zhi Wan; Junchao Cao; Jishan Shi; Jiayi Xue; Xirui Liu; Hongzhuo Wu; Chuanxiao Xiao; Meng Li; Meng Li; Chao Zhang; Zhen Li; Zhen Li. Nature Communications (2025). Cited by 107. Found by: openalex. DOI: 10.1038/s41467-025-57102-3

Abstract: Flexible perovskite solar cells (FPSCs) with high stability in moist air are required for their practical applications. However, the poor mechanical stability under high humidity air remains a critical challenge for flexible perovskite devices. Herein, inspired by the exceptional wet adhesion of mussels via dopamine groups, we propose a multidentate-cross-linking strategy, which combine multibranched structure and adequate dopamine anchor sites in three-dimensional hyperbranched polymer to directly chelate perovskite materials in multiple directions, therefore construct a vertical scaffold across the bulk of perovskite films from the bottom to the top interfaces, intimately bind to the perovskite grains and substrates with a strong adhesion ability, and enhance mechanical durability under high humidity. Consequently, the modified rigid PSCs achieve superior PCE up to 25.92%, while flexible PSCs exhibit a PCE of 24.43% and maintain 94.1% of initial PCE after 10,000 bending cycles with a bending radius of 3 mm under exposed to 65% humidity.

Dual Additive for Simultaneous Improvement of Photovoltaic Performance and Stability of Perovskite Solar Cell

In Seok Yang; Nam-Gyu Park. Advanced Functional Materials (2021). Cited by 105. Found by: openalex. openalex_2yr: 14.975 (2026). DOI: 10.1002/adfm.202100396

Abstract: Abstract Additive engineering is one of the most efficient approaches to improve not only photovoltaic performance but also phase stability of formamidinium (FA)-based perovskite. Chlorine-based additives, such as methylammonium chloride (MACl), have been in general used to improve phase stability of FAPbI3, which however often leads to loss of open-circuit voltage Voc, accompanied by instability of the perovskite phase due to the volatile nature of the MA cation. A dual additive strategy for improving Voc and thereby the overall efficiency are reported here. The mixing ratio of MACl to CsCl is varied from [MACl]/[CsCl] = 4 to 1, where Voc increases with decreasing the ratio and best performance is achieved from [MACl]/[CsCl] = 2. As compared to the single source of MACl, the addition of CsCl reduces trap density and increases resistance against charge recombination, which is responsible for the increased Voc. Moreover, defect passivation achieved by dual additive enables better stability than the single additive MACl as confirmed by long-term stability tests with unencapsulated devices for 50 days under relative humidity of about 40% at room temperature. The best power conversion efficiency of 23.22% is achieved by dual additive, which is higher than that for single additive of MACl or CsCl.

Solution processed pristine PDPP3T polymer as hole transport layer for efficient perovskite solar cells with slower degradation

Ashish Dubey; Nirmal Adhikari; Swaminathan Venkatesan; Shaopeng Gu; Devendra Khatiwada; Qi Wang; Lal Mohammad; Mukesh Kumar; Qiquan Qiao. Solar Energy Materials and Solar Cells (2015). Cited by 105. Found by: openalex. DOI: 10.1016/j.solmat.2015.10.008

Spray reaction prepared FA 1−x Cs x PbI 3 solid solution as a light harvester for perovskite solar cells with improved humidity stability

Xiang Xia; Wenyi Wu; Hongcui Li; Bo Zheng; Yebin Xue; Jing Xu; Dawei Zhang; Chunxiao Gao; Xizhe Liu. RSC Advances (2016). Cited by 104. Found by: openalex. DOI: 10.1039/c5ra23359c

Abstract: High efficiency and humidity stability perovskite solar cells based on FA 1−x Cs x PbI 3 solid solution are prepared by spray reaction technology.

Amino acid salt-driven planar hybrid perovskite solar cells with enhanced humidity stability

Seong-Cheol Yun; Sunihl Ma; Hyeok-Chan Kwon; Kyung-Mi Kim; Gyumin Jang; Hyunha Yang; Jooho Moon. Nano Energy (2019). Cited by 103. Found by: openalex. DOI: 10.1016/j.nanoen.2019.02.064

Degradation and Self-Healing in Perovskite Solar Cells

Blake P. Finkenauer; Akriti Akriti; Ke Ma; Letian Dou. ACS Applied Materials & Interfaces (2022). Cited by 101. Found by: openalex. openalex_2yr: 8.382 (2026). DOI: 10.1021/acsami.2c01925

Abstract: Organic-inorganic halide perovskites are well-known for their unique self-healing ability. In the presence of strong external stimuli, such as light, temperature, and moisture, high-energy defects are created which can be healed by removing the perovskite from the degradation source. This self-healing ability has been showcased in devices with recoverable performance and day-and-night cycling operation to dramatically extend the device lifetime and even mechanical durability. However, to date, the mechanistic details and theory around this captivating trait are sparse and convoluted by the complex nature of perovskites. With a clear understanding of the intrinsic self-healing property, perovskite solar cells with extended lifetimes and durability can be designed to realize the large-scale commercialization of perovskite solar cells. Here, we spotlight the relevant degradation and self-healing literature and then propose design strategies to help conceptualize future research.

Recent Progress of All-Bromide Inorganic Perovskite Solar Cells

Guoqing Tong; Luis K. Ono; Yabing Qi. Energy Technology (2019). Cited by 100. Found by: openalex. openalex_2yr: 3.823 (2026). DOI: 10.1002/ente.201900961

Abstract: Inorganic perovskite solar cells (PSCs) have attracted enormous attention during the past 5 years. Many advanced strategies and techniques have been developed for fabricating inorganic PSCs with improved efficiency and stability to realize commercial applications. CsPbBr3 is one of the representative materials of inorganic perovskites and has demonstrated excellent stability against thermal and high humidity environmental conditions. The power conversion efficiency of CsPbBr3-based PSCs has increased significantly from 5.95% in 2015 to 10.91%, and the storage stability under moisture (80% relative humidity) and heat (80 °C) is more than 2000 h. The outstanding performance of CsPbBr3 PSCs shows great potential in light-to-electricity conversion applications. In this review, recent developments of CsPbBr3-based PSCs including the physico-chemical as well as optoelectronic properties, processing techniques for fabricating CsPbBr3 films, derivative phase structures, efficiency, and stability of devices are reviewed and discussed. Finally, the challenges and outlook of CsPbBr3 PSCs for future research directions are outlined.

Enhancing stability for organic-inorganic perovskite solar cells by atomic layer deposited Al2O3 encapsulation

Eun Young Choi; Jincheol Kim; Sean Lim; Ekyu Han; Anita Ho-Baillie; Nochang Park. Solar Energy Materials and Solar Cells (2018). Cited by 99. Found by: openalex. DOI: 10.1016/j.solmat.2018.08.016

Enhanced Ambient Stability of Efficient Perovskite Solar Cells by Employing a Modified Fullerene Cathode Interlayer

Zonglong Zhu; Chu-Chen Chueh; Francis Lin; Alex K.-Y. Jen. Advanced Science (2016). Cited by 97. Found by: openalex. openalex_2yr: 8.903 (2026). DOI: 10.1002/advs.201600027

Abstract: is employed to facilitate the fabrication of stable and efficient perovskite solar cells. This modified fullerene surfactant significantly increases air stability of the derived devices due to its hydrophobic characteristics to enable 80% of the initial PCE to be retained after being exposed in ambient condition with 20% relative humidity for 14 days.

Photodecomposition and Morphology Evolution of Organometal Halide Perovskite Solar Cells

Fukashi Matsumoto; Sarah M. Vorpahl; Jannel Q. Banks; Esha Sengupta; David S. Ginger. The Journal of Physical Chemistry C (2015). Cited by 95. Found by: openalex. DOI: 10.1021/acs.jpcc.5b06269

Abstract: We study the photoinduced degradation of hybrid organometal perovskite photovoltaics under illumination and ambient atmosphere using UV–vis absorption, atomic force microscopy, and device performance. We correlate the structural changes in the surface of the perovskite film with changes in the optical and electronic properties of the devices. The photodecomposition of the methylammonium lead triiodide perovskite layer itself proceeds much more slowly than the photodegradation of the performance of devices with fullerene/bathocuproine/aluminum top contacts, indicating that the active layer alone is more stable than the interface with the electrodes in this geometry. The evolution of the perovskite active layer performance proceeded through several phases: (1) an initial improvement in device characteristics, (2) a plateau with very slow degradation, and (3) a catastrophic decline in material performance accompanied by marked changes in film morphology. The rapid increase in surface roughness of the active perovskite semiconductor associated with sudden failure also correlates with decreased absorption at the perovskite band edge and growth of a lead iodide absorption feature. We find that degradation requires both light and moisture, is accelerated at increased humidity, and scales linearly with light intensity, depending primarily on total photon dose.

Humidity-Assisted Chlorination with Solid Protection Strategy for Efficient Air-Fabricated Inverted CsPbI 3 Perovskite Solar Cells

Sheng Fu; Wenxiao Zhang; Xiaodong Li; Jianming Guan; Weijie Song; Junfeng Fang. ACS Energy Letters (2021). Cited by 91. Found by: openalex. openalex_2yr: 16.712 (2026). DOI: 10.1021/acsenergylett.1c01817

Abstract: Humidity-assisted chlorination with solid protection via 1,2-bis(chlorodimethylsilyl)ethane (Si–Cl) incorporation is demonstrated as an effective strategy to address water erosion for efficient air-fabricated inverted CsPbI 3 perovskite solar cells (PSCs). Si–Cl molecules can rapidly react with water to reduce moisture erosion during deposition in air. In addition, the hydrolysis production of hydrogen chloride (HCl) can chlorinate CsPbI 3 while the spontaneous polymerization of 1,2-bis(hydroxydimethylsilyl)ethane (Si–OH) builds a solid protection, which would improve the crystallization and phase stability of CsPbI 3 . Therefore, a champion efficiency of 18.93% with high open-circuit voltage of 1.176 V is achieved, which is the highest efficiency among the inverted inorganic PSCs. Furthermore, the moisture tolerance of both the CsPbI 3 films and air-fabricated PSCs gets significant improvement. In addition, the nonencapsulated device shows good operational stability without efficiency drop after 1 sun illumination at 65 °C for 900 h.

Achieving Long-Term Operational Stability of Perovskite Solar Cells with a Stabilized Efficiency Exceeding 20% after 1000 h

Tae-Youl Yang; Nam Joong Jeon; Hee-Won Shin; Seong Sik Shin; Young Yun Kim; Jangwon Seo. Advanced Science (2019). Cited by 91. Found by: openalex. openalex_2yr: 8.903 (2026). DOI: 10.1002/advs.201900528

Abstract: Abstract Perovskite solar cells (PSCs) with mesoporous TiO2 (mp-TiO2) as the electron transport material attain power conversion efficiencies (PCEs) above 22%; however, their poor long-term stability is a critical issue that must be resolved for commercialization. Herein, it is demonstrated that the long-term operational stability of mp-TiO2 based PSCs with PCE over 20% is achieved by isolating devices from oxygen and humidity. This achievement attributes to systematic understanding of the critical role of oxygen in the degradation of PSCs. PSCs exhibit fast degradation under controlled oxygen atmosphere and illumination, which is accompanied by iodine migration into the hole transport material (HTM). A diffusion barrier at the HTM/perovskite interface or encapsulation on top of the devices improves the stability against oxygen under light soaking. Notably, a mp-TiO2 based PSC with a solid encapsulation retains 20% efficiency after 1000 h of

1 sun (AM1.5G including UV) illumination in ambient air.

Graphdiyne-Based Bulk Heterojunction for Efficient and Moisture-Stable Planar Perovskite Solar Cells

Hongshi Li; Rui Zhang; Yusheng Li; Yusheng Li; Yiming Li; Yiming Li; Huibiao Liu; Jiangjian Shi; Huiyin Zhang; Huijue Wu; Yanhong Luo; Dongmei Li; Yuliang Li; Yuliang Li; Qingbo Meng. Advanced Energy Materials (2018). Cited by 90. Found by: openalex. openalex_2yr: 18.63 (2026). DOI: 10.1002/aenm.201802012

Abstract: Abstract Graphdiyne (GDY) is introduced into the FA0.85MA0.15Pb(I0.85Br0.15)3 perovskite film to construct a perovskite/graphdiyne (PVSK/GDY) bulk heterojunction for planar perovskite solar cells (PSCs). This PVSK/GDY bulk heterojunction is suggested to provide an extra channel to favor exciton separation and facilitate the photogenerated electron extraction ability. The electron transportation ability has been improved for higher Jsc, and the introduction of GDY can passivate grain boundaries and interfaces to effectively suppress photogenerated carrier recombination, leading to relatively higher fill factor. Up to 20.54% power conversion efficiency is achieved. In addition, the ability of this perovskite film with PVSK/GDY bulk heterojunctions to function in humid conditions is also improved, leading to good stability of PSCs toward moisture. This work provides a new way to fabricate highly efficient, stable PSCs by constructing bulk heterojunctions simultaneously for accelerating the exciton separation and photogenerated electron transportation and improving the device stability.

Long-term stability of organic–inorganic hybrid perovskite solar cells with high efficiency under high humidity conditions

Yue Sun; Yihui Wu; Xiang Fang; Linjun Xu; Zhijie Ma; Yongting Lu; Wen-Hua Zhang; Qiang Yu; Ningyi Yuan; Jiangning Ding. Journal of Materials Chemistry A (2016). Cited by 89. Found by: openalex. DOI: 10.1039/c6ta08117g

Abstract: Perovskite solar cells with superior tolerance to humidity (85–95% RH) and long-term stability have been achieved via adding a certain amount of a cost-effective and available water soluble additive, polyvinyl alcohol (PVA).

Tetrabutylammonium cations for moisture-resistant and semitransparent perovskite solar cells

Isabella Poli; Salvador Eslava; Petra J. Cameron. Journal of Materials Chemistry A (2017). Cited by 87. Found by: openalex. DOI: 10.1039/c7ta06735f

Abstract: Tetra-butylammonium cations have been partially substituted for methylammonium cations in perovskite thin films. The stability of devices stored under ambient conditions was enhanced by the presence of TBA and cells with high mol% TBA were found to have reasonable efficiencies while being semi-transparent.

Textured CH3NH3PbI3 thin film with enhanced stability for high performance perovskite solar cells

Mingzhu Long; Tiankai Zhang; Houyu Zhu; Guixia Li; Feng Wang; Wenyue Guo; Yang Chai; Wei Chen; Qiang Li; Kam Sing Wong; Jianbin Xu; Keyou Yan. Nano Energy (2017). Cited by 86. Found by: openalex. DOI: 10.1016/j.nanoen.2017.02.002

Stability Enhancement in Perovskite Solar Cells with Perovskite/Silver–Graphene Composites in the Active Layer

Tahmineh Mahmoudi; Yousheng Wang; Yoon-Bong Hahn. ACS Energy Letters (2018). Cited by 86. Found by: openalex. openalex_2yr: 16.712 (2026). DOI: 10.1021/acsenergylett.8b02201

Abstract: For practical use of perovskite solar cells (PSCs) the long-term stability of devices should be ensured. The device stability is attributed to degradation of the perovskite molecule due to moisture and ion migration and thermal and light instability under sunlight illumination. To solve the stability issues, we proposed a new approach, i.e., fabrication of PSCs based on perovskite/silver nanoparticle-anchored reduced graphene oxide (perovskite/Ag-rGO) composites. The Ag-rGO in the composite not only suppresses the ion migration but improves the thermal and light stability. The composite-based solar cells thermally aged at 90 °C for 90 h sustained over 94% of the initial value of the power conversion efficiency (PCE), while a pristine PSC showed a 39% decrease in its initial PCE. More importantly, the long-term stability of composite-based PSCs was exceptional, retaining almost 100% of the initial values of performance parameters over 330 days under ambient conditions (i.e., 25–30 °C, 45–57% humidity).

Reviewing and understanding the stability mechanism of halide perovskite solar cells

Caixin Zhang; Tao Shen; Dan Guo; Li-Ming Tang; Kaike Yang; Hui-Xiong Deng. InfoMat (2020). Cited by 84. Found by: openalex. DOI: 10.1002/inf2.12104

Abstract: Abstract Finding sustainable and renewable energy to replace traditional fossil fuel is critical for reducing greenhouse gas emission and avoiding environment pollution. Solar cells that convert energy of sunlight into electricity offer a viable route for solving this issue. At present, halide perovskites are the most potential candidate materials for solar cell with considerable power conversion efficiency, whereas their stability remains a challenge. In this work, we summarize four different key factors that influence the stability of halide perovskites: (a) effect of environmental moisture on the degradation of halide perovskites. The performance of halide perovskite solar cells is reduced due to hydrated crystal hinders the diffusion of photo-generated carriers, which can be solved by materials encapsulation technique; (b) photo-induced instability. Through uncovering the underlying physical mechanism, we note that materials engineering or novel device structure can extend the working life of halide perovskites under continuous light exposure; (c) thermal stability. Halide perovskites are rapidly degraded into PbI2 and volatile substances as heating due to lower formation energy, whereas hybrid perovskite is little changed; (d) electric field effect in the degradation of halide perovskites. The electric field impacts significantly on the carrier separation, changes direction of photo-induced currents and generates switchable photovoltaic effect. For each key factor, we have shown in detail the underlying physical mechanisms and discussed the strategies to overcome this stability difficulty. We expect this review from both theoretical and experimental points of view can be beneficial for development of perovskite solar cell materials and promotes practical applications. image

Double Barriers for Moisture Degradation: Assembly of Hydrolysable Hydrophobic Molecules for Stable Perovskite Solar Cells with High Open-Circuit Voltage

Pengfei Guo; Qian Ye; Chen Liu; Fengren Cao; Xiaokun Yang; Linfeng Ye; Wenhao Zhao; Hongyue Wang; Hongyue Wang; Liang Li; Hongqiang Wang; Hongqiang Wang. Advanced Functional Materials (2020). Cited by 84. Found by: openalex. openalex_2yr: 14.975 (2026). DOI: 10.1002/adfm.202002639

Abstract: Abstract The rapid growth in the device efficiency of perovskite solar cells (PSCs) has raised great demands for tackling their long-term stability upon external environmental stimuli that restricts the commercialization of PSCs, in which the instability upon exposure to moisture has been one of the major obstacles. Herein, an effective way of building up double barriers for moisture degradation of the perovskite films is demonstrated by modifying them with rationally selected hydrolyzable hydrophobic molecules (1H,1H,2H,2H-perfluorooctyl trichlorosilane, PFTS). The layer of oligomer derived from the hydrolyzed PFTS at the surface that increases the hydrophobicity of perovskite film could serve as an efficient wall preventing the moisture invasion. The long-term exposure of the film upon moisture allows for the formation of a secondary wall that employs the hydrolyzation of PFTS at grain boundaries, favoring defects passivation to further improve the humidity stability. Such gradual hydrolyzation is encouragingly helpful for the enhancement of the open-circuit voltage of the PSCs from the original 1.136 up to 1.205 V. The PSCs constructed with the double barriers demonstrate excellent stability upon moisture and improved thermal and light stabilities, as well as a champion power conversion efficiency up to 21.34%.

Spontaneous grain polymerization for efficient and stable perovskite solar cells

Xiaodong Li; Wenxiao Zhang; Wenjun Zhang; Hai-Qiao Wang; Junfeng Fang. Nano Energy (2019). Cited by 84. Found by: openalex. DOI: 10.1016/j.nanoen.2019.02.009

Glass-to-glass encapsulation with ultraviolet light curable epoxy edge sealing for stable perovskite solar cells

Easwaramoorthi Ramasamy; Vaithinathan Karthikeyan; Kadusu Rameshkumar; Ganapathy Veerappan. Materials Letters (2019). Cited by 83. Found by: openalex. DOI: 10.1016/j.matlet.2019.04.082

Composite Encapsulation Enabled Superior Comprehensive Stability of Perovskite Solar Cells

Yifan Lv; Hui Zhang; Ruqing Liu; Yanan Sun; Wei Huang. ACS Applied Materials & Interfaces (2020). Cited by 82. Found by: openalex. openalex_2yr: 8.382 (2026). DOI: 10.1021/acsami.0c06823

Abstract: Solar cells based on organometal hybrid perovskites have exhibited promising commercialization potential owing to their high efficiency and low-cost manufacturing. However, the poor outdoor operational stability of perovskite solar cells restricted their practical application, and moisture permeation and organic compounds volatilization are realized as the main factors accelerating performance degradation. Herein, we developed a composite encapsulation, by sequentially depositing a compact Al 2 O 3 layer and a hydrophobic 1 H,1 H,2 H,2 H -perfluorodecyltrichlorosilane layer on the completed device, to efficiently circumvent vapor permeability. Thus, the stability of the encapsulated perovskite solar cells was systematically investigated under simulated operational conditions. It was found that the MAPbI 3 perovskite was prone to decay into solid PbI 2 and organic vapor at high temperature or upon light illumination, and the decomposition was reversible in a well-encapsulated environment, resulting in reversible performance degradation and recovery. The enhanced thermal stability was ascribed to the competition between the perovskite decomposition and reverse synthesis. The as-prepared high-quality, multilayered encapsulation scheme demonstrated superior sealing property, and no obvious performance decline was observed when the device was stored under ambient air, continuous light illumination, double 85 condition (85 °C, 85% humidity), or even water immersion. Therefore, this work paves the way for a scalable and robust encapsulation strategy feasible to hybrid perovskite optoelectronics in a reproducible manner.

Degradation mechanism of planar-perovskite solar cells: correlating evolution of iodine distribution and photocurrent hysteresis

Riski Titian Ginting; Mi-Kyoung Jeon; Kwang-Jae Lee; Won-Yong Jin; Tae-Wook Kim; Jae-Wook Kang. Journal of Materials Chemistry A (2017). Cited by 80. Found by: openalex. DOI: 10.1039/c6ta09202k

Abstract: The penetration of moisture/O 2 into the perovskite solar cells (PSCs) causes shifts of iodine distribution within the perovskite layer and facilitates diffusion of iodine towards the hole transporting layer. Interestingly, these negative effects can be prevented by a simple encapsulation process and further prolong the stability of PSCs.

Stability Issues of Perovskite Solar Cells: A Critical Review

Shahriyar Safat Dipta; Ashraf Uddin. Energy Technology (2021). Cited by 79. Found by: openalex. openalex_2yr: 3.823 (2026). DOI: 10.1002/ente.202100560

Abstract: The efficiency of perovskite solar cells (PSCs) has risen rapidly over the past decade, and it has already crossed the 25% mark. However, stability has long been the bottleneck toward the commercialization of these devices. Perovskite is inherently vulnerable to moisture, high temperature, UV light, and other environmental factors, which naturally come in contact during operation. Moreover, degradation of the device is also associated with the hole transport layer (HTL), electron transport layer (ETL), and buffer layers. The mechanisms for PSCs' physical, chemical, structural, and environmental instabilities are discussed critically herein, along with recent efforts made by various groups to overcome these stability issues. Comparison is made among different engineering techniques to stabilize the devices. Moreover, the lack of unified criteria for stability tests of PSCs is discussed. Different degradation mechanisms are collated and compared and recent approaches of different groups on stability analysis from a neutral point of view are critically evaluated. Finally, this review urges future research to focus on novel materials for different layers which are reasonably lattice matched and stable with perovskite layer and use suitable encapsulation techniques for proper sealing of the device against degrading substances.

Advancement in CsPbBr3 inorganic perovskite solar cells: Fabrication, efficiency and stability

Naveen Kumar; Jyoti Rani; Rajnish Kurchania. Solar Energy (2021). Cited by 79. Found by: openalex. DOI: 10.1016/j.solener.2021.04.042

3D/2D multidimensional perovskites: Balance of high performance and stability for perovskite solar cells

Fei Zhang; Dong Hoe Kim; Kai Zhu. Current Opinion in Electrochemistry (2018). Cited by 79. Found by: openalex. openalex_2yr: 5.342 (2026). DOI: 10.1016/j.coelec.2018.10.001

Electrospray-Assisted Fabrication of Moisture-Resistant and Highly Stable Perovskite Solar Cells at Ambient Conditions

Shalinee Kavadiya; Dariusz M. Niedzwiedzki; Su Huang; Pratim Biswas. Advanced Energy Materials (2017). Cited by 76. Found by: openalex. openalex_2yr: 18.63 (2026). DOI: 10.1002/aenm.201700210

Abstract: An electrospray deposition technique to fabricate a perovskite (CH3NH3PbI3) layer for highly stable and efficient perovskite solar cells at ambient humidity (30%–50% relative humidity) conditions is demonstrated. A detailed study is conducted to determine the effect of different electrospray parameters on the device performance and to provide a mechanistic explanation of the superior stability of the films. Due to the controlled reactivity that results in the formation of a smooth perovskite film, these cells exhibit stability exceeding 4000 h, in contrast to much lower stability of those fabricated by conventional spin coating methods. Furthermore, the perovskite film deposited by electrospray methods exhibits a self-healing behavior when exposed to moisture. The authors hypothesize the formation of an intermediate metastable phase and smooth morphology of the film as the reason for this enhanced stability. Electrospray is a scalable technique that provides precise control over the amount of material required for deposition, reducing significant material loss that occurs in conventional solution-based methods. Overall, this work shows that stability of perovskite solar cells can be improved by fabrication using a well controlled and optimized electrospray technique, without the use of any additives or cell encapsulants.

In Situ Identification of Photo- and Moisture-Dependent Phase Evolution of Perovskite Solar Cells

Bo-An Chen; Jin-Tai Lin; Nian-Tzu Suen; Che-Wei Tsao; Tzu-Chi Chu; Ying-Ya Hsu; Ting-Shan Chan; Yi-Tsu Chan; Jye-Shane Yang; Ching-Wen Chiu; Hao Ming Chen. ACS Energy Letters (2017). Cited by 75. Found by: openalex. openalex_2yr: 16.712 (2026). DOI: 10.1021/acsenergylett.6b00698

Abstract: Photovoltaic performance of a perovskite solar cell is observed to decrease with increasing humidity, which might be attributed to the formation of hydrated intermediates leading to a decrease in extraction of photocarriers. However, direct evidence of the interplay between the perovskite layer and the photovoltaic performance under operating conditions (consecutive illumination) has not yet been reported. Herein, we investigated the degradation of perovskite solar cells under operating situations through in situ X-ray diffraction and in situ X-ray absorption spectroscopies, which revealed that lead hydroxide iodide (PbIOH), a new phase that has not previously been identified as the degradation product of perovskite solar cells, was formed as an end decomposition product inside the cell. The formation of PbIOH could break the interface inside and be the key reason behind the problem of reduced cell life. This work illustrates that in operando direct observation of the photo- and moisture-induced effects can provide needed insights toward realizing stable perovskite solar cells.

Printing High-Efficiency Perovskite Solar Cells in High-Humidity Ambient Environment—An In Situ Guided Investigation

W.K. Fong; Hanlin Hu; Zhiwei Ren; Kuan Liu; Li Cui; Tao Bi; Qiong Liang; Zehan Wu; Jianhua Hao; Gang Li. Advanced Science (2021). Cited by 75. Found by: openalex. openalex_2yr: 8.903 (2026). DOI: 10.1002/advs.202003359

Abstract: Abstract Extensive studies are conducted on perovskite solar cells (PSCs) with significant performance advances (mainly spin coating techniques), which have encouraged recent efforts on scalable coating techniques for the manufacture of PSCs. However, devices fabricated by blade coating techniques are inferior to state-of-the-art spin-coated devices because the power conversion efficiency (PCE) is highly dependent on the morphology and crystallization kinetics in the controlled environment and the delicate solvent system engineering. In this study, based on the widely studied perovskite solution system dimethylformamide–dimethyl sulfoxide, air-knife-assisted ambient fabrication of PSCs at a high relative humidity of 55 $\pm$ 5% is reported. In-depth time-resolved UV–vis spectrometry is carried out to investigate the impact of solvent removal and crystallization rate, which are critical factors influencing the crystallization kinetics and morphology because of adventitious moisture. UV–vis spectrometry enables accurate determination of the thickness of the wet precursor film. Anti-solvent-free, high-humidity ambient coatings of hysteresis-free PSCs with PCE s of 21.1% and 18.0% are demonstrated for 0.06 and 1 cm 2 devices, respectively. These PSCs exhibit comparable stability to those fabricated in a glovebox, thus demonstrating their high potential.

Self-healing perovskite solar cells

Yue Yu; Fu Zhang; Hua Yu. Solar Energy (2020). Cited by 75. Found by: openalex. DOI: 10.1016/j.solener.2020.09.018

Effect of relative humidity during the preparation of perovskite solar cells: Performance and stability

Isabel Mesquita; Luísa Andrade; Adélio Mendes. Solar Energy (2020). Cited by 72. Found by: openalex. DOI: 10.1016/j.solener.2020.02.052

Enhanced Moisture Stability by Butyldimethylsulfonium Cation in Perovskite Solar Cells

Bo-Hyung Kim; Maengsuk Kim; Jun Hee Lee; Sang Il Seok. Advanced Science (2019). Cited by 71. Found by: openalex. openalex_2yr: 8.903 (2026). DOI: 10.1002/advs.201901840

Abstract: Many organic cations in halide perovskites have been studied for their application in perovskite solar cells (PSCs). Most organic cations in PSCs are based on the protic nitrogen cores, which are susceptible to deprotonation. Here, a new candidate of fully alkylated sulfonium cation (butyldimethylsulfonium; BDMS) is designed and successfully assembled into PSCs with the aim of increasing humidity stability. The BDMS-based perovskites retain the structural and optical features of pristine perovskite, which results in the comparable photovoltaic performance. However, the fully alkylated aprotic nature of BDMS shows a much more pronounced effect on the increase in humidity stability, which emphasizes a generic electronic difference between protic ammonium and aprotic sulfonium cation. The current results would pave a new way to explore cations for the development of promising PSCs.

Excellent Moisture Stability and Efficiency of Inverted All-Inorganic CsPbIBr2 Perovskite Solar Cells through Molecule Interface Engineering

Shuzhang Yang; Liang Wang; Liguo Gao; Junmei Cao; Qianji Han; Fengyang Yu; Yusuke Kamata; Chu Zhang; Meiqiang Fan; Guoying Wei; Tingli Ma. ACS Applied Materials & Interfaces (2020). Cited by 70. Found by: openalex. openalex_2yr: 8.382 (2026). DOI: 10.1021/acsami.9b23532

Abstract: All-inorganic lead halide perovskite solar cells (PSCs) have drawn widespread interest because of its excellent thermal stability compared to its organic–inorganic hybrid counterpart. Poor phase stability caused by moisture, however, has thus far limited their commercial application. Herein, by modifying the interface between the hole-transport layer (HTL) and the perovskite light absorption layer, and by optimizing the HTL for better energy alignment, we controlled the growth of perovskite, reduced carrier recombination, facilitated carrier injection and transport, and improved the PSC's power conversion efficiency (PCE) and moisture stability. When testing using a positive bias scan, we obtained a significant improvement in PCE, 9.49%, which is the champion efficiency of CsPbIBr 2 -based inverted PSC at present. The stability measurement shows that the passivated CsPbIBr 2 -based inverted PSCs can retain 86% of its initial efficiency after 1000 h preserved in ambient air with 65% relative humidity. This study paves a new way for enhancing the moisture stability and power conversion efficiency of CsPbIBr 2 -based PSCs.

Amphiphilic Fullerenes Employed to Improve the Quality of Perovskite Films and the Stability of Perovskite Solar Cells

Qingxia Fu; Shuqin Xiao; Xianglan Tang; Yiwang Chen; Ting Hu. ACS Applied Materials & Interfaces (2019). Cited by 70. Found by: openalex. openalex_2yr: 8.382 (2026). DOI: 10.1021/acsami.9b07149

Abstract: Fullerene end-capped polyethylene glycol (C60-PEG) was introduced via an antisolvent method to fabricate the perovskite films. C60-PEG could enlarge the perovskite crystal size and passivate the defects of perovskite films, facilitating the carrier transport and hindering the carrier recombination. In consequence, the superior optoelectronic properties were attained with an improved power conversion efficiency of 17.71% for the perovskite device with C60-PEG treatment. Meanwhile, amphiphilic C60-PEG enhanced the resistance of perovskite films to moisture. After 40 days, the C60-PEG-based devices without encapsulation remained 93 and 86% of the original power conversion efficiency value under nitrogen and ambient conditions (25 °C temperature, 60% humidity), respectively.

From Exceptional Properties to Stability Challenges of Perovskite Solar Cells

Somayeh Gholipour; Michael Saliba. Small (2018). Cited by 69. Found by: openalex. DOI: 10.1002/smll.201802385

Abstract: The discovery and development of organic-inorganic halide perovskites with exceptional properties has become an active research area in the field of photovoltaics. Perovskite solar cells (PSCs) have attracted much attention in recent years due to various attractive advantages, such as simple solution processing, low manufacturing cost, and high performances with power conversion efficiencies now reaching certified values close to 23% within a very short time frame of five years. Despite this rapid progress, the inferior device stability remains a great challenge. This review focuses on the factors limiting the stability of PSCs, such as humidity, heat, and irradiation, summarizing recent strategies to overcome stability and fabrication obstacles in order to open new perspectives to achieve highly durable perovskite devices toward future industrialization.

Aromatic Alkylammonium Spacer Cations for Efficient Two-Dimensional Perovskite Solar Cells with Enhanced Moisture and Thermal Stability

Deepak Thrithamarassery Gangadharan; Yujie Han; Ashish Dubey; Xinyu Gao; Baoquan Sun; Qiquan Qiao; Ricardo Izquierdo; Dongling Ma. Solar RRL (2018). Cited by 68. Found by: openalex. DOI: 10.1002/solr.201700215

Abstract: Three-dimensional (3D) perovskite solar cells are prone to degradation in the presence of moisture, heat, and light. Recently, two-dimensional (2D) perovskites are synthesized by isolating metal halide perovskite layers using aliphatic or aromatic alkylammonium spacer cation, which can retain their performance under ambient humidity levels due to the hydrophobic property of the spacer cation. However, the best 2D perovskite thus far, using aliphatic short butylammonium (BA) cation as spacer cation, shows only a modest tolerance against moisture and heat due to the inferior hydrophobicity as well as the relatively smaller size of the BA cation. Here, a bulkier aromatic phenylethylammonium (PEA) used as a spacer cation to synthesis 2D perovksite in order to achieve highly stable solar cells. By modifying the crystallization process, an average power conversion efficiency (PCE) of 5.50% is achieved, which is the highest reported PCE for aromatic alkylammonium-based lower dimensional perovskite solar cells. Importantly, unencapsulated (PEA)2(MA)3Pb4I13 devices show enhanced moisture stability compared to other reported perovskite solar cells in harsh moisture environment (72 ± 2% relative humidity). Moreover, the use of organic materials in p-i-n type device, instead of metal oxides, as electron and hole extraction layers also paves the way toward constructing flexible perovskite solar cells.

Thermal and Humidity Stability of Mixed Spacer Cations 2D Perovskite Solar Cells

Huayang Yu; Yulin Xie; Jia Zhang; Jiashun Duan; Xu Chen; Yu-Dong Liang; Kai Wang; Ling Xu. Advanced Science (2021). Cited by 67. Found by: openalex. openalex_2yr: 8.903 (2026). DOI: 10.1002/advs.202004510

Abstract: Abstract In this article, two different types of spacer cations, 1,4-butanediamonium (BDA2+) and 2-phenylethylammonium (PEA+) are co-used to prepare the perovskite precursor solutions with the formula of (BDA)1-a(PEA2)aMA4Pb5X16. By simply mixing the two spacer cations, the self-assembled polycrystalline films of (BDA)0.8(PEA2)0.2MA4Pb5X16 are obtained, and BDA2+ is located in the crystal grains and PEA+ is distributed on the surface. The films display a small exciton binding energy, uniformly distributed quantum wells and improved carrier transport. Besides, utilizing mixed spacer cations also induces better crystallinity and vertical orientation of 2D perovskite (BDA)0.8(PEA2)0.2MA4Pb5X16 films. Thus, a power conversion efficiency (PCE) of 17.21% is achieved in the optimized perovskite solar cells with the device structure of ITO/PEDOT:PSS/Perovskite/PCBM/BCP/Ag. In addition, the complementary humidity and thermal stability are obtained, which are ascribed to the enhanced interlayer interaction by BDA2+ and improved moisture resistance by the hydrophobic group of PEA+. The encapsulated devices are retained over 95% or 75% of the initial efficiency after storing 500 h in ambient air under 40 ± 5% relative humidity or 100 h in nitrogen at 60 °C.

Chlorine-terminated MXene quantum dots for improving crystallinity and moisture stability in high-performance perovskite solar cells

Xiao Liu; Zhiang Zhang; Jinkun Jiang; Congcong Tian; Xin Wang; Luyao Wang; Zhanfei Zhang; Xueyun Wu; Yiting Zheng; Jianghu Liang; Chun-Chao Chen. Chemical Engineering Journal (2021). Cited by 64. Found by: openalex. openalex_2yr: 10.953 (2026). DOI: 10.1016/j.cej.2021.134382

Stability and Performance Enhancement of Perovskite Solar Cells: A Review

Maria Khalid; Tapas K. Mallick. Energies (2023). Cited by 62. Found by: openalex. openalex_2yr: 4.847 (2026). DOI: 10.3390/en16104031

Abstract: Perovskite solar cells (PSCs) have seen a rapid increase in power conversion efficiencies (PCEs) over just a few years and are already competing against other photovoltaic (PV) technologies. The PCE of hybrid PSCs exhibiting distinct properties has increased from 3.8% in 2009 to 30% in 2023, making it a strong contender for the next generation of PV devices. However, their long-term stability is a critical issue that must be addressed before these devices can be commercialised. This review begins with a discussion of the evolution of different generations of solar cells, and the following part presents details of perovskite characteristics and prospective strategies to improve their performance. Next, the relationship of stability of PSCs with different environmental conditions, including moisture, UV light, and temperature, is discussed. Besides the development of PSC–silicon tandem solar cells, an efficient way to improve PCE is also discussed. Towards the end, we discuss a novel idea of implementing PSCs with a concentrated PV application in order to achieve higher efficiency and compete with other PV technologies by catching incident high-proton density. This review offers perspectives on the future development of emerging PSC technologies in terms of device performance enhancement and improved stability, which are central to tandem and concentrated PSC technology.

Progress on the stability and encapsulation techniques of perovskite solar cells

Ling Xiang; Fangliang Gao; Yunxuan Cao; Dongyang Li; Qing Liu; Hongliang Liu; Shuti Li. Organic Electronics (2022). Cited by 61. Found by: openalex. DOI: 10.1016/j.orgel.2022.106515

Encapsulating perovskite solar cells for long-term stability and prevention of lead toxicity

Shahriyar Safat Dipta; Md. Arifur Rahim; Ashraf Uddin. Applied Physics Reviews (2024). Cited by 61. Found by: openalex. openalex_2yr: 11.309 (2026). DOI: 10.1063/5.0197154

Abstract: Lead halide perovskite solar cells (PSCs) have achieved remarkable efficiencies comparable to those of their established silicon counterparts at a very fast pace. Moreover, solution-processable facile technologies offer low-cost, low-temperature, scalable fabrication of these solar cells. Numerous studies have focused on improving the performance, stability, and processing of PSCs. However, potential lead toxicity and poor long-term stability impede their commercialization. In recent years, several studies have developed novel encapsulants for PSCs that can simultaneously improve stability and impede potential lead leakage. Although improvements have been made on both fronts, no solution to date could achieve a level of stability and leakage prevention that could result in a market breakthrough. Here, we analyze PSC encapsulation and lead leakage prevention techniques undertaken in recent years. While most of the related studies focused on improving either stability or toxicity, we note that both can be solved together with a suitable encapsulant that is impermeable to both moisture and Pb2+ ions. In addition, the lack of a unified standard stability testing protocol has led to stability testing under a variety of temperatures, humidities, and environmental conditions. Therefore, the urgency for a standard protocol for stability and lead toxicity testing cannot be overlooked.

Recent Progress on the Stability of Perovskite Solar Cells in a Humid Environment

Mengying Li; Haibo Li; Jing Fu; Tianyu Liang; Wei Ma. The Journal of Physical Chemistry C (2020). Cited by 60. Found by: openalex. DOI: 10.1021/acs.jpcc.0c08019

Abstract: In recent years, photovoltaic devices based on inorganic–organic hybrid perovskite materials have become one of the most promising research subjects in the field of energy conversion. CH 3 NH 3 PbI 3 stands out among a wide variety of perovskite structural materials with the advantages of a suitable band structure, small exciton binding energy, long carrier diffusion distance, and so on. Unfortunately, the CH 3 NH 3 PbI 3 perovskite undergoes severe degradation under humid conditions, which limits the service life of the device. To address this issue, researchers have recently discovered that the humidity stability of perovskite solar cells (PSCs) can be improved through doping or interfacial engineering. Here, we have reviewed the state of the research progress in improving the humidity stability of PSCs, including (a) improving the structural stability of perovskite material itself by doping or element substitutions; (b) interface engineering between the hole transport layer and the perovskite active layer; (c) interface modification between the electron transport layer and the perovskite layer; and (d) encapsulation. The strategy of improving the humidity stability of CH 3 NH 3 PbI 3 by optimizing the device structure and developing new materials is summarized. We also make constructive suggestions for improving the stability of PSCs in humid environments.

Light enhanced moisture degradation of perovskite solar cell material CH 3 NH 3 PbI 3

Ying-Bo Lu; Wei-Yan Cong; ChengBo Guan; Hui Sun; Yanqing Xin; Kunlun Wang; Shumei Song. Journal of Materials Chemistry A (2019). Cited by 60. Found by: openalex. DOI: 10.1039/c9ta10443g

Abstract: We reveal the relationship between the light enhanced moisture degradation and the photostriction effect in MAPbI 3 perovskite.

Enhanced moisture stability of metal halide perovskite solar cells based on sulfur–oleylamine surface modification

Yu Hou; Zi Ren Zhou; Tian Wen; Hong Qiao; Ze Qing Lin; Bing Ge; Hua Gui Yang. Nanoscale Horizons (2018). Cited by 57. Found by: openalex. DOI: 10.1039/c8nh00163d

Abstract: As one of the most promising light-harvesting materials, perovskites have drawn tremendous attention for their unique advantages, such as high efficiency, low cost and facile fabrication compared with other photovoltaic materials. Nevertheless, poor moisture tolerance of the perovskites greatly hampers the operation of such devices and hinders their commercialization. Herein, we demonstrate a facile dipping treatment using sulfur-oleylamine solution for surface atomic modulation of perovskite films. Oleylammonium polysulfides (OPs) would be self-assembled on the etched perovskite film as an ultrathin outer layer. This layer could passivate the surface chemical activity of the outer perovskite layers. Moreover, the hydrophobic OPs significantly enhance moisture stability of such devices. As a result, the obtained device without encapsulation retains more than 70% of its initial power conversion efficiency (PCE) after 14 days of exposure to a relative humidity of 40 $\pm$ 10%.

Homeopathic Perovskite Solar Cells: Effect of Humidity during Fabrication on the Performance and Stability of the Device

Lidia Contreras-Bernal; Clara Aranda; Marta Vallés-Pelarda; Thi Tuyen Ngo; Susana Ramos-Terrón; Juan Jesús Gallardo; Javier Navas; Antonio Guerrero; Iván Mora-Seró; Jesús Idígoras; Juan A. Anta. The Journal of Physical Chemistry C (2018). Cited by 56. Found by: openalex. DOI: 10.1021/acs.jpcc.8b01558

Abstract: Rapid degradation in humid environments is a major drawback of methylammonium lead iodide (CH 3 NH 3 PbI 3), which is the archetypical component of perovskite solar cells. In this work, we have investigated the aging and degradation kinetics of CH 3 NH 3 PbI 3 films and devices fabricated under controlled conditions as a function of relative humidity (RH) and compared their performance with those that were prepared under dry conditions. The aging and degradation kinetics is monitored by optical absorption and impedance spectroscopy measurements under monochromatic illumination at two different wavelengths. Aged devices show a substantial difference between the recombination rate under red and blue light illumination, which is attributed to the enhancement of local recombination routes upon aging. Interestingly, we observe that this feature is less pronounced in devices prepared under conditions of the highest RH of 50%. In general, we found that these devices keep their original electric properties and withstand a humid environment better. Chemical analysis by X-ray photoelectron spectroscopy reveals the presence of coordinating water in the CH 3 NH 3 PbI 3 crystalline structure. This indicates that the presence of a small amount of water has a beneficial effect against degradation in a humid environment.

Nonionic Sc3N@C80 Dopant for Efficient and Stable Halide Perovskite Photovoltaics

Kai Wang; Xiaoyang Liu; Rong Huang; Congcong Wu; Dong Yang; Xiaowen Hu; Xiaofang Jiang; James C. Duchamp; Harry C. Dorn; Shashank Priya. ACS Energy Letters (2019). Cited by 56. Found by: openalex. openalex_2yr: 16.712 (2026). DOI: 10.1021/acsenergylett.9b01042

Abstract: One major challenge in the commercialization of halide perovskite solar cells is the device instability against moisture. Efficient perovskite solar cells usually incorporate the ionic dopants within the charge transfer material to secure their electrical conductivity. Typical ionic dopants are hygroscopic, resulting in significant moisture adsorption and accelerated perovskite degradation. Herein, we report a nonionic dopant, Sc 3 N@C 80, consisting of a hydrophobic fullerene cage that encapsulates the metal salt to achieve a moisture resistive and highly electrically conductive hole transfer layer (HTL). The direct electronic transaction between Sc 3 N@C 80 and spiro-OMeTAD renders a drastically improved conductivity and a lower Fermi-level of the HTL to minimize the Schottky barrier. The hydrophobicity of Sc 3 N@C 80 also decreases the moisture wettability. Perovskite solar cells using the Sc 3 N@C 80 -doped HTL exhibit an efficiency surge from 18.15 to 20.77% (champion cell exhibiting 21.09%) and improved device stability (survival efficiency over 17% after continuous

illumination for 800 h). The bifunctional hydrophobic Sc 3 N@C 80 dopant provides a pathway toward efficient and stable perovskite photovoltaics.

Improving the moisture stability of perovskite solar cells by using PMMA/P3HT based hole-transport layers

Soumya Kundu; Timothy L. Kelly. Materials Chemistry Frontiers (2017). Cited by 56. Found by: openalex. DOI: 10.1039/c7qm00396j

Abstract: In recent years, the performance of lead halide perovskite solar cells has increased dramatically, setting a record efficiency of 22.1%; however, their sensitivity towards water limits their utility and still needs to be addressed.

Moisture-tolerant supermolecule for the stability enhancement of organic–inorganic perovskite solar cells in ambient air

Dong Wei; Hao Huang; Peng Cui; Jun Ji; Shangyi Dou; Endong Jia; Sajid Sajid; Mengqi Cui; Lihua Chu; Yingfeng Li; Bing Jiang; Meicheng Li. Nanoscale (2018). Cited by 53. Found by: openalex. DOI: 10.1039/c8nr07638c

Abstract: Instability of the perovskite materials, especially in high humidity, is one of the major limitations that hinders the development of perovskite devices. Herein, to eliminate the degradation of perovskite solar cells in humid air, a water-resistant perovskite absorption layer is proposed by introducing a macrocycle-type cyclodextrin material (-CD) into the films. The -CD was proved to be capable of facilitating the crystallization of grains and enhancing the stability of the perovskite by forming supramolecular interactions with organic cations through the hydrogen bonding in the perovskite films. Consequently, the average efficiency of the PSCs remarkably increased from 16.19% to 19.98%. The champion solar cell even delivered an efficiency of 20.09%. The PSCs with -CD exhibited superior long-term stability in ambient air without encapsulation, which retained 90% of the initial efficiency after continuous AM 1.5 illumination in ambient air with 80% humidity for 300 h.

Enhanced moisture stability of MAPbI3 perovskite solar cells through Barium doping

Jitendra Bahadur; Amir H. Ghahremani; Sunil Gupta; Thad Druffel; Mahendra K. Sunkara; Kaushik Pal. Solar Energy (2019). Cited by 51. Found by: openalex. DOI: 10.1016/j.solener.2019.08.033

Reduced Graphene Oxide Improves Moisture and Thermal Stability of Perovskite Solar Cells

Hui-Seon Kim; Bowen Yang; Minas M. Stylianakis; Emmanuel Kymakis; Shaik M. Zakeeruddin; Michaël Grätzel; Anders Hagfeldt. Cell Reports Physical Science (2020). Cited by 50. Found by: openalex. openalex_2yr: 5.933 (2026). DOI: 10.1016/j.xcrp.2020.100053

Abstract: Reduced graphene oxide (rGO) is a very stable material with tunable electronic properties dependent on its reduction level. Nevertheless, the incorporation of rGO into perovskite solar cells (PSCs) frequently demonstrates inconsistent results, and it plays an ambiguous role. Here, we systematically explore rGO as an additive in perovskite and 2,2 ,7,7 -tetrakis[N,N-di(4-methoxyphenyl)amino]-9,9 -spirobifluorene (spiro-MeOTAD) using different reduction levels (O18.42%, O9.67%, and O7.14%). Perovskite with rGOO9.67% results in the most stable shelf stability, demonstrating a negligible performance degradation over 430 h, while the one without rGO shows a 6.4% drop, mainly due to fill-factor deterioration. Moreover, the improved stability is more pronounced when rGOO9.67% is employed with spiro-MeOTAD where crystallization is effectively retarded. The integrated effect is also observed in PSCs based on rGOO9.67% in both perovskite and spiro-MeOTAD, leading to enhanced stability at 85°C and 45% relative humidity.

A simple method to evaluate the effectiveness of encapsulation materials for perovskite solar cells

Timothy J. Wilderspin; Francesca De Rossi; Trystan Watson. Solar Energy (2016). Cited by 48. Found by: openalex. DOI: 10.1016/j.solener.2016.09.038

Simultaneously enhanced moisture tolerance and defect passivation of perovskite solar cells with cross-linked grain encapsulation

Ke Xiao; Qiaolei Han; Yuan Gao; Shuai Gu; Xin Luo; Renxing Lin; Jia Zhu; Jun Xu; Hairen Tan. Journal of Energy Chemistry (2020). Cited by 46. Found by: openalex. DOI: 10.1016/j.jechem.2020.08.020

Protecting Perovskite Solar Cells against Moisture-Induced Degradation with Sputtered Inorganic Barrier Layers

Ramez Hosseinian Ahangharnejhad; Zhaoning Song; Tamanna Mariam; Jayla J. Gardner; Geethika K. Liyanage; Zahrah S. Almutawah; Bhuiyan M. Anwar; Maxwell M. Junda; Nikolas J. Podraza; Adam B. Phillips; Yanfa Yan; Michael J. Heben. ACS Applied Energy Materials (2021). Cited by 46. Found by: openalex. openalex_2yr: 5.576 (2026). DOI: 10.1021/acsaem.1c00816

Abstract: We describe a general approach for protecting metal halide perovskite solar cells against degradation in high-humidity environments using a sputtered barrier coating. A SiO2 protective layer, applied to two different types of perovskite solar cells, was deposited without strongly impacting the initial device performance. The degradation of the cells was imaged in real time using laser beam-induced current (LBIC) measurements in an accelerated test. We show that SiO2 barrier films can improve the tolerance of the devices to extreme humidity conditions and extend lifetimes by a factor of ~60 when a 45 nm SiO2 barrier layer is applied to CH3NH3PbI3 and by a factor of ~600 when a 300 nm barrier is applied to a triple-cation perovskite material. The LBIC data revealed that the approach promises protection against degradation initiation at the edges of scribed lines, which will be necessary for the fabrication of monolithically integrated modules.

Role of interface in stability of perovskite solar cells

Chris Manspeaker; Swaminathan Venkatesan; Alex Zakhidov; Karen S. Martirosyan. Current Opinion in Chemical Engineering (2016). Cited by 45. Found by: openalex. openalex_2yr: 8.446 (2026). DOI: 10.1016/j.coche.2016.08.013

Hysteresis and Instability Predicted in Moisture Degradation of Perovskite Solar Cells

Kelvin J. Xu; Ryan Taoran Wang; Fan Xu; Jason Y. Chen; Gu Xu. ACS Applied Materials & Interfaces (2020). Cited by 38. Found by: openalex. openalex_2yr: 8.382 (2026). DOI: 10.1021/acsami.0c17323

Abstract: The degradation of the perovskite solar cell structure was expected recently to be reversible, which opened a new gate to the enhancement of the device lifetime by reversing the process. However, the kinetic details of the structural collapse and recovery are still missing, without which the perovskite reversibility cannot be further explored. Due to the experimental difficulty, a purposeful numerical model was conducted in this report, to simulate the water diffusion process in the perovskite structure in both directions. It was found that the moisture diffusion needs to be initiated by a certain level of structural imperfection and is non-Fickian, as assisted by the collapse of the perovskite into the 1D chains. The reversibility was verified by the back diffusion, but accompanied by hysteresis, stagnancy, and even surprising instability, which initiated the water flow under initial equilibrium, due possibly to the imbalance during the reconstruction of the perovskite lattice. These observations offer new insights to form strategies of improvement, for example, via the possible self-healing perovskite devices.

Morphological Insights into the Degradation of Perovskite Solar Cells under Light and Humidity

Kun Sun; Renjun Guo; Yuxin Liang; Julian E. Heger; Shangpu Liu; Shanshan Yin; Manuel A. Reus; Lukas V. Spanier; Felix Deschler; Sigrid Bernstorff; Peter Müller-Buschbaum. ACS Applied Materials & Interfaces (2023). Cited by 36. Found by: openalex. openalex_2yr: 8.382 (2026). DOI: 10.1021/acsami.3c05671

Abstract: Perovskite solar cells (PSCs) have achieved competitive power conversion efficiencies compared with established solar cell technologies. However, their operational stability under different external stimuli is limited, and the underlying mechanisms are not fully understood. In particular, an understanding of degradation mechanisms from a morphology perspective during device operation is missing. Herein, we investigate the operational stability of PSCs with CsI bulk modification and a CsI-modified buried interface under AM 1.5G illumination and 75 ± 5% relative humidity, respectively, and concomitantly probe the morphology evolution with grazing-incidence small-angle X-ray scattering. We find that volume expansion within perovskite grains, induced by water incorporation, initiates the degradation of PSCs under light and humidity and leads to the degradation of device performance, in particular, the fill factor and short-circuit current. However, PSCs with modified buried interface degrade faster, which is ascribed to grain fragmentation and increased grain boundaries. In addition, we reveal a slight lattice expansion and PL redshifts in both PSCs after exposure to light and humidity. Our detailed insights from a buried microstructure perspective on the degradation mechanisms under light and humidity are essential for extending the operational stability of PSCs.

Moisture stability in nanostructured perovskite solar cells

Mahdi Malekshahi Byranvand; Ali Nemati Kharat; Nima Taghavinia. Materials Letters (2018). Cited by 31. Found by: openalex. DOI: 10.1016/j.matlet.2018.10.029

Encapsulation of Perovskite Solar Cells for High Humidity Conditions

Qi Dong; Fangzhou Liu; Man Kwong Wong; Ho Won Tam; Aleksandra B. Djurišić; Annie Ng; C. Surya; Wai Kin Chan; Alan Man Ching Ng. ChemSusChem (2016). Cited by 23. Found by: openalex. openalex_2yr: 5.476 (2026). DOI: 10.1002/cssc.201601090

Abstract: Abstract Invited for this month s cover is the group of Aleksandra B. Djurišić at the University of Hong Kong along with collaborators from the University of Hong Kong, Hong Kong Polytechnic University, and South University of Science & Technology of China. The image shows outdoor testing performance of perovskite solar cells during summer in Hong Kong. The Full Paper itself is available at 10.1002/cssc.201600868 .

Degradation Kinetics of Inverted Perovskite Solar Cells

Mejd Alsari; Andrew J. Pearson; Jacob Tse-Wei Wang; Zhiping Wang; Augusto Montisci; Neil C. Greenham; Henry J. Snaith; Samuele Lilliu; Richard H. Friend. arXiv preprint (2018). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/1801.09761v1

Abstract: We explore the degradation behaviour under continuous illumination and direct oxygen exposure of inverted unencapsulated formamidinium(FA)0.83Cs0.17Pb(I0.8Br0.2)3, CH3NH3PbI3, and CH3NH3PbI3-xClx perovskite solar cells. We continuously test the devices in-situ and in-operando with current-voltage sweeps, transient photocurrent, and transient photovoltage measurements, and find that degradation in the CH3NH3PbI3-xClx solar cells due to oxygen exposure occurs over shorter timescales than FA0.83Cs0.17Pb(I0.8Br0.2)3 mixed-cation devices. We attribute these oxygen-induced losses in the power conversion efficiencies to the formation of electron traps within the perovskite photoactive layer. Our results highlight that the formamidinium-caesium mixed-cation perovskites are much less sensitive to oxygen-induced degradation than the methylammonium-based perovskite cells, and that further improvements in perovskite solar cell stability should focus on the mitigation of trap generation during ageing.

Degradation Dynamics of Perovskite Solar Cells Under Fixed Reverse Current Injection

Fangyuan Jiang; Haruka Koizumi; Hannah Contreras; Rajiv Giridharagopal; Akash Dasgupta; Zixu Huang; Ryan A. DeCrescent; Kell Fremouw; Michael D. McGehee; Neal R. Armstrong; David S. Ginger. arXiv preprint (2026). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/2603.20516v1

Abstract: Previous studies of reverse-bias stability in perovskite solar cells have focused primarily on voltage controlled reverse-bias tests. Here we instead present an investigation of perovskite solar cell degradation under well-defined, constant reverse-current stress. We show that the choice of hole-transport layer dictates the dominant degradation pathway: cells using thick poly(triphenylamine) (PTAA) layers with better indium-doped tin oxide (ITO) coverage can tolerate high reverse bias but quickly undergo catastrophic breakdown under fixed reverse current near their one-sun maximum power-point. In contrast, cells modified with the phosphonic-acid interface layer MeO-2PACz, with poorer ITO coverage compared to PTAA, exhibit soft, gradual, and largely recoverable degradation, regardless of the shading conditions. For MeO-2PACz devices, degradation increases with both current magnitude and duration. Importantly, when normalized by injected charge (current times duration), lower currents applied over longer times cause more severe degradation than higher currents over shorter periods. Combining electrical measurements with spatially resolved photoluminescence imaging, we argue against shunt formation and instead support an ion- and charge-mediated interfacial electrochemical degradation mode.

Effect of Ion Migration Induced Electrode Degradation on the Operational Stability of Perovskite Solar Cells

Boris Rivkin; Paul Fassl; Qing Sun; Alexander D. Taylor; Zhuoying Chen; Yana Vaynzof. arXiv preprint (2019). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/1904.11731v1

Abstract: Perovskite-based solar cells are promising due to their rapidly improving efficiencies, but suffer from instability issues. Recently it has been claimed that one of the key contributors to the instability of perovskite solar cells is ion migration induced electrode degradation, which can be avoided by incorporating inorganic hole

blocking layers (HBL) in the device architecture. In this work, we investigate the operational environmental stability of methylammonium lead iodide (MAPbI3) perovskite solar cells that contain either an inorganic or organic HBL, with only the former effectively blocking ions from migrating to the metal electrode. This is confirmed by X-ray photoemission spectroscopy measured on electrodes of degraded devices, where only electrodes of devices with an organic HBL show a significant iodine signal. Despite this, we show that when these devices are degraded under realistic operational conditions (i.e. constant illumination in a variety of atmospheric conditions), both types of devices exhibit nearly identical degradation behavior. These results demonstrate that contrary to prior suggestions, ion-induced electrode degradation is not the dominant factor in perovskite environmental instability under operational conditions.

Predictive modeling of ion migration induced degradation in perovskite solar cells

Vikas Nandal; Pradeep R. Nair. arXiv preprint (2017). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/1704.06603v1

Abstract: With excellent efficiencies being reported from multiple labs across the world, device stability and the degradation mechanisms have emerged as the key aspects that could determine the future prospects of perovskite solar cells. However, the related experimental efforts remain scattered due to the lack of any unifying theoretical framework. In this context, here we provide a comprehensive analysis of ion migration effects in perovskite solar cells. Specifically, we show, for the first time, that (a) the effect of ionic charges is almost indistinguishable from that of dopant ions, (b) ion migration could lead to simultaneous improvement in Voc and degradation in Jsc - a unique observation which is beyond the realm of mere parametric variation in carrier mobility and lifetime, (c) champion devices are more resilient towards the ill effects of ion migration, and finally (d) we propose unique characterization schemes to determine both magnitude and polarity of ionic species. Our results, supported by detailed numerical simulations and direct comparison with experimental data, are of broad interest and provide a much needed predictive capability towards the research on performance degradation mechanisms in perovskite solar cells

Spontaneous Radiative Cooling to Enhance the Operational Stability of Perovskite Solar Cells via a Black-body-like Full Carbon Electrode

Bingcheng Yu; Jiangjian Shi; Yiming Li; Shan Tan; Yuqi Cui; Fanqi Meng; Huijue Wu; Yanhong Luo; Dongmei Li; Qingbo Meng. arXiv preprint (2022). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/2205.13103v1

Abstract: Operational stability of perovskite solar cells is remarkably influenced by the device temperature, therefore, decreasing the interior temperature of the device is one of the most effective approaches to prolong the service life. Herein, we introduce the spontaneous radiative cooling effect into the perovskite solar cell and amplified this effect via functional structure design of a full-carbon electrode (F-CE). Firstly, with interface engineering, >19% and >23% power conversion efficiencies of F-CE based inorganic CsPbI3 and hybrid perovskite solar cells have been achieved, respectively, both of which are the highest reported efficiencies based on carbon electrode and are comparative to the results for metal electrodes. Highly efficient thermal radiation of this F-CE can reduce the temperature of the operating cell by about 10 °C. Compared with the conventional metal electrode-based control cells, the operational stability of the above two types of cells have been significantly improved due to this cooling effect. Especially, the CsPbI3 PSCs exhibited no efficiency degradation after 2000 hours of continuous operational tracking.

Quantifying Perovskite Solar Cell Degradation via Machine Learning from Spatially Resolved Multimodal Luminescence Time Series

Giulio Barletta; Simon Ternes; Saif Ali; Zohair Abbas; Chiara Ostendi; Marialucia D'Addio; Erica Magliano; Pietro Asinari; Eliodoro Chiavazzo; Aldo Di Carlo. arXiv preprint (2026). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/2603.12857v2

Abstract: Perovskite solar cells achieve remarkable power conversion efficiencies, yet operational stability remains a major barrier to large-scale deployment. Reliable and rapid assessment of device state of health is therefore essential. Conventional electrical diagnostics, such as illuminated current-voltage (J–V) sweeps, provide accurate performance metrics but are time-consuming and do not resolve spatially localized degradation, motivating non-invasive imaging-based alternatives. A deep-learning framework is introduced to estimate PSC efficiency retention, $R_\mathrm{PCE}=\mathrm{PCE}_t/\mathrm{PCE}_0$, directly from multimodal luminescence imaging acquired during device aging. Each sample combines electroluminescence (EL), open-circuit photoluminescence (PLoc), and short-circuit photoluminescence (PLsc) at an aged state with device-specific reference images at $t=0$, enabling learning of degradation-relevant spatial changes. LumPerNet, a compact convolutional neural network, is benchmarked against a spatially homogenized control in which each luminescence channel is replaced by its spatial average while retaining the same learning framework and leakage-aware protocol. The comparison indicates that global luminescence evolution contains most of the predictive signal, while spatial information provides a secondary contribution to robustness. These results establish spatially resolved luminescence imaging as a practical route for accelerated stability testing and non-invasive degradation monitoring in perovskite photovoltaics.

Thermal Degradation Mechanisms and Stability Enhancement Strategies in Perovskite Solar Cells: A Review

Arghya Paul; Kanak Raj; Prince Raj Lawrence Raj; Pratim Kumar. arXiv preprint (2025). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/2509.13700v1

Abstract: Perovskite Solar Cells (PSCs) have garnered global research interest owing to their superior photovoltaic (PV) performance. The future of photovoltaic technology lies in PSCs since they can produce power with performance on par with the best silicon solar cells while being less expensive. PSCs have enormous potential; in just ten years, their efficiency increased from 3.8% to 25.2%, and research into new developments is still ongoing. Thermal instability is PSCs' main disadvantage, despite their high efficiency, flexibility, and lightweight nature. This paper looks at how temperature affects the ways that hole transport layers (HTLs) like spiro-OMeTAD and perovskite layers, especially MAPbI3, degrade. Elevated temperatures cause MAPbI3 to degrade into PbI2, CH3I, and NH3, with decomposition rates affected by moisture, oxygen, and environmental factors. Mixed cation compositions, such as Cs-MA-FA, have higher thermal stability, whereas MA+ cations break-down faster under heat stress. HTLs deteriorate due to morphological changes and the hydrophilicity of dopant additions like Li-TFSI and t-BP. Alternative dopant-free HTMs, such as P3HT and inorganic materials including CuSCN, NiOx, and Cu2O, have shown improved thermal stability and efficiency. Hybrid HTLs, dopant-free designs, and interface tweaks are all viable solutions for increasing the stability of PSC. Addressing thermal stability issues remains crucial for the development of more reliable and efficient PSC technology.

Architecture Optimization Dramatically Improves Reverse Bias Stability in Perovskite Solar Cells: A Role of Polymer Hole Transport Layers

Fangyuan Jiang; Yangwei Shi; Tanka R. Rana; Daniel Morales; Isaac Gould; Declan P. McCarthy; Joel Smith; Grey Christoforo; Hannah Contreras; Stephen Barlow; Aditya D. Mohite; Henry Snaith; Seth R. Marder; J. Devin MacKenzie; Michael D. McGehee; David S. Ginger. arXiv preprint (2023). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/2308.08084v1

Abstract: We report that device architecture engineering has a substantial impact on the reverse bias instability that has been reported as a critical issue in commercializing perovskite solar cells. We demonstrate breakdown voltages exceeding -15 V in typical pin structured perovskite solar cells via two steps: i) using polymer hole transporting materials; ii) using a more electrochemically stable gold electrode. While device degradation can be exacerbated by higher reverse bias and prolonged exposure, our as-fabricated perovskite solar cells completely recover their performance even after stressing at -7 V for 9 hours both in the dark and under partial illumination. Following these observations, we systematically discuss and compare the reverse bias driven degradation pathways in perovskite solar cells with different device architectures. Our model highlights the role of electrochemical reaction rates and species in dictating the reverse bias stability of perovskite solar cells.

Improving efficiency and stability for perovskite solar cell with diethylene glycol dimethacrylate modification

Xiaodan Wei; Shuyuan Zhang; Boyang Ma; Zehua Feng; chunhan Yu; Linyuan Peng; Zijian Hua; Jie Xu. arXiv preprint (2025). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/2504.15169v1

Abstract: The humidity resistance is the key challenges that hinder the commercial application of perovskite solar cells (PSCs). Herein, we propose an ultra-thin acrylate polymer (diethylene glycol dimethacrylate, DGDMA) into perovskite films to investigate the influence of polymerized networks on stability. The monomer molecules containing acrylate and carbonyl groups were selected, and the effects of the polymerized were quantified with different concentration. The experimental results show that, when the concentration of DGDMA is 1 mg/ml, the PCE increases from 18.06% to 21.82%, which is optimum. The monomer molecules with carbonyl groups polymerize, they can chelate with uncoordinated Pb2+ in perovskite films to improve the film quality, reduce the surface defect density to decrease non-radiative recombination, and also significantly enhance the humidity stability of PSCs.

Influence of the Cathodes Microstructure on the Stability of Inverted Planar Perovskite Solar Cells

Svetlana Sirotinskaya; Roland Schmechel; Niels Benson. arXiv preprint (2019). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/1909.12679v1

Abstract: One of the main challenges for perovskite solar cells (PSC) is their stability, due to environment-induced perovskite decomposition. The resulting decomposition compounds are mobile and may, therefore, react with charge carrier extraction layers or the contact metallization, in addition to enhancing the recombination rate in the absorber layer. In this contribution, the influence of different contact metallization layers, such as aluminum (Al), silver (Ag), gold (Au) and nickel (Ni) on the storage stability of inverted planar methylammonium lead iodide (MAPI)-based perovskite solar cells without encapsulation has been investigated. For this study current-voltage (J-V) and impedance measurements in combination with scanning electron microscope (SEM) and Energy-dispersive X-ray spectroscopy (EDX) analysis were used to examine and correlate structural device information with the development of PSC electrical properties. While a strong perovskite decomposition and further iodide diffusion to the contacts were detected for devices using Al, Ag or Au as cathode electrodes, the microstructure of Ni cathodes inhibits such decomposition process. This experiment has allowed for the realization of MAPI based solar cells with Ni contacts, which exhibit no efficiency decrease below as-fabricated values for up to one month of storage and select AM1.5 testing in ambient atmosphere.

Boosting Perovskite Solar Cell Stability: Dual Protection with Ultrathin Plasma Polymer Passivation Layers

Mahmoud Nabil; Lidia Contreras-Bernal; Gloria P. Moreno-Martinez; Jose Obrero-Perez; Javier Castillo-Seoane; Juan A. Anta; Gerko Oskam; Paul Pistor; Ana Borrás; Juan R. Sanchez-Valencia; Angel Barranco. arXiv preprint (2024). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/2412.19863v1

Abstract: Metal halide perovskite solar cells (MHPSCs) hold great promise related to their high efficiency and low fabrication costs, but their long-term stability under environmental conditions remains a major challenge. In this study, we demonstrate an effective protection strategy to enhance the stability of MHPSCs through the incorporation of a double passivation layer based on an adamantane-based plasma polymer (ADA) at both the electron transport layer (ETL)/perovskite and perovskite/hole transport layer (HTL) interfaces. Our results show that the implemented ADA deposition technique is compatible with delicate substrates such as perovskites thin films, as their optical, morphological and optoelectronic properties are unaltered upon ADA deposition. At the same time, it provides effective protection to the perovskite material in high humidity environments. The ADA-double passivation not only reduces the formation of mobile ionic defects that cause additional recombination, but also significantly reduces humidity-induced degradation and mitigates the photocatalytic degradation caused by TiO2 under UV exposure. Stability tests performed under 100% relative humidity and continuous AM 1.5G illumination (ISOS-L-1) show that ADA dual passivated devices retained 80% of their initial efficiency after 4000 minutes, while reference samples dropped to 30%. The enhanced performance of ADA-passivated cells is attributed to protective nature of the plasma polymer layer resulting in a preservation of photocurrent and the prevention of new recombination routes. This dual passivation strategy offers a promising route to improve the environmental stability of PSCs and extend their operational lifetime.

Disentangling the origin of degradation in perovskite solar cells via optical imaging and Bayesian inference

Akash Dasgupta; Robert D. J. Oliver; Manuel Kober-Czerny; Charlie H. G. Nicholls; Xueli Cao; Yen-Hung Lin; Alexandra J. Ramadan; Henry J. Snaith. arXiv preprint (2026). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/2606.13114v1

Abstract: Machine learning and computational inference, coupled with experimental data, promise to significantly accelerate our rate of learning in most scientific disciplines. In this study, we develop tools that connect microscopic observations to macroscopic device behaviour, a capability that is essential for accelerating the design of durable energy materials. To this end, we introduce a novel approach that integrates photoluminescence imaging with drift diffusion simulations to understand operation and degradation in fully fabricated perovskite solar cells. By employing Bayesian inference, we generate "inferred maps" of parameters that govern recombination processes present in devices. We track these parameter maps while the devices are aged (70 °C, full spectrum sunlight) to analyse their temporal evolution during degradation. Notably, our approach allows us to distinguish between degradation occurring at the hole or electron transporting layer interface, or within the bulk. Our analysis reveals pronounced spatially non-uniform degradation, with significant macroscopic heterogeneity observed in the optoelectronic parameter maps. We pinpoint the greatest degradation observed in

specific regions to stem from the perovskite/transport layer interfaces. Finally, we demonstrate that an aminosilane molecular passivation treatment suppresses this degradation, highlighting its specific role in enhancing device stability. Our approach offers valuable insights for future device fabrication and is a clear exemplification of how advanced Bayesian inference can significantly increase the value of experimental data.

Analysis of degradation in perovskite solar cells through physics-based machine learning

Kjeld O. Jensen; Gemma Giliberti; Aldo Di Carlo; Will Clarke; Giles Richardson; Taylor Blackwell; Petra J. Cameron; Alison B. Walker. arXiv preprint (2026). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/2608.10691v1

Abstract: Degradation in lead halide perovskite solar cells is analysed by inverse modelling of published measurements of characteristics of a single solar cell at ages 0, 90, 280, 480 minutes. We employ machine learning to deduce distributions of material parameter values and hence the physics linked to measured changes. Bayesian parameter estimation is coupled with drift diffusion simulations using the IonMonger code combined with an optical model. We accurately replicated measured changes in device performance with age through variations in model input parameters. Our key result is that degradation is influenced by correlated changes in the concentrations and diffusion coefficients of mobile ions and by interface recombination at large mobile ion concentrations. This study demonstrates the power of machine learning combined with simulations to reliably interpret experimental results, a task which is problematic if using simulation models with only manual exploration of the input parameter space.

Critical role of water in defect aggregation and chemical degradation of perovskite solar cells

Yun-Hyok Kye; Chol-Jun Yu; Un-Gi Jong; Yue Chen; Aron Walsh. arXiv preprint (2017). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/1708.07606v3

Abstract: The chemical stability of methylammonium lead iodide (\ce{MAPbI3}) under humid conditions remains the primary challenge facing halide perovskite solar cells. We investigate defect processes in the water-intercalated iodide perovskite (\ce{MAPbI3}_\ce{H2O}) and monohydrated phase (\ce{MAPbI3}$\cdot$\ce{H2O}) within a first-principles thermodynamic framework. We consider the formation energies of isolated and aggregated vacancy defects with different charge states under I-rich and I-poor conditions. It is found that a \ce{PbI2} (partial Schottky) vacancy complex can be formed readily, while the \ce{MAI} vacancy complex is difficult to form in the hydrous compounds. Vacancies in the hydrous phases create deep charge transition levels, indicating the degradation of halide perovskite upon exposure to moisture. Electronic structure analysis supports a novel mechanism of water-mediated vacancy-pair formation.

Ionic Charge Imbalance in Perovskite Solar Cells

Dhyana Sivadas; Abhimanyu Singareddy; Chittiboina vinod; Pradeep Nair. arXiv preprint (2023). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/2301.10973v1

Abstract: Ion migration in perovskite solar cells is usually analyzed and understood in terms of charge neutrality condition. However, several recent reports indicate possibility of ionic imbalance in the active layer due to external ion migration and/or chemical reactions. In this context, here we explore the influence of ionic charge neutrality on the performance of perovskite solar cells. Our results indicate that ionic imbalance leads to an asymmetry in the device electrostatics, which have interesting implications on the impact of material/interface degradation, hysteresis, and finally on the long-term stability and influence of optimal device architecture (NIP vs. PIN).

First-principles studies of oxygen interstitial dopants in RbPbI$_3$ halide for perovskite solar cells

Chongyao Yang; Wei Wu; Kwang-Leong Choy. Modelling and Simulation in Materials Science and Engineering, 32, 015007 (2023) (2023). Cited by 0. Found by: arxiv. DOI: 10.1088/1361-651X/ad104f

Abstract: Recent research on perovskite solar cells has caught much attention for the application in renewable energy materials. However, the effect of external dopants on the performance of perovskite solar cell is yet to be understood properly. Oxygen atom or molecule is important dopant to influence the stability of structural, electronic and optical properties as well as the performance of perovskite solar cells. RbPbX3-type perovskites have fantastic chemical stability and good power conversion efficiency. Here for the first time, we have studied the effect of interstitial oxygen atom (O1) and molecule (O2) on the structural properties, and hence the electronic structure of RbPbI3 from first principles. A significant reduction of the band gap from ~2.6 eV to ~ 1.0 eV, which is close to the optimal band gap, has been predicted when incorporating oxygen. This could in turn be applied to improve the optical properties for harvesting light if we can control the oxygen level appropriately. In addition, an exotic metallic state has been found in our calculations for interstitial oxygen molecule when there are strong O-O, O-Pb, and O-I bonds, indicating the complex nature of oxygen-doped perovskite solar cells. The comparison between oxygen atom and molecules is consistent with the previous report about oxygen-molecule passivation of perovskite solar cells. This indicated oxygen incorporation can not only improve efficiency and stability but also facilitate the optimal band-gap engineering. Our work has therefore provided an important and timely theoretical insight to the effect of oxygen dopants in perovskite solar cells. Moreover, these results also provide theoretical foundation for further simulations such as molecular dynamics.

Efficient indoor p-i-n hybrid perovskite solar cells using low temperature solution processed NiO as hole extraction layers

Lethy Krishnan Jagadamma; Oskar Blaszczyk; Muhammad T. Sajjad; Arvydas Ruseckas; Ifor D. W. Samuel. Solar Energy Materials and Solar Cells, 201, 110071, 2019 (2019). Cited by 0. Found by: arxiv. DOI: 10.1016/j.solmat.2019.110071

Abstract: Hybrid perovskites have received tremendous attention due to their exceptional photovoltaic and optoelectronic properties. Among the two widely used perovskite solar cell device architectures of n-ip and p-i-n, the latter is interesting in terms of its simplicity of fabrication and lower energy input. However this structure mostly uses PEDOT:PSS as a hole transporting layer which can accelerate the perovskite solar cell degradation. Hence the development of stable, inorganic hole extraction layers (HEL), without compromising the simplicity of device fabrication is crucial in this fast-growing photovoltaic field. Here we demonstrate a low temperature (~100 oC) solution - processed and ultrathin (~ 6 nm) NiO nanoparticle thin films as an efficient HEL for CH3NH3PbI3 based perovskite solar cells. We measure a power conversion efficiency (PCE) of 13.3 % on rigid glass substrates and 8.5 % on flexible substrates. A comparison with PEDOT:PSS based MAPbI3 solar cells (PCE ~ 7.9 %) shows that NiO based solar cells have higher short circuit current density and improved open circuit voltage (1.03V). Apart from the photovoltaic performance under 1 Sun, the efficient hole extraction property of NiO is demonstrated for indoor lighting as well with a PCE of 23.0 % for NiO based CH3NH3PbI2.9Cl0.1 p-i-n solar cells under compact fluorescent lighting. Compared to the perovskite solar cells fabricated on PEDOT:PSS HEL, better shelf-life stability is observed for perovskite solar cells fabricated on NiO HEL. Detailed microstructural and photophysical investigations imply uniform morphology, lower recombination losses, and improved charge transfer properties for CH3NH3PbI3 grown on NiO HEL.

Enhancing Intrinsic Stability of Hybrid Perovskite Solar Cell by Strong, yet Balanced, Electronic Coupling

Fedwa El-Mellouhi; El Tayeb Bentria; Sergey N Rashkeev; Sabre Kais; Fahhad H Alharbi. arXiv preprint (2016). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/1604.06875v1

Abstract: In the past few years, the meteoric development of hybrid organic–inorganic perovskite solar cells (PSC) astonished the community. The efficiency has already reached the level needed for commercialization; however, the instability hinders its deployment on the market. Here, we report a mechanism to chemically stabilize PSC absorbers. We propose to replace the widely used methylammonium cation (\ce{CH3NH3+}) by alternative molecular cations allowing an enhanced electronic coupling between the cation and the \ce{PbI6} octahedra while maintaining the band gap energy within the suitable range for solar cells. The mechanism exploits establishing a balance between the electronegativity of the materials' constituents and the resulting ionic electrostatic interactions. The calculations demonstrate the concept of enhancing the electronic coupling, and hence the stability, by exploring the stabilizing features of \ce{CH3PH3+}, \ce{CH3SH2+}, and \ce{SH3+} cations, among several other possible candidates. Chemical stability enhancement hence results from a strong, yet balanced, electronic coupling between the cation and the halides in the octahedron. This shall unlock the hindering instability problem for PSCs and allow them to hit the market as a serious low-cost competitor to silicon based solar cell technologies.

Charge transport layer dependent electronic band bending in perovskite solar cells and its correlation to device degradation

Junseop Byeon; Jutae Kim; Ji-Young Kim; Gunhee Lee; Kijoon Bang; Namyoung Ahn; Mansoo Choi. arXiv preprint (2019). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/1910.12451v1

Abstract: Perovskite solar cells (PSCs) have shown remarkably improved power-conversion efficiency of around 25%. However, their working principle remains arguable and the stability issue has not been solved yet. In this report, we revealed that the working mechanism of PSCs is explained by a dominant pn junction occurring at the different interface depending on electron transport layer, and charges are accumulated at the corresponding dominant junction initiating device degradation. Locations of a dominant pn junction, the electric field, and carrier-density distribution with respect to electron-transport layers in the PCS devices were investigated by using the electron-beam-induced current measurement and Kelvin probe force microscopy. The amount of accumulated charges in the devices was analyzed using the charge-extraction method and the degradation process of devices was confirmed by SEM measurements. From these observations, we identified that the dominant pn junction appears at the interface where the degree of band bending is higher compared to the other interface, and charges are accumulated at the corresponding junction where the device degradation is initiated, which suggests that there exists a strong correlation between PSC working principle and device degradation. We highlight that an ideal pin PSC that can minimize the degree of band bending should be designed for ensuring long-term stability, via using proper selective contacts

Nitrobenzene as Additive to Improve Reproducibility and Degradation Resistance of Highly Efficient Methylammonium-Free Inverted Perovskite Solar Cells

Apostolos Ioakeimidis; Stelios. A. Choulis. Materials 2020, 13, 3289 (2020). Cited by 0. Found by: arxiv. DOI: 10.3390/ma13153289

Abstract: We show that the addition of 1 % (v/v) nitrobenzene within the perovskite formulation can be used as a method to improve the power conversion efficiency and reliability performance of methylammonium-free (CsFA) inverted perovskite solar cells. Addition of nitrobenzene increased PCE due to defect passivation and provides smoother films resulting in PVSCs with narrower PCE distribution. Moreover, the nitrobenzene additive methylammonium-free hybrid PVSCs exhibit prolonged lifetime compare to additive free PVSCs due to enhanced air and moisture degradation resistance.

Ultrathin plasma polymer passivation of perovskite solar cells for improved stability and reproducibility

Jose M. Obrero-Perez; Lidia Contreras-Bernal+; Fernando Nunez-Galvez; Javier Castillo-Seoane; Karen Valadez-Villalobos; Francisco J. Aparicio; Juan A. Anta; Ana Borras; Juan R. Sanchez-Valencia; Angel Barranco+. arXiv preprint (2022). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/2202.08188v1

Abstract: Despite the youthfulness of hybrid halide perovskite solar cells, their efficiencies are currently comparable to commercial silicon and have surpassed quantum-dots solar cells. Yet, the scalability of these devices is a challenge due to their low reproducibility and stability under environmental conditions. However, the methods reported to date to tackle such issues recurrently involve the use of solvent methods that would further complicate their transfer to industry. Herein we present a reliable alternative relaying in the implementation of an ultrathin plasma polymer as passivation interface between the electron transport material and the hybrid perovskite layer. Such nanoengineering interface provides solar devices with increased long-term stability under ambient conditions. Thus, without consideringr any additional encapsulation step, the cells retain more than 80 % of their efficiency after being exposed to the ambient atmosphere for more than 1000 h. Moreover, this plasma polymer passivation strategy significantly improves the coverage of the mesoporous scaffold by the perovskite layer, providing the solar cells with enhanced performance as well as improved reproducibility.

Material selection method for a perovskite solar cell design based on the genetic algorithm

Eungkyun Kim; Indranil Bhattacharya. arXiv preprint (2020). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/2011

Abstract: In this work, we propose a method of selecting the most desirable combinations of material for a perovskite solar cell design utilizing the genetic algorithm. Solar cells based on the methylammonium lead halide, CH3NH3PbX3, attract researchers due to the benefits of their high absorption coefficient and sharp Urbach tail, long diffusion length and carrier lifetime, and high carrier mobility. However, their poor stability under exposure to moisture still poses a challenge. In our work, we assigned stability index, power conversion efficiency index, and cost-effectiveness index for each material based on the available experimental data in the literature, and our algorithm determined the TiO2/CH3NH3PbI2.1Br0.9/Spiro-OMeTAD as the most well balanced solution in terms of cost, efficiency, and stability. The proposed method can be extended further to aid the material selection in all-perovskite multijunction solar cell design as more data on perovskite materials become available in the future.

Low Frequency Carrier Kinetics in Perovskite Solar Cells

Vinod K. Sangwan; Menghua Zhu; Sarah Clark; Kyle A. Luck; Tobin J. Marks; Mercouri G. Kanatzidis; Mark C. Hersam. arXiv preprint (2019). Cited by 0. Found by: arxiv. DOI: 10.1021/acsami.9b03884

Abstract: Hybrid organic-inorganic halide perovskite solar cells have emerged as leading candidates for third-generation photovoltaic technology. Despite the rapid improvement in power conversion efficiency (PCE) for perovskite solar cells in recent years, the low-frequency carrier kinetics that underlie practical roadblocks such as hysteresis and degradation remain relatively poorly understood. In an effort to bridge this knowledge gap, we perform here correlated low-frequency noise (LFN) and impedance spectroscopy (IS) characterization that elucidates carrier kinetics in operating perovskite solar cells. Specifically, we focus on planar cell geometries with a SnO2 electron transport layer and two different hole transport layers, namely, poly(triarylamine) (PTAA) and Spiro-OMeTAD. PTAA and Sprio-OMeTAD cells with moderate PCEs of 5 to 12 percent possess a Lorentzian feature at 200 Hz in LFN measurements that corresponds to a crossover from electrode to dielectric polarization. In comparison, Spiro-OMeTAD cells with high PCEs (15 percent) show four orders of magnitude lower LFN amplitude and are accompanied by a cyclostationary process. Through a systematic study of more than a dozen solar cells, we establish a correlation with noise amplitude, power conversion efficiency, and fill factor. Overall, this work establishes correlated LFN and IS as an effective methodology for quantifying low frequency carrier kinetics in perovskite solar cells, thereby providing new physical insights that can rationally guide ongoing efforts to improve device performance, reproducibility, and stability.

Multimodal operando microscopy reveals that interfacial chemistry and nanoscale performance disorder dictate perovskite solar cell stability

Kyle Frohna; Cullen Chosy; Amran Al-Ashouri; Florian Scheler; Yu-Hsien Chiang; Milos Dubajic; Julia E. Parker; Jessica M. Walker; Lea Zimmermann; Thomas A. Selby; Yang Lu; Bart Roose; Steve Albrecht; Miguel Anaya; Samuel D. Stranks. arXiv preprint (2024). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/2403.16988v1

Abstract: Next-generation low-cost semiconductors such as halide perovskites exhibit optoelectronic properties dominated by nanoscale variations in their structure, composition and photophysics. While microscopy provides a proxy for ultimate device function, past works have focused on neat thin-films on insulating substrates, missing crucial information about charge extraction losses and recombination losses introduced by transport layers. Here we use a multimodal operando microscopy toolkit to measure nanoscale current-voltage curves, recombination losses and chemical composition in an array of state-of-the-art perovskite solar cells before and after extended operational stress. We apply this toolkit to the same scan areas before and after extended operation to reveal that devices with the highest performance have the lowest initial performance spatial heterogeneity - a crucial link that is missed in conventional microscopy. We find that subtle compositional engineering of the perovskite has surprising effects on local disorder and resilience to operational stress. Minimising variations in local efficiency, rather than compositional disorder, is predictive of improved performance and stability. Modulating the interfaces with different contact layers or passivation treatments can increase initial performance but can also lead to dramatic nanoscale, interface-dominated degradation even in the presence of local performance homogeneity, inducing spatially varying transport, recombination, and electrical losses. These operando measurements of full devices act as screenable diagnostic tools, uniquely unveiling the microscopic mechanistic origins of device performance losses and degradation in an array of halide perovskite devices and treatments. This information in turn reveals guidelines for future improvements to both performance and stability.

Bayesian parameter estimation for characterising mobile ion vacancies in perovskite solar cells

Samuel G. McCallum; Oliver Nicholls; Kjeld O. Jensen; Matthew V. Cowley; James E. Lerpinière; Alison B. Walker. arXiv preprint (2023). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/2309.14302v1

Abstract: To overcome the challenges associated with poor temporal stability of perovskite solar cells, methods are required that allow for fast iteration of fabrication and characterisation, such that optimal device performance and stability may be actively pursued. Currently, establishing the causes of underperformance is both complex and time-consuming, and optimisation of device fabrication thus inherently slow. Here, we present a means of computational device characterisation of mobile halide ion parameters from room temperature current-voltage (J-V) measurements only, requiring ~ 2 hours of computation on basic computing resources. With our approach, the physical parameters of the device may be reverse modelled from experimental J-V measurements. In a drift-diffusion model, the set of coupled drift-diffusion partial differential equations cannot be inverted explicitly, so a method for inverting the drift-diffusion simulation is required. We show how Bayesian Parameter Estimation (BPE) coupled with a drift-diffusion perovskite solar cell model can determine the extent to which device parameters affect performance measured by J-V characteristics. Our method is demonstrated

by investigating the extent to which device performance is influenced by mobile halide ions for a specific fabricated device. The ion vacancy density N_0 and diffusion coefficient D_I were found to be precisely characterised for both simulated and fabricated devices. This result opens up the possibility of pinpointing origins of degradation by finding which parameters most influence device J-V curves as the cell degrades.

Enhanced Power Point Tracking for High Hysteresis Perovskite Solar Cells: A Galvanostatic Approach

Emilio J. Juarez-Perez; Cristina Momblona; Roberto Casas; Marta Haro. Cell Reports Physical Science, 2024, 5, 101885 (2023). Cited by 0. Found by: arxiv. DOI: 10.1016/j.xcrp.2024.101885

Abstract: This article introduces a novel Maximum Power Point Tracking (MPPT) algorithm and cost-effective hardware for long-term operational stability measurements in perovskite solar cells (PSCs). Harnessing the untapped potential of solar energy sources is crucial for achieving a sustainable future, and accurate MPPT is vital to maximizing power generation. However, existing MPPT algorithms for classical photovoltaic technology lead to suboptimal performance and decreased energy efficiency conversion when applied to the most stable perovskite devices, the so-called triple mesoscopic hole transport material (HTM)-free metal halide PSCs. To address this challenge, our research focuses on developing an innovative low-cost hardware solution for research purposes that enables massive long-term stability measurements, eliminating the need for expensive and complex stability monitoring systems. Our galvanostatic MPPT algorithm ensures continuous and precise tracking achieving superior operational performance for high hysteresis PSCs. The suggested enhancements bear significant implications for the extensive integration of perovskite solar cell technologies, particularly those dependent on power optimizer devices.

Quantification of mobile ions in perovskite solar cells with thermally activated ion current measurements

Moritz C. Schmidt; Agustin O. Alvarez; Riccardo Pallotta; Biruk A. Seid; Jeroen J. de Boer; Jarla Thiesbrummel; Felix Lang; Giulia Grancini; Bruno Ehrler. arXiv preprint (2025). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/2508.19403v1

Abstract: Mobile ions play a key role in the degradation of perovskite solar cells, making their quantification essential for enhancing device stability. Various electrical measurements have been applied to characterize mobile ions. However, discerning between different ionic migration processes can be difficult. Furthermore, multiple measurements at different temperatures are usually required to probe different ions and their activation energies. Here, we demonstrate a new characterization technique based on measuring the thermally activated ion current (TAIC) of perovskite solar cells. The method reveals density, diffusion coefficient, and activation energy of mobile ions within a single temperature sweep and offers an intuitive way to distinguish mobile ion species. We apply the TAIC technique to quantify mobile ions of MAPbI3 and triple-cation perovskite solar cells. We find a higher activation energy and a lower diffusion coefficient in the triple-cation devices. TAIC measurements are a simple yet powerful tool to better understand ion migration in perovskite solar cells.

Photo Stabilization of p-i-n Perovskite Solar Cells with Bathocuproine: MXene

Anastasia Yusheva; Danila Saranin; Dmitry Muratov; Pavel Gostishchev; Hanna Pazniak; Alessia Di Vito; Son Thai Le; Lev Luchnikov; Anton Vasiliev; Dmitry Podgorny; Denis Kuznetsov; Sergey Didenko; Aldo Di Carlo. Small, 2022 (2023). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/2311.09695v1

Abstract: Interface engineering is one of the promising strategies for the long-term stabilization of perovskite solar cells, preventing chemical decomposition induced by external agents and promoting fast charge transfer. Recently, MXenes-2D structured transition metal carbides and nitrides with various functionalization (=O,-F,-OH) demonstrated high potential for mastering the work function in halide perovskite absorbers and significantly improved the n-type charge collection in solar cells. This work demonstrates that MXenes allow for efficient stabilization of perovskite solar cells besides improving their performances. We introduce a new mixed composite bathocuproine:MXene, i.e., (BCP:MXene) interlayer at the interface between an electron-transport layer (ETL) and a metal cathode in the p-in device structure. Our investigation demonstrates that the use of BCP:MXene interlayer slightly increases the power conversation efficiency (PCE) for PSCs (from 16.5 for reference to 17.5%) but dramatically improves the out of Glove-Box stability. Under ISOS-L-2 light soaking stress at 63$\pm$ 1.5{\textdegree}C, The T80 (time needed to reduce efficiency down to 80% of the initial one) period increased from 460 h to > 2300 h.

Water-resistant hybrid perovskite solar cell – drop triboelectric energy harvester

Fernando Nunez-Galvez; Xabier Garcia-Casas; Lidia Contreras Bernal; Alejandro Descalzo; Jose Manuel Obrero-Perez; javier Castillo-Seoane; Antonio Gines; Gildas Leger; Juan Carlos Sanchez-Lopez; Juan Pedro Espinos; Angel Barranco; Ana Borras; Juan Ramon Sanchez-Valencia; Carmen Lopez-Santos. Nano Energy, vol. 148, 2026, 111678 (2024). Cited by 0. Found by: arxiv. DOI: 10.1016/j.nanoen.2025.111678

Abstract: Hybrid energy-harvesting systems combining perovskite solar cells (PSCs) with drop-driven triboelectric nanogenerators (D-TENGs) provide continuous power under various weather conditions. However, halide perovskites' vulnerability to moisture hampers widespread use. We present plasma-deposited fluorinated polymers (CFX) as multifunctional encapsulation layers offering water resistance, triboelectric functionality, and > 90 % optical transparency. These conformal, room-temperature, solvent-free coatings protect PSCs without reducing performance; encapsulated cells maintained a PCE of 17.9 %, and devices kept over 50 % of initial PCE after 10 days in high humidity and temperature. CFx layers also enabled compatibility with UV-curable resins, creating a hybrid PSC/D-TENG capable of harvesting solar and rain energy. This device retained 80 % of its performance after 300 hours of humid operation and stayed stable under continuous dripping and illumination for over 5 hours. Optimizing CFx's chemical composition improved triboelectric performance. Using the same CFx layer for encapsulation and triboelectric function, the device achieved 11.6 mA/cm2 short-circuit current under 0.5 sun and 12 V peak voltage per raindrop, enabling simultaneous solar and rain energy harvesting. A self-charging prototype powered LED arrays via a boost converter, demonstrating practical multisource energy harvesting.

Deposition of Reduced Graphene Oxide Thin Film by Spray Pyrolysis Method for Perovskite Solar Cell

Manoj Pandey; Dipendra Hamal; Deepak Subedi; Bijaya Basnet; Rajaram Sah; Santosh K. Tiwari; Bhim Kafle. arXiv preprint (2022). Cited by 0. Found by: arxiv. DOI: 10.3126/jnphyssoc.v7i3.42193

Abstract: The Perovskite absorber layer, the electron transport layer (ETL), the hole transport layer (HTL), and the transparent conducting oxide layer (TCO) are the major components that make up a Perovskite solar cell. Between ETL and HTL, the absorber layer is sandwiched, on which electron-hole pairs are created after absorption of solar radiation. Despite substantial progress toward efficiency, long-term stability still remains a serious concern. Present work focuses toward contributing on the later issue by adopting Titanium dioxide (TiO2) as ETL and reduced graphene oxide (rGO) as HTL. Specifically, in the present work, we report our efforts on the preparation of compact titanium dioxide (C-TiO2) and mesoporous titanium dioxide (M-TiO2) layers as an ETL and a reduced graphene oxide thin film as a HTL. The C-TiO2 film was spin casted on FTO glass followed by casting of M-TiO2 film using Doctor Blading technique. Similarly, the rGO film was produced by spray casting over the glass substrate. The as-prepared ETL and HTL layers were characterized by measuring their optical properties (transmittance and reflectance of thin films). Then, the bandgap, Eg was extracted from reflectance and transmittance curves for ETL and HTL respectively. In the case of rGO, we found the value of Eg to be 2.1 eV, which varies between 2.7eV and 0.02eV depending upon its reduction level based on the previously reported values. Similarly, the bandgap of the C-TiO2 was 4.51 eV which was reduced to 4.12 eV after the addition of M-TiO2, which are 0.9 to 1.1 eV higher than previously reported values. However, bandgap shows decreasing trend after employing M- TiO2 over C-TiO2. In a Perovskite solar cell, both ETL and HTL will be investigated.

Degradation Analysis of Perovskite Solar Cells via Short-Circuit Impedance Spectroscopy: A case study on NiOx passivation

Osbel Almora; Pilar López-Varo; Renán Escalante; John Mohanraj; Luis F Marsal; Selina Olthof; Juan A. Anta. arXiv preprint (2024). Cited by 0. Found by: arxiv. DOI: 10.1063/5.0216983

Abstract: Perovskite solar cells (PSCs) continue to be the front runner technology among emerging photovoltaic devices in terms of power conversion efficiency and application versatility. However, not only the stability but also the understanding of their ionic-electronic transport mechanisms continues to be challenging. In this work, the case study of NiOx-based inverted PSCs and the effect of different interface passivating treatments on the device performance are approached. Our experiments include impedance spectroscopy (IS) measurements in short-circuit under different illumination intensities and operational stability tests under constant illumination intensity. It is found that certain surface treatments lead to more stable performance. However, protic anion donors can induce, both, an initial performance decrease and a subsequent reactivity during light exposure which apparently improves the cells performance. Our drift-diffusion simulations suggest that the modification of the interface with the hole transport material may have decrease the conductivity, as well as the ion and

electron mobilities at the perovskite and the NiOx, respectively. Importantly, capacitance and resistance are shown to peak maximum and minimum values, respectively, around specific ranges of mobile ion concentration. Our results introduce a general route for characterization of degradation paths in PSCs via IS in short-circuit.

Simultaneous Optimization of Efficiency and Degradation in Tunable HTL-Free Perovskite Solar Cells with MWCNT-Integrated Back Contact Using a Machine Learning-Derived Polynomial Regressor

Ihtesham Ibn Malek; Hafiz Imtiaz; Samia Subrina. arXiv preprint (2025). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/2505.18693v1

Abstract: Perovskite solar cells (PSCs) without a hole transport layer (HTL) offer a cost-effective and stable alternative to conventional architectures, utilizing only an absorber layer and an electron transport layer (ETL). This study presents a machine learning (ML)-driven framework to optimize the efficiency and stability of HTL-free PSCs by integrating experimental validation with numerical simulations. Excellent agreement is achieved between a fabricated device and its simulated counterpart at a molar fraction \(x = 68.7\% \) in \(\mathrm{MAPb}_{1-x}\mathrm{Sb}_{2x/3}\mathrm{I}_3\), where MA is methylammonium. A dataset of 1650 samples is generated by varying molar fraction, absorber defect density, thickness, and ETL doping, with corresponding efficiency and 50-hour degradation as targets. A fourth-degree polynomial regressor (PR-4) shows the best performance, achieving RMSEs of 0.0179 and 0.0117, and \(R^2 \) scores of 1 and 0.999 for efficiency and degradation, respectively. The derived model generalizes beyond the training range and is used in an L-BFGS-B optimization algorithm with a weighted objective function to maximize efficiency and minimize degradation. This improves device efficiency from 13.7\% to 16.84\% and reduces degradation from 6.61\% to 2.39\% over 1000 hours. Finally, the dataset is labeled into superior and inferior classes, and a multilayer perceptron (MLP) classifier achieves 100\% accuracy, successfully identifying optimal configurations.

Influence of Phase Segregation on the Hysteresis of Perovskite Solar Cells

Abhimanyu Singareddy; Pradeep R. Nair. arXiv preprint (2024). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/2412.15558v1

Abstract: Organic-inorganic hybrid perovskite solar cells (PSC) have demonstrated impressive performance improvement. Among the various characteristics, the time-dependent current-voltage (J-V) hysteresis allows a direct exploration of various critical phenomena that affect the stability of PSCs. The hysteresis is associated with various spatial heterogeneity-related phenomena, including lifetime, bandgap, and phase segregation. We investigate these phenomena through numerical simulations and quantify how the spatial non-uniformity in the perovskite active layer impacts the hysteresis. Further, we correlate the time dependent device degradation with the hysteresis trends in terms of ion density and effective carrier lifetime.

Pulsatile therapy for perovskite solar cells

Kiwan Jeong; Junseop Byeon; Jihun Jang; Namyoung Ahn; Mansoo Choi. arXiv preprint (2020). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/2007.06190v1

Abstract: The current utmost challenge for commercialization of perovskite solar cells is to ensure long-term operation stability. Here, we developed the pulsatile therapy which can prolong device lifetime by addressing accumulation of both charges and ions in the middle of maximum power point tracking (MPPT). In the technique, reverse biases are repeatedly applied for a very short time without any pause of operation, leading to stabilization of the working device. The observed efficacies of our pulsatile therapy are delaying irreversible degradation as well as restoring degraded photocurrent during MPPT operation. We suggest an integrated mechanism underlying the therapy, in which harmful deep-level defects can be prevented to form and already formed defects can be cured by driving charge-state transition. We demonstrated the therapy to maintain defect-tolerance continuously, leading to outstanding improvement of lifetime and harvesting power. The unique technique will open up new possibility to commercialize perovskite materials into a real market.

LLM tools in the prediction of the stability of perovskite solar cells

S. Frenkel; V. Zakharov; E. A. Katz. arXiv preprint (2025). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/2512.116

Abstract: Predicting degradation rates is an important task in the development of new perovskite solar cells (PSCs). In this paper, we explore the feasibility of solving this problem using Machine Learning models supported by LLM tools. We consider both the "lifetime" prediction of the device and the prediction of degree

of its degradation at specific time intervals. We demonstrate the ability of common LLM tools (ChatGPT, DeepSeek) to suggest and justify prediction methods in a dialogue with the user under conditions of incomplete information about the physical models of PSC degradation and the influence of the environment, providing rather accurate prediction. The results cover the formation of time series of efficiency with a given architecture, calculated using various available mathematical models together with environmental characteristics archived in various meteorological databases (illumination, temperature, humidity, UV level). We conclude that ChatGPT currently has sufficient access to training samples, can suggest to PSC designer various PSC models in the literature, and has adequate solutions for predicting degradation trends.

Efficiency limits of Perovskite Solar Cells with Transition Metal Oxides as Hole Transport Layers

Dhyana Sivadas; Swasti Bhatia; Pradeep Nair. arXiv preprint (2021). Cited by 0. Found by: arxiv. DOI: 10.1063/5.0059221

Abstract: Transition metal oxides (TMOs) like MoOx are increasingly explored as hole transport layers for perovskite-based solar cells. Due to their large work function, the hole collection mechanism of such solar cells are fundamentally different from other materials like PEDOT: PSS, and the associated device optimizations are not well elucidated. In addition, the prospects of such architectures against the challenges posed by ion migration are yet to be explored - which we critically examine in this contribution through detailed numerical simulations. Curiously, we find that, for similar ion densities and interface recombination velocities, ion migration is more detrimental for Perovskite solar cells with TMO contact layers with much lower achievable efficiency limits (21%). The insights shared by this work should be of broad interest to the community in terms of long-term stability, efficiency degradation and hence could help critically evaluate the promises and prospects of TMOs as hole contact layers for perovskite solar cells.

Enhanced thermal stability of inverted perovskite solar cells by bulky passivation with pyridine-functionalized triphenylamine

Ekaterina A. Ilicheva; Irina A. Chuyko; Lev O. Luchnikov; Polina K. Sukhorukova; Nikita S. Saratovsky; Anton A. Vasilev; Luiza Alexanyan; Anna A. Zarudnyaya; Dmitri Yu. Dorofeev; Sergey S. Kozlov; Andrey P. Morozov; Dmitry S. Muratov; Yuriy N. Luponosov; Danila S. Saranin. arXiv preprint (2025). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/2510.08401v1

Abstract: Despite competitive efficiency compared to Si solar cells and relevant stability at near room temperatures the rapid degradation at elevated temperatures remains the critical obstacle for exploitation of perovskite photovoltaics. In this work, a 4-(pyridin-4-yl)triphenylamine (TPA-Py) with pyridine anchor group was employed for inter-grain bulk modification of double-cation CsCH(NH2)2PbI3 perovskite absorbers to enhance thermal stability. Through coordination and dipole-dipole interactions, nitrogen-containing fragments (diphenylamine and pyridine) of TPA-Py passivate uncoordinated cations and improve the phase resilience of perovskite films against segregation. This resulted in a power conversion efficiency of 21.3% with a high open-circuit voltage of 1.14 V. Notable impact of self-assembled monolayer incorporated into the bulk of perovskite film manifested in huge improvement of thermal stability at 85°C (ISOS-D-2). TPA-Py modification improved extended the T80 lifetime to ~600 h compared to only 200 h for the reference under harsh heating stress in ambient conditions. In-depth analysis using photoinduced voltage transients and admittance spectroscopy after different stress periods revealed the screening of ion migration (0.45 eV) for devices with TPA-Py. This work offers an important understanding of the bulk modification of microcrystalline perovskite absorbers and guide for robust design of bulk and buried interfaces in highly efficient perovskite solar cells.

Employing surfactant-assisted hydrothermal synthesis to control CuGaO2 nanoparticle formation and improved carrier selectivity of perovskite solar cells

Ioannis T. Papadas; Achilleas Savva; Apostolos Ioakeimidis; Polyvios Eleftheriou; Gerasimos S. Armatas; Stelios A. Choulis. Materials Today Energy, Volume 8, Pages 57-64, 2018 (2018). Cited by 0. Found by: arxiv. DOI: 10.1016/j.mtener.2018.03.003

Abstract: Delafossites like CuGaO2 have appeared as promising p-type semiconductor materials for optoelectronic applications mainly due to their high optical transparency and electrical conductivity. However, existing synthetic efforts usually result in particles with large diameter limiting their performance relevant to functional electronic applications. In this article, we report a novel surfactant-assisted hydrothermal synthesis method, which allows the development of ultrafine (~5 nm) monodispersed p-type CuGaO2 nanoparticles (NPs). We show that DMSO can be used as a ligand and dispersing solvent for stabilizing the CuGaO2 NPs. The resulting dispersion is used for the fabrication of dense, compact functional CuGaO2 electronic layer with

properties relevant to advanced optoelectronic applications. As a proof of concept, the surfactant-assisted hydrothermal synthesized CuGaO2 is incorporated as a hole transporting layer (HTL) in the inverted p-i-n perovskite solar cell device architecture providing improved hole carrier selectivity and power conversion efficiency compared to conventional PEDOT:PSS HTL based perovskite solar cells.

Experimental demonstration of ions induced electric field in perovskite solar cells

Zeguo Tang; Takashi Minemoto. arXiv preprint (2015). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/1508.05491v

Abstract: The anomalous hysteresis is reported in photo current density-voltage (J-V) curves of perovskite solar cells. The origins responsible for hysteresis are primarily attributed to ferroelectricity, trapping/de-trapping, and ions migration. Meanwhile, a switchable photovoltaic effect is disclosed in lateral structure perovskite solar cells, where a p-i-n structure is formed after poling the device with a reverse bias. Considering the normal sandwiched structure of perovskite solar cells, i.e. the intrinsic perovskite layer is situated in an electric field built by p-type (Spiro-OMeTAD) and n-type (TiO2) layers, the poling process also works in such devices. The migration of ions will induce a new electric field with opposite direction as inherent built-in electric field. As a consequence, less collection of photogenerated carriers is predicted due to the built-in electric field is partially screened. Here, the ions induced electric field is experimentally demonstrated by exploring the external quantum efficiency (EQE) spectra under various forward voltage biases. The anomalous hysteresis observed on J-V curves is explained based on the insight of ions migrations under bias voltage. The external applied voltage presents significant influence on the stability of solar cells. Meanwhile, a novel phenomenon of negative capacitance is initially detected at low frequency, which is ascribed to the position reversal of negatively and positively charged ions under forward bias.

Quantification of Ion Migration in CH3NH3PbI3 Perovskite Solar Cells by Transient Capacitance Measurements

Moritz H. Futscher; Ju Min Lee; Lucie McGovern; Loreta A. Muscarella; Tianyi Wang; Muhammad Irfan Haider; Azhar Fakharuddin; Lukas Schmidt-Mende; Bruno Ehrler. arXiv preprint (2018). Cited by 0. Found by: arxiv. DOI: 10.1039/c9mh00445a

Abstract: Solar cells based on organic-inorganic metal halide perovskites show efficiencies close to highly-optimized silicon solar cells. However, ion migration in the perovskite films leads to device degradation and impedes large scale commercial applications. We use transient ion-drift measurements to quantify activation energy, diffusion coefficient, and concentration of mobile ions in methylammonium lead triiodide (MAPbI3) perovskite solar cells, and find that their properties change close to the tetragonal-to-orthorhombic phase transition temperature. We identify three migrating ion species which we attribute to the migration of iodide (I-) and methylammonium (MA+). We find that the concentration of mobile MA+ ions is one order of magnitude higher than the one of mobile I- ions, and that the diffusion coefficient of mobile MA+ ions is three orders of magnitude lower than the one for mobile I- ions. We furthermore observe that the activation energy of mobile I- ions (0.29 eV) is highly reproducible for different devices, while the activation energy of mobile MA+ depends strongly on device fabrication. This quantification of mobile ions in MAPbI3 will lead to a better understanding of ion migration and its role in operation and degradation of perovskite solar cells.

Enhanced Performance and Stability of Perovskite Solar Cells with Ag-Cu-Zn Alloy Electrodes

Keshav Kumar Sharma; Ashutosh Ujjwal; Rohit Saini; Ramesh Karuppannan. Energy Technology, 2025, e202501392 (2025). Cited by 0. Found by: arxiv. DOI: 10.1002/ente.202501392

Abstract: Though the common metal electrode-based perovskite solar cells have achieved a power conversion efficiency of >25%, they also play a crucial role in accelerating the degradation of the cells. In this study, we investigated phase transition engineering in Ag electrodes via Cu and Zn alloying, transforming from a cubic to a tetragonal phase. These alloyed electrodes are then thermally deposited as back electrodes in perovskite solar cells. We conducted a comprehensive analysis of the pure Ag and Ag-Cu-Zn alloys deposited atop a hole-transport layer for use in Cs0.05(FA0.83MA0.17)0.95Pb(I0.83Br0.17)3-based solar cells. Our findings reveal that solar cells developed with pure Ag electrodes demonstrate a power conversion efficiency (PCE) of 18.71%, characterized by a fill factor (FF) of 74.8%, an open-circuit voltage (VOC) of 1.08 V, and a short-circuit current density (JSC) of 23.17 mA/cm2. Conversely, solar cells fabricated with optimized Ag0.875Cu0.120Zn0.005 electrodes exhibit enhanced performance metrics, with an FF of 72.5%, VOC of 1.12 V, and JSC of 23.39 mA/cm2, culminating in an elevated PCE of 19.02%. Moreover, this electrode demonstrates remarkable durability, sustaining operational integrity for 460 hours for the PSCs stored in the N2 glove box, in contrast to the 320 hours for cells with Ag electrodes. The Ag-Cu-Zn alloys exhibited high resistance to corrosion and good adhesion on

the hole-transport material layer compared to a layer of Ag. These advancements may lead to the realization of cost-effective, durable, and efficient solar energy conversion systems.

The hysteresis-free behavior of perovskite solar cells from the perspective of the measurement conditions

George Alexandru Nemnes; Cristina Besleaga; Andrei Gabriel Tomulescu; Lucia Nicoleta Leonat; Viorica Stancu; Mihaela Florea; Andrei Manolescu; Ioana Pintilie. Journal of Materials Chemistry C 2019 (2018). Cited by 0. Found by: arxiv. DOI: 10.1039/C8TC05999C

Abstract: We investigate in how far the hysteresis-free behavior of perovskite solar cells can be reproduced using particular pre-conditioning and measurement conditions. As there are currently more and more reports of perovskite solar cells without J-V hysteresis it is crucial to distinguish between genuine performance improvements and measurement artifacts. We focus on two of the parameters that influence the dynamic J-V scans, namely the bias scan rate and and the bias poling voltage, and point out measurement conditions for achieving a hysteresis-free behavior. In this context we discuss the suitability of defining a hysteresis index (HI) for the characterization of dynamic J-V scans. Using HI, aging effects are also investigated, establishing a potential connection between the sample degradation and the variation of the maximal hysteresis on one hand, and relaxation time scale of the slow process, on the other hand. Pre-poling induced recombinations effects are identified. In addition, our analysis based on sample pre-biasing reveals potential indication regarding two types of slow processes, with two different relaxation time scales, which provides further insight regarding ionic migration.

A Design Based on Stair-case Band Alignment of Electron Transport Layer for Improving Performance and Stability in Planar Perovskite Solar Cells

Shang-Hsuan Wu; Ming-Yi Lin; Sheng-Hao Chang; Wei-Chen Tu; Chih-Wei Chu; Yia-Chung Chang. arXiv preprint (2017). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/1708.03153v1

Abstract: Among the n-type metal oxide materials used in the planar perovskite solar cells, zinc oxide (ZnO) is a promising candidate to replace titanium dioxide (TiO2) due to its relatively high electron mobility, high transparency, and versatile nanostructures. Here, we present the application of low temperature solution processed ZnO/Al-doped ZnO (AZO) bilayer thin film as electron transport layers (ETLs) in the inverted perovskite solar cells, which provide a stair-case band profile. Experimental results revealed that the power conversion efficiency (PCE) of perovskite solar cells were significantly increased from 12.25 to 16.07% by employing the AZO thin film as the buffer layer. Meanwhile, the short-circuit current density (Jsc), open-circuit voltage (Voc), and fill factor (FF) were improved to 20.58 mA/cm2, 1.09V, and 71.6%, respectively. The enhancement in performance is attributed to the modified interface in ETL with stair-case band alignment of ZnO/AZO/CH3NH3PbI3, which allows more efficient extraction of photogenerated electrons in the CH3NH3PbI3 active layer. Thus, it is demonstrated that the ZnO/AZO bilayer ETLs would benefit the electron extraction and contribute in enhancing the performance of perovskite solar cells.

Improved evaluation of deep-level transient spectroscopy on perovskite solar cells reveals ionic defect distribution

Sebastian Reichert; Jens Flemming; Qingzhi An; Yana Vaynzof; Jan-Frederik Pietschmann; Carsten Deibel. Phys. Rev. Applied 13, 034018 (2020) (2019). Cited by 0. Found by: arxiv. DOI: 10.1103/PhysRevApplied.13.034018

Abstract: One of the key challenges for future development of efficient and stable metal halide perovskite solar cells is related to the migration of ions in these materials. Mobile ions have been linked to the observation of hysteresis in the current–voltage characteristics, shown to reduce device stability against degradation and act as recombination centers within the band gap of the active layer. In the literature one finds a broad spread of reported ionic defect parameters (e.g. activation energies) for seemingly similar perovskite materials, rendering the identification of the nature of these species difficult. In this work, we performed temperature dependent deep-level transient spectroscopy (DLTS) measurements on methylammonium lead iodide perovskite solar cells and developed a extended regularization algorithm for inverting the Laplace transform. Our results indicate that mobile ions form a distribution of emission rates (i.e. a distribution of diffusion constants) for each observed ionic species, which may be responsible for the differences in the previously reported defect parameters. Importantly, different DLTS modes such as optical and current DLTS yield the same defect distributions. Finally the comparison of our results with conventional boxcar DLTS and impedance spectroscopy (IS) verifies our evaluation algorithm.

Automated and Scalable SEM Image Analysis of Perovskite Solar Cell Materials via a Deep Segmentation Framework

Jian Guo Pan; Lin Wang; Xia Cai. arXiv preprint (2025). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/2509.2654

Abstract: Scanning Electron Microscopy (SEM) is indispensable for characterizing the microstructure of thin films during perovskite solar cell fabrication. Accurate identification and quantification of lead iodide and perovskite phases are critical because residual lead iodide strongly influences crystallization pathways and defect formation, while the morphology of perovskite grains governs carrier transport and device stability. Yet current SEM image analysis is still largely manual, limiting throughput and consistency. Here, we present an automated deep learning-based framework for SEM image segmentation that enables precise and efficient identification of lead iodide, perovskite and defect domains across diverse morphologies. Built upon an improved YOLOv8x architecture, our model named PerovSegNet incorporates two novel modules: (i) Adaptive Shuffle Dilated Convolution Block, which enhances multi-scale and fine-grained feature extraction through group convolutions and channel mixing; and (ii) Separable Adaptive Downsampling module, which jointly preserves fine-scale textures and large-scale structures for more robust boundary recognition. Trained on an augmented dataset of 10,994 SEM images, PerovSegNet achieves a mean Average Precision of 87.25% with 265.4 Giga Floating Point Operations, outperforming the baseline YOLOv8x-seg by 4.08%, while reducing model size and computational load by 24.43% and 25.22%, respectively. Beyond segmentation, the framework provides quantitative grain-level metrics, such as lead iodide/perovskite area and count, which can serve as reliable indicators of crystallization efficiency and microstructural quality. These capabilities establish PerovSegNet as a scalable tool for real-time process monitoring and data-driven optimization of perovskite thin-film fabrication.The source code is available at:https://github.com/wlyyj/PerovSegNet/tree/master.

Double-side Integration of the Fluorinated Self-Assembling Monolayers for Enhanced Stability of Inverted Perovskite Solar Cells

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Abstract: Traps and structural defects at the hole and electron transport interfaces of the microcrystalline absorber limits the efficiency and long-term stability of perovskite solar cells (PSCs) due to accumulation of the ionic clusters, non-radiative recombination and electrochemical corrosion. Surface engineering using self-assembled monolayers (SAM) was considered as an effective strategy for modification of charge-collection junctions. In this work, we demonstrate the first report about complex integration of a SAM for double-side passivation in p-i-n PSCs. Integrating the novel 5-(4-[bis(4-fluorophenyl)amino]phenyl)thiophene-2-carboxylic acid (FTPATC) as a fluorinated SAM at the hole-transport interface reduced potential barriers and lattice stresses in the absorber. At the electron-transport side, FTPATC interacted with the A-site cations of the perovskite molecule (Cs, formamidinium), inducing a dipole for defect compensation. Using the passivation approach with fluorinated SAM demonstrated benefits in the gain of the output performance up to 22.2%. The key-advantage of double-side passivation was confirmed by the enhanced stability under continuous light-soaking (1-sun equivalent, 65 C, ISOS-L-2), maintaining 88% of the initial performance over 1680 hours and thermal stabilization under harsh heating at 90 C.

Unraveling UV Stability in Metal Halide Perovskites: From Degradation Mechanisms to Molecular Passivation

Xin Wen; Zhiyi Yao; Wenzhuo Li; Zhijun Ning; Fan Zheng. arXiv preprint (2025). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/2511.22136v1

Abstract: Understanding the mechanisms of UV-induced degradation is crucial for enhancing the UV stability of perovskite solar cells. The UV-driven structural dynamics of CH3NH3PbI3 (MAPbI3) are investigated using real-time TDDFT simulations, revealing that under the electron and hole excitation, the distortion of the inorganic framework (PbI) is primarily driven by the electron occupation of Pb-p and I-p antibonding states, whereas in the hole case, it is mainly governed by the direct cooling induced distortion. We also find that UV accelerates the rotation of MA+ molecules. Further, a BDO molecule is introduced as a passivant, which suppresses structural distortions and provides multi-phonon channels to dissipate carrier cooling energy. Experimental results confirm the UV-protective role of BDO, with suppressed PbI2 formation and improved device stability. These results clarify the mechanism of the UV-induced degradation in the MAPbI3 perovskite and further elucidate how passivation molecules enhance UV stability.

Impact of Doping Spiro-OMeTAD with Li-TFSI, FK209, and tBP on the Performance of Perovskite Solar Cells

Mehran Hosseinzadeh Dizaj. arXiv preprint (2024). Cited by 0. Found by: arxiv. DOI: 10.22036/org.chem.2024.494577.1370

Abstract: This study delves into the effects of doping Spiro-OMeTAD, a widely employed hole transport material (HTM) in perovskite solar cells, with three distinct impurities: lithium bis(trifluoromethanesulfonyl)imide (Li-TFSI), FK209, and tertbutylpyridine (tBP). The focus lies on their influence on critical photovoltaic parameters, including Fill Factor (FF), power conversion efficiency (eta), short-circuit current density (Jsc), and open-circuit voltage (Voc). Employing advanced simulation tools, we systematically compare the performance of undoped and doped Spiro-OMeTAD structures. Among the examined dopants, Li-TFSI exhibits the most pronounced enhancements, significantly improving both FF and efficiency. In contrast, FK209 and tBP yield moderate gains in device performance. These results underscore the superior charge transport properties imparted by Li-TFSI doping and its capacity to optimize HTM functionality, thereby presenting a promising approach to advancing the efficiency and stability of perovskite solar cells

Touching is believing: interrogating organometal halide perovskite solar cells at the nanoscale via scanning probe microscopy

Jiangyu Li; Boyuan Huang; Ehsan Nasr Esfahani; Linlin Wei; Jianjun Yao; Jinjin Zhao; Wei Chen. npj Quantum Materials 2, Article number: 56, 2017 (2017). Cited by 0. Found by: arxiv. DOI: 10.1038/s41535-017-0061-4

Abstract: Halide perovskite solar cells based on CH3NH3PbI3 and related materials have emerged as the most exciting development in the next generation photovoltaic technologies, yet the microscopic phenomena involving photo-carriers, ionic defects, spontaneous polarization, and molecular vibration and rotation interacting with numerous grains, grain boundaries, and interfaces are still inadequately understood. In fact, there is still need for an effective method to interrogate the local photovoltaic properties of halide perovskite solar cells that can be directly traced to their microstructures on one hand and linked to their device performance on the other hand. In this perspective, we propose that scanning probe microscopy techniques have great potential to realize such promises at the nanoscale, and highlight some of the recent progresses and challenges along this line of investigation toward local probing of photocurrent, work function, ionic activities, polarization switching, and chemical degradation. We also emphasize the importance of multi-modality imaging, in-operando scanning, big data analysis, and multidisciplinary collaboration for further studies toward fully understanding of these complex systems.

Dopant-free molecular hole transport material that mediates a 20% power conversion efficiency in a perovskite solar cell

Yang Cao; Yunlong Li; Thomas Morrissey; Brian Lam; Brian O. Patrick; David J. Dvorak; Zhicheng Xia; Timothy L. Kelly; Curtis P. Berlinguette. arXiv preprint (2019). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/1908.04439v1

Abstract: Organic molecular hole-transport materials (HTMs) are appealing for the scalable manufacture of perovskite solar cells (PSCs) because they are easier to reproducibly prepare in high purity than polymeric and inorganic HTMs. There is also a need to construct PSCs without dopants and additives to avoid formidable engineering and stability issues. We report here a power conversion efficiency (PCE) of 20.6% with a molecular HTM in an inverted (p-i-n) PSC without any dopants or interlayers. This new benchmark was made possible by the discovery that annealing a spiro-based dopant-free HTM (denoted DFH) containing redox-active triphenyl amine (TPA) units undergoes preferential molecular organization normal to the substrate. This structural order, governed by the strong intermolecular interactions of the DFH dioxane groups, affords high intrinsic hole mobility (1x10-3 cm2 V-1 s-1). Annealing films of DFH also enables the growth of large perovskite grains (up to 2 um) that minimize charge recombination in the PSC. DFH can also be isolated at a fraction of the cost of any other organic HTM.

Comparison of Electroluminescence and Photoluminescence Imaging of Mixed-Cation Mixed-Halide Perovskite Solar Cells at Low Temperatures

Hurriyet Yuce-Cakir; Haoran Chen; Isaac Ogunniranye; Susanna M. Thon; Yanfa Yan; Zhaoning Song; Behrang H. Hamadani. arXiv preprint (2025). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/2510.14213v1

Abstract: Halide perovskites have emerged as promising candidates for high-performance solar cells. This study investigates the temperature-dependent optoelectronic properties of mixed-cation mixed-halide perovskite solar cells using electroluminescence (EL) and photoluminescence (PL) hyperspectral imaging, along with current-voltage analysis. Luminescence images, which were converted to EL and PL external radiative efficiency (ERE) maps, revealed significant changes in the optoelectronic behavior of these devices at low temperatures. Specifically, we found that a significant source of heterogeneity in the low-temperature EL ERE maps below 240 K is related to local charge injection and extraction bottlenecks, whereas PL ERE maps show suppressed non-radiative recombination and significant improvements in efficiency throughout the investigated temperature range. The spatial distribution of ERE and its variation with applied current were analyzed, offering insights into charge-carrier dynamics and defect behavior. Our results reveal that while the perovskite layer exhibits enhanced ERE at low temperatures, charge injection barriers at the interfaces of the perovskite solar cells significantly suppress EL and degrade the fill factor below 240 K. These findings reveal that a deeper understanding of the performance of perovskite solar cells under low-temperature conditions is an essential step toward their potential application in space power systems and advanced semiconductor devices.

Operando multidimensional spectroscopy reveals A-site-dependent carrier cooling in perovskite solar cells

Edoardo Amarotti; Luca Bolzonello; Qi Shi; Sun-Ho Lee; Torbjörn Pascher; Donatas Zigmantas; Niek van Hulst; Nam-Gyu Park; Tönu Pullerits. arXiv preprint (2026). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/2608.17577v

Abstract: Understanding how photogenerated carriers dissipate excess energy in operating perovskite solar cells is essential for connecting ultrafast photophysics with photovoltaic function and provides design principles for engineering next-generation cell architectures. Here we use photocurrent-detected two-dimensional electronic spectroscopy (PC-2DES) to resolve energy-dependent carrier relaxation in fully encapsulated, functioning metal halide perovskite solar cells. By comparing devices with identical architecture but different absorber compositions - MAPbI3, mixed FAMA, and FAPbI3 - we directly follow the redistribution of photoexcited carriers from initially populated high-energy states toward lower-energy band-edge states. The multidimensional photocurrent response reveals a cascade-like intraband cooling process whose rate depends strongly on absorber composition, with the slowest relaxation in MA-based devices, intermediate behaviour in mixed-cation devices, and fastest relaxation in FA-based devices. A reduced kinetic model incorporating phonon-mediated intraband scattering, supplemented by a phenomenological many-body contribution, captures the main energy-dependent trends. These results establish action-detected multidimensional spectroscopy as a device-level probe of ultrafast energy dissipation and show that subtle changes in perovskite composition can substantially reshape the carrier relaxation pathways that precede charge extraction.

La-Doped BaSnO$_3$ Electron Transport Layer for Perovskite Solar Cells

Chang Woo Myung; Geunsik Lee; Kwang S. Kim. arXiv preprint (2018). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/1808.04979v2

Abstract: Due to the photo-instability and hysteresis of TiO$_2$ electron transport layer (ETL) in perovskite solar cells (PSCs), novel electron transport materials are highly demanded. Here, we show ideal band alignment between La-doped BaSnO$_3$ (LBSO) and methyl ammonium (MA) lead iodide perovskite (MAPbI$_3$). The CH$_3$NH$_3$PbI$_3$/La$_x$Ba$_{(1-x)}$SnO$_3$ interface forms a stable all-perovskite heterostructure. The selective band alignment is manipulated with band gap renormalization by La-doping on the Ba site. LBSO shows high mobility, photo-stability, and structural stability, promising the next generation ETL materials.

NiOx passivation in perovskite solar cells: from surface reactivity to device performance

John Mohanraj; Bipasa Samanta; Osbel Almora; Renán Escalante; Lluis F. Marsal; Sandra Jenatsch; Arno Gadola; Beat Ruhstaller; Juan A. Anta; Maytal Caspary Toroker; Selina Olthof. arXiv preprint (2024). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/2402.10286v1

Abstract: Non-stoichiometric nickel oxide (NiOx) is the only metal oxide successfully used as hole transport material in p-i-n type perovskite solar cells (PSCs). Its favorable opto-electronic properties and facile large-scale preparation methods are potentially relevant for future commercialization of PSCs, though currently low operational stability of PSCs containing NiOx hole transport layers are reported. Poorly understood degradation reactions at the interface to the perovskite are seen as cause for the inferior stability and a variety of interface passivation approaches have been shown to be effective in improving the overall solar cell performance. To gain a better understanding of the processes happening at this interface, we systematically passivated possible specific defects on NiOx with three different categories of organic/inorganic compounds. The effects on the NiOx and the perovskite (MAPbI3) were investigated using x-ray photoelectron spectroscopy (XPS), X-ray diffraction

(XRD), and scanning electron microscopy (SEM) where we find that the structural stability and film formation can be significantly affected. In combination with Density Functional Theory (DFT) calculations, a likely origin of NiOx-perovskite degradation interactions is proposed. The surface passivated NiOx was incorporated into MAPbI3 based PSCs and its influence on overall performance, particularly operational stability, was investigated by current-voltage (J-V), impedance spectroscopy (IS), and open circuit voltage decay (OCVD) measurements. Interestingly, we find that a superior structural stability due to an interface passivation must not relate to high operational stability. The discrepancy comes from the formation of excess ions at the interface which negatively impacts all solar cell parameters.

Phenothiazine-Based Self-Assembled Monolayer with Thiophene Head Groups Minimizes Buried Interface Losses in Tin Perovskite Solar Cells

Valerio Stacchini; Madineh Rastgoo; Mantas Marčinskas; Chiara Frasca; Kazuki Morita; Lennart Frohloff; Antonella Treglia; Orestis Karalis; Vytautas Getautis; Annamaria Petrozza; Norbert Koch; Hannes Hempel; Tadas Malinauskas; Antonio Abate; Artem Musiienko. arXiv preprint (2025). Cited by 0. Found by: arxiv. DOI: 10.1002/aenm.202500841

Abstract: Self-assembled monolayers (SAMs) have revolutionized the fabrication of lead-based perovskite solar cells, but they remain underexplored in tin perovskite systems. PEDOT is the material of choice for hole-selective layers in tin perovskite solar cells (TPSCs), but presents challenges for both performance and stability. MeO-2PACz, the only SAM reported for Sn perovskites, enables device fabrication but consistently underperforms when compared to PEDOT. In this work, we identify that MeO-2PACz's limitations arise from excessively strong interactions with perovskite surface and poor lattice matching, leading to poor interface quality. To overcome these issues, we design, synthesize, and characterize a novel SAM-forming molecule called Th-2EPT. Th-2EPT optimizes coordination strength and improves lattice compatibility, contributing to the creation of a high-quality buried interface and dramatically suppressing non-radiative recombination. We used Density Functional Theory (DFT) to evaluate coordination strength and lattice compatibility, complemented by nanosecond-resolution optical characterization techniques to confirm significantly reduced interfacial recombination and enhanced carrier lifetimes in Th-2EPT-Perovskite films. With Th-2EPT, we demonstrated the first SAM-based tin perovskite solar cells to outperform PEDOT-based devices, delivering a record power conversion efficiency (PCE) of 8.2% with a DMSO-free solvent system.

Exploring the Way to Approach the Efficiency Limit of Perovskite Solar Cells by Drift-Diffusion Model

Xingang Ren; Zishuai Wang; Wei E. I. Sha; Wallace C. H. Choy. ACS Photonics, 2017, 4(4), 934-942 (2017). Cited by 0. Found by: arxiv. DOI: 10.1021/acsphotonics.6b01043

Abstract: Drift-diffusion model is an indispensable modeling tool to understand the carrier dynamics (transport, recombination, and collection) and simulate practical-efficiency of solar cells (SCs) through taking into account various carrier recombination losses existing in multilayered device structures. Exploring the way to predict and approach the SC efficiency limit by using the drift-diffusion model will enable us to gain more physical insights and design guidelines for emerging photovoltaics, particularly perovskite solar cells. Our work finds out that two procedures are the prerequisites for predicting and approaching the SC efficiency limit. Firstly, the intrinsic radiative recombination needs to be corrected after adopting optical designs which will significantly affect the open-circuit voltage at its Shockley-Queisser limit. Through considering a detailed balance between emission and absorption of semiconductor materials at the thermal equilibrium, and the Boltzmann statistics at the non-equilibrium, we offer a different approach to derive the accurate expression of intrinsic radiative recombination with the optical corrections for semiconductor materials. The new expression captures light trapping of the absorbed photons and angular restriction of the emitted photons simultaneously, which are ignored in the traditional Roosbroeck-Shockley expression. Secondly, the contact characteristics of the electrodes need to be carefully engineered to eliminate the charge accumulation and surface recombination at the electrodes. The selective contact or blocking layer incorporated nonselective contact that inhibits the surface recombination at the electrode is another important prerequisite. With the two procedures, the accurate prediction of efficiency limit and precise evaluation of efficiency degradation for perovskite solar cells are attainable by the drift-diffusion model.

Triphenylamine-based interlayer with carboxyl anchoring group for tuning of charge collection interface in stabilized p-i-n perovskite solar cells and modules

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Abstract: A novel triphenylamine-based hole transport material (HTM) with a carboxyl anchoring group (TPATC) was developed for tuning the interface between nanocrystalline NiO and double cation CsCH3(NH2)2PbI3-xClx absorber in p-i-n device architectures. We present a unique comprehensive confinement study of PSCs with TPATC as the self-assembled HTM, including analysis of numerical defect parameters, phase composition evolution, and up-scaling capabilities. Our investigation shows that the ultrathin TPATC interlayer effectively passivates traps, increases work-function of HTL by about ~0.2 eV, and enhances the charge carrier extraction efficiency. Advanced transient spectroscopy measurements revealed that modification of the NiO surface with TPATC in perovskite solar cells (PSCs) reduces the concentration of ionic defects by an order of magnitude. Interface engineering with TPATC allowed to reach power conversion efficiency of 20.58% for small area devices (0.15 cm2) under standard AM 1.5 G conditions. Using TPATC interlayer also provided stabilized performance of PSCs under operation conditions and improved sustainability of the perovskite absorber to decomposition. After continuous light-soaking (1000 h, ISOS-L-2 protocol), NiO/TPATC devices showed a slight decrease of 2% in maximum power. In contrast, NiO PSCs demonstrated decrease in power output (>20%) after 400 h. We explored the potential of TPATC to modify interfaces in large-area perovskite solar modules (PSM, active area-64.8 cm2, 12 sub-cells). By applying slot-die-coated TPATC, the PCE at AM 1.5 G conditions increased from 13.22% for NiO PSM to 15.64% for NiO/TPATC ones. This study provides new insights into the interface engineering for p-i-n perovskite solar cells, behavior of the ionic defects and their contribution to the long-term stability.

Ion Induced Passivation of Grain Boundaries in Perovskite Solar Cells

Vikas Nandal; Pradeep R. Nair. arXiv preprint (2018). Cited by 0. Found by: arxiv. DOI: 10.1063/1.5082967

Abstract: Demonstration of high-efficiency large area cells with excellent stability is an important requirement towards commercialization of perovskite solar cells (PSC). With reports of high-quality perovskite grains, it is evident that the performance of such large area cells will be strongly influenced by phenomena like carrier recombination and ion migration at grain boundaries (GBs). Here, we develop a modeling framework to address performance limitation due to GBs in large area PSCs. Through detailed numerical simulations, we show that photo-carrier recombination has a non-trivial dependence on the orientation of GBs. Interestingly, we find that ions at GBs lead to significant performance recovery through field effect passivation, which is influenced by critical parameters like density and polarity of ions, and the location of GB. These results have interesting implications towards long-term stability and hence are relevant for the performance optimization of large area polycrystalline based thin film solar cells such as PSCs, CIGS, CZTS, etc.

Super-droplet-repellent carbon-based printable perovskite solar cells

Cuc Thi Kim Mai; Janne Halme; Heikki A. Nurmi; Aldeliane M. da Silva; Gabriela S. Lorite; David Martineau; Stéphanie Narbey; Naeimeh Mozaffari; Robin H. A. Ras; Syed Ghufran Hashmi and; Maja Vuckovac. arXiv preprint (2024). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/2401.07621v1

Abstract: Despite attractive cost-effectiveness, scalability, and superior stability, carbon-based printable perovskite solar cells (CPSCs) still face moisture-induced degradation that limits their lifespan and commercial potential. Here, we investigate the moisture-preventing mechanisms of thin nanostructured super-repellent coating (advancing contact angle $>$167° and contact angle hysteresis 7° integrated into CPSCs for different moisture forms (falling water droplets vs water vapor vs condensed water droplets). We show that unencapsulated super-repellent CPSCs have superior performance under continuous droplet impact for 12h (rain simulation experiments) compared to unencapsulated pristine (uncoated) CPSCs that degrade within seconds. Contrary to falling water droplets, where super-repellent coating serves as a shield, we found water vapor to physisorb through porous super-repellent coating (room temperature and relative humidity, RH 65\% and 85\%) that increased the CPSCs performance for 21\% during ~43 days similarly to pristine CPSCs. We further showed that, water condensation forms within or below the super-repellent coating (40° C and RH 85\%), followed by chemisorption and degradation of CPSCs. Because different forms of water have distinct effect on CPSC, we suggest that future standard tests for repellent CPSCs should include rain simulation and condensation tests. Our findings will thus inspire the development of super-repellent coatings for moisture prevention.

An eco-friendly passivation strategy of resveratrol for highly efficient and antioxidative perovskite solar cells

Xianhu Wu; Jieyu Bi; Guanglei Cui; Nian Liu; Gaojie Xia; Ping Li; Chunyi Zhao; Zewen Zuo; Min Gu. arXiv preprint (2024). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/2405.01058v1

Abstract: The stability of perovskite solar cells is closely related to the defects in perovskite crystals, and there are a large number of crystal defects in the perovskite thin films prepared by the solution method, which is not conducive to the commercial production of PSCs. In this study, resveratrol(RES), a green natural antioxidant abundant in knotweed and grape leaves, was introduced into perovskite films to passivate the defect. RES achieves defect passivation by interacting with uncoordinated Pb2+ in perovskite films. The results show that the quality of the perovskite film is significantly improved, and the energy level structure of the device is optimized, and the power conversion efficiency of the device is increased from 21.62% to 23.44%. In addition, RES can hinder the degradation of perovskite structures by O2- and CO2- free radicals, and the device retained 88% of its initial PCE after over 1000 hours in pure oxygen environment. The device retains 91% of the initial PCE after more than 1000 hours at 25°C and 50+5% relative humidity. This work provides a strategy for the use of natural and environmentally friendly additives to improve the efficiency and stability of devices, and provides an idea for the development of efficient, stable and environmentally friendly PSCs.

Normal and inverted hysteresis in perovskite solar cells

George Alexandru Nemnes; Cristina Besleaga; Viorica Stancu; Lucia Nicoleta Leonat; Lucian Pintilie; Kristinn Torfason; Marjan Ilkov; Andrei Manolescu; Ioana Pintilie. arXiv preprint (2017). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/1704.03300v1

Abstract: Hysteretic effects are investigated in perovskite solar cells in the standard FTO/TiO$_2$/CH$_3$NH$_3$PbI$_{3-x}$Cl$_x$/spiro-OMeTAD/Au configuration. We report normal (NH) and inverted hysteresis (IH) in the J-V characteristics occurring for the same device structure, the behavior strictly depending on the pre-poling bias. NH typically appears at pre-poling biases larger than the open circuit bias, while pronounced IH occurs for negative bias pre-poling. The transition from NH to IH is marked by a intermediate mixed hysteresis behavior characterized by a crossing point in the J-V characteristics. The measured J-V characteristics are explained quantitatively by the dynamic electrical model (DEM). Furthermore, the influence of the bias scan rate on the NH/IH hysteresis is discussed based on the time evolution of the non-linear polarization. Introducing a three step measurement protocol, which includes stabilization, pre-poling and measurement, we put forward the difficulties and possible solutions for a correct PCE evaluation.

Improved Performance and Reliability of p-i-n Perovskite Solar Cells via Doped Metal Oxides

Achilleas Savva; Ignasi Burgues-Ceballos; Stelios A. Choulis. Advanced Energy Materials 2016, 1600285 (2016). Cited by 0. Found by: arxiv. DOI: 10.1002/aenm.201600285

Abstract: Perovskite photovoltaics (PVs) have attracted attention because of their excellent power conversion efficiency (PCE). Critical issues related to large area PV performance, reliability and lifetime need to be addressed. Here, we show that doped metal oxides can provide ideal electron selectivity, improved reliability and stability for perovskite PVs. We report p-i-n perovskite PVs with device areas ranging from 0.09cm2 to 0.5cm2 incorporating a thick aluminum doped zinc oxide (AZO) electron selective contact with hysteresis-free PCE of over 13% and high fill factor values in the range of 80%. AZO provides suitable energy levels for carrier selectivity, neutralizes the presence of pinholes and provides intimate interfaces. Devices using AZO exhibit an average PCE increase of over 20% compared with the devices without AZO and maintain the high PCE for the larger area devices reported. Furthermore, the device stability of p-i-n perovskite solar cells under the ISOS-D-1 is enhanced when AZO is used, and maintains 100% of the initial PCE for over 1000 hours of exposure when AZO/Au is used as the top electrode. Our results indicate the importance of doped metal oxides as carrier selective contacts to achieve reliable and high performance long lived large area perovskite solar cells.

Influences of Dielectric Constant and Scan Rate to Hysteresis Effect in Perovskite Solar Cell: Simulation and Experimental Analyses

Jun-Yu Huang; You-Wei Yang; Wei-Hsuan Hsu; En-Wen Chang; Mei-Hsin Chen; Yuh-Renn Wu. Scientific Reports, 12, 7927, 2022 (2021). Cited by 0. Found by: arxiv. DOI: 10.1038/s41598-022-11899-x

Abstract: In this work, perovskite solar cells (PSCs) with different transport layers were fabricated to understand the hysteresis phenomenon under a series of scan rates. The experimental results show that the hysteresis

phenomenon would be affected by the dielectric constant of transport layers and scan rate significantly. To explain this, a modified Poisson and drift-diffusion solver coupled with a fully time-dependent ion migration model is developed to analyze how the ion migration affects the performance and hysteresis of PSCs. The simulation model was optimized for carrier transportation of organic materials, which can simulate the organic transport layer correctly without using heavy doping in simulating the organic transport layer. The modeling results show that the most crucial factor in the hysteresis behavior is the built-in electric field of the perovskite. The non-linear hysteresis curves are demonstrated under different scan rates, and the mechanism of the hysteresis behavior is explained. The findings reveal why the change in hysteresis degree with scan rate is Gaussian shaped rather than monotonic. Additionally, other factors contributing to the degree of hysteresis are determined to be the degree of degradation in the perovskite material, the quality of the perovskite crystal, and the materials of the transport layer, which corresponds to the total ion density, carrier lifetime of perovskite, and the dielectric constant of the transport layer, respectively. Finally, it was found that the dielectric constant of the transport layer is a key factor affecting hysteresis in perovskite solar cells; a lower dielectric constant corresponds to a higher electric field of the transport layer. Hence, if the electric field of the perovskite material is small, the degree of hysteresis is small and vice versa.

Optimization and Performance Evaluation of Cs$_2$CuBiCl$_6$ Double Perovskite Solar Cell for Lead-Free Photovoltaic Applications

Syeda Anber Urooj Wasti; Sundus Naz; Ammara Sattar; Asad Yaqoob; Ateeq ul Rehman; Fawad Ali; Shahbaz Afzal. arXiv preprint (2025). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/2502.16850v1

Abstract: In the previous decade, there has been a significant advancement in the performance of perovskite solar cells (PSCs), characterized by a notable increase in efficiency from 3.8% to 25%. Nonetheless, PSCs face many problems when we commercialize them because of their toxicity and stability. Consequently, lead-PSCs need an alternative solar cell with high performance and low processing cost; lead-free inorganic perovskites have been explored. Recent research showcased Cs$_2$CuBiCl$_6$, a lead-free inorganic double perovskite material with remarkable photoelectric characteristics and exceptional environmental robustness. To investigate the potential of Cs$_2$CuBiCl$_6$ material, the solar cell structure FTO/ETL/Cs$_2$CuBiCl$_6$/HTL/Au was used and analyzed through a solar cell capacitance simulator (SCAPS-1D). CeO$_2$ is used as the Electron transport layer (ETL), and CuI is the Hole transport layer (HTL). Furthermore, the research examined the optimization of different parameters of the absorber layer (AL), such as thickness, defect density, electron affinity, band gap, and operational temperature. In the end, it has been noticed that by setting the temperature at 300 K and an electron affinity of 4.3 eV of the absorber layer, the PSCs achieve the highest efficiency of 24.51 %, FF of 43.01 %, Voc of 1.73V, and Jsc of 32.82mA/cm^2. This is the highest Cs$_2$CuBiCl$_6$ double PSCs efficiency we've reached yet. In theoretical studies, 17.03% of PCE was achieved using Cs$_2$CuBiCl$_6$ as an active layer. The analysis underscores the significant potential of Cs$_2$CuBiCl$_6$ as an absorbing layer in developing highly efficient lead-free all-inorganic PSCs.

Remarkable performance recovery in highly defective perovskite solar cells by photo-oxidation

Katelyn P. Goetz; Fabian T. F. Thome; Qingzhi An; Yvonne J. Hofstetter; Tim Schramm; Aymen Yangui; Alexander Kiligaridis; Markus Loeffler; Alexander D. Taylor; Ivan G. Scheblykin; Yana Vaynzof. arXiv preprint (2024). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/2404.11635v1

Abstract: Exposure to environmental factors is generally expected to cause degradation in perovskite films and solar cells. Herein, we show that films with certain defect profiles can display the opposite effect, healing upon exposure to oxygen under illumination. We tune the iodine content of methylammonium lead triiodide perovskite from understoichiometric to overstoichiometric and expose them to oxygen and light prior to the addition of the top layers of the device, thereby examining the defect dependence of their photooxidative response in the absence of storage-related chemical processes. The contrast between the photovoltaic properties of the cells with different defects is stark. Understoichiometric samples indeed degrade, demonstrating performance at 33% of their untreated counterparts, while stoichiometric samples maintain their performance levels. Surprisingly, overstoichiometric samples, which show low current density and strong reverse hysteresis when untreated, heal to maximum performance levels (the same as untreated, stoichiometric samples) upon the photooxidative treatment. A similar, albeit smaller-scale, effect is observed for triple cation and methylammonium-free compositions, demonstrating the general application of this treatment to state-of-the-art compositions. We examine the reasons behind this response by a suite of characterization techniques, finding that the performance changes coincide with microstructural decay at the crystal surface, reorientation of the bulk crystal structure for the understoichiometric cells, and a decrease in the iodine-to-lead ratio of all films. These results indicate that defect engineering is a powerful tool to manipulate the stability of perovskite solar cells.

Towards low-temperature processing of efficient $ $-CsPbI$_3$ perovskite solar cells

Zongbao Zhang; Ran Ji; Yvonne J. Hofstetter; Marielle Deconinck; Julius Brunner; Yanxiu Li; Qingzhi An; Yana Vaynzof. arXiv preprint (2024). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/2404.11636v1

Abstract: Inorganic cesium lead iodide (CsPbI$_3$) perovskite solar cells (PSCs) have attracted enormous attention due to their excellent thermal stability and optical bandgap (~1.73 eV), well-suited for tandem device applications. However, achieving high-performing photovoltaic devices processed at low temperatures is still challenging. Here we reported a new method to fabricate high-efficiency and stable $ $-CsPbI$_3$ PSCs at lower temperatures than was previously possible by introducing the long-chain organic cation salt ethane-1,2-diammonium iodide (EDAI2) and regulating the content of lead acetate (Pb(OAc)2) in the perovskite precursor solution. We find that EDAI2 acts as an intermediate that can promote the formation of $ $-CsPbI$_3$, while excess Pb(OAc)2 can further stabilize the $ $-phase of CsPbI$_3$ perovskite. Consequently, improved crystallinity and morphology and reduced carrier recombination are observed in the CsPbI$_3$ films fabricated by the new method. By optimizing the hole transport layer of CsPbI$_3$ inverted architecture solar cells, we demonstrate up to 16.6% efficiencies, surpassing previous reports examining $ $-CsPbI$_3$ in inverted PSCs. Notably, the encapsulated solar cells maintain 97% of their initial efficiency at room temperature and dim light for 25 days, demonstrating the synergistic effect of EDAI2 and Pb(OAc)2 on stabilizing $ $-CsPbI$_3$ PSCs.

Polyaniline as a conductive polymer and its role in improving the efficiency and conductivity of perovskite solar cells

Mehran Hosseinzadeh Dizaj. arXiv preprint (2026). Cited by 0. Found by: arxiv. DOI: 10.11591/ijpeds.v16.i4.pp2731-2743

Abstract: This article investigates the role of polyaniline as a conductive polymer in the active layer of perovskite solar cells. Samples were created by incorporating polyaniline into the transport layers to assess its impact on enhancing efficiency and conductivity. The application of this polymer across various layers of the cell structure led to improved stability and performance. Given its high doping capability, polyaniline was examined in detail, particularly focusing on two types of oxidation doping and its integration into the hole transport layer. Graphene oxide and reduced graphene oxide were chosen as comparative models, and their performance was evaluated against the standard polyaniline configuration. Laboratory results revealed that power conversion efficiency increased by 17.5% with graphene oxide and by 36.8% with reduced graphene oxide. Furthermore, short-circuit current density improved by 9.8% and 23.1%, respectively. These findings are consistent with existing studies in the field and support the validity of the approach.

Gefitinib-Induced Interface Engineering Enhances the Defect Formation Energy for Highly Efficient and Stable Perovskite Solar Cells

Xianhu Wu; Guanglei Cui; Jieyu Bi; Gaojie Xia; Zewen Zuo; Min Gu. arXiv preprint (2025). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/2506.03611v1

Abstract: Poly(3,4-ethylenedioxythiophene):polystyrenesulfonate (PEDOT:PSS) has been widely used as a hole transport layer in perovskite solar cells (PSCs). However, the high interface defect density and energy level mismatch between PEDOT:PSS and perovskite can lead to significant open-circuit voltage loss. Additionally, the free PSS chains on the surface of PEDOT:PSS can absorb water molecules, promoting the degradation of perovskite at the PEDOT:PSS/perovskite interface. Here, gefitinib is used to modify the surface of PEDOT:PSS, removing a portion of the free PSS chains from the surface, reducing the PSS/PEDOT ratio, and enhancing the conductivity of PEDOT:PSS. Gefitinib has altered the energy level structure of PEDOT:PSS, facilitating hole transport at the interface. The Cl, F, and NH groups in gefitinib also passivated defects in the perovskite, reducing the defect density at the interface and significantly enhancing the stability of PSCs. This modification increased the open-circuit voltage from 1.077 to 1.110 V and the power conversion efficiency (PCE) from 17.01% to 19.63%. When gefitinib was used to modify the interface between SnO2 and perovskite, the PCE of PSCs (ITO/SnO2/perovskite/Spiro-OMETAD/Au) increased from 22.46% to 23.89%. This approach provides new perspectives and strategies for improving the efficiency and stability of PSCs.

Imaging real-time amorphization of hybrid perovskite solar cells under electrical biasing

Min-cheol Kim; Namyoung Ahn; Diyi Cheng; Mingjie Xu; Xiaoqing Pan; Suk Jun Kim; Yanqi Luo; David P. Fenning; Darren H. S. Tan; Minghao Zhang; So-Yeon Ham; Kiwan Jeong; Mansoo Choi; Ying Shirley Meng. arXiv preprint (2020). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/2010.12509v1

Abstract: Perovskite solar cells have drawn much attention in recent years, owing to its world-record setting photovoltaic performances. Despite its promising use in tandem applications and flexible devices, its practicality is still limited by its structural instability often arising from ion migration and defect formation. While it is generally understood that ion instability is a primary cause for degradation, there is still a lack of direct evidence of structural transformation at the atomistic scale. Such an understanding is crucial to evaluate and pin-point how such instabilities are induced relative to external perturbations such as illumination or electrical bias with time, allowing researchers to devise effective strategies to mitigate them. Here, we designed an in-situ TEM setup to enable real-time observation of amorphization in double cation mixed perovskite materials under electrical biasing at 1 V. It is found that amorphization occurs along the (001) and (002) planes, which represents the observation of in-situ facet-dependent amorphization of a perovskite crystal. To reverse the degradation, the samples were heated at 50 oC and was found to recrystallize, effectively regaining its performance losses. This work is vital toward understanding fundamental ion-migration phenomena and address instability challenges of perovskite optoelectronics.

Surface Treatment of Cu:NiOx Hole-Transporting Layer Using \b{eta}-Alanine for Hysteresis-Free and Thermally Stable Inverted Perovskite Solar Cells

Fedros Galatopoulos; Ioannis T. Papadas; Apostolos Ioakeimidis; Polyvios Eleftheriou; Stelios A. Choulis. Nanomaterials 2020, 10(10), 1961 (2020). Cited by 0. Found by: arxiv. DOI: 10.3390/nano10101961

Abstract: Inverted perovskite solar cells (PSCs) using a Cu:NiOx hole transporting layer (HTL) often exhibit stability issues and in some cases J/V hysteresis. In this work, we developed a \b{eta}-alanine surface treatment process on Cu:NiOx HTL that provides J/V hysteresis-free, highly efficient, and thermally stable inverted PSCs. The improved device performance due to \b{eta}-alanine-treated Cu:NiOx HTL is attributed to the formation of an intimate Cu:NiOx/perovskite interface and reduced charge trap density in the bulk perovskite active layer. The \b{eta}-alanine surface treatment process on Cu:NiOx HTL eliminates major thermal degradation mechanisms, providing 40 times increased lifetime performance under accelerated heat lifetime conditions. By using the proposed surface treatment, we report optimized devices with high power conversion efficiency (PCE) (up to 15.51%) and up to 1000 h lifetime under accelerated heat lifetime conditions (60 C, N2).

Exploring Lead Free Mixed Halide Double Perovskites Solar Cell

Md Yekra Rahman; Sharif Mohammad Mominuzzaman. arXiv preprint (2024). Cited by 0. Found by: arxiv. DOI: 10.1109/ICECE64886.2024.11024609

Abstract: The significant surge in energy use and escalating environmental concerns have sparked worldwide interest towards the study and implementation of solar cell technology. Perovskite solar cells (PSCs) have garnered remarkable attention as an emerging third-generation solar cell technology. This paper presents an in-depth analysis of lead-free mixed halide double perovskites in the context of their potential uses in solar cell technology. Through the previous studies of various mixed halide double perovskite materials as potential absorber layer materials, it has been observed that Cs$_2$TiI$_{6-x}$Br$_x$ possesses promising characteristics. In this study, simulations were conducted using SCAPS-1D software to explore all possible combinations for x= 0 to 6. The materials for the Hole Transport Layer (HTL) and Electron Transport Layer (ETL) and absorber layer, along with their optimal thicknesses, are selected to yield the most promising results in terms of open-circuit voltage (V$_{oc}$), short-circuit current density (J$_{sc}$), fill factor (FF), and power conversion efficiency. A novel tolerance factor is employed to assess structural stability of perovskites. FTO/Cu$_2$O/Cs$_2$TiI$_5$Br$_1$/WS$_2$ emerges as the best combination in terms of the efficiency with 19.03% but FTO/Cu$_2$O/Cs$_2$TiI$_2$Br$_4$/WS$_2$ shows good stability with 13.31% efficiency.

A green solvent system for precursor phase-engineered sequential deposition of stable formamidinium lead triiodide for perovskite solar cells

Benjamin M. Gallant; Philippe Holzhey; Joel A. Smith; Saqlain Choudhary; Karim A. Elmestekawy; Pietro Caprioglio; Igal Levine; Alex Sheader; Fengning Yang; Daniel T. W. Toolan; Rachel C. Kilbride; Augustin K. A. Zaininger; James M. Ball; M. Greyson Christoforo; Nakita Noel; Laura M. Herz; Dominik J. Kubicki; Henry J. Snaith. arXiv preprint (2024). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/2406.08441v2

Abstract: Perovskite solar cells (PSCs) offer an efficient, inexpensive alternative to current photovoltaic technologies, with the potential for manufacture via high-throughput coating methods. However, challenges for commercial-scale solution-processing of metal-halide perovskites include the use of harmful solvents, the expense of maintaining controlled atmospheric conditions, and the inherent instabilities of PSCs under operation. Here, we address these challenges by introducing a high volatility, low toxicity, biorenewable solvent system to fabricate a range of 2D perovskites, which highly effective precursor phases for subsequent transformation to alpha-formamidinium lead triiodide (FAPbI3), fully processed under ambient conditions. PSCs utilising our FAPbI3 reproducibly show remarkable stability under illumination and elevated temperature (ISOS-L-2) and ”damp heat” (ISOS-D-3) stressing, surpassing other state-of-the-art perovskite compositions. We determine that this enhancement is a consequence of the 2D precursor phase crystallisation route, which simultaneously avoids retention of residual low-volatility solvents (such as DMF and DMSO) and reduces the rate of degradation of FA+ in the material. Our findings highlight both the critical role of the initial crystallisation process in determining the operational stability of perovskite materials, and that neat FA+-based perovskites can be competitively stable despite the inherent metastability of the alpha-phase.

Impact of interface defects on the band alignment and performance of TiO$_2$/MAPI/Cu$_2$O perovskite solar cells

Nicolae Filipoiu; Marina Cuzminschi; Mihaela Cosinschi; Calin-Andrei Pantis-Simut; Kristinn Torfason; Rachel Elizabeth Brophy; Andrei Manolescu; Roxana E. Patru; Cristina Besleaga; George E. Stan; Ioana Pintilie; George Alexandru Nemnes. Physica Scripta 101, 265906 (2026) (2024). Cited by 0. Found by: arxiv. DOI: 10.1088/1402-4896/ae7cef

Abstract: Optimizing the interfaces in perovskite solar cells (PSCs) is essential for enhancing their performance, improving their stability, and making them commercially viable for large-scale deployment in solar energy harvesting applications. Point defects, like vacancies, have a dual role, as they can inherently provide a proper doping, but they can also reduce the collected current by trap-assisted recombination. Moreover, they can play an active role in ion migration and degradation. Using ab initio density functional theory (DFT) calculations we investigate the changes in the band alignment induced by interfacial vacancy defects in a TiO$_2$/MAPI/Cu$_2$O based PSC. Depending on the type of the vacancy (Ti, Cu, O, Pb, I) in the oxide and perovskite materials, additional doping is superimposed on the already existing background. Their effect on the performance of the PSCs becomes visible, as shown by SCAPS simulations. The most significant impact is observed for p type doping of TiO$_2$ and n type doping of Cu$_2$O, while the effective doping of the perovskite layer affects one of the two interfaces. We discuss these results based on modifications of the band structure near the active interfaces and provide further insights concerning the optimization of electron and hole collection.

Efficient Passivation of Surface Defects by Lewis Base in Lead-free Tin-based Perovskite Solar Cells

Hejin Yan; Bowen Wang; Xuefei Yan; Qiye Guan; Hongfei Chen; Zheng Shu; Dawei Wen; Yongqing Cai. Materials Today Energy (2022) (2022). Cited by 0. Found by: arxiv. DOI: 10.1016/j.mtener.2022.101038

Abstract: Lead-free tin-based perovskites are highly appealing for the next generation of solar cells due to their intriguing optoelectronic properties. However, the tendency of Sn2+ oxidation to Sn4+ in the tin-based perovskites induces serious film degradation and performance deterioration. Herein, we demonstrate, through the density functional theory based first-principle calculations in a surface slab model, that the surface defects of the Sn-based perovskite FASnI3 (FA = NH2CHNH2+) could be effectively passivated by the Lewis base molecules. The passivation performance of Lewis base molecules in tin-based perovskite is tightly correlated with their molecular hardness. We reveal that the degree of hardness of Lewis adsorbate governs the stabilization via dual effects: first, changing the stubborn spatial distribution of tin vacancy (VSn) by triggering charge redistribution; second, saturating the dangling states while simultaneously reducing the amounts of deep band gap states. Specifically, the hard Lewis base molecules like edamine (N-donor group) and Isatin-Cl (Cl-donor group) would show a better healing effect than other candidates on the defects-contained tin-based perovskite surface with a somehow hard Lewis acid nature. Our research provides a general strategy for additive engineering and fabricating stable and high-efficiency lead-free Sn-based perovskite solar cells.

Capacitive and inductive effects in perovskite solar cells: the different roles of ionic current and ionic charge accumulation

Nicolae Filipoiu; Amanda Teodora Preda; Dragos Victor Anghel; Roxana Patru; Rachel Elizabeth Brophy; Movaffaq Kateb; Cristina Besleaga; Andrei Gabriel Tomulescu; Ioana Pintilie; Andrei Manolescu; George Alexandru Nemnes. Phys. Rev. Applied 18, 064087 (2022) (2022). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/2208.10199v2

Abstract: Dynamic hysteresis effects have been long known to occur in the J-V characteristics of perovskite solar cells (PSCs), with the ionic migration being identified as the primary factor. The hysteretic effects impacted

early studies by the uncertainty in the evaluation of the power conversion efficiency, while currently, potential links to degradation mechanisms are in the focus. Therefore, understanding ion migration is a central goal, typically addressed by performing a combined large- and small signal analysis. The reported large capacitive and inductive effects created controversies with respect to the underlying mechanisms, yielding essentially two classes of models, one based on large accumulation capacitances and the other based on ionic modulation of the collected current. We introduce here an equivalent circuit model and interpret these phenomena in terms of recombination current modulation, identifying the distinct contributions from ion current and ionic charge accumulations. These contributions to the recombination current are associated with capacitive and inductive effects, respectively, and we corroborate the numerical simulations with electrochemical impedance spectroscopy (EIS) measurements. These show the role of the recombination currents of photogenerated carriers in producing both capacitive and inductive effects as the illumination is varied. Moreover, we provide a bridging point between the two classes of models and suggest a framework of investigation of defect states based on the observed inductive behavior, which would further aid the mitigation of the degradation effects.

First-Principles Study on Material Properties and Stability of Inorganic Halide Perovskite Solid Solutions CsPb(I$_{1-x}$Br$_x$)$_3$ towards High Performance Perovskite Solar Cells

Chol-Jun Yu; Un-Hyok Ko; Suk-Gyong Hwang; Yun-Sim Kim; Un-Gi Jong; Yun-Hyok Kye; Chol-Hyok Ri. Phys. Rev. Materials 4, 045402 (2020) (2019). Cited by 0. Found by: arxiv. DOI: 10.1103/PhysRevMaterials.4.045402

Abstract: All-inorganic halide perovskites have attracted a great interest as a promising light harvester of perovskite solar cells due to their enhanced chemical stability. In this work we investigate the material properties of solid solutions CsPb(I$_{1-x}$Br$_x$)$_3$ in cubic phase by applying the virtual crystal approximation approach within a density functional theory framework. First we check the validity of constructed pseudopotentials of the virtual atoms (X = I$_{1-x}$Br$_x$) by verifying that the lattice constants follow the linear function of mixing ratio. We then suggest an idea of using the hybrid HSE functional with linear increasing value of exact exchange term as increasing the Br content x, which produces the band gaps of CsPbX3 in good agreement with the available experimental data. The calculated light absorption coefficients and reflectivity show the systematic varying tendency to the Br content. We calculate the phonon dispersions of CsPbX3, CsX and PbX2 as slightly changing their volumes, revealing the phase instability of CsPbX3 and calculating the thermodynamic potential function differences. By projecting Gibbs free energy differences onto the plane of G = 0, we determine the P - T diagram for CsPbX3 to be stable against the chemical decomposition, highlighting that the area of being stable extends gradually as the Br content increases.

An eco-friendly universal strategy via ribavirin to achieve highly efficient and stable perovskite solar cells

Xianhu Wu; Gaojie Xia; Guanglei Cui; Jieyu Bi; Nian Liu; Jiaxin Jiang; Jilong Sun; Luyang Liu; Ping Li; Ning Lu; Zewen Zuo; Min Gu. arXiv preprint (2025). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/2507.10557v1

Abstract: The grain boundaries of perovskite films prepared by the solution method are highly disordered, with a large number of defects existing at the grain boundaries. These defect sites promote the decomposition of perovskite. Here, we use ribavirin obtained through bacillus subtilis fermentation to regulate the crystal growth of perovskite, inducing changes in the work function and energy level structure of perovskite, which significantly reduces the defect density. Based on density functional theory calculations, the defect formation energies of VI, VMA, VPb, and PbI in perovskite are improved. This increases the open-circuit voltage of perovskite solar cells (PSCs) (ITO/PEDOT:PSS/perovskite/PCBM/BCP/Ag) from 1.077 to 1.151 V, and the PCE increases significantly from 17.05% to 19.86%. Unencapsulated PSCs were stored in the environment (humidity approximately 35+-5%) for long-term stability testing. After approximately 900 hours of storage, the PCE of the ribavirin-based device retains 84.33% of its initial PCE, while the control-based device retains only 13.44% of its initial PCE. The PCE of PSCs (ITO/SnO2/perovskite/Spiro-OMETAD/Ag) is increased from 20.16% to 22.14%, demonstrating the universality of this doping method. This universal doping strategy provides a new approach for improving the efficiency and stability of PSCs using green molecular doping strategies.

(111) Facet-engineered SnO2 as Electron Transport Layer for Efficient and Stable Triple-Cation Perovskite Solar Cells

Keshav Kumar Sharma; Rohit; Sochannao Machinao; Ramesh Karuppannan. Sustainable Energy Fuels, 2025,9, 3102-3109 (2025). Cited by 0. Found by: arxiv. DOI: 10.1039/D5SE00339C

Abstract: We report the (111) facet-engineered cubic phase SnO2 (C-SnO2) as a novel electron transport layer (ETL) for triple-cation mixed-halide Cs0.05(FA0.83MA0.17)0.95Pb(I0.83Br0.17)3 perovskite solar cells (PSCs). The C-SnO2 layer was prepared via a normal sol-gel process followed by the spin-coating technique. The (111) facet C-SnO2 layer provides a larger surface contact area with an adjacent perovskite layer, enhancing charge transfer dynamics at the interface. In addition, the well-matched overlapping band structures improve the charge extraction efficiency between the two layers. Using (111) facet C-SnO2 as ETLs, we obtain PSCs with a higher power conversion efficiency of 20.34% (0.09 cm2) than those employing tetragonal phase SnO2 ETL. The PSCs with C-SnO2 ETL retain over 81% of their initial efficiency even after 480 h. This work concludes with a brief discussion on recombination and charge transport mechanisms, providing ways to optimize C-SnO2 ETL to improve the PSCs' performance and stability.

Transparent Multispectral Photonic Electrode for All-Weather Stable and Efficient Perovskite Solar Cells

George Perrakis; Anna C. Tasolamprou; George Kakavelakis; Konstantinos Petridis; Michael Graetzel; George Kenanakis; Stelios Tzortzakis; Maria Kafesaki. arXiv preprint (2023). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/2308.03386v1

Abstract: Perovskite solar cells (PSCs) are the most promising technology for advancing current photovoltaic performance. However, the main challenge for their practical deployment and commercialization is their operational stability, affected by solar illumination and heating, as well as the electric field that is generated in the PV device by light exposure. Here, we propose a transparent multispectral photonic electrode placed on top of the glass substrate of solar cells, which simultaneously reduces the device solar heating and enhances its efficiency. Specifically, the proposed photonic electrode, composed of a low-resistivity metal and a conductive layer, simultaneously serves as a highly-efficient infrared filter and an ultra-thin transparent front contact, decreasing devices' solar heating and operating temperature. At the same time, it simultaneously serves as an anti-reflection coating, enhancing the efficiency. We additionally enhance the device cooling by coating the front glass substrate side with a visibly transparent film (PDMS), which maximizes substrate's thermal radiation. To determine the potential of our photonic approach and fully explore the cooling potential of PSCs, we first provide experimental characterizations of the absorption properties (in both visible and infrared wavelengths) of state-of-the-art PSCs among the most promising ones regarding the efficiency, stability, and cost. We then numerically show that applying our approach to promising PSCs can result in lower operating temperatures by over 9.0 oC and an absolute efficiency increase higher than 1.3%. These results are insensitive to varying environmental conditions. Our approach is simple and only requires modification of the substrate; it therefore points to a feasible photonic approach for advancing current photovoltaic performance with next-generation solar cell materials.

Atomistic mechanism for trapped-charge driven degradation of perovskite solar cells

Kwisung Kwak; Eunhak Lim; Namyoung Ahn; Jiyoung Heo; Kijoon Bang; Seong Keun Kim; Mansoo Choi. arXiv preprint (2017). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/1709.04130v1

Abstract: It is unmistakably paradoxical that the weakest point of the photoactive organic-inorganic hybrid perovskite is its instability against light. Why and how perovskites break down under light irradiation and what happens at the atomistic level during the degradation still remains unanswered. In this paper, we revealed the fundamental origin and mechanism for irreversible degradation of hybrid perovskite materials from our new experimental results and ab initio molecular dynamics (AIMD) simulations. We found that the photo-generated charges trapped along the grain boundaries of the perovskite crystal result in oxygen-induced irreversible degradation in air even in the absence of moisture. The present result, together with our previous experimental finding on the same critical role of trapped charges in the perovskite degradation under moisture, suggests that the trapped charges are the main culprit in both the oxygen- and moisture-induced degradation of perovskite materials. More detailed roles of oxygen and water molecules were investigated by tracking the atomic motions of the oxygen- or water-covered CH3NH3PbI3(MAPbI3) perovskite crystal surface with trapped charges via AIMD simulation. In the first few picoseconds of our simulation, trapped charges start disrupting the crystal structure, leading to a close-range interaction between oxygen or water molecules and the compositional ions of MAPbI3. We found that there are different degradation pathways depending on both the polarity of the trapped charge and the kind of gas molecule. Especially, the deprotonation of organic cations was theoretically predicted for the first time in the presence of trapped anionic charges and water molecules. We confirmed that a more structurally stable, multi-component perovskite material(MA0.6FA0.4PbI2.9Br0.1) exhibited a much longer lifespan than MAPbI3 under light irradiation even in 100% oxygen ambience.

A simple all-inorganic hole-only structure for trap density measurement in perovskite solar cells

Atena mohamadnezhad; Alireza Fathi-Beiraghvandi; Mahmoud Samadpour. arXiv preprint (2023). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/2306.07136v2

Abstract: One of the critical challenges in enhancing the performance of perovskite solar cells is reducing the density of trap states in the light-absorbing perovskite layer. These trap states lead to increased charge carrier recombination, thus dropping device efficiency. Space charge limited current (SCLC) analysis serves as a valuable method to study trap density, requiring structures capable of selectively transporting either electrons or holes. By analyzing current-voltage (I-V) characteristics and identifying the voltage at which the slope changes, trap density can be calculated effectively. Traditional organic polymer hole transport layers such as Spiro-OMeTAD, PEDOT: PSS, and PTAA face challenges, including moisture instability, low charge mobility, low conductivity, and high costs. This work introduces a novel hole-only device structure utilizing inorganic materials, offering improved stability, straightforward fabrication, and reduced costs compared to conventional structures. This device comprises a nanostructured NiOx layer, a perovskite layer, a copper indium selenide (CIS) layer, and an Au electrode on an ITO substrate. The performance of this structure is assessed by fabricating various perovskite layers under different experimental conditions. The trap density was successfully determined using the proposed hole-only device structure. Analysis of the photovoltaic properties revealed a clear correlation between trap density in the perovskite layers and the overall performance of the solar cells.

Deep-level transient spectroscopy of the charged defects in p-i-n perovskite solar cells induced by light-soaking

A. A. Vasilev; D. S. Saranin; P. A. Gostishchev; M. P. Tuhova; S. I. Didenko; A. Y. Polyakov; A. Di Carlo. arXiv preprint (2022). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/2210.11176v1

Abstract: The long-term stability of halide perovskite solar cells (PSCs) remains the critical problem of this photovoltaic technology. Different structural defects formed in the thin-film perovskite films were considered as a main trigger for the decomposition of the absorber and corrosion of the interfaces in the device structure. The changes in the stability performance of the PSCs require a detailed analysis of the defects generated under external stress (light and heat). Using admittance, deep-level transient spectroscopy (DLTS) and reverse DLTS we determined the evolution of the defect energy levels in p-i-n PCS under continuous light soaking stress. We compared the impact of the charged defects on the performance and long-term stability of the CsFAPbI3 based devices with and without Cl-doping. Despite the gain in the output performance of the PCSs, the devices with CsFAPbI3-xClx showed improved light soaking stability. The T80 (time required to reduce initial efficiency by 20%) for Cl-doped PSCs was 1280h, while for pure CsFAPbI3 based devices only 650h. Three different defect energy levels were determined for different device configurations. We found that Cl-doping suppressed the formation of the antisite defects (IPb, IFA) and iodine interstitials (Ii). The changes in the defect's energy levels after continuous light soaking stress were analyzed and discussed. The present work provides new insights for the defect behavior of PSCs under continuous external stress, revealing the physical-chemical impact of the Cl-additive strategy.

Turning bad into good: a water-splitting-active hole transporting material to preserve the performance of perovskite solar cells in humid environments

Min Kim; Antonio Alfano; Giovanni Perotto; Michele Serri; Nicola Dengo; Alessandro Mezzetti; Silvia Gross; Mirko Prato; Marco Salerno; Roberto Sorrentino; Gaudenzio Meneghesso; Fabio Di Fonzo; Annamaria Petrozza; Teresa Gatti; Francesco Lamberti. arXiv preprint (2020). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/2005.0970

Abstract: Lead halide perovskite-based photoactive layers are nowadays employed for a large number of optoelectronic applications, from solar cells to photodetectors and light-emitting diodes, because of their excellent absorption, emission and charge-transport properties. Unfortunately, their commercialization is still hindered by an intrinsic instability towards classical environmental conditions. Water in particular promotes fast decomposition, leading to a drastic decrease in device performance. An innovative functional approach to overcome this major issue could derive from integrating water-splitting active species within charge extracting layers adjacent to the perovskite photoactive layer, converting incoming water molecules into molecular oxygen and hydrogen before they reach this last one, thus preserving device performance in time. In this work we report for the first time on a perovskite-ancillary layer based on CuSCN nanoplateletes dispersed in a p-type semiconducting polymeric matrix, combining hole extraction/transport properties with good water-oxidation activity, that transforms incoming water molecules and further triggers the in situ p-doping of the conjugated polymer by means of the produced dioxygen, further improving transport of photogenerated charges. This composite layer enables the long-term stabilization of a mixed cation lead halide perovskite within a direct solar cell architecture,

maintaining a stable performance for 28 days in high-moisture simulated conditions. Our findings demonstrate that the engineering of a hole extraction layer with water-splitting active additives represent a valuable strategy to mitigate the degradation of perovskite solar cells exposed to atmospheric humidity. A similar approach could be employed in the future to improve stabilities of other optoelectronic devices based on water-sensitive species.

AI-supported Degradation Study of Carbon-based Perovskite Solar Cells: Learning the Device Physics of Perovskite Solar Cells: A Drift-Diffusion Guided Autoencoder Approach

Oliver Zbinden; Sharun Parayil Shaji; Wolfgang Tress. arXiv preprint (2026). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/2603.22520v1

Abstract: Carbon-electrode-based PSC devices are stressed under 1 Sun equivalent illumination in a stability setup, and different scan-speed dependent current-voltage (J-V) curves are measured during aging. The collected data is used to estimate several physical parameters that contain information about charge transport and recombination using Machine Learning (ML), which allows for in situ tracking of possible signs of degradation. These results are compared to what can be classically interpreted by analysing changes in J-V curves, and the evolution of the predicted parameters is studied. The predictions are then used to simulate a digital twin of the measured devices, and their physical implications and the differences between measurements and devices are discussed.

Separation of Hoke and Schottky effects for improvement of mixed halide perovskite solar cell stability

Vladimir Ivanov; Eduard Ageev; Denis Danilov; Eduard Danilovskiy; Dmitry Gets. arXiv preprint (2025). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/2512.07523v1

Abstract: Ion migration in halide perovskites is a key factor limiting the operational stability of solar cells due to formation of halogen ion enriched domains and accumulation layers. The present work demonstrates the manifestation of ion migration in two ways via Hoke and Schottky effects. Both effects are induced by external exposure but have its peculiar way of solar cell performance degradation. We demonstrate the effects of ion migration on the device performance by measuring time dependent short-circuit current and different impedance characteristics that allow to see how charge-carrier separation property degrades. The Schottky effect leads to the rapid decrease of charge-carrier separation characteristic of solar cell while Hoke effect leads to the slow defect accumulation in the perovskite layer leading to the enhanced Shockley-Read-Hall recombination. Separation of these two effects can be realized by simple increase of transport layer thickness. A thick transport layer blocks losses of charge-carrier selectivity in a solar cell and leads to an enormous increase of T80 time, from 15 seconds up to 60 minutes.

Local Nanoscale Defective Phase Impurities Are the Sites of Degradation in Halide Perovskite Devices

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Abstract: Halide perovskites excel in the pursuit of highly efficient thin film photovoltaics, with power conversion efficiencies reaching 25.5% in single junction and 29.5% in tandem halide perovskite/silicon solar cell configurations. Operational stability of perovskite solar cells remains a barrier to their commercialisation, yet a fundamental understanding of degradation processes, including the specific sites at which failure mechanisms occur, is lacking. Recently, we reported that performance-limiting deep sub-bandgap states appear in nanoscale clusters at particular grain boundaries in state-of-the-art $Cs_{0.05}FA_{0.78}MA_{0.17}Pb(I_{0.83}Br_{0.17})_{3}$ (MA=methylammonium, FA=formamidinium) perovskite films. Here, we combine multimodal microscopy to show that these very nanoscale defect clusters, which go otherwise undetected with bulk measurements, are sites at which degradation seeds. We use photoemission electron microscopy to visualise trap clusters and observe that these specific sites grow in defect density over time under illumination, leading to local reductions in performance parameters. Scanning electron diffraction measurements reveal concomitant structural changes at phase impurities associated with trap clusters, with rapid conversion to metallic lead through iodine depletion, eventually resulting in pinhole formation. By contrast, illumination in the presence of oxygen reduces defect densities and reverses performance degradation at these local clusters, where phase impurities instead convert to amorphous and electronically benign lead oxide. Our work shows that the trapping of charge carriers at sites associated with phase impurities, itself reducing performance, catalyses redox reactions that compromise device longevity. Importantly, we reveal that both performance losses and intrinsic degradation can be mitigated by eliminating these defective clusters.

Combined DFT, SCAPS-1D, and wxAMPS frameworks for design optimization of efficient Cs2BiAgI6-based perovskite solar cells with different charge transport layers

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Abstract: In this study, combined DFT, SCAPS-1D, and wxAMPS frameworks are used to investigate the optimized designs of Cs2BiAgI6 double perovskite-based solar cells. The first-principle calculation is employed to investigate the structural stability, optical responses, and electronic contribution of the constituent elements in Cs2BiAgI6 absorber material, where SCAPS-1D and wxAMPS simulators are used to scrutinize different configurations of Cs2BiAgI6 solar cells. Here, PCBM, ZnO, TiO2, C60, IGZO, SnO2, WS2, and CeO2 are used as ETL, and Cu2O, CuSCN, CuSbS2, NiO, P3HT, PEDOT: PSS, Spiro-MeOTAD, CuI, CuO, V2O5, CBTS, CFTS are used as HTL, and Au is used as a back contact. About ninety-six combinations of Cs2BiAgI6-based solar cell structures are investigated, in which eight sets of solar cell structures are identified as the most efficient structures. Besides, holistic investigation on the effect of different factors such as the thickness of different layers, series and shunt resistances, temperature, capacitance, Mott-Schottky and generation-recombination rates, and J-V (current-voltage density) and QE (quantum efficiency) characteristics is performed. The results show CBTS as the best HTL for Cs2BiAgI6 with all eight ETLs used in this work, resulting in a power conversion efficiency (PCE) of 19.99%, 21.55%, 21.59%, 17.47%, 20.42%, 21.52%, 14.44%, 21.43% with PCBM, TiO2, ZnO, C60, IGZO, SnO2, CeO2, WS2, respectively. The proposed strategy may pave the way for further design optimization of lead-free double perovskite solar cells.

A Comprehensive Computational Photovoltaic Study of Lead-free Inorganic NaSnCl$_3$-based Perovskite Solar Cell: Effect of Charge Transport Layers and Material Parameters

Md Tashfiq Bin Kashem; Sadia Anjum Esha. arXiv preprint (2025). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/2503.02845v1

Abstract: Lead-free all-inorganic halide perovskite solar cells (PSCs) have emerged as a promising alternative to toxic lead-based solar cells and organic solar cells, which have limited stability. This work explores such a PSC with sodium tin chloride (NaSnCl$_3$) as the absorber, due to its significant potential for optoelectronic applications. To investigate this potential, a comprehensive computational analysis of NaSnCl$_3$-based solar cells is performed using the one-dimensional solar cell capacitance simulator (SCAPS -1D). Simulations are performed for device structures with front contact/Indium Tin Oxide (ITO)/electron transport layer (ETL)/NaSnCl$_3$/hole transport layer (HTL)/back contact configuration, where TiO$_2$, SnS$_2$, IGZO, ZnSe, CdS, GaSe, ZnSnN$_2$, WS$_2$, PCBM, STO, and CSTO are utilized as ETLs and CNTS, GO, Mg-CuCrO$_2$, Spiro-OMeTAD, CdTe, GaAs, MoTe$_2$, BaSi$_2$, and P3HT are utilized as HTLs. Based on the obtained power conversion efficiency (PCE), six best ETL-HTL combinations with SnS$_2$, STO, WS$_2$, IGZO, ZnSe and CSTO as ETLs and MoTe$_2$ as HTL are chosen for further analysis. The effects of different material and device parameters, such as thickness and doping density; effective density of states; bulk and interface defects; series and shunt resistance; and operating conditions, such as temperature and light intensity are investigated. Using the optimized material parameters, SnS$_2$ ETL and MoTe$_2$ HTL-based solar cell show the best performance with open circuit voltage, Voc = 1.196V, short circuit current density, Jsc = 35.82 mA/cm^2, fill factor, FF = 89.72% and PCE = 38.42%. This detailed study provides valuable insights for the fabrication of high efficiency NaSnCl$_3$-based solar cells.

Ab Initio Thermodynamic Study of PbI$_2$ and CH$_3$NH$_3$PbI$_3$ Surfaces in Reaction with CH$_3$NH$_2$ Gas for Perovskite Solar Cells

Yun-Sim Kim; Chol-Hyok Ri; Yun-Hyok Kye; Un-Gi Jong; Chol-Jun Yu. arXiv preprint (2021). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/2109.14553v1

Abstract: Hybrid organic-inorganic halide perovskites for photovoltaics have attracted research interest due to their unique material properties, but suffered from poor material stability. In this work, we investigated the surface phase diagrams of PbI$_2$ and cubic CH$_3$NH$_3$PbI$_3$ (MAPbI$_3$) for a better understanding of precursor effect on perovskite synthesis via solid-gas reaction, by density functional theory calculations combined with thermodynamics. Using the devised slab models of PbI$_2$(001) and MAPbI$_3$(100), (110) and (111) surfaces with perfect and various vacant defect terminations, we calculated their formation energies and adsorption energies of CH$_3$NH$_2$ molecule on the PbI$_2$(001) surfaces under different synthesis conditions of temperature, pressure and pH via chemical potentials of species. Our calculations revealed that the adsorption can be facilitated by including HI or NH$_4$I molecule by dissociation of this additive and formation of CH$_3$NH$_3^+$ cation or NH$_3$-CH$_3$NH$_3^+$ complex, which is beneficial for conversion of PbI$_2$ to MAPbI$_3$ via solid-gas reaction. Furthermore, we found that among different perovskite MAPbI$_3$ surfaces, the MA-terminated (110) and MAI-terminated (100) surfaces are placed on the thermodynamically stable region of chemical potentials at pH values of 1 and 6, being agreed well with the experimental findings. We believe this work gives a fundamental understanding of solid-gas reaction for high-crystallinity perovskite synthesis towards perovskite solar cells with improved stability.

Structural Stability and Optoelectronic Properties of Tetragonal MAPbI3 under Strain

Lei Guo; Gao Xu; Gang Tang; Daining Fang; Jiawang Hong. Nanotechnology 31, 225204 (2020) (2020). Cited by 0. Found by: arxiv. DOI: 10.1088/1361-6528/ab7679

Abstract: In recent years, organic-inorganic hybrid perovskites have attracted wide attention due to their excellent optoelectronic properties in the application of optoelectronic devices. In the manufacturing process of perovskite solar cells, perovskite films inevitably have residual stress caused by non-stoichiometry components and the external load. However, their effects on the structural stability and photovoltaic performance of perovskite solar cells are still not clear. In this work, we investigated the effects of external strain on the structural stability and optoelectronic properties of tetragonal MAPbI3 by using the first-principles calculations. We found that the migration barrier of I- ion increases in the presence of compressive strain and decreases with tensile strain, indicating that the compressive strain can enhance the structural stability of halide perovskites. In addition, the light absorption and electronic properties of MAPbI3 under compressive strain are also improved. The variations of the band gap under triaxial and biaxial strains are consistent within a certain range of strain, resulting from the fact that the band edge positions are mainly influenced by the Pb-I bond in the equatorial plane. Our results provide useful guidance for realizing the commercial applications of MAPbI3-based perovskite solar cells.

Stability Engineering of Halide Perovskite via Machine Learning

Zhenzhu Li; Qichen Xu; Qingde Sun; Zhufeng Hou; Wan-Jian Yin. arXiv preprint (2018). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/1803.06042v1

Abstract: Perovskite stability is of the core importance and difficulty in current research and application of perovskite solar cells. Nevertheless, over the past century, the formability and stability of perovskite still relied on simplified factor based on human knowledge, such as the commonly used tolerance factor t. Combining machine learning (ML) with first-principles density functional calculations, we proposed a strategy to firstly calculate the decomposition energies, considered to be closely related to thermodynamic stability, of 354 kinds halide perovskites, establish the machine learning relationship between decomposition energy and compositional ionic radius and investigate the stabilities of 14,190 halide double perovskites. The ML-predicted results enable us to rediscover a series of stable rare earth metal halide perovskites (up to ~1000 kinds), indicating the generalization of this model and further provide elemental and concentration suggestion for improving the stability of mixed perovskite.

Nanoparticulate Metal Oxide Top Electrode Interface Modification Improves the Thermal Stability of Inverted Perovskite Photovoltaics

Ioannis T. Papadas; Fedros Galatopoulos; Gerasimos S. Armatas; Nir Tessler; Stelios A. Choulis. Nanomaterials 2019, 9(11), 1616 (2019). Cited by 0. Found by: arxiv. DOI: 10.3390/nano9111616

Abstract: Solution processed -Fe2O3 nanoparticles via the solvothermal colloidal synthesis in conjunction with ligand-exchange method are used for interface modification of the top electrode in inverted perovskite solar cells. In comparison to more conventional top electrodes such as PC(70)BM/Al and PC(70)BM/AZO/Al, we show that incorporation of a -Fe2O3 provides an alternative solution processed top electrode (PC(70)BM/ -Fe2O3/Al) that not only results in comparable power conversion efficiencies but also improved thermal stability of inverted perovskite photovoltaics. The origin of improved stability of inverted perovskite solar cells incorporating PC(70)BM/ -Fe2O3/Al under accelerated heat lifetime conditions is attributed to the acidic surface nature of -Fe2O3 and reduced charge trapped density within PC(70)BM/ -Fe2O3/Al top electrode interfaces.

Long Thermal Stability of Inverted Perovskite Photovoltaics Incorporating Fullerene-based Diffusion Blocking Layer

Fedros Galatopoulos; Ioannis T. Papadas; Gerasimos S. Armatas; Stelios A. Choulis. Adv. Mater. Interfaces 2018, 5, 1800280 (2019). Cited by 0. Found by: arxiv. DOI: 10.1002/admi.201800280

Abstract: In this article, the stability of p-i-n perovskite solar cells is studied under accelerated heat lifetime conditions (60 oC ,85oC and N2 atmosphere). By using a combination of buffer layer engineering, impedance spectroscopy and other characterization techniques, we propose the interaction of the perovskite active layer with the top Al metal electrode through diffusion mechanisms as the major thermal degradation pathway for planar inverted perovskite photovoltaics (PVs) under 85oC heat conditions. We show that by using thick solution processed fullerene buffer layer the perovskite active layer can be isolated from the top metal electrode and improve the lifetime performance of the inverted perovskite photovoltaics at 85 oC. Finally, we present an optimized solution processed inverted perovskite PV device using thick fullerene-based diffusion blocking layer with over 1000 hours accelerated heat lifetime performance at 60oC.

Revealing the stability and efficiency enhancement in mixed halide perovskites MAPb(I$_{1-x}$Cl$_x$)$_3$ with ab initio calculations

Un-Gi Jong; Chol-Jun Yu; Yong-Man Jang; Gum-Chol Ri; Song-Nam Hong; Yong-Hyon Pae. Journal of Power Sources, Volume 350, 15 May 2017, Pages 65-72 (2016). Cited by 0. Found by: arxiv. DOI: 10.1016/j.jpowsour.2017.03.038

Abstract: A little addition of Cl to \ce{MAPbI3} has been reported to improve the material stability as well as light harvesting and carrier conducting properties of organometal trihalide perovskites, the key component of perovskite solar cell (PSC). However, the mechanism of performance enhancement of PSC by Cl addition is still unclear. Here, we apply the efficient virtual crystal approximation method to revealing the effects of Cl addition on the structural, electronic, optical properties and material stability of \ce{MAPb(I_{1-x}Cl_x)3}. Our {\it ab initio} calculations present that as the increase of Cl content cubic lattice constants and static dielectric constants decrease linearly, while band gaps and exciton binding energies increase quadratically. Moreover, we find the minimum of exciton binding energy at the Cl content of 7\%, at which the chemical decomposition reaction changes coincidentally to be from exothermic to endothermic. Interactions among constituents of compound and electronic charge transferring during formation are carefully discussed. This reveals new prospects for understanding and designing of stable, high efficiency PSCs.

Stabilization of interfaces for double-cation halide perovskites with AVA2FAPb2I7 additives

Lev O. Luchnikov; Ekaterina A. Ilicheva; Victor A. Voronov; Prokhor A. Alekseev; Mikhail S. Dunaevskiy; Vladislav Kalinichenko; Vladimir Ivanov; Aleksandra Furasova; Daria A. Krupanova; Ekaterina V. Tekshina; Sergey A. Kozyukhin; Dmitry S. Muratov; Maria I. Voronova; Danila S. Saranin; Eugene I. Terukov. arXiv preprint (2025). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/2502.20903v1

Abstract: The use of mixed cation absorber composition was considered as an efficient strategy to mitigate the degradation effects in halide perovskite solar cells. Despite the reports about partial stabilization at elevated temperatures, unfavorable phase transition after thermocycling and electric field-driven corrosion remains critical bottlenecks of perovskite thin-films semiconductors. In this work, we developed stabilized heterostructures based on CsFAPbI3, modified with mechanically synthesized quasi-2D perovskite incorporating the 5-ammonium valeric acid cation (AVA2FAPb2I7). We found that integration of AVA2FAPb2I7 into grain boundaries boosts phase resilience under harsh thermocycling from -10 up to 100 C and suppresses transitions, as well as decomposition to PbI2. The rapid oxidation of metal contacts in the multi-layer stacks with non-passivated CsFAPbI3 was effectively suppressed in the fabricated heterostructure. A comprehensive interface study of the copper electrode contact revealed that the incorporation of AVA2FAPb2I7 stabilized the lead and iodine states and suppressed contamination of FA cation in ambient conditions. Meanwhile, the metal-perovskite interface remained predominantly in the Cu(0)-Cu(I) state. The observed stabilization in perovskite heterostructure was attributed to an increased activation energy for delta-phase accumulation at the grain boundaries combined with reduced ionic diffusion. The obtained results opened important highlights for the mechanisms of the improved phase stability after thermal cycling and mitigation of the interface corrosion and under an applied electric field.

Impermeable Inorganic Walls Sandwiching Photoactive Layer toward Inverted Perovskite Solar and Indoor-Photovoltaic Devices

Jie Xu; Jun Xi; Hua Dong; Namyoung Ahn; Zonglong Zhu; Jinbo Chen; Peizhou Li; Xinyi zhu; Jinfei Dai; Ziyang Hu; Bo Jiao; Xun Hou; Jingrui Li; Zhaoxin Wu. arXiv preprint (2020). Cited by 0. Found by: arxiv.

URL: http://arxiv.org/abs/2010.05679v1

Abstract: Interfaces between the perovskite active layer and the charge-transport layers (CTLs) play a critical role in both efficiency and stability of halide-perovskite photovoltaics. One of the major concerns is that surface defects of perovskite could cause detrimental nonradiative recombination and material degradation. In this work, we addressed this challenging problem by inserting ultrathin alkali-fluoride (AF) films between the tri-cation lead-iodide perovskite layer and both CTLs. This bilateral inorganic walls strategy makes use of both physical-blocking and chemical-anchoring functionalities of the continuous, uniform and compact AF framework: on the one hand, the uniformly distributed alkali-iodine coordination at the perovskite-AF interfaces effectively suppresses the formation of iodine-vacancy defects at the surfaces and grain boundaries of the whole perovskite film, thus reducing the trap-assisted recombination at the perovskite-CTL interfaces and therewith the open-voltage loss; on the other hand, the impermeable AF buffer layers effectively prevent the bidirectional ion migration at the perovskite-CTLs interfaces even under harsh working conditions. As a result, a power-conversion efficiency (PCE) of 22.02% (certified efficiency 20.4%) with low open-voltage deficit (< 0.4V) was achieved for the low-temperature processed inverted planar perovskite solar cells. Exceptional operational stability (500 h, ISOS-L-2) and thermal stability (1000 h, ISOS-D-2) were obtained. Meanwhile, a 35.7% PCE was obtained under dim-light source (1000 lux white LED light) with the optimized device, which is among the best records in perovskite indoor photovoltaics.

Impact of Grain Boundaries on Efficiency and Stability of Organic-Inorganic Trihalide Perovskites

Zhaodong Chu; Mengjin Yang; Philip Schulz; Di Wu; Xin Ma; Edward Seifert; Liuyang Sun; Kai Zhu; Xiaoqin Li; Keji Lai. arXiv preprint (2016). Cited by 0. Found by: arxiv. DOI: 10.1038/s41467-017-02331-4

Abstract: Organic-inorganic perovskite solar cells have attracted tremendous attention because of their remarkably high power conversion efficiencies (PCEs). To further improve the device performance, however, it is imperative to obtain fundamental understandings on the photo-response and long-term stability down to the microscopic level. Here, we report the first quantitative nanoscale photoconductivity imaging on two methylammonium lead triiodide (MAPbI3) thin films with different PCEs by light-stimulated microwave impedance microscopy. The intrinsic photo-response is largely uniform across grains and grain boundaries, which is direct evidence on the inherently benign nature of microstructures in the perovskite thin films. In contrast, the carrier mobility and lifetime are strongly affected by bulk properties such as the sample crystallinity. As visualized by the spatial evolution of local photoconductivity, the degradation due to water diffusion through the capping layer begins with the disintegration of large grains rather than the nucleation and propagation from grain boundaries. Our findings provide new insights to improve the electro-optical properties of MAPbI3 thin films towards large-scale commercialization.

Computational Prediction of Structural, Electronic, Optical Properties and Phase Stability of Double Perovskites K2SnX6 (X = I, Br, Cl)

Un-Gi Jong; Chol-Jun Yu; Yun-Hyok Kye. arXiv preprint (2019). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/1908.05236v3

Abstract: Vacancy-ordered double perovskites K2SnX6 (X = I, Br, Cl) attract significant research interest due to their potential application as light-absorbing materials in perovskite solar cells. However, a deep insight into their material properties at the atomic scale is yet scarce. Here we present a systematic investigation on their structural, electronic, optical properties and phase stabilities in cubic, tetragonal, and monoclinic phases based on density functional theory calculations. Quantitatively reliable prediction of lattice constants, band gaps, effective masses of charge carriers, exciton binding energies is provided in comparison with the available experimental data, revealing the increasing tendency of band gap and exciton binding energy as lowering the crystallographic symmetry from cubic to monoclinic and going from I to Cl. We highlight that cubic K2SnBr6 and monoclinic K2SnI6 are suitable for the application as a light-absorber for solar cell devices due to their proper band gaps of 1.65 and 1.16 eV and low exciton binding energies of 59.4 and 15.3 meV, respectively. The constant-volume Helmholtz free energies are determined through phonon calculations, giving a prediction of their phase transition temperatures as 449, 433 and 281 K for cubic-tetragonal and 345, 301 and 210 K for tetragonal-monoclinic transitions for X = I, Br and Cl. Our calculations provide an understanding of material properties of vacancy-ordered double perovskite K2SnX6, helping to devise a low-cost and high performance perovskite solar cell.

Charting the thermodynamic stability of hybrid perovskite alloys with machine learning

Jarno Laakso; Armi Tiihonen; Patrick Rinke. arXiv preprint (2026). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/2605.30012v1

Abstract: Alloy-based perovskite solar cells offer tunable properties and improved stability, but their complexity has impeded accurate modeling, hindering development. We present a machine-learning (ML) accelerated atomistic modeling approach for the phase stability of (Cs/FA)Pb(Br/I)3 and (Cs/FA)Sn(Br/I)3 perovskites, with FA being formamidinium. To make such quaternary alloys tractable, we adopt a two-level ML strategy, combining 1) graph neural network interatomic potentials trained on density functional theory data for efficient structure relaxations with 2) secondary ML models for direct energy prediction from unrelaxed structures. These models enable computations of free energy landscapes across compositions and phases, capturing alloy disorder and FA molecular orientations. Our results reveal narrower stable composition regions for the Sn-based system compared to its Pb-based counterpart, limiting options for compositional engineering. Maximum stability occurs at high I content, and no stabilization is observed near the center of the composition space. Our results guide the design of stable perovskites.

Crystal structure, stability and optoelectronic properties of the organic-inorganic wide bandgap perovskite CH3NH3BaI3: Candidate for transparent conductor applications

Akash Kumar; K. R. Balasubramaniam; Jiban Kangsabanik; Vikram; Aftab Alam. Phys. Rev. B 94, 180105 (2016) (2016). Cited by 0. Found by: arxiv. DOI: 10.1103/PhysRevB.94.180105

Abstract: Structural stability, electronic structure and optical properties of CH3NH3BaI3 hybrid perovskite is examined both from theory as well as experiment. Solution-processed thin films of CH3NH3BaI3 exhibited a high band gap of approximately 3.87 eV, which is in excellent agreement with the theoretical estimate of 4 eV. Also, the XRD patterns of the thin films match well with the l-peaks of the simulated pattern obtained from the relaxed unit cell of CH3NH3BaI3, crystallizing in the I4/mcm space group, with lattice parameters, a = 9.30 A, c = 13.94 A. Atom projected density of state and band structure calculations reveal the conduction and valence band edges to be comprised primarily of Barium d-orbitals and Iodine p-orbitals, respectively. The larger band gap of CH3NH3BaI3 compared to CH3NH3PbI3 can be attributed to the lower electro-negativity coupled with the lack of d-orbitals in the valence band of Ba{2+}. A more detailed analysis reveals the excellent chemical and mechanical stability of CH3NH3BaI3 against humidity, unlike its lead halide counterpart, which degrades under such conditions. The dopability of the CH3NH3BaI3 compound e.g. by doping La on the Ba site combined with its structural and mechanical stability under the ambient conditions, suggests this compound as a promising candidate for transparent conductor applications, especially for all perovskite solar cells.

Eliminating the Electric Field Response in a Perovskite Heterojunction Solar Cell to Improve Operational Stability

Jiangjian Shi; Yiming Li; Yusheng Li; Huijue Wu; Yanhong Luo; Dongmei Li; Qingbo Meng. arXiv preprint (2020). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/2008.01956v1

Abstract: Intrinsic and extrinsic ion migration is a very large threat to the operational stability of perovskite solar cells and is difficult to completely eliminate due to the low activation energy of ion migration and the existence of internal electric field. We propose a heterojunction route to help suppress ion migration, thus improving the operational stability of the cell from the perspective of eliminating the electric field response in the perovskite absorber. A heavily doped p-type (p+) thin layer semiconductor is introduced between the electron transporting layer (ETL) and perovskite absorber. The heterojunction charge depletion and electric field are limited to the ETL and p+ layers, while the perovskite absorber and hole transporting layer remain neutral. The p+ layer has a variety of candidate materials and is tolerant of defect density and carrier mobility, which makes this heterojunction route highly feasible and promising for use in practical applications.

Improvement of both performance and stability of photovoltaic devices by in situ formation of a sulfur-based 2D perovskite

Milon Kundar; Sahil Bhandari; Sein Chung; Kilwon Cho; Satinder K. Sharma; Ranbir Singh; Suman Kalyan Pal. arXiv preprint (2022). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/2211.15236v1

Abstract: Perovskite solar cells (PSCs) with superior performance have been recognized as a potential candidate in photovoltaic technologies. However, the defects in active perovskite layer induce non-radiative recombination which restricts the performance and stability of the PSCs. The construction of thiophene-based

2D structure is one of the significant approaches for surface passivation of hybrid PSCs that may combine the benefits of the stability of 2D perovskite with the high performance of 3D perovskite. Here, a sulfur-rich spacer cation 2-thiopheneethylamine iodide (TEAI) is synthesized as a passivation agent for the construction of three-dimensional/two-dimensional (3D/2D) perovskite bilayer structure. TEAI-treated PSCs possess a much higher efficiency (20.06%) compared to the 3D perovskite (MAFAPbI3) devices (17.42%). Time-resolved photoluminescence (TRPL) and femtosecond transient absorption (TA) spectroscopy are employed to investigate the effect of surface passivation on the charge carrier dynamics of the 3D perovskite. Additionally, the stability test of TEAI-treated perovskite devices reveals significant improvement in humid (RH ~ 56%) and thermal stability as the sulfur-based 2D (TEA)2PbI4 material self-assembles on the 3D surface making the perovskite surface hydrophobic. Our findings provide a reliable approach to improve device stability and performance successively, paving the way for industrialization of PSCs.

Enhanced stability of 2D organic-inorganic halide perovskites by doping and heterostructure engineering

Rahul Singh; Prashant Singh; Ganesh Balasubramanian. arXiv preprint (2018). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/1807.01221v2

Abstract: Organic-inorganic halide perovskite solar cells have recently attracted much attention due to their low-cost fabrication, flexibility, and high-power conversion efficiency. The reduction from three- to two-dimension (2D) promises an exciting opportunity to tune the electronic properties of organic-inorganic halide perovskites. Here, we propose first-principles density-functional theory based route to study the effect of reduced dimensionality, impurity doping, and heterostructure engineering on energy stability, band-gap and transport properties of 2D hybrid organic-inorganic halide perovskites. We show that the energetic stability of two-dimensional organic-inorganic halide perovskites can be significantly enhanced by chemically depositing MoS2 monolayer as a precursor in the system by heterostructure engineering. While on one hand, the structures have similar and excellent transport properties as their bulk counterparts, on the other, they possess the advantage of a broad range of tunable band gaps. Our predictions will expedite future efforts (both theoretical and experimental) in the synthesis, measurements and applications of high performance perovskites.

Influence of halide composition on the structural, electronic, and optical properties of mixed CH$_3$NH$_3$Pb(I$_{1-x}$Br$_x$)$_3$ perovskites calculated using the virtual crystal approximation method

Un-Gi Jong; Chol-Jun Yu; Nam-Hyok Kim; Guk-Chol Ri. Phys. Rev. B 94, 125139 (2016) (2016). Cited by 0. Found by: arxiv. DOI: 10.1103/PhysRevB.94.125139

Abstract: We investigate the structural, electronic and optical properties of mixed bromide-iodide lead perovskite solar cell CH$_3$NH$_3$Pb(I$_{1-x}$Br$_x$)$_3$ by means of the virtual crystal approximation (VCA) within density functional theory (DFT). Optimizing the atomic positions and lattice parameters increasing the bromide content x from 0.0 to 1.0, we fit the calculated lattice parameter and energy band gap to the linear and quadratic function of Br content, respectively, which are in good agreement with the experiment, respecting the Vegard's law. With the calculated exciton binding energy and light absorption coefficient, we make sure that VCA gives consistent results with the experiment, and the mixed halide perovskites are suitable for generating the charge carriers by light absorption and conducting the carriers easily due to their strong photon absorption coefficient, low exciton bindign energy, and high carrier mobility at low Br contents. Furthermore analyzing the bonding lengths between Pb and X (I$_{1-x}$Br$_x$: virtual atom) as well as C and N, we stress that the stability of perovskite solar cell is definitely improved at x=0.2.

Predicting Organic-Inorganic Halide Perovskite Photovoltaic Performance from Optical Properties of Constituent Films through Machine Learning

Ruiqi Zhang; Brandon Motes; Shaun Tan; Yongli Lu; Meng-Chen Shih; Yilun Hao; Karen Yang; Shreyas Srinivasan; Moungi G. Bawendi; Vladimir Bulovic. arXiv preprint (2024). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/2412.09638v1

Abstract: We demonstrate a machine learning (ML) approach that accurately predicts the current-voltage behavior of 3D/2D-structured (FAMA)Pb(IBr)3/OABr hybrid organic-inorganic halide perovskite (HOIP) solar cells under AM1.5 illumination. Our neural network algorithm is trained on measured responses from several hundred HOIP solar cells, using three simple optical measurements of constituent HOIP films as input: optical transmission spectrum, spectrally-resolved photoluminescence, and time-resolved photoluminescence, from which we predict the open-circuit voltage (Voc), short-circuit current (Jsc), and fill factors (FF) values of solar cells that contain the HOIP active layers. Determined average prediction accuracies for 95 % of the predicted Voc, Jsc, and FF values are 91%, 94% and 89%, respectively, with R2 coefficients of determination of 0.47, 0.77, and 0.58, respectively. Quantifying the connection between ML predictions and physical parameters extracted from the measured HOIP films optical properties, allows us to identify the most significant parameters influencing the prediction results. With separate ML-classifying algorithms, we identify degraded solar cells using the same optical input data, achieving over 90% classification accuracy through support vector machine, cross entropy loss, and artificial neural network algorithms. To our knowledge, the demonstrated regression and classification work is the first to use ML to predict device photovoltaic properties solely from the optical properties of constituent materials.

Protective Coating Interfaces for perovskite Solar Cell Materials: A first Principles Study

Azimatu Fangnon; Marc Dvorak; Ville Havu; Milica Todorovic; Jingrui Li; Patrick Rinke. arXiv preprint (2022). Cited by 0. Found by: arxiv. DOI: 10.1021/acsami.1c21785

Abstract: The protection of halide perovskites is important for the performance and stability of emergent perovskite-based optoelectronic technologies. In this work, we investigate the potential inorganic protective coating materials ZnO, SrZrO3, and ZrO2 for the CsPbI3perovskite. The optimal interface registries are identified with Bayesian optimization. We then use semi-local density-functional theory (DFT) to determine the atomic structure at the interfaces of each coating material with the clean CsI-terminated surface and three reconstructed surface models with added PbI2and CsI complexes. For the final structures, we explore the level alignment at the interface with hybrid DFT calculations. Our analysis of the level alignment at the coating-substrate interfaces reveals no detrimental mid-gap states, but substrate-dependent valence and conduction band offsets. While ZnO and SrZrO3act as insulators on CsPbI3, ZrO2 might be suitable as electron transport layer with the right interface engineering.

Microscopic Intricacies of Self-Healing in Halide Perovskite-Charge Transport Layer Heterostructures

Tejmani Behera; Boris Louis; Lukas Paesen; Roel Vanden Brande; Koki Asano; Martin Vacha; Maarten Roeffaers; Elke Debroye; Johan Hofkens; Sudipta Setha. arXiv preprint (2025). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/2510.11948v1

Abstract: The stability and performance of halide perovskite photovoltaic devices are critically limited by progressive defect generation and associated local non-radiative losses during operation. Self-healing of defects provides a promising pathway to prolong device functionality, yet the underlying microscopic mechanisms remain poorly understood, particularly the role of interfacial chemistry on trap dynamics and healing kinetics. Here, we elucidate self-healing and defect evolution in triple-cation mixed halide (TCMH) perovskite films and their device-relevant charge transport layer heterostructures subjected to photo-induced damage. Using correlation clustering imaging (CLIM), our recently developed local functional imaging tool, we map spatiotemporal photoluminescence heterogeneity to track defect dynamics in pristine and heterostructure films. The defect healing follows bi-phasic kinetics, with an initial electronic relaxation (tens of minutes) and a subsequent slower phase (~ hours) associated to ionic and lattice rearrangement. Most importantly, our results demonstrate that the chemical nature of charge-transport layers modulates trap activity, healing kinetics, and halide redistribution, with heterostructures exhibiting faster recovery than pristine films, a boon for device resilience. These findings provide new insights into the dynamic interaction between defects, interfaces, and ion migration, and establish a framework for rational design of durable, next-generation perovskite optoelectronic devices.

Multi-site mixing and entropy stabilization of CsPbI$_{3}$ with potential application in photovoltaics

Namitha Anna Koshi; Krishnamohan Thekkepat; Doh-Kwon Lee; Seung-Cheol Lee; Satadeep Bhattacharjee. arXiv preprint (2025). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/2504.18272v1

Abstract: Metal halide perovskite solar cells have achieved dramatic improvements in their power conversion efficiency in the recent past. Since compositional engineering plays an important role in optimizing material properties, we investigate the effect of alloying at Cs and Pb sites on the energetics and electronic structure of CsPbI$_{3}$ using cluster expansion method in combination with first-principles calculations. For Ge-mixing at Pb-site, the $ $ and $ $-phases are considered with emphasis on the electronic structure, transition probability, absorption coefficient, efficiency, and carrier mobility of higher-symmetry configurations. CsPb$_{0.50}$Ge$_{0.50}$I$_{3}$ (Cs$_{2}$PbGeI$_{6}$) which takes up a double perovskite (elpasolite) structure has a direct band gap with no parity-forbidden transitions. Further, we utilize the alloy entropic effect to improve the material stability and optoelectronic properties of CsPbI$_{3}$ by multi-element mixing. For the proposed mixed compositions, the Fr{ö}hlich electron-phonon coupling constant is determined. Scattering rates and electron mobility are obtained from first-principles inputs. These lower Pb-content inorganic perovskites offer great promise as efficient solar cell materials for photovoltaic applications.

Photo-effect on ion transport in mixed cation and halide perovskites and implications for photo de-mixing

Gee Yeong Kim; Alessandro Senocrate; Yaru Wang; Davide Moia; Joachim Maier. arXiv preprint (2019). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/1912.09777v1

Abstract: Organic-inorganic hybrid perovskites are considered to be most promising photovoltaic materials. Highest efficiencies of perovskite solar cells have been achieved by using appropriate cation and anion mixtures. Mixed perovskite solar cells also show an improved stability. For both performance as well as stability, experimental information on electronic and ionic charge carriers is key, an information that so far has only been provided for methylammonium lead iodide; there we also found that light can enhance not only electronic but also ionic conductivities by more than one order of magnitude. We also proposed a mechanism for this surprising photo-ionic effect and explained its impact on photo-decomposition. Here we quantitatively deconvolute ionic and electronic transport properties for the practically relevant substitutions and mixtures. Specifically, we investigate various cation and anion substitutions (Cs; FA; Br) with a special eye on their photo-ionic effect. The results are not only of importance for light-induced degradation but also for light-induced demixing. As far as the photo-ionic effect is concerned, we find that the choice of the halide is of crucial importance, while the cationic substitutions are less relevant. The huge ionic conductivity enhancement found for iodide perovskites, is weakened by bromide substitution and eventually becomes insignificant for the pure bromide. Based on these experimental results, we provide a rationale for the experimentally observed photo-demixing.

A microstructural analysis of 2D halide perovskites: Stability and functionality

Susmita Bhattacharya; Goutam Kumar Chandra; P. Predeep. Frontiers in Nanotechnology 2021 (2021). Cited by 0. Found by: arxiv. DOI: 10.3389/fnano.2021.657948

Abstract: Recent observations indicated that the photoelectric conversion properties of perovskite materials are intimately related to the presence of superlattice structures and other unusual nanoscale features in them. The low dimensional or mixed dimensional halide perovskite family are found to be more efficient materials for device application compared to 3-dimensional halide perovskites. The emergence of perovskite solar cell has revolutionized the solar cell industry because of their flexible architecture and rapidly increased efficiency. Tuning the dielectric constant, charge separation are the main objective in designing a photovoltaic device that can be explored using 2-dimensional perovskite family. Thus, revisiting the fundamental properties of perovskite crystals could reveal further possibilities for recognizing these improvements towards device functionality. In this context, this review discusses the material properties of 2-dimensional halide perovskite and related optoelectronic devices aiming particularly for solar cell application.

Experimentally observed defect tolerance in the electronic structure of lead bromide perovskites

Gabriel J. Man; Aleksandr Kalinko; Dibya Phuyal; Pabitra K. Nayak; Håkan Rensmo; Sergei M. Butorin. arXiv preprint (2023). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/2305.04014v1

Abstract: Point defect tolerance in materials, which extends operational lifetime, is essential for societal sustainability, and the creation of a framework to design such properties is a grand challenge in the material sciences. Using three prototypical lead bromide perovskites in single crystal form and high-resolution synchrotron-based X-ray spectroscopy, we reveal the unexpectedly pivotal role of the A-cation in mediating the influence of photoinduced defects. Organic A-cation hydrogen bonding facilitates chemical flexing of the lead-bromide bond that mitigates the self-doping effect of bromide vacancies. The contribution of partially ionic lead-bromide bonding to the electronic band edges, where the bonding becomes more ionic upon the formation of defects, mitigates re-hybridization of the electronic structure upon degradation. These findings reveal two new general design principles for defect tolerance in materials. Our findings uncover the foundations of defect tolerance in halide perovskites and have implications for defect calculations, all beam-based measurements of photophysical properties and perovskite solar cell technology.

What Happens at Surfaces and Grain Boundaries of Halide Perovskites: Insights from Reactive Molecular Dynamics Simulations of CsPbI$_{3}$

Mike Pols; Tobias Hilpert; Ivo Filot; Adri C. T. van Duin; Sofía Calero; Shuxia Tao. ACS Appl. Mater. Interfaces 2022, 14, 36, 40841-40850 (2022). Cited by 0. Found by: arxiv. DOI: 10.1021/acsami.2c09239

Abstract: The commercialization of perovskite solar cells is hindered by the poor long-term stability of the metal halide perovskite (MHP) light absorbing layer. Solution processing, the common fabrication method for MHPs, produces polycrystalline films with a wide variety of defects, such as point defects, surfaces, and grain boundaries. Although the optoelectronic effects of such defects have been widely studied, the evaluation of their impact on the long-term stability remains challenging. In particular, an understanding of the dynamics of degradation reactions at the atomistic scale is lacking. In this work, using reactive force field (ReaxFF) molecular dynamics simulations, we investigate the effects of defects, in the forms of surfaces, surface defects and grain boundaries, on the stability of the inorganic halide perovskite CsPbI$_{3}$. Our simulations establish a stability trend for a variety of surfaces, which correlates well with the occurrence of these surfaces in experiments. We find that a perovskite surface degrades by progressively changing the local geometry of PbI$_{\mathrm{x}}$ octahedra from corner- to edge- to face-sharing. Importantly, we find that Pb dangling bonds and the lack of steric hindrance of I species are two crucial factors that induce degradation reactions. Finally, we show that the stability of these surfaces can be modulated by adjusting their atomistic details, either by creating additional point defects or merging them to form grain boundaries. While in general additional defects, particularly when clustered, have a negative impact on the material stability, some grain boundaries have a stabilizing effect, primarily because of the additional steric hindrance.

Influence of interstitial Li on the electronic properties of Li$_{x}$CsPbI$_{3}$ for photovoltaic and battery applications

Wei Wei; Julian Gebhardt; Daniel F. Urban; Christian Elsässer. Physical Review B 112, 245103 (2025) (2024). Cited by 0. Found by: arxiv. DOI: 10.1103/nvxh-kwzt

Abstract: The integrated device of a perovskite solar cell with a Li-ion battery is an innovative solution for decentralized energy storage in smart electronic devices. In this study, we examine the stability of Li ions intercalated in a CsPbI$_3$ perovskite and their effect on the electronic structure of Li$_x$CsPbI$_3$ compounds using first-principles density functional theory. Our simulations demonstrate that the insertion of Li at concentrations up to x = 1 into CsPbI$_3$ is energetically possible. Moreover, we identify that the distortion of the Pb-I octahedra has the strongest impact on the change in the electronic band gap. Specifically, an increase in the amount of intercalated Li causes larger structural distortions, which in turn lead to an increasing band gap as function of the Li content.

Transmission electron microscopy of organic-inorganic hybrid perovskites: myths and truths

Shulin Chen; Ying Zhang; Jinjin Zhao; Zhou Mi; Jingmin Zhang; Jian Cao; Jicai Feng; Guanglei Zhang; Junlei Qi; Jiangyu Li; Peng Gao. Science Bulletin. 65, 1643-1649 (2020) (2020). Cited by 0. Found by: arxiv. DOI: 10.1016/j.scib.2020.05.020

Abstract: Organic-inorganic hybrid perovskites (OIHPs) have attracted extensive research interest as a promising candidate for efficient and inexpensive solar cells. Transmission electron microscopy characterizations that can benefit the fundamental understanding and the degradation mechanism are widely used for these materials. However, their sensitivity to the electron beam illumination and hence structural instabilities usually prevent us from obtaining the intrinsic information or even lead to significant artifacts. Here, we systematacially investigate the structural degradation behaviors under different experimental factors to reveal the optimized conditions for TEM characterizations of OIHPs by using low-dose electron diffraction and imaging techniques. We find that a low temperature does not slow down the beam damage but instead induces a rapid amorphization for OIHPs. Moreover, a less severe damage is observed at a higher accelerating voltage. The beam-sensitivity is found to be facet-dependent that a (100) exposed MAPbI3 surface is more stable than (001) surface. With these guidance, we successfully acquire the atomic structure of pristine MAPbI3 and identify the characterization window that is very narrow. These findings are helpful to guide future electron microscopy characterization of these beam-sensitive materials, which are also useful for finding strategies to improve the stability and the performance of the perovskite solar cells.

Hexagonal MASnI$_3$ exhibiting strong absorption of ultraviolet photons

Qiaoqiao Li; Wenhui Wan; Yanfeng Ge; Busheng Wang; Yingmei Li; Chuang Wang; Yong-Hong Zhao; Yong Liu. Appl. Phys. Lett. 114, 101906 (2019); (2019). Cited by 0. Found by: arxiv. DOI: 10.1063/1.5087649

Abstract: MASnI$_3$, an organometallic halide, has great potential in the field of lead-free perovskite solar cells. Ultraviolet photons have been shown to generate deep trapping electronic defects in mesoporous TiO$_2$-based perovskite, affecting its performance and stability. In this study, the structure, electronic properties, and optical properties of the cubic, tetragonal, and hexagonal phases of MASnI$_3$ were studied using first-principles calculations. The results indicate that the hexagonal phase of MASnI$_3$ possesses a larger indirect band gap and larger carrier effective mass along the \emph{c}-axis compared with the cubic and tetragonal phases. These findings were attributed to the enhanced electronic coupling and localization in the hexagonal phase. Moreover, the hexagonal phase exhibited high absorption of ultraviolet photons and high transmission of visible photons, particularly along the \emph{c}-axis. These characteristics demonstrate the potential of hexagonal MASnI$_3$ for application in multijunction perovskite tandem solar cells or as coatings in mesoporous TiO$_2$-based perovskite solar cells to enhance ultraviolet stability and photon utilization.

First-principles prediction into robust high-performance photovoltaic double perovskites A$_{2}$SiI$_{6}$ (A = K, Rb, Cs)

Qiaoqiao Li; Liujiang Zhou; Yanfeng Ge; Yulu Ren; Jiangshan Zhao; Wenhui Wan; Kaicheng Zhang; Yong Liu. arXiv preprint (2019). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/1908.02187v2

Abstract: Despite the exceeding 23\% photovoltaic efficiency achieved in organic-inorganic hybrid perovskite solar cells obtaining, the stable materials with desirable band gap are rare and are highly desired. With the aid of first-principles calculations, we predict a new promising family of nontoxic inorganic double perovskites (DPs), namely, silicon (Si)-based halides A$_{2}$SiI$_{6}$ (A = K, Rb, Cs; X = Cl, Br, I). This family containing the earth-abundant Si could be applied for perovskite solar cells (PSCs). Particularly A$_{2}$SiI$_{6}$ exhibits superb physical traits, including suitable band gaps of 0.84-1.15 eV, dispersive lower conduction bands, small carrier effective masses, wide photon absorption in the visible range. Importantly, the good stability at high temperature renders them as promising optical absorbers for solar cells.

Multiscale Models For Perovskite Optimisation

Philippe Baranek; James P. Connolly; Antoine Gissler; Philip Schulz; Michel Rérat; Roberto Dovesi. EUPVSEC 2025 42nd European Photovoltaic Solar Energy Conference and Exhibition, WIP Renewable Energies, Sep 2025, Bilbao, Spain (2025). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/2510.14396v1

Abstract: This paper presents a multiscale approach to evaluate perovskite solar cell performance which determines material properties at the atomistic scale with first-principles calculations, and applies them in macro-scale device models. This work focuses on the MAPbI3 (MA = CH3NH3) perovskite and how its phase transitions impact on its optical, electronic, and structural properties which are investigated at the first-principles level. The obtained data are coupled to a numerical drift-diffusion device model enabling evaluation of the performance of corresponding single junction devices. The first-principles simulation applies a hybrid exchange-correlation functional adapted to the studied family of compounds. Validation by available experimental data is presented from materials properties to device performance, justifying the use of the approach for predictive evaluation of existing and novel perovskites. The coupling between atomistic and device models is described in terms of a framework for exchange of optical, vibrational, and electronic parameters between the two scales. The result of this theoretical investigation is a methodology for designing and optimising perovskite materials for both cell performance and stability, the key obstacle in the societal implementation of these record-breaking new materials.

Ionically gated perovskite solar cell with tunable carbon nanotube interface at thick fullerene electron transporting layer: comparison to gated OPV

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Abstract: We demonstrate an ionically gated planar PS-PV solar cell with ultra-thick fullerene ETL with a porous CNT electron collector on top of it. Perovskite photovoltaic devices usually have undoped electron transport layers, usually thin like C60 due to its high resistance. Metallic low work function cathodes are extremely unstable in PS-PV due to reaction with halogens I-/Br-, and it would be desirable to have stable carbon cathodes on top of thick low resistance ETL for enhancing the stability of PS-PVs. We show that gating such top CNT cathode in ionic liquid, as part of a supercapacitor charged by Vg tunes the Fermi level of CNT by EDL charging, and causes lowering of a barrier at of C60/C70 ETL. Moreover, at higher gating voltage ions further propagates into fullerene by electrochemical n-doping, which increases dramatically PV performance by raising mostly two parameters: Isc and FF, resulting in PCE efficiency raised from 3 % to 11 %. N-doping of ETL strongly enhances charge collection by ETL and CNT raising Isc and lowering series resistance and thus increasing strongly PCE. Surprisingly Voc is not sensitive in PS-PV to external Vg gating, on the contrary, to strongly enhanced Voc in ionically gated organic PV, where it is the main gating effect. This insensitivity of Voc to lowering of the work function of Vg gated CNT electrode is a clear indication that Voc in PS-PV is determined by inner p-i-n junction formation in PS itself, via accumulation of its intrinsic mobile ionic species halogens and cations and their vacancies.

Molecular dynamics simulations of nucleation of hexagonal($ $) and cubic($ $)-FAPbI$_3$ perovskites from solution

Paramvir Ahlawat. arXiv preprint (2023). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/2306.14172v1

Abstract: Solar to power conversion certified efficiencies of formamidinium lead iodide based single-junction perovskite solar cell is now 26%, all perovskite-perovskite tandem $\sim$28%, and the perovskite-silicon tandem solar cell is $\sim$34% going beyond gallium arsenide solar cells. Therefore, it is now one of the most promising materials for generating cheaper sunlight-based electricity. This material has two commonly known polymorphs; one is a thermodynamically stable yellow hexagonal phase. The other one is a metastable black perovskite phase: a powerful photo-active material. Thousands of experiments have been performed to produce the highly crystalline photoactive phase of formamidinium lead iodide. Despite that, PSCs often suffer from poor reproducibility and stability. One of the root cause is the lack of control over their synthesis, i.e. crystallization process, where one of the main quests is to synthesize and stabilize corner-sharing defects-free photoactive perovskite form of pure or doped black perovskite form of formamidinium lead iodide and prevent the formation of hexagonal phases. Thus, for the rapid industrialization of perovskite based solar farms to combat global rising temperatures: it is all-important to understand the polymorph selective nucleation of formamidinium lead iodide from its precursors. Towards this ultimate goal, here we perform molecular simulations of the nucleation of formamidinium lead iodide from solution. This study aims to take the primary steps for the all-atoms simulations of the polymorph selective crystallization of halide perovskites.

Stable Perovskite Solar Cells via exfoliated graphite as an ion diffusion-blocking layer

Abdullah S. Alharbi; Miqad S. Albishi; Temur Maksudov; Tariq F. Alhuwaymel; Chrysa Aivalioti; Kadi S. AlShebl; Naif R. Alshamrani; Furkan H. Isikgor; Mubarak Aldosari; Majed M. Aljomah; Konstantinos Petridis; Thomas D. Anthopoulos; George Kakavelakis; Essa A. Alharbi. arXiv preprint (2024). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/2407.21662v1

Abstract: Ion and metal diffusion in metal halide perovskites, charge-transporting layers, and electrodes are detrimental to the performance and stability of perovskite-based photovoltaic devices. As a result, there is an intense research interest in developing novel defect and ion diffusion mitigation strategies. We present a simple, low-cost, scalable, and highly effective method that uses spray-coated exfoliated graphite interlayers to block ion and metal diffusion and humidity ingress within the perovskite, the hole transport material, and metal electrodes. The influence of inserting the exfoliated graphite films on the structural, surface morphology, and optoelectronic properties were examined through various methods, including X-ray diffraction, Time-of-Flight Secondary Ion Mass Spectrometry, Scanning electron microscope, atomic force microscopy, Current-voltage (J-V) characteristics, Transient photocurrent, and transient photovoltage. Our comprehensive investigation found that exfoliated graphite films reduced the I- and Li+ diffusion among the layers, leading to defect mitigation, reducing non-radiative recombination, and enhancing the device stability. Consequently, the best-performing device demonstrated a power conversion efficiency of 25% and a fill factor exceeding 80%. Additionally, these devices were subjected to different lifetime tests, which significantly enhanced the operational stability.

Generalised Framework for Controlling and Understanding Ion Dynamics with Passivated Lead Halide Perovskites

Tomi K. Baikie; Philip Calado; Krzysztof Galkowski; Zahra Andaji-Garmaroudi; Yi-Chun Chin; Joel Luke; Charlie Henderson; Tom Dunlop; James McGettrick; Ji-Seon Kim; Akshay Rao; Jenny Nelson; Samuel D. Stranks; Piers R. B. Barnes. arXiv preprint (2023). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/2305.14942v1

Abstract: Metal halide perovskite solar cells have gained widespread attention due to their high efficiency and high defect tolerance. The absorbing perovskite layer is as a mixed electron-ion conductor that supports high rates of ion and charge transport at room temperature, but the migration of mobile defects can lead to degradation pathways. We combine experimental observations and drift-diffusion modelling to demonstrate a new framework to interpret surface photovoltage (SPV) measurements in perovskite systems and mixed electronic ionic conductors more generally. We conclude that the SPV in mixed electronic ionic conductors can be understood in terms of the change in electric potential at the surface associated with changes in the net charge within the semiconductor system. We show that by modifying the interfaces of perovskite bilayers, we may control defect migration behaviour throughout the perovskite bulk. Our new framework for SPV has broad implications for developing strategies to improve the stability of perovskite devices by controlling defect accumulation at interfaces. More generally, in mixed electronic conductors our framework provides new insights into the behaviour of mobile defects and their interaction with photoinduced charges, which are foundational to physical mechanisms in memristivity, logic, impedance, sensors and energy storage.

Stabilizing Solution-Substrate Interaction of Perovskite Ink on PEDOT:PSS for Scalable Blade Coated Narrow Bandgap Perovskite Solar Modules by Gas Quenching

Severin Siegrist; Johnpaul K. Pious; Huagui Lai; Radha K. Kothandaraman; Jincheng Luo; Vitor Vlnieska; Ayodhya N. Tiwari; Fan Fu. arXiv preprint (2024). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/2406.06088v1

Abstract: The development of scalable 1.25 eV mixed Pb-Sn perovskite solar modules by blade coating lags behind Pb-based perovskites due to limited understanding of solution-substrate interaction of the perovskite ink on PEDOT:PSS and subsequent gas quenching. To address this challenge, we systematically studied the wet film deposition and quenching process to better understand narrow bandgap perovskite film formation on PEDOT:PSS. We found, the wetting of Pb-Sn perovskite ink on PEDOT:PSS is highly unstable over relevant coating time scales, causing the contact angles to decrease rapidly from 42° to 16° within seconds. This instability leads to localized irregularities in the wet film, resulting in uneven solvent extraction and inhomogeneous nuclei density. As a result, rough perovskite films with voids at the buried interface are obtained. To overcome this problem, we developed a quasi-static wetting process by reducing the blade coating speed, thereby stabilizing the wetting behavior of Pb-Sn perovskite precursor ink on PEDOT:PSS. This optimized process facilitates the deposition of high-quality, void-free Pb-Sn perovskite films with uniform thickness over 8 cm of coating length using moderate (1.4 bar) N2 quenching. We achieved 20 % efficient narrow bandgap perovskite solar cells and mini-modules with 15.8 % active area efficiency on 15.9 cm2.

Design of Lead-Free Inorganic Halide Perovskites for Solar Cells via Cation-Transmutation

Xin-Gang Zhao; Ji-Hui Yang; Yuhao Fu; Dongwen Yang; Qiaoling Xu; Liping Yu; Su-Huai Wei; Lijun Zhang. J. Am. Chem. Soc. 139, 2630 (2017) (2017). Cited by 0. Found by: arxiv. DOI: 10.1021/jacs.6b09645

Abstract: Hybrid organic-inorganic halide perovskites with the prototype material of $CH_{3}NH_{3}PbI_{3}$ have recently attracted intense interest as low-cost and high-performance photovoltaic absorbers. Despite the high power conversion efficiency exceeding 20% achieved by their solar cells, two key issues – the poor device stabilities associated with their intrinsic material instability and the toxicity due to water soluble Pb^{2+} – need to be resolved before large-scale commercialization. Here, we address these issues by exploiting the strategy of cation-transmutation to design stable inorganic Pb-free halide perovskites for solar cells. The idea is to convert two divalent Pb^{2+} ions into one monovalent M^{+} and one trivalent M^{3+} ions, forming a rich class of quaternary halides in double-perovskite structure. We find through first-principles calculations this class of materials have good phase stability against decomposition and wide-range tunable optoelectronic properties. With photovoltaic-functionality-directed materials screening, we identify eleven optimal materials with intrinsic thermodynamic stability, suitable band gaps, small carrier effective masses, and low excitons binding energies as promising candidates to replace Pb-based photovoltaic absorbers in perovskite solar cells. The chemical trends of phase stabilities and electronic properties are also established for this class of materials, offering useful guidance for the development of perovskite solar cells fabricated with them.

Unraveling the water degradation mechanism of $CH_3NH_3PbI_3$

Chao Zheng; Oleg Rubel. J. Phys. Chem. C 123, 19385-19394 (2019) (2019). Cited by 0. Found by: arxiv. DOI: 10.1021/acs.jpcc.9b05516

Abstract: Instability of perovskite photovoltaics is still a topic which is currently under intense debate, especially the role of water environment. Unraveling the mechanism of this instability is urgent to enable practical application of perovskite solar cells. Here, ab initio metadynamics is employed to investigate the initial phase of a dissolution process of CH$_3$NH$_3$PbI$_3$ (MAPbI$_3$) in explicit water. It is found that the initial dissolution of MAPbI$_3$ is a complex multi-step process triggered by the departure of I$^-$ ion from the CH$_3$NH$_3$I-terminated surface. Reconstruction of the free energy landscape indicates a low energy barrier for water dissolution of MAPbI$_3$. In addition, we propose a two-step thermodynamic cycle for MAPbI$_3$ dissolution in water at a finite concentration that renders a spontaneity of the dissolution process. The low energy barrier for the initial dissolution step and the spontaneous nature of MAPbI$_3$ dissolution in water explain why the water immediately destroys pristine MAPbI$_3$. The dissolution thermodynamics of all-inorganic CsPbI$_3$ perovskite is also analyzed for comparison. Hydration enthalpies and entropies of aqueous ions play an important role for the dissolution process. Our findings provide a comprehensive understanding to the current debate on water instability of MAPbI$_3$.

Physics-based material parameters extraction from perovskite experiments via Bayesian optimization

Hualin Zhan; Viqar Ahmad; Azul Mayon; Grace Tabi; Anh Dinh Bui; Zhuofeng Li; Daniel Walter; Hieu Nguyen; Klaus Weber; Thomas White; Kylie Catchpole. arXiv preprint (2024). Cited by 0. Found by: arxiv. DOI: 10.1039/D4EE00911H

Abstract: The ability to extract material parameters of perovskite from quantitative experimental analysis is essential for rational design of photovoltaic and optoelectronic applications. However, the difficulty of this analysis increases significantly with the complexity of the theoretical model and the number of material parameters for perovskite. Here we use Bayesian optimization to develop an analysis platform that can extract up to 8 fundamental material parameters of an organometallic perovskite semiconductor from a transient photoluminescence experiment, based on a complex full physics model that includes drift-diffusion of carriers and dynamic defect occupation. An example study of thermal degradation reveals that the carrier mobility and trap-assisted recombination coefficient are reduced noticeably, while the defect energy level remains nearly unchanged. The reduced carrier mobility can dominate the overall effect on thermal degradation of perovskite solar cells by reducing the fill factor, despite the opposite effect of the reduced trap-assisted recombination coefficient on increasing the fill factor. In future, this platform can be conveniently applied to other experiments or to combinations of experiments, accelerating materials discovery and optimization of semiconductor materials for photovoltaics and other applications.

Influence of water intercalation and hydration on chemical decomposition and ion transport in methylammonium lead halide perovskites

Un-Gi Jong; Chol-Jun Yu; Gum-Chol Ri; Andrew P. McMahon; Nicholas M. Harrison; Piers R. F. Barnes; Aron Walsh. arXiv preprint (2017). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/1708.07608v3

Abstract: The use of methylammonium (MA) lead halide perovskites \ce{CH3NH3PbX3} (X=I, Br, Cl) in perovskite solar cells (PSCs) has made great progress in performance efficiency during recent years. However, the rapid decomposition of \ce{MAPbI3} in humid environments hinders outdoor application of PSCs, and thus, a comprehensive understanding of the degradation mechanism is required. To do this, we investigate the effect of water intercalation and hydration of the decomposition and ion migration of \ce{CH3NH3PbX3} using first-principles calculations. We find that water interacts with \ce{PbX6} and MA through hydrogen bonding, and the former interaction enhances gradually, while the latter hardly changes when going from X=I to Br and to Cl. Thermodynamic calculations indicate that water exothermically intercalates into the perovskite, while the water intercalated and monohydrated compounds are stable with respect to decomposition. More importantly, the water intercalation greatly reduces the activation energies for vacancy-mediated ion migration, which become higher going from X=I to Br and to Cl. Our work indicates that hydration of halide perovskites must be avoided to prevent the degradation of PSCs upon moisture exposure.

Terahertz Time-Domain Spectroscopy as a Universal Defect Fingerprinting Tool for Organic Halide Perovskite Solar Cells

Inhee Maeng; Young Mi Lee; Jinwoo Park; Seung Jae Oh; Min-Cherl Jung. arXiv preprint (2026). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/2607.15538v1

Abstract: Organic-inorganic hybrid perovskites (OHPs) deliver certified single-junction power conversion efficiencies (PCEs) exceeding 26% and perovskite-silicon tandem values surpassing 34%, yet a substantial gap with the Shockley-Queisser (S-Q) limit persists-primarily due to grain-boundary (GB) defects that drive non-radiative recombination, ion migration, and degradation. Rational passivation demands a non-contact tool capable of

identifying and quantifying specific defect species in device-relevant thin films, a capability absent from conventional probes. This short review demonstrates that terahertz time-domain spectroscopy (THz-TDS, 0.3-3.0 THz) fulfills this role. Across four OHP compositions fabricated by sequential vacuum evaporation (SVE)-MAPbI3, MAPbBr3, FAPbI3, and CsPbI3-the THz spectral window captures both intrinsic phonon modes and GB-localized molecular defect vibrations, enabling species-specific, quantitative characterization at room temperature. Notably, the oscillator strength of the SVE-specific 1.58 THz absorption in MAPbI3 scales linearly with XPS-quantified CH3NH2 defect concentration, establishing THz-TDS as a direct, non-destructive defect meter. Building on these findings, we propose a three-pillar framework for THz-guided defect engineering: (I) quantitative defect measurement via oscillator-strength analysis, (II) material-specific fingerprint identification from a systematically constructed THz library, and (III) fingerprint-guided defect elimination with real-time feedback-together defining a closed-loop quality-control cycle that connects spectroscopic diagnosis to passivation strategy and, ultimately, to enhanced solar-cell efficiency.

Post-Transition Metal Sn-Based Chalcogenide Perovskites: A Promising Lead-Free and Transition Metal Alternative for Stable, High-Performance Photovoltaics

Surajit Adhikari; Sankhasuvra Das; Priya Johari. arXiv preprint (2024). Cited by 0. Found by: arxiv. DOI: 10.1039/D4TC04701J

Abstract: Chalcogenide perovskites (CPs) have emerged as promising materials for optoelectronic applications due to their stability, non-toxicity, small bandgaps, high absorption coefficients, and defect tolerance. Although transition metal-based CPs, particularly those incorporating Zr and Hf, have been well-studied, they often exhibit higher bandgaps, lower charge carrier mobility, and reduced efficiency compared to lead-based halide perovskites (HPs). Tin (Sn), a post-transition metal with a similar oxidation state (+4) as Zr and Hf in ABX$_{3}$ structures but with different valence characteristics, remains underexplored in CPs. Given the influence of valence states on material properties, Sn-based CPs are of great interest. This study employs density functional theory (DFT), density functional perturbation theory (DFPT), and many-body perturbation theory (GW and BSE) to investigate a series of distorted Sn-based CPs (ASnX$_{3}$, A = Ca, Sr, Ba; X = S, Se). Our results demonstrate that these perovskites are mechanically stable and exhibit lower direct G$_{0}$W$_{0}$ bandgaps (0.79-1.50 eV) compared to their Zr- and Hf-based counterparts. Analysis of carrier-phonon interactions reveals that the charge-separated polaronic state is less stable than the bound exciton state in these materials. Additionally, polaron-assisted charge carrier mobilities for electrons (21.33-416.02 $cm^{2}V^{-1}s^{-1}$) and holes (7.02-260.69 $cm^{2}V^{-1}s^{-1}$) are comparable to or higher than those in lead-based HPs and significantly exceed those of Zr- and Hf-based CPs, owing to reduced carrier-phonon coupling. The estimated spectroscopic limited maximum efficiency (24.2%-31.2%)-confirmed through perovskite solar cell (PSC) simulations using SCAPS-1D software-indicates that these materials are promising candidates for photovoltaic applications.

Light Intensity Modulated Impedance Spectroscopy (LIMIS) in All-Solid-State Solar Cells at Open Circuit

Osbel Almora; Yicheng Zhao; Xiaoyan Du; Thomas Heumueller; Gebhard J. Matt; Germà Garcia-Belmonte; Christoph J. Brabec. arXiv preprint (2019). Cited by 0. Found by: arxiv. DOI: 10.1016/j.nanoen.2020.104982

Abstract: Potentiostatic impedance spectroscopy (IS) is a well stablished characterization technique for elucidating the electric resistivity and capacitive features of materials and devices. In the case of solar cells, by applying a small voltage perturbation the current signal is recorded and the recombination processes and defect distributions are among the typical outcomes in IS studies. In this work a photo-impedance approach, named light intensity modulated impedance spectroscopy (LIMIS), is first tested in all-solid-state photovoltaic cells by recording the individual photocurrent (IMPS) and photovoltage (IMVS) responsivity signals due to a small light perturbation at open-circuit (OC), and combining them: LIMIS=IMVS/IMPS. The experimental LIMIS spectra from silicon, organic, and perovskite solar cells are presented and compared with IS. An analysis of the equivalent circuit numerical models for total resistive and capacitive features is discussed. Our theoretical findings show a correction to the lifetimes evaluations by obtaining the total differential resistances and capacitances combining IS and LIMIS measurements. This correction addresses the discrepancies among different techniques, as shown with transient photovoltage. The experimental differences between IS and LIMIS (i) proves the unviability of the superposition principle, (ii) suggest a bias-dependent photo-current correction to the empirical Shockley equation of the steady-state current at different illumination intensities around OC and (iii) are proposed as a potential figure of merit for characterizing performance and stability of solar cells. In addition, new features are reported for the low-frequency capacitance of perovskite solar cells, measured by IS and LIMIS.

Performance Analysis of Double Perovskite-Based Solar Cells Using SCAPS-1D Simulation: A brief review

H. Laltlanmawii; Lalrem Kima; Mahabur Rahman; Md. Ferdous Rahman; Dilshod Nematov; S. Bhattarai; C. V. M. Chaturvedi; Yazen M. Alawaideh; A. Laref; D. P. Rai. arXiv preprint (2026). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/2608.04736v1

Abstract: Lead-free double perovskites are among the rapidly developing next-generation solar cell technologies, providing the required low toxicity, stability, as well as high optoelectronic potential. So far, experimentally prepared lead-free perovskite solar cell devices are reported to have low power conversion efficiency (PCE) for practical application as compared to the lead-based perovskites. In recent years, numerical simulations have emerged as a cost-effective approach that plays a crucial role in expediting scientific research, can bridge the gap between experiment and theory, and provide predictive information regarding the preparation of solar cells and their PCEs without undergoing real-time experiments. The tools, such as 1D numerical simulation software SCAPS-1D, are now needed to test newer architectures and determine what exactly is holding them back. So far in the field of solar cell research, SCAPS-1D has been extensively used and looks like a powerful software due to its user-friendliness and simulation of results in a few seconds. The speed and ease of simulation make SCAPS-1D a very popular tool; as a result, it enables rapid optimization of a large number of photovoltaic devices and their performances without undergoing any experimental work, which can save time and money. However, one serious drawback is that the SCAPS-1D simulator works only for 1D configurations. It is ineffective in incorporating atomistic interactions and 3D effects. Hence, the efficacy of the SCAPS-1D simulator solely relies on the accuracy of the input parameters that the user provides, failing which may give wrong results and large deviations from accuracy.

Intrinsic Instability of the Hybrid Halide Perovskite Semiconductor CH3NH3PbI3

Yue-Yu Zhang; Shiyou Chen; Peng Xu; Hongjun Xiang; Xin-Gao Gong; Aron Walsh; Su-Huai Wei. Chin. Phys. Lett. 35, 036104 (2018) (2015). Cited by 0. Found by: arxiv. DOI: 10.1088/0256-307X/35/3/036104

Abstract: The organic-inorganic hybrid perovskite CH3NH3PbI3 has attracted significant interest for its high performance in converting solar light into electrical power with an efficiency exceeding 20%. Unfortunately, chemical stability is one major challenge in the development of the CH3NH3PbI3 solar cells. It was commonly assumed that moisture or oxygen in the environment causes the poor stability of hybrid halide perovskites, however, here we show from the first-principles calculations that the room-temperature tetragonal phase of CH3NH3PbI3 is thermodynamically unstable with respect to the phase separation into CH3NH3I + PbI2, i.e., the disproportionation is exothermic, independent of the humidity or oxygen in the atmosphere. When the structure is distorted to the low-temperature orthorhombic phase, the energetic cost of separation increases, but remains small. Contributions from vibrational and configurational entropy at room temperature have been considered, but the instability of CH3NH3PbI3 is unchanged. When I is replaced by Br or Cl, Pb by Sn, or the organic cation CH3NH3 by inorganic Cs, the perovskites become more stable and do not phase-separate spontaneously. Our study highlights that the poor chemical stability is intrinsic to CH3NH3PbI3 and suggests that element-substitution may solve the chemical stability problem in hybrid halide perovskite solar cells.

Effect of heterostructure engineering on electronic structure and transport properties of two-dimensional halide perovskites

Rahul Singh; Prashant Singh; Ganesh Balasubramania. Computational Materials Science, Volume 200, December 2021, 110823 (2021). Cited by 0. Found by: arxiv. DOI: 10.1016/j.commatsci.2021.110823

Abstract: Organic-inorganic halide perovskite solar cells have attracted much attention due to their low-cost fabrication, flexibility, and high-power conversion efficiency. More recent efforts show that the reduction from three- to two-dimensions (2D) of organic-inorganic halide perovskites promises an exciting opportunity to tune their electronic properties. Here, we explore the effect of reduced dimensionality and heterostructure engineering on the intrinsic material properties, such as energy stability, bandgap and transport properties of 2D hybrid organic-inorganic halide perovskites using first-principles density functional theory. We show that the energetic stability is significantly enhanced by engineered perovskite heterostructures that also possess excellent transport properties similar to their bulk counterparts. These layered chemistries also demonstrate the advantage of a broad range of tunable bandgaps and high-absorption coefficient in the visible spectrum. The proposed 2D heterostructured material holds potential for nano-optoelectronic devices as well as for effective photovoltaics.

Molecular Bond Engineering and Feature Learning for the Design of Hybrid Organic-Inorganic Perovskites Solar Cells with Strong Non-Covalent Halogen-Cation Interactions

Johannes Teunissen; Fabiana da Pieve. arXiv preprint (2021). Cited by 0. Found by: arxiv. DOI: 10.1021/acs.jpcc.1c07295

Abstract: Hybrid organic-inorganic perovskites are exceedingly interesting candidates for new solar energy technologies, for both ground-based and space applications. However, their large-scale production is hampered by the lack of long-term stability, mostly associated to ion migration. The specific role of non-covalent bonds in contributing to the stability remains elusive, and in certain cases controversial. Here, we perform an investigation on a large perovskite chemical space via a combination of first-principles calculations for the bond strengths and the recently developed Sure Independent Screening and Sparsifying Operator (SISSO) algorithm. The latter is used to formulate mathematical descriptors that, by highlighting the importance of specific non-covalent molecular bonds, can guide the design of perovskites with suppressed ion migration. The results unveil the distinct nature of different non-covalent interactions, with remarkable differences compared to previous arguments and interpretations in the literature on the basis of smaller chemical spaces. In particular, we clarify the origin of the higher stability offered by FA compared to MA, which shows to be different from previous arguments in the literature, the reasons of the improved stability given by the halogen F, and explain the exceptional case of overall stronger bonds for Guanidiunium. The found descriptors reveal the criteria that, within the stability boundaries given by the Goldschmidt factor, give an all-in-one picture of non-covalent interactions which provide the more stable configurations, also including interactions other than H-bonds. Such descriptors are more informative than previously used quantities and can be used as universal input to better inform new machine learning studies.

Inverted Perovskite Photovoltaics using Flame Spray Pyrolysis Solution based CuAlO2/Cu-O Hole Selective Contact

Achilleas Savva; Ioannis T. Papadas; Dimitris Tsikritzis; Apostolos Ioakeimidis; Fedros Galatopoulos; Konstantinos Kapnisis; Roland Fuhrer; Benjamin Hartmeier; Marek F. Oszajca; Norman A. Luechinger; Stella Kennou; Gerasimos Armatas; Stelios A. Choulis. ACS Applied Energy Materials,2,3,2276-2287,2019 (2019). Cited by 0. Found by: arxiv. DOI: 10.1021/acsaem.9b00070

Abstract: We present the functionalization process of a conductive and transparent copper aluminum oxide, copper oxide alloy. The copper aluminum oxide, copper oxide powders were developed by flame spray pyrolysis and their stabilized dispersions were treated by sonication and centrifugation methods. We show that when the supernatant part of the treated copper aluminum oxide, copper oxide dispersions is used for the development of hole transporting layers the corresponding inverted perovskite solar cells show improved functionality and power conversion efficiency with negligible hysteresis effect.

Unraveling the varied nature and roles of defects in hybrid halide perovskites with time-resolved photoemission electron microscopy

Sofiia Kosar; Andrew J. Winchester; Tiarnan A. S. Doherty; Stuart Macpherson; Christopher E. Petoukhoff; Kyle Frohna; Miguel Anaya; Nicholas S. Chan; Julien Madéo; Michael K. L. Man; Samuel D. Stranks; Keshav M. Dani. arXiv preprint (2021). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/2107.12572v1

Abstract: With rapidly growing photoconversion efficiencies, hybrid perovskite solar cells have emerged as promising contenders for next generation, low-cost photovoltaic technologies. Yet, the presence of nanoscale defect clusters, that form during the fabrication process, remains critical to overall device operation, including efficiency and long-term stability. To successfully deploy hybrid perovskites, we must understand the nature of the different types of defects, assess their potentially varied roles in device performance, and understand how they respond to passivation strategies. Here, by correlating photoemission and synchrotron-based scanning probe X-ray microscopies, we unveil three different types of defect clusters in state-of-the-art triple cation mixed halide perovskite thin films. Incorporating ultrafast time-resolution into our photoemission measurements, we show that defect clusters originating at grain boundaries are the most detrimental for photocarrier trapping, while lead iodide defect clusters are relatively benign. Hexagonal polytype defect clusters are only mildly detrimental individually, but can have a significant impact overall if abundant in occurrence. We also show that passivating defects with oxygen in the presence of light, a previously used approach to improve efficiency, has a varied impact on the different types of defects. Even with just mild oxygen treatment, the grain boundary defects are completely healed, while the lead iodide defects begin to show signs of chemical alteration. Our findings highlight the need for multi-pronged strategies tailored to selectively address the detrimental impact of the different defect types in hybrid perovskite solar cells.

More than just light management – The multiple advantages of nano- and micro-textures in perovskite solar cells

Guillermo Martínez-Denegri; Christiane Becker. arXiv preprint (2026). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/2605.30182v2

Abstract: Perovskite-based solar cells have undergone rapid improvements over the last decade enabling highest power conversion efficiencies of single-junction and multi-junction devices. The implementation of nano- or micro-textures has played a major role in this development due to their ability to minimize reflection losses, maximize light trapping, and hence increase light harvesting in the active layer. In the slipstream of these advances, it has become apparent that nano- and micro-textures can also have many other collateral benefits beyond light management. In this article, very recently reported texture-related benefits for perovskite-based solar cells different from light management are reviewed. These are namely (1) improved film wetting from perovskite-solution, (2) enhanced perovskite crystallinity in terms of grain size, crystal orientation and phase homogeneity, (3) enhanced carrier extraction leading to higher open-circuit voltages, and potentially (4) increased mechanical stability upon bending in flexible devices and reduced residual stress. This survey helps to understand textures as a holistic concept to further improve perovskite-based solar cells, photodetectors and light emitting diodes.

Lattice matched heterogeneous nucleation eliminate defective buried interface in halide perovskites

Paramvir Ahlawat; Cecilia Clementi; Felix Musil; Maria-Andreea Filip. arXiv preprint (2024). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/2405.11599v1

Abstract: Metal halide perovskite-based semi-conducting hetero-structures have emerged as promising electronics for solar cells, light-emitting diodes, detectors, and photo-catalysts. Perovskites' efficiency, electronic properties and their long-term stability directly depend on their morphology [1-24]. Therefore, to manufacture stable and higher efficiency perovskite solar cells and electronics, it is now crucial to understand their microstructure evolution. In this study, we perform molecular dynamics simulations to investigate the formation of cesium lead bromide perovskite on interfaces. Our simulations reveal that perovskite crystallizes in a heteroepitaxial manner on widely employed oxide interfaces. This could introduce the formation of dislocations, voids and defects in the buried interface, and grain boundaries in the bulk crystal. From simulations, we find that lattice-matched interfaces could enable epitaxial ordered growth of perovskites and may prevent defect formation in the buried interface.

Enhanced Electron Extraction in Co-Doped TiO2 Quantified by Drift-Diffusion Simulation for Stable CsPbI3 Solar Cells

Thomas W. Gries; Davide Regaldo; Hans Koebler; Titan Noor Hartono Putri; Gennaro V. Sannino; Emilio Gutierrez Partida; Roberto Felix; Elif Huesam; Ahmed Saleh; Regan G. Wilks; Zafar Iqbal; Zahra Loghman Nia; Florian Ruske; Martin Stolterfoht; Dieter Neher; Marcus Baer; Stefan A. Weber; Paola Delli Veneri; Philip Schulz; Jean-Baptiste Puel; Jean-Paul Kleider; Qiong Wang; Eva Unger; Artem Musiienko; Antonio Abate. Small Sci. 2025, 5, 2400578 (2024). Cited by 0. Found by: arxiv. DOI: 10.1002/smsc.202400578

Abstract: Solar cells based on inorganic perovskite CsPbI3 are promising candidates to resolve the challenge of operational stability in the field of perovskite photovoltaics. For stable operation, however, it is crucial to thoroughly understand the extractive and recombinative processes occurring at the interfaces of perovskite and the charge-selective layers. In this study, we focus on the electronic properties of (doped) TiO2 as an electron-selective contact. We show via KPFM that co-doping of TiO2 with Nb(V) and Sn(IV) reduces the materials work function by 270 meV, giving it stronger n-type characteristics compared to Nb(V) mono-doped TiO2. The altered electronic alignment with CsPbI3 translates to enhanced electron extraction, as demonstrated with ssPL, trPL and trSPV in triad. Importantly, we extract crucial parameters, such as the concentration of extracted electrons and the interface hole recombination velocity, from the SPV transients via 2D drift-diffusion simulations. When implementing the co-doped TiO2 into full n-i-p solar cells, the operational stability is enhanced to 32000 h of projected TS80 lifetime. This study provides fundamental understanding of interfacial charge extraction and its correlation with operational stability of perovskite solar cells, which can be transferred to other charge-selective contacts.

Resolving the Hydrophobicity of Me-4PACz Hole Transport Layer for High-Efficiency Inverted Perovskite Solar Cells

Kashimul Hossain; Ashish Kulkarni; Urvashi Bothra; Benjamin Klingebiel; Thomas Kirchartz; Michael Saliba; Dinesh Kabra. arXiv preprint (2023). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/2304.13788v1

Abstract: [4-(3,6-dimethyl-9H-carbazole-9-yl)butyl]phosphonic acid (Me-4PACz) self-assembled monolayer (SAM) has been employed in perovskite single junction and tandem devices demonstrating high efficiencies. However, a uniform perovskite layer does not form due to the hydrophobicity of Me-4PACz. Here, we tackle this challenge by adding a conjugated polyelectrolyte poly(9,9-bis(3'-(N,N-dimethyl)-N-ethylammonium-propyl-2,7-fluorene)-alt-2,7-(9,9dioctylfluorene)dibromide (PFN-Br) to the Me-4PACz in a specific ratio, defines as Pz:PFN. With this mixing engineering strategy of Pz:PFN, the PFN-Br interacts with the A-site cation and is confirmed via solution-state nuclear magnetic resonance studies. The narrow full width at half maximum (FWHM) of diffraction peaks of perovskite film revealed improved crystallization on the optimal mixing ratio of Pz:PFN. Interestingly, the mixing of PFN-Br additionally tunes the work function of the Me-4PACz as revealed by the Kelvin probe force microscopy and built-in-voltage estimation in solar cells. Devices employing optimized Pz:PFN mixing ratio deliver open-circuit voltage (Voc)of 1.16 V and efficiency >20% for perovskites with a bandgap of 1.6 eV with high reproducibility and concomitant stability. Considering significant research on Me-4PACz SAM, our work highlights the importance of obtaining a uniform perovskite layer with improved yield and performance.

First-principles study on the chemical decomposition of inorganic perovskites \ce{CsPbI3} and \ce{RbPbI3} at finite temperature and pressure

Un-Gi Jong; Chol-Jun Yu; Yun-Hyok Kye; Chol-Ho Kim; Son-Guk Ri; Yue Chen. arXiv preprint (2018). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/1807.02277v1

Abstract: Inorganic halide perovskite \ce{Cs(Rb)PbI3} has attracted significant research interest in the application of light-absorbing material of perovskite solar cells (PSCs). Although there have been extensive studies on structural and electronic properties of inorganic halide perovskites, the investigation on their thermodynamic stability is lack. Thus, we investigate the effect of substituting Rb for Cs in \ce{CsPbI3} on the chemical decomposition and thermodynamic stability using first-principles thermodynamics. By calculating the formation energies of solid solutions \ce{Cs$_{1-x}$Rb$_x$PbI3} from their ingredients \ce{Cs$_{1-x}$Rb$_x$I} and \ce{PbI2}, we find that the best match between efficiency and stability can be achieved at the Rb content $x\approx$ 0.7. The calculated Helmholtz free energy of solid solutions indicates that \ce{Cs$_{1-x}$Rb$_x$PbI3} has a good thermodynamic stability at room temperature due to a good miscibility of \ce{CsPbI3} and \ce{RbPbI3}. Through lattice-dynamics calculations, we further highlight that \ce{RbPbI3} never stabilize in cubic phase at any temperature and pressure due to the chemical decomposition into its ingredients \ce{RbI} and \ce{PbI2}, while \ce{CsPbI3} can be stabilized in the cubic phase at the temperature range of 0$-$600 K and the pressure range of 0$-$4 GPa. Our work reasonably explains the experimental observations, and paves the way for understanding material stability of the inorganic halide perovskites and designing efficient inorganic halide PSCs.

Compound Defects in Halide Perovskites: A First-Principles Study of CsPbI$_3$

Haibo Xue; José Manuel Vicent-Luna; Shuxia Tao; Geert Brocks. arXiv preprint (2022). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/2209.03484v1

Abstract: Lattice defects affect the long-term stability of halide perovskite solar cells. Whereas simple point defects, i.e., atomic interstitials and vacancies, have been studied in great detail, here we focus on compound defects that are more likely to form under crystal growth conditions, such as compound vacancies or interstitials, and antisites. We identify the most prominent defects in the archetype inorganic perovskite CsPbI$_3$, through first-principles density functional theory (DFT) calculations. We find that under equilibrium conditions at room temperature, the antisite of Pb substituting Cs forms in a concentration comparable to those of the most prominent point defects, whereas the other compound defects are negligible. However, under nonequilibrium thermal and operating conditions, other complexes also become as important as the point defects. Those are the Cs substituting Pb antisite, and, to a lesser extent, the compound vacancies of PbI$_2$ or CsPbI$_3$ units, and the I substituting Cs antisite. These compound defects only lead to shallow or inactive charge carrier traps, which testifies to the electronic stability of the halide perovskites. Under operating conditions with a quasi Fermi level very close to the valence band, deeper traps can develop.

Quantum optoelectronics in semiconductor solar cell materials and devices

Xi Liu; Wenxi Fang. arXiv preprint (2026). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/2608.04328v2

Abstract: We analyze the integration of quantum optical phenomena, such as cavity quantum electrodynamics (CQED), Fabry Perot resonances, and strong light-matter coupling, into the design and engineering of next generation photovoltaic systems. We examine how these phenomena can be harnessed through photonic structures including optical cavities, plasmonic materials, and metasurfaces to improve light trapping, absorption, and carrier dynamics in future solar cell devices. Specific focus is given to semiconductor materials such as perovskites, organics, transition metal dichalcogenides (TMD), cadmium telluride (CdTe), and silicon. For perovskite solar cells, we analyze device architectures, interfacial engineering with hyperbranched polymers, and additive optimization using molecular dopants and nanosheets to enhance film morphology and stability. We further examine laser-based metrology for thin-film characterization and coherent spectroscopy techniques involving frequency combs and high-harmonic generation. The paper also shows how machine learning (ML), combined with density functional theory (DFT), accelerates material screening and performance prediction for next-generation solar cell absorbers. These developments demonstrate how quantum optoelectronic design principles are transforming photovoltaic research and enabling higher efficiency, stability, and functionality in solar energy devices.

Recent advances in La2NiMnO6 Double Perovskites for various applications; Challenges and opportunities

Suresh Chandra Baral; P. Maneesha; E. G. Rini; Somaditya Sen. arXiv preprint (2023). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/2304.06951v1

Abstract: Double perovskites R2NiMnO6 (R= Rare earth element) (RNMO) are a significant class of materials owing to their Multifunctional properties with structural modifications. In particular, multifunctional double perovskite oxides La2NiMnO6 (LNMO) which possess both electric and magnetic orderings, chemical flexibility, versatility, and indispensable properties like high ferromagnetic curie temperature, high absorption rates, dielectrics, etc. have drawn a lot of attention due their rich physics and diverse applications in various technology. This justifies the intense research in this class of materials, and the keen interest they are subject to both the fundamental and practical side. In view of the demands of this material in lead-free perovskite solar cells, photocatalytic degradation of organic dyes, clean hydrogen production, electric tuneable devices, Fuel cells, gas sensing, and Biomedical applications, there is a need for an overview of all the literature so far, the ongoing research and the future prospective. This review summarised all the Physical and Structural Properties of LNMO such as electric, magnetic, catalytic, and dielectric properties with their underlying mechanisms. This review article provides insight into the scope of studies in LNMO material for exploring unexposed properties in new material research and to identify areas of future investigation of the materials in the double perovskite family.

Interface Engineering in Hybrid Iodide CH3NH3PbI3 Perovskite Using Lewis Base and Graphene towards High Performance Solar Cells

Chol-Jun Yu; Yun-Hyok Kye; Un-Gi Jong; Song-Guk Ko; Kum-Chol Ri; Song-Hyok Choe; Jin-Song Kim; Gwon-Il Ryu; Byol Kim. arXiv preprint (2019). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/1908.05237v2

Abstract: Perovskite solar cells have achieved a substantial breakthrough via advanced interface engineerings. Reports have emphasized that combining the hybrid perovskites with Lewis base and graphene improve the performance; the underlying mechanisms are not yet fully understood. Here, using density functional theory, we show that upon the formation of CH3NH3PbI3 interfaces with three different Lewis base molecules and graphene, the binding strength with S-donors thiocarbamide and thioacetamide is higher than with O-donor dimethyl sulfoxide, while the interface dipole and work function reduction tend to increase from S-donors to O-donor. Furthermore, we provide evidences of deep trap states elimination in the S-donor perovskite interfaces through the analysis of defect formation on CH3NH3PbI3(110) surface, and of stability enhancement by estimating activation barriers for iodine atom migrations. These theoretical predictions are in line with the experimental observation of performance enhancement in the perovskites prepared using thiocarbamide.

Ultra-stable 2D/3D hybrid perovskite photovoltaic module

Giulia Grancini; Cristina Roldán-Carmona; Iwan Zimmermann; David Martineau; Stéphanie Narbey; Frédéric Oswald; Mohammad Khaja Nazeeruddin. arXiv preprint (2016). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/1609.09846v1

Abstract: Hybrid perovskite solar cells, with their power conversion efficiency (PCE) exceeding 22%, have been representing a revolutionary concept for future energy power generation. Although listed among the Top 10 Emerging Technologies of 2016, device longevity is the actual bottleneck for their real uptake in the market. Here we design an ultra-stable molecular junction of two/three dimensional (2D/3D) perovskites. It consists of a 2D (HOOC(CH2)2NH3)2PbI4, anchored at the oxide substrate, that templates the growth of a highly ordered 3D CH3NH3PbI3 perovskite stabilizing in the orthorhombic phase, even at room temperature. The unique and exceptional 2D/3D structure yields 14.6% PCE in solar cells with Spiro-OMeTAD and Au, and 12.9% PCE in hole-conductor free architecture. Aiming at the up-scaling of this technology, we realize 10x10 cm2 large-area photovoltaic modules by a low-cost, fully printable, industrial-scale process delivering 11.2% PCE. We demonstrate a record stability in the PCE of 5,000 hours, setting the direction for the new generation of carbon free energy.

Comprehensive numerical analysis of doping controlled efficiency in lead free Cs(SnGe)I3 perovskites solar cell

Nazmul Hasan; M. Hussayeen Khan Anik; Mohammed Mehedi Hasan; Sharnali Islam; Alamgir Kabir. arXiv preprint (2025). Cited by 0. Found by: arxiv. DOI: 10.1007/s00339-024-08125-y

Abstract: One effective way to prevent toxicity and improve the stability of materials for photovoltaic applications is to exclude lead and organic molecules from perovskite materials. Specifically, the CsSn1-xGexI3 appears to be a promising contender; nonetheless, it requires optimization, particularly bandgap tuning by doping concentration modifications. In this study, density functional theory (DFT) was employed to comprehensively analyze the electronic properties of CsSn1-xGexI3 that influenced light-matter interactions tuning of the perovskite materials by varying composition in B site atoms. We use the solar cell capacitance (SCAPS-1D) simulator to compute device performance; however, it computes the absorption spectrum using a simplified mathematical function that approximates the actual spectrum. To achieve a quantum-mechanical level of accuracy DFT extracted parameters like absorption spectra and bandgap were fed into SCAPS-1D. We find that increasing the Ge concentration leads to a higher bandgap and improved absorption profile, thereby enhancing solar energy conversion efficiency. Thermal and field distribution analyses were also done for the optimized device through a finite-difference time-domain (FDTD) framework. By optimizing the absorber layer with a 75% Ge concentration, we achieve a remarkable PCE of 23.80%. Our findings guide future research in designing high-performance non-leaded halide PSCs, paving the way for low-cost, stable, and highly efficient solar cells through atomic doping-tuned perovskite absorber layers.

Methylamine Vapor Exposure for Improved Morphology and Stability of Cesium-Methylammonium Lead Halide Perovskite Thin-Films

Akash Singh; Arun Singh Chouhan; Sushobhan Avasthi. arXiv preprint (2017). Cited by 0. Found by: arxiv. DOI: 10.1007/978-3-319-97604-4_60

Abstract: Mixed-cation Cesium-Methylammonium lead halide perovskite (CsxMA1-xPbI3-xBrx) thin-films have been used to demonstrate stable and efficient perovskite devices. However, a systematic study of the Cs incorporation on the properties of the perovskite films has not been reported. In this report, Impact of Cesium incorporation on the minority carrier recombination lifetime of Cesium-Methylammonium lead halide perovskite thin-films is studied. The lifetime for the as-deposited perovskite films decreases with increasing concentration of cesium. However, mixed cation perovskite film is more stable, showing higher lifetime (15-20 micro seconds) after 9 hours of ambient exposure than just after deposition (6-13 micro seconds). Methylamine Vapor Exposure (MVE) technique was used to improve the morphology of the as-deposited film. MVE treated films are more oriented along (110) direction and were even more stable in ambient, with Cs0.10MA0.90PbI2.90Br0.10 films showing lifetime of almost 50 micro seconds after 9 hours of ambient exposure, twice the lifetime of a comparable MAPbI3 film. These results throw light on why mixed-cation cesium-methylamine lead halide perovskite films are better for highly efficient and stable perovskite solar cells.

A Review on Improving PSC Performance through Charge Carrier Management: Where We Stand and What's Next?

Dilshod Nematov. Next Energy, 2026, 13 (2025). Cited by 0. Found by: arxiv. DOI: 10.1016/j.nxener.2026.100763

Abstract: Perovskite solar cells (PSCs) represent a breakthrough in photovoltaic technology, combining high power conversion efficiencies (PCEs), ease of fabrication, and tunable optoelectronic properties. However, their commercial viability is limited by critical issues such as charge carrier recombination, interfacial defects, instability under environmental stress, and toxicity of lead-based components. This review systematically examines recent advancements in charge carrier management strategies aimed at overcoming these limitations. Initially, fundamental mechanisms governing carrier generation, separation, transport, and recombination are outlined to provide a clear foundation. The study then delves into an in-depth analysis of carrier lifetime and mobility, evaluating recent methodologies for their enhancement through compositional engineering and structural optimization. Subsequently, trap state passivation techniques and interface engineering approaches are reviewed, with a particular focus on their impact on device stability and efficiency. The review also discusses long-term stability strategies and emerging trends in lead-free and scalable PSC technologies. In this work, recent strategies for charge carrier management are systematically categorized, comparative analyses are provided and synergistic solutions with high potential for real-world implementation are highlighted. By synthesizing data and perspectives from over thirty recent studies, this article offers a comprehensive roadmap for researchers seeking to optimize PSC performance and accelerate their transition toward commercial application.

Organic-inorganic Copper(II)-based Material: a Low-Toxic, Highly Stable Light Absorber beyond Organolead Perovskites

Xiaolei Li; Xiangli Zhong; Yue Hu; Bochao Li; Yusong Sheng; Yang Zhang; Chao Weng; Ming Feng; Hongwei Han; Jinbin Wang. arXiv preprint (2017). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/1703.02401v2

Abstract: Lead halide perovskite solar cells have recently emerged as a very promising photovoltaic technology due to their excellent power conversion efficiencies; however, the toxicity of lead and the poor stability of perovskite materials remain two main challenges that need to be addressed. Here, for the first time, we report a lead-free, highly stable C6H4NH2CuBr2I compound. The C6H4NH2CuBr2I films exhibit extraordinary hydrophobic behavior with a contact angle of approximately 90 degree, and their X-ray diffraction patterns remain unchanged even after four hours of water immersion. UV-Vis absorption spectrum shows that C6H4NH2CuBr2I compound has an excellent optical absorption over the entire visible spectrum. We applied this copper-based light absorber in printable mesoscopic solar cell for the initial trial and achieved a power conversion efficiency of 0.5%. Our study represents an alternative pathway to develop low-toxic and highly stable organic-inorganic hybrid materials for photovoltaic application.

Pressure-induced reversible phase transition and amorphization of CH$_3$NH$_3$PbI$_3$

Kai Wang; Ran Liu; Yuancun Qiao; Jinxing Cui; Bo Song; Bingbing Liu; Bo Zou. arXiv preprint (2015). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/1509.03717v1

Abstract: Recent advances of highly efficient solar cells based on organic-inorganic halide perovskites have triggered intense research efforts to establish the fundamental properties of these materials. In this work, we utilized diamond anvil cell to investigate the pressure-induced structural and electronic transformations in methylammonium lead iodide (CH$_3$NH$_3$PbI$_3$) up to 7 GPa at room temperature. The synchrotron X-ray diffraction experiment show that the sample transformed from tetragonal to orthorhombic phase at 0.3 GPa and amorphized above 4 GPa. Further high pressure IR spectroscopy experiments illustrated the high pressure behavior of organic (CH$_3$NH$_3$)$^+$ cations. We also analyzed the pressure dependence of the band gap energy based on the optical absorption and photoluminescence (PL) results. Moreover, all the observed changes were fully reversible when the pressure was completely released. Our in situ high pressure studies provide essential information for the intrinsic properties and stability of organic-inorganic halide perovskites, which significantly affect the performance of perovskite solar cells.

Size dependent solid-solid crystallization of halide perovskites

Paramvir Ahlawat. arXiv preprint (2024). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/2404.05644v1

Abstract: The efficiency and stability of halide perovskite-based solar cells and light-emitting diodes directly depend on the intricate dynamics of solid-solid crystallization[1-23]. In this study, we employ a multi-scale approach using random phase approximation, density functional theory, machine learning potentials, reduced charge force fields, and both enhanced sampling biased and brute-force unbiased molecular dynamics simulations to understand the solid-solid phase transitions in cesium lead iodide perovskite. Our simulations uncover that the direct phase transition from the non-perovskite to the perovskite involves the formation of stacked-faulted and low-dimensional intermediate structures. Through extensive large-scale all-atom simulations encompassing up to 650,000 atoms, we observe that solid-solid crystallization may require the formation of a sufficiently large critical nucleus to grow into a faceted perovskite crystal. Based on simulations, we determine that utilizing (100)-faceted seeded crystallization could offer a promising path for manufacturing high-performance and stable perovskite solar cells.

Efficient and ultra-stable perovskite light-emitting diodes

Bingbing Guo; Runchen Lai; Sijie Jiang; Yaxiao Lian; Zhixiang Ren; Puyang Li; Xuhui Cao; Shiyu Xing; Yaxin Wang; Weiwei Li; Chen Zou; Mengyu Chen; Cheng Li; Baodan Zhao; Dawei Di. Nature Photonics 16, 637-643 (2022) (2022). Cited by 0. Found by: arxiv. DOI: 10.1038/s41566-022-01046-3

Abstract: Perovskite light-emitting diodes (PeLEDs) have emerged as a strong contender for next-generation display and information technologies. However, similar to perovskite solar cells, the poor operational stability remains the main obstacle toward commercial applications. Here we demonstrate ultra-stable and efficient PeLEDs with extraordinary operational lifetimes (T50) of 1.0x10^4 h, 2.8x10^4 h, 5.4x10^5 h, and 1.9x10^6 h at initial radiance (or current densities) of 3.7 W/sr/m2 (~5 mA/cm2), 2.1 W/sr/m2 (~3.2 mA/cm2), 0.42 W/sr/m2 (~1.1 mA/cm2), and 0.21 W/sr/m2 (~0.7 mA/cm2) respectively, and external quantum efficiencies of up to 22.8%. Key to this breakthrough is the introduction of a dipolar molecular stabilizer, which serves two critical roles simultaneously. First, it prevents the detrimental transformation and decomposition of the alpha-phase FAPbI3 perovskite, by inhibiting the formation of lead and iodide intermediates. Secondly, hysteresis-free device operation and microscopic luminescence imaging experiments reveal substantially suppressed ion migration in the emissive perovskite. The record-long PeLED lifespans are encouraging, as they now satisfy the stability requirement for commercial organic LEDs (OLEDs). These results remove the critical concern that halide perovskite devices may be intrinsically unstable, paving the path toward industrial applications.

Advances in Modelling and Simulation of Halide Perovskites for Solar Cell Applications

Chol-Jun Yu. arXiv preprint (2018). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/1811.00702v1

Abstract: Perovskite solar cells (PSCs) are attracting great attention as the most promising candidate for the next generation solar cells. This is due to their low cost and high power conversion efficiency in spite of their relatively short period of development. Key components of PSCs are a variety of halide perovskites with ABX3 stoichiometry used as a photoabsorber, which brought the factual breakthrough in the field of photovoltaic (PV) technology with their outstanding optoelectronic properties. To commercialize PSCs in the near future, however, these materials need to be further improved for a better performance, represented by high efficiency and high stability. As in other materials development, atomistic modelling and simulation can play a significant role in finding new functional halide perovskites as well as revealing the underlying mechanisms of their material processes and properties. In this sense, computational works for the halide perovskites, mostly focusing on first-principles works, are reviewed with an eye looking for an answer how to improve the performance of PSCs. Specific modelling and simulation techniques to quantify material properties of the halide perovskites are also presented. Finally, the outlook for the challenges and future research directions in this field is provided.

Lessons from Chalcopyrites for Scaling Thin Film Perovskite Photovoltaic Technology

Mirjana Dimitrievska; Edgardo Saucedo; Stefaan De Wolf; Billy J. Stanbery; Veronica Bermudez Benito. arXiv preprint (2025). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/2508.11638v2

Abstract: The growing demand for photovoltaic (PV) technologies that are lightweight, flexible, and seamlessly integrated into diverse applications has propelled interest in thin-film solar cells. Among these, Cu(In,Ga)(S,Se)2 (CIGS) and metal halide perovskites have garnered significant attention in the past and present, respectively. While CIGS reached commercial readiness after decades of refinement, their large-scale deployment was hindered by manufacturing complexity, scale-up challenges, and a lack of coordination between materials, device design, and production systems. Perovskite solar cells, despite setting record efficiencies at an unprecedented pace, now face similar challenges on their path to commercialization: ensuring long-term stability, translating laboratory performance to scalable architectures, and aligning with industrial realities. In this perspective, we revisit the CIGS experience not as a benchmark, but as a blueprint, highlighting how its successes and failures can inform a more deliberate and durable trajectory for perovskite PV. Bridging this historical perspective with the current frontier, we propose that the future of perovskites depends not only on continued innovation, but on learning from past thin-film PV experience to avoid its repetition.

Lead-Free Halide Double Perovskites via Heterovalent Substitution of Noble Metals

George Volonakis; Marina R. Filip; Amir Abbas Haghighirad; Nobuya Sakai; Bernard Wenger; Henry J. Snaith; Feliciano Giustino. arXiv preprint (2016). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/1603.01585v1

Abstract: Lead-based halide perovskites are emerging as the most promising class of materials for next generation optoelectronics. However, despite the enormous success of lead-halide perovskite solar cells, the issues of stability and toxicity are yet to be resolved. Here we report on the computational design and the experimental synthesis of a new family of Pb-free inorganic halide double-perovskites based on bismuth or antimony and noble metals. Using first-principles calculations we show that this hitherto unknown family of perovskites exhibits very promising optoelectronic properties, such as tunable band gaps in the visible range and low carrier effective masses. Furthermore, we successfully synthesize the double perovskite Cs2BiAgCl6, we perform structural refinement using single-crystal X-ray diffraction, and we characterize its optical properties via optical absorption and photoluminescence measurements. This new perovskite belongs to the Fm-3m space group, and consists of BiCl6 and AgCl6 octahedra alternating in a rock-salt face-centered cubic structure. From UV-Vis and PL measurements we obtain an indirect gap of 2.2 eV. The new compound is very stable under ambient conditions.

Fully printed flexible perovskite solar modules with improved energy alignment by tin oxide surface modification

Lirong Dong; Shudi Qiu; Jose Garcia Cerrillo; Michael Wagner; Olga Kasian; Sarmad Feroze; Dongju Jang; Chaohui Li; Vincent M. Le Corre; Kaicheng Zhang; Heiko Peisert; Felix U Kosasih; Charline Arrive; Tian Du; Fu Yang; Christoph J. Brabec; Hans-Joachim Egelhaaf. arXiv preprint (2024). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/2406.03123v1

Abstract: Fully printed flexible perovskite solar cells (f-PSCs) show great potential for the commercialization of perovskite photovoltaics owing to their compatibility with high-throughput roll-to-roll (R2R) production. However, the challenge remains in the deficiency in controlling interfacial recombination losses of the functional layer, causing remarkable loss of power conversion efficiency (PCE) in industrial production. Here, a fullerene-substituted alkylphosphonic acid dipole layer is introduced between the R2R-printed tin oxide electron transport layer and the perovskite active layer to reduce the energetic barrier and to suppress surface recombination at the buried interface. The resulting f-PSCs exhibit a PCE of 17.0% with negligible hysteresis, retain 95% of their initial PCE over 3000 bending cycles and achieve a T95 lifetime of 1200 h under 1 sun and 65 degreeC in nitrogen atmosphere. Moreover, the fully printed flexible perovskite solar mini-modules (f-PSMs) with a 20.25 cm2 aperture area achieve a PCE of 11.6%. The encapsulated f-PSMs retain 90% of their initial PCE after 500 h damp-heat testing at 65 degreeC and 85% relative humidity (ISOS-D3). This work marks an important progress toward the realization of efficient and stable flexible perovskite photovoltaics for commercialization.

Perovskite-R1: a domain-specialized large language model for intelligent discovery of precursor additives and experimental design

Xin-De Wang; Zhi-Rui Chen; Peng-Jie Guo; Ze-Feng Gao; Cheng Mu; Zhong-Yi Lu. Communications Materials 7, 86 (2026) (2025). Cited by 0. Found by: arxiv. DOI: 10.1038/s43246-026-01099-9

Abstract: Perovskite solar cells (PSCs) have rapidly emerged as a leading contender in next-generation photovoltaic technologies, owing to their exceptional power conversion efficiencies and advantageous material properties. Despite these advances, challenges such as long-term stability, environmental sustainability, and scalable manufacturing continue to hinder their commercialization. Precursor additive engineering has shown promise in addressing these issues by enhancing both the performance and durability of PSCs. However, the explosive growth of scientific literature and the complex interplay of materials, processes, and device architectures make it increasingly difficult for researchers to efficiently access, organize, and utilize domain knowledge in this rapidly evolving field. To address this gap, we introduce Perovskite-R1, a specialized large language model (LLM) with advanced reasoning capabilities tailored for the discovery and design of PSC precursor additives. By systematically mining and curating 1,232 high-quality scientific publications and integrating a comprehensive library of 33,269 candidate materials, we constructed a domain-specific instruction-tuning dataset using automated question-answer generation and chain-of-thought reasoning. Fine-tuning the QwQ-32B model on this dataset resulted in Perovskite-R1, which can intelligently synthesize literature insights and generate innovative and practical solutions for defect passivation and the selection of precursor additives. Experimental validation of several model-proposed strategies confirms their effectiveness in improving material stability and performance. Our work demonstrates the potential of domain-adapted LLMs in accelerating materials discovery and provides a closed-loop framework for intelligent, data-driven advancements in perovskite photovoltaic research.

Precise Control of Process Parameters for >23% Efficiency Perovskite Solar Cells in Ambient Air Using an Automated Device Acceleration Platform

Jiyun Zhang; Anastasia Barabash; Tian Du; Jianchang Wu; Vincent M. Le Corre; Yicheng Zhao; Shudi Qiu; Kaicheng Zhang; Frederik Schmitt; Zijian Peng; Jingjing Tian; Chaohui Li; Chao Liu; Thomas Heumueller;

Larry Lüer; Jens A. Hauch; Christoph J. Brabec. arXiv preprint (2024). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/2404.00106v1

Abstract: Achieving high-performance perovskite photovoltaics, especially in ambient air relies heavily on optimizing process parameters. However, traditional manual methods often struggle to effectively control the key variables. This inherent challenge requires a paradigm shift toward automated platforms capable of precise and reproducible experiments. Herein, we use a fully automated device acceleration platform (DAP) to optimize the process parameters for preparing full perovskite devices using a two-step method in ambient air. Eight process parameters that have the potential to significantly influence device performance are systematically optimized. Specifically, we delve into the impact of the dispense speed of organic ammonium halide, a parameter that is difficult to control manually, on both perovskite film and device performance. Through the targeted design of experiments, we reveal that the dispense speed significantly affects device performance primarily by adjusting the residual PbI2 content in the films. We find that moderate dispense speeds, e.g., 50 l/s, contribute to top-performance devices. Conversely, too fast or too slow speeds result in devices with relatively poorer performance and lower reproducibility. The optimized parameter set enables us to establish a Standard Operation Procedure (SOP) for additive-free perovskite processing under ambient conditions, which yield devices with efficiencies surpassing 23%, satisfactory reproducibility, and state-of-the-art photo-thermal stability. This research underscores the importance of understanding the causality of process parameters in enhancing perovskite photovoltaic performance. Furthermore, our study highlights the pivotal role of automated platforms in discovering innovative workflows and accelerating the development of high-performing perovskite photovoltaic technologies.

Dimethylammonium additives alter both vertical and lateral composition in halide perovskite semiconductors

Sarthak Jariwala; Rishi Kumar; Giles E. Eperon; Yangwei Shi; David Fenning; David S. Ginger. arXiv preprint (2021). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/2110.11458v1

Abstract: Adding a large A-site cation, such as dimethylammonium (DMA), to the perovskite growth solution has been shown to improve performance and long-term operational stability of halide perovskite solar cells. To better understand the origins of these improvements, we explore the changes in film structure, composition, and optical properties of a formamidinium (FA), Cs, Pb, mixed halide perovskite following addition of DMA to the perovskite growth solution in the ratio DMA0.1FA0.6Cs0.3Pb(I0.8Br0.2)3. Using time-of-flight secondary-ion mass spectrometry (TOF-SIMS) we show that DMA is indeed incorporated into the perovskite, with a higher DMA concentration at the surface. Using a combination of PL microscopy and photo-induced force microscopy-based (PiFM) nanoinfrared (nanoIR), we demonstrate that incorporating DMA into the film leads to increased local heterogeneity in the local bandgap, and clustering of the local formamidinium (CH5N2+) composition. In addition, using nano-X-ray diffraction, we demonstrate that DMA incorporation also alters the local structural composition by changing the local d-spacing distribution and grain size. Our results suggest that compositional variations in the organic cations at the A-site drive the structural heterogeneity observed in case of DMA incorporated films. Our results also suggest that while current-DMA-additive based approaches do have benefits to operational stability and device performance, process optimization to achieve local compositional and structural homogeneity could further boost both of these gains in performance, bringing further gains to solar cells using DMA additives.

The effect of B-site alloying on the electronic and opto-electronic properties of RbPbI3: A DFT study

Anupriya Nyayban; Subhasis Panda; Avijit Chowdhury. arXiv preprint (2022). Cited by 0. Found by: arxiv. DOI: 10.1016/j.physb.2022.414384

Abstract: Divalent cations mixed lead halide perovskites with enhanced performances, high stabilities, and reduced toxicity are requisite to make persistent progress in perovskite solar cells. However, the mixing strategy is not reported extensively in search of a lead reduced structure. Herein, we report the structural, electronic and optical properties of RbPb{1-x}MxI3 (where, M={Sn,Ge} and x={0.25, 0.50, 0.75}) by alloying the B-site with Sn and Ge, using the density functional theory. The formation enthalpy is estimated for all RbPb{1-x}MxI3 (with x= 0.25, 0.50, 0.75), which confirms stability for all the structures. The energy bandgap and density of states (DOS) have been thoroughly investigated. The energy bandgap decreases with the increasing Sn/Ge contents, the lowest bandgap of 1.850 eV is observed at x = 0.50 in the case of RbPb{1-x}GexI3 systems. Further, the effective masses and the binding energy of excitons and spectroscopic limited maximum efficiency (SLME) are also estimated for all the mixed systems. The exciton type is observed to change from Mott-Wannier to Frenkel type with increasing the contents of both Sn and Ge at the B-site. The maximum efficiency of 23%

is achieved using an active layer containing an equal admixture of Sn/Ge and Pb. The estimated parameters of both the mixed systems are consistent with the available literature of similar types.

Strong Performance Enhancement in Lead-Halide Perovskite Solar Cells through Rapid, Atmospheric Deposition of n-type Buffer Layer Oxides

Ravi D. Raninga; Robert A. Jagt; Solène Béchu; Tahmida N. Huq; Mark Nikolka; Yen-Hung Lin; Mengyao Sun; Zewei Li; Wen Li; Muriel Bouttemy; Mathieu Frégnaux; Henry J. Snaith; Philip Schulz; Judith L. MacManus-Driscoll; Robert L. Z. Hoye. arXiv preprint (2019). Cited by 0. Found by: arxiv. DOI: 10.1016/j.nanoen.2020.104946

Abstract: Thin (approximately 10 nm) oxide buffer layers grown over lead-halide perovskite device stacks are critical for protecting the perovskite against mechanical and environmental damage. However, the limited perovskite stability restricts the processing methods and temperatures (<=110 C) that can be used to deposit the oxide overlayers, with the latter limiting the electronic properties of the oxides achievable. In this work, we demonstrate an alternative to existing methods that can grow pinhole-free TiOx (x = 2.00+/-0.05) films with the requisite thickness in <1 min without vacuum. This technique is atmospheric pressure chemical vapor deposition (AP-CVD). The rapid but soft deposition enables growth temperatures of >=180 °C to be used to coat the perovskite. This is >=70 °C higher than achievable by current methods and results in more conductive TiOx films, boosting solar cell efficiencies by >2%. Likewise, when AP-CVD SnOx (x ~ 2) is grown on perovskites, there is also minimal damage to the perovskite beneath. The SnOx layer is pinhole-free and conformal, which reduces shunting in devices, and increases steady-state efficiencies from 16.5% (no SnOx) to 19.4% (60 nm SnOx), with fill factors reaching 84%. This work shows AP-CVD to be a versatile technique for growing oxides on thermally-sensitive materials.

Clever Materials: When Models Identify Good Materials for the Wrong Reasons

Kevin Maik Jablonka. arXiv preprint (2026). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/2602.17730v1

Abstract: Machine learning can accelerate materials discovery. Models perform impressively on many benchmarks. However, strong benchmark performance does not imply that a model learned chemistry. I test a concrete alternative hypothesis: that property prediction can be driven by bibliographic confounding. Across five tasks spanning MOFs (thermal and solvent stability), perovskite solar cells (efficiency), batteries (capacity), and TADF emitters (emission wavelength), models trained on standard chemical descriptors predict author, journal, and publication year well above chance. When these predicted metadata ("bibliographic fingerprints") are used as the sole input to a second model, performance is sometimes competitive with conventional descriptor-based predictors. These results show that many datasets do not rule out non-chemical explanations of success. Progress requires routine falsification tests (e.g., group/time splits and metadata ablations), datasets designed to resist spurious correlations, and explicit separation of two goals: predictive utility versus evidence of chemical understanding.

PeroMAS: A Multi-agent System of Perovskite Material Discovery

Yishu Wang; Wei Liu; Yifan Li; Shengxiang Xu; Xujie Yuan; Ran Li; Yuyu Luo; Jia Zhu; Shimin Di; Min-Ling Zhang; Guixiang Li. arXiv preprint (2026). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/2602.13312v1

Abstract: As a pioneer of the third-generation photovoltaic revolution, Perovskite Solar Cells (PSCs) are renowned for their superior optoelectronic performance and cost potential. The development process of PSCs is precise and complex, involving a series of closed-loop workflows such as literature retrieval, data integration, experimental design, and synthesis. However, existing AI perovskite approaches focus predominantly on discrete models, including material design, process optimization,and property prediction. These models fail to propagate physical constraints across the workflow, hindering end-to-end optimization. In this paper, we propose a multi-agent system for perovskite material discovery, named PeroMAS. We first encapsulated a series of perovskite-specific tools into Model Context Protocols (MCPs). By planning and invoking these tools, PeroMAS can design perovskite materials under multi-objective constraints, covering the entire process from literature retrieval and data extraction to property prediction and mechanism analysis. Furthermore, we construct an evaluation benchmark by perovskite human experts to assess this multi-agent system. Results demonstrate that, compared to single Large Language Model (LLM) or traditional search strategies, our system significantly enhances discovery efficiency. It successfully identified candidate materials satisfying multi-objective constraints. Notably, we verify PeroMAS's effectiveness in the physical world through real synthesis experiments.

Interface Modeling of Perovskite Polymer Heterostructures for Enhanced Charge Transfer Efficiency in Hybrid Photovoltaic Materials

Somayyeh Alidoust; V. Ongun Özçelik. arXiv preprint (2025). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/2512.

Abstract: Perovskite solar cells (PSCs) based on methylammonium lead iodide (MAPbI3) exhibit remarkable photovoltaic performance, where interface engineering with hole transport layers (HTLs) is crucial for optimizing charge transfer and device efficiency. In this work, we present a density functional theory (DFT) study of the MAPbI3/poly(3-hexylthiophene) (P3HT) hybrid interface, focusing on the role of perovskite surface termination in determining interfacial stability and electronic structure. We model MAI- and PbI-terminated MAPbI3 surfaces interfaced with P3HT and compare their interfacial electronic properties. Electronic structure calculations reveal distinct differences in orbital hybridization and band alignment: the MAI/m-P3HT interface exhibits weak coupling, whereas the PbI/m-P3HT interface shows stronger hybridization and enhanced charge transfer. Band alignment confirms type-II, hole-selective character in both cases, with more pronounced valence band maximum adjustment for PbI. Charge difference maps, Bader analysis, and local density of states consistently indicate higher charge transfer and stronger electronic coupling for PbI termination. Electrostatic potential offsets and transport parameters further highlight termination-dependent differences, with lighter effective masses at PbI/m-P3HT and higher hole velocity at MAI/m-P3HT. These findings provide theoretical insight into interfacial charge transport mechanisms and offer guidelines for optimizing perovskite-organic hybrid solar cells.

Antiphase boundary in CH$_3$NH$_3$PbI$_3$ repels charge carriers while promotes fast ion migrations

Shulin Chen; Changwei Wu; Qiuyu Shang; Caili He; Wenke Zhou; Jinjin Zhao; Jingmin Zhang; Junlei Qi; Qing Zhang; Xiao Wang; Jiangyu Li; Peng Gao. Acta Materialia, 2022 (2022). Cited by 0. Found by: arxiv. DOI: 10.1016/j.actamat.2022.118010

Abstract: Defects in organic-inorganic hybrid perovskites (OIHPs) greatly influence their optoelectronic properties. Identification and better understanding of defects existing in OIHPs is an essential step towards fabricating high-performance perovskite solar cells. However, direct visualizing the defects is still a challenge for OIHPs due to their sensitivity during electron microscopy characterizations. Here, by using low dose scanning transmission electron microscopy techniques, we observe the common existence of antiphase boundary (APB) in CH$_3$NH$_3$PbI$_3$ (MAPbI$_3$), resolve its atomic structure, and correlate it to the electrical/ionic activities and structural instabilities. Such an APB is caused by the half-unit-cell shift of [PbI$_6$]$_4$-octahedron along the [100]/[010] direction, leading to the transformation from corner-sharing [PbI$_6$]$_4$-octahedron in bulk MAPbI$_3$ into edge-sharing ones at the APB. Based on the identified atomic-scale configuration, we further carry out density functional theory calculations and reveal that the APB in MAPbI$_3$ repels both electrons and holes while serves as a fast ion-migration channel, causing a rapid decomposition into PbI$_2$ that is detrimental to optoelectronic performance. These findings provide valuable insights into the relationships between structures and optoelectronic properties of OIHPs and suggest that controlling the APB is essential for their stability.

Substitution of Lead with Tin Suppresses Ionic Transport in Halide Perovskite Optoelectronics

Krishanu Dey; Dibyajyoti Ghosh; Matthew Pilot; Samuel R Pering; Bart Roose; Priyanka Deswal; Satyaprasad P Senanayak; Petra J Cameron; M Saiful Islam; Samuel D Stranks. arXiv preprint (2023). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/2305.02014v1

Abstract: Despite the rapid rise in the performance of a variety of perovskite optoelectronic devices with vertical charge transport, the effects of ion migration remain a common and longstanding Achilles heel limiting the long-term operational stability of lead halide perovskite devices. However, there is still limited understanding of the impact of tin (Sn) substitution on the ion dynamics of lead (Pb) halide perovskites. Here, we employ scan-rate-dependent current-voltage measurements on Pb and mixed Pb-Sn perovskite solar cells to show that short circuit current losses at lower scan rates, which can be traced to the presence of mobile ions, are present in both kinds of perovskites. To understand the kinetics of ion migration, we carry out scan-rate-dependent hysteresis analyses and temperature-dependent impedance spectroscopy measurements, which demonstrate suppressed ion migration in Pb-Sn devices compared to their Pb-only analogues. By linking these experimental observations to first-principles calculations on mixed Pb-Sn perovskites, we reveal the key role played by Sn vacancies in increasing the iodide ion migration barrier due to local structural distortions. These results highlight the beneficial effect of Sn substitution in mitigating undesirable ion migration in halide perovskites, with potential implications for future device development.

Understanding and exploiting interfacial interactions between phosphonic acid functional groups and co-evaporated perovskites

Thomas Feeney; Julian Petry; Abderrezak Torche; Dirk Hauschild; Benjamin Hacene; Constantin Wansorra; Alexander Diercks; Michelle Ernst; Lothar Weinhardt; Clemens Heske; Ganna Grynova; Ulrich W. Paetzold; Paul Fassl. arXiv preprint (2024). Cited by 0. Found by: arxiv. DOI: 10.1016/j.matt.2024.02.004

Abstract: Interfacial engineering has fueled recent development of p-i-n perovskite solar cells (PSCs), with self-assembled monolayer-based hole-transport layers (SAM-HTLs) enabling almost lossless contacts for solution-processed PSCs, resulting in the highest achieved power conversion efficiency (PCE) to date. Substrate interfaces are particularly crucial for the growth and quality of co-evaporated PSCs. However, adoption of SAM-HTLs for co-evaporated perovskite absorbers is complicated by the underexplored interaction of such perovskites with phosphonic acid functional groups. In this work, we highlight how exposed phosphonic acid functional groups impact the initial phase and final bulk crystal structures of co-evaporated perovskites and their resultant PCE. The explored surface interaction is mediated by hydrogen bonding with interfacial iodine, leading to increased formamidinium iodide adsorption, persistent changes in perovskite structure, and stabilization of bulk -FAPbI3, hypothesized as being due to kinetic trapping. Our results highlight the potential of exploiting substrates to increase control of co-evaporated perovskite growth.

Illuminating the Future: Nanophotonics for Future Green Technologies, Precision Healthcare, and Optical Computing

Osama M. Halawa; Esraa Ahmed; Malk M. Abdelrazek; Yasser M. Nagy; Omar A. M. Abdelraouf. arXiv preprint (2025). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/2507.06587v1

Abstract: Nanophotonics, an interdisciplinary field merging nanotechnology and photonics, has enabled transformative advancements across diverse sectors including green energy, biomedicine, and optical computing. This review comprehensively examines recent progress in nanophotonic principles and applications, highlighting key innovations in material design, device engineering, and system integration. In renewable energy, nanophotonic allows light-trapping nanostructures and spectral control in perovskite solar cells, concentrating solar power, and thermophotovoltaics. That have significantly enhanced solar conversion efficiencies, approaching theoretical limits. For biosensing, nanophotonic platforms achieve unprecedented sensitivity in detecting biomolecules, pathogens, and pollutants, enabling real-time diagnostics and environmental monitoring. Medical applications leverage tailored light-matter interactions for precision photothermal therapy, image-guided surgery, and early disease detection. Furthermore, nanophotonics underpins next-generation optical neural networks and neuromorphic computing, offering ultra-fast, energy-efficient alternatives to von Neumann architectures. Despite rapid growth, challenges in scalability, fabrication costs, and material stability persist. Future advancements will rely on novel materials, AI-driven design optimization, and multidisciplinary approaches to enable scalable, low-cost deployment. This review summarizes recent progress and highlights future trends, including novel material systems, multidisciplinary approaches, and enhanced computational capabilities, to pave the way for transformative applications in this rapidly evolving field.

A Morphotropic Phase Boundary in MA$_{1-x}$FA$_x$PbI$_3$: Linking Structure, Dynamics, and Electronic Properties

Tobias Hainer; Erik Fransson; Sangita Dutta; Julia Wiktor; Paul Erhart. arXiv preprint (2025). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/2503.22372v1

Abstract: Understanding the phase behavior of mixed-cation halide perovskites is critical for optimizing their structural stability and optoelectronic performance. Here, we map the phase diagram of MA$_{1-x}$FA$_x$PbI$_3$ using a machine-learned interatomic potential in molecular dynamics simulations. We identify a morphotropic phase boundary (MPB) at approximately 27% FA content, delineating the transition between out-of-phase and in-phase octahedral tilt patterns. Phonon mode projections reveal that this transition coincides with a mode crossover composition, where the free energy landscapes of the M and R phonon modes become nearly degenerate. This results in nanoscale layered structures with alternating tilt patterns, suggesting minimal interface energy between competing phases. Our results provide a systematic and consistent description of this important system, complementing earlier partial and sometimes conflicting experimental assessments. Furthermore, density functional theory calculations show that band edge fluctuations peak near the MPB, indicating an enhancement of electron-phonon coupling and dynamic disorder effects. These findings establish a direct link between phonon dynamics, phase behavior, and electronic structure, providing a further composition-driven pathway for tailoring the optoelectronic properties of perovskite materials. By demonstrating that phonon overdamping serves as a hallmark of the MPB, our study offers new insights into the design principles for stable, high-performance perovskite solar cells.

Efficient dataset generation for machine learning perovskite alloys

Henrietta Homm; Jarno Laakso; Patrick Rinke. Physical Review Materials, 9(5), 053802 (2025) (2025). Cited by 0. Found by: arxiv. DOI: 10.1103/PhysRevMaterials.9.053802

Abstract: Lead-based perovskite solar cells have reached high efficiencies, but toxicity and lack of stability hinder their wide-scale adoption. These issues have been partially addressed through compositional engineering of perovskite materials, but the vast complexity of the perovskite materials space poses a significant obstacle to exploration. We previously demonstrated how machine learning (ML) can accelerate property predictions for the CsPb(Cl/Br)$_3$ perovskite alloy. However, the substantial computational demand of density functional theory (DFT) calculations required for model training prevents applications to more complex materials. Here, we introduce a data-efficient scheme to facilitate model training, validated initially on CsPb(Cl/Br)$_3$ data and extended to the ternary alloy CsSn(Cl/Br/I)$_3$. Our approach employs clustering to construct a compact yet diverse initial dataset of atomic structures. We then apply a two-stage active learning approach to first improve the reliability of the ML-based structure relaxations and then refine accuracy near equilibrium structures. Tests for CsPb(Cl/Br)$_3$ demonstrate that our scheme reduces the number of required DFT calculations during the different parts of our proposed model training method by up to 20% and 50%. The fitted model for CsSn(Cl/Br/I)$_3$ is robust and highly accurate, evidenced by the convergence of all ML-based structure relaxations in our tests and an average relaxation error of only 0.5 meV/atom.

Dynamic Nanodomains Dictate Macroscopic Properties in Lead Halide Perovskites

Milos Dubajic; James R. Neilson; Johan Klarbring; Xia Liang; Stephanie A. Boer; Kirrily C. Rule; Josie E. Auckett; Leilei Gu; Xuguang Jia; Andreas Pusch; Ganbaatar Tumen-Ulzii; Qiyuan Wu; Thomas A. Selby; Yang Lu; Julia C. Trowbridge; Eve M. Mozur; Arianna Minelli; Nikolaj Roth; Kieran W. P. Orr; Arman Mahboubi Soufiani; Simon Kahmann; Irina Kabakova; Jianning Ding; Tom Wu; Gavin J. Conibeer; Stephen P. Bremner; Aron Walsh; Michael P. Nielsen; Samuel D. Stranks. arXiv preprint (2024). Cited by 0. Found by: arxiv. DOI: 10.1038/s41565-025-01917-0

Abstract: Empirical A-site cation substitution has advanced the stability and efficiency of hybrid organic-inorganic lead halide perovskites solar cells and the functionality of X-ray detectors. Yet, the fundamental mechanisms underpinning their unique performance remain elusive. This multi-modal study unveils the link between nanoscale structural dynamics and macroscopic optoelectronic properties in these materials by utilising X-ray diffuse scattering, inelastic neutron spectroscopy and optical microscopy complemented by state-of-the-art machine learning-assisted molecular dynamics simulations. Our approach uncovers the presence of dynamic, lower-symmetry local nanodomains embedded within the higher-symmetry average phase in various perovskite compositions. The properties of these nanodomains are tunable via the A-site cation selection: methylammonium induces a high density of anisotropic, planar nanodomains of out-of-phase octahedral tilts, while formamidinium favours sparsely distributed isotropic, spherical nanodomains with in-phase tilting, even when crystallography reveals cubic symmetry on average. The observed variations in the properties of dynamic nanodomains are in agreement with our simulations and are directly linked to the differing macroscopic opto-electronic and ferroelastic behaviours of these compositions. By demonstrating the influence of A-site cation on local nanodomains and consequently, on macroscopic properties, we propose leveraging this relationship to engineer the optoelectronic response of these materials, propelling further advancements in perovskite-based photovoltaics, optoelectronics, and X-ray imaging.

Influence of Disorder and Anharmonic Fluctuations on the Dynamical Rashba Effect in Purely Inorganic Lead-Halide Perovskites

Arthur Marronnier; Guido Roma; Marcelo Carignano; Yvan Bonnassieux; Claudine Katan; Jacky Even; Edoardo Mosconi; Filippo De Angelis. arXiv preprint (2018). Cited by 0. Found by: arxiv. DOI: 10.1021/acs.jpcc.8b11288

Abstract: Doping organic metal-halide perovskites with cesium could be the best solution to stabilize highly-efficient perovskite solar cells. The understanding of the respective roles of the organic molecule, on one hand, and the inorganic lattice, on the other, is thus crucial in order to be able to optimize the physical properties of the mixed-cation structures. In particular, the study of the recombination mechanisms is thought to be one of the key challenges towards full comprehension of their working principles. Using molecular dynamics and frozen phonons, we evidence sub-picosecond anharmonic fluctuations in the fully inorganic $CsPbI_3$ perovskite. We reveal the effect of these fluctuations, combined with spin-orbit coupling, on the electronic band structure, evidencing a dynamical Rashba effect. Our study show that under certain conditions space disorder can quench the Rashba effect. As for time disorder, we evidence a dynamical Rashba effect which is similar to what was found for $MAPbI_3$ and which is still sizable despite temperature disorder, the large investigated supercell, and the absence of the organic cations' motion. We show that the spin texture associated to the Rashba splitting cannot be deemed responsible for a consistent reduction of recombination rates, although the spin mismatch between valence and conduction band increases with the ferroelectric distortion causing the Rashba splitting.

A closed-loop AI framework for hypothesis-driven and interpretable materials design

Kangyu Ji; Tianran Liu; Fang Sheng; Shaun Tan; Moungi Bawendi; Tonio Buonassisi. arXiv preprint (2025). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/2509.18604v1

Abstract: Scientific hypothesis generation is central to materials discovery, yet current approaches often emphasize either conceptual (idea-to-data) reasoning or data-driven (data-to-idea) analysis, rarely achieving an effective integration of both. Here, we present a generalizable active learning workflow that integrates top-down, theory-driven hypothesis generation, guided by a large language model. This is complemented by bottom-up, data-driven hypothesis testing through a root-cause association study. We demonstrate this approach through the design of equimolar quinary-cation two-dimensional perovskite, a chemically complex system with over 850,000 possible cation combinations. In the top-down component, the large language model drives closed-loop optimization by proposing candidates that are likely to achieve phase purity, leveraging domain knowledge and chain-of-thought reasoning. With each iteration, the model identifies an increasing number of near phase-pure compositions, sampling less than 0.004% of the design space. In parallel, the bottom-up association study identifies molecular features with statistically significant influences on phase purity. The integration of these approaches enables the convergence of conceptual and statistical hypotheses, leading to generalizable and rational design rules for phase-pure quinary-cation two-dimensional perovskites. As a proof of concept, we applied the optimized phase-pure quinary-cation two-dimensional perovskite film as a surface capping layer in perovskite solar cells, achieving good performance and stability. Our framework enables the development of interpretable and generalizable design rules that are applicable to a wide range of optimization processes within complex design spaces, providing a foundational strategy for rational, scalable, and efficient materials discovery.

Tailoring wetting properties of organic hole-transport interlayers for slot-die coated perovskite solar modules

T. S. Le; I. A. Chuykoa; L. O. Luchnikova; K. A. Ilicheva; P. O. Sukhorukova; D. O. Balakirev; N. S. Saratovsky; A. O. Alekseev; S. S. Kozlov; D. S. Muratov; V. V. Voronov; P. A. Gostishchev; D. A. Kiselev; T. S. Ilina; A. A. Vasilev; A. Y. Polyakov; E. A. Svidchenko; O. A. Maloshitskaya; Yu. N. Luponosov; D. S. Saranin. arXiv preprint (2024). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/2406.08415v1

Abstract: The use of self-assembled monolayers (SAMs) with anchoring groups was considered as an effective approach for interface engineering in perovskite solar cells with metal oxide charge transporting layers. However, the coating of SAM layers in PSMs by means of a slot-die is a challenging process due to the low viscosity of the solutions and the low wettability of the films. In this study, we integrate a triphenylamine-based polymer, pTPA-TDP, blended with SAM based on 5-[4-[4-(diphenylamino) phenyl] thiophene-2-carboxylic acid (TPATC), to address the challenges of uniform slot-die coating and interface passivation in large-area modules. We fabricated p-i-n oriented PSMs on 50x50 mm2 substrates (12-sub-cells) with NiO hole transport layer (HTL) and organic interlayers for surface modification. Wetting angle mapping demonstrated that ununiform regions of the slot-die coated SAM have hydrophobicity with contact angle values up to 90°, causing fluctuations in absorber thickness and the presence of macro-defects at buried interfaces. The incorporation of the blended interlayer to NiO/perovskite junction homogenized the surface wettability (contact angle=40°) and mitigated lattice strain in the absorber. This enabled the effective use of SAM properties on a large-area surface, improving energy level alignment and enhancing the power conversion efficiency (PCE) of the modules from 13.98% to 15.83% and stability (ISOS-L-2, T80 period) from 500-1000 hours to 1630 hours. Investigation of PSMs upon cooling till -5 °C showed that the PCE increased by +0.19%/°C for samples with NiO HTL, while using SAM and blended interlayers raised the coefficient to ~0.40%/°C due to changes in activation energy and trap contributions to device performance across a wide temperature range.

Block MLP (313 unique)

Atom-centered symmetry functions for constructing high-dimensional neural network potentials

Jörg Behler. The Journal of Chemical Physics (2011). Cited by 1644. Found by: openalex. DOI: 10.1063/1.3553717

Abstract: Neural networks offer an unbiased and numerically very accurate approach to represent high-dimensional ab initio potential-energy surfaces. Once constructed, neural network potentials can provide the energies and forces many orders of magnitude faster than electronic structure calculations, and thus enable molecular dynamics simulations of large systems. However, Cartesian coordinates are not a good choice to represent the atomic positions, and a transformation to symmetry functions is required. Using simple benchmark systems, the properties of several types of symmetry functions suitable for the construction of high-dimensional neural network potential-energy surfaces are discussed in detail. The symmetry functions are general and can be applied to all types of systems such as molecules, crystalline and amorphous solids, and liquids.

Realistic Atomistic Structure of Amorphous Silicon from Machine-Learning-Driven Molecular Dynamics

Volker L. Deringer; Noam Bernstein; Albert P. Bartók; Matthew J. Cliffe; Rachel N. Kerber; Lauren E. Marbella; Clare P. Grey; Stephen R. Elliott; Gábor Cśanyi. The Journal of Physical Chemistry Letters (2018). Cited by 269. Found by: openalex. DOI: 10.1021/acs.jpclett.8b00902

Abstract: Abstract Amorphous silicon (a-Si) is a widely studied noncrystalline material, and yet the subtle details of its atomistic structure are still unclear. Here, we show that accurate structural models of a-Si can be obtained using a machine-learning-based interatomic potential. Our best a-Si network is obtained by simulated cooling from the melt at a rate of 1011 K/s (that is, on the 10 ns time scale), contains less than 2% defects, and agrees with experiments regarding excess energies, diffraction data, and 29Si NMR chemical shifts. We show that this level of quality is impossible to achieve with faster quench simulations. We then generate a 4096-atom system that correctly reproduces the magnitude of the first sharp diffraction peak (FSDP) in the structure factor, achieving the closest agreement with experiments to date. Our study demonstrates the broader impact of machine-learning potentials for elucidating structures and properties of technologically important amorphous materials.

Machine-learned interatomic potentials by active learning: amorphous and liquid hafnium dioxide

Ganesh Sivaraman; Anand Narayanan Krishnamoorthy; Matthias Baur; Christian Holm; Marius Stan; Gábor Csányi; Chris Benmore; Álvaro Vázquez-Mayagoitia. npj Computational Materials (2020). Cited by 185. Found by: openalex. DOI: 10.1038/s41524-020-00367-7

Abstract: Abstract We propose an active learning scheme for automatically sampling a minimum number of uncorrelated configurations for fitting the Gaussian Approximation Potential (GAP). Our active learning scheme consists of an unsupervised machine learning (ML) scheme coupled with a Bayesian optimization technique that evaluates the GAP model. We apply this scheme to a Hafnium dioxide (HfO 2) dataset generated from a "melt-quench" ab initio molecular dynamics (AIMD) protocol. Our results show that the active learning scheme, with no prior knowledge of the dataset, is able to extract a configuration that reaches the required energy fit tolerance. Further, molecular dynamics (MD) simulations performed using this active learned GAP model on 6144 atom systems of amorphous and liquid state elucidate the structural properties of HfO 2 with near ab initio precision and quench rates (i.e., 1.0 K/ps) not accessible via AIMD. The melt and amorphous X-ray structural factors generated from our simulation are in good agreement with experiment. In addition, the calculated diffusion constants are in good agreement with previous ab initio studies.

Machine-learned interatomic potentials by active learning: amorphous and liquid hafnium dioxide

Ganesh Sivaraman; Anand Narayanan Krishnamoorthy; Matthias Baur; Christian Holm; Marius Stan; Gábor Cśanyi; Chris J. Benmore; Álvaro Vázquez-Mayagoitia. Cambridge University Engineering Department Publications Database (2019). Cited by 184. Found by: openalex. openalex_2yr: 1 (2026). DOI: 10.17863/cam.55408

Abstract: Abstract: We propose an active learning scheme for automatically sampling a minimum number of uncorrelated configurations for fitting the Gaussian Approximation Potential (GAP). Our active learning scheme consists of an unsupervised machine learning (ML) scheme coupled with a Bayesian optimization technique that evaluates the GAP model. We apply this scheme to a Hafnium dioxide (HfO2) dataset generated from a "melt-quench" ab initio molecular dynamics (AIMD) protocol. Our results show that the active learning scheme, with no prior knowledge of the dataset, is able to extract a configuration that reaches the required energy fit tolerance. Further, molecular dynamics (MD) simulations performed using this active learned GAP model on 6144 atom systems of amorphous and liquid state elucidate the structural properties of HfO2 with near ab initio precision and quench rates (i.e., 1.0 K/ps) not accessible via AIMD. The melt and amorphous X-ray structural factors generated from our simulation are in good agreement with experiment. In addition, the calculated diffusion constants are in good agreement with previous ab initio studies.

Growth Mechanism and Origin of High sp3 Content in Tetrahedral Amorphous Carbon

Miguel A. Caro; Volker L. Deringer; Jari Koskinen; Tomi Laurila; Gábor Cśanyi. Physical Review Letters (2018). Cited by 178. Found by: openalex. DOI: 10.1103/physrevlett.120.166101

Abstract: We study the deposition of tetrahedral amorphous carbon (ta-C) films from molecular dynamics simulations based on a machine-learned interatomic potential trained from density-functional theory data. For the first time, the high sp^{3} fractions in excess of 85% observed experimentally are reproduced by means of computational simulation, and the deposition energy dependence of the film's characteristics is also accurately described. High confidence in the potential and direct access to the atomic interactions allow us to infer the microscopic growth mechanism in this material. While the widespread view is that ta-C grows by "subplantation," we show that the so-called "peening" model is actually the dominant mechanism responsible for the high sp^{3} content. We show that pressure waves lead to bond rearrangement away from the impact site of the incident ion, and high sp^{3} fractions arise from a delicate balance of transitions between three- and fourfold coordinated carbon atoms. These results open the door for a microscopic understanding of carbon nanostructure formation with an unprecedented level of predictive power.

Constructing first-principles phase diagrams of amorphous LixSi using machine-learning-assisted sampling with an evolutionary algorithm

Nongnuch Artrith; Alexander Urban; Gerbrand Ceder. The Journal of Chemical Physics (2018). Cited by 155. Found by: openalex. DOI: 10.1063/1.5017661

Abstract: The atomistic modeling of amorphous materials requires structure sizes and sampling statistics that are challenging to achieve with first-principles methods. Here, we propose a methodology to speed up the sampling of amorphous and disordered materials using a combination of a genetic algorithm and a specialized machine-learning potential based on artificial neural networks (ANNs). We show for the example of the amorphous LiSi alloy that around 1000 first-principles calculations are sufficient for the ANN-potential assisted sampling of low-energy atomic configurations in the entire amorphous LixSi phase space. The obtained phase diagram is validated by comparison with the results from an extensive sampling of LixSi configurations using molecular dynamics simulations and a general ANN potential trained to 45 000 first-principles calculations. This demonstrates the utility of the approach for the first-principles modeling of amorphous materials.

Study of Li atom diffusion in amorphous Li3PO4 with neural network potential

Wenwen Li; Yasunobu Ando; Emi Minamitani; Satoshi Watanabe. The Journal of Chemical Physics (2017). Cited by 154. Found by: openalex. DOI: 10.1063/1.4997242

Abstract: To clarify atomic diffusion in amorphous materials, which is important in novel information and energy devices, theoretical methods having both reliability and computational speed are eagerly anticipated. In the present study, we applied neural network (NN) potentials, a recently developed machine learning technique, to the study of atom diffusion in amorphous materials, using Li3PO4 as a benchmark material. The NN potential was used together with the nudged elastic band, kinetic Monte Carlo, and molecular dynamics methods to characterize Li vacancy diffusion behavior in the amorphous Li3PO4 model. By comparing these results with corresponding DFT calculations, we found that the average error of the NN potential is 0.048 eV in calculating energy barriers of diffusion paths, and 0.041 eV in diffusion activation energy. Moreover, the diffusion coefficients obtained from molecular dynamics are always consistent with those from ab initio molecular dynamics simulation, while the computation speed of the NN potential is 3–4 orders of magnitude faster than DFT. Lastly, the structure of amorphous Li3PO4 and the ion transport properties in it were studied with the NN potential using a large supercell model containing more than 1000 atoms. The formation of P2O7 units was observed, which is consistent with the experimental characterization. The Li diffusion activation energy was estimated to be 0.55 eV, which agrees well with the experimental measurements.

Impact of the Local Environment on Li Ion Transport in Inorganic Components of Solid Electrolyte Interphases

Taiping Hu; Jianxin Tian; Fuzhi Dai; Xiaoxu Wang; Rui Wen; Shenzhen Xu. Journal of the American Chemical Society (2022). Cited by 107. Found by: openalex. DOI: 10.1021/jacs.2c11521

Abstract: The spontaneously formed passivation layer, the solid electrolyte interphase (SEI) between the electrode and electrolyte, is crucial to the performance and durability of Li ion batteries. However, the Li ion transport mechanism in the major inorganic components of the SEI (Li 2 CO 3 and LiF) is still unclear. Partic-

ularly, whether introducing an amorphous environment is beneficial for improving the Li ion diffusivity is under debate. Here, we investigate the Li ion diffusion mechanism in amorphous LiF and Li 2 CO 3 via machine-learning-potential-assisted molecular dynamics simulations. Our results show that the Li ion diffusivity in LiF at room temperature cannot be accurately captured by the Arrhenius extrapolation from the high temperature (>600 K) diffusivities (difference of 2 orders of magnitude). We reveal that the spontaneous formation of Li–F regular tetrahedrons at low temperatures (<500 K) leads to an extremely low Li ion diffusivity, suggesting that designing an amorphous bulk LiF-based SEI cannot help with the Li ion transport. We further show the critical role of Li 2 CO 3 in suppressing the Li–F regular tetrahedron formation when these two components of SEIs are mixed. Overall, our work provides atomic insights into the impact of the local environment on Li ion diffusion in the major SEI components and suggests that suppressing the formation of large-sized bulk-phase LiF might be critical to improve battery performance.

Thermal conductivity modeling using machine learning potentials: application to crystalline and amorphous silicon

Xin Qian; Shiyu Peng; Xiaobo Li; Yong Wei; Ronggui Yang. Materials Today Physics (2019). Cited by 103. Found by: openalex. DOI: 10.1016/j.mtphys.2019.100140

A unified deep neural network potential capable of predicting thermal conductivity of silicon in different phases

Ruiyang Li; Eungkyu Lee; Tengfei Luo. Materials Today Physics (2020). Cited by 102. Found by: openalex. DOI: 10.1016/j.mtphys.2020.100181

Unraveling the crystallization kinetics of the Ge2Sb2Te5 phase change compound with a machine-learned interatomic potential

Omar Abou El Kheir; Luigi Bonati; Michele Parrinello; Marco Bernasconi. npj Computational Materials (2024). Cited by 62. Found by: openalex. DOI: 10.1038/s41524-024-01217-6

Abstract: Abstract The phase change compound Ge 2 Sb 2 Te 5 (GST225) is exploited in advanced non-volatile electronic memories and in neuromorphic devices which both rely on a fast and reversible transition between the crystalline and amorphous phases induced by Joule heating. The crystallization kinetics of GST225 is a key functional feature for the operation of these devices. We report here on the development of a machine-learned interatomic potential for GST225 that allowed us to perform large scale molecular dynamics simulations (over 10,000 atoms for over 100 ns) to uncover the details of the crystallization kinetics in a wide range of temperatures of interest for the programming of the devices. The potential is obtained by fitting with a deep neural network (NN) scheme a large quantum-mechanical database generated within density functional theory. The availability of a highly efficient and yet highly accurate NN potential opens the possibility to simulate phase change materials at the length and time scales of the real devices.

Development of a neuroevolution machine learning potential of Pd-Cu-Ni-P alloys

Rui Zhao; Shucheng Wang; Zhuangzhuang Kong; Yunlei Xu; Kuan Fu; Ping Peng; Cuilan Wu. Materials & Design (2023). Cited by 43. Found by: openalex. DOI: 10.1016/j.matdes.2023.112012

Abstract: Pd-Cu-Ni-P alloy is an ideal model system of metallic glass known for its exceptional glass-forming ability. However, few correlation of structures with properties was systematically investigated owing to a lack of interatomic potential. In this work, a neuroevolution machine learning potential (NEP) with efficiency close to embedded atom method (EAM) potentials is developed. Its accuracy has been compared to density functional theory (DFT) calculations. For energy, force and virial, the training errors are 6.0 meV/atom, 111.1 meV/Å and 21.5 meV/atom, respectively. By means of this NEP, several thermodynamic parameters such as glass transition temperatures and pair distribution functions of Pd40Cu30Ni10P20 and Pd40Ni40P20 liquid and glassy alloys as well as their short-range orders, tensile and compression strengths, transport properties etc. have been evaluated by a series of molecular dynamics simulations. A good agreement with DFT calculations and previous experiments indicates this NEP provides an accurate and efficient scheme in the analysis and exploration of microstructures, thermodynamic and kinetic properties of Pd-Cu-Ni-P alloys.

Training machine-learning potentials for crystal structure prediction using disordered structures

Chang-Ho Hong; Jeong Min Choi; Wonseok Jeong; Sungwoo Kang; Suyeon Ju; Kyeongpung Lee; Jisu Jung; Yong Youn; Seungwu Han. Physical review. B./Physical review. B (2020). Cited by 40. Found by: openalex. DOI: 10.1103/physrevb.102.224104

Abstract: Prediction of the stable crystal structure for multinary (ternary or higher) compounds with unexplored compositions demands fast and accurate evaluation of free energies in exploring the vast configurational space. The machine-learning potential such as the neural network potential (NNP) is poised to meet this requirement but a dearth of information on the crystal structure poses a challenge in choosing training sets. Herein we propose constructing the training set from density functional theory (DFT)–based dynamical trajectories of liquid and quenched amorphous phases, which does not require any preceding information on material structures except for the chemical composition. To demonstrate suitability of the trained NNP in the crystal structure prediction, we compare NNP and DFT energies for ${\mathrm{Ba}}_{2}{\mathrm{AgSi}}_{3}$, ${\mathrm{Mg}}_{2}{\mathrm{SiO}}_{4}$, ${\mathrm{LiAlCl}}_{4}$, and ${\mathrm{InTe}}_{2}{\mathrm{O}}_ over experimental phases as well as low-energy crystal structures that are generated theoretically. For every material, we find strong correlations between DFT and NNP energies, ensuring that the NNPs can properly rank energies among low-energy crystalline structures. We also find that the evolutionary search using the NNPs can identify low-energy metastable phases more efficiently than the DFT-based approach. By proposing a way to developing reliable machine-learning potentials for the crystal structure prediction, this work paves the way to identifying unexplored multinary phases efficiently.

Crystallization of amorphous GeTe simulated by neural network potential addressing medium-range order

Dong-Heon Lee; Kyeongpung Lee; Dongsun Yoo; Wonseok Jeong; Seungwu Han. Computational Materials Science (2020). Cited by 40. Found by: openalex. openalex_2yr: 3.958 (2026). DOI: 10.1016/j.commatsci.2020.109725

Ultralow and glass-like lattice thermal conductivity in crystalline BaAg2Te2: Strong fourth-order anharmonicity and crucial diffusive thermal transport

Zezhu Zeng; Cunzhi Zhang; Hulei Yu; Wen Li; Yanzhong Pei; Yue Chen. Materials Today Physics (2021). Cited by 37. Found by: openalex. DOI: 10.1016/j.mtphys.2021.100487

Dependence of a cooling rate on structural and vibrational properties of amorphous silicon: A neural network potential-based molecular dynamics study

Wenwen Li; Yasunobu Ando. The Journal of Chemical Physics (2019). Cited by 35. Found by: openalex. DOI: 10.1063/1.5114652

Abstract: Amorphous materials have variable structural order, which has a significant influence on their electronic, transport, and thermal properties. However, this difference in structure has rarely been investigated by atomistic modeling. In this study, a high-quality machine-learning-based interatomic potential was used to generate a series of atomic structures of amorphous silicon with different degrees of disorder by simulated cooling from the melt with different cooling rates (1011–1015 K/s). We found that the short- and intermediate-range orders are enhanced with decreasing cooling rate, and the influence of the structural order change is in excellent agreement with the experimental annealing process in terms of the structural, energetic, and vibrational properties. In addition, by comparing the excess energies, structure factors, radial distribution functions, phonon densities of states, and Raman spectra, it is possible to determine the corresponding theoretical model for experimental samples prepared with a certain method and thermal history.

Experiment-Driven Atomistic Materials Modeling: A Case Study Combining X-Ray Photoelectron Spectroscopy and Machine Learning Potentials to Infer the Structure of Oxygen-Rich Amorphous Carbon

Tigany Zarrouk; Rina Ibragimova; Albert P. Bartók; Miguel A. Caro. Journal of the American Chemical Society (2024). Cited by 35. Found by: openalex. DOI: 10.1021/jacs.4c01897

Abstract: High Resolution Image Download MS PowerPoint Slide An important yet challenging aspect of atomistic materials modeling is reconciling experimental and computational results. Conventional approaches involve generating numerous configurations through molecular dynamics or Monte Carlo structure optimization and selecting the one with the closest match to experiment. However, this inefficient process is not guaranteed to succeed. We introduce a general method to combine atomistic machine learning (ML) with experimental observables that produces atomistic structures compatible with experiment by design . We use this approach in combination with grand-canonical Monte Carlo within a modified Hamiltonian formalism, to generate configurations that agree with experimental data and are chemically sound (low in energy). We apply our approach to understand the atomistic structure of oxygenated amorphous carbon (a-CO x), an intriguing carbon-based material, to answer the question of how much oxygen can be added to carbon before it fully decomposes into CO and CO 2 . Utilizing an ML-based X-ray photoelectron spectroscopy (XPS) model trained from GW and density functional theory (DFT) data, in conjunction with an ML interatomic potential, we identify a-CO x structures compliant with experimental XPS predictions that are also energetically favorable with respect to DFT. Employing a network analysis, we accurately deconvolve the XPS spectrum into motif contributions, both revealing the inaccuracies inherent to experimental XPS interpretation and granting us atomistic insight into the structure of a-CO x . This method generalizes to multiple experimental observables and allows for the elucidation of the atomistic structure of materials directly from experimental data, thereby enabling experiment-driven materials modeling with a degree of realism previously out of reach.

Accurate and Transferable Machine Learning Potential for Molecular Dynamics Simulation of Sodium Silicate Glasses

Marco Bertani; Thibault Charpentier; Francesco Faglioni; Alfonso Pedone. Journal of Chemical Theory and Computation (2024). Cited by 33. Found by: openalex. DOI: 10.1021/acs.jctc.3c01115

Abstract: An accurate and transferable machine learning (ML) potential for the simulation of binary sodium silicate glasses over a wide range of compositions (from 0 to 50% Na 2 O) was developed. The potential energy surface is approximated by the sum of atomic energy contributions mapped by a neural network algorithm from the local geometry comprising information on atomic distances and angles with neighboring atoms using the DeePMD code [Wang, H. Comput. Phys. Commun. 2018, 228, 178–184]. Our model was trained on a large data set of total energies and atomic forces computed at the density functional theory level on structures extracted from classical molecular dynamics (MD) simulations performed at several temperatures from 300 to 3000 K. This allows for the generation of a robust and transferable ML potential applicable over the full compositional range of glass formability at different temperatures that outperforms the empirical potentials available in the literature in reproducing structures and properties such as bond angle distribution, total distribution functions, and vibrational density of state. The generality of the approach enables the future training of a potential with other or more elements allowing for simulations of structures, properties, and behavior of ternary and multicomponent oxide glasses with nearly ab initio accuracy at a fraction of the computational cost.

Applications of machine-learning interatomic potentials for modeling ceramics, glass, and electrolytes: A review

Shingo Urata; Marco Bertani; Alfonso Pedone. Journal of the American Ceramic Society (2024). Cited by 32. Found by: openalex. DOI: 10.1111/jace.19934

Abstract: Abstract The emergence of artificial intelligence has provided efficient methodologies to pursue innovative findings in material science. Over the past two decades, machine-learning potential (MLP) has emerged as an alternative technology to density functional theory (DFT) and classical molecular dynamics (CMD) simulations for computational modeling of materials and estimation of their properties. The MLP offers more efficient computation compared to DFT, while providing higher accuracy compared to CMD. This enables us to conduct more realistic simulations using models with more atoms and for longer simulation times. Indeed, the number of research studies utilizing MLPs has significantly increased since 2015, covering a broad range of materials and their structures, ranging from simple to complex, as well as various chemical and physical phenomena. As a result, there are high expectations for further applications of MLPs in the field of material science and industrial development. This review aims to summarize the applications, particularly in ceramics and glass science, and fundamental theories of MLPs to facilitate future progress and utilization. Finally, we provide a summary and discuss perspectives on the next challenges in the development and application of MLPs.

Vibrational anharmonicity results in decreased thermal conductivity of amorphous HfO2 at high temperature

Honggang Zhang; Xiaokun Gu; Zheyong Fan; Hua Bao. Physical review. B./Physical review. B (2023). Cited by 31. Found by: openalex. DOI: 10.1103/physrevb.108.045422

Abstract: While the high-temperature thermal transport in crystalline materials has been recently carefully addressed, it is much less explored for amorphous materials. Most of the existing studies have focused on the low-/mid-temperature range and have generally found the increasing trend of thermal conductivity with temperature and converging to a constant value, mainly due to the temperature dependence of heat capacity. In this work, we investigate the temperature-dependent thermal conductivity of amorphous ${\mathrm{HfO}}_{2}$ with three different methods, including molecular dynamics, the Allen-Feldman theory, and the quasiharmonic Green-Kubo method, with the forces extracted from a machine-learning potential parametrized from first principles. While the Allen-Feldman theory and the quasiharmonic Green-Kubo method show the same temperature dependence trend as the previous expectation even at high temperatures, molecular dynamics simulations show a clear decreasing trend of thermal conductivity at high temperatures. By comparing the results from these approaches, we identify that two anharmonic effects, i.e., thermal expansion and vibrational mode softening, are the mechanisms of the decreased thermal conductivity of amorphous ${\mathrm{HfO}}_{2}$ at high temperatures.

AENET–LAMMPS and AENET–TINKER: Interfaces for accurate and efficient molecular dynamics simulations with machine learning potentials

Michael S. Chen; Tobias Morawietz; Hideki Mori; Thomas E. Markland; Nongnuch Artrith. The Journal of Chemical Physics (2021). Cited by 29. Found by: openalex. DOI: 10.1063/5.0063880

Abstract: Machine-learning potentials (MLPs) trained on data from quantum-mechanics based first-principles methods can approach the accuracy of the reference method at a fraction of the computational cost. To facilitate efficient MLP-based molecular dynamics and Monte Carlo simulations, an integration of the MLPs with sampling software is needed. Here, we develop two interfaces that link the atomic energy network (ænet) MLP package with the popular sampling packages TINKER and LAMMPS. The three packages, ænet, TINKER, and LAMMPS, are free and open-source software that enable, in combination, accurate simulations of large and complex systems with low computational cost that scales linearly with the number of atoms. Scaling tests show that the parallel efficiency of the ænet-TINKER interface is nearly optimal but is limited to shared-memory systems. The ænet-LAMMPS interface achieves excellent parallel efficiency on highly parallel distributed-memory systems and benefits from the highly optimized neighbor list implemented in LAMMPS. We demonstrate the utility of the two MLP interfaces for two relevant example applications: the investigation of diffusion phenomena in liquid water and the equilibration of nanostructured amorphous battery materials.

The Roadmap of Graphene: From Fundamental Research to Broad Applications

Jin Zhang; Hailin Peng; Hua Zhang; Shigeo Maruyama; Li Lin. Advanced Functional Materials (2022). Cited by 27. Found by: openalex. openalex_2yr: 14.975 (2026). DOI: 10.1002/adfm.202208378

Abstract: Owing to its atomically thin structure and fascinating properties, graphene has emerged as an important material in modern physics and chemistry. Tremendous breakthroughs have been made in the mass production of graphene materials and the development of a wealth of applications in electronics, photonics, batteries, wearables, and sensors. Currently, commercially available graphene products can be divided into graphene films, graphene oxide (GO), and reduced graphene oxide (rGO), and the mainstream preparation techniques include chemical vapor deposition (CVD) and chemical exfoliation. The global graphene market size exceeded USD 4 billion by 2021, and over 20000 companies are currently engaged in graphene-related businesses. Scientists, engineers, and enterprises are working in tandem to find scalable approaches to producing graphene with reliable quality and performance, and to unveiling novel and revolutionary applications. The success of graphene industrialization and commercialization will be promising through joint academic and industrial efforts. This special issue organized by Profs. Jin Zhang, Hailin Peng, Hua Zhang, Shigeo Maruyama, and Li Lin, provides a body of high-impact reviews and research articles in the fields of synthesis, engineering, and application of graphene materials. We proposed this special issue in Advanced Functional Materials on the occasion of the 60th birthday of Prof. Zhongfan Liu, a world-famous expert in graphene research. This special issue consists of 16 reviews and 14 research articles, focusing on the latest summary and progress from fundamental research to broad applications of graphene. This special issue will include a range of sub-topics, such as the controllable synthesis of graphene films, powders, and fibers, innovative production techniques for specific applications, structural and functional engineering of graphene architectures, novel characterization of graphene materials by synchrotron radiation, and advanced applications of graphene materials and composites. We believe that the topics in this special issue of Advanced Functional Materials would provide pivotal viewpoints relevant for promoting the mass production and commercialization of graphene, and are valuable for a broad audience of academia and industry professionals engaged in graphene-related research and business. The commercial success of graphene materials should stand on the reliable technique for the mass production of graphene products with

fine control over the quality, cost, and reproducibility. The CVD approach can produce graphene films and powders with excellent scalability and quality at high temperatures. In the last decade, great progress has been made in the controllable CVD synthesis of graphene products, exemplified by wafer-scale single-crystal graphene, wrinkle-free graphene, and the successful removal of surface contamination. Prof. Feng Ding et al. (article number 2203191) provide a detailed summary of the achievements and challenges of graphene CVD synthesis from a theoretical perspective. Future focus is also detailed on the growth of graphene at relatively low temperatures, insulating substrates, and fine control over the thickness and interlayer stacking order. Prof. Yanwu Zhu (article number 2203894) systematically investigates the structural evolution from C60 fullerene crystal to other nanocarbon materials, such as graphene, by combining neural network potential and stochastic surface walking methods. Prof. Haofei Shi et al. (article number 2202415) reveal that graphene domains are subject to different compressive strains when growing across the twin boundaries of copper substrates during high-temperature CVD. A new understanding of the effect of twin grain boundaries on graphene epitaxy is provided as a guideline for growing high-quality graphene on twinned metal foil. In another study by Prof. Shi (article number 2202376), a selective-area reconstruction method is proposed for the patterned growth of high-quality graphene, in which selective-area oxidation and high-temperature reduction are capable of regulating the surface characteristics of copper substrates, thereby enabling control over nucleation and epitaxial growth. Profs. Bin Wu and Yunqi Liu et al. (article number 2202584) focus on the growth, characterization, and properties of multilayered graphene. Various state-of-the-art synthetic approaches for multilayered graphene and twisted bilayer graphene are detailed in this review, as well as the underlying growth mechanisms. Innovation in graphene synthesis techniques that provide new graphene products for specific applications has become the theme of some reviews and research articles. Prof. Liangti Qu et al. (article number 202203164) provide a systematic review of laser-processing technologies for producing graphene with high spatial precision, fast reaction rate, and high controllability. The interaction principle and processing strategies of functional graphene treated by lasers are presented in detail, as well as the regulation of graphene optoelectronic properties for advanced photo/electro-devices. Prof. Qingkai Yu et al. (article number 2202381) propose the synthesis of carbon black and 2D graphene in molten salt at high temperatures by bubbling CVD, and reveal that the formation of transition metal chloride anion complexes in NaCl-KCl can significantly enhance the pyrolysis efficiency of methane. As a new phase, bubbles can be employed as the driving force to cleave the graphite layers to produce graphene nanoplates or separate graphene from the growth substrates. Profs. Qingkai Yu, Guqiao Ding, and Xiaoming Xie (article number 2203124) also introduces bubble-based mass production of graphene in an efficient and controllable manner, covering both “top-down” and “bottom-up” routines. The functional integration of graphene with traditional materials is expected to uncover new applications with outstanding performance that cannot be achieved using individual components. Optical fibers provide the highest-quality optical waveguides for information communication and photon manipulation, and the integration of optical fibers with graphene has enabled significant advances in all-fiber photonics and optoelectronics. Profs. Kaihui Liu and Zhongfan Liu (article number 2202282) systematically introduce various graphene optical fibers and related fabrication methods, and summarize graphene optical fiber devices, including ultrafast fiber lasers, polarizers, modulators, detectors, and sensors. Profs. Jingyu Sun and Zhongfan Liu (article number 2200428) report the direct preparation of ultra-flat graphene on economical quartz substrates at the wafer scale without the formation of folds, multilayer islands, or metallic impurities. The obtained graphene enables atomically smooth growth of GaN films with (001) single-crystallinity over amorphous silica substrates, facilitating the cost-effective integration of emerging III-nitrides for exciting device applications. A wealth of engineering or modifying strategies from the perspective of morphological and electronic structures have been applied for multifunctional applications, such as compact energy storage, ion/molecular separation, gas barrier films, and flexible electronics. Prof. Hua Zhang (article number 2202319) summarize strategies for hybridizing 2D nanomaterials with a 3D graphene architecture as well as related applications in electrochemical energy storage and conversion. Prof. Guqiao Ding (article number 2202697) report the strategy to fabricate ultra-large graphene oxide by directly oxidizing the electrochemically derived porous graphene networks, and the graphene networks could enable fast diffusion channels for oxidant. The oxidation process consumes only 40 minutes and ultralow oxidant dosage. Profs. Yanfeng Zhang and Zhongfan Liu (article number 2202026) review the recent advances in direct plasma-enhanced CVD growth of vertically oriented graphene films on functional insulating substrates, with special emphasis on the design of practical synthetic strategies for improving material qualities and developing high-end applications. Prof. Richard Kaner et al. (article number 2203101) report a simple electrochemical approach for the direct deposition of a functional graphene framework using electrostatic interactions between graphene oxide and a cationic surfactant. The reported approach leads to supercapacitors with high specific capacitance and areal capacitance, comparable to those of commercial activated carbon supercapacitors. Prof. Dan Li (article number 2201535) produce a graphene-based membrane by reducing graphene oxide with hydrazine and discover that the resultant densely packed rGO membrane retains the interconnected network nanochannels and shows good capacitive performance. Profs. Andreeva and Novoselov (article number 2201904) propose a novel 2D hydrogel material consisting of the self-assembly of abundant and biodegradable thermo-responsive hydroxypropyl cellulose hydrogel and 2D rGO nanolayers. The composite was developed to produce thermal stimuli and construct an electro-thermo-controlled valve for regulating water transfer. A review by Profs. Li Lin and Zhongfan Liu (article number 2203179) provides a detailed summary of graphene doping strategies for modulating the electronic properties of graphene and a perspective on the applications of doped graphene in optoelectronics. Some reviews have advanced characterization techniques for understanding the structure–property relationship as their main topic. Prof. Lei Fu and Prof. Mengqi Zeng (article number 2202469) introduce how synchrotron radiation techniques can monitor the structural evolution of graphene at different interfaces, such as layer number, interlayer spacing, and stacking order. Prof. Robert Young and Dr. Zheling Li (article number 2202373) monitor the evolution of stress concentration ahead of crack tips in single-crystal graphene flakes based on Raman spectroscopy and, in particular, Raman mapping. Profs. Wanlin Guo and Zhuhua Zhang (article number 2203866) report colossal mechanocaloric effects in superelastic graphene architectures, and the estimated adiabatic temperature change can reach a record value of 155 K under a compressive stress of 0.7 GPa in a wide temperature range. Finding new "killer" applications by virtue of irreplaceable functionalities and performance is the driving force of graphene industrialization in the future. The design of graphene materials with high electrical conductivity and high permeability, such as graphene films, fabrics, and composites, for applications in electromagnetic wave shielding and absorption is discussed in the aforementioned review by Chao Gao et al. (article number 2204591). Prof. Quan-Hong Yang et al. (article number 2204272) presents an overview of graphene assemblies, membranes, and powders for advanced batteries, and summarize the applications of graphene in compact energy storage, thermal management, and conductive networks for practical energy devices. Prof. Qiang Zhang et al. (article number 2204755) summarize the recent progress in emerging graphene derivatives and analogs for electrocatalysis, and the material requirements for efficient electrocatalysis and graphene advantages are detailed in this review. Prof. Novoselov, Dr. Zheling Li, and Dr. Boyang Mao (article number 2205934) provide a summary of the progress on the graphene-based smart textiles with special emphasis on thermal management and flame retardancy, covering mechanics, material developments, fabric designs, and on-body applications. Profs. Wencai Ren and Huiming Cheng (article number 2203115) systematically review recent advances and significant developments in CVD-grown graphene films for flexible optoelectronics, including photodetectors, organic solar cells, and light-emitting diodes. The ability to fabricate large-area graphene devices on flexible substrates, work-function tuning, and interfacial control of CVD graphene for high-performance devices are discussed in the aforementioned review. Profs. Il Jeon and Shigeo Maruyama (article number 2204594) provides a summary of the applications of graphene and carbon nanotubes in perovskite solar cells. Versatile roles and the synthetic method of the nanocarbons in perovskite solar cells are provided in the review, covering the material growth methods, and functions as transparent electrodes, charge-transporting layers, interfacial layers, additives, and encapsulants. The application of suspended graphene membranes in multidimensional electron microscopy (EM) imaging is highlighted in a review by Prof. Hailin Peng et al. (article number 2202502). Various approaches to producing suspended graphene membranes and their applications in EM characterization, including high-resolution 2D imaging, cryo-EM 3D reconstruction, and 4D in situ liquid EM, are summarized in this review. Profs. Jin Zhang, Zhong Zhang, and Shuzhou Li (article number 2200937), develop a novel holey rGO/poly(p-phenylene-benzimidazole-terephthalamide) (PBIA) composite fiber with a scaffolded structure. Holey rGO functions as a clamp to effectively band plentiful PBIA chains through in-plane holes, which is capable of enhancing the mechanical performance of graphene-based fibers. Prof. Milo Shaffer (article number 2202596) reveals that polyethylene glycol-grafted graphene, which can be stabilized in water, can provide dispensable nucleant systems for protein crystallization. Prof. Yingying Zhang et al. (article number 2200162) propose a hydrophilic, breathable, biocompatible, and washable graphene-decorated electronic textile that is realized with the assistance of silk sericin; such a graphene-based textile enables the fabrication of comfortable and integrated multi-sensing textiles. Finally, we would like to express our sincere thanks to the editors of Advanced Functional Materials, in particular Dr. Muxian Shen, for initiating this special issue and handling all the papers. We would also like to thank all authors and reviewers for their great efforts in making this special issue possible. Any comment, suggestion, and feedback are always welcome at , , , , and . The authors declare no conflict of interest. Jin Zhang obtained his B.S. and M.S. degrees at Lanzhou University in 1992 and 1995, respectively, and completed his Ph.D. degree with Prof. Hulin Li and Prof. Zhongfan Liu at Lanzhou University and Peking University in 1997. After a two-year postdoctoral fellowship at the University of Leeds, UK, he returned to Peking University where he was appointed Associate Professor (2000) and promoted to Full Professor in 2006. In 2013, he was appointed as Changjiang professor. His research focuses on the controlled synthesis and spectroscopic characterization of carbon nanomaterials. Hailin Peng received his B.S. degree in chemistry from Jilin University in 2000 and his Ph.D. degree in physical chemistry from Peking University in 2005. After postdoctoral study at Stanford University from 2005 to 2009, he became an associate professor (2009) and then full professor (2014) at Peking University. Currently, his research interests focus on the synthesis and device applications of 2D materials such as graphene, nanostructured topological insulators, metal oxychalcogenides, and other novel 2D crystals. Hua Zhang is the Herman Hu Chair Professor of Nanomaterials at the City University of Hong Kong. He completed his Ph.D. at Peking University (1998). As a postdoctoral fellow, he joined Katholieke Universiteit Leuven (1999) and moved to Northwestern University (2001). After working at NanoInk Inc. (USA) and the Institute of Bioengineering and Nanotechnology (Singapore), he joined Nanyang Technological University in 2006 and moved to the City University of Hong Kong in 2019. His current research interests focus on the phase engineering of nanomaterials (PEN), especially for preparing novel metallic and 2D nanomaterials with unconventional phases, and epitaxially growing heterostructures for various applications. Shigeo Maruyama, FRSC, received his Ph.D. from the University of Tokyo in 1988. Since 2014, he has been a distinguished professor at the Department of Mechanical Engineering, the University of Tokyo. His research interests include nanoscale thermal engineering, molecular dynamics simulation, nanocarbon materials, and solar cell applications. He developed the alcohol catalytic chemical vapor deposition process for the synthesis of single-walled carbon nanotubes. He served as the president of "The Fullerenes, Nanotubes and Graphene Research Society" during 2011–2020 and is the co-chair of the steering committee of "International Conference on Science and Application of Nanotubes and Low-Dimensional Materials". Li Lin received his Chemistry B.S. degree from Shandong University in 2012 and Ph.D. degree in Physical Chemistry from Peking University in 2017. After postdoctoral study at the University of Manchester and National University of Singapore from 2018–2021, he became an assistant professor (2022) of Peking University. Currently, his research interests focus on synthesis, transfer, and applications of graphene and other novel 2D crystals.

Modeling Short-Range and Three-Membered Ring Structures in Lithium Borosilicate Glasses Using a Machine-Learning Potential

Shingo Urata. The Journal of Physical Chemistry C (2022). Cited by 26. Found by: openalex. DOI: 10.1021/acs.jpcc.2c07597

Abstract: Lithium borosilicate (LBS) glass is a prototypical lithium-ion conducting oxide glass available for an all-solid-state buttery. Nevertheless, the atomistic modeling of LBS glass using ab initio (AIMD) and classical molecular dynamics (CMD) simulations has critical limitations due to computational cost and inaccuracy in reproducing the glass microstructures, respectively. To overcome these difficulties, a machine-learning potential (MLP) was examined in this work for modeling LBS glasses using DeepMD. The glass structures obtained by this MLP possessed 4-fold coordinated boron (4 B) confirmed well with the experimental data and abundance of three-membered rings. The models were energetically more stable compared with those constructed with a functional force field even though both the models included reasonable 4 B. The results confirmed MLP to be superior to model the boron-containing glasses and address the inherent shortcomings of the AIMD and CMD. This study also discusses some limitations of MLP for modeling glasses.

Atomistic Simulation of HF Etching Process of Amorphous Si 3 N 4 Using Machine Learning Potential

Chang-Ho Hong; Sangmin Oh; Hyungmin An; Purun-hanul Kim; Yaeji Kim; Jae-hyeon Ko; J.A. Sue; Dongyean Oh; Sung-Kye Park; Seungwu Han. ACS Applied Materials & Interfaces (2024). Cited by 24. Found by: openalex. openalex_2yr: 8.382 (2026). DOI: 10.1021/acsami.4c07949

Abstract: An atomistic understanding of dry-etching processes with reactive molecules is crucial for achieving geometric integrity in highly scaled semiconductor devices. Molecular dynamics (MD) simulations are instrumental, but the lack of reliable force fields hinders the widespread use of MD in etching simulations. In this work, we develop an accurate neural network potential (NNP) for simulating the etching process of amorphous Si 3 N 4 with HF molecules. The surface reactions in diverse local environments are considered by incorporating several types of training sets: baseline structures, reaction-specific data, and general-purpose training sets. Furthermore, the NNP is refined through iterative comparisons with the density functional theory results. Using the trained NNP, we carry out etching simulations, which allow for detailed observation and analysis of key processes such as preferential sputtering, surface modification, etching yield, threshold energy, and the distribution of etching products. Additionally, we develop a simple continuum model, built from the MD simulation results, which effectively reproduces the surface composition obtained with MD simulations. By establishing a computational framework for atomistic etching simulation and scale bridging, this work will pave the way for more accurate and efficient design of etching processes in the semiconductor industry, enhancing device performance and manufacturing precision.

Structure of disordered TiO2 phases from ab initio based deep neural network simulations

Marcos F. Calegari Andrade; Annabella Selloni. Physical Review Materials (2020). Cited by 24. Found by: openalex. DOI: 10.1103/physrevmaterials.4.113803

Abstract: Amorphous ${\mathrm{TiO}}_{2}$ (a-${\mathrm{TiO}}_{2}$) is widely used in many fields, ranging from photoelectrochemistry to bioengineering, hence detailed knowledge of its atomic structure is of scientific and technological interest. Here we use an ab initio-based deep neural network potential (DP) to simulate large scale atomic models of crystalline and disordered ${\mathrm{TiO}}_{2}$ with molecular dynamics. Our DP reproduces the structural properties of all 11 ${\mathrm{TiO}}_{2}$ crystalline phases, predicts the densities and structure factors of molten and amorphous ${\mathrm{TiO}}_{2}$ with only a few percent deviation from experiments, and describes the pressure dependence of the amorphous structure in agreement with recent observations. It can be extended to model additional structures and compositions, and can be thus of great value in the study of ${\mathrm{TiO}}_{2}$-based nanomaterials.

Proton Transport in Perfluorinated Ionomer Simulated by Machine-Learned Interatomic Potential

Ryosuke Jinnouchi; Saori Minami; Ferenc Karsai; Carla Verdi; Georg Kresse. The Journal of Physical Chemistry Letters (2023). Cited by 23. Found by: openalex. DOI: 10.1021/acs.jpclett.3c00293

Abstract: Polymers are a class of materials that are highly challenging to deal with using first-principles methods. Here, we present an application of machine-learned interatomic potentials to predict structural and dynamical properties of dry and hydrated perfluorinated ionomers. An improved active-learning algorithm using a small number of descriptors allows to efficiently construct an accurate and transferable model for this multielemental amorphous polymer. Molecular dynamics simulations accelerated by the machine-learned potentials accurately reproduce the heterogeneous hydrophilic and hydrophobic domains formed in this material as well as proton and water diffusion coefficients under a variety of humidity conditions. Our results reveal pronounced contributions of Grotthuss chains consisting of two to three water molecules to the high proton mobility under strongly humidified conditions.

Machine learning interatomic potentials for amorphous zeolitic imidazolate frameworks

Nicolas Castel; Dune André; Connor Edwards; Jack D. Evans; François-Xavier Coudert. Digital Discovery (2024). Cited by 23. Found by: openalex. openalex_2yr: 5.263 (2026). DOI: 10.1039/d3dd00236e

Abstract: Accurate microscopic models of amorphous metal–organic frameworks (MOFs) are difficult to create. Machine learning potentials based on data from ab initio molecular dynamics offer a novel way to achieve this goal.

Simulation of Phase-Change-Memory and Thermoelectric Materials using Machine-Learned Interatomic Potentials: Sb2Te3

Konstantinos Konstantinou; Juraj Mavračić; Felix C. Mocanu; Stephen R. Elliott. physica status solidi (b) (2020). Cited by 21. Found by: openalex. DOI: 10.1002/pssb.202000416

Abstract: Density-functional-theory (DFT)-based, ab initio molecular dynamics (AIMD) simulations of amorphous materials generally suffer from three computer-resource-related limitations due to their O(N3) cubic scaling with model system size, N. They are limited to a maximum model size of N 500 atoms; they are limited to time scales <1 ns; and, usually, only a single model can be simulated in any one investigation. This article discusses a machine-learned, linear-scaling (O(N)), DFT-accurate interatomic potential (a Gaussian approximation potential, GAP), originally developed by Mocanu et al. [J. Phys. Chem. B 2018, 122, 8998] using a Gaussian process regression method for the ternary phase-change-memory material Ge2Sb2Te5 (GST). The chemical transferability of this GAP potential is explored in an application to the case of simulating amorphous models of the phase-change-memory and thermoelectric material Sb2Te3, an end-member of the GST compositional tie-line GeTe–Sb2Te3. The GAP-model results are compared with those obtained from conventional DFT-based AIMD simulations.

Suppression of Rayleigh Scattering in Silica Glass by Codoping Boron and Fluorine: Molecular Dynamics Simulations with Force-Matching and Neural Network Potentials

Shingo Urata; Nobuhiro Nakamura; Tomofumi Tada; Aik Rui Tan; Rafael Gómez-Bombarelli; Hideo Hosono. The Journal of Physical Chemistry C (2022). Cited by 21. Found by: openalex. DOI: 10.1021/acs.jpcc.1c10300

Abstract: Rayleigh scattering attributed to the density fluctuation of silica glass is considered as the intrinsic origin of optical loss in glass fiber. Therefore, minimizing the density fluctuation is key to improving the information and telecommunications networks. In this study, classical molecular dynamics (MD) simulations were employed to theoretically examine the effectiveness of codoping boron and fluorine for ameliorating the homogeneity of silica glass. For the MD simulations, the force-matching potential (FMP) with a Buckingham

formula was developed by optimizing the parameters to reproduce the force and energy calculated by the density functional theory (DFT). The accuracy of the FMP was confirmed via comparisons with available experimental data as well as glass models constructed using the neural network potential, which was superior in reproducing the force and energy of the DFT data to the FMP. As a result, the small amount of boron and fluorine added to the silica glass was found not to deteriorate the density fluctuation of silica glass. The additives reduce the viscosity of silica glass, which leads to a lower fictive temperature and, thus, to a better homogeneity. Consequently, the codoping of boron and fluorine was suggested as a possible solution to suppress the Rayleigh scattering of optical glass fiber.

Nature of the Amorphous–Amorphous Interfaces in Solid-State Batteries Revealed Using Machine-Learned Interatomic Potentials

Chuhong Wang; Muratahan Aykol; Tim Mueller. Chemistry of Materials (2023). Cited by 20. Found by: openalex. openalex_2yr: 6.387 (2026). DOI: 10.1021/acs.chemmater.3c00993

Abstract: Non-crystalline solid materials have attracted growing attention in energy storage for their desirable properties such as ionic conductivity, stability, and processability. However, compared to bulk crystalline materials, fundamental understanding of these highly complex metastable systems is hindered by the scale limitations of density functional theory (DFT) calculations and resolution limitations of experimental methods. To fill the knowledge gap and guide the rational design of amorphous battery materials and interfaces, we present a molecular dynamics (MD) framework based on machine-learned interatomic potentials trained on the fly to study the amorphous solid electrolyte Li 3 PS 4 and its protective coating, amorphous Li 3 B 11 O 18 . The use of machine-learned potentials allows us to simulate the materials at time and length scales that are not accessible to DFT while maintaining a near-DFT level of accuracy. This approach allows us to calculate amorphization energies, amorphous–amorphous interface energies, and the impact of the interface on lithium ion conductivity. This study demonstrates the promising role of actively learned interatomic potentials in extending the application of ab initio modeling to more complex and realistic systems such as amorphous materials and interfaces.

Ab initio molecular dynamics and high-dimensional neural network potential study of VZrNbHfTa melt

I. A. Balyakin; A. A. Yuryev; . . ; . . Rempel. Journal of Physics Condensed Matter (2020). Cited by 20. Found by: openalex. DOI: 10.1088/1361-648x/ab6f87

Abstract: The structural and dynamics properties of melts are directly related to their solidification processes, and consequently to the properties of as-cast solid alloys. Ab initio molecular dynamics (AIMD) is a powerful tool that can study both of these factors. However, the main disadvantage of this method is its low performance which is critical for simulation of the multicomponent liquids. At the same time the atomistic simulation of multicomponent liquids has found its application for prediction of the formation of high-entropy alloys-a novel class of materials with enhanced mechanical properties. An effective method to solve the problem of AIMD low performance may be the design of pair or many-body potentials for classical molecular dynamics. One of the promising approaches is high-dimensional neural networks-the method of constructing many-body potentials for classical molecular dynamics from ab initio data. Thus, in this work, the high-dimensional neural network potential for multicomponent liquid VZrNbHfTa melt was constructed. The structure of this melt obtained by AIMD and high-dimensional neural network potential was compared by analyzing partial radial distribution functions. Dynamics of the melt obtained by both methods was also compared analyzing velocity autocorrelation functions and mean-square displacement for each type of atom in multicomponent VZrNbHfTa melt. It was shown that structure and dynamics are reproduced well by high-dimensional neural network potential (HDNNP). Some differences between HDNNP- and AIMD-obtained structure and dynamics are explained by finite-size effect and lack of statistics in AIMD simulation along with inherent errors in energy and force estimations made by high-dimensional neural network potentials. Analysis of melt structure via partial radial distribution functions and chemical short range order parameters led to the conclusion that vanadium atoms are repulsed from all the atoms of another type in liquid VZrNbHfTa system, which lowers the probability of single phase disordered solid solution formation. Diffusivity in multicomponent melt was found to decrease with increasing mass and size of an atom.

Boron coordination and three-membered ring formation in sodium borate glasses: a machine-learning molecular dynamics study

Takeyuki Kato; Federica Lodesani; Shingo Urata. Journal of the American Ceramic Society (2023). Cited by 20. Found by: openalex. DOI: 10.1111/jace.19629

Abstract: Abstract Classical molecular dynamics (CMD) simulations that employ analytical force fields have been commonly utilized to investigate mechanical, chemical, and thermal properties of oxide glasses owing to their superior computational efficiency. Conversely, simple functional forms limit the accuracy in modeling complicated glass structures, specifically, in alkaline borate glasses, which exhibit boron coordination numbers that vary nonlinearly with changes in glass composition and temperature. Machine-learning potentials (MLPs), which are trained using datasets on energy and force evaluated via the density functional theory (DFT), are garnering significant attention as a novel simulation technology for enhancing the accuracy in modeling materials. Therefore, this study applied a universal MLP, PreFerred Potential (PFP) (trade-name: Matlantis), to model sodium borate glasses, and its accuracy was verified in reproducing boron coordination and ring structures by comparing its results to available experimental data. We found that PFP can quantitatively reproduce the boron coordination change with glass composition without any empirical correction, while the boron coordination in the melts at high temperatures is overestimated, even though the qualitative variation was better estimated than CMD simulations. Furthermore, the MLP could generate many 3-rings, unlike the analytical force-field. Accordingly, we demonstrated superior accuracy of the MLP in modeling alkaline borate glasses, while discussing the challenges faced in reproducing the elaborated microstructures in borate glasses.

Modeling the aqueous interface of amorphous TiO2 using deep potential molecular dynamics

Zhutian Ding; Annabella Selloni. The Journal of Chemical Physics (2023). Cited by 19. Found by: openalex. DOI: 10.1063/5.0157188

Abstract: Amorphous titanium dioxide (a-TiO2) is widely used as a coating material in applications such as electrochemistry and self-cleaning surfaces where its interface with water has a central role. However, little is known about the structures of the a-TiO2 surface and aqueous interface, particularly at the microscopic level. In this work, we construct a model of the a-TiO2 surface via a cut-melt-and-quench procedure based on molecular dynamics simulations with deep neural network potentials (DPs) trained on density functional theory data. After interfacing the a-TiO2 surface with water, we investigate the structure and dynamics of the resulting system using a combination of DP-based molecular dynamics (DPMD) and ab initio molecular dynamics (AIMD) simulations. Both AIMD and DPMD simulations reveal that the distribution of water on the a-TiO2 surface lacks distinct layers normally found at the aqueous interface of crystalline TiO2, leading to an 10 times faster diffusion of water at the interface. Bridging hydroxyls (Ti2-ObH) resulting from water dissociation decay several times more slowly than terminal hydroxyls (Ti-OwH) due to fast Ti-OwH2 → Ti-OwH proton exchange events. These results provide a basis for a detailed understanding of the properties of a-TiO2 in electrochemical environments. Moreover, the procedure of generating the a-TiO2-interface employed here is generally applicable to studying the aqueous interfaces of amorphous metal oxides.

Assessing the Reactivity of the Na 3 PS 4 Solid-State Electrolyte with the Sodium Metal Negative Electrode Using Total Trajectory Analysis with Neural-Network Potential Molecular Dynamics

Lieven Bekaert; Suzuno Akatsuka; Naoto Tanibata; Frank De Proft; Annick Hubin; Mesfin Haile Mamme; Masanobu Nakayama. The Journal of Physical Chemistry C (2023). Cited by 19. Found by: openalex. DOI: 10.1021/acs.jpcc.3c02379

Abstract: Rechargeable batteries play a central role in the global shift from fossil fuels to renewable energy. Since the commercial introduction of lithium-ion batteries in the early 1990s, recent progress is focused on the development of solid-state materials and new battery chemistries. Specifically, solid-state sodium ion batteries are an attractive alternative alongside lithium-based rechargeable batteries, with improvements in safety, lifespan, sustainability, and price. Critical to the battery performance are electrode–electrolyte interfaces since undesirable side-reactions often proceed between electrode and electrolyte materials. In addition, atomistic or electronic-level knowledge on the reactions at the interface is limited due to technical difficulties of experimental observation. Computational studies on interfacial reactivity, such as first-principles techniques, have also long been limited by computational limitations. Recent advances using neural-network potential molecular dynamics simulations are allowing significantly larger systems and longer timescales to be simulated at a significantly reduced computational cost without loss of accuracy. In this study, the chemical stability of the glass–ceramic Na 3 PS 4 solid-state electrolyte with the sodium metal electrode is investigated through a combined total trajectory analysis computational and experimental approach. PS 4 groups in the Na 3 PS 4 material were

found to decompose sequentially into PS 3, PS 2, PS, and phosphide and sulfide species through the insertion of sodium atoms. Whereas the decomposition is thermodynamically favored, it is kinetically hindered due to steric effects in the PS 3 intermediate. Machine learning-assisted analysis was found to be able to visualize the reactivity tendencies of individual element types. The formed SEI layer exhibited a good chemical stability and a low electronic conductivity. These findings provide new design principles to optimize and develop new solid-state electrolytes with an increased chemical stability toward the sodium metal electrode.

Binding energies and sticking coefficients of H2 on crystalline and amorphous CO ice

Germán Molpeceres; Viktor Zaverkin; Naoki Watanabe; Johannes Kästner. Astronomy and Astrophysics (2021). Cited by 18. Found by: openalex. openalex_2yr: 5.028 (2026). DOI: 10.1051/0004-6361/202040023

Abstract: Context. Molecular hydrogen (H2) is the most abundant interstellar molecule and plays an important role in the chemistry and physics of the interstellar medium. The interaction of H2 with interstellar ices is relevant for several processes (e.g., nuclear spin conversion and chemical reactions on the surface of the ice). To model surface processes, quantities such as binding energies and sticking coefficients are required. Aims. We provide sticking coefficients and binding energies for the H2/CO system. These data are absent in the literature so far and could help modelers and experimentalists to draw conclusions on the H2/CO interaction in cold molecular clouds. Methods. Ab initio molecular dynamics simulations, in combination with neural network potentials, were employed in our simulations. Atomistic neural networks were trained against density functional theory calculations on model systems. We sampled a wide range of H2 internal energies and three surface temperatures. Results. Our results show that the binding energy for the H2/CO system is low on average, − 157 K for amorphous CO and −266 K for crystalline CO. This carries several implications for the rest of the work. H2 binding to crystalline CO is stronger by 109 K than to amorphous CO, while amorphous CO shows a wider H2 binding energy distribution. Sticking coefficients are never unity and vary strongly with surface temperature, but less so with ice phase, with values between 0.95 and 0.17. With the values of this study, between 17 and 25% of a beam of H2 molecules at room temperature would stick to the surface, depending on the temperature of the surface and the ice phase. Residence times vary by several orders of magnitude between crystalline and amorphous CO, with the latter showing residence times on the order of seconds at 5 K. H2 may diffuse before desorption in amorphous ices, which might help to accommodate it in deeper binding sites. Conclusions. Based on our results, a significant fraction of H2 molecules will stick on CO ice under experimental conditions, even more so under the harsh conditions of prestellar cores. However, with the low H2–CO binding energies, residence times of H2 on CO ice before desorption are too short to consider a significant population of H2 molecules on pure CO ices. Diffusion is possible in a time window before desorption, which might help accommodate H2 on deeper binding sites, which would increase residence times on the surface.

Density dependence of thermal conductivity in nanoporous and amorphous carbon with machine-learned molecular dynamics

Yanzhou Wang; Zheyong Fan; Ping Qian; A. Miguel; Tapio Ala-Nissilä. Physical review. B./Physical review. B (2025). Cited by 17. Found by: openalex. DOI: 10.1103/physrevb.111.094205

Abstract: Disordered forms of carbon are an important class of materials for applications such as thermal management. However, a comprehensive theoretical understanding of the structural dependence of thermal transport and the underlying microscopic mechanisms is lacking. Here we study the structure-dependent thermal conductivity of disordered carbon by employing molecular dynamics (MD) simulations driven by a machine-learned interatomic potential based on the efficient neuroevolution potential approach. Using large-scale MD simulations, we generate realistic nanoporous carbon (NP-C) samples with densities varying from 0.3 to 1.5 g ${\mathrm{cm}}^{\ensuremath{-}3}$ dominated by ${sp}^{2}$ motifs, and amorphous carbon (a-C) samples with densities varying from 1.5 to 3.5 g ${\mathrm{cm}}^{\ensuremath{-}3}$ exhibiting mixed ${sp}^{2}$ and ${sp}^{3}$ motifs. Structural properties including short- and medium-range order are characterized by the atomic coordination, pair correlation function, angular distribution function, and structure factor. Using the homogeneous nonequilibrium MD method and the associated quantum-statistical correction scheme, we predict a linear and a superlinear density dependence of thermal conductivity for NP-C and a-C, respectively, in good agreement with relevant experiments. The distinct density dependences are attributed to the different impacts of the ${sp}^{2}$ and ${sp}^{3}$ motifs on the spectral heat capacity, vibrational mean free paths, and group velocity. We additionally highlight the significant role of structural order in regulating the thermal conductivity of disordered carbon.

PotentialMind: Graph Convolutional Machine Learning Potential for Sb–Te Binary Compounds of Multiple Stoichiometries

Guanjie Wang; Yuqi Sun; Jian Zhou; Zhimei Sun. The Journal of Physical Chemistry C (2023). Cited by 17. Found by: openalex. DOI: 10.1021/acs.jpcc.3c07110

Abstract: Machine learning potential (MLP) has emerged as a powerful tool in materials research and design. However, most MLP methods rely only on a single descriptor generated by mathematical functions instead of mapping the three-dimensional space of the materials structure, and thus this type of potential is typically limited to specific compositions. In this research, we present graph convolutional machine learning potential (GCMLP) software, termed PotentialMind, which can transform three-dimensional atomic structures into vectors comprising nodes, edges, and weights based on multiple descriptors. Using Sb–Te phase change materials as examples, a model named GCMLP-ST suitable for 12 stoichiometries of Sb–Te compounds has been constructed, whose root-mean-square errors for energy and forces are, respectively, 4.51 and 73.13 meV/Å for training data sets and are, respectively, 4.97 and 76.25 meV/Å for unfamiliar testing data sets. Moreover, for the energy-volume curves and radius distribution function by molecular dynamics, the GCMLP-ST model with 10,000 atoms exhibits good agreement with the ab initio molecular dynamics (AIMD) results across crystalline, liquid, and amorphous phases for the six representative Sb–Te material systems, which also exhibit 50 times the computational efficiency of AIMD. With this framework, the architecture of the machine learning model can be customized by deep and transfer learning, extending to other material systems. In addition, benefiting from the high efficiency of PotentialMind molecular dynamics (PMMD), it can be used for real devices, spanning tens of nanoseconds and comprising millions of atoms under different programming conditions that are impossible with AIMD simulations.

Crystal-like thermal transport in amorphous carbon

Jaeyun Moon; Zhiting Tian. npj Computational Materials (2025). Cited by 17. Found by: openalex. DOI: 10.1038/s41524-025-01625-2

Abstract: Thermal transport in amorphous carbon has attracted immense attention due to its extreme thermal properties: It has been reported to have among the highest thermal conductivity for bulk amorphous solids up to ~37 W m −1 K −1 , comparable to crystalline sapphire (-Al 2 O 3). However, mechanism behind the high thermal conductivity remains elusive due to many variables at play. In this work, we perform large-scale (~10 5 atoms) molecular dynamics simulations utilizing a machine learning potential based on neural networks with first-principles accuracy. Through spectral decomposition of thermal conductivity which enables a quantum correction to classical heat capacity, we find that propagating vibrational excitations govern thermal transport in amorphous carbon (~100 % of thermal conductivity) in sharp contrast to the convention that diffusive vibrational excitations dominate thermal transport in amorphous solids. This remarkable behavior resembles thermal transport in simple crystals. Our work, therefore, provides a perspective that deepens our understanding of intermediate thermal transport mechanisms between the two ends of spectrum of solids: crystalline and amorphous solids.

Plasma Oxidation of Copper: Molecular Dynamics Study with Neural Network Potentials

Yantao Xia; Philippe Sautet. ACS Nano (2022). Cited by 16. Found by: openalex. openalex_2yr: 16.894 (2026). DOI: 10.1021/acsnano.2c07712

Abstract: The formation of thin oxide films is of significant scientific and practical interest. In particular, the semiconductor industry is interested in developing a plasma atomic layer etching process to pattern copper, replacing the dual Damascene process. Using a nonthermal oxygen plasma to convert the metallic copper into copper oxide, followed by a formic acid organometallic reaction to etch the copper oxide, this process has shown great promise. However, the current process is not optimal because the plasma oxidation step is not self-limiting, hampering the degree of thickness control. In the present study, a neural network potential for the binary interaction between copper and oxygen is developed and validated against first-principles calculations. This potential covers the entire range of potential energy surfaces of metallic copper, copper oxides, atomic oxygen, and molecular oxygen. The usable kinetic energy ranges from 0 to 20 eV. Using this potential, the plasma oxidation of copper surfaces was studied with large-scale molecular dynamics at atomic resolution, with an accuracy approaching that of the first principle calculations. An amorphous layer of CuO is formed on Cu, with thicknesses reaching 2.5 nm. Plasma is found to create an intense local heating effect that rapidly dissipates across the thickness of the film. The range of this heating effect depends on the kinetic energy of the ions. A higher ion energy leads to a longer range, which sustains faster-than-thermal rates for longer periods of time for the oxide growth. Beyond the range of this agitation, the growth is expected to be limited to the thermally activated rate. High-frequency, repeated ion impacts result in a microannealing effect that leads to a quasicrystalline oxide beneath the amorphized layer. The crystalline layer slows down oxide growth. Growth rate is fitted to the temperature gradient due to ion-induced thermal agitations, to obtain an apparent activation energy of 1.0 eV. A strategy of lowering the substrate temperature and increasing plasma power is proposed as being favorable for more self-limited oxidation.

Generalized modeling of carbon film deposition growth via hybrid MD/MC simulations with machine-learning potentials

Yutao Liu; Tinghong Gao; Qingquan Xiao; Yunjun Ruan; Qian Chen; Bei Wang; Jin Huang. npj Computational Materials (2025). Cited by 16. Found by: openalex. DOI: 10.1038/s41524-025-01781-5

Abstract: Theoretical investigations into the controlled growth of carbon films are essential for guiding the experimental fabrication of carbon-based devices. However, accurately simulating the deposition process remains a significant challenge. In this work, we developed an active learning workflow to construct a machine learning-based neuroevolution potential (NEP) for investigating carbon atoms deposition growth on various substrates. By integrating molecular dynamics and time-stamped force-biased Monte Carlo simulations, we studied the growth of amorphous carbon films on Si(111) and found that deposition energy strongly influenced bonding topology and film morphology. The NEP reliably captured the surface diffusion of carbon atoms, the formation of carbon chains and rings. We revealed a new growth mechanism of adhesion-driven growth at low energies and peening-induced densification at high energies of carbon atoms on Si(111) substrates. To evaluate the transferability of fitting workflow, we extended the NEP to simulate carbon deposition on Cu(111) and Al 2 O 3 (0001) surface. Simulation results demonstrate that the NEP can reproduce the subprocesses of graphene formation during carbon growth on the Cu(111) substrate. In contrast, only disordered carbon chains are observed on the Al 2 O 3 (0001) substrate. This work provides atomistic insights into the growth mechanisms of carbon films on representative substrates and establishes a robust computational framework for synthesis of diverse carbon nanostructures.

The ab initio non-crystalline structure database: empowering machine learning to decode diffusivity

Hui Zheng; Eric Sivonxay; Rasmus Christensen; Max C. Gallant; Ziyao Luo; Matthew J. McDermott; Patrick Huck; Morten M. Smedskjær; Kristin A. Persson. npj Computational Materials (2024). Cited by 16. Found by: openalex. DOI: 10.1038/s41524-024-01469-2

Abstract: Abstract Non-crystalline materials exhibit unique properties that make them suitable for various applications in science and technology, ranging from optical and electronic devices and solid-state batteries to protective coatings. However, data-driven exploration and design of non-crystalline materials is hampered by the absence of a comprehensive database covering a broad chemical space. In this work, we present the largest computed non-crystalline structure database to date, generated from systematic and accurate ab initio molecular dynamics (AIMD) calculations. We also show how the database can be used in simple machine-learning models to connect properties to composition and structure, here specifically targeting ionic conductivity. These models predict the Li-ion diffusivity with speed and accuracy, offering a cost-effective alternative to expensive density functional theory (DFT) calculations. Furthermore, the process of computational quenching non-crystalline structures provides a unique sampling of out-of-equilibrium structures, energies, and force landscape, and we anticipate that the corresponding trajectories will inform future work in universal machine learning potentials, impacting design beyond that of non-crystalline materials. In addition, combining diffusion trajectories from our dataset with models that predict liquidus viscosity and melting temperature could be utilized to develop models for predicting glass-forming ability.

Thermal conductivity of Li 3 PS 4 solid electrolytes with ab initio accuracy

Davide Tisi; Federico Grasselli; Lorenzo Gigli; Michele Ceriotti. Physical Review Materials (2024). Cited by 15. Found by: openalex. DOI: 10.1103/physrevmaterials.8.065403

Abstract: The vast amount of computational studies on electrical conduction in solid-state electrolytes is not mirrored by comparable efforts addressing thermal conduction, which has been scarcely investigated despite its relevance to thermal management and (over)heating of batteries. The reason for this lies in the complexity of the calculations: on one hand, the diffusion of ionic charge carriers makes lattice methods formally unsuitable, due to the lack of equilibrium atomic positions needed for normal-mode expansion. On the other hand, the prohibitive cost of large-scale molecular dynamics (MD) simulations of heat transport in large systems at ab initio levels has hindered the use of MD-based methods. In this paper, we leverage recently developed machine-learning potentials targeting different ab initio functionals (PBEsol, ${\mathrm{r}}^{2}\text{SCAN}$, PBE0) and a state-of-the-art formulation of the Green-Kubo theory of heat transport in multicomponent systems to compute the thermal conductivity of a promising solid-state electrolyte, ${\mathrm{Li}}_{3}{\mathrm{PS}}_{4}$, in all its polymorphs ($\ensuremath{\alpha}, \ensuremath{\beta}$, and $\ensuremath{\gamma}$). By comparing MD estimates with lattice methods on the low-temperature, nondiffusive $\ensuremath{\gamma}\ensuremath{-}{\mathrm{Li}}_{3}{\mathrm{PS}}_{4}$, we highlight strong anharmonicities and negligible nuclear quantum effects, hence further justifying MD-based methods even for nondiffusive phases. Finally, for the ion-conducting $\ensuremath{\alpha}$ and $\ensuremath{\beta}$ phases, where the multicomponent Green-Kubo MD approach is mandatory, our simulations indicate a weak temperature dependence of the thermal conductivity, a glass-like behavior due to the effective local disorder characterizing these Li-diffusing phases.

Structural Evolution of C 60 Molecular Crystal Predicted by Neural Network Potential

Kun Ni; Fei Pan; Yanwu Zhu. Advanced Functional Materials (2022). Cited by 14. Found by: openalex. openalex_2yr: 14.975 (2026). DOI: 10.1002/adfm.202203894

Abstract: Abstract The preparation of fullerene and a wide range of derivatives has attracted much attention in past decades. Understanding the structural evolution starting from fullerene is critical to guide the experimental exploration but has been paid less attention yet. Using ab initio molecular dynamics or global structural search algorithm accompanied with density functional theory calculations can give a glance of the potential energy surface (PES), but suffers high cost and low efficiency. Herein, by using neural network potentials and stochastic surface walking global structural search method, an accelerated energy calculation, and the higher efficient PES mapping starting from C 60 molecular crystal are reported. The structural evolution is found to follow an order of molecular crystal → polymers → opened caged ordered structures → 3D) curved carbon and graphite under zero pressure. Under high pressures, e.g., 50 GPa, the evolution follows a pathway of fullerene → sp 2 and sp 3 hybrid carbon → sp 3 amorphous carbon and crystal diamond. The electronic property calculation of the obtained structures in the evolution pathway shows a band gap depending on the order parameter of the generated structures, suggesting that more novel carbons can be potentially prepared in experiments, starting from C 60 .

Role of hydrogen-doping for compensating oxygen-defect in non-stoichiometric amorphous In2O3—x: Modeling with a machine-learning potential

Shingo Urata; Nobuhiro Nakamura; Junghwan Kim; Hideo Hosono. Journal of Applied Physics (2023). Cited by 13. Found by: openalex. DOI: 10.1063/5.0149199

Abstract: Transparent amorphous oxide semiconductors (TAOSs) are essential materials and ushering in information and communications technologies. The performance of TAOS depends on the microstructures relating to the defects and dopants. Density functional theory (DFT) is a powerful tool to understand the structure–property relationship relating to electronic state; however, the computation of DFT is expensive, which often hinders appropriate structural modeling of amorphous materials. This study, thus, applied machine-learning potential (MLP) to reproduce the DFT level of accuracy with enhanced efficiency, to model amorphous In2O3 (a-In2O3), instead of expensive molecular dynamics (MD) simulations with DFT. MLP-MD could reproduce a-In2O3 structure closer to the experimental data in comparison with DFT-MD and classical MD simulations with an analytical force field. Using the relatively large models obtained by the MLP-MD simulations, it was unraveled that the anionic hydrogen atoms bonding to indium atoms attract electrons instead of the missing oxygen and remedy the optical transparency of the oxygen deficient a-In2O3. The preferential formation of metal–H bonding through the reaction of oxygen vacancy was demonstrated as analogous to InGaZnOx thin films [Joonho et al., Appl. Phys. Lett. 110, 232105 (2017)]. The present simulation suggests that the same mechanism works in a-In2O3, and our finding on the structure–property relationship is informative to clarify the factors affecting the optical transparency of In-based TAOS thin films.

Nuclear quantum effects in the acetylene:ammonia plastic co-crystal

Atul C. Thakur; Richard C. Remsing. The Journal of Chemical Physics (2024). Cited by 13. Found by: openalex. DOI: 10.1063/5.0179161

Abstract: Organic molecular solids can exhibit rich phase diagrams. In addition to structurally unique phases, translational and rotational degrees of freedom can melt at different state points, giving rise to partially disordered solid phases. The structural and dynamic disorder in these materials can have a significant impact on the physical properties of the organic solid, necessitating a thorough understanding of disorder at the atomic scale. When these disordered phases form at low temperatures, especially in crystals with light nuclei, the prediction of material properties can be complicated by the importance of nuclear quantum effects. As an example, we investigate nuclear quantum effects on the structure and dynamics of the orientationally disordered, translationally ordered plastic phase of the acetylene:ammonia (1:1) co-crystal that is expected to exist on the surface of Saturn's moon Titan. Titan's low surface temperature (90 K) suggests that the quantum mechanical behavior of nuclei may be important in this and other molecular solids in these environments. By using neural network potentials combined with ring polymer molecular dynamics simulations, we show that nuclear quantum effects increase orientational disorder and rotational dynamics within the acetylene:ammonia (1:1) co-crystal by weakening hydrogen bonds. Our results suggest that nuclear quantum effects are important to accurately model molecular solids and their physical properties in low-temperature environments.

Development and Validation of Neural Network Potentials for Multicomponent Oxide Glasses

Ryuki Kayano; Yaohiro Inagaki; Ryuta Matsubara; Keisuke Ishida; Takahiro Ohkubo. The Journal of Physical Chemistry C (2024). Cited by 12. Found by: openalex. DOI: 10.1021/acs.jpcc.4c04604

Abstract: A neural network potential (NNP) for molecular dynamics simulation (MD) of multicomponent oxide glasses was developed with a particular focus on structural reproducibility. The NNP was constructed through pretraining using a density functional theory (DFT) data set provided by the Open Catalysis project and fine-tuning based on a data set of nine-component glasses. Thorough validation was performed by comparing a glass structure derived using neural network molecular dynamics simulation (NNMD) and the NNP developed here with previous experimental and DFT-MD data. The accuracy of the NNMD was investigated in terms of the local structure of the glass, in comparison with a glass derived from MD with conventional potentials. The composition dependence of the local structure in Na 2 O–SiO 2 and Na 2 O–B 2 O 3 glass systems was well-reproduced for the NNMD-derived glass. The ability to reproduce the glass structure was demonstrated in the four-coordinated boron population, by formation of superstructures in alkali borate glasses, and by the Al local structure in a novel Al-rich binary aluminoborosilicate glass. The importance of pretraining was investigated by comparing NNMD results obtained using the NNP developed with and without pretraining. Although better metric scores were obtained for the NNP without pretraining, the resulting structure was not realistic. This is an important lesson, showing that the metric score alone is inadequate to determine the accuracy of an NNP for glasses. Finally, the developed NNMD was used to model a reference nuclear waste glass (60.1SiO 2 –3.84Al 2 O 3 –15.97B 2 O 3 –12.65Na 2 O–2.87CaO–2.86MgO–1.72ZrO 2), and the charge compensation mechanism of the cations was investigated.

Atomic Structure of Na 4 P 2 S 7 Glass Solid Electrolyte: Fine-Tuning Machine Learning Potentials for Enhanced Accuracy

Marco Bertani; Alfonso Pedone. The Journal of Physical Chemistry C (2025). Cited by 12. Found by: openalex. DOI: 10.1021/acs.jpcc.5c01857

Abstract: High Resolution Image Download MS PowerPoint Slide Understanding the atomic structure of glassy solid electrolytes (GSEs) is essential for optimizing their performance in all-solid-state sodium batteries (ASSSBs). In this study, we investigate the structural organization of Na 4 P 2 S 7 glass, a representative composition of sulfide-based GSEs, using ab initio molecular dynamics (AIMD) and machine learning interatomic potentials (MLIPs). We assess the accuracy of the pretrained MACE foundation models (MP0-small, MP0-medium, and MP0-large) in reproducing the short- and medium-range order and demonstrate that fine-tuning with system-specific density functional theory (DFT) data significantly improves structural predictions. Our analysis of pair distribution functions (PDFs) and bond angle distributions (BADs) reveals that the pre-trained models overestimate the presence of homonuclear P–P bonds and predict unrealistic edge-sharing P 2 S 6 units, artifacts that are corrected upon fine-tuning. The refined MLIPs also provide a better description of sulfur chain formation, which strongly influences sodium ion transport. Additionally, we validate the structural accuracy through 31 P NMR spectra calculations, showing improved agreement with experimental data. The vibrational density of states (VDOS) and elastic properties further highlight the advantages of fine-tuned models in capturing the structural dynamics of Na 4 P 2 S 7 glass. This work establishes a robust methodology for simulating sulfide-based GSEs and paves the way for future investigations into composition-structure–property relationships using MLIPs.

Enhanced heat transport in amorphous silicon via microstructure modulation

Youtian Li; Yangyu Guo; Shiyun Xiong; Hong-Liang Yi. International Journal of Heat and Mass Transfer (2024). Cited by 12. Found by: openalex. DOI: 10.1016/j.ijheatmasstransfer.2023.125167

Elaboration of a neural-network interatomic potential for silica glass and melt

Salomé Trillot; Julien Lam; Simona Ispas; Akshay Krishna Ammothum Kandy; Mark E. Tuckerman; Nathalie Tarrat; Magali Benoit. Computational Materials Science (2024). Cited by 12. Found by: openalex. openalex_2yr: 3.958 (2026). DOI: 10.1016/j.commatsci.2024.112848

Abstract: Because of its importance in various aspects of everyday life, silica is a material that has been the subject of extensive research. Studies on its amorphous phase have particularly benefited from the contribution of atomistic simulations to understand the close relationships between its structure and properties. In this context, the main difficulty lies in the compromise that had to be made between the precision of the interactions that need to be computed at an ab initio level and the important statistics required to describe disorder. With the advent of machine learning approaches, it is now possible to couple accuracy and statistics by using interatomic potentials trained on ab initio databases. This opens up unprecedented prospects for studies where calculation accuracy, system size and trajectory length are critical. In this work, we propose a machine learning potential for silica obtained from a neural network trained on a database consisting of a few hundred configurations extracted from an ab initio molecular dynamics trajectory at the Density Functional Theory (DFT) level of a silica liquid at high temperature and under pressure. We show that this potential is sufficiently accurate to describe the liquid and amorphous phases of silica, and that it is also transferable to glasses under moderate pressure and, more surprisingly, to certain crystalline phases.

Machine Learning Interatomic Potential for Modeling the Mechanical and Thermal Properties of Naphthyl-Based Nanotubes

Hugo X. Rodrigues; Hudson R. Armando; Daniel A. da Silva; João Paulo C. L. da Costa; Luiz Antônio Ribeiro; Marcelo Lopes Pereira. Journal of Chemical Theory and Computation (2025). Cited by 12. Found by: openalex. DOI: 10.1021/acs.jctc.4c01578

Abstract: Two-dimensional (2D) nanomaterials are at the forefront of potential technological advancements. Carbon-based materials have been extensively studied since synthesizing graphene, which revealed properties of great interest for novel applications across diverse scientific and technological domains. New carbon allotropes continue to be explored theoretically, with several successful synthesis processes for carbon-based materials recently achieved. In this context, this study investigates the mechanical and thermal properties of DHQ-based monolayers and nanotubes, a carbon allotrope characterized by 4-, 6-, and 10-membered carbon rings, with a potential synthesis route using naphthalene as a molecular precursor. A machine-learned interatomic potential (MLIP) was developed to explore this nanomaterial's mechanical and thermal behavior at larger scales than those accessible through the first-principles calculations. The MLIP was trained on data derived from the DFT/PBE (density functional theory/Perdew-Burke-Ernzerhof) level using ab initio molecular dynamics (AIMD). Classical molecular dynamics (CMD) simulations, employing the trained MLIP, revealed that Young's modulus of DHQ-based nanotubes ranges from 127 to 243 N/m, depending on chirality and diameter, with fracture occurring at strains between 13.6 and 17.4% of the initial length. Regarding thermal response, a critical temperature of 2200 K was identified, marking the onset of a transition to an amorphous phase at higher temperatures.

Neural Network Force Fields for Metal Growth Based on Energy Decompositions

Qin Hu; Mouyi Weng; Xin Chen; Shucheng Li; Feng Pan; Lin-Wang Wang. The Journal of Physical Chemistry Letters (2020). Cited by 12. Found by: openalex. DOI: 10.1021/acs.jpclett.9b03780

Abstract: A method using machine learning (ML) is proposed to describe metal growth for simulations, which retains the accuracy of ab initio density functional theory (DFT) and results in a thousands-fold reduction in the computational time. This method is based on atomic energy decomposition from DFT calculations. Compared with other ML methods, our energy decomposition approach can yield much more information with the same DFT calculations. This approach is employed for the amorphous sodium system, where only 1000 DFT molecular dynamics images are enough for training an accurate model. The DFT and neural network potential (NNP) are compared for the dynamics to show that similar structural properties are generated. Finally, metal growth experiments from liquid to solid in a small and larger system are carried out to demonstrate the ability of using NNP to simulate the real growth process.

Dynamic mesophase transition induces anomalous suppressed and anisotropic phonon thermal transport

Linfeng Yu; Kexin Dong; Qi Yang; Yi Zhang; Zheyong Fan; Xiong Zheng; Huimin Wang; Zhenzhen Qin; Guangzhao Qin. npj Computational Materials (2024). Cited by 12. Found by: openalex. DOI: 10.1038/s41524-

024-01442-z

Abstract: The physical/chemical properties undergo significant transformations in the different states arising from phase transition. However, due to the lack of a dynamic perspective, transitional mesophases are largely underexamined, constrained by the high resource burden of first principles. Here, using molecular dynamics (MD) simulations empowered by the machine-learning potential, we proffer an innovative paradigm for phase transition: regulating the thermal transport properties via the transitional mesophase triggered by a uniaxial force field. We investigate the mechanical, electrical, and thermal transport properties of the two-dimensional carbon allotrope of Janus-graphene with strain-engineered phase transition. Notably, we found that the transitional mesophase significantly suppresses the thermal conductivity and induces strong anisotropy near the phase transition point. Through machine-learning-driven MD simulations, we achieved high-precision atomic-level simulations of Janus-graphene. The results show that thermal vibration-induced intermediate amorphous or interfacial phases induce strong and anisotropic interfacial thermal resistance. The investigation not only endows us with a novel perspective on mesophases during phase transitions but also enhances our holistic comprehension of the evolution of material properties.

Accurate prediction of structural and mechanical properties on amorphous materials enabled through machine-learning potentials: A case study of silicon nitride

Ganesh Kumar Nayak; Prashanth Srinivasan; Juraj Todt; R. Daniel; Paolo Nicolini; David Holec. Computational Materials Science (2025). Cited by 11. Found by: openalex. openalex_2yr: 3.958 (2026). DOI: 10.1016/j.commatsci.2024.113629

Abstract: Ab initio calculations represent the technique of election to study material system, however, they present severe limitations in terms of the size of the system that can be simulated. Often, the results in the simulation of amorphous materials depend dramatically on the size of the system. Here, we overcome this limitation for the specific case of mechanical properties of amorphous silicon nitride (a-Si 3 N 4) by training a machine learning (ML) interatomic model. Our strategy is based on the generation of targeted training sets, which also include deliberately stressed structures. Using this dataset, we trained a moment tensor potential (MTP) for a-Si 3 N 4 . We show that molecular dynamics simulations using the ML model on much larger systems yield elastically isotropic response and can reproduce experimental measurement. To do so, models containing at least 3 , 500 atoms are necessary. The Young's modulus calculated from the MTP at room temperature is 220 GPa , which is very well in agreement with the nanoindentation measurement. Our study demonstrates the broader impact of machine learning potentials for predicting structural and mechanical properties, even for complex amorphous structures.

Efficient Training of Machine Learning Potentials by a Randomized Atomic-System Generator

Young-Jae Choi; Seung-Hoon Jhi. The Journal of Physical Chemistry B (2020). Cited by 11. Found by: openalex. DOI: 10.1021/acs.jpcb.0c05075

Abstract: Machine learning potentials provide an efficient and comprehensive tool to simulate large-scale systems inaccessible by conventional first-principles methods still in a similar level of accuracy. One critical issue in constructing machine learning potentials is to build training data sets cost-effectively that can represent the potential energy surface in a wide range of configurations. We develop a scheme named randomized atomic-system generator (RAG) to produce the training sets that widely cover the potential energy surface by combining the random sampling and structural optimization. We apply the scheme to construct the machine learning potentials for simulation of chalcogen-based phase change materials. Constructed machine learning potentials successfully simulate the dynamics of melting and crystallization processes of binary GeTe at a level comparable to first-principles simulations. The visual analysis shows that the RAG-generated training set represents the crystallization process including the amorphous phases. From the velocity autocorrelation function obtained from the molecular dynamics simulations, we calculate the phonon density of states to analyze the vibrational properties during crystallization.

Structural and mechanical properties of monolayer amorphous carbon and boron nitride

Xi Zhang; Yutian Zhang; Yun-Peng Wang; Yun-Peng Wang; Shixuan Du; Shixuan Du; Yuyang Zhang; Yuyang Zhang; Sokrates T. Pantelides. Physical review. B./Physical review. B (2024). Cited by 11. Found by: openalex. DOI: 10.1103/physrevb.109.174106

Abstract: Amorphous materials exhibit various characteristics that are not featured by crystals and can sometimes be tuned by their degree of disorder (DOD). Here, we report results on the mechanical properties of monolayer amorphous carbon (MAC) and monolayer amorphous boron nitride (maBN) with different DOD. The pertinent structures are obtained by kinetic Monte Carlo (kMC) simulations using machine-learning potentials with density-functional theory-level accuracy. An intuitive order parameter, namely, the areal fraction ${F}_{\mathrm{x}}$ occupied by crystallites within the continuous random network, is proposed to describe the DOD. We find that ${F}_{\mathrm{x}}$ captures the essence of the DOD: Samples with the same ${F}_{\mathrm{x}}$ but different sizes and arrangements of crystallites, obtained using two distinct kMC procedures, have virtually identical radial distribution functions as well as bond-length and bond-angle distributions. Furthermore, by simulating the fracture process with molecular dynamics, we found that the mechanical responses of MAC and maBN before fracture are mainly determined by ${F}_{\mathrm{x}}$ and are insensitive to the sizes and specific arrangements and to some extent the numbers and area distributions of the crystallites. The behavior of cracks in the two materials is analyzed and found to mainly propagate in meandering paths in the continuous random network region and to be influenced by crystallites in distinct ways that toughen the material. The present results reveal the relation between structure and mechanical properties in amorphous monolayers and may provide a universal toughening strategy for two-dimensional materials.

Structure and Crystallization Kinetics of As-Deposited Films of the GeTe Phase Change Compound from Atomistic Simulations

Simone Perego; Daniele Dragoni; Silvia Gabardi; Davide Campi; Marco Bernasconi. physica status solidi (RRL) - Rapid Research Letters (2022). Cited by 11. Found by: openalex. DOI: 10.1002/pssr.202200433

Abstract: Models of the amorphous phase of the GeTe compound have been generated by depositing individual atoms on a substrate by molecular dynamics simulations. This material is of interest for applications in phase change memories. The kinetic energy of the atoms impinging on the surface is in the range 1.5–10 eV which is typical of magnetron sputtering growth. The simulations with up to 8000 atoms exploit a machine learning potential devised previously by the group. The structural properties of the films depend very little on the energy of the impinging atoms in the range 1.5–10 eV and they are also very similar to those of amorphous models generated by quenching from the melt, but for a larger fraction of Ge Ge bonds and Ge atoms in tetrahedral geometries in the films. The crystal nucleation and growth are then simulated at 600 K which is close to the temperature of maximal crystal growth velocity exploited in the memory devices. Despite the structural differences mentioned above, the kinetics of crystallization at 600 K is very similar for the as-deposited and melt-quenched amorphous models.

High-Accuracy Neural Network Interatomic Potential for Silicon Nitride

Hui Xu; Zeyuan Li; Zhaofu Zhang; Sheng Liu; Shengnan Shen; Yuzheng Guo. Nanomaterials (2023). Cited by 11. Found by: openalex. DOI: 10.3390/nano13081352

Abstract: In the field of machine learning (ML) and data science, it is meaningful to use the advantages of ML to create reliable interatomic potentials. Deep potential molecular dynamics (DEEPMD) are one of the most useful methods to create interatomic potentials. Among ceramic materials, amorphous silicon nitride (SiNx) features good electrical insulation, abrasion resistance, and mechanical strength, which is widely applied in industries. In our work, a neural network potential (NNP) for SiNx was created based on DEEPMD, and the NNP is confirmed to be applicable to the SiNx model. The tensile tests were simulated to compare the mechanical properties of SiNx with different compositions based on the molecular dynamic method coupled with NNP. Among these SiNx, Si3N4 has the largest elastic modulus (E) and yield stress (s), showing the desired mechanical strength owing to the largest coordination numbers (CN) and radial distribution function (RDF). The RDFs and CNs decrease with the increase of x; meanwhile, E and s of SiNx decrease when the proportion of Si increases. It can be concluded that the ratio of nitrogen to silicon can reflect the RDFs and CNs in micro level and macro mechanical properties of SiNx to a large extent.

Crystallization kinetics of nanoconfined GeTe slabs in GeTe/TiTe$$_2$$-like superlattices for phase change memories

Debdipto Acharya; Omar Abou El Kheir; Davide Campi; Marco Bernasconi. Scientific Reports (2024). Cited by 11. Found by: openalex. DOI: 10.1038/s41598-024-53192-z

Abstract: Superlattices made of alternating blocks of the phase change compound Sb[Formula: see text]Te[Formula: see text] and of TiTe[Formula: see text] confining layers have been recently proposed for applications in neuromorphic devices. The Sb[Formula: see text]Te[Formula: see text]/TiTe[Formula: see text] heterostructure allows for a better control of multiple intermediate resistance states and for a lower drift with time of the electrical resistance of the amorphous phase. However, Sb[Formula: see text]Te[Formula: see text] suffers from

a low data retention due to a low crystallization temperature T[Formula: see text]. Substituting Sb[Formula: see text]Te[Formula: see text] with a phase change compound with a higher T[Formula: see text], such as GeTe, seems an interesting option in this respect. Nanoconfinement might, however, alters the crystallization kinetics with respect to the bulk. In this work, we investigated the crystallization process of GeTe nanoconfined in geometries mimicking GeTe/TiTe[Formula: see text] superlattices by means of molecular dynamics simulations with a machine learning potential. The simulations reveal that nanoconfinement induces a mild reduction in the crystal growth velocities which would not hinder the application of GeTe/TiTe[Formula: see text] heterostructures in neuromorphic devices with superior data retention.

Graph-EAM: An Interpretable and Efficient Graph Neural Network Potential Framework

Jun Yang; Zhitao Chen; Hong Sun; Amit Samanta. Journal of Chemical Theory and Computation (2023). Cited by 9. Found by: openalex. DOI: 10.1021/acs.jctc.3c00344

Abstract: The development of deep learning interatomic potentials has enabled efficient and accurate computations in quantum chemistry and materials science, circumventing computationally expensive ab initio calculations. However, the huge number of learnable parameters in deep learning models and their complex architectures hinder physical interpretability and affect the robustness of the derived potential. In this work, we propose graph-EAM, a lightweight graph neural network (GNN) inspired by the empirical embedded atom method to model the interatomic potential of single-element structures. Four material systems: platinum, niobium, silicon, and amorphous-carbon, for which quantum simulation data sets are publicly available, are examined to demonstrate that graph-EAM can achieve high energy and force prediction accuracy comparable or better than existing state-of-the-art machine learning models with much fewer parameters. It is also shown that the explicit inclusion of the angular information via three-body atomic density increases the prediction accuracy. The accuracy and efficiency of potentials obtained from graph-EAM can help accelerate the molecular dynamics simulation.

Exploration of Amorphous V 2 O 5 as Cathode for Magnesium Batteries

Vijay Choyal; Debsundar Dey; Gopalakrishnan Sai Gautam. Small (2025). Cited by 9. Found by: openalex. DOI: 10.1002/smll.202505851

Abstract: Abstract Development of energy storage technologies that can exhibit higher energy densities, better safety, and lower supply-chain constraints than the current state-of-the-art Li-ion batteries is crucial for our transition into sustainable energy use. In this context, Mg batteries offer a promising pathway to achieve superior volumetric energy densities than Li-ion but require the development of positive electrodes (cathodes) that exhibit high energy densities at a reasonable power performance. Notably, amorphous materials that lack long range order can exhibit "flatter" potential energy surfaces than crystalline frameworks, possibly resulting in faster Mg 2+ motion. Here, we use a combination of ab initio molecular dynamics (AIMD), and machine learned interatomic potential based calculations is used to explore amorphous-V 2 O 5 as a potential cathode for Mg batteries. Using an AIMD-generated dataset, we train and validate moment tensor potentials that can accurately model amorphous-V 2 O 5 Importantly, we find a 7 (5) orders of magnitude higher Mg 2+ diffusivity in amorphous-MgV 2 O 5 than crystalline-Mg x V 2 O 5 (thiospinel-Mg x Ti 2 S 4), which is directly attributable to the amorphization of the structure, along with a 10-14% drop in the average Mg intercalation voltage. Our work highlights the potential of amorphous-V 2 O 5 as a cathode that can exhibit both high energy and power densities, resulting in the practical deployment of Mg batteries.

Atomistic Study of the Configurational Entropy and the Fragility of Supercooled Liquid GeTe

Yuhan Chen; Davide Campi; Marco Bernasconi; Riccardo Mazzarello. Advanced Functional Materials (2024). Cited by 9. Found by: openalex. openalex_2yr: 14.975 (2026). DOI: 10.1002/adfm.202314264

Abstract: Abstract Phase-change materials (PCMs) are an important family of alloys employed in non-volatile memories and neuromorphic devices. These devices exploit the ability of PCMs to undergo rapid transitions between amorphous and crystalline phases at moderately high temperature. At ambient conditions, both phases are stable. These properties imply a very strong temperature dependence of the crystallization kinetics, which has been attributed to the high fragility of the supercooled liquid phase. In this work, the fragility index of GeTe, an important PCM, is extracted from a computational exploration of the potential energy landscape of the system by molecular dynamics simulations. For this purpose, a neural-network potential is used to describe the interatomic interactions, which enables the evaluation of the configurational entropy in the deeply supercooled regime. The viscosity and the relaxation times are also calculated in the same temperature range. Finally, the Adam-Gibbs relation is used to extrapolate the data to low temperatures and estimate the glass transition temperature and the fragility index. The latter quantity turns out to be of the order of 135–140, which shows that liquid GeTe is a highly fragile system.

Neural network potential for Zr–Rh system by machine learning

Kun Xie; Chong Qiao; Hong Shen; Riyi Yang; Ming Xu; Chao Zhang; Yuxiang Zheng; Rongjun Zhang; Liangyao Chen; Kai-Ming Ho; Cai-Zhuang Wang; Songyou Wang. Journal of Physics Condensed Matter (2021). Cited by 8. Found by: openalex. DOI: 10.1088/1361-648x/ac37dc

Abstract: Abstract Zr–Rh metallic glass has enabled its many applications in vehicle parts, sports equipment and so on due to its outstanding performance in mechanical property, but the knowledge of the microstructure determining the superb mechanical property remains yet insufficient. Here, we develop a deep neural network potential of Zr–Rh system by using machine learning, which breaks the dilemma between the accuracy and efficiency in molecular dynamics simulations, and greatly improves the simulation scale in both space and time. The results show that the structural features obtained from the neural network method are in good agreement with the cases in ab initio molecular dynamics simulations. Furthermore, we build a large model of 5400 atoms to explore the influences of simulated size and cooling rate on the melt-quenching process of Zr77Rh23. Our study lays a foundation for exploring the complex structures in amorphous Zr77Rh23, which is of great significance for the design and practical application.

Investigating Ionic Diffusivity in Amorphous LiPON using Machine-Learned Interatomic Potentials

Aqshat Seth; Rutvij Pankaj Kulkarni; Gopalakrishnan Sai Gautam. ACS Materials Au (2025). Cited by 7. Found by: openalex. openalex_2yr: 5.184 (2026). DOI: 10.1021/acsmaterialsau.4c00117

Abstract: High Resolution Image Download MS PowerPoint Slide Due to its immense importance as an amorphous solid electrolyte in thin-film devices, lithium phosphorus oxynitride (LiPON) has garnered significant scientific attention. However, investigating Li + transport within the LiPON framework, especially across a Li||LiPON interface, has proven challenging due to its amorphous nature and varying stoichiometry, necessitating large supercells and long time scales for computational models. Notably, machine-learned interatomic potentials (MLIPs) can combine the computational speed of classical force fields with the accuracy of density functional theory (DFT), making them the ideal tool for modeling such amorphous materials. Thus, in this work, we train and validate the neural equivariant interatomic potential (NequIP) framework on a comprehensive DFT-based data set consisting of 13,454 chemically relevant structures to describe LiPON. With optimized training (validation) energy and force mean absolute errors of 5.5 (6.1) meV/atom and 13.6 (13.2) meV/Å, respectively, we employ the trained potential to model Li transport in both bulk LiPON and across Li||LiPON interfaces. Amorphous LiPON structures generated by the optimized potential resemble those generated by ab initio molecular dynamics, with N being incorporated on nonbridging apical and bridging sites. Subsequent analysis of Li + diffusivity in the bulk LiPON structures indicates broad agreement with prior computational and experimental literature. Further, we investigate the anisotropy in Li + transport across the Li(110)||LiPON and Li(111)||LiPON interface, where we observe Li transport across the interface to be one order of magnitude slower than Li motion within the bulk Li and LiPON phases. Nevertheless, we note that this anisotropy of Li transport across the interface is minor, and we do not expect it to cause any significant impedance buildup. Finally, our work highlights the efficiency of MLIPs in enabling high-fidelity modeling of complex noncrystalline systems over large length and time scales.

Machine-Learning Molecular Dynamics Study on the Structure and Glass Transition of Calcium Aluminosilicate Glasses

Takeyuki Kato; Ryuki Kayano; Takahiro Ohkubo. The Journal of Physical Chemistry B (2025). Cited by 7. Found by: openalex. DOI: 10.1021/acs.jpcb.5c03404

Abstract: Calcium aluminosilicate (CAS) glass systems represent an important class of materials for industrial applications due to their superior thermal and mechanical properties. Although CAS glasses have been extensively studied, the structure–property relationships, particularly in the peraluminous region (Al 2 O 3 /CaO > 1), remain insufficiently understood. Experimental studies have identified the presence of five-coordinated aluminum (Al 5) depending on the Al 2 O 3 content; however, classical molecular dynamics simulations have struggled to accurately reproduce the aluminum coordination environment. To address this limitation, we developed a machine-learning potential tailored for CAS systems, trained on a comprehensive molecular dynamics simulation based on density functional theory data set and refined using a fine-tuning approach. Melt-quench simulations were then carried out to model CAS glass structures. The resulting structures from machine

learning-based molecular dynamics (MLMD) accurately reproduced both experimental glass densities and the fractions of Al 5, including the observed increase in Al 5 and oxygen triclusters (TBOs) in the peraluminous region. In addition, we performed heating simulations to evaluate enthalpy changes and structural evolution as a function of temperature. Analogous to differential scanning calorimetry experiments, the glass transition temperature (T g) was determined from the MLMD data. The compositional dependence of Al 5 and TBO near the T g was quantitatively analyzed, providing insights into the role of aluminum in structural rearrangements. These findings demonstrate that MLMD enables the accurate modeling of CAS glass structures and yields valuable insights into their thermal behavior. This approach offers a robust framework for understanding structure–property relationships in complex glass systems.

Umbrella sampling with machine learning potentials applied for solid phase transition of GeSbTe

Yanliang Zhao; Jikai Sun; Li Yang; Dong Zhai; Lei Sun; Wei Deng. Chemical Physics Letters (2022). Cited by 7. Found by: openalex. openalex_2yr: 2.847 (2026). DOI: 10.1016/j.cplett.2022.139813

Realistic Atomistic Structure of Amorphous Silicon from Machine-Learning-Driven Molecular Dynamics.

Volker L. Deringer; Noam Bernstein; Albert P. Bartók; Matthew J. Cliffe; Rachel N. Kerber; Lauren E. Marbella; Clare P. Grey; Stephen R. Elliott; Gábor Cśanyi. arXiv (Cornell University) (2018). Cited by 6. Found by: openalex. DOI: 10.17863/cam.28044

Abstract: Amorphous silicon (a-Si) is a widely studied noncrystalline material, and yet the subtle details of its atomistic structure are still unclear. Here, we show that accurate structural models of a-Si can be obtained using a machine-learning-based interatomic potential. Our best a-Si network is obtained by simulated cooling from the melt at a rate of 1011 K/s (that is, on the 10 ns time scale), contains less than 2% defects, and agrees with experiments regarding excess energies, diffraction data, and 29Si NMR chemical shifts. We show that this level of quality is impossible to achieve with faster quench simulations. We then generate a 4096-atom system that correctly reproduces the magnitude of the first sharp diffraction peak (FSDP) in the structure factor, achieving the closest agreement with experiments to date. Our study demonstrates the broader impact of machine-learning potentials for elucidating structures and properties of technologically important amorphous materials.

Basic Stability Tests of Machine Learning Potentials for Molecular Simulations in Computational Drug Discovery

Kavindri Ranasinghe; Adam L. Baskerville; Geoffrey P. F. Wood; Gerhard König. Journal of Chemical Information and Modeling (2025). Cited by 6. Found by: openalex. DOI: 10.1021/acs.jcim.5c01150

Abstract: Neural network potentials trained on quantum-mechanical data can calculate molecular interactions with relatively high speed and accuracy. However, not all neural network potentials are suitable for molecular simulations, as they might exhibit instabilities, nonphysical behavior, or lack accuracy. To assess the reliability of neural network potentials, a series of tests is conducted during model training, in the gas phase, and in the condensed phase. The testing procedure is performed for eight in-house neural network potentials based on the ANI-2x data set, using both the ANI-2x and MACE architectures. This consistent framework allows an evaluation of the effect of the model architecture on its performance. For comparison, we also perform stability tests of the publicly available neural network potentials: ANI-2x, ANI-1ccx, MACE-OFF23, and AIMNet2. The results show that the different models have different weaknesses. A normal-mode analysis of 14 simple benchmark molecules with large displacements from the energy minima revealed that the published MACE-OFF23-S model shows large deviations from the reference quantum-mechanical energy surface. Also, some MACE models with a reduced number of parameters failed to produce stable molecular dynamics simulations in the gas phase, and all MACE models exhibit unfavorable behavior during steric clashes. In addition, the published ANI-2x and one of the in-house MACE models are not able to reproduce the structure of liquid water at ambient conditions, forming an amorphous solid phase instead. For the ANI-1ccx model, the multibody interactions in the condensed water phase lead to nonphysical additional energy minima in bond length and bond angle space, which caused a phase transition to an amorphous solid. Out of all 13 considered public and in-house models, only one in-house model based on the ANI-2x B97–3c data set shows better agreement with the experimental radial distribution function of water than the simple molecular mechanics TIP3P and OPC models. Protein–ligand interaction energies for the four benchmark systems TYK2, CDK2, JNK1, and P38 show that almost all models exhibit a higher correlation with experimental binding affinities than the Chemgauss4 docking score (average R 2 > 0.16). With an average R 2 of 0.43, the ANI-2x model outperforms molecular mechanics calculations with the GAFF2 force field and DFTB3 semiempirical calculations (average R 2 of 0.39 and 0.38), approaching the accuracy of absolute binding free energy calculations (average R 2 of 0.52). However, the rather mixed results for the different machine learning potentials show that great care must be taken during model training and when selecting a neural network potential for real-world applications.

Sampling rare events using nanostructures for universal Pt neural network potential

Joonhee Kang; Byung-Hyun Kim; Min Ho Seo; Jehyun Lee. Current Applied Physics (2024). Cited by 6. Found by: openalex. openalex_2yr: 3.341 (2026). DOI: 10.1016/j.cap.2024.07.005

Study of vacancy ordering and the boson peak in metastable cubic Ge-Sb-Te using machine learning potentials

Young-Jae Choi; Minjae Ghim; Seung-Hoon Jhi. Physical Review Materials (2024). Cited by 6. Found by: openalex. DOI: 10.1103/physrevmaterials.8.013606

Abstract: The mechanism of the vacancy ordering in metastable cubic Ge-Sb-Te (c-GST) that underlies the ultrafast phase-change dynamics and prominent thermoelectric properties remains elusive. Achieving a comprehensive understanding of the vacancy-ordering process at an atomic level is challenging because of enormous computational demands required to simulate disordered structures on large temporal and spatial scales. In this study, we investigate the vacancy ordering in c-GST by performing large-scale molecular dynamics simulations using machine learning potentials. The initial c-GST structure with randomly distributed vacancies rearranges to develop a semiordered cubic structure with layerlike ordered vacancies after annealing at 700 K for 100 ns. The vacancy ordering significantly affects the lattice dynamical properties of c-GST. In the initial structure with fully disordered vacancies, we observe a boson peak, usually associated with amorphous solids, that consists of localized modes at \ensuremath{\sim}0.575 THz. The boson peak modes are highly localized around specific atomic arrangements of straight vacancy-Te-vacancy trios. As vacancies become ordered, the boson peak disappears, and the Debye-Waller thermal B factor of Te decreases substantially. This finding indicates that the c-GST undergoes a transition from amorphous-like to crystalline-like solid state by thermal annealing in low-frequency dynamics.

Atomistic Simulations of the Crystallization of Amorphous GeTe Nanoparticles

Debdipto Acharya; Omar Abou El Kheir; Simone Perego; Davide Campi; Marco Bernasconi. The Journal of Physical Chemistry C (2024). Cited by 6. Found by: openalex. DOI: 10.1021/acs.jpcc.4c05157

Abstract: The effect of dimensionality reduction on the crystallization kinetics of phase change materials is relevant for the operation of ultrascale memory devices. Therefore, the crystallization of amorphous nanoparticles (NPs) of the prototypical phase change compounds, GeTe and Ge 2 Sb 2 Te 5, has been addressed by several experimental works in recent years. In this work, we performed molecular dynamics simulations of the crystallization process of amorphous GeTe NPs with diameters in the range 3–6 nm (512–4096 atoms) by exploiting a machine-learned interatomic potential. We saw a few crystal nucleation events in the larger NPs but no crystallization in the smallest NP, 3 nm in diameter, in simulations lasting up to 80 ns in the temperature range 500–750 K. The analysis of the crystallization kinetics suggests that the nucleation rate per volume decreases with the NP size to an extent that prevents us from seeing crystallization in the smallest NP on our simulation time scale. This result is consistent with the large increase in crystallization temperature observed experimentally for NPs with diameters shorter than 3.5 nm.

Multikernel similarity-based clustering of amorphous systems and machine-learned interatomic potentials by active learning

Firas Shuaib; Guido Ori; Philippe Thomas; Olivier Masson; Assil Bouzid. Journal of the American Ceramic Society (2024). Cited by 5. Found by: openalex. DOI: 10.1111/jace.20128

Abstract: Abstract We present a hybrid similarity kernel that exemplifies the integration of short- and long-range descriptors via the use of an average kernel approach. This technique allows for a direct measure of the similarity between amorphous configurations, and when combined with an active learning (AL) spectral clustering approach, it leads to a classification of the amorphous configurations into uncorrelated clusters. Subsequently, a minimum size database is built by considering a small fraction of configurations belonging to each cluster and a machine learning interatomic potential (MLIP), within the Gaussian approximation scheme, is fitted by relying on a Bayesian optimization of the potential hyperparameters. This step is embedded within an AL

loop that allows to sequentially increase the size of the learning database whenever the MLIP fails to meet a predefined energy convergence threshold. As such, MLIP are fitted in an almost fully automatized fashion. This approach is tested on two diverse amorphous systems that were previously generated using first-principles molecular dynamics. Accurate potentials with less than 2 meV/atom root mean square energy error compared to the reference data are obtained. This accuracy is achieved with only 175 configurations sampling the studied systems at various temperatures. The robustness of these potentials is then confirmed by producing models with several thousands of atoms featuring a good agreement with reference ab initio and experimental data.

Crystallization kinetics in Ge-rich Ge x Te alloys from large scale simulations with a machine-learned interatomic potential

Dario Baratella; Omar Abou El Kheir; Marco Bernasconi. Acta Materialia (2024). Cited by 5. Found by: openalex. openalex_2yr: 10.171 (2026). DOI: 10.1016/j.actamat.2024.120608

Abstract: A machine-learned interatomic potential for Ge-rich Ge x Te alloys has been developed aiming at uncovering the kinetics of phase separation and crystallization in these materials. The results are of interest for the operation of embedded phase change memories which exploits Ge-enrichment of GeSbTe alloys to raise the crystallization temperature. The potential is generated by fitting a large database of energies and forces computed within Density Functional Theory with the neural network scheme implemented in the DeePMD-kit package. The potential is highly accurate and suitable to describe the structural and dynamical properties of the liquid, amorphous and crystalline phases of the wide range of compositions from pure Ge and stoichiometric GeTe to the Ge-rich Ge x Te alloy. Large scale molecular dynamics simulations have suggested a crystallization mechanism which depends on temperature. At 600 K, segregation of most of Ge in excess was observed to occur on the ns time scale followed by crystallization of nearly stoichiometric GeTe regions. At 500 K, nucleation of crystalline GeTe was observed to occur before phase separation, followed by a slow crystal growth due to the concurrent expulsion of Ge in excess.

Amorphous Zirconia-doped Tantala modeling and simulations using explicit multi-element spectral neighbor analysis machine learning potentials (EME-SNAP)

Jun Jiang; Xiangguo Li; A. Mishkin; R. Zhang; R. Bassiri; J. N. Fry; M. M. Fejer; Hai-Ping Cheng. Physical Review Materials (2023). Cited by 5. Found by: openalex. DOI: 10.1103/physrevmaterials.7.045602

Abstract: We model amorphous Zirconia-doped Tantala with machine learning interactomc potentials based on explicit multielement spectral neighbor analysis (EME-SNAP). These atomic structure models can reproduce partial radial distribution functions obtained from first-principles calculations and elastic moduli found from experimental measurements. The two-body pair forces calculated from EME-SNAP further affirm that the potentials capture the atomic interactions well. Molecular dynamics simulations of simulated annealing with EME-SNAP show that the final density of the amorphous models depends on the thermal history even when the annealing rate is kept constant, which captures experimental observations of history-dependent densities. Mechanical spectroscopy is also simulated using both Morse-Beest-Kramer-Santen pair potentials and EME-SNAP. The success in applying the EME-SNAP to amorphous Zirconia-doped Tantala pushes the boundaries of simulation accuracy and system size and enables better and more realistic atomistic modeling for amorphous systems. There are still some limitations in applying the potentials generated in this paper. They are only optimized for trained amorphous phases; high-temperature stability and transferability need to be further investigated.

Temperature-density dependent hidden order of amorphous carbon

Yutao Liu; Tinghong Gao; Qingquan Xiao; Yunjun Ruan; Qian Chen; Bei Wang; Jin Huang. Communications Physics (2025). Cited by 5. Found by: openalex. openalex_2yr: 5.555 (2026). DOI: 10.1038/s42005-025-02320-w

Abstract: The microscopic structure of amorphous carbon is inherently complex, and its underlying topological order remains poorly understood. Here, we developed a high-precision machine learning potential and performed large-scale molecular dynamics simulations of the annealing process over a wide range of temperatures and densities. Nine representative amorphous carbon structures were identified and characterized through X-ray diffraction and structure factor analysis, enabling detailed comparison of their topological differences. From these results, we constructed a temperature–density phase diagram that reveals two distinct phase boundaries associated with discontinuous structural transitions, with the critical temperature increasing approximately linearly with density. The diagram further indicates two transformation pathways, where high-density disordered graphene networks and paracrystalline diamond structures act as intermediate states. Additional pressure-controlled simulations demonstrate asymmetric nucleation during diamond formation and decomposition. These findings provide a clearer understanding of hidden order in amorphous carbon and offer a theoretical framework to guide its controlled synthesis. Carbon exhibits several known allotropes and the amorphous equivalents exhibit an even wider range of structural variations that can be difficult to identify due to their lack of crystallinity. Here, the authors report large-scale machine-learned molecular dynamics simulations that maps the hidden topological orders of amorphous carbon as a function of temperature and density.

The ab initio amorphous materials database: Empowering machine learning to decode diffusivity

Hui Zheng; Eric Sivonxay; Max C. Gallant; Ziyao Luo; Matthew J. McDermott; Patrick Huck; Kristin A. Persson. arXiv (Cornell University) (2024). Cited by 5. Found by: openalex. DOI: 10.48550/arxiv.2402.00177

Abstract: Amorphous materials exhibit unique properties that make them suitable for various applications in science and technology, ranging from optical and electronic devices and solid-state batteries to protective coatings. However, data-driven exploration and design of amorphous materials is hampered by the absence of a comprehensive database covering a broad chemical space. In this work, we present the largest computed amorphous materials database to date, generated from systematic and accurate \textit{ab initio} molecular dynamics (AIMD) calculations. We also show how the database can be used in simple machine-learning models to connect properties to composition and structure, here specifically targeting ionic conductivity. These models predict the Li-ion diffusivity with speed and accuracy, offering a cost-effective alternative to expensive density functional theory (DFT) calculations. Furthermore, the process of computational quenching amorphous materials provides a unique sampling of out-of-equilibrium structures, energies, and force landscape, and we anticipate that the corresponding trajectories will inform future work in universal machine learning potentials, impacting design beyond that of non-crystalline materials.

Mechanisms of nanodiamond and amorphous diamond-like carbon formation following room temperature compression of C60

Hendrik Heimes; Ethan P. Turner; Alan Salek; Jessica Wierbik; Xingshuo Huang; William R. Dunstan; Nigel A. Marks; Dougal G. McCulloch; J. E. Bradby. Diamond and Related Materials (2025). Cited by 5. Found by: openalex. openalex_2yr: 5.623 (2026). DOI: 10.1016/j.diamond.2025.112776

Abstract: The transformation of carbon into sp 3 -rich phases is of broad interest for developing superhard materials, yet the role of non-hydrostatic pressures during such transformations remains poorly understood. While buckminsterfullerene (C 60) is known to collapse under high pressures and temperatures, the structural evolution pathways under non-hydrostatic conditions have not been fully established. Here, we report the formation of transparent, amorphous diamond-like carbon (a-D) from C 60 compressed to 49 GPa at room temperature under non-hydrostatic stress. Transmission electron microscopy of the recovered material reveals a predominantly sp 3 -bonded network with (85 ± 5)% sp 3 bonds and a density of (3.05 ± 0.10) g cm −3 , interlaced with residual C 60 molecules and narrow bands of nanocrystalline diamond. To understand the transformation mechanism, we performed molecular dynamics simulations using an atomic cluster expansion potential. The simulations show that under uniaxial stress, fullerene cages collapse at lower pressures (29–32 GPa) than under hydrostatic conditions (42–45 GPa), forming a structure consistent with amorphous diamond-like carbon. Annealing simulations on the amorphous structure at 50 GPa result in nanocrystalline diamond formation, suggesting that localised heat spikes generated by an adiabatic shear band process drive the formation of observed nanodiamond regions. • Formation of recoverable sp 3 -bonded carbon at room temperature. • Formation of nanodiamonds in shear bands. • Non-hydrostatic stresses promote C 60 cage collapse. • Machine learning potentials are powerful tools to model high pressure responses in carbon.

Non-Monotonic Ion Conductivity in Lithium-Aluminum-Chloride Glass Solid-State Electrolytes Explained by Cascading Hopping

Beomgyu Kang; Jinqiu Yu; Shinji Saito; Ji-Hyun Jang; Bong June Sung. Advanced Science (2025). Cited by 5. Found by: openalex. openalex_2yr: 8.903 (2026). DOI: 10.1002/advs.202509205

Abstract: Abstract Inorganic glass solid-state electrolytes (IGSSEs) exhibit superionic conductivity at ambient temperature. Understanding their ion conduction mechanism remains challenging but is essential for the development of next-generation all-solid-state batteries. The coupling between lithium ion diffusion and the rotation of neighboring polyanions, known as the paddlewheel effect, has been proposed as a possible mechanism, though its existence remains controversial. Herein, a systematic and extendible approach is proposed to explore the ion conduction mechanism of IGSSEs using large-scale machine learning molecular dynamics (MLMD) simulations and hop function analysis. A machine learning potential is constructed for model IGSSEs of Li x AlCl 3 + x (x = 0.25 to 3). MLMD simulation results reproduce the experimentally observed non-monotonic composition dependence of lithium-ion conductivity, with a maximum at x = 1. Hop function analysis reveals that lithium ion diffusion occurs mainly via cascading hopping events rather than paddlewheel motions. The cascading hops are composition-dependent and account for the observed non-monotonic composition dependence. The non-monotonic composition dependence arises from a delicate balance between the local concentrations of lithium ions and the lithium vacancies.

A high-efficiency neuroevolution potential for tobermorite and calcium silicate hydrate systems with ab initio accuracy

Xiao Xu; Shijie Wang; Haifeng Qin; Zhiqiang Zhao; Zheyong Fan; Zhuhua Zhang; Hang Yin. Cement and Concrete Research (2025). Cited by 5. Found by: openalex. openalex_2yr: 13.05 (2026). DOI: 10.1016/j.cemconres.2025.108091

Softening of Vibrational Modes and Anharmonicity Induced Thermal Conductivity Reduction in a-Si:H at High Temperatures

Zhuo Chen; Yuejin Yuan; Yanzhou Wang; Penghua Ying; Shouhang Li; Cheng Shao; Wenyang Ding; Gang Zhang; Meng An. Advanced Electronic Materials (2025). Cited by 5. Found by: openalex. openalex_2yr: 4.957 (2026). DOI: 10.1002/aelm.202500104

Abstract: Abstract Hydrogenated amorphous silicon (a-Si:H) has garnered considerable attention in the semiconductor industry, particularly for its use in solar cells and passivation layers for high-performance silicon solar cells, owing to its exceptional photoelectric properties and scalable manufacturing processes. A comprehensive understanding of thermal transport mechanism in a-Si:H is essential for optimizing thermal management and ensuring the reliable operation of these devices. In this study, we developed a neuroevolution machine learning potential based on first-principles calculations of energy, forces, and virial, which enables accurate modeling of interatomic interaction in both a-Si:H and a-Si systems. Using the homogeneous nonequilibrium molecular dynamics (HNEMD) method, this study systematically investigated the thermal conductivity of a-Si:H and a-Si across a temperature range of 300–1000 K and hydrogen concentrations ranging from 6 to 12 at%. The simulation results showed that thermal conductivity of a-Si:H with 12 at% hydrogen is significantly reduced by 41.2% compared to that of a-Si at 300 K. The spectral thermal conductivity, vibrational density of states and lifetimes of vibrational modes are analyzed, which revealed both the softening of vibrational modes and anharmonicity effects contribute to the reduction of thermal conductivity as temperature and hydrogen concentration increase. Furthermore, the influence of hydrogen concentration and temperature on diffuson and propagon contribution to thermal conductivity of a-Si:H is quantatively revealed. This study provides valuable insight for developing thermal management strategies in silicon-based semiconducting devices and advancing the understanding of thermal transport in amorphous systems.

Machine learning potential for predicting thermal conductivity of -phase and amorphous tantalum nitride

Zhicheng Zong; Yangjun Qin; Jiahong Zhan; Haisheng Fang; Nuo Yang. International Journal of Heat and Mass Transfer (2025). Cited by 4. Found by: openalex. DOI: 10.1016/j.ijheatmasstransfer.2025.128155

Molecular Dynamics Study of Silicon Carbide Using an Ab Initio-Based Neural Network Potential: Effect of Composition and Temperature on Crystallization Behavior

Jinyoung Lim; Youngseon Shim; Jaehong Park; Hongkee Yoon; Mun-Bo Shim; Young-Gu Kim; Dae Sin Kim. The Journal of Physical Chemistry C (2023). Cited by 4. Found by: openalex. DOI: 10.1021/acs.jpcc.3c04224

Abstract: Structure and diffusion dynamics of silicon carbide (Si 1− x C x) are investigated via molecular dynamics computer simulations with ab initio-based neural network potentials, exploring the effect of composition and temperature on crystallization behaviors. A neural network potential is developed to describe high-dimensional potential energy surfaces of silicon carbide (SiC) systems, reproducing first-principles results on their potential energies and forces. The phase behavior of amorphous Si 1− x C x below its experimental melting point is systematically demonstrated by analyzing the structural and dynamic properties as a function of temperature and carbon concentration x in the composition range 0 x 0.5 and the temperature range T = 2000–2600 K, compared to available experiments. The phase of Si 1− x C x is characterized by analyzing the pair correlation function, coordination number, tetrahedral order parameter, SiC tetrahedron fraction, Si

disordered fraction, and excess entropy. Our results indicate that the system undergoes the crystallization by organizing the short- and medium-range order as the carbon content increases, where the critical carbon fraction for crystallization increases with temperature. The addition of carbon to silicon results in the phase separation into liquid Si and crystal SiC as well as the partial crystallization of Si 1− x C x . The self-diffusivity of Si 1− x C x is also evaluated to understand how the structural change caused by the crystallization works on diffusion dynamics. The diffusion dynamics of Si 1− x C x becomes slower with increasing carbon content and decreasing temperature, which significantly slows down with onset of the crystallization.

Slowly quenched, high pressure glassy B2O3 at DFT accuracy

Debendra Meher; Nikhil V. S. Avula; Sundaram Balasubramanian. The Journal of Chemical Physics (2025). Cited by 4. Found by: openalex. DOI: 10.1063/5.0240030

Abstract: Modeling inorganic glasses requires an accurate representation of interatomic interactions, large system sizes to allow for intermediate-range structural order, and slow quenching rates to eliminate kinetically trapped structural motifs. Neither first principles-based nor force field-based molecular dynamics (MD) simulations satisfy these three criteria unequivocally. Herein, we report the development of a machine learning potential (MLP) for a classic glass, B2O3, which meets these goals well. The MLP is trained on condensed phase configurations whose energies and forces on the atoms are obtained using periodic quantum density functional theory. Deep potential MD simulations based on this MLP accurately predict the equation of state and the densification of the glass with slower quenching from the melt. At ambient conditions, quenching rates larger than 1011 K/s are shown to lead to artifacts in the structure. Pressure-dependent x-ray and neutron structure factors from the simulations compare excellently with experimental data. High-pressure simulations of the glass show varied coordination geometries of boron and oxygen, which concur with experimental observations.

Nature of the amorphous-amorphous interfaces in solid-state batteries revealed using machine-learned interatomic potentials

Chuhong Wang; Muratahan Aykol; Tim Mueller. ChemRxiv (2023). Cited by 3. Found by: openalex. openalex_2yr: 0.467 (2026). DOI: 10.26434/chemrxiv-2023-frr79-v2

Abstract: Non-crystalline solid materials have attracted growing attention in energy storage for their desirable properties such as ionic conductivity, stability and processability. However, compared to bulk crystalline materials, fundamental understanding of these highly complex metastable systems is hindered by the scale limitations of density functional theory (DFT) calculations and resolution limitations of experimental methods. To fill the knowledge gap and guide the rational design of amorphous battery materials and interfaces, we present a molecular dynamics (MD) framework based on machine-learned interatomic potentials trained on the fly to study the amorphous solid electrolyte Li3PS4 and its protective coating, amorphous Li3B11O18. The use of machine-learned potentials allows us to simulate the materials at time and length scales that are not accessible to DFT while maintaining a near-DFT level of accuracy. This approach allows us to calculate amorphization energies, amorphous-amorphous interface energies, and the impact of the interface on lithium ion conductivity. This study demonstrates the promising role of actively-learned interatomic potentials in extending the application of ab-initio modeling to more complex and realistic systems such as amorphous materials and interfaces.

Viscosity, breakdown of Stokes–Einstein relation, and dynamical heterogeneity in supercooled liquid Ge2Sb2Te5 from simulations with a neural network potential

Simone Marcorini; Rocco Pomodoro; Omar Abou El Kheir; Marco Bernasconi. The Journal of Chemical Physics (2025). Cited by 3. Found by: openalex. DOI: 10.1063/5.0282855

Abstract: Phase change materials are exploited in non-volatile electronic memories and photonic devices that rely on a fast and reversible transformation between the amorphous and crystalline phases upon heating. Recrystallization of the amorphous phase under the operation conditions of the memories occurs in the supercooled liquid phase above the glass transition temperature Tg. The dynamics of the supercooled liquid is thus of great relevance for the operation of the devices and, close to Tg, also for the structural relaxations of the glass that affect the performances of the memories. Information on the atomic dynamics is provided by the diffusion coefficient (D) and by the viscosity ( ), which are, however, both difficult to be measured experimentally under the operation conditions of the devices due to fast crystallization. In this work, we leverage a machine learning interatomic potential for the flagship phase change compound Ge2Sb2Te5 to compute , D, and the -relaxation time in a wide temperature range from 1200 K to about 100 K above Tg. Large-scale molecular dynamics simulations allowed quantifying the fragility of the liquid and the occurrence of a breakdown of the Stokes-Einstein relation between and D in the supercooled phase. Isoconfigurational analysis provided a visualization of the

emergence of dynamical heterogeneities responsible for the breakdown of the Stokes-Einstein relation. The analysis revealed that the regions of most mobile atoms are related to the presence of Ge atoms with particular local environments.

Anisotropic response under shock compression in refractory high-entropy alloy via machine-learning potential

Hongyang Liu; Rong Chen; Bo Chen; Jingzhi He; Dongdong Kang; Yu Tang; Jiayu Dai. Materials & Design (2025). Cited by 3. Found by: openalex. DOI: 10.1016/j.matdes.2025.115222

Abstract: • Trained a DP potential model that can be used to study the anisotropy of shock compression in TiZrNbTa. • TiZrNbTa exhibits strong orientation-dependent deformation during shock loading, unloading, and spallation. • Spall strength anisotropy is linked to void nucleation timing, with premature nucleation causing the weakest spall. The anisotropic shock response of BCC-structured refractory high-entropy alloys (RHEAs) single crystals fundamentally governs their dynamic deformation and failure. Given experimental challenges in observing orientation-dependent shock behavior, we employed molecular dynamics with a Deep Potential machine-learning potential to simulate TiZrNbTa RHEA under 800 m/s impact. The [1 1 0] orientation exhibited twinning-induced plasticity during compression and formed residual strain-induced distortion zones that triggered severe amorphization upon unloading. Premature void nucleation (32.2 ps) in amorphous regions before peak stress (35.2 ps) caused thermal softening and catastrophic void proliferation, yielding minimal spall strength (12.20 GPa). Conversely, [1 0 0] underwent elastic compression followed by BCC → FCC transformation during unloading, suppressing amorphization and achieving intermediate spall strength (15.12 GPa). The structurally stable [1 1 1] orientation showed minimal amorphization and maximal spall strength (15.25 GPa), revealing unloading-induced microstructural evolution as the dominant factor in anisotropic spall failure for impact-resistant RHEA design.

Comparison of intermediate-range order in GeO2 glass: Molecular dynamics using machine-learning interatomic potential vs reverse Monte Carlo fitting to experimental data

Kenta Matsutani; Shusuke Kasamatsu; Takeshi Usuki. The Journal of Chemical Physics (2024). Cited by 3. Found by: openalex. DOI: 10.1063/5.0240087

Abstract: The short-range order and intermediate-range order in GeO2 glass are investigated by molecular dynamics using machine-learning interatomic potential trained on ab initio calculation data and compared with the reverse Monte Carlo fitting of neutron diffraction data. To characterize the structural differences in each model, the total/partial structure factors, coordination number, ring size and shape distributions, and persistent homology analysis were performed. These results show that although the two approaches yield similar two-body correlations, they can lead to three-dimensional models with different short- and intermediate-range ordering. A clear difference was observed especially in the ring distributions; RMC models exhibit a broad distribution in the ring size distribution, while neural network potential molecular dynamics yield much narrower ring distributions. This confirms that the density functional approximation in the ab initio calculations determines the preferred network assembly more strictly than RMC with simple coordination constraints even when using multiple diffraction data.

A Generalized Bond Switching Monte Carlo method for amorphous structure generation

Zhengneng Zheng; Fan Zheng; Yuning Wu; Shiyou Chen; Lin-Wang Wang. Computational Materials Today (2025). Cited by 3. Found by: openalex. openalex_2yr: 4.183 (2026). DOI: 10.1016/j.commt.2025.100031

Abstract: Amorphous semiconductors are used in many electronic devices. For Si and other highly covalent bond systems, the amorphous structures can be generated using the bond switching method. However, for other non-covalent amorphous materials, it becomes challenging to generate their atomic structures. In this work, we extend the bond switching method to various complex and ionic systems. This method can be used to transform a general ionic crystalline structure to an amorphous structure (such as HfO 2), or generate the amorphous structures without the crystalline counterparts such as carbon black. We also introduce additional terms in the Bond Switching Monte Carlo process to control the porosity of the amorphous system, such as the amorphous In 2 O 3 . In addition, the amorphous structure can be generated to reproduce the experimental pair correlation function. We have investigated the changes in coordination numbers of the Bond Switching Monte Carlo process, and analyzed the usability of this method depending on the ionicity of the system. • We have introduced a new atomic repulsing term to control the uniformity of the amorphous structure, and successfully eliminated the voids in In 2 O 3 , making the structure generated by BSMC more reasonable. • Our method has a faster running speed compared to molecular dynamics (MD) methods like DFT MD and

MD with machine learning potential , especially for materials with strong covalent properties. MD usually requires a long simulation time to break covalent bonds , however, this process is quite easy in our BSMC program. In addition, we also provide a series of universal parameters to generate amorphous materials for most systems, while machine learning methods require model retraining for different systems • In our program, the experimental pair correlation function can be used as input to reproduce the desired amorphous structure. Therefore, the structure generated by our program can match the experimental results . • Our algorithm is not only applied to covalent bond systems , but also tested for more ionic systems like HfO 2 , and In 2 O 3 . We have not seen such ionic systems constructed by BSMC in the literature. According to our calculations, the structure generated by our program has a local bonding number similar to MD , a bandgap and dielectric constant consistent with experiments, and does not contain any deep defect levels. • The algorithm can generate amorphous system in specific shaped structures, like slab, bending structure, interface and device structures like GAA FET, thus pave the way for realistic nanosized electronic device simulations.

Atomic-scale factors that control the rate capability of nanostructured amorphous Si for high-energy-density batteries

Nongnuch Artrith; Alexander S. Urban; Yan Wang; Gerbrand Ceder. arXiv (Cornell University) (2019). Cited by 3. Found by: openalex. DOI: 10.48550/arxiv.1901.09272

Abstract: Nanostructured Si is the most promising high-capacity anode material to substantially increase the energy density of Li-ion batteries. Among the remaining challenges is its low rate capability as compared to conventional materials. To understand better what controls the diffusion of Li in the amorphous Li-Si alloy, we use a novel machine-learning potential trained on more than 40,000 ab-initio calculations and nanosecond-scale molecular dynamics simulations, to visualize for the first time the delithiation of entire LiSi nanoparticles. Our results show that the Si host is not static but undergoes a dynamic rearrangement from isolated atoms, to chains, and clusters, with the Li diffusion strongly governed by this Si rearrangement. We find that the Li diffusivity is highest when Si segregates into clusters, so that Li diffusion proceeds via hopping between the Si clusters. The average size of Si clusters and the concentration range over which Si clustering occurs can thus function as design criteria for the development of rate-improved anodes based on modified Si.

Machine learning metallic glass critical cooling rates through elemental and molecular simulation based featurization

Lane E. Schultz; Benjamin Afflerbach; Paul M. Voyles; Dane Morgan. Journal of Materiomics (2024). Cited by 3. Found by: openalex. DOI: 10.1016/j.jmat.2024.100964

Abstract: We have developed a machine learning model for critical cooling rates for metallic glasses based on computational properties, supporting in-silico screening for desired R c values and significantly reducing reliance on time-consuming laboratory work. We compare results for features derived from easy-to-compute functions of elemental properties to more complex physically motivated properties using ab initio , machine-learning potentials, and empirical potential molecular dynamics methods. The established approach enables property acquisition across a diverse range of alloys. Analysis of various features for 34 alloys from 20 chemical systems shows that the best model for critical cooling rates was learned from one elemental property-based feature and three simulated features. The elemental property based feature is an ideal entropy value based on alloy stoichiometry. The simulated features were acquired from estimates of energies above the convex hull, changes in heat capacity, and the fraction of icosahedra-like Voronoi polyhedra. Models were assessed through a demanding cross validation test based on repeatedly leaving out full chemical systems as test sets and had an R 2 of 0.78 and a mean average error of 0.76 in units of lg(K/s). We demonstrate with Shapley additive explanation analysis that the most impactful features have physically reasonable influence on model predictions. The established methodology can be applied to other high-throughput studies of material properties of diverse compositions. • Utilized machine learning potentials to efficiently simulate properties across diverse alloy compositions. • Developed machine learning model for critical cooling rates of metallic glasses using elemental properties and molecular simulation features. • The best model achieved an R 2 of 0.78 with a cross validation based on grouping chemical systems. • Shapley additive explanation analysis showed most impactful features have physically reasonable influence on model predictions. • Methodology integrating elemental and molecular simulation features is broadly applicable to high-throughput studies of diverse material properties.

When more data hurts: Optimizing data coverage while mitigating diversity-induced underfitting in an ultrafast machine-learned potential

Jason Gibson; Tesia D. Janicki; Ajinkya C. Hire; Chris Bishop; J. Matthew D. Lane; Richard G. Hennig. Physical Review Materials (2025). Cited by 3. Found by: openalex. DOI: 10.1103/physrevmaterials.9.043802

Abstract: Machine-learned interatomic potentials (MLIPs) are becoming an essential tool in materials modeling. However, optimizing the generation of training data used to parametrize the MLIPs remains a significant challenge. This is because MLIPs can fail when encountering local enviroments too different from those present in the training data. The difficulty of determining a priori the environments that will be encountered during molecular dynamics simulation necessitates diverse, high-quality training data. This study investigates how training data diversity affects the performance of MLIPs using the Ultra-Fast force field (${\mathrm{UF}}^{3}$) to model amorphous silicon nitride. We employ expert and autonomously generated data to create the training data and fit four force field variants to subsets of the data. Our findings reveal a critical balance in training data diversity: insufficient diversity hinders generalization, while excessive diversity can exceed the MLIP's learning capacity, reducing simulation accuracy. Specifically, we found that the ${\mathrm{UF}}^{3}$ variant trained on a subset of the training data, in which nitrogen-rich structures were removed, offered vastly better prediction and simulation accuracy than any other variant. By comparing these ${\mathrm{UF}}^{3}$ variants, we highlight the nuanced requirements for creating accurate MLIPs, emphasizing the importance of application-specific training data to achieve optimal performance in modeling complex material behaviors.

Insights into radiation damage in YBa2Cu3O7− from machine learned interatomic potentials

Ashley Dickson; Niccolò Di Eugenio; Federico Ledda; Daniele Torsello; F. Laviano; Flyura Djurabekova; Jesper Byggmästar; Mark R. Gilbert; D. Nguyen-Manh; Erik Gallo; Antonio Trotta; Davide Gambino; Samuel T. Murphy. Superconductivity (2026). Cited by 2. Found by: openalex. DOI: 10.1016/j.supcon.2026.100264

Abstract: Accurate prediction of radiation damage in YBa 2 Cu 3 O 7 − (YBCO) is essential for assessing the performance of high-temperature superconducting (HTS) tapes in compact fusion reactors. Existing empirical interatomic potentials have been used to model radiation damage in stoichiometric YBCO, but fail to describe oxygen-deficient compositions, which are ubiquitous in industrial Rare-Earth Barium Copper Oxide conductors and strongly influence superconducting properties. In this work, we demonstrate that modern machine-learned interatomic potentials enable predictive modelling of radiation damage in YBCO across a broad range of oxygen stoichiometries, at a greater fidelity than previous empirical models. We employ two recently developed approaches: an Atomic Cluster Expansion (ACE) potential and a tabulated Gaussian Approximation Potential (tabGAP). Both machine-learned models are shown to accurately reproduce Density Functional Theory (DFT) energies, forces, and threshold displacement energy distributions, providing a quantitatively reliable description of atomic-scale collision processes. Molecular dynamics simulations of 5 keV cascades predict significantly enhanced peak defect production and recombination relative to a previous empirical potential, indicating qualitatively different cascade evolution. Moreover, these new machine learning models generate increased proportions of copper and oxygen vacancies compared to previous models, in direct agreement with experimental observations. By explicitly varying oxygen deficiency, we further show that total defect production exhibits only a weak dependence on stoichiometry, providing new insight into the robustness of radiation damage processes in oxygen-deficient YBCO. Finally, fusion-relevant 300 keV cascade simulations reveal amorphous regions with characteristic dimensions comparable to the superconducting coherence length, consistent with electron microscopy observations of neutron-irradiated HTS tapes. These results establish machine-learned interatomic potentials as powerful, computationally efficient tools for uncovering radiation-damage physics in YBCO, enabling predictive simulations across technologically relevant compositions and irradiation conditions.

Critical Composition Window for Co–P Bulk Metallic Glasses Revealed by Machine-Learned Interatomic Potentials

Md Sharif Khan; Bourgeois Biova Irénée Gadjagboui; Oliviero Andreussi. The Journal of Physical Chemistry C (2025). Cited by 2. Found by: openalex. DOI: 10.1021/acs.jpcc.5c05928

Abstract: Bulk metallic glasses are a distinct class of alloys whose disordered atomic structure imparts exceptional mechanical strength, corrosion resistance, and catalytic activity. Nevertheless, the optimization of their compositional ratios remains predominantly empirical, relying largely on qualitative and nonpredictive computational models. In this work, we developed a machine interatomic potential for Co–P binary alloys, trained on approximately 32,000 density functional theory-labeled configurations spanning a wide range of compositions (5–40% P) and temperatures (300–1800 K). The model accurately reproduced reference energies and forces and remained reliable in high-temperature molecular dynamics simulations. Analysis of structural motifs revealed that high phosphorus content promotes medium-range ordering, while very low phosphorus content favors partial crystallinity. Intermediate compositions (20, 15, and 10% P) exhibited structural signatures of bulk metallic glasses, including pronounced icosahedral clustering, split second-neighbor peaks in radial distribution functions, and coordination environments resembling distorted bicapped square antiprisms. Thermal and dynamical assessments showed that these compositions form stable amorphous phases with favorable glass transition temperatures. The developed potential enables large-scale simulations and offers guidance for optimizing Co–P-based metallic glasses for structural and energy applications.

The structural evolution of SiBCNZr amorphous ceramics analyzed by machine learning potential

Meng Zhang; Siyan Deng; Jianghong Zhang; Daxin Li; Dechang Jia; Yu Zhou. Ceramics International (2024). Cited by 2. Found by: openalex. openalex_2yr: 5.869 (2026). DOI: 10.1016/j.ceramint.2024.05.296

Unraveling the Crystallization Kinetics of the Ge$_2$Sb$_2$Te$_5$ Phase Change Compound with a Machine-Learned Interatomic Potential

Omar Abou El Kheir; Luigi Bonati; Michele Parrinello; Marco Bernasconi. arXiv (Cornell University) (2023). Cited by 2. Found by: openalex. DOI: 10.48550/arxiv.2304.03109

Abstract: The phase change compound Ge$_2$Sb$_2$Te$_5$ (GST225) is exploited in advanced non-volatile electronic memories and in neuromorphic devices which both rely on a fast and reversible transition between the crystalline and amorphous phases induced by Joule heating. The crystallization kinetics of GST225 is a key functional feature for the operation of these devices. We report here on the development of a machine-learned interatomic potential for GST225 that allowed us to perform large scale molecular dynamics simulations (over 10000 atoms for over 100 ns) to uncover the details of the crystallization kinetics in a wide range of temperatures of interest for the programming of the devices. The potential is obtained by fitting with a deep neural network (NN) scheme a large quantum-mechanical database generated within Density Functional Theory. The availability of a highly efficient and yet highly accurate NN potential opens the possibility to simulate phase change materials at the length and time scales of the real devices.

Thermal Conductivity Modeling using Machine Learning Potentials: Application to Crystalline and Amorphous Silicon

Xin Qian; Shenyou Peng; Xiaobo Li; Yujie Wei; Ronggui Yang. arXiv (Cornell University) (2019). Cited by 2. Found by: openalex. DOI: 10.48550/arxiv.1907.09088

Abstract: First-principles based modeling on phonon dynamics and transport using density functional theory and Boltzmann transport equation has proven powerful in predicting thermal conductivity of crystalline materials, but it remains unfeasible for modeling complex crystals and disordered solids due to the prohibitive computational cost to capture the disordered structure, especially when the quasiparticle "phonon" model breaks down. Recently, machine-learning regression algorithms show great promises for building high-accuracy potential fields for atomistic modeling with length and time scales far beyond those achievable by first-principles calculations. In this work, using both crystalline and amorphous silicon as examples, we develop machine learning based potential fields for predicting thermal conductivity. The machine learning based interatomic potential is derived from density functional theory calculations by stochastically sampling the potential energy surface in the configurational space. The thermal conductivities of both amorphous and crystalline silicon are then calculated using equilibrium molecular dynamics, which agree well with experimental measurements. This work documents the procedure for training the machine-learning based potentials for modeling thermal conductivity, and demonstrates that machine-learning based potential can be a promising tool for modeling thermal conductivity of both crystalline and amorphous materials with strong disorder.

Density dependence of thermal conductivity in nanoporous and amorphous carbon with machine-learned molecular dynamics

Yanzhou Wang; Zheyong Fan; Ping Qian; Miguel A. Caro; Tapio Ala-Nissilä. Aaltodoc (Aalto University) (2024). Cited by 2. Found by: openalex. openalex_2yr: 0.023 (2026). DOI: 10.48550/arxiv.2408.12390

Abstract: Disordered forms of carbon are an important class of materials for applications such as thermal management. However, a comprehensive theoretical understanding of the structural dependence of thermal transport and the underlying microscopic mechanisms is lacking. Here we study the structure-dependent thermal

conductivity of disordered carbon by employing molecular dynamics (MD) simulations driven by a machine-learned interatomic potential based on the efficient neuroevolution potential approach. Using large-scale MD simulations, we generate realistic nanoporous carbon (NP-C) samples with density varying from 0.3 to 1.5 g cm^{-3} dominated by sp^2 motifs, and amorphous carbon (a-C) samples with density varying from 1.5 to 3.5 g cm^{-3} exhibiting mixed sp^2 and sp^3 motifs. Structural properties including short- and medium-range order are characterized by atomic coordination, pair correlation function, angular distribution function and structure factor. Using the homogeneous nonequilibrium MD method and the associated quantum-statistical correction scheme, we predict a linear and a superlinear density dependence of thermal conductivity for NP-C and a-C, respectively, in good agreement with relevant experiments. The distinct density dependences are attributed to the different impacts of the sp^2 and sp^3 motifs on the spectral heat capacity, vibrational mean free paths and group velocity. We additionally highlight the significant role of structural order in regulating the thermal conductivity of disordered carbon.

Strong lattice anharmonicity and glass-like lattice thermal conductivity in nitrohalide double antiperovskites: A case study based on machine-learning potentials

Yuan Li; Weidong Zheng; Jin Huan Pu; Qi Wang; Jian-Hua Jiang. Thermo-X (2025). Cited by 2. Found by: openalex. DOI: 10.70401/tx.2025.0001

Abstract: Antiperovskites have attracted significant interest in the field of energy conversion in recent years. While extensive research has focused on the magnetism, ionic conductivity and superconductivity of antiperovskites, their thermal properties including lattice anharmonicity and thermal transport remain less explored compared to their well-studied perovskite counterparts. Recently, nitrohalide double antiperovskites have been successfully synthesized. In this work, we investigate the thermal transport properties of nitrohalide double antiperovskites Li6NII2 and Li6NBrBr2 using first-principles machine-learning potentials. Our results reveal that within the perturbation theory framework, imaginary phonons appear throughout the entire Brillouin zone in both the harmonic regime and at elevated temperatures. Atomic vibrational analysis indicates that stochastic Li-ion movements confined within a single conventional unit cell are responsible for the presence of these imaginary phonons. Furthermore, homogeneous nonequilibrium molecular dynamics and equilibrium molecular dynamics simulations demonstrate that Li6NII2 and Li6NBrBr2 exhibit ultralow glass-like lattice thermal conductivities. Spectral thermal conductivity analysis shows that the dominant contributions arise from phonons with frequencies below 5 THz and around 11 THz. The substantial phonon contribution near 11 THz is attributed to the confined stochastic motions of Li ions. This work uncovers the unconventional microscopic cation dynamics and strong lattice anharmonicity in double antiperovskites Li6NII2 and Li6NBrBr2, thereby advancing the understanding of phonon transport in these materials.

Crystal-like thermal transport in amorphous carbon

Jaeyun Moon; Zhiting Tian. arXiv (Cornell University) (2024). Cited by 2. Found by: openalex. DOI: 10.48550/arxiv.2405.07298

Abstract: Thermal transport properties of amorphous carbon has attracted increasing attention due to its extreme thermal properties: It has been reported to have among the highest thermal conductivity for bulk amorphous solids up to $\sim$ 37 Wm^{-1}K^{-1}, comparable to crystalline sapphire ($ $-Al\textsubscript{2}O\textsubscript{3}). Further, large density dependence in thermal conductivity demonstrates a potential for largely tunable thermal conductivity. However, mechanism behind the high thermal conductivity and its large density dependence remains elusive due to many variables at play. In this work, we perform large-scale ($\sim$ 10^{5} atoms) molecular dynamics simulations utilizing a machine learning potential based on neural networks. Through spectral decomposition of thermal conductivity which enables a quantum correction to classical heat capacity, we find that propagating vibrational excitations govern thermal transport in amorphous carbon ($\sim$ 100 \% of thermal conductivity) in sharp contrast to the conventional wisdom that diffusive vibrational excitations dominate thermal transport in amorphous solids. Instead, this remarkable behavior resembles thermal transport in simple crystals. Moreover, our temperature dependent spectral diffusivity and velocity current correlation analyses reveal that the density dependent thermal conductivity originates from anharmonicity sensitive propagating excitations. Our work suggests a novel insight and design principle into developing mechanically hard, thermally conductive amorphous solids.

Universal Interatomic Potentials with DFT for Understanding Orbital Localization in Polydimethylsiloxane-Amorphous Silica Nanocomposites

Carson Farmer; Hector Medina. ACS Omega (2025). Cited by 2. Found by: openalex. openalex_2yr: 4.692 (2026). DOI: 10.1021/acsomega.5c09273

Abstract: High Resolution Image Download MS PowerPoint Slide To investigate molecular orbital localization in polydimethylsiloxane (PDMS) amorphous silica nanocomposites, an approach that integrates first-principles with data-driven methods is introduced: Integrated Modeling and Prediction using Ab initio and Combined Trained potentials for orbital localization (IMPACT4OL). This approach is used to accelerate the identification of localized orbitals near the polymer–nanoparticle interface in polymer nanocomposites (PNC). To accelerate the structure generation, machine-learned interatomic potentials (MLIPs) are utilized to perform molecular dynamics studies for studying the interfacial region with quantum mechanical methods. To investigate the adsorption behavior, various cross-linking densities of PDMS are utilized. While cross-links have been shown to induce defect sites for deep traps, for a small degree of polymerization of PDMS, they hinder the surface adsorption and, thereby, the locations of localized orbitals, which, in turn, can serve as trap sites. The acceleration of orbital localization, especially in large model systems, using IMPACT4OL facilitates the elucidation of intricate mechanisms and could help advance the development of novel PNC-based insulators and electrets. Furthermore, our method introduces a new path to advance understanding of the conformational dynamics for polymer–nanoparticle interface science and engineering.

Ballistic transport from propagating vibrational modes in amorphous silicon dioxide: Thermal experiments and atomistic-machine learning modeling

Man Li; Lingyun Dai; Huan Wu; Yan Yan; Joon Sang Kang; Sophia C. King; Patricia E. McNeil; Danielle M. Butts; Tiphaine Galy; Michal Marszewski; Esther H. Lan; Bruce Dunn; Sarah H. Tolbert; Laurent Pilon; Yongjie Hu. Materials Today Physics (2025). Cited by 2. Found by: openalex. DOI: 10.1016/j.mtphys.2025.101659

Abstract: Understanding thermal transport in amorphous materials is critical for a wide range of applications, including buildings, vehicles, aerospace, and acoustic technologies. Despite its importance, the fundamental behavior of heat carriers in amorphous structures remains poorly understood and is often attributed to localized vibrational modes with mean free paths of about 1 nm, posing significant challenges for engineering their thermal functionalities. In this study, we present experimental measurements on mesoporous silica and atomistic analyses using Monte Carlo simulations and machine learning models to quantify the relationship between nanoarchitecture and effective thermal conductivity. Through rational chemical synthesis and ultrafast spectroscopy measurements, a strong size dependence within the sub-10 nm regime is observed, where the classical Fourier heat conduction theory fails to account for the effects of porosity and pore size. This deviation from diffusive transport is attributed to the significant contribution of propagating vibrational modes, in addition to non-propagating modes, revealing unexpectedly long mean free paths and ballistic thermal transport for heat carriers in amorphous silica. The fundamental vibrational modes in amorphous silica are further investigated using spectral-dependent Boltzmann transport equation simulations and molecular dynamics with machine learning potentials, showing good agreement with experimental results. This study provides valuable insights into nanoscale-modulated thermal transport properties in mesoporous silica and opens new opportunities for the rational design of thermally insulating materials.

Advanced active learning techniques for precision-driven machine-learned potentials in Sb2S3

V Priyadarshini; Pengfei Zou; Debashish Dash; Sanskriti Mahata; Manwendra Chauhan; Kumar Akash. Results in Engineering (2025). Cited by 2. Found by: openalex. DOI: 10.1016/j.rineng.2025.106626

Abstract: • A machine-learned interatomic potential for amorphous and crystalline Sb S was developed using active learning and DFT reference data. • The trained moment tensor potential accurately reproduces DFT-level energies, forces, and stresses with high R^2 values, demonstrating excellent predictive capability. • Structural features such as pair distribution functions and bond environments from MTP-generated configurations closely match experimental and ab initio references. • Ab initio molecular dynamics simulations were employed to benchmark thermal and dynamic behavior, guiding accurate MTP fitting. The model effectively captures the medium-range order and coordination in disordered Sb S , making it suitable for simulating amorphous phases at low computational cost. • This work enables large-scale simulations of Sb S for electronic and thermomechanical applications, bridging accuracy and efficiency through ML potentials. We present a powerful and efficient strategy for modeling amorphous antimony trisulfide (Sb S) using a Machine-Learned Interatomic Potential (MLIP) developed via an active learning approach based on the Moment Tensor Potential (MTP) framework. Beginning with bulk geometries of the orthorhombic, triclinic, and monoclinic phases of Sb S , we

extracted essential elemental and structural features to construct a robust training dataset. The active learning loop systematically identified high-uncertainty configurations, which were refined through Density Functional Theory (DFT) calculations to interact with improving model accuracy. This iterative, data-efficient learning process not only significantly reduced computational expense but also yielded remarkable accuracy in predicting energies, atomic forces, and stress tensors. Our findings reveal a striking concordance between MTP predictions and DFT results, affirming the potential's reliability. Furthermore, the resulting MTP model is benchmarked against experimental and DFT/PBE (density functional theory/Perdew—Burke—Ernzerhof) level using ab initio molecular dynamics (AIMD) in terms of structural metrics such as radial and angular distribution functions. Our results show that the MTP model reliably captures the short- and medium-range ordering in amorphous Sb S and can serve as an efficient surrogate for DFT in large-scale atomistic simulations of disordered materials. By advancing the ML training of interatomic potentials tailored to complex, disordered systems like amorphous Sb S , this study lays the groundwork for a scalable and generalizable approach to simulating next-generation amorphous materials with high precision and efficiency.

Atomic armor for thermal stability in nanoporous structures

Rui Yang; Qiaoling Si; Qiang Sheng; Mingyang Yang; Mu Du; Hu Zhang; Xiao-Lei Shi; G.H. Tang; Zhi-Gang Chen. Proceedings of the National Academy of Sciences (2025). Cited by 2. Found by: openalex. DOI: 10.1073/pnas.2510746122

Abstract: Nanoporous structures play a critical role in a wide range of applications, including catalysis, thermoelectrics, energy storage, gas adsorption, and thermal insulation. However, their thermal instability remains a persistent challenge. Inspired by the extraordinary resilience of tardigrades, an "atomic armor" strategy is introduced to enhance the stability of nanoporous structures. Applied to mesoporous silica at parts-per-million levels, the atomic armor provides thermal resistance exceeding that of existing stabilization techniques. Thermal treatment at 1,000 °C for 168 h results in a fivefold increase in specific surface area, 66% lower thermal conductivity, and a sixfold increase in pore volume compared to untreated samples. Surface viscosity is linked to sintering resistance, and glass transition temperature and fragility are introduced as design parameters. Machine-learned interatomic potentials and metabasin escape algorithm-assisted molecular dynamics simulations are employed to reveal that materials traditionally classified as nonglass formers can exhibit glass transition temperatures and display intrinsic fragility. Alumina is identified as having a record-high glass transition temperature. By modulating the surface viscosity of nanoparticles, this approach stabilizes nanoporous structures effectively. The proposed method offers a simple and universal posttreatment process for improving the thermal stability of nanoporous structures.

Physics-informed machine learning exploration of Na storage mechanisms in disordered carbon

Nikhil Rampal; Stephen E. Weitzner; Fredrick Omenya; Marissa Wood; David Reed; Xiaolin Li; Jonathan R. I. Lee; Liwen F. Wan. Energy storage materials (2026). Cited by 2. Found by: openalex. openalex_2yr: 13.488 (2026). DOI: 10.1016/j.ensm.2026.104967

Thermal conductivity of Li$_3$PS$_4$ solid electrolytes with ab initio accuracy

Davide Tisi; Federico Grasselli; Lorenzo Gigli; Michele Ceriotti. arXiv (Cornell University) (2024). Cited by 2. Found by: openalex. DOI: 10.48550/arxiv.2401.12936

Abstract: The vast amount of computational studies on electrical conduction in solid-state electrolytes is not mirrored by comparable efforts addressing thermal conduction, which has been scarcely investigated despite its relevance to thermal management and (over)heating of batteries. The reason for this lies in the complexity of the calculations: on one hand, the diffusion of ionic charge carriers makes lattice methods formally unsuitable, due to the lack of equilibrium atomic positions needed for normal-mode expansion. On the other hand, the prohibitive cost of large-scale molecular dynamics (MD) simulations of heat transport in large systems at ab initio levels has hindered the use of MD-based methods. In this paper, we leverage recently developed machine-learning potentials targeting different ab initio functionals (PBEsol, r^{2}SCAN, PBE0) and a state-of-the-art formulation of the Green-Kubo theory of heat transport in multicomponent systems to compute the thermal conductivity of a promising solid-state electrolyte, Li$_3$PS$_4$, in all its polymorphs ($ $, $ $, and $ $). By comparing MD estimates with lattice methods on the low-temperature, nondiffusive $ $-Li$_3PS_4$, we highlight strong anharmonicities and negligible nuclear quantum effects, hence further justifying MD-based methods even for nondiffusive phases. Finally, for the ion-conducting $ $ and $ $ phases, where the multicomponent Green-Kubo MD approach is mandatory, our simulations indicate a weak temperature dependence of the thermal conductivity, a glass-like behavior due to the effective local disorder characterizing these Li-diffusing phases.

Unveiling the crystallization kinetics in Ge-rich Ge$_x$Te alloys by large scale simulations with a machine-learned interatomic potential

Dario Baratella; Omar Abou El Kheir; Marco Bernasconi. arXiv (Cornell University) (2024). Cited by 1. Found by: openalex. DOI: 10.48550/arxiv.2404.15128

Abstract: A machine-learned interatomic potential for Ge-rich Ge$_x$Te alloys has been developed aiming at uncovering the kinetics of phase separation and crystallization in these materials. The results are of interest for the operation of embedded phase change memories which exploits Ge-enrichment of GeSbTe alloys to raise the crystallization temperature. The potential is generated by fitting a large database of energies and forces computed within Density Functional Theory with the neural network scheme implemented in the DeePMD-kit package. The potential is highly accurate and suitable to describe the structural and dynamical properties of the liquid, amorphous and crystalline phases of the wide range of compositions from pure Ge and stoichiometric GeTe to the Ge-rich Ge$_2$Te alloy. Large scale molecular dynamics simulations revealed a crystallization mechanism which depends on temperature. At 600 K, segregation of most of Ge in excess occurs on the ns time scale followed by crystallization of nearly stoichiometric GeTe regions. At 500 K, nucleation of crystalline GeTe occurs before phase separation, followed by a slow crystal growth due to the concurrent expulsion of Ge in excess.

On the boroxol ring fraction in melt-quenched B2O3 glass: Insights from machine learning potentials

Debendra Meher; Nikhil V. S. Avula; Sundaram Balasubramanian. The Journal of Chemical Physics (2026). Cited by 1. Found by: openalex. DOI: 10.1063/5.0321441

Abstract: An atomistic structural model for melt-quenched B2O3 glass has eluded the simulation community so far. The difficulty lies in the abundance of six-membered boroxol rings-an intermediate-range order motif suggested by Raman and NMR spectroscopy-which is challenging to capture in atomistic molecular dynamics simulations. Here, we report the development of a density functional theory-accurate machine-learned potential and employ quench rates as low as 109 K/s to obtain B2O3 glasses with more than 30% of boron atoms in boroxol rings. Additionally, we show that the pressure, and consequently the boroxol fraction, in the deep potential molecular dynamics simulations critically depends on the range of the geometry descriptor used in the embedding neural network, and it converges beyond a value of 7 Å. The boroxol ring fraction increases with decreasing quench rate. Finally, amorphous B2O3 configurations display a minimum in energy at a boroxol fraction of 75%, remarkably close to the experimental estimate in B2O3 glass.

Accurate prediction of structural and mechanical properties on amorphous materials enabled through machine-learning potentials: a case study of silicon nitride

Ganesh Kumar Nayak; Prashanth Srinivasan; Juraj Todt; R. Daniel; Paolo Nicolini; David Holec. DESY Publication Database (PUBDB) (Deutsches Elektronen-Synchrotron) (2024). Cited by 1. Found by: openalex. openalex_2yr: 0.03 (2026). DOI: 10.48550/arxiv.2408.05782

Abstract: Amorphous silicon nitride (a-SiN) is a material which has found wide application due to its excellent mechanical and electrical properties. Despite the significant effort devoted in understanding how the microscopic structure influences the material performance, many aspects still remain elusive. If on the one hand \textit{ab initio} calculations respresent the technique of election to study such a system, they present severe limitations in terms of the size of the system that can be simulated. Such an aspect plays a determinant role, particularly when amorphous structure are to be investigated, as often results depend dramatically on the size of the system. Here, we overcome this limitation by training a machine-learning (ML) interatomic model to \textit{ab initio} data. We show that molecular dynamics simulations using the ML model on much larger systems can reproduce experimental measurements of elastic properties, including elastic isotropy. Our study demonstrates the broader impact of machine-learning potentials for predicting structural and mechanical properties, even for complex amorphous structures.

Developing a Neural Network potential to investigate interface phenomena in solid-phase epitaxy

Ruggero Lot; Layla Martin-Samos; Stefano de Gironcoli; Anne Hémeryck. (2021). Cited by 1. Found by: openalex. DOI: 10.1109/nmdc50713.2021.9677541

Abstract: In this work, we develop a new neural network potential for silicon and perform accurate molecular dynamics simulations of the liquid, amorphous and diamond phases. The potential is tested against several physical properties and the solid phase epitaxy process is simulated.

A Unified Deep Neural Network Potential Capable of Predicting Thermal Conductivity of Silicon in Different Phases

Ruiyang Li; Eungkyu Lee; Tengfei Luo. arXiv (Cornell University) (2019). Cited by 1. Found by: openalex. DOI: 10.48550/arxiv.1912.05044

Abstract: Molecular dynamics simulations have been extensively used to predict thermal properties, but simulating different phases with similar precision using a unified force field is often difficult, due to the lack of accurate and transferrable interatomistic potential fields. As a result, this issue has become a major barrier to predicting the phase change of materials and their transport properties with atomistic-level modeling techniques. Recently, machine learning based algorithms have emerged as promising tools to develop accurate potentials for molecular dynamics simulations. In this work, we approach the problem of predicting the thermal conductivity of silicon in different phases by performing molecular dynamics simulations with a deep neural network potential. This neural network potential is trained with ab-initio data of silicon in the crystalline, liquid and amorphous phases. The accuracy of our potential is first validated through reproducing the atomistic structures during the phase transition, where other empirical potentials usually fail. The thermal conductivity of different phases is then calculated, showing a good agreement with the experimental results and ab-initio calculation results. Our work shows that a unified neural network-based potential can be a promising tool for studying phase change and thermal transport of materials with high accuracy.

Simulation of structure and dynamics of phosphate glasses with fluoride additives: A hybrid approach combining universal and specialized neural network potentials

I. A. Balyakin; K. A. Arslanov; Maxim I. Vlasov. Computational Materials Science (2025). Cited by 1. Found by: openalex. openalex_2yr: 3.958 (2026). DOI: 10.1016/j.commatsci.2025.114065

Nature of adsorption of amino acids and precursors on interstellar amorphous solid water

Natsuki Watanabe; Yuta Hori; Kazuki Okazawa; Yasuteru Shigeta; Mitsuo Shoji. Monthly Notices of the Royal Astronomical Society (2025). Cited by 1. Found by: openalex. DOI: 10.1093/mnras/staf1913

Abstract: ABSTRACT Hundreds of interstellar molecules have been identified in interstellar molecular clouds by radio telescope observations. Although aminoacetonitrile, one of the glycine precursors, has been identified in molecular clouds, amino acids have not been detected. Therefore, the presence of amino acids in molecular clouds has not been confirmed yet. This study elucidates the nature of the adsorption of amino acids and their precursors on interstellar amorphous solid water (ASW) from the perspective of the binding energy. To calculate the binding energies of the adsorbed molecules on model ASW, machine learning potential-based molecular dynamics simulations and quantum chemical calculations were performed. For amino nitriles, precursors of amino acids, we found that the amino group forms stronger hydrogen bonds with ASW than the nitrile group, owing to the larger negative partial charge on the nitrogen in the amino group. We also revealed that the presence of methyl groups enhances the stability of adsorption, since methyl groups act as stronger electron donors than hydrogen atoms. Additionally, our results suggest that the adsorption strength of amino acids is larger than that of typical interstellar complex organic molecules, such as methanol, indicating that the detection of amino acids using radio telescopes will be challenging. In contrast, 2-aminopropionitrile, an alanine precursor, has binding energies comparable to those of aminoacetonitrile. These results suggest that 2-aminopropionitrile could be desorbed into the gas phase and has a high probability of being detected in molecular clouds by telescopic observations.

AI-Aided Mapping of the Structure-Composition-Conductivity Relationships of Glass-Ceramic Lithium Thiophosphate Electrolytes

Haoyue Guo; Qian Wang; Alexander E. Urban; Nongnuch Artrith. arXiv (Cornell University) (2022). Cited by 1. Found by: openalex. DOI: 10.48550/arxiv.2201.11203

Abstract: Lithium thiophosphates (LPS) with the composition (Li$_2$S)$_x$(P$_2$S$_5$)$_{1-x}$ are among the most promising prospective electrolyte materials for solid-state batteries (SSBs), owing to their superionic conductivity at room temperature ($>10^{-3}$ S cm^{-1}), soft mechanical properties, and low grain boundary resistance. Several glass-ceramic (gc) LPS with different compositions and good Li conductivity have been previously reported, but the relationship between composition, atomic structure, stability, and Li conductivity remains unclear due to the challenges in characterizing non-crystalline phases in experiments or simulations. Here, we mapped the LPS phase diagram by combining first principles and artificial intelligence (AI) methods, integrating density functional theory, artificial neural network potentials, genetic-algorithm sampling, and ab initio molecular dynamics simulations. By means of an unsupervised structure-similarity analysis, the glassy/ceramic phases were correlated with the local structural motifs in the known LPS crystal structures, showing that the energetically most favorable Li environment varies with the composition. Based on the discovered trends in the LPS phase diagram, we propose a candidate solid-state electrolyte composition, (Li$_{2}$S)$_{x}$(P$_{2}$S$_{5}$)$_{1-x}$ ($x\sim{}0.725$), that exhibits high ionic conductivity ($>10^{-2}$ S cm^{-1}) in our simulations, thereby demonstrating a general design strategy for amorphous or glassy/ceramic solid electrolytes with enhanced conductivity and stability.

Liquid anomalies and fragility of supercooled antimony

Flavio Giuliani; Francesco Guidarelli Mattioli; Yuhan Chen; Dario Baratella; Daniele Dragoni; Marco Bernasconi; John Russo; Lilia Boeri; Riccardo Mazzarello. Proceedings of the National Academy of Sciences (2026). Cited by 1. Found by: openalex. DOI: 10.1073/pnas.2531605123

Abstract: Phase-change materials (PCMs) based on group IV, V, and VI elements, such as Ge, Sb, and Te, exhibit distinctive liquid-state features, including thermodynamic anomalies and unusual dynamical properties, which are believed to play a key role in their fast and reversible crystallization behavior. Antimony (Sb), a monoatomic PCM with ultrafast switching capabilities, stands out as the only elemental member of this group for which the properties of the liquid and supercooled states have so far remained unknown. In this work, we use large-scale molecular dynamics simulations with a neural network potential trained on first-principles data to investigate the liquid, supercooled, and amorphous phases of Sb across a broad pressure-temperature range. We uncover clear signatures of anomalous behavior, including a density maximum and nonmonotonic thermodynamic response functions, which are described by a two-state model based on the structural evolution of the liquid. Moreover, extrapolation of the viscosity to the glass transition, based on configurational and excess entropies, indicates that Sb is a highly fragile material. Our results present a compelling case for the connection between the liquid-state properties of PCMs and their unique ability to combine high amorphous-phase stability with ultrafast crystallization.

Origin of Doping-Induced Structural Amorphization and Improved Cycling Performance in Aluminum Anodes for Lithium-Ion Batteries

Jiwon Yu; Gyeong S. Hwang. The Journal of Physical Chemistry C (2025). Cited by 1. Found by: openalex. DOI: 10.1021/acs.jpcc.5c06267

Abstract: Aluminum (Al) has been considered a promising anode material for advanced lithium-ion batteries (LIBs); however, it is prone to degradation after multiple electrochemical cycles. Using first-principles-based simulations, we emphasize the critical role of structural amorphization and demonstrate how doping influences structural disorder and, in turn, the cyclic stability of Al anodes. The amorphous phase of Al is found to be readily lithiated, whereas its crystalline counterpart exhibits resistance during the initial stage of lithiation. Our work reveals that silicon (Si) doping effectively stabilizes the Al amorphous structure, while iron (Fe) and copper (Cu) are ineffective in suppressing its recrystallization. This behavior is largely attributed to the strong Al–Si bonding interaction. In addition, our neural network potential-based molecular dynamics simulations illustrate that Li atoms preferentially incorporate around Si dopants, leading to facilitated and more spatially uniform lithiation in Si-doped Al compared to undoped Al. This, in turn, enhances structural integrity and promotes reversible volume changes during electrochemical cycling. Our findings highlight that the distinct lithiation behavior of Si-doped Al anodes may offer advantages over conventional alloy-type anodes that suffer from structural disintegration arising from large volume changes.

Softening of Vibrational Modes and Anharmonicity Induced Thermal Conductivity Reduction in a-Si:H at High Temperatures

Zhuo Chen; Yuejin Yuan; Yanzhou Wang; Penghua Ying; Shouhang Li; Cheng Shao; Wenyang Ding; Gang Zhang; Meng An. arXiv (Cornell University) (2025). Cited by 1. Found by: openalex. DOI: 10.48550/arxiv.2502.05584

Abstract: Hydrogenated amorphous silicon (a-Si:H) has garnered considerable attention in the semiconductor industry, particularly for its use in solar cells and passivation layers for high performance silicon solar cells, owing to its exceptional photoelectric properties and scalable manufacturing processes. A comprehensive understanding of thermal transport mechanism in a-Si:H is essential for optimizing thermal management and ensuring the reliable operation of these devices. In this study, we developed a neuroevolution machine learning potential based on first-principles calculations of energy, forces, and virial, which enables accurate modeling of interatomic interactions in both a-Si:H and a-Si systems. Using the homogeneous nonequilibrium molecular dynamics (HNEMD) method, we systematically investigated the thermal conductivity of a-Si:H and a-Si across a temperature range of 300-1000 K and hydrogen concentrations ranging from 6 to 12 at%. Our simulation results found that thermal conductivity of a-Si:H with 12 at% hydrogen was significantly reduced by 12% compared to that of a-Si at 300 K. We analyzed the spectral thermal conductivity, vibrational density of states and lifetimes of vibrational modes, and revealed the softening of vibrational modes and anharmonicity effects contribute to the reduction of thermal conductivity as temperature and hydrogen concentration increase. Furthermore, the influence of hydrogen concentration and temperature on diffuson and propagon contribution to thermal conductivity of a-Si:H was revealed. This study provides valuable insights for developing thermal management strategies in silicon-based semiconducting devices and advances the understanding of thermal transport in amorphous systems.

Atomistic Insights into Stress-Driven Lithiation at Silicon Anode Crack Tips

Bowen Zhang; Peiyao Zhang; Changguo Wang; Huifeng Tan; Liwei Dong; Jia-Yan Liang; Yuanpeng Liu. ACS Applied Materials & Interfaces (2025). Cited by 1. Found by: openalex. openalex_2yr: 8.382 (2026). DOI: 10.1021/acsami.5c11237

Abstract: Silicon is a leading candidate for next-generation lithium-ion battery anodes due to its high theoretical capacity. However, large volume changes during lithiation and delithiation generate significant mechanical stress, leading to particle fracture and crack formation, which degrade electrode performance. In this work, we investigate the lithiation dynamics at the tip of the crack along the [112] direction using molecular dynamics simulations driven by a machine learning potential trained with the NEP framework. By applying a range of tensile strains, we systematically explored how crack-tip stress fields influence the atomic-scale lithiation process. Under zero stress, lithium insertion proceeds uniformly, generating a flat amorphous-crystalline interface. Moderate tensile stress leads to the formation of a stepped interface morphology, which is consistent with a ledge-mediated amorphization mechanism. Quantitative kinetic analysis reveals that tensile strain reduces the activation energy for lithiation, thereby accelerating interface propagation. At higher stress levels, the lithiation front becomes unstable and advances via narrow, stress-guided channels that penetrate deeply into the crystalline silicon. These results demonstrate that mechanical stress fields not only modulate lithiation kinetics but also dictate the morphological evolution of the reaction front. This study provides fundamental atomistic insights into the chemo-mechanical coupling that governs lithiation behavior in silicon and may inform the design of more durable high-capacity battery electrodes.

Form and Function of Disorder

Parthapratim Biswas; Gang Chen; Serge Nakhmanson; Jianjun Dong. physica status solidi (b) (2021). Cited by 1. Found by: openalex. DOI: 10.1002/pssb.202100366

Abstract: David A. Drabold Viewing the subject in the light of its development over a period of the past five decades, it is evident that the field of disordered materials continues to present significant challenges and problems of outstanding interest to date. While the generic behavior of many disordered systems is now more or less understood, our knowledge and understanding of the properties of real disordered materials are still evolving and far from being complete. From a theoretical point of view, the main difficulty arises from the lack of knowledge of atomic positions in disordered solids. Diffraction experiments, while provide useful structural information through atomic pair-correlation functions, alone cannot determine the three-dimensional structure of disordered solids via inversion of pair-correlation data. In view of this, the great majority of current theoretical methods for structural inference of disordered solids are limited to Monte Carlo and molecular-dynamics simulations, using a variety of total-energy functionals from classical and semi-classical to quantum-mechanical approaches. The last approach mostly relies on the density-functional theory (DFT), developed by Kohn and Sham in the sixties. While these approaches have been highly successful in determining structures and properties of many disordered solids, such as structural glasses, they are nonetheless limited to either small system sizes (via DFT calculations) or relatively simple amorphous/glassy solids for which accurate classical or semi-classical force-fields are available. The computational complexity associated with DFT calculations for large disordered systems continues to present a major obstacle in studying complex multi-component glasses

of technological importance. Metallic glasses are classic examples in this category, the properties of which are often characterized by medium-range ordering on the nanometer length scale. Likewise, the interplay between magnetism and disorder remains largely an uncharted territory. While sophisticated theories of magnetism (for disordered materials) are already in place, a unified approach that simultaneously takes into account structural disorder and magnetic interactions between atoms is still missing. In recent years, the emergence of a new breed of computational methods, which rely on experimental data and relevant structural information, brings fresh impetus to address the problem of materials modeling from a data-centric point of view. These data-driven methods either construct a machine-learning (ML) potential, using a set of training data from a small number of high-quality DFT simulations and experiments, or generate realistic models of disordered solids by directly incorporating a set of experimental data and (approximate) total-energy functionals in an augmented solution space. Although it remains to be seen to what extent ML approaches can address some of the most difficult problems in materials modeling, involving multi-component glasses in a non-stoichiometric environment with associated length- and time-scale issues, there is a cautious optimism in the air that ML approaches can bring a paradigm shift in computational materials discovery. While we wait for the future to deliver a definitive answer, it is undoubtedly an exciting time for studying disordered materials when new data-driven methods and experimental results are reported regularly. The current issue "Form and Function of Disorder" originates from our desire to honor and celebrate the sixtieth birthday of Professor David A. Drabold and his contributions in this field. Although we had to cancel the accompanying conference (May 30–31, 2020 in Athens, Ohio, USA), due to the outbreak of Covid19 pandemic in 2020, we are delighted that the special issue has finally come to fruition irrespective of the pandemic-related difficulties that we have experienced recently. The issue consists of twenty-two (22) articles of which fourteen (14) are directly related to the study of disordered systems/materials. The topics covered are diverse in nature and they range from applications of classical, quantum-mechanical, and ML approaches to glasses of technological interest, as well as generic understanding of disordered systems from geometric and topological points of view. Following our discussion on emerging methods and their applications, and noting the limited availability of space, we briefly mention here a small number of articles that are thematically related to the title of the issue. The article by Konstantinou et al. (pssb.202000416) illustrates the use of machine-learning potentials with DFT accuracy in studying a phase-change material, Sb2Te3, whereas the application of a data-assisted FEAR approach is presented by Thapa et al. (pssb.202000415) to study an important metallic glassy system, Cu46–Zr46–Al8. The transition from glasses to crystalline structures is discussed by Sun et al. (pssb.202000427), where the authors examine the evolution of the interface between glassy and crystalline structures in lithium disilicate systems. The work by Dahal et al. (pssb.202000447) explores the origin of the fast sharp diffraction (FSDP) peak in amorphous silicon and provides some new insights on the relationship between the position of the FSDP in the wavevector space and a radial length scale in real space. Subedi et al. (pssb.202000438) discuss a fascinating approach to employ the Kubo-Greenwood formula for studying conductivity in amorphous solids that leads to identification of conductivity paths/regions in real space via construction of a conducting matrix. The article by Biswas (pssb.202000610) addresses the structure of energy landscapes of some highly degenerate model spin-glass systems. The author presents a new classification scheme by providing a two-dimensional description of high-dimensional potential-energy surfaces, showing the energy minima and barriers that separate them. The nature of atomic vibrations in models of vitreous silica has been studied by Shcheblanov et al. (pssb.202000422) with an emphasis on the low- frequency vibrational modes. The authors show that the modes are quasi-localized in nature and resulted from a mixing of non-localized and localized vibrational states. The latter have been found to exhibit both exponential and power-law decays. The article by Sadjadi et al. (pssb.202000555), on isostatic material frameworks, provides a brilliant discussion on how topological and geometric constraints in random networks can play a decisive role in understanding some generic properties of real disordered materials. The remaining articles are equally fascinating and enjoyable to read. We thank the authors for delivering a high standard of scientific content in their articles for this special issue. Finally, we are grateful to the editorial team of physica status solidi (b), and in particular Dr. Sabine Bahrs, who worked alongside us for the past year to make this issue finally appear in print. Parthapratim Biswas Gang Chen Serge Nakhmanson Jianjun Dong July 27, 2021 Parthapratim Biswas is Professor of Physics in the University of Southern Mississippi (Hattiesburg, MS, USA). He obtained his Ph.D. from S. N. Bose National Center for Basic Sciences in Kolkata (India) and did his postdoctoral research in the Netherlands, UK, USA, and Germany. The primary area of his research interest is electronic-structure theory of disordered solids. His current research involves development of information-based inverse and hybrid approaches to material structure determination, using geometric, topological, and experimental information. Gang Chen received his Ph.D. in Materials Science and Engineering at Lehigh University (Bethlehem, PA, USA) in 2004. He is currently an Associate Professor in the Department of Physics and Astronomy at Ohio University (USA). His research is focused on fundamental understanding of the structure–property relations in amorphous electronic materials. Serge Nakhmanson's research interests involve modeling of functional materials properties using quantum-mechanical and finite-element based computational tools. He is one of the original creators and a developer of the 'Ferret' code for simulating ferroic functionalities at mesoscale. Specific areas of interest include novel multifunctional materials and predicting their behavior; influence of shape, size and orientation on the material properties, as well as occasional excursions into "soft" ferroelectrics, such as polymers, oligomers and molecular crystals. Jianjun (JJ) Dong is a Professor of Physics at Auburn University (Auburn, AL, USA). He received his Ph.D. degree from Ohio University in 1998, under the supervision of Prof. David Drabold. His research focuses on theories and simulations of real materials. A current research thrust is to develop new theories and computational methodologies to understand heat transfer properties of materials, such as materials at extreme low or high temperatures, very anharmonic crystals, disordered/glassy solids, and material interfaces.

Nature of the amorphous-amorphous interfaces in solid-state batteries revealed using machine-learned interatomic potentials

Chuhong Wang; Muratahan Aykol; Tim Mueller. ChemRxiv (2023). Cited by 0. Found by: openalex. openalex_2yr: 0.467 (2026). DOI: 10.26434/chemrxiv-2023-frr79

Abstract: Non-crystalline solid materials have attracted growing attention in energy storage for their desirable properties such as ionic conductivity, stability and processability. However, compared to bulk crystalline materials, fundamental understanding of these highly complex metastable systems is hindered by the scale limitations of density functional theory (DFT) calculations and resolution limitations of experimental methods. To fill the knowledge gap and guide the rational design of amorphous battery materials and interfaces, we present a molecular dynamics (MD) framework based on machine-learned interatomic potentials trained on the fly to study the amorphous solid electrolyte Li3PS4 and its protective coating, amorphous Li3B11O18. The use of machine-learned potentials allows us to simulate the materials at time and length scales that are not accessible to DFT while maintaining a near-DFT level of accuracy. This approach allows us to calculate amorphization energies, amorphous-amorphous interface energies, and the impact of the interface on lithium ion conductivity. This study demonstrates the promising role of actively-learned interatomic potentials in extending the application of ab-initio modeling to more complex and realistic systems such as amorphous materials and interfaces.

Investigating Ionic Diffusivity in Amorphous Solid Electrolytes using Machine Learned Interatomic Potentials

Aqshat Seth; Rutvij Pankaj Kulkarni; Gopalakrishnan Sai Gautam. arXiv (Cornell University) (2024). Cited by 0. Found by: openalex. DOI: 10.48550/arxiv.2409.06242

Abstract: Investigating Li$^+$ transport within the amorphous lithium phosphorous oxynitride (LiPON) framework, especially across a Li||LiPON interface, has proven challenging due to its amorphous nature and varying stoichiometry, necessitating large supercells and long timescales for computational models. Notably, machine learned interatomic potentials (MLIPs) can combine the computational speed of classical force fields with the accuracy of density functional theory (DFT), making them the ideal tool for modelling such amorphous materials. Thus, in this work, we train and validate the neural equivariant Interatomic potential (NequIP) framework on a comprehensive DFT-based dataset consisting of 13,454 chemically relevant structures to describe LiPON. With an optimized training (validation) energy and force mean absolute errors of 5.5 (6.1) meV/atom and 13.6 (13.2) meV/Å, respectively, we employ the trained potential in model Li-transport in both bulk LiPON and across a Li||LiPON interface. Amorphous LiPON structures generated by the optimized potential do resemble those generated by ab initio molecular dynamics, with N being incorporated on non-bridging apical and bridging sites. Subsequent analysis of Li$^+$ diffusivity in the bulk LiPON structures indicates broad agreement with computational and experimental literature so far. Further, we investigate the anisotropy in Li$^+$ transport across the Li(110)||LiPON interface, where we observe Li-transport across the interface to be one order-of-magnitude slower than Li-motion within the bulk Li and LiPON phases. Nevertheless, we note that this anisotropy of Li-transport across the interface is minor and do not expect it to cause any significant impedance buildup. Finally, our work highlights the efficiency of MLIPs in enabling high-fidelity modelling of complex non-crystalline systems over large length and time scales.

Machine-learned interatomic potentials for modelling nanoscale fracturing in silica and basalt

Marthe Grønlie Guren; Henrik Anderson Sveinsson; Anders Malthe-Sørenssen; Razvan Caracas; François Renard. (2023). Cited by 0. Found by: openalex. DOI: 10.5194/egusphere-egu23-6256

Abstract: At the nanoscale, fracturing creates surface area and flow pathways, which control the rates of fluid-rock interactions. However, how fractures form at the nanoscale remains enigmatic. Here, we implement molecular dynamics simulations to reproduce fracture propagation in quartz and basalt. These simulations require large systems and long simulation times and are therefore currently depending on interatomic potentials. In the recent years, machine learning approaches have been established as a way to fit interatomic potentials,

where the potentials are trained with quantum-mechanical data obtained from ab initio molecular dynamics simulations. We have developed machine-learned interatomic potentials for silica and basalt that allow using molecular dynamics simulations to simulate fracture propagation at the nanoscale. The interatomic potentials reproduce the mechanical properties of bulk silica and basalt sand have also been trained to account for fracture propagation. First, we trained a potential on silica to verify the fitting procedure, and then we used the same procedure to train an interatomic potential for basalt. By training the potential with water and carbon dioxide as fluids, we aim to study how a dynamic fracture damage basaltic glass and how the water and carbon dioxide enter these fractures in the wake of rupture. Our results are relevant for carbon mineralization where a coupling between dissolution of the basalt and precipitation of carbonate minerals can lead to nanofracturing of the rock.

Dynamic Metrics to Validate the Use of Machine-Learned Interatomic Potential for Dynamic Sampling

Bourgeois Biova Irénée Gadjagboui; Md Sharif Khan; Oliviero Andreussi. (2026). Cited by 0. Found by: openalex. DOI: 10.1145/3773966.3785507

Abstract: Machine learning and data-driven approaches are increasingly being used to predict the energies and forces of chemical compounds and materials, enabling in silico design and the construction of digital twins. Machine-learning interatomic potentials (MLIPs), typically trained on large databases of quantum-mechanical calculations for small systems, are widely employed to sample configurations of realistic, large-scale atomistic systems. However, these models are usually evaluated only on static benchmark datasets, which does not reveal how their accuracy evolves along dynamical trajectories or under changes in chemical composition. Here, we introduce a simple and general validation protocol for a MLIP of an amorphous material that tracks prediction error along molecular-dynamics (MD) trajectories and across compositions. For the specific system under study, the designed protocol shows that the model exhibits consistently low energy errors, with no systematic drift over time and no loss of accuracy as composition varies. This type of diagnostic provides a compact complement to standard validation procedures and can be readily applied to other models and materials systems.

Dependence of a cooling rate on structural and vibrational properties of amorphous silicon: A neural network potential-based molecular dynamics study

; Yasunobu Ando. Institutional Repositories DataBase (IRDB) (2019). Cited by 0. Found by: openalex. URL: https://openalex.org/W7145405791

Critical Composition Window for Co–P Bulk Metallic Glasses Revealed by Machine-Learned Interatomic Potentials

Md Sharif Khan; Bourgeois Biova Irénée Gadjagboui; Oliviero Andreussi. ChemRxiv (2025). Cited by 0. Found by: openalex. openalex_2yr: 0.467 (2026). DOI: 10.26434/chemrxiv-2025-vl9mj

Abstract: Bulk metallic glasses are a distinct class of alloys whose disordered atomic structure imparts exceptional mechanical strength, corrosion resistance, and catalytic activity. Nevertheless, the optimization of their compositional ratios remains predominantly empirical, relying largely on qualitative and non-predictive computational models. In this work we developed a machine interatomic potential for Co–P binary alloys, trained on approximately 32,000 density functional theory–labeled configurations spanning a wide range of compositions (5–40% P) and temperatures (300–1800 K). The model accurately reproduced reference energies and forces and remained reliable in high-temperature molecular dynamics simulations. Analysis of structural motifs revealed that high phosphorus content promotes medium-range ordering, while very low phosphorus content favors partial crystallinity. Intermediate compositions (20, 15, and 10% P) exhibited structural signatures of bulk metallic glasses, including pronounced icosahedral clustering, split second-neighbor peaks in radial distribution functions, and coordination environments resembling distorted bicapped square antiprisms. Thermal and dynamical assessments showed that these compositions form stable amorphous phases with favorable glass transition temperatures. The developed potential enables large-scale simulations and offers guidance for optimizing Co–P-based metallic glasses for structural and energy applications.

Suppression of Rayleigh Scattering in Silica Glass by Codoping Boron and Fluorine: Molecular Dynamics Simulations with ForceMatching and Neural Network Potentials

; Nobuhiro Nakamura; ; Tomofumi Tada (1674688); ; HIDEO HOSONO. Institutional Repositories DataBase (IRDB) (2022). Cited by 0. Found by: openalex. URL: https://openalex.org/W7145446861

Insights Into Radiation Damage in YBa$_2$Cu$_3$O$_{7- }$ From Machine-Learned Interatomic Potentials

Ashley Dicksona; Niccolò Di Eugenio; Federico Ledda; Daniele Torsello; F. Laviano; Flyura Djurabekova; Jesper Byggmästar; Mark R. Gilbert; D. Nguyen-Manh; Erik Gallo; Antonio Trotta; Davide Gambino; Samuel T. Murphy. arXiv (Cornell University) (2025). Cited by 0. Found by: openalex. URL: https://openalex.org/W7118052361

Abstract: Accurate prediction of radiation damage in YBa$_2$Cu3O${7- }$ (YBCO) is essential for assessing the performance of high-temperature superconducting (HTS) tapes in compact fusion reactors. Existing empirical interatomic potentials have been used to model radiation damage in stoichiometric YBCO, but fail to describe oxygen-deficient compositions, which are ubiquitous in industrial Rare-Earth Barium Copper Oxide conductors and strongly influence superconducting properties. In this work, we demonstrate that modern machine-learned interatomic potentials enable predictive modelling of radiation damage in YBCO across a wide range of oxygen stoichiometries, with higher fidelity than previous empirical models. We employ two recently developed approaches: an Atomic Cluster Expansion (ACE) potential and a tabulated Gaussian Approximation Potential (tabGAP). Both models accurately reproduce Density Functional Theory (DFT) energies, forces, and threshold displacement energy distributions, providing a reliable description of atomic-scale collision processes. Molecular dynamics simulations of 5 keV cascades predict enhanced peak defect production and recombination relative to a widely used empirical potential, indicating different cascade evolution. By explicitly varying oxygen deficiency, we show that total defect production depends only weakly on stoichiometry, offering insight into the robustness of radiation damage processes in oxygen-deficient YBCO. Finally, fusion-relevant 300 keV cascade simulations reveal amorphous regions with dimensions comparable to the superconducting coherence length, consistent with electron microscopy observations of neutron-irradiated HTS tapes. These results establish machine-learned interatomic potentials as efficient and predictive tools for investigating radiation damage in YBCO across relevant compositions and irradiation conditions.

Modeling Carbon Nanodomain Formation in Amorphous SiOC Using MACE Neural Network Potentials

Tuuli Tettenborn. Digital Library of the University of Innsbruck (University of Innsbruck) (2026). Cited by 0. Found by: openalex. openalex_2yr: 0.006 (2026). URL: https://openalex.org/W7153957297

Abstract: In this work, the temperature-dependent structural evolution of amorphous silicon oxycar bide (SiOC) was investigated using molecular dynamics simulations based on the equiv ariant machine-learned interatomic potential MACE (Multi-Atomic Cluster Expansion). Within the framework of theoretical chemistry, particular emphasis was placed on as sessing the capability of symmetry-preserving neural network potentials to describe com plex bond reorganization processes in disordered ceramic systems. Starting from melt quenched amorphous SiO2, SiOC model systems with experimentally motivated compo sitions were generated and simulated between 300 and 1000 K for up to 12 ns. Carbon segregation and nanodomain formation were quanti ed using graph-based cluster analysis with modi ed Voronoi neighbor detection. A systematic increase in carbon cluster size with temperature was observed across all model variants and initial con gurations. Radial distribution function analysis reveals a progressive transformation of carbon bonding mo tifs toward sp2-dominated con gurations, accompanied by decreasing SiC correlations, indicating temperature-driven carbon segregation from the hybrid SiOC network. Sim ulated powder X-ray di raction patterns con rm the predominantly amorphous character of the material while re ecting enhanced medium-range ordering at elevated temperatures. The results demonstrate that equivariant neural network potentials provide a physically consistent and numerically stable description of temperature-driven carbon nanodomain formation in amorphous SiOC, highlighting their suitability for investigating complex reactive processes in disordered materials.

Insights Into Radiation Damage in YBa$_2$Cu$_3$O$_{7- }$ From Machine-Learned Interatomic Potentials

Ashley Dicksona; Niccolò Di Eugenio; Federico Ledda; Daniele Torsello; F. Laviano; Flyura Djurabekova; Jesper Byggmästar; Mark R. Gilbert; D. Nguyen-Manh; Erik Gallo; Antonio Trotta; Davide Gambino; Samuel T. Murphy. arXiv (Cornell University) (2025). Cited by 0. Found by: openalex. DOI: 10.48550/arxiv.2512.24430

Abstract: Accurate prediction of radiation damage in YBa$_2$Cu3O${7- }$ (YBCO) is essential for assessing the performance of high-temperature superconducting (HTS) tapes in compact fusion reactors. Existing empirical interatomic potentials have been used to model radiation damage in stoichiometric YBCO, but fail to describe oxygen-deficient compositions, which are ubiquitous in industrial Rare-Earth Barium Copper Oxide conductors and strongly influence superconducting properties. In this work, we demonstrate that modern machine-learned interatomic potentials enable predictive modelling of radiation damage in YBCO across a wide

range of oxygen stoichiometries, with higher fidelity than previous empirical models. We employ two recently developed approaches: an Atomic Cluster Expansion (ACE) potential and a tabulated Gaussian Approximation Potential (tabGAP). Both models accurately reproduce Density Functional Theory (DFT) energies, forces, and threshold displacement energy distributions, providing a reliable description of atomic-scale collision processes. Molecular dynamics simulations of 5 keV cascades predict enhanced peak defect production and recombination relative to a widely used empirical potential, indicating different cascade evolution. By explicitly varying oxygen deficiency, we show that total defect production depends only weakly on stoichiometry, offering insight into the robustness of radiation damage processes in oxygen-deficient YBCO. Finally, fusion-relevant 300 keV cascade simulations reveal amorphous regions with dimensions comparable to the superconducting coherence length, consistent with electron microscopy observations of neutron-irradiated HTS tapes. These results establish machine-learned interatomic potentials as efficient and predictive tools for investigating radiation damage in YBCO across relevant compositions and irradiation conditions.

Ion Dynamics in Amorphous Solid Electrolytes Studied Using Neural Network Potentials

Kōji Shimizu. The Materials Research Society series (2024). Cited by 0. Found by: openalex. DOI: 10.1007/978-981-97-6039-8_35

On the Boroxol Ring Fraction in Melt-Quenched B2O3 Glass: Insights from Machine Learning Potentials

Nikhil Avula; Sundaram Balasubramanian; Debendra Meher. AIP Publishing (2026). Cited by 0. Found by: openalex. openalex_2yr: 0 (2026). DOI: 10.60893/figshare.jcp.c.8338831.v1

Abstract: An atomistic structural model for melt-quenched B2O3 glass has eluded the simulation community so far. The difficulty lies in the abundance of six-membered boroxol rings- an intermediate-range order motif suggested by Raman and NMR spectroscopy- which is challenging to capture in atomistic molecular dynamics simulations. Here, we report the development of a DFT-accurate machine-learned potential and employ quench rates as low as 109 K/s to obtain B2O3sses with more than 30% of boron atoms in boroxol rings. Also, we show that the pressure, and consequently the boroxol fraction, in the deep potential molecular dynamics (DPMD) simulations critically depends on the range of the geometry descriptor used in the embedding neural network, and it converges beyond a value of 7 Å. The boroxol ring fraction increases with decreasing quench rate. Finally, amorphous B2O3 configurations display a minimum in energy at a boroxol fraction of 75%, remarkably close to the experimental estimate in B2O3 glass.

A neural network potential for the phase change material gete: large scale molecular dynamics simulations with close to ab initio accuracy

Gabriele C. Sosso. BOA (University of Milano-Bicocca) (2013). Cited by 0. Found by: openalex. openalex_2yr: 0.014 (2026). URL: https://openalex.org/W2341698921

Abstract: Phase change materials based on chalcogenide alloys are attracting an increasing interest worldwide due to their ability to undergo reversible and fast transitions between the amorphous and crystalline phases upon heating. This property is exploited in rewritable optical media (CD, DVD, Blu-Ray Discs) and electronic nonvolatile memories of new concept, the Phase Change Memories (PCM). The strong optical and electronic contrast between the crystal and the amorphous allows discriminating between the two phases that correspond to the two bits of binary information zero and one. The material of choice for applications is the ternary compound Ge2Sb2Te5 (GST). However, the related binary alloy GeTe has also been thoroughly investigated because of its higher crystallization temperature and better data retention at high temperature with respect to GST. PCM devices, born thanks to the work of Ovshinsky in the late 1960s, offer extremely fast programming, extended cycling endurance, good reliability and inexpensive, easy integration. A PCM is essentially a resistor of a thin film of the chalcogenide alloy with a low field resistance that changes by several orders of magnitude across the phase change. In memory operations, cell read out is performed at low bias. Programming the memory requires instead a relatively large current to heat up the chalcogenide and induce the phase change, either the melting of the crystal and subsequent amorphization (RESET) or the recrystallization of the amorphous (SET). In the last few years, atomistic simulations based on density functional theory (DFT) have provided useful insights into the properties of phase change materials. However, several key issues such as the thermal conductivity at the nanoscale or the origin of the fast crystallization, just to name a few, are presently beyond the reach of ab initio simulations due to the high computational cost. In fact, first principles simulations can deal at most with 10^2 atoms on the timescale of 10^2 ps, while many properties like thermal conductivity or direct simulations of the crystallization process require at least 10^3 atoms on the timescale of 10^3 ps. The development of

reliable classical interatomic potentials is a possible route to overcome the limitations in system size and time scale of ab initio molecular dynamics. However, traditional approaches based on the fitting of simple functional forms for the interatomic potentials are very challenging due to the complexity of the chemical bonding in the crystalline and amorphous phases revealed by the ab initio simulations. A possible solution has been proposed recently by Behler and Parrinello, who developed highdimensional interatomic potentials with close to ab initio accuracy employing artificial neural networks (NN). In this thesis work, we developed a classical interatomic potential for the bulk phases of GeTe employing this NN technique. The potential was validated by comparing results on the structural and dynamical properties of liquid, amorphous, and crystalline GeTe derived from NN-based simulations with the ab initio data obtained here and in previous works [10]. The NN potential displays an accuracy close to that of the underlying DFT framework at a much reduced computational load that scales linearly with the size of the system. It allows us to simulate several thousands of atoms for tens of ns, which is well beyond present-day capabilities of DFT MD. The development of a reliable interatomic potential with close to DFT accuracy is thus a breakthrough in the modeling of phase change materials. To date, we employed our NNP in order to investigate three issues: 1. the thermal conductivity of the amorphous phase 2. the viscosity and atomic mobility in the supercooled liquid and overheated amorphous phases 3. the dynamics of homogeneous crystallization of the liquid and amorphous phases Thermal conductivity is a key property for the PCM operation, as the phase changes corresponding to the memory writing/erasing processes strongly depend upon heat dissipation and transport. Moreover, thermal cross-talks among the different bits is a crucial reliability issue in PCM. Although experimental data on thermal conductivity are available for few materials in this class it is unclear whether the value measured in the bulk phase could also describe the behavior of the material in nanoscaled devices (10-20 nm) which might be smaller than the phonon mean free path. This is particularly relevant for PCM architectures based on nanostructures employing nanowires, colloidal nanoparticles, thin bridges and nanotubes. In fact, amorphous materials can also display propagating phonons with mean free path as long as 0.5 m. This has been demonstrated for amorphous Si where propagating modes with long mean free path contribute to half of the total thermal conductivity. It is therefore of great technological relevance to assess whether similar propagating modes with long mean free path might be present in amorphous phase change materials as well. Atomistic simulations can provide crucial insights into the thermal transport properties of phase change materials suitable to aid a reliable modelling of the device operation. However, the calculation of the thermal conductivity in an amorphous system requires very long simulations (on the ns scale) of large models (thousands of atoms) that are presently beyond the reach of fully DFT simulations. The use of NN potentials allowed us to compute the mean free path of phonons in a-GeTe and to assess that ballistic transport is inhibited by disorder even at the nanoscale. The key property that makes some chalcogenide alloys suitable for applications in PCM is the high speed of the transformation which leads to full crystallization on the time scale of 10-100 ns upon Joule heating. What makes some compounds alloys so special in this respect and so different from most amorphous semiconductors is, however, still a matter of debate. The driving force for crystallization of the supercooled liquid is actually the free energy difference between the crystal and the supercooled liquid. However, the crystallization is controlled both by the driving force and by the atomic mobility. The driving force vanishes at melting and increases upon cooling. A large atomic mobility at high supercooling can thus boost the crystallization speed. These conditions can actually be met by fragile liquids. In fact, supercooled liquids are classified as fragile or strong on the basis of the temperature dependence of their viscosity [16]. An ideal strong liquid shows an Arrhenius behavior of the viscosity from the melting temperature Tm down to the glass transition temperature Tg. On the contrary, in a fragile liquid is very low down to a crossover temperature T$*$, below which a steep rise of the viscosity (and thus a steep decrease in the mobility) is observed. If T$*$ is sufficiently far from the melting temperature (Tm), high supercooling and large atomic mobility can be met at the same time. The question is thus whether phase change materials are fragile liquids or not. Due to the high crystallization speed it is unfortunately not possible to measure below Tm experimentally. We have thus addressed this problem by MD simulations and we have demonstrated that indeed GeTe is a very fragile liquid (fragility index 100). Moreover a breakdown of the Stokes-Einstein relation between the viscosity and the diffusion coefficient is found, which further increase the atomic mobility down to temperatures very close to Tg. Hysteretic effects in the heating of the amorphous phase above Tg have been addressed as well. Finally, we have performed direct simulations of the homogeneous crystallization of the supercooled liquid and amorphous phases with 4000-atom models and simulation times of several ns. Although similar simulations have been previously performed by fully DFT simulations, the limitations in size (about 200 atoms) and time scale (500 ps) prevented a reliable estimate of the size of the critical nucleus and of the crystallization speed, which are instead accessible by our large scale simulations. This thesis is organised as follows: in the introductory chapter 1 I provide essential information about phase change materials and phase change memories. Chapter 2 is dedicated to theory and methods, while chapters 3,4,5 and 6 are devoted to the results on the thermal conductivity of the amorphous phase of GeTe, on the properties of its supercooled liquid phase and on the homogeneous crystallization from the melt and the overheated amorphous phase.

On the Boroxol Ring Fraction in Melt-Quenched B2O3 Glass: Insights from Machine Learning Potentials

Nikhil Avula; Sundaram Balasubramanian; Debendra Meher. AIP Publishing (2026). Cited by 0. Found by: openalex. openalex_2yr: 0 (2026). DOI: 10.60893/figshare.jcp.c.8338831

Abstract: An atomistic structural model for melt-quenched B2O3 glass has eluded the simulation community so far. The difficulty lies in the abundance of six-membered boroxol rings- an intermediate-range order motif suggested by Raman and NMR spectroscopy- which is challenging to capture in atomistic molecular dynamics simulations. Here, we report the development of a DFT-accurate machine-learned potential and employ quench rates as low as 109 K/s to obtain B2O3sses with more than 30% of boron atoms in boroxol rings. Also, we show that the pressure, and consequently the boroxol fraction, in the deep potential molecular dynamics (DPMD) simulations critically depends on the range of the geometry descriptor used in the embedding neural network, and it converges beyond a value of 7 Å. The boroxol ring fraction increases with decreasing quench rate. Finally, amorphous B2O3 configurations display a minimum in energy at a boroxol fraction of 75%, remarkably close to the experimental estimate in B2O3 glass.

Revealing Phonon Bridge Effect for Amorphous vs Crystalline Metal–Silicide Layers at Si/Ti Interfaces by a Machine Learning Potential

Mayur Pratap Singh; Lokanath Patra; Cheng Zhang; G. J. MacDougall; Suman Datta; David G. Cahill; Sudhir Kumar. ACS Applied Electronic Materials (2026). Cited by 0. Found by: openalex. openalex_2yr: 4.884 (2026). DOI: 10.1021/acsaelm.5c02592

Abstract: Metal–semiconductor interfaces play a central role in micro- and nanoelectronic devices, as heat dissipation or temperature drop across these interfaces can significantly affect device performance. Prediction of accurate thermal boundary resistance (TBR) across these interfaces, considering realistic structures and their correlation with underlying thermal transport, remains challenging. In this work, we develop a unified Neuroevolution Potential (NEP) for the Si–Ti system that accurately reproduces energies, forces, and phonon properties of bulk Si, Ti, and TiSi2 and extends naturally to interfacial environments to analyze interfacial transport. An important development over current machine-learned interatomic potentials is the capability to model complex structures at metal–semiconductor interfaces, as the NEP enables large-scale nonequilibrium molecular dynamics simulations of epitaxial Si/Ti interfaces to elucidate the effect of amorphous or crystalline silicide interfacial layers. Simulated TBRs show excellent agreement with our time-domain thermoreflectance (TDTR) measurements, validating the robustness of our predictions. Spectral analyses reveal that the amorphous TiSi2 interfacial layer helps in efficient interfacial transport when the thickness is less than 1.5 nm compared to the crystalline TiSi2 layer, due to the high spectral conductance in the 3–6 THz frequency range and also due to the opening of channels for anharmonic transport, but this trend reverses when the interfacial layer thickness increases beyond 1.5 nm. Comparison of TBRs at the Si/TiSi2 interface for different crystalline phases of TiSi2 establishes that the C54 phase has reduced TBR compared to the C49 phase, which is correlated with the difference in their phonon density of states (PDOS) overlap with Si. These results provide atomistic insight into the role of crystalline versus amorphous silicides in interfacial heat transport and demonstrate a transferable machine-learned potential for studying heat dissipation in advanced semiconductor devices.

Investigation of medium-range order in amorphous GeTe and Ge2Sb2Te5 using density functional theory and neural network potential

. Seoul National University Open Repository (Seoul National University) (2020). Cited by 0. Found by: openalex. URL: https://openalex.org/W3138260678

Abstract: Phase change memory (PCM) is a promising non-volatile memory.Among emerging memories, PCM has been successfully commercialized and mature technology.However, there is still a lack of understanding of the phase transition process at the atomic scale.Since molecular dynamics simulation can provide insight into crystallization kinetics of phase change materials, we perform the crystallization simulations and show that the medium-range orders in amorphous phase change materials are critical in crystallization kinetics.We develop neural network potentials (NNP) for GeTe as a representative phase change material and investigate the crystallization process of amorphous GeTe.With the accuracy of density functional theory (DFT) level and much cheaper computational cost, we achieve the realistic simulations using the NNP.In developing the NNP, we find that overly flattened fourfold rings in the amorphous structure exaggerate the crystallization process, especially for nucleation.By explicitly including relaxation paths from flat to puckered fourfold rings, we obtain a modified NNP, which produces medium-range orders that are more consistent with DFT.This structural change increases interfacial energy between crystalline and amorphous phases and suppresses the nucleation.Using

the modified NNP, we perform crystallization simulations at two densities (equilibrium density and crystalline density) and temperatures ranging from 500 to 650 K.We observe finite incubation times at both densities.In particular, the incubation time at the equilibrium density is found to be 7 or 17 ns, which is consistent with experiments.In practice, properties of the phase change materials are tuned by doping and Ge-Sb-Te alloys are mainly used.However, developing NNP for the multi-component systems is challenging at the present, we study the effects of Al and Ga dopants using ab initio calculations.We find that the two dopants behave similarly in amorphous i Ge 2 Sb 2 Te 5 (GST), and they are mostly coordinated by Te atoms in a tetrahedral geometry, which is similar to those in crystalline M x Te y (M=Al or Ga).The number of wrong bonds increases as dopant atoms predominantly bond with Te atoms, which affects the medium-range order structures.The number of fourfold ring structures, especially ABAB-type, decreases significantly and the number of odd-numbered rings is increased, explaining the enhanced thermal stability and slow crystallization speed of doped amorphous GST in the experiment.

Charge integrated graph neural network-based machine learning potential for amorphous and non-stoichiometric hafnium oxide

Hyo Gyeong Shin; Seong Hun Kim; Eun Ho Kim; Jun Hyeong Gu; Jaeseon Kim; Seon-Gyu Kim; Shin Hyun Kim; Hyo Kim; Sunghyun Kim; Duk-Hyun Choe; Donghwa Lee. npj Computational Materials (2025). Cited by 0. Found by: openalex. DOI: 10.1038/s41524-025-01864-3

Abstract: Amorphous and non-stoichiometric hafnium oxide (a-HfO x) systems are essential for advanced electronic applications due to their superior electrical properties. Simulating their atomic behaviors under electric fields (\({E}_{{field}}\)) is critical but challenging. Ab-initio molecular dynamics (AIMD) offer high accuracy but is computationally expensive, while classical MD lacks precision. To address this, we develop a charge equilibration integrated graph neural network (CIGNN) model that predicts atomic charge, energy, and force under \({E}_{{field}}\) conditions. Using the CIGNN model and AIMD datasets, we develop a CIGNN-based machine learning potential (CNMP) optimized for a-HfO x systems. The CNMP achieves quantum mechanical accuracy and effectively captures the atomic behaviors and dynamic properties of these systems across varying temperatures, densities, and \({E}_{{field}}\) conditions. We expect the CNMP to serve as a valuable tool for studying field-induced phenomena in complex systems and to provide a foundation for advancing innovations in electronic applications.

Application of Neural Network Potentials to Materials Science Analysis of Partial Crystallization in Glass Structures and Ion Conduction Mechanisms

Kōji Shimizu. Vacuum and Surface Science (2025). Cited by 0. Found by: openalex. DOI: 10.1380/vss.68.350

Abstract: This study introduces the application of neural network potentials (NNPs) in materials science, with a focus on their use in analyzing partial crystallization and ion conduction mechanisms in glass structures, while also highlighting recent advancements in NNPs. Specifically, molecular dynamics (MD) simulations employing NNPs were performed to investigate the crystallization process of Li3PS4 glass under heat treatment. The simulations revealed the nucleation and growth of crystalline phases within the glass matrix, providing atomic-level insights into the crystallization mechanism. Furthermore, an assessment of the impact of crystallization on lithium transport properties demonstrated that the precipitated crystalline phase corresponded to the high-temperature -phase. Additionally, the formation of new ion conduction pathways through the interconnection of crystal nuclei was identified as a key factor in enhancing ionic conductivity in glass-ceramic materials.

Desorption dynamics of interstellar molecule on amorphous solid water investigated by machine learning potential-based PaCS-MD simulation

Watanabe Natsuki; Kästner Johannes; Yuta Hori; Yasuteru Shigeta; Mitsuo Shoji; Harada Ryuhei. Institutional Repositories DataBase (IRDB) (2026). Cited by 0. Found by: openalex. URL: https://openalex.org/W7169694995

Abstract: Molecular adsorption, diffusion, reaction, and desorption on interstellar amorphous solid water (ASW) dominate the chemical processes in interstellar molecular clouds. Elucidating the chemical evolution in molecular clouds requires understanding the interactions between molecules and the ASW surface. Investigating the dynamics on ASW surfaces requires highly accurate and efficient molecular dynamics (MD) simulations. This study proposes a computational scheme that combines a rare-event sampling method, parallel cascade selection MD (PaCS-MD), with machine learning potential-based MD (MLP-MD), referred to as PaCS-MLP-MD. Because PaCS-MLP-MD explores rare-event dynamics without applying external forces or increasing the temperature, it provides a suitable framework for investigating astrochemically relevant molecular processes. Using PaCS-MLP-MD simulations, we simulated desorption of aminoacetonitrile (AAN) from an ASW surface

model. Two potential desorption processes were identified: one in which the amino group desorbs first, followed by the nitrile group, and another in which the nitrile group desorbs first, followed by the amino group. Additional unbiased MLP-MD simulations, initiated from representative PaCS-MLP-MD snapshots, were performed to reveal desorption and re-adsorption behaviors of AAN on the ASW surface. These new findings from the PaCS-MLP-MD simulations may contribute to a more detailed understanding of processes occurring on ASW surfaces in molecular cloud environments.

Desorption Dynamics of Interstellar Molecule on Amorphous Solid Water Investigated by Machine Learning Potential-based PaCS-MD Simulation

Yuta Hori; Natsuki Watanabe; Johannes Kästner; Ryuhei Harada; Yasuteru Shigeta; Mitsuo Shoji. AIP Publishing (2026). Cited by 0. Found by: openalex. openalex_2yr: 0 (2026). DOI: 10.60893/figshare.jcp.c.8384920

Abstract: Molecular adsorption, diffusion, reaction, and desorption on interstellar amorphous solid water (ASW) dominate the chemical processes in interstellar molecular clouds. Elucidating the chemical evolution in molecular clouds requires understanding the interactions between molecules and the ASW surface. Investigating the dynamics on ASW surfaces requires highly accurate and efficient molecular dynamics (MD) simulations. This study proposes a computational scheme that combines a rare-event sampling method, Parallel Cascade Selection MD (PaCS-MD), with machine learning potential-based MD (MLP-MD), referred to as PaCS-MLP-MD. Because PaCS-MLP-MD explores rare-event dynamics without applying external forces or increasing the temperature, it provides a suitable framework for investigating astrochemically relevant molecular processes. Using PaCS-MLP-MD simulations, we simulated the desorption of aminoacetonitrile (AAN) from an ASW surface model. Two potential desorption processes were identified: one in which the amino group desorbs first, followed by the nitrile group, and another in which the nitrile group desorbs first, followed by the amino group. Additional unbiased MLP-MD simulations, initiated from representative PaCS-MLP-MD snapshots, were performed to reveal the desorption and re-adsorption behaviors of AAN on the ASW surface. These new findings from the PaCS-MLP-MD simulations may contribute to a more detailed understanding of processes occurring on ASW surfaces in molecular cloud environments.

Desorption dynamics of interstellar molecule on amorphous solid water investigated by machine learning potential-based PaCS-MD simulation

Natsuki Watanabe; Johannes Kästner; Yuta Hori; Yasuteru Shigeta; Mitsuo Shoji; Ryuhei HARADA. The Journal of Chemical Physics (2026). Cited by 0. Found by: openalex. DOI: 10.1063/5.0325531

Abstract: Molecular adsorption, diffusion, reaction, and desorption on interstellar amorphous solid water (ASW) dominate the chemical processes in interstellar molecular clouds. Elucidating the chemical evolution in molecular clouds requires understanding the interactions between molecules and the ASW surface. Investigating the dynamics on ASW surfaces requires highly accurate and efficient molecular dynamics (MD) simulations. This study proposes a computational scheme that combines a rare-event sampling method, parallel cascade selection MD (PaCS-MD), with machine learning potential-based MD (MLP-MD), referred to as PaCS-MLP-MD. Because PaCS-MLP-MD explores rare-event dynamics without applying external forces or increasing the temperature, it provides a suitable framework for investigating astrochemically relevant molecular processes. Using PaCS-MLP-MD simulations, we simulated desorption of aminoacetonitrile (AAN) from an ASW surface model. Two potential desorption processes were identified: one in which the amino group desorbs first, followed by the nitrile group, and another in which the nitrile group desorbs first, followed by the amino group. Additional unbiased MLP-MD simulations, initiated from representative PaCS-MLP-MD snapshots, were performed to reveal desorption and re-adsorption behaviors of AAN on the ASW surface. These new findings from the PaCS-MLP-MD simulations may contribute to a more detailed understanding of processes occurring on ASW surfaces in molecular cloud environments.

Development and Validation of Neural Network Potentials for Multicomponent Oxide Glasses

Ryuki Kayano; Yaohiro Inagaki; Ryuta Matsubara; Keisuke Ishida; Takahiro Ohkubo. ChemRxiv (2024). Cited by 0. Found by: openalex. openalex_2yr: 0.467 (2026). DOI: 10.26434/chemrxiv-2024-hb50k-v2

Abstract: A neural network potential (MLP) for molecular dynamics simulation (MD) of multicomponent oxide glasses was developed with special consideration of structural reproducibility. The MLP was constructed through pre-training using the dataset of density functional theory (DFT) data provided by the Open Catalysis project and fine-tuning for the dataset of nine-component glasses. A thorough validation based on previous experimental and DFT-MD data was performed for the glass structure derived from neural network molecular dynamics simulation ({\MLMD}) with the developed {\MLP}. The accuracy of {\MLMD} was investigated in terms of the local structure of the glass, comparing it to the glass derived from MD with conventional potentials. The composition dependence of the local structure in \ce{Na2O}–\ce{SiO2} and \ce{Na2O}–\ce{B2O3} glass systems was well reproduced for the {\MLMD}-derived glass. The ability to reproduce the glass structure was demonstrated in the population of four-coordinated boron population and formation of superstructures in alkali borate glasses, and the Al local structure in the novel Al-rich binary aluminoborosilicate glass. The importance of pre-training was revealed by comparing {\MLMD} results using {\MLP} developed with and without pre-training. Although better metric scores for {\MLP} without pre-training can be achieved, the resultant structure was not realistic. This is an important lesson that the metric score alone is inadequate to construct accurate {\MLP} for glasses. Finally, the developed {\MLMD} was applied to the modeling of the reference nuclear waste glass (60.1{\Si}–3.84{\Al}–15.97{\B}–12.65{\Na}–2.87{\Ca}–2.86{\Mg}–1.72{\Zr}), and the charge compensation mechanism of the cations was discussed.

Development and Validation of Neural Network Potentials for Multicomponent Oxide Glasses

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Neutron and X-ray Diffraction Reveal the Limits of Long-Range Machine Learning Potentials for Medium-Range Order in Silica Glass

Sai Harshit Balantrapu; Atul C. Thakur; Chris J. Benmore; Ganesh Sivaraman. arXiv (Cornell University) (2026). Cited by 0. Found by: openalex. URL: https://openalex.org/W7155654491

Abstract: Glassy silica is a foundational material in optics and electronics, yet accurately predicting its medium-range order (MRO) remains a major challenge for machine-learning interatomic potentials (MLIPs). While local MLIPs reproduce the short-range SiO4 tetrahedral network well, it remains unclear whether locality alone is sufficient to recover the first sharp diffraction peak (FSDP), the principal experimental signature of MRO. Here, we combine neutron and X-ray diffraction measurements with large-scale molecular dynamics driven by two MACE-based models: a short-range (SR) potential and a long-range (LR) extension incorporating reciprocal-space gated attention. The SR model systematically over-structures the network, producing an overly intense FSDP in both the liquid and glassy states. Incorporating long-range interactions improves agreement with experiment for the liquid structure by reducing this excess ordering, but the LR model still fails to recover the experimental amorphous MRO after quenching. Ring-statistics and bond-angle analyses reveal that SR model exhibits an artificially narrow distribution dominated by six-membered rings, while the LR model produces a broader but still biased ring population. Despite preserving the correct tetrahedral geometry, both models show limited variability in Si-O-Si angles, indicating constrained network flexibility. These structural signatures demonstrate that both models retain excessive memory of the parent liquid network, leading to kinetically trapped and nonphysical medium-range configurations during vitrification. These results show that explicit long-range interactions are necessary but not sufficient for predictive modelling of disordered silica and suggest that accurate MRO further requires training data and sampling strategies that adequately represent the liquid-to-glass transition.

Neutron and X-ray Diffraction Reveal the Limits of Long-Range Machine Learning Potentials for Medium-Range Order in Silica Glass

Sai Harshit Balantrapu; Atul C. Thakur; Chris J. Benmore; Ganesh Sivaraman. arXiv (Cornell University) (2026). Cited by 0. Found by: openalex. DOI: 10.48550/arxiv.2604.21222

Abstract: Glassy silica is a foundational material in optics and electronics, yet accurately predicting its medium-range order (MRO) remains a major challenge for machine-learning interatomic potentials (MLIPs). While local MLIPs reproduce the short-range SiO4 tetrahedral network well, it remains unclear whether locality alone is sufficient to recover the first sharp diffraction peak (FSDP), the principal experimental signature of MRO. Here, we combine neutron and X-ray diffraction measurements with large-scale molecular dynamics driven by two MACE-based models: a short-range (SR) potential and a long-range (LR) extension incorporating reciprocal-space gated attention. The SR model systematically over-structures the network, producing an overly intense FSDP in both the liquid and glassy states. Incorporating long-range interactions improves agreement with experiment for the liquid structure by reducing this excess ordering, but the LR model still fails to recover the experimental amorphous MRO after quenching. Ring-statistics and bond-angle analyses reveal that SR model exhibits an artificially narrow distribution dominated by six-membered rings, while the LR model produces a broader but still biased ring population. Despite preserving the correct tetrahedral geometry, both models show limited variability in Si-O-Si angles, indicating constrained network flexibility. These structural signatures demonstrate that both models retain excessive memory of the parent liquid network, leading to kinetically trapped and nonphysical medium-range configurations during vitrification. These results show that explicit long-range interactions are necessary but not sufficient for predictive modelling of disordered silica and suggest that accurate MRO further requires training data and sampling strategies that adequately represent the liquid-to-glass transition.

Neutron and x-ray diffraction reveal the limits of long-range machine learning potentials for medium-range order in silica glass

Sai Harshit; Atul C. Thakur; Chris J. Benmore; Ganesh Sivaraman. Journal of Physics Materials (2026). Cited by 0. Found by: openalex. DOI: 10.1088/2515-7639/ae8643

Abstract: Abstract Glassy silica is a foundational material in optics and electronics, yet accurately predicting its medium-range order (MRO) remains a major challenge for machine-learning interatomic potentials (MLIPs). While local MLIPs reproduce the short-range (SR) SiO 4 tetrahedral network well, it remains unclear whether locality alone is sufficient to recover the first sharp diffraction peak (FSDP), the principal experimental signature of MRO. Here, we combine neutron and x-ray diffraction measurements with large-scale molecular dynamics driven by two MACE-based models: a SR potential and a long-range (LR) extension incorporating reciprocal-space gated attention. The SR model systematically over-structures the network, producing an overly intense FSDP in both the liquid and glassy states. Incorporating LR interactions improves agreement with experiment for the liquid structure by reducing this excess ordering, but the LR model still fails to recover the experimental amorphous MRO after quenching. Ring-statistics and bond-angle analyses reveal that SR model exhibits an artificially narrow distribution dominated by six-membered rings, while the LR model produces a broader but still biased ring population. Despite preserving the correct tetrahedral geometry, both models show limited variability in Si–O–Si angles, indicating constrained network flexibility. These structural signatures demonstrate that both models retain excessive memory of the parent liquid network, leading to kinetically trapped and nonphysical medium-range configurations during vitrification. These results show that explicit LR interactions are necessary but not sufficient for predictive modeling of disordered silica and suggest that accurate MRO further requires training data and sampling strategies that adequately represent the liquid-to-glass transition.

A systematic approach to generating accurate neural network potentials: the case of carbon

Yusuf Shaidu; Emine Küçükbenli; Ruggero Lot; Franco Pellegrini; Efthimios Kaxiras; Stefano de Gironcoli. npj Computational Materials (2021). Cited by 0. Found by: openalex. DOI: 10.1038/s41524-021-00508-6

Abstract: Abstract Availability of affordable and widely applicable interatomic potentials is the key needed to unlock the riches of modern materials modeling. Artificial neural network-based approaches for generating potentials are promising; however, neural network training requires large amounts of data, sampled adequately from an often unknown potential energy surface. Here we propose a self-consistent approach that is based on crystal structure prediction formalism and is guided by unsupervised data analysis, to construct an accurate, inexpensive, and transferable artificial neural network potential. Using this approach, we construct an interatomic potential for carbon and demonstrate its ability to reproduce first principles results on elastic and vibrational properties for diamond, graphite, and graphene, as well as energy ordering and structural properties

of a wide range of crystalline and amorphous phases.

High-dimensional neural-network potentials for phase change materials for data storage

Marco Bernasconi. BOA (University of Milano-Bicocca) (2012). Cited by 0. Found by: openalex. openalex_2yr: 0.014 (2026). URL: https://openalex.org/W2325852788

Abstract: In this seminar I will review our recent study of the properties of phase change materials by means of molecular dynamics simulations based on Neural Network potentials [1]. The materials we are dealing with are chalcogenide alloys (typically Ge2Sb2Te5 or GeTe) which are the subject of extensive experimental and theoretical research because of their use in optical (digital versatile disc, DVD) and electronic (phase change memories, PCM) storage devices [2]. Both applications rely on the fast and reversible transformation between the crystalline and amorphous phases induced by heating either via laser irradiation (DVD) or Joule effect (PCM). The two states of the memory can be discriminated because of the large difference in optical reflectivity and electronic conductivity of the two phases. In the last few years atomistic simulations based on density functional theory have provided useful insights on the properties of materials in this class [3]. However, several key issues such as the crystallization dynamics occurring at the ns time scale, the properties of the crystalline/amorphous interface and the thermal conductivity at the nanoscale, just to name a few, are presently beyond the reach of fully ab-initio simulations. A route to overcome the limitations in system size and time scale of ab-initio molecular dynamics is the development of empirical interatomic potentials. However, traditional approaches based on the fitting of simple functional forms are very challenges due to the complexity and variability of the chemical bonding in the crystal and amorphous phases of these materials as revealed by ab-initio simulations. A possible solution was demonstrated few years ago by Behler and Parrinello [2] who developed classical interatomic potentials with close to ab-initio accuracy by fitting large ab-initio databases by means of a NN scheme. By means of this technique, we have developed a interatomic potential for GeTe [4] which is one of the compounds under scrutiny for applications in phase change memories. The NN potential with a close to ab-initio accuracy has allowed us to address several key issues for applications in data storage such as the homogeneous and heterogeneous crystallization of the amorphous, the glass transition of the supercooled liquid and the thermal conductivity of the amorphous phase. [1] J. Behler and M. Parrinello, Phys. Rev. Lett. 14, 146401 (2007); J. Behler. Chemical Modelling, 7, 1 (2010). [2] M. Wuttig and N. Yamada, Nature Mater. 6, 824 (2007) [3] S. Caravati, M. Bernasconi, T.D. Kuehne, M. Krack, and M. Parrinello, Appl. Phys. Lett. 91, 171906 (2007); ibidem, Phys. Rev. Lett. 102, 205502 (2009); J. Phys. Cond. Matt. 23, 265801 (2011); J. Phys. Cond. Matt. 21, 255501 (2009); D. Lencer, M. Salinga and M. Wuttig, Adv. Mater. 23, 2030 (2011). [4] G. C. Sosso, G. Miceli, S. Caravati, J. Behler, M. Bernasconi, arXiv:1201.2026v1.

ALDo 2.0: Scalable thermal transport from first principles and machine learning potentials

Giuseppe Barbalinardo; Zekun Chen; Dylan A. Folkner; Bohan Li; Nicholas W. Lundgren; Nathaniel Troup; Alfredo Fiorentino; Davide Donadio. Mendeley Data (2026). Cited by 0. Found by: openalex. DOI: 10.17632/t3f42nv4fw.1

Abstract: We introduce ALDo 2.0, an open-source Python package for computing vibrational, elastic, and thermal transport properties of crystalline and disordered solids from first principles and machine-learned interatomic potentials. Building on the anharmonic lattice dynamics (ALD) framework, ALDo 2.0 provides efficient CPU and GPU-accelerated implementations of the Boltzmann transport equation (BTE) for crystals and the quasi-harmonic Green-Kubo (QHGK) method. The QHGK formalism extends thermal transport predictions beyond translationally-invariant crystals to materials lacking long-range order, including glasses, alloys, and complex nanostructures. ALDo 2.0 introduces native integration with modern machine-learned potentials (MLPs), enabling thermal transport workflows that combine the accuracy of first-principles methods with the scalability of classical force fields. It also features comprehensive support for temperature-dependent effective potentials (TDEP) workflows, flexible storage backends for large-scale calculations, and advanced quantification of anharmonicity. The software seamlessly interfaces with electronic structure codes (Quantum ESPRESSO, VASP), molecular dynamics packages (LAMMPS), and state-of-the-art MLPs (ACE, NEP, MACE, MatterSim, Orb), enabling thermal transport studies from 0 K to finite temperatures. ALDo 2.0 implements multiple BTE solution strategies (relaxation time approximation, self-consistent iteration, full matrix inversion, and eigendecomposition) and supports essential physical corrections, including isotopic scattering and non-analytical terms for polar materials. A modular Python architecture with lazy evaluation and multiple storage formats (formatted text, NumPy, HDF5) enables simulations of systems containing more than 10,000 atoms. This paper describes the theoretical framework, implementation details, software architecture, and validation examples demonstrating ALDo 2.0's capabilities for studying complex materials, including halide perovskites with strong anharmonicity and polar oxides requiring long-range electrostatic corrections.

ALDo 2.0: Scalable thermal transport from first principles and machine learning potentials

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Dynamic heterogeneity in sodium silicate melts via machine-learning potential

Kumpei Shiraishi; Rikuta Nozawa; Emi Minamitani. arXiv (Cornell University) (2026). Cited by 0. Found by: openalex. URL: https://openalex.org/W7166400150

Abstract: We present a comprehensive characterisation of dynamic heterogeneity in sodium silicate melts using molecular dynamics simulation with machine-learning potentials. By studying sodium disilicate, tetrasilicate, and hexasilicate melts across a range of temperatures, mean squared displacement and a time-correlation function computed up to the nanosecond timescale provide a detailed account of how spatial mobility disparities emerge in a realistic multicomponent oxide glass. Within these timescales, the self-part of the van Hove function for sodium displays a bimodality, demonstrating that alkali transport is mediated by discrete displacement events consistent with a hopping mechanism. This distinct hopping allows sodium ions to decouple from the sluggish relaxation of the silicate matrix. Furthermore, evaluation of the non-Gaussian parameter reveals that, although all constituent species exhibit dynamic heterogeneity, the non-Gaussian behaviour is most pronounced for oxygen atoms. This trend reflects the intermittency of structural rearrangements, where framework atoms undergo rare and stochastic events compared to the frequent displacements of mobile ions. Our findings elucidate the microscopic mechanism of ion transport and its connection to dynamic heterogeneity in silicate melts, offering a new avenue to study fundamental glassy physics in realistic vitreous materials.

Dynamic heterogeneity in sodium silicate melts via machine-learning potential

Kumpei Shiraishi; Rikuta Nozawa; Emi Minamitani. arXiv (Cornell University) (2026). Cited by 0. Found by: openalex. DOI: 10.48550/arxiv.2606.26646

Abstract: We present a comprehensive characterisation of dynamic heterogeneity in sodium silicate melts using molecular dynamics simulation with machine-learning potentials. By studying sodium disilicate, tetrasilicate, and hexasilicate melts across a range of temperatures, mean squared displacement and a time-correlation function computed up to the nanosecond timescale provide a detailed account of how spatial mobility disparities emerge in a realistic multicomponent oxide glass. Within these timescales, the self-part of the van Hove function for sodium displays a bimodality, demonstrating that alkali transport is mediated by discrete displacement events consistent with a hopping mechanism. This distinct hopping allows sodium ions to decouple from the sluggish relaxation of the silicate matrix. Furthermore, evaluation of the non-Gaussian parameter reveals that, although all constituent species exhibit dynamic heterogeneity, the non-Gaussian behaviour is most pronounced for oxygen atoms. This trend reflects the intermittency of structural rearrangements, where framework atoms undergo rare and stochastic events compared to the frequent displacements of mobile ions. Our findings elucidate the microscopic mechanism of ion transport and its connection to dynamic heterogeneity in silicate melts, offering a new avenue to study fundamental glassy physics in realistic vitreous materials.

A Systematic Approach to Generating Accurate Neural Network Potentials:\n the Case of Carbon

Yusuf Shaidu; Emine Küçükbenli; Ruggero Lot; Franco Pellegrini; Efthimios Kaxiras; Stefano de Gironcoli. arXiv (Cornell University) (2020). Cited by 0. Found by: openalex. DOI: 10.48550/arxiv.2011.04604

Abstract: Availability of affordable and widely applicable interatomic potentials is\nthe key needed to unlock the riches of modern materials modelling. Artificial\nneural network based approaches for generating potentials are promising;\nhowever neural network training requires large amounts of data, sampled\nadequately from an often unknown potential energy surface. Here we propose a\nself-consistent approach that is based on crystal structure prediction\nformalism and is guided by unsupervised data analysis, to construct an\naccurate, inexpensive and transferable artificial neural network potential.\nUsing this approach, we construct an interatomic potential for Carbon and\ndemonstrate its ability to reproduce first principles results on elastic and\nvibrational properties for diamond, graphite and graphene, as well as energy\nordering and structural properties of a wide range of crystalline and amorphous\nphases.\n

ALDo 2.0: Scalable thermal transport from first principles and machine learning potentials

Giuseppe Barbalinardo; Zekun Chen; Dylan A. Folkner; Bohan Li; Nicholas W. Lundgren; Nathaniel Troup; Alfredo Fiorentino; Davide Donadio. Computer Physics Communications (2026). Cited by 0. Found by: openalex. openalex_2yr: 3.452 (2026). DOI: 10.1016/j.cpc.2026.110282

Deformation mechanisms and compressive response of NbTaTiZr alloy via machine learning potentials

LIU Hongyang; Bo Chen; Rong Chen; Kang Dongdong; Dai Jiayu. Acta Physica Sinica (2025). Cited by 0. Found by: openalex. openalex_2yr: 0 (2026). DOI: 10.7498/aps.74.20250738

Abstract: Refractory multi-principal element alloys (RMPEAs)have become a hotspot in materials science research in recent years due to their excellent high-temperature mechanical properties and broad application prospects. However, the unique deformation mechanisms and mechanical behaviors of the NbTaTiZr quaternary RMPEA under extreme conditions such as high temperature and high strain rate are still unclear, limiting its further design and engineering applications. In order to reveal in depth the dynamic response of this alloy on an atomic scale, this study develops a high-accuracy machine learning potential (MLP) for the NbTaTiZr quaternary alloy and combines it with large-scale molecular dynamics (MD) simulations to systematically investigate the effects of crystallographic orientation, strain rate, temperature, and chemical composition on the mechanical properties and microstructural evolution mechanisms of the alloy under compressive loading. The results show that the NbTaTiZr alloy exhibits significant mechanical and structural anisotropy during uniaxial compression. The alloy exhibits the highest yield strength when loaded along the [111] crystallographic direction, while it shows the lowest yield strength when compressed along the [110] direction, where twinning is more likely to occur. Under compression along the [100] direction, the primary deformation mechanisms include local disordering transitions and dislocation slip, with 1/2 \begin{document}$ \left\langle{111}\right\rangle $\end{document} dislocations being the dominant type. When the strain rate increases to 1010 s–1, the yield strength of the alloy is significantly enhanced, accompanied by a notable increase in the proportion of amorphous or disordered structures, indicating that high strain rate loading suppresses dislocation nucleation and motion while promoting disordering transitions. Simulations at varying temperatures indicate that the alloy maintains a high strength level even at temperatures as high as 2100 K. Compositional analysis further indicates that increasing the atomic percentage of Nb or Ta effectively enhances the yield strength of the alloy, whereas an increase in Ti or Zr content adversely affects the strength. By combining MLP with MD methods, this study elucidates the anisotropic characteristics of the mechanical behavior and the strain rate dependence of disordering transitions in the NbTaTiZr RMPEA under combination of high strain rate and high temperature, providing an important theoretical basis and simulation foundation for optimizing and designing novel material under extreme environments.

Density functional and neural-network potential simulations of Ag migration in disordered GeS 2 electrolytes

Jaakko Akola; Sondre Dahl; A Götz; R. Jones. Physical Review Materials (2026). Cited by 0. Found by: openalex. DOI: 10.1103/75rr-pqqp

Abstract: Density functional/molecular dynamics (DF/MD) simulations on ${\mathrm{GeS}}_{2}$ with four compositions of silver $(2.5%\text{–}20%)$ provide valuable information on the structures and energetics of the amorphous material. Ag fills the available empty volume (cavities, voids) and interacts mainly with sulfur. The number of Ag-Ag contacts increases with increasing Ag composition, leading mainly to branched chains that partially disrupt the covalent Ge-S network. We use DF/MD simulations to study the energetics of Ag dissolution, the structures of $\mathrm{Ag}\text{\ensuremath{-}}{\mathrm{GeS}}_{2}$ alloys for differing Ag content, migration paths in the energy surfaces at 0 K, and liquid dynamics at 1100 K, close to the experimental melting point. We discuss the electronic structures of the optimized glass structures. DF/MD simulations can describe the dynamics of these systems at liquid temperatures, but are too short to allow the study of Ag migration at lower temperatures. We have used DF/MD trajectories and generated snapshots of high-T liquids to develop a neural network potential (NNP) that allows simulations of these systems for tens of nanoseconds. These simulations are used for migration dynamics at $500–800\phantom{\rule{0.28em}{0ex}}\mathrm{K}$, leading to a linear behavior of the mean-square displacement on a logarithmic scale; the effective migration energy of Ag ($0.33\ensuremath{-}0.43\phantom{\rule{0.28em}{0ex}}\mathrm{eV}$, with little composition dependence) agrees well with the static DF migration barriers.

Development of a TaN-Ce Machine Learning Potential and Its Application to Solid–Liquid Interface Simulations

Yunhan Zhang; Jianfeng Cai; Hongjian Chen; Xuming Lv; Bowen Huang. Metals (2025). Cited by 0. Found by: openalex. DOI: 10.3390/met15090972

Abstract: This study develops a machine learning potential (MLP) based on the Moment Tensor Potential (MTP) method for the TaN-Ce system. This potential is employed to investigate the interfacial structure and wetting behavior between liquid Ce and solid TaN. Molecular dynamics (MDs) simulations reveal that liquid Ce exhibits significant wetting on the TaN surface at high temperatures. The interfacial region undergoes pre-melting and component interdiffusion, forming an amorphous transition layer. Nitrogen atoms display high diffusivity, leading to surface mass loss, while tantalum atoms demonstrate excellent thermal stability and penetration resistance. These findings provide theoretical support for the design of interfacial materials and corrosion control in high-temperature metallurgy.

Modeling Short-Range and Three-Membered Ring Structures in Lithium Borosilicate Glasses using Machine Learning Potential

Shingo Urata. arXiv (Cornell University) (2022). Cited by 0. Found by: openalex. DOI: 10.48550/arxiv.2209.00316

Abstract: Lithium borosilicate (LBS) glass is a prototypical lithium-ion conducting oxide glasses available for an all-solid state buttery. Nevertheless, the atomistic modeling of LBS glass using ab $initio$ (AIMD) and classical molecular dynamics (CMD) simulations have critical limitations due to computational cost and inaccuracy in reproducing the glass microstructures, respectively. To overcome these difficulties, a machine-learning potential (MLP) was examined in this work for modeling LBS glasses using DeepMD. The glass structures obtained by this MLP possessed fourhold-coordinated boron (^{4}B) confirmed well with the experimental data and abundance of three-membered rings. The models were energetically more stable compared with those constructed with a functional force-field even though both the models included reasonable ^{4}B. The results confirmed MLP to be superior to model the boron-containing glasses and address the inherent shortcomings of the AIMD and CMD. This study also discusses some limitations of MLP for modeling glasses.

Neural Network Potentials with Effective Charge Separation for Non-Equilibrium Dynamics of Ionic Solids: A ZnO Case Study

Gang Seob Jung; Lei Cheng. ChemRxiv (2025). Cited by 0. Found by: openalex. openalex_2yr: 0.467 (2026). DOI: 10.26434/chemrxiv-2025-fm5t6

Abstract: Developing neural network potentials (NNPs) that remain accurate under non-equilibrium dynamics is challenging, as such systems require extensive sampling of diverse atomic environments beyond equilibrium phases. Here we construct high-fidelity NNPs for zinc oxide (ZnO), a polymorphic ionic solid with wurtzite (W), rock-salt (RS), zinc-blende (ZB), and amorphous states, using density functional theory (DFT) reference data. To efficiently capture transitional configurations, we combine enhanced-sampling molecular dynamics with empirical potentials, data distillation, and pretraining on short-range atomic energies (A-Train), followed by transfer learning with DFT-relabeled datasets. Compared with conventional total-energy training (T-Train) directly on DFT data, this hierarchical approach improves transferability across polymorphs and stress states.

We further introduce effective charge separation, treating long-range Coulombic terms analytically while short-range residual interactions are learned by the NNP. The optimal effective charges fall in the range 0.5–1.0 qe, consistent with dielectric-screened values derived from formal charges but distinct from Bader estimates. This strategy can significantly improve accuracy and transferability in DFT-level predictions of energies, forces, and stress. Together, these results establish a broadly applicable framework for robust NNP development in ionic solids, enabling reliable simulation of polymorphic phase transformations and non-equilibrium dynamics.

Study of vacancy ordering and the boson peak in metastable cubic Ge-Sb-Te using machine learning potentials

Young-Jae Choi; Minjae Ghim; Seung-Hoon Jhi. arXiv (Cornell University) (2023). Cited by 0. Found by: openalex. DOI: 10.48550/arxiv.2309.01089

Abstract: The mechanism of the vacancy ordering in metastable cubic Ge-Sb-Te (c-GST) that underlies the ultrafast phase-change dynamics and prominent thermoelectric properties remains elusive. Achieving a comprehensive understanding of the vacancy-ordering process at an atomic level is challenging because of enormous computational demands required to simulate disordered structures on large temporal and spatial scales. In this study, we investigate the vacancy ordering in c-GST by performing large-scale molecular dynamics simulations using machine learning potentials. The initial c-GST structure with randomly distributed vacancies rearranges to develop a semi-ordered cubic structure with layer-like ordered vacancies after annealing at 700~K for 100~ns. The vacancy ordering significantly affects the lattice dynamical properties of c-GST. In the initial structure with fully disordered vacancies, we observe a boson peak, usually associated with amorphous solids, that consists of localized modes at \$\sim\$0.575~THz. The boson peak modes are highly localized around specific atomic arrangements of straight vacancy-Te-vacancy trios. As vacancies become ordered, the boson peak disappears and the Debye-Waller thermal \textit{B} factor of Te decreases substantially. This finding indicates that the c-GST undergoes a transition from amorphous-like to crystalline-like solid state by thermal annealing in low-frequency dynamics.

Dynamic mesophase transition induces anomalous suppressed and anisotropic phonon transport revealed by unified machine learning potential

; Linfeng Yu; ; . Research Square (2024). Cited by 0. Found by: openalex. DOI: 10.21203/rs.3.rs-4082274/v1

Abstract: Abstract The physical/chemical properties undergo significant transformation in the different states arising from phase transition. However, owing to the lack of a dynamic perspective, transitional mesophases are largely underexamined, which is limited by the high resources burden of first-principles. Here, using molecular dynamics (MD) simulations empowered by advanced unified machine learning (ML) potential, we proffer an innovative paradigm for phase transition: regulating the thermal transport properties via the transitional mesophase triggered by a uniaxial force field. We investigate the mechanical, electrical, and thermal transport properties of the novel two-dimensional carbon allotrope of Janus-graphene with strain engineered phase transition. Notably, we found that the transitional mesophase significantly suppresses the thermal conductivity and induces strong anisotropy near the phase transition point. ML-driven MD simulations meticulously recapitulate the atomic-scale dynamic metamorphosis exhibited in Janus-graphene, where thermal vibration-induced intermediate amorphous or interfacial phases induce strong and anisotropic interfacial thermal resistance, which eludes capture from traditional first-principles methods. The investigation not only endows us with a novel perspective on mesophases during phase transitions but also augment our holistic comprehension of the evolution of material properties.

Viscosity, breakdown of Stokes-Einstein relation and dynamical heterogeneity in supercooled liquid Ge\$_2\$Sb\$_2\$Te\$_5\$ from simulations with a neural network potential

Simone Marcorini; Rocco Pomodoro; Omar Abou El Kheir; Marco Bernasconi. arXiv (Cornell University) (2025). Cited by 0. Found by: openalex. DOI: 10.48550/arxiv.2506.13668

Abstract: Phase change materials are exploited in non-volatile electronic memories and photonic devices that rely on a fast and reversible transformation between the amorphous and crystalline phase upon heating. The recrystallization of the amorphous phase at the operation conditions of the memories occurs in the supercooled liquid phase above the glass transition temperature \$T_g\$. The dynamics of the supercooled liquid is thus of great relevance for the operation of the devices and, close to \$T_g\$, also for the structural relaxations of the glass that affect the performances of the memories. Information on the atomic dynamics is provided by the diffusion coefficient (\$D\$) and by the viscosity (\$ \$) which are, however, both difficult to be measured experimentally

at the operation conditions of the devices due to the fast crystallization. In this work, we leverage a machine learning interatomic potential for the flagship phase change compound compound Ge$_2$Sb$_2$Te$_5$ to compute $ $, D and the $ $-relaxation time in a wide temperature range from 1200 K to about 100 K above T_g. Large scale molecular dynamics simulations allowed quantifying the fragility of the liquid and the occurrence of a breakdown of the Stokes-Einstein relation between $ $ and D in the supercooled phase. Isoconfigurational analysis provided a visualization of the emergence of dynamical heterogeneities responsible for the breakdown of the Stokes-Einstein relation. The analysis revealed that the regions of most mobile atoms are related to the presence of Ge atoms with particular local environments.

Comparison of intermediate-range order in GeO$_2$ glass: molecular dynamics using machine-learning interatomic potential vs.\ reverse Monte Carlo fitting to experimental data

Kenta Matsutani; Shusuke Kasamatsu; Takeshi Usuki. arXiv (Cornell University) (2024). Cited by 0. Found by: openalex. DOI: 10.48550/arxiv.2409.06982

Abstract: The short and intermediate-range order in GeO$_2$ glass are investigated by molecular dynamics using machine-learning interatomic potential trained on ab initio calculation data and compared with reverse Monte Carlo fitting of neutron diffraction data. To characterize the structural differences in each model, the total/partial structure factors, coordination number, ring size and shape distributions, and persistent homology analysis were performed. These results show that although the two approaches yield similar two-body correlations, they can lead to three-dimensional models with very different short and intermediate-range ordering. A clear difference was observed especially in the ring distributions; RMC models exhibit a broad distribution in the ring size distribution, while neural network potential molecular dynamics yield much narrower ring distributions. This confirms that the density functional approximation in the ab initio calculations determines the preferred network assembly more strictly than RMC with simple coordination constraints and neutron diffraction data with isotope substitution.

Study of the thermodynamics and kinetic anomalies of Antimony-based phase-change materials by machine-learned molecular dynamics

FLAVIO GIULIANI. IRIS Research product catalog (Sapienza University of Rome) (2026). Cited by 0. Found by: openalex. URL: https://openalex.org/W7130263246

Abstract: Energy-intensive data centers and large-scale machine learning models, combined with the exponential growth of global data generation, are unsustainable in the long term unless more efficient data storage and computing technologies are developed. Promising next-generation devices include ultrafast non-volatile memories and brain-inspired computing units. The core of these technologies is enabled by phase-change materials, alloys of group-IV to -VI elements exhibiting unique, reversible switching properties between a resistive amorphous (glass) phase and a conducting crystal on a timescale of hundreds of nanoseconds. The design and optimization of phase-change materials are closely linked to the physics of supercooled liquids, i.e., metastable liquids below their melting points. The elucidation of the kinetics and thermodynamics of complex materials under deep supercooling is now within reach of atomistic simulations, thanks to the advent of machine-learned interatomic potentials trained on ab-initio data. In this work, we focus on antimony (Sb), a recent candidate for monoatomic phase-change materials, and Sb-rich Ge-Sb alloys. Signs of a liquid-liquid transition in Ge$_{15}$Sb$_{85}$ have been reported in femtosecond X-ray experiments supported by ab-initio simulations. Motivated by this, we investigate possible signatures of a liquid-liquid transition in elemental Sb using a machine-learning potential taken from the literature. By systematically exploring the structure of the crystals spontaneously nucleating from the supercooled liquid phase, we discover an unreported phase of Sb: a bulk A17 (black phosphorus–type) layered structure emerging upon decompression, which is the only phase to nucleate below −1 GPa. Within the supercooled liquid, we find water-like thermodynamic and structural anomalies, suggesting the presence of a liquid-liquid transition. We introduce a custom octahedral order parameter — analogous to the tetrahedral parameter for water — to describe these anomalies in terms of the emergence of a structured liquid at low temperature, and we perform a two-state model fit. Direct computation of the configurational entropy provides no evidence for a kinetic crossover, indicating that Sb is a highly fragile system. Finally, we train an machine-learning potential for Sb using an in-house architecture - \alphanes, proposed by Guidarelli Mattioli, Sciortino and Russo in 2023 -, which at ambient pressure achieves comparable accuracy to the existing potential. Building on this experience, we generate a dedicated ab initio dataset for Ge$_{15}$Sb$_{85}$, comprising nearly 200k atoms in total, on which we train a machine-learning potential via the NeuroEvolution Potential framework. Validation confirms its robustness against overfitting and its suitability for multi-phase modeling in future studies of this complex alloy.

Electronic structure and reactive molecular dynamics simulations for modelling solid-state batteries

; . The HKU Scholars Hub (University of Hong Kong) (2025). Cited by 0. Found by: openalex. URL: https://openalex.org/W7119437731

Abstract: Solid-state batteries exhibit superior theoretical energy density, significantly enhanced safety characteristics, and prolonged cycling stability, establishing them as a pivotal focus in next-generation energy materials research. Compared to conventional liquid electrolyte batteries, solid-state batteries eliminate safety hazards associated with flammable liquid electrolytes while promising higher energy densities and extended operational lifespans—attributes of paramount importance for meeting the future demands of electric vehicles and renewable energy storage applications. Nevertheless, their commercialization trajectory remains impeded by multiple challenges, including interface stability, ionic transport mechanisms, and materials compatibility issues. In solid-state battery research, computational methodologies have emerged as powerful instruments for elucidating fundamental mechanisms and accelerating technological advancement. Electronic structure calculations provide precise insights into band structures and interfacial properties of materials, while molecular dynamics simulations capture atomic-scale dynamic processes. This thesis presents innovative computational modelling techniques focused on these two critical domains, offering novel perspectives for solid-state battery investigation. In the realm of electronic structure methods, this research establishes an innovative density-functional tight-binding parameterization scheme for accurately modelling band alignment at solid-state battery interfaces. Through the integration of genetic algorithms with explicit band alignment constraints, this approach successfully reproduces the electronic structure of a typical all solid-state battery system (Li-Li PO N-LiCoO) in accordance with density functional theory calculations. The developed parameter set demonstrates excellent transferability across diverse lithium-containing materials and various battery charging states, providing a computational tool that balances efficiency and accuracy for interface design. In the domain of molecular dynamics simulations, this thesis develops a universal machine learning potential framework that incorporates equivariant graph neural networks with polarizable long-range interactions. This methodological innovation overcomes a critical limitation of existing machine learning potentials by accurately capturing electrostatic and dispersion effects beyond the typical cutoff radius. The pretrained universal model, encompassing elements up to plutonium, exhibits exceptional performance in predicting various materials properties, including bulk modulus, ionic diffusivity, and phase transitions. This framework enables large-scale reactive molecular dynamics simulations of solid-state battery interfaces, revealing unprecedented insights into the formation mechanisms of the solid-electrolyte interphase. Simulations of Li PS /Li and Li PS Cl/Li interfaces elucidate the atomic-level processes involved in interface stabilization, including the crystallization of Li S and the formation of amorphous Li P/Li S regions, findings that align closely with experimental observations. These methodological developments provide computational tools for investigating critical phenomena in solid-state batteries across relevant length and time scales, contributing to the theoretical design of improved energy storage technologies.

Elucidating Lithium Transport Mechanisms in Disordered LiF from Machine-Learning Molecular Dynamics Simulations

; Nikhil Rampal; Nicole Adelstein; Liwen F. Wan. ACS Materials Letters (2026). Cited by 0. Found by: openalex. openalex_2yr: 9.392 (2026). DOI: 10.1021/acsmaterialslett.6c00029

Abstract: Abstract Lithium fluoride (LiF) is a ubiquitous component of solid- and cathode–electrolyte interphases, yet its functional role remains unclear under the structural and chemical heterogeneity typical of cycling batteries. Here, we systematically quantify how structural disorder, off-stoichiometry, and strain govern Li-ion transport in LiF. Using a machine-learning potential to enable extensive molecular-dynamics sampling, we compare crystalline and amorphous LiF, Li0.95F, and LiF0.95, and evaluate the impact of small homogeneous deformations. Defect-free crystalline LiF is effectively ion-blocking at 300−500 K, whereas amorphization generates free-volume–assisted percolation pathways that facilitates Li-ion diffusion. At elevated temperatures, thermodynamically driven crystallization disrupts these pathways, leading to non-Arrhenius behavior. In crystalline phases, Li deficiency activates vacancy-mediated diffusion, while in amorphous LiF, transport is governed primarily by network connectivity. Strain is found to have only a marginal effect on Li mobility in both crystalline and amorphous structures.

Machine Learning Guided Exploration of Amorphous Electrodes and Electrolytes

Sai Gautam Gopalakrishnan; Aqshat Seth; Debsundar Dey; Vijay Choyal. ECS Meeting Abstracts (2025). Cited by 0. Found by: openalex. openalex_2yr: 0.091 (2026). DOI: 10.1149/ma2025-0271034mtgabs

Abstract: The state of the art in energy storage technologies, namely lithium-ion and solid-state batteries,

are made viable due to the use of robust crystalline materials that are used as electrodes and/or solid electrolytes. However, there are fundamental limitations in the properties that can be accessed within crystalline materials, owing to bonding, symmetry, and ordering constraints. For example, crystalline materials typically do not exhibit good ionic conductivity for multivalent ions (such as Mg, Ca, etc.), which is a critical bottleneck in the development of multivalent batteries. Similarly, the defective regions that form across a metallic anode||crystalline solid electrolyte interfaces typically cause a spike in the electric field, eventually resulting in metal dendrite nucleation and propagation. Such limitations of crystalline materials can be mitigated, to an extent, by the use of (metastable) amorphous materials. Thus, in my talk, I will present some of our recent work on modelling and understanding possible amorphous electrodes, electrolytes, and their interfaces that are relevant for energy storage systems. Using a combination of density functional theory based calculations, ab initio molecular dynamics, and trained machine learned interatomic potentials, I will highlight our work on characterising ionic transport and reaction energies within bulk electrodes, solid electrolytes, and their interfaces, where the bulk materials can also adopt an amorphous form. I hope that our study will kickstart further work on utilising amorphous materials to a greater extent in battery systems.

Large-scale first-principle simulations of amorphous indium oxide

Matthew Bousquet; Francois Gygi; Giulia Galli. arXiv (Cornell University) (2026). Cited by 0. Found by: openalex. URL: https://openalex.org/W7168054795

Abstract: Amorphous indium oxide (a-In$_2$O$_3$) is a high-electron-mobility semiconductor of central importance in thin-film transistors and a promising photoanode for solar-driven water oxidation. Despite sustained experimental and computational investigations, the structural motifs underlying its unusual transport properties and the existence of O-O peroxide-like bonds within its network have remained unresolved. Here we develop a MACE-based machine-learned interatomic potential trained on first-principles molecular dynamics trajectories and use it to generate and analyze amorphous structures containing up to 5120 atoms, two orders of magnitude larger than those adopted in typical ab initio studies. We find X-ray structure factors in excellent quantitative agreement with experiment and we confirm that In$_2$O$_3$ is a poor glass former, with the likely presence of quasi-crystalline regions in amorphous samples. Our large-scale structural analysis reveals extended chains of edge-sharing InO$_k$ polyhedra providing a concrete structural basis for the high electron mobility of a-In$_2$O$_3$. Our results strongly support the formation of O-O peroxide-like bonds in the amorphous network, with a mean length of 1.5 Å. We show that these bonds introduce localized in-gap states near the conduction band minimum, acting as a source of intrinsic n-type self-doping and enhancing sub-gap optical absorption. These effects are detectable via a distinct Raman feature near 850 cm^{-1} that is absent in the IR spectrum. Overall, our results establish a comprehensive structure-property picture of a-In$_2$O$_3$, provide directly testable experimental predictions, and suggest that controlled amorphization is a viable strategy for improving the photoelectrochemical activity of a-In$_2$O$_3$.

Large-scale first-principle simulations of amorphous indium oxide

Matthew Bousquet; Francois Gygi; Giulia Galli. arXiv (Cornell University) (2026). Cited by 0. Found by: openalex. DOI: 10.48550/arxiv.2607.08617

Abstract: Amorphous indium oxide (a-In$_2$O$_3$) is a high-electron-mobility semiconductor of central importance in thin-film transistors and a promising photoanode for solar-driven water oxidation. Despite sustained experimental and computational investigations, the structural motifs underlying its unusual transport properties and the existence of O-O peroxide-like bonds within its network have remained unresolved. Here we develop a MACE-based machine-learned interatomic potential trained on first-principles molecular dynamics trajectories and use it to generate and analyze amorphous structures containing up to 5120 atoms, two orders of magnitude larger than those adopted in typical ab initio studies. We find X-ray structure factors in excellent quantitative agreement with experiment and we confirm that In$_2$O$_3$ is a poor glass former, with the likely presence of quasi-crystalline regions in amorphous samples. Our large-scale structural analysis reveals extended chains of edge-sharing InO$_k$ polyhedra providing a concrete structural basis for the high electron mobility of a-In$_2$O$_3$. Our results strongly support the formation of O-O peroxide-like bonds in the amorphous network, with a mean length of 1.5 Å. We show that these bonds introduce localized in-gap states near the conduction band minimum, acting as a source of intrinsic n-type self-doping and enhancing sub-gap optical absorption. These effects are detectable via a distinct Raman feature near 850 cm^{-1} that is absent in the IR spectrum. Overall, our results establish a comprehensive structure-property picture of a-In$_2$O$_3$, provide directly testable experimental predictions, and suggest that controlled amorphization is a viable strategy for improving the photoelectrochemical activity of a-In$_2$O$_3$.

Amorphous boron nitride as an ultrathin copper diffusion barrier for advanced interconnects

Onurcan Kaya; Hyeongjoon Kim; Byeongkyu Kim; Thomas Galvani; Luigi Colombo; Mario Lanza; Hyeon-Jin Shin; Ivan Cole; Hyeon Suk Shin; Stephan Roche. 2D Materials (2025). Cited by 0. Found by: openalex. openalex_2yr: 4.053 (2026). DOI: 10.1088/2053-1583/ae2521

Abstract: Abstract This study focuses on amorphous boron nitride (-BN) as a novel diffusion barrier for advanced semiconductor technology, particularly addressing the critical challenge of copper diffusion in back-end-of-line (BEOL) interconnects. Owing to its ultralow dielectric constant and robust barrier properties, -BN is examined as an alternative to conventional low-k dielectrics. The investigation primarily employs theoretical modelling, using a Gaussian approximation potential, to simulate and understand the atomic-level interactions. This machine-learning-based potential enables realistic simulations of amorphous -BN structures and allows us to examine how different film morphologies affect barrier performance. Furthermore, we studied the electronic and optical properties of the films using a simple Tight-Binding model. In addition to the theoretical work, we performed copper diffusion experiments through PECVD-grown -BN on Si substrates. Theoretical and experimental results indicate that -BN films can suppress Cu diffusion at nanometre thicknesses. Together, molecular dynamics simulations based on a machine-learned interatomic potential and PECVD experiments support the use of -BN as a Cu diffusion barrier for BEOL interconnects.

Large scale molecular modelling of basalt surfaces and fracturing in basaltic glass

Marthe Grønlie Guren; Henrik Andersen Sveinsson; Razvan Caracas; Anders Malthe-Sørenssen; François Renard. (2024). Cited by 0. Found by: openalex. DOI: 10.5194/egusphere-egu24-11861

Abstract: How fracture initiate and propagate at the nanoscale controls the specific surface area available for fluid-rock interactions. Fracturing creates surface area and flow pathways, which control the flow mixing properties and reactivity. However, how fractures form at the nanoscale in basaltic glasses remains enigmatic. Here, we implement molecular dynamics simulations to reproduce fracture propagation in amorphous basalt. These simulations require large systems and long simulation times and are therefore currently depending on interatomic potentials rather than ab initio calculations. We have developed a machine-learned interatomic potential for basaltic glass that allow using molecular dynamics simulations to simulate fracture propagation at the nanoscale. This interatomic potential reproduces the mechanical properties of bulk solid and molten basalt over a wide range of temperatures and pressures. During a molecular dynamics simulation, bonds are formed and broken as the atoms and ions move. As a result, various species may form and migrate into the glass or towards the surface. In order to study how a basalt surface changes over time we looked at the cation-anion species on the surface and measured how long each species lived on the surface before the coordination changes. Our results show that the process zone around propagating cracks in basaltic glasses at the nanoscale is much larger than for instance when quartz breaks, and that cavities open ahead of the crack tip, and grow with time until they coalesce. A similar propagation process has been observed in fracture experiments on silica glasses at nanometer scale using atomic force microscopy and is reminiscent of the ductile fracturing process observed in metals. By training the interatomic potentials with water and carbon dioxide as fluids, we also aim to study how dynamic fractures may damage a basaltic glass and how the water and carbon dioxide enter these fractures in the wake of rupture. These simulations are relevant for carbon mineralization where a coupling between dissolution of the basalt and precipitation of carbonate minerals may lead to nanoscale fracturing of the rock.

Structure and mechanical properties of monolayer amorphous carbon and boron nitride

Xi Zhang; Yutian Zhang; Yunpeng Wang; Yiran Li; Shixuan Du; Yuyang Zhang; Sokrates T. Pantelides. arXiv (Cornell University) (2023). Cited by 0. Found by: openalex. DOI: 10.48550/arxiv.2309.15352

Abstract: Amorphous materials exhibit various characteristics that are not featured by crystals and can sometimes be tuned by their degree of disorder (DOD). Here, we report results on the mechanical properties of monolayer amorphous carbon (MAC) and monolayer amorphous boron nitride (maBN) with different DOD. The pertinent structures are obtained by kinetic-Monte-Carlo (kMC) simulations using machine-learning potentials (MLP) with density-functional-theory (DFT)-level accuracy. An intuitive order parameter, namely the areal fraction Fx occupied by crystallites within the continuous random network, is proposed to describe the DOD. We find that Fx captures the essence of the DOD: Samples with the same Fx but different sizes and distributions of crystallites have virtually identical radial distributions functions as well as bond-length and bond-angle distributions. Furthermore, by simulating the fracture process with molecular dynamics, we found that the mechanical responses of MAC and maBN before fracture are solely determined by Fx and are insensitive to the sizes and specific arrangements of the crystallites. The behavior of cracks in the two materials is analyzed and found to mainly propagate in meandering paths in the CRN region and to be influenced by

crystallites in distinct ways that toughen the material. The present results reveal the relation between structure and mechanical properties in amorphous monolayers and may provide a universal toughening strategy for 2D materials.

Large scale simulations of amorphous phase change materials for data storage

Marco Bernasconi. BOA (University of Milano-Bicocca) (2012). Cited by 0. Found by: openalex. openalex_2yr: 0.014 (2026). URL: https://openalex.org/W1151744864

Abstract: In this seminar I will review our recent study of the properties of phase change materials by means of molecular dynamics simulations based on Neural Network potentials [1]. The materials we are dealing with are chalcogenide alloys (typically Ge2Sb2Te5 or GeTe) which are the subject of extensive experimental and theoretical research because of their use in optical (digital versatile disc, DVD) and electronic (phase change memories, PCM) storage devices [2]. Both applications rely on the fast and reversible transformation between the crystalline and amorphous phases induced by heating either via laser irradiation (DVD) or Joule effect (PCM). The two states of the memory can be discriminated because of the large difference in optical reflectivity and electronic conductivity of the two phases. In the last few years atomistic simulations based on density functional theory have provided useful insights on the properties of materials in this class [3]. However, several key issues such as the crystallization dynamics occurring at the ns time scale, the properties of the crystalline/amorphous interface and the thermal conductivity at the nanoscale, just to name a few, are presently beyond the reach of fully ab-initio simulations. A route to overcome the limitations in system size and time scale of ab-initio molecular dynamics is the development of empirical interatomic potentials. However, traditional approaches based on the fitting of simple functional forms are very challenges due to the complexity and variability of the chemical bonding in the crystal and amorphous phases of these materials as revealed by ab-initio simulations. A possible solution was demonstrated few years ago by Behler and Parrinello [2] who developed classical interatomic potentials with close to ab-initio accuracy by fitting large ab-initio databases by means of a NN scheme. By means of this technique, we have developed an interatomic potential for GeTe [4] which is one of the compounds under scrutiny for applications in phase change memories. The NN potential with a close to ab-initio accuracy has allowed us to address several key issues for applications in data storage such as the homogeneous and heterogeneous crystallization of the amorphous, the glass transition of the supercooled liquid and the thermal conductivity of the amorphous phase.

The ab initio amorphous materials database: Empowering machine learning to decode diffusivity

Hui Zheng; Patrick Huck. OSTI OAI (U.S. Department of Energy Office of Scientific and Technical Information) (2024). Cited by 0. Found by: openalex. DOI: 10.17188/mpcontribs/2281590

Abstract: Amorphous materials exhibit unique properties that make them suitable for various applications in science and technology, ranging from optical and electronic devices and solid-state batteries to protective coatings. However, data-driven ex- ploration and design of amorphous materials is hampered by the absence of a com- prehensive database covering a broad chemical space. In this work, we present the largest computed amorphous materials database to date, generated from sys- tematic and accurate ab initio molecular dynamics (AIMD) calculations. We also show how the database can be used in simple machine-learning models to connect properties to composition and structure, here specifically targeting ionic conductiv- ity. These models predict the Li-ion diffusivity with speed and accuracy, offering a cost-effective alternative to expensive density functional theory (DFT) calculations. Furthermore, the process of computational quenching amorphous materials provides a unique sampling of out-of-equilibrium structures, energies, and force landscape, and we anticipate that the corresponding trajectories will inform future work in uni- versal machine learning potentials, impacting design beyond that of non-crystalline materials.

Energy dissipation at the atomic scale explains how fracture energy depends on crack velocity in silica glass

Marthe Grønlie Guren; Sigbjørn Løland Bore; François Renard; Henrik Andersen Sveinsson. arXiv (Cornell University) (2026). Cited by 0. Found by: openalex. DOI: 10.48550/arxiv.2605.03457

Abstract: The fracture energy of brittle materials rises with crack velocity, and this effect is typically attributed to surface roughening from path instabilities. Here we show, using molecular dynamics simulations of silica glass with a first-principles machine learned interatomic potential, that the structural fracture energy rises by up to 33 % already below the branching threshold, showing that fracture energy is not a constant material property. This rise in fracture energy is roughly equally partitioned between an increase in the intrinsic surface energy density and nanoscale roughening that increases the real fracture surface area. Results demonstrate that

dynamic fracture in silica glass increases the fracture energy not merely by creating more apparent surface, but also by creating a fundamentally different surface at the nanoscale.

Energy dissipation at the atomic scale explains how fracture energy depends on crack velocity in silica glass

Marthe Grønlie Guren; Sigbjørn Løland Bore; François Renard; Henrik Andersen Sveinsson. arXiv (Cornell University) (2026). Cited by 0. Found by: openalex. URL: https://openalex.org/W7160565694

Abstract: The fracture energy of brittle materials rises with crack velocity, and this effect is typically attributed to surface roughening from path instabilities. Here we show, using molecular dynamics simulations of silica glass with a first-principles machine learned interatomic potential, that the structural fracture energy rises by up to 33 % already below the branching threshold, showing that fracture energy is not a constant material property. This rise in fracture energy is roughly equally partitioned between an increase in the intrinsic surface energy density and nanoscale roughening that increases the real fracture surface area. Results demonstrate that dynamic fracture in silica glass increases the fracture energy not merely by creating more apparent surface, but also by creating a fundamentally different surface at the nanoscale.

Role of Characteristic Length Scale in Interface Graphitization-Induced Wear Resistance of Diamond and Amorphous Carbon

Wenliang Shi; YanYu Zhang; WenZheng Liao; Kai Xu; HongYu Wu; ZhiCheng Zhong; KeKe Chang. arXiv (Cornell University) (2026). Cited by 0. Found by: openalex. URL: https://openalex.org/W7163597948

Abstract: The evolution of interfacial atomic structures critically influences the friction and wear behavior of carbon-based materials. However, how the characteristic length scale of friction-induced sp^{2} reconstruction governs macroscopic wear remains poorly understood, particularly for diamond and amorphous carbon where the interfacial graphitization modes differ fundamentally. In this work, we develop a machine learning potential for these carbon systems and investigate the structural evolution at interfaces in both diamond/diamond and amorphous/amorphous carbon systems using molecular dynamics simulations. Our results reveal distinct atomic-scale characteristics of graphitization at the two interfaces. Diamond interfaces develop a laterally continuous sp^{2} reconstruction layer with a characteristic length of 30–45~Å, while amorphous carbon interfaces form only fully isolated sp^{2} patches of 8–12~Å. This disparity in characteristic length scale determines the density of weakly bonded interfacial atoms left outside the reconstruction layer, thereby directly dictating the macroscopic wear rate. Based on these insights, we propose a strategy to regulate friction-induced graphitization in diamond coatings by protecting specific crystallographic orientations, such as the (111) close-packed planes. This work bridges the gap between atomic-scale interfacial structure and macroscopic tribological performance, offering mechanistic guidelines for the rational design of wear-resistant carbon-based coatings.

Role of Characteristic Length Scale in Interface Graphitization-Induced Wear Resistance of Diamond and Amorphous Carbon

Wenliang Shi; YanYu Zhang; WenZheng Liao; Kai Xu; HongYu Wu; ZhiCheng Zhong; KeKe Chang. arXiv (Cornell University) (2026). Cited by 0. Found by: openalex. DOI: 10.48550/arxiv.2606.03325

Abstract: The evolution of interfacial atomic structures critically influences the friction and wear behavior of carbon-based materials. However, how the characteristic length scale of friction-induced sp^{2} reconstruction governs macroscopic wear remains poorly understood, particularly for diamond and amorphous carbon where the interfacial graphitization modes differ fundamentally. In this work, we develop a machine learning potential for these carbon systems and investigate the structural evolution at interfaces in both diamond/diamond and amorphous/amorphous carbon systems using molecular dynamics simulations. Our results reveal distinct atomic-scale characteristics of graphitization at the two interfaces. Diamond interfaces develop a laterally continuous sp^{2} reconstruction layer with a characteristic length of 30–45~Å, while amorphous carbon interfaces form only fully isolated sp^{2} patches of 8–12~Å. This disparity in characteristic length scale determines the density of weakly bonded interfacial atoms left outside the reconstruction layer, thereby directly dictating the macroscopic wear rate. Based on these insights, we propose a strategy to regulate friction-induced graphitization in diamond coatings by protecting specific crystallographic orientations, such as the (111) close-packed planes. This work bridges the gap between atomic-scale interfacial structure and macroscopic tribological performance, offering mechanistic guidelines for the rational design of wear-resistant carbon-based coatings.

Neural Network-Based Simulation Method to Examine Ion Behaviors under Electric Fields: Application to Ion Migration in Amorphous Li3PO4

Kōji Shimizu; Ryuji Otsuka; Satoshi Watanabe. Journal of the Japan Society of Powder and Powder Metallurgy (2023). Cited by 0. Found by: openalex. DOI: 10.2497/jjspm.23-00043

Abstract: We developed a neural network-based model to predict the Born effective charges from atomic structures. By combining forces due to an applied electric field, expressed as a product of the Born effective charge and the electric field, and forces evaluated by a neural network potential (NNP), a simulation scheme of ion dynamics under an electric field was proposed. Taking Li3PO4 as a prototype, we demonstrated the validity of our computation scheme. Using the constructed model of the Born effective charge predictor and NNP based on density functional (perturbation) theory calculation data, molecular dynamics (MD) simulations under a uniform electric field of 0.1 V/Å were performed. We obtained an enhanced mean square displacement of Li along the electric field, which seems physically reasonable. In addition, we found that the external forces along the direction perpendicular to the electric field, which originated from the off-diagonal components of the Born effective charges, had a non-negligible effect on the Li motion. Furthermore, we observed a more susceptive response of Li to the electric field in an amorphous structure.

Machine-Learned Potential–Driven Insights into Li S–P S Glass Electrolytes and Li PS Cl Interfacial Chemistry for Solid-State Batteries

Byungju Lee; C S Kim; Sojeong Yang. ECS Meeting Abstracts (2026). Cited by 0. Found by: openalex. openalex_2yr: 0.091 (2026). DOI: 10.1149/ma2026-016700mtgabs

Abstract: Machine-learned interatomic potentials (MLPs) provide an avenue for exploring complex structural motifs and chemical reactions in solid-state battery materials that are challenging to access via purely first-principles approaches. In this presentation, I introduce our recent studies employing MLP-based simulations to investigate two key classes of sulfide solid-state electrolytes (SSEs): Li S–P S glass for composition and density optimization, and Li PS Cl argyrodite for interfacial and surface reactivity. For the Li S–P S glass system , we develop an MLP-enabled framework that systematically maps the relationships among composition, densification behavior, structural motifs, and ionic transport. By performing extensive sampling of glass structures across a wide composition range, we identify optimal glass formulations and density regimes that enhance Li-ion mobility and thermodynamic stability. This approach allows efficient navigation of the broad design space inherent to multicomponent amorphous SSEs. In complement, we investigate Li metal | Li PS Cl interfaces and the surface chemistry of Li PS Cl under oxygen and moisture exposure. MLP-accelerated molecular dynamics reveals the initial reaction pathways, interphase formation, and structural evolution at the Li–solid interface, providing insights into stability and ion-transport implications. Additionally, we analyze degradation reactions triggered by O /H O on Li PS Cl surfaces, elucidating mechanisms relevant to processing conditions and long-term performance. Together, these results demonstrate how machine-learned potentials enable multiscale understanding—from bulk amorphous structure to interfacial chemistry—crucial for the design of robust and high-conductivity sulfide solid-state electrolytes.

Suppressed glass transition behavior through high configurational entropy in high-entropy metallic glasses

Jinyue Wang; Fuzhi Dai; Yihuan Cao; Yuan Wu; Hui Wang; Suihe Jiang; Xiaobin Zhang; Xianzhen Wang; Xiongjun Liu; Zhaoping Lu. Acta Materialia (2026). Cited by 0. Found by: openalex. openalex_2yr: 10.171 (2026). DOI: 10.1016/j.actamat.2026.122361

Visualizing a Li-depleted amorphous cathode–electrolyte interphase in sulfide solid-state batteries using in situ cryogenic electron microscopy

Yuki Nomura; Ryoma Sasaki; Huu Duc Luong; Misaki Hasegawa; Willy Shun Kai Bong; Koji Hiraoka; Kazuo Yamamoto; Yutaka Ito; Yoshiya Fujiwara; Yoshitaka Tateyama; Takuhiro Miyuki. ChemRxiv (2026). Cited by 0. Found by: openalex. openalex_2yr: 0.467 (2026). DOI: 10.26434/chemrxiv.15003253/v1

Abstract: Electrochemical degradation at the cathode/solid-electrolyte interface critically limits the performance of sulfide-based solid-state batteries; however, its nanoscale origin remains unclear. Here, we directly visualize the structural and chemical evolutions of the LiNi 0.5 Co 0.2 Mn 0.3 O 2 /Li 6 PS 5 Cl interface during charging using in situ scanning transmission electron microscopy combined with electron energy-loss spectroscopy and energy-filtered nanobeam electron diffraction. A potential-controlled sample-preparation protocol is developed to preserve the native interphase structure during sample preparation, and low-dose-rate

cryogenic electron microscopy suppresses electron-beam-induced damage. At the uncoated interface, charging induced the formation of a ~50-nm-thick Li-depleted cathode–electrolyte interphase. In regions within ~20 nm of the interface, the Li concentration decreased to below x = 4 in Li x PS 5 Cl, leading to complete amorphization of the solid electrolyte. Machine-learning-potential molecular dynamics simulations reveal that Li depletion destabilizes the argyrodite Li 6 PS 5 Cl framework and increases the Li-ion migration barrier, thereby identifying the origin of interfacial resistance. Conversely, LiNbO y -coated interfaces exhibited neither pronounced Li depletion nor significant amorphization, even with coating thicknesses as small as 5 nm, thereby elucidating the buffering mechanism of LiNbO y at the cathode/sulfide-solid-electrolyte interface. These results establish Li-depletion-induced structural disorder as a fundamental transport-limiting mechanism at cathode interfaces employing sulfide solid electrolytes, providing a framework for interface engineering.

Quantitative assessment of the structure of (Na2O)0.1-(V2O5)0.6-(TeO2)0.3 glass: An ab initio and machine learning interatomic potential study of oxide systems with mixed-valence

Firas Shuaib; Rémi Piotrowski; Gaëlle Delaizir; Pierre-Marie Geffroy; Steve Dave Wansi Wendji; Carlo Massobrio; Mauro Boero; Guido Ori; Philippe Thomas; Olivier Masson; Assil Bouzid. Artificial Intelligence Chemistry (2026). Cited by 0. Found by: openalex. openalex_2yr: 9.034 (2026). DOI: 10.1016/j.aichem.2026.100123

Abstract: This study investigates the atomic-scale structure of a glassy system using first-principles molecular dynamics (FPMD) and machine-learned interatomic potentials (MLIP). The structural models generated using MLIP demonstrate excellent agreement with FPMD results while offering substantial computational efficiency, enabling the exploration of much larger system sizes that are prohibitive for FPMD while retaining comparable accuracy. Validation against experimental X-ray total structure factors and pair distribution functions indicates the reliability of both the FPMD- and MLIP-generated structures. The glass network is found to be composed of a mix of VO 4 , VO 5 and VO 6 polyhedra, TeO 3 , TeO 4 and TeO 5 units. Local environments are studied through maximally localized Wannier function (MLWF)-based bonding analysis, coordination number analysis, and bond angle distributions. The MLIP model is further used to study glass compositions with varying amounts of V 5 + and V 4 + to investigate oxidation-state-dependent structural changes. Our results demonstrate that melt-quench glass models produced through MLIP-based molecular dynamics and subsequently relaxed via a single-shot DFT calculation can capture changes in local topology, coordination, and V–O bonding motifs with high accuracy, revealing the reliability of the MLIP model in describing mixed-valence oxide glass systems. This finding can be explained by charge self-regulation in transition-metal oxides, where changes in formal oxidation state are accommodated by local metal–O electronic redistribution and accompanying structural rearrangements. This work highlights the capability of MLIPs to accurately describe complex oxide glasses with multicomponent compositions and variable oxidation states, extending their applicability to chemically diverse disordered systems.

Thermochemical properties and glass transition of Na 2 O–Al 2 O 3 –P 2 O 5 system: a deep learning potential approach and its experimental validation

Dmitry Zakiryanov; Maxim I. Vlasov; Victor Bykov. Modelling and Simulation in Materials Science and Engineering (2026). Cited by 0. Found by: openalex. DOI: 10.1088/1361-651x/ae39f7

Abstract: Abstract Observation of glass properties including the glass transition temperature T g is one of the most challenging problems for atomistic simulations due to the complexity of composition and slow time evolution rates. To address the issue, robust machine learning potentials trained with accurate ab initio data can be employed. In this paper, a prospective glass for immobilization of nuclear waste with 0.25 Na 2 O–0.25 Al 2 O 3 –0.5 P 2 O 5 (mass fraction) composition was studied via a neural network potential. We used 42 independent ab initio calculations as a dataset. The glass was obtained by slow cooling during 96 ns from the molten state down to room temperature. We used an extrapolation technique to assess the true glass transition temperature for infinitely slow cooling and found that the resulting T g of 715 K agrees well with the experimentally observed value 682 K. The specific heat was studied for a temperature range up to T = 1500 K. Also, thermal conductivity was calculated using the non-equilibrium molecular dynamics. Among other properties, Young's modulus, thermal diffusivity, and vibrational density of states were calculated. The reliability of the developed model was further confirmed by comparison with the experimentally measured density and Raman spectrum.

Study on Cu 50 Zr 50 Metallic Glass Based on Machine Learning Force Field: Correlation between Virial and Equipartition Theorem Violations and Atomic-Scale Dynamic Behaviors

Wenliang Shi; YanYu Zhang; HongYu Wu; KeKe Chang; ZhiCheng Zhong. Chinese Physics B (2026). Cited by 0. Found by: openalex. openalex_2yr: 1.149 (2026). DOI: 10.1088/1674-1056/ae42ba

Abstract: Abstract The thermodynamic properties of amorphous materials are crucial. However, their lack of long-range order makes both vibrational theory and experimental characterization methods ineffective in studying the related microscopic behaviors. This fundamental limitation hinders a mechanistic understanding of how relaxation behaviors govern macroscopic properties. Traditional simulation methods such as empirical potential-based molecular dynamics and density functional theory can link relaxation behaviors and macroscopic statistical properties, but are often limited by trade-offs between scale and accuracy. In this work, we use a machine learning potential with Density Functional Theory-level accuracy to investigate the microscopic relaxation behavior and their influence on kinetic energy and Virial in large-scale Cu 50 Zr 50 metallic glass. Our simulations reveal two distinct types of atomic relaxation behaviors with different temperature dependencies, which influence the dynamics of the glass above and below the glass transition temperature. Furthermore, we observe a breaking of equipartition theorem and the Virial theorem during the evolution of Cu 50 Zr 50 metallic glass. The deviation from this relation gradually decays to zero over time as microscopic atomic relaxation processes are activated. These findings can provide a more accurate method for understanding the atomic-level mechanisms underlying the formation and evolution of glassy materials.

Zero-Dimensional Ice Ring Topological Proton Glass: Gapless Texture Soft Modes, Anharmonic Wandering, and Three Conserved Invariants

Zhongzheng Miao; Miao Zhongzheng. ChemRxiv (2026). Cited by 0. Found by: openalex. openalex_2yr: 0.467 (2026). DOI: 10.26434/chemrxiv.15004437/v1

Abstract: Traditional studies of confined ice have heavily relied on exogenous pore confinement such as carbon nanotubes and porous materials. Through a strategy of stepwise dimensionality reduction and geometric topological closure, and without any chemical doping or external substrate confinement, this work systematically constructs a full-curvature-gradient family of zerodimensional closed-loop ice rings, denoted PNH-M-Ring. Starting from two-dimensional intrinsic ice and one-dimensional free-standing multi-helix ice nanotubes (PNH-INT, N = 3–7) and employing a self-developed topological curling program, we generate these structures. Using two first-principles software packages (CASTEP, DMol3) and the semi-empirical tight-binding method DFTB+ in a hierarchical and complementary computational framework, together with stepwise convergence tests of phonons, supercell size evolution analyses, and multi-temperature ab initio molecular dynamics (AIMD) simulations, we demonstrate from four perspectives (atomic geometry, electronic structure, harmonic lattice vibrations, and finite-temperature dynamics) that closing a one-dimensional helical ice axially via two S1 ends to form a zero-dimensional ice ring generates a new state of matter − the topological proton glass (TPG). Unlike KOH-doped proton glasses where disorder is impurity-induced, the present system relies solely on topological constraints to achieve a perfect satisfaction of the Bernal-Fowler ice rules, yet becomes locked into a manifold of nearly degenerate ground states because of the closed-loop topology ("perfect but trapped"). High-precision harmonic phonon calculations consistently yield low-frequency imaginary frequencies, which signal a cooperative, gapless texture soft mode throughout the ring and do not indicate structural mechanical instability. Finite-temperature AIMD shows that at low temperatures the soft mode evolves into an anharmonic wandering dominated by global elliptical breathing and helical sliding; the oxygen skeleton remains fully connected, and any transient hydrogen-bond breakage heals rapidly. We rigorously derive three topological invariants − enforced zero polarization flux, the Gauss linking number Lk, and the global 2 chiral charge C0 − thereby establishing a quantitative criterion for TPG. Leveraging the complete research spectrum from 2D through 1D to 0D, we elucidate the evolutionary law of progressive stripping of translational degrees of freedom and the concomitant enhancement of topological frustration. We systematically clarify the role of nuclear quantum effects (NQE), propose two decisive experiments (deuteration and inelastic neutron scattering), and outline potential device applications. All configurations and input scripts are open-source, and we invite the community to re-examine our findings using PIMD or machine-learning potentials.

Nanoparticules d'argent enrobées dans une matrice de silice pour des dispositifs antibactériens : synthèse en phase gazeuse et modélisation assistée par apprentissage automatique

Marie-Salomé Trillot. HAL (Le Centre pour la Communication Scientifique Directe) (2025). Cited by 0. Found by: openalex. openalex_2yr: 0.03 (2026). URL: https://openalex.org/W7153401716

Abstract: This study focuses on the development of antibacterial nanocomposite materials composed of silver nanoparticles (AgNPs) embedded in a silica matrix, and located a few nanometers below the free surface. The AgNPs act as reservoirs of Ag ions which, once released, diffuse through the matrix to reach the surface and exert antibacterial action. The silica serves both as protective barrier for the metallic nanoparticles and for the environment, and as ion release regulator, in line with the "nano-safer by design " principle. A combined experimental and theoretical approach was developed to understand the atomic-scale mechanisms of ionic dif-

fusion in amorphous silica and to optimize the structural parameters of the materials.On the experimental side, gas-phase synthesis was selected for its ability to produce nanoparticles under well-controlled conditions, free of organic contamination. This method is coupled with radio-frequency magnetron sputtering to deposit a thin silica layer over the AgNPs. The main objective was to achieve independent control over the size and surface density of deposited AgNPs, with optimal coverage. A key result concerns the effect of oxygen on the nucleation mechanisms. The addition of oxygen at very low concentration ($< 0.2\%$) significantly increases the nucleation rate and surface density of nanoparticles. Optical emission spectroscopy (OES) measurements led to the hypothesis that oxygen may trigger ion-induced nucleation, accounting for the observed increase in the nucleation rate. Preliminary release tests were performed in aqueous solution. Future work will focus on embedding AgNPs in a silica matrix to investigate the influence of nanoparticle and matrix properties on controlled ion release and antibacterial efficiency.On the theoretical side, Ag ion diffusion in amorphous silica was studied using molecular dynamics simulations with machine-learned interatomic potentials. Two neural network potentials (NNPs) were developed from DFT databases. The first, dedicated to pure silica, accurately reproduces the structural and vibrational properties of amorphous silica. The second, extended to systems containing a single Ag ion, enabled the simulation of its diffusion through the matrix. The computed molecular dynamics trajectories revealed sub-diffusive behaviour, with strong ion trapping effects related to the disordered topology of the network. In particular, under-coordinated oxygen atoms (O1 defects) form stable Ag-O bonds, which slow down transport. The glass density also affects network connectivity and the accessible diffusion pathways.These results show that Ag ion diffusion can be modulated by fine-tuning the local structure of the matrix and pave the way for precise control of ion release, directly linked to the antibacterial performance of the surfaces.

Influence of crystal orientation and polymorphism on the shock response of boron carbide

Kimia Ghaffari; Salil Bavdekar; Douglas E. Spearot; Ghatu Subhash. Acta Materialia (2025). Cited by 0. Found by: openalex. openalex_2yr: 10.171 (2026). DOI: 10.1016/j.actamat.2025.121305

Phonon Transport in Disordered Two-Dimensional Graphene

Hassan Shoaib. McGill-DEV (2025). Cited by 0. Found by: openalex. DOI: 10.82308/27920

Abstract: We employ molecular dynamics (MD) simulations, equipped with a machine-learned interatomic potential, to examine the impact of structural disorder on phonon transport in two-dimensional graphene. By generating amorphous structures through Monte Carlo methods, we systematically investigate how defects modify thermal conductivity and vibrational properties. Our findings demonstrate that a minimal defect concentration of 0.2% significantly degrades thermal conductivity: the out-of-plane component declines by over an order of magnitude, while the in-plane component decreases by half. Spectral analyses reveal that low-frequency flexural acoustic (ZA) modes, which predominantly govern heat transport in pristine graphene, are substantially suppressed in defective samples. In defective graphene, the mean free paths of ZA phonons are reduced by approximately half an order of magnitude compared to pristine graphene, with this reduction exceeding that of in-plane phonon modes; however, the decrease is not preferentially biased toward low frequencies as initially hypothesized. Furthermore, only minor localization effects are observed at low frequencies (PPR 0.9). These observations suggest a more complex scenario, implying that defects may preferentially disrupt normal scattering processes at low frequencies, potentially necessitating a deeper examination of hydrodynamic effects

Atomic structure at the surface of warm basaltic glasses

Marthe Grønlie Guren; Henrik Anderson Sveinsson; Razvan Caracas; Anders Malthe-Sørenssen; François Renard. (2025). Cited by 0. Found by: openalex. DOI: 10.5194/egusphere-egu25-12944

Abstract: Silicate melts exist as lava flows which form when molten or partially molten magma erupts, and when they cool, magmas evolve into solid rocks. Depending on the cooling rate, they can evolve into fully crystalline rocks, partially-crystallized rocks or even glassy rocks. The composition of the glass at the cooling interface with air or water may be different than in the bulk. Here we study how some major elements could be concentrated or depleted at the surface of a cooling basaltic melt. This may have effects on how glass will interact with water at the onset of weathering. To model silicate melts, we have trained a machine-learned interatomic potential for basaltic glass, which we use to run molecular dynamics simulations of molten basalt and a basalt surface at temperatures consistent with fresh deposits of basalt during eruption. We have studied the difference between bulk molten basalt and a free surface of molten basalt by comparing the diffusion coefficient, lifetime of species and the spatial distribution of atoms between the two domains. We show that the diffusion at the surface is higher than in the bulk, indicating a higher rearrangement of the surfaces compared to the bulk, and the

coordination numbers are generally lower at the surface than in the bulk. When studying the composition of a surface and bulk, our results show that most of the cations on the surface are iron, magnesium and calcium, i.e. the cations that can react with CO2 to precipitate as carbonate minerals. These simulations are relevant for the initial weathering of silicate melt, and knowledge of the composition of the surface are relevant for the potential reactions with CO2 and carbon mineralization.

Through Rainbow-Tinted Glasses: Machine Learning-Driven Modeling of Chromophores

Eric Lindgren. (2026). Cited by 0. Found by: openalex. DOI: 10.63959/chalmers.dt/5890

Abstract: Chromophores are a class of molecules that give color to the world around us, from the chlorophyll in plants that enables photosynthesis to the retinal molecules in our eyes that allow us to see. Chromophores are also fundamental for developing a range of technologies crucial for a transition to a sustainable society, including photovoltaics and energy storage. While chromophores have been widely studied experimentally, we still lack a sufficient understanding of their structure and dynamics on the atomic scale. In particular, many chromophores are glass formers or form supramolecular aggregates due to intermolecular interactions, leading to a complicated landscape of interactions spanning many time and length scales. Addressing this limitation requires atomistic simulations capable of connecting microscopic dynamics to experimentally measurable quantities. In this spirit, this thesis presents a simulation framework that links electronic structure calculations via molecular dynamics simulations to experiments, with a focus on neutron scattering.The key ingredient in this framework is machine-learned interatomic potentials, enabling simulations with the accuracy of quantum mechanical calculations for large systems of chromophores. Methodological developments focus on the neuroevolution potential framework implemented in the GPUMD package. A major contribution is the development of the calorine package, which is a companion software for GPUMD that interfaces with the broader scientific software ecosystem.The framework is applied to three challenging applications: glass formation, optical response, and neutron scattering. First, glass formation occurs beyond timescales accessible to molecular dynamics simulations, and this limitation is circumvented by extrapolating relaxation processes from nanoseconds to experimental timescales using Bayesian regression. Second, optical properties are obtained by using the neuroevolution potential framework to predict tensorial properties such as the dipole moment, or spectral quantities such as the electronic dielectric function. Finally, instrument-specific inelastic neutron scattering signatures are predicted using electronic structure calculations, machine learning, and correlation functions. Together, these developments establish a framework for connecting atomistic simulations with experimental observables, enabling modeling of chromophores over multiple time and length scales. The framework is transferable and directly applicable to other systems.

When More Data Hurts: Optimizing Data Coverage While Mitigating Diversity Induced Underfitting in an Ultra-Fast Machine-Learned Potential

Jason B. Gibson; Tesia Janicki; Ajinkya C. Hire; Chris Bishop; J. Matthew D. Lane; Richard G. Hennig. arXiv (Cornell University) (2024). Cited by 0. Found by: openalex. DOI: 10.48550/arxiv.2409.07610

Abstract: Machine-learned interatomic potentials (MLIPs) are becoming an essential tool in materials modeling. However, optimizing the generation of training data used to parameterize the MLIPs remains a significant challenge. This is because MLIPs can fail when encountering local enviroments too different from those present in the training data. The difficulty of determining \textit{a priori} the environments that will be encountered during molecular dynamics (MD) simulation necessitates diverse, high-quality training data. This study investigates how training data diversity affects the performance of MLIPs using the Ultra-Fast Force Field (UF\$^3\$) to model amorphous silicon nitride. We employ expert and autonomously generated data to create the training data and fit four force-field variants to subsets of the data. Our findings reveal a critical balance in training data diversity: insufficient diversity hinders generalization, while excessive diversity can exceed the MLIP's learning capacity, reducing simulation accuracy. Specifically, we found that the UF\$^3\$ variant trained on a subset of the training data, in which nitrogen-rich structures were removed, offered vastly better prediction and simulation accuracy than any other variant. By comparing these UF\$^3\$ variants, we highlight the nuanced requirements for creating accurate MLIPs, emphasizing the importance of application-specific training data to achieve optimal performance in modeling complex material behaviors.

Dynamic Anion Sublattices, Cooperative Ion Migration and Metastable Pathway Engineering for Next-Generation Solid Electrolytes

Rajab Abbas; Rabia Shahzad; Aasma Bibi; Muhammad Adeel; Muhammad Usman; Rizwan Haider; Waqar Yousaf; Hunza Afzal; Junaid Zaman. Saudi Journal of Engineering and Technology (2026). Cited by 0. Found by: openalex. DOI: 10.36348/sjet.2026.v11i08.003

Abstract: Solid electrolytes are commonly designed using static descriptors such as crystallographic bottlenecks, vacancy concentrations, and migration barriers, yet emerging evidence shows that anion translation, rotation, vibration, and disorder can actively reorganize ionic energy landscapes. This review establishes a unified materials framework linking dynamic anion sublattices, cooperative ion migration, and metastable pathway engineering across lithium-, sodium-, proton-, and multivalent-ion conductors. A literature synthesis will compare sulfides, halides, oxyhalides, complex hydrides, solid acids, antiperovskites, and amorphous or partially crystalline electrolytes. Particular emphasis is placed on distinguishing independent hopping from correlated, concerted, paddle-wheel, and phonon-assisted transport; identifying when anion motion is causal rather than merely coincident; and determining how mixed-anion chemistry, defects, strain, mechanochemistry, quenching, and controlled amorphization stabilize transport-active configurations. Mode-resolved spectroscopy, neutron methods, solid-state nuclear magnetic resonance, total scattering, ab initio molecular dynamics, enhanced sampling, and machine-learned interatomic potentials are critically evaluated for resolving coupled sublattice dynamics across time and length scales. Quantitative structure–dynamics–transport relationships are proposed using conductivity, activation energy, correlation factors, rotational timescales, phonon characteristics, disorder metrics, and electrochemical stability. The review concludes with experimentally testable design rules and standardized reporting priorities for discovering room-temperature solid electrolytes that combine rapid ion conduction, metastable retention, interfacial compatibility, and device-level durability.

Structural origin of disorder-induced ion conduction in NaFePO4 cathode materials

Rasmus Christensen; Kristin A. Persson; Morten M. Smedskjær. ChemRxiv (2025). Cited by 0. Found by: openalex. openalex_2yr: 0.467 (2026). DOI: 10.26434/chemrxiv-2025-9sf4n

Abstract: Most modern battery technologies depend on solid-state crystalline cathode materials. However, some of these materials are constrained by the low ionic conductivity of their most stable phases. An example of this is maricite (NaFePO4). Interestingly, experiments have shown that maricite can improve its rate capability through disordering (amorphization). However, experimental characterization of amorphous cathode materials remains a major challenge, hindering a clear understanding of the structural origin of the disorder-induced improvement in sodium-ion mobility. Here, we employ molecular dynamics simulations by first training a machine learning potential for NaFePO4 based on the atomic cluster expansion approach and a batch active learning potential parameterization scheme. This potential is then applied to explore the structural and dynamical properties of NaFePO4 glasses as cathode materials. Specifically, we investigate the effect of glass structure on sodium-ion diffusion, revealing the relative influences of short-range and medium-range order features. We find significant heterogeneity in sodium-ion diffusivity in the glass, with fast-conducting ions residing in less constrained atomic environments with fewer P and Fe neighbors. These more mobile ions are also surrounded by larger ring-type structures. Overall, the results and developed approach present promising avenues for developing high-performance glassy cathodes for next-generation batteries.

MOLECULAR SIMULATION METHODS FOR MODELING

Dylan David Fortney. Purdue (2026). Cited by 0. Found by: openalex. DOI: 10.25394/pgs.32048331.v1

Abstract: Molecular dynamics is a computational tool that is critical to modern methods of understanding materials across a wide range of applications. Molecular dynamics simulations calculate the motions of atoms through small time steps using Newton's laws of motion and potentials that approximate the forces. This allows for the prediction of a variety of physical properties with a fundamentally simple method. This method is used for the two main projects discussed here: the development of a genetic algorithm that parametrizes coarse-grain models and a framework to computationally probe the underwater adhesion of a range of molecules.Deep neural networks have become popular model architectures for fitting coarse-grained molecular dynamics potentials (CGMD) owing to their ability to describe complex features and ease of training against large datasets. However, such architectures are much more complicated than traditional functional forms. This raises the question of whether a similar data-driven approach that used a simpler functional form would have any advantages. In in the first part of this defense, we develop a genetic algorithm that optimizes Lennard-Jones potentials for coarse-grained models of various crystal and liquid crystal forming materials based on both structural and thermodynamic information. A detailed description of the genetic algorithm, its loss function, and hyperparameters is presented. The models developed by the algorithm reproduce a much broader range of physical properties than achieved by simpler parametrization schemes used in more general methods, such as Martini. The models also show surprising transferability in reproducing properties that were not directly trained against. Simulations of larger systems with these models demonstrate stabilization of the reference crystal structure on long time scales, preserve melting-point trends, and reproduce liquid crystalline phase transitions, despite this information being absent from training. The genetic algorithm is then expanded to parametrize amorphous systems. The algorithm is used to parametrize a set of oligomers, originally modeled in all-atom simulations with developed machine learned interatomic potentials. The developed models show excellent performance in reproducing the radial distribution function, pressure, and interaction energy of these oligomers, showing that this parametrization methodology can be extended to a wide variety of systems. Despite the rush to adopt neural network potentials, these case-studies show that simpler functional forms retain untapped potential for CGMD when coupled with data-driven training algorithms.Adhesives are critical materials across numerous applications, yet most commercial adhesives fail in aqueous environments. Catechol-based adhesives, inspired by marine mussels and other aquatic organisms, are distinctive in providing both strong surface adhesion and bulk cohesion underwater; however, the molecular mechanism enabling this performance remains poorly understood, limiting the rational design of improved materials. In the second part of this defense, steered molecular dynamics (SMD) simulations and density functional theory (DFT) calculations were used to systematically screen several hypothesis-driven series of candidate molecules for underwater adhesive performance on an alumina surface. Bidentate molecules consistently outperform unidentate molecules, and increasing hydroxyl substitution monotonically improves adhesion up to pentahydroxybenzene. Binding group geometry is also critical, with ortho-positioned groups outperforming meta and para isomers. Binding motif analysis revealed two mechanisms: direct surface binding and binding to a surface-bound water layer, with the highest-performing molecules engaging in both simultaneously. Oxidation energy–a surrogate for crosslinking capacity–shows no correlation with adhesive performance, suggesting that adhesive and cohesive properties can be independently optimized. Alternative core groups were also studied, finding that 1,8-dihydroxynaphthalene can outperform the canonical benzene ring found in catechol. This framework was then extended to polymeric systems, where the adhesive small molecules are attached to a selection of experimentally known backbones and linkers. With the catechol binding group it was found that the best performing linkers tended to be less bulky or more hydrophobic, indicating that competing with water for physical space is a major limitation in underwater adhesion for the polymer case. When testing several of the small molecules with a consistent backbone and linker, it was found the attaching to polymers greatly exaggerated the existing trends in adhesive performance, and diminished some of the performance of higher substituted binding groups. These results establish structure-based design rules for underwater adhesives and identify promising candidates beyond catechol for future experimental validation.

MOLECULAR SIMULATION METHODS FOR MODELING

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Abstract: Molecular dynamics is a computational tool that is critical to modern methods of understanding materials across a wide range of applications. Molecular dynamics simulations calculate the motions of atoms through small time steps using Newton's laws of motion and potentials that approximate the forces. This allows for the prediction of a variety of physical properties with a fundamentally simple method. This method is used for the two main projects discussed here: the development of a genetic algorithm that parametrizes coarse-grain models and a framework to computationally probe the underwater adhesion of a range of molecules.Deep neural networks have become popular model architectures for fitting coarse-grained molecular dynamics potentials (CGMD) owing to their ability to describe complex features and ease of training against large datasets. However, such architectures are much more complicated than traditional functional forms. This raises the question of whether a similar data-driven approach that used a simpler functional form would have any advantages. In in the first part of this defense, we develop a genetic algorithm that optimizes Lennard-Jones potentials for coarse-grained models of various crystal and liquid crystal forming materials based on both structural and thermodynamic information. A detailed description of the genetic algorithm, its loss function, and hyperparameters is presented. The models developed by the algorithm reproduce a much broader range of physical properties than achieved by simpler parametrization schemes used in more general methods, such as Martini. The models also show surprising transferability in reproducing properties that were not directly trained against. Simulations of larger systems with these models demonstrate stabilization of the reference crystal structure on long time scales, preserve melting-point trends, and reproduce liquid crystalline phase transitions, despite this information being absent from training. The genetic algorithm is then expanded to parametrize amorphous systems. The algorithm is used to parametrize a set of oligomers, originally modeled in all-atom simulations with developed machine learned interatomic potentials. The developed models show excellent performance in reproducing the radial distribution function, pressure, and interaction energy of these oligomers, showing that this parametrization methodology can be extended to a wide variety of systems. Despite the rush to adopt neural network potentials, these case-studies show that simpler functional forms retain untapped potential for CGMD when coupled with data-driven training algorithms.Adhesives are critical materials across numerous applications, yet most commercial adhesives fail in aqueous environments. Catechol-based adhesives, inspired by marine mussels and other aquatic organisms, are distinctive in providing both strong surface adhesion and bulk cohesion underwater; however, the molecular mechanism enabling this performance remains poorly understood, limiting the rational design of improved materials. In the second part of this defense, steered molecular dy-

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Nuclear Quantum Effects in the Acetylene:Ammonia Plastic Co-crystal

Atul C. Thakur; Richard C. Remsing. arXiv (Cornell University) (2023). Cited by 0. Found by: openalex. DOI: 10.48550/arxiv.2310.00480

Abstract: Organic molecular solids can exhibit rich phase diagrams. In addition to structurally unique phases, translational and rotational degrees of freedom can melt at different state points, giving rise to partially disordered solid phases. The structural and dynamic disorder in these materials can have a significant impact on the physical properties of the organic solid, necessitating a thorough understanding of disorder at the atomic scale. When these disordered phases form at low temperatures, especially in crystals with light nuclei, the prediction of materials properties can be complicated by the importance of nuclear quantum effects. As an example, we investigate nuclear quantum effects on the structure and dynamics of the orientationally-disordered, translationally-ordered plastic phase of the acetylene:ammonia (1:1) co-crystal that is expected to exist on the surface of Saturn's moon Titan. Titan's low surface temperature (~90 K) suggests that the quantum mechanical behavior of nuclei may be important in this and other molecular solids in these environments. By using neural network potentials combined with ring polymer molecular dynamics simulations, we show that nuclear quantum effects increase orientational disorder and rotational dynamics within the acetylene:ammonia (1:1) co-crystal by weakening hydrogen bonds. Our results suggest that nuclear quantum effects are important to accurately model molecular solids and their physical properties in low temperature environments.

Beyond Boltzmann transport: Green–Kubo prediction of lattice thermal conductivity with machine-learned potentials

Xiaoying Zhang; Ning Liu; Qian Guo; Guo-Yong Shi; Yue Wang. The Journal of Chemical Physics (2026). Cited by 0. Found by: openalex. DOI: 10.1063/5.0297462

Abstract: We investigate the microscopic origin of the ultralow lattice thermal conductivity in Rb2ZnTe by combining anharmonic lattice dynamics based on first-principles density functional theory with the Boltzmann transport equation (BTE). Full cubic and quartic anharmonicity is included, accounting for phonon renormalization and both three-phonon and four-phonon scattering processes. In addition, due to the unusually strong coherence effects in Rb2ZnTe, we consider off-diagonal contributions beyond the conventional BTE framework. The calculated phonon transport approaches the Ioffe-Regel limit, indicating the breakdown of the quasiparticle picture. To obtain an accurate thermal conductivity, we employ a machine-learning potential and perform Green-Kubo molecular dynamics simulations using GPUMD. At 300 K, the computed thermal conductivity increases by 38% compared to previous estimates, yet remains close to the amorphous limit, highlighting Rb2ZnTe as a promising thermoelectric material. The methodology presented here provides a robust framework for understanding lattice thermal transport in strongly anharmonic crystals and guiding the rational design of low-thermal-conductivity materials.

Revisit Many-body Interaction Heat Current and Thermal Conductivity Calculation in Moment Tensor Potential/LAMMPS Interface

Tai, Siu Ting; Chen Wang; Ruihuan Cheng; Yue Chen. arXiv (Cornell University) (2024). Cited by 0. Found by: openalex. DOI: 10.48550/arxiv.2411.01255

Abstract: The definition of heat current operator for systems for non-pairwise additive interactions and its impact on related lattice thermal conductivity ($ _{L}$) via molecular dynamics simulation (MD) are ambiguous and controversial when migrating from conventional empirical potential models to machine learning potential (MLP) models. Empirical model descriptions are often limited to three- to four-body interaction while a sophisticated representation of the many-body physics could be resembled in MLPs. Herein, we study and compare the significance of many-body interaction to the heat current computation in one of the most popular MLP models, the Moment Tensor Potential (MTP). Non-equilibrium MD simulations and equilibrium MD simulations among four different materials, $PbTe$, amorphous $Sc_{0.2}Sb_{2}Te_{3}$, graphene, and BAs, were performed. We found inconsistency between the simulation thermostat and its implemented heat current operator in our non-equilibrium MD results which violate law of energy conservation and suggest a need for revision. We revisit the virial stress tensor expression within the calculator and identified the lack of a generalised many-body heat current description in it. We uncover the influence of the modified heat current formula that could alter the $ _{L}$ results 29% to 64% using the equilibrium MD computational approach. Our work demonstrates the importance of a many-body description during thermal analysis in MD simulations when MLPs are in concern. This work sheds light on a better understanding of the relationship between interatomic interaction and its heat transport mechanism.

Probing the Formation Mechanisms and Vibrational Spectra of Silicate Cloud Particles on WASP-17b using Molecular Simulation

Stephen Ingram; Yifei Chen; Hanna Vehkamäki. (2026). Cited by 0. Found by: openalex. DOI: 10.5194/egusphere-egu26-9272

Abstract: Grant et al (2023) recently detected an 8.6 micron absorption in the atmosphere of the exoplanet WASP-17b using the James Webb Space Telescope (JWST), an infrared feature proposed to be caused by the presence of ~10nm radius “Quartz Cloud” droplets. WASP-17b is a so-called “hot Jupiter” exoplanet: its upper atmosphere has a temperature of 1250 K and pressure of 10-3 bar. Such conditions are below the melting point of quartz but also within the region of the silica phase diagram where -tridymite should be the most stable phase. In this work, the vibrational densities of states of silica nanoparticles (quartz, -tridymite and amorphous silica) have been calculated using both traditional force field molecular dynamics (Heinz et al 2013) and extended tight binding (Bannwarth et al 2019) methods, showing generally good agreement with the JWST data. The degree of hydrogenation of dangling Si-O bonds is found to affect the absorption wavelength (and therefore the overlap with the planetary spectrum) more significantly than the phase of the nanoparticles, confirming solid Silica is likely present in the WASP-17b atmosphere, but is not necessarily quartz.Under the assumption that such ”droplets” form through a nucleation and growth mechanism akin to that of terrestrial aerosol, atomistic simulations were conducted using the new generation neural network potential MACE-MH-1 (Batatia et al 2025). Both molecular SiO2 nucleation and the proposed oxidation of silicon monoxide by water were studied. We report a novel exoplanetary aerosol formation mechanism, involving clusters that initially form as polymeric chains with tetrahedral arrangements of Silica units, before transitioning to larger interconnected rings as they grow.ReferencesDavid Grant et al, The Astrophysical Journal Letters, 2023, 956, L29Hendrik Heinz et al, Langmuir, 2013, 29(6), 1754Christoph Bannwarth et al , J. Chem. Theory Comput., 2019, 15(3), 1652Ilyes Batatia et al, arXiv, 2025, 2510.2538

Competitive Hydrogen-Bond Partitioning in Deep Eutectic Solvents: From Cooperative Charge Spreading to Structure–Property Design Rules

Sergio de-la-Huerta-Sainz; Valentín Diez-Cabanes; Alberto Gutiérrez; Sara Santamaria; María A. Escobedo; Pedro A. Marcos; Alfredo Bol; Jose L. Trenzado; Mert Atilhan; Santiago Aparício. ACS Omega (2026). Cited by 0. Found by: openalex. openalex_2yr: 4.692 (2026). DOI: 10.1021/acsomega.6c02376

Abstract: High Resolution Image Download MS PowerPoint Slide Deep eutectic solvents (DESs) owe their remarkable melting-point depression, high viscosity, and tunable solvation to a hydrogen-bond network far richer than the binary donor–acceptor picture suggests. This study advances the framework of competitive hydrogen-bond partitioning: ionic $Cl^- \cdots H–X$ interactions, neutral donor–donor self-association, cation-mediated contacts, and water-competitive motifs coexist and continuously redistribute as a function of composition, temperature, and interfacial confinement. Evidence is synthesized from vibrational spectroscopy, multinuclear

NMR, neutron and X-ray scattering, dielectric relaxation, classical and ab initio molecular dynamics, DFT cluster calculations, and machine-learning potentials, establishing that no single technique can fully characterize the network a triangulation criterion requiring at least two independent method categories is essential. A quantitative structure–property framework is developed linking six hydrogen-bond descriptors motif population, persistence distribution, network connectivity, competitive hydration index, dynamic heterogeneity, and interfacial partitioning to viscosity, conductivity, diffusion, and glass transition across Type III, Type V, Natural DES (NADES), and hydrophobic DES. A central finding is the cooperativity–mobility tradeoff: cooperative charge spreading at Cl – simultaneously drives eutectic depression and network rigidity, defining a design axis along which DES can be rationally positioned. Water is analyzed as both competitive and cooperative partner across four hydration regimes, and interfacial hydrogen-bond reorganization at electrodes largely neglected in prior studies is critically examined. An integrated characterization workflow with standardized reporting criteria, validated force-field benchmarks, and data-driven descriptors for predictive screening is proposed.

Thermal Transport Mechanisms of Phonons, Diffusons, Electrons, and Ions in Functional Materials

Yatian Zhang. Media (https://www.suub.uni-bremen.de/) (2026). Cited by 0. Found by: openalex. DOI: 10.26092/elib/5520

Abstract: The design of advanced functional materials relies on a fundamental understanding of heat transport processes, which are governed by multiple thermal carriers including phonons, diffusons, electrons, and mobile ions. Each of these carriers interacts with the underlying atomic structure through mechanisms closely related to lattice dynamics and anharmonicity. In this thesis, I systematically investigate thermal transport behavior in solid-state materials—ranging from two-dimensional crystals to layered and bulk compounds—with a focus on identifying the distinct contributions from phononic, diffusonic, electronic, and ionic channels. Employing a theoretical framework which incorporates first-principles anharmonic lattice dynamics into a unified heat transport theory, I explore how intrinsic structural features and anharmonic interactions influence thermal conductivity. The findings provide insights into carrier scattering processes, transport anisotropy, and temperature-dependent conductivity trends. Firstly, I focus on the impact of phonon transport on thermal properties by employing the two-channel model, which distinguishes between propagating phonon modes and diffusive (locally confined or non-propagating) vibrational modes. In Chapter 3, I examine the potential utilization of the diverse crystal structure found in Cu2Te for on/off thermal conductivity switching. In Chapter 4, I find that the nitride perovskite LaWN3 displays strong anharmonic lattice dynamics manifested into a low lattice thermal conductivity and a non-standard temperature dependence. In Chapter 5, I investigate the thermal properties of Ag8SnSe6. Despite being a crystal, Ag8SnSe6 exhibits an exceptionally-low and nearly temperature independent lattice thermal conductivity. Secondly, another key heat carrier, electrons, is introduced and investigated. In Chapters 6, I focus on the effects of high order phonon scattering and electron-phonon scattering on thermal properties in two-dimensional 2H-TaS2. In Chapters 7, the early concept of a "phonon-glass-electron-crystal" for enhancing the thermoelectric figure of merit (ZT) is explored theoretically in layered Ge-Se crystals, where phonon transport exhibits wave-like behavior. Finally, I employ a hybrid approach combining the Green-Kubo formalism and molecular dynamics based on machine-learning potential to predict the thermal conductivity of Li3OCl. I systematically analyze the contributions of atomic vibrations, ion transport, and vibrationsions couplings to its thermal conductivity. It is found that as temperatures increase, Li3OCl transitions into a quasi-liquid state, resulting in a substantial ion migration that contributes non-negligibly to thermal conductivity. These findings are expected to enhance the fundamental understanding of thermal properties in solids and promote the practical applications of functional materials.

Accelerating Amorphous Alloy Discovery: Data-Driven Property Prediction via General-Purpose Machine Learning Interatomic Potential

Xuhe Gong; Hengbo Zhao; Xiao Fu; Jingchen Lian; Qifan Yang; Ran Li; Ruijuan Xiao; Tao Zhang; Hong Li. arXiv preprint (2025). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/2508.11989v1

Abstract: While traditional trial-and-error methods for designing amorphous alloys are costly and inefficient, machine learning approaches based solely on composition lack critical atomic structural information. Machine learning interatomic potentials, trained on data from first-principles calculations, offer a powerful alternative by efficiently approximating the complex three-dimensional potential energy surface with near-DFT accuracy. In this work, we develop a general-purpose machine learning interatomic potential for amorphous alloys by using a dataset comprising 20400 configurations across representative binary and ternary amorphous alloys systems. The model demonstrates excellent predictive performance on an independent test set, with a mean absolute error of 5.06 meV/atom for energy and 128.51 meV/Å for forces. Through extensive validation, the model is shown to reliably capture the trends in macroscopic property variations such as density, Young's modulus and glass transition temperature across both the original training systems and the compositionally modified systems derived from them. It can be directly applied to composition-property mapping of amorphous alloys. Furthermore, the developed interatomic potential enables access to the atomic structures of amorphous alloys, allowing for microscopic analysis and interpretation of experimental results, particularly those deviating from empirical trends.This work breaks the long-standing computational bottleneck in amorphous alloys research by developing the first general-purpose machine learning interatomic potential for amorphous alloy systems. The resulting framework provides a robust foundation for data-driven design and high-throughput composition screening in a field previously constrained by traditional simulation limitations.

Unveiling defect motifs in amorphous GeSe using machine learning interatomic potentials

Minseok Moon; Seungwoo Hwang; Jaesun Kim; Yutack Park; Changho Hong; Seungwu Han. arXiv preprint (2025). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/2506.15934v1

Abstract: Ovonic threshold switching (OTS) selectors play a critical role in non-volatile memory devices because of their nonlinear electrical behavior and polarity-dependent threshold voltages. However, the atomic-scale origins of the defect states responsible for these properties are not yet fully understood. In this study, we use molecular dynamics simulations accelerated by machine-learning interatomic potentials to investigate defects in amorphous GeSe. We begin by benchmarking several potential architectures-including descriptor-based models and graph neural network (GNN) models-and show that faithfully representing amorphous GeSe requires capturing higher-order interactions (at least four-body correlations) and medium-range structural order. We find that GNN architectures with multiple interaction layers successfully capture these correlations and structural motifs, preventing the spurious defects that less expressive models introduce. With our optimized GNN potential, we examine twenty independent 960-atom amorphous GeSe structures and identify two distinct defect motifs: aligned Ge chains, which give rise to defect states near the conduction band, and overcoordinated Ge chains, which produce defect states near the valence band. We further correlate these electronic defect levels with specific structural features-namely, the average alignment of bond angles in the aligned chains and the degree of local Peierls distortion around overcoordinated Ge atoms. These findings provide a theoretical framework for interpreting experimental observations and deepen our understanding of defect-driven OTS phenomena in amorphous GeSe.

Li-P-S Electrolyte Materials as a Benchmark for Machine-Learned Interatomic Potentials

Natascia L. Fragapane; Volker L. Deringer. arXiv preprint (2025). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/2511.16569v1

Abstract: With the growing availability of machine-learned interatomic potential (MLIP) models for materials simulations, there is an increasing demand for robust, automated, and chemically insightful benchmarking methodologies. In response, we here introduce LiPS-25, a curated benchmark dataset for a canonical series of solid-state electrolyte materials from the Li2S-P2S5 pseudo-binary compositional line, including crystalline and amorphous configurations. Together with the dataset, we present a suite of performance tests that range from conventional numerical error metrics to physically motivated evaluation tasks. With a focus on graph-based MLIP architectures, we run numerical experiments that assess (i) the effect of hyperparameters and (ii) the fine-tuning behavior of selected pre-trained ("foundational") MLIP models. Beyond the Li-P-S solid-state electrolytes, we expect that such benchmarks and their code implementations can be readily adapted to other material systems.

Persistent homology-based descriptor for machine-learning potential of amorphous structures

Emi Minamitani; Ippei Obayashi; Koji Shimizu; Satoshi Watanabe. arXiv preprint (2022). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/2206.13727v3

Abstract: High-accuracy prediction of the physical properties of amorphous materials is challenging in condensed-matter physics. A promising method to achieve this is machine-learning potentials, which is an alternative to computationally demanding ab initio calculations. When applying machine-learning potentials, the construction of descriptors to represent atomic configurations is crucial. These descriptors should be invariant to symmetry operations. Handcrafted representations using a smooth overlap of atomic positions and graph neural networks (GNN) are examples of methods used for constructing symmetry-invariant descriptors. In this study, we propose a novel descriptor based on a persistence diagram (PD), a two-dimensional representation of persistent homology (PH). First, we demonstrated that the normalized two-dimensional histogram obtained from PD could predict the average energy per atom of amorphous carbon (aC) at various densities, even when using a simple model. Second, an analysis of the dimensional reduction results of the descriptor spaces revealed that PH can be used to construct descriptors with characteristics similar to those of a latent space in a GNN. These results indicate that PH is a promising method for constructing descriptors suitable for machine-learning potentials without hyperparameter tuning and deep-learning techniques.

Unveiling the crystallization kinetics in Ge-rich Ge$_x$Te alloys by large scale simulations with a machine-learned interatomic potential

Dario Baratella; Omar Abou El Kheir; Marco Bernasconi. arXiv preprint (2024). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/2404.15128v1

Abstract: A machine-learned interatomic potential for Ge-rich Ge$_x$Te alloys has been developed aiming at uncovering the kinetics of phase separation and crystallization in these materials. The results are of interest for the operation of embedded phase change memories which exploits Ge-enrichment of GeSbTe alloys to raise the crystallization temperature. The potential is generated by fitting a large database of energies and forces computed within Density Functional Theory with the neural network scheme implemented in the DeePMD-kit package. The potential is highly accurate and suitable to describe the structural and dynamical properties of the liquid, amorphous and crystalline phases of the wide range of compositions from pure Ge and stoichiometric GeTe to the Ge-rich Ge$_2$Te alloy. Large scale molecular dynamics simulations revealed a crystallization mechanism which depends on temperature. At 600 K, segregation of most of Ge in excess occurs on the ns time scale followed by crystallization of nearly stoichiometric GeTe regions. At 500 K, nucleation of crystalline GeTe occurs before phase separation, followed by a slow crystal growth due to the concurrent expulsion of Ge in excess.

Melt-Quench Failures and Practical Solutions for Universal Machine-Learning Interatomic Potentials in Amorphous Structure Generation

Shuwei Li; Yuqi An; Xingyu Guo; Wenqiang Yang; Zhenbin Wang. arXiv preprint (2026). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/2606.16385v2

Abstract: Generating experimentally relevant amorphous structures via melt-quench molecular dynamics is prohibitively expensive at the first-principles level. Universal machine-learning interatomic potentials (uMLIPs) could accelerate such simulations, but their reliability under non-equilibrium conditions remains unclear. Here, we examine eight leading uMLIPs for generating amorphous IrO2, using this electrocatalytically relevant oxide as a diagnostic case. Under the conventional melt-quench protocol, all models yield unphysically expanded structures with densities of 1-4 g/cm3, far below the ab initio molecular dynamics (AIMD) reference value of 10.04 g/cm3. Comparisons against ab initio references show that accurate energies and forces alone do not ensure stable NPT dynamics; correct energy-volume responses and pressure predictions are also essential. We identify two practical remedies: pressure-targeted fine-tuning and a revised NVT-quench/NPT-equilibration protocol that avoids unphysical volume expansion without additional ab initio training data. Both recover IrO2 densities and local structures consistent with AIMD. Across 30 chemically diverse materials, the volume-expansion failure proves general, and the revised protocol substantially improves density predictions, reducing the AIMD-referenced MAE from 2.46 to 0.35 g/cm3. This work establishes practical validation criteria and simulation strategies for robust uMLIP-driven amorphous structure generation.

Comparison of intermediate-range order in GeO$_2$ glass: molecular dynamics using machine-learning interatomic potential vs.\ reverse Monte Carlo fitting to experimental data

Kenta Matsutani; Shusuke Kasamatsu; Takeshi Usuki. arXiv preprint (2024). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/2409.06982v2

Abstract: The short and intermediate-range order in GeO$_2$ glass are investigated by molecular dynamics using machine-learning interatomic potential trained on ab initio calculation data and compared with reverse Monte Carlo fitting of neutron diffraction data. To characterize the structural differences in each model, the total/partial structure factors, coordination number, ring size and shape distributions, and persistent homology analysis were performed. These results show that although the two approaches yield similar two-body correlations, they can lead to three-dimensional models with very different short and intermediate-range ordering. A clear difference was observed especially in the ring distributions; RMC models exhibit a broad distribution in the ring size distribution, while neural network potential molecular dynamics yield much narrower ring distributions. This confirms that the density functional approximation in the ab initio calculations determines the preferred network assembly more strictly than RMC with simple coordination constraints and neutron diffraction data with isotope substitution.

Machine-learned potential for amorphous Indium-Tin-Oxide alloys

Shuaiyang Guo; Yuan Wang; Wei Zhang. arXiv preprint (2026). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/2601.00145v1

Abstract: Machine-learned potential-driven molecular dynamics (MLMD) simulations are of great value in guiding the design and optimization of memory devices. Amorphous indium-tin-oxide (ITO) is widely used as transparent conducting oxide for flat-panel display and solar cell applications, and also as a capping layer in phase-change-materials-based reconfigurable color display devices. However, atomistic simulations of ITO using ab initio molecular dynamics (AIMD) are limited to systems of a few hundred atoms due to expensive computational costs, which prevents the device-scale modelling of real-world applications. In this work, we develop a machine-learned potential for ITO and its parent phase In2O3 based on the Gaussian approximation potential (GAP) framework. We generate a comprehensive training dataset using an iterative training protocol. Our MLMD simulations of crystalline, liquid and melt-quenched amorphous ITO models are in great agreement with the AIMD reference. In particular, the ML potential well captures the minority atomic interaction, such as Sn-Sn bonds, which have poor statistics in small-scale AIMD simulations. We demonstrate that the MLMD simulations are 3-4 orders of magnitude faster than AIMD. The training dataset and the fitted GAP potentials for ITO and In2O3 are openly accessible.

Modeling Short-Range and Three-Membered Ring Structures in Lithium Borosilicate Glasses using Machine Learning Potential

Shingo Urata. arXiv preprint (2022). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/2209.00316v2

Abstract: Lithium borosilicate (LBS) glass is a prototypical lithium-ion conducting oxide glasses available for an all-solid state buttery. Nevertheless, the atomistic modeling of LBS glass using ab $initio$ (AIMD) and classical molecular dynamics (CMD) simulations have critical limitations due to computational cost and inaccuracy in reproducing the glass microstructures, respectively. To overcome these difficulties, a machine-learning potential (MLP) was examined in this work for modeling LBS glasses using DeepMD. The glass structures obtained by this MLP possessed fourhold-coordinated boron (^{4}B) confirmed well with the experimental data and abundance of three-membered rings. The models were energetically more stable compared with those constructed with a functional force-field even though both the models included reasonable ^{4}B. The results confirmed MLP to be superior to model the boron-containing glasses and address the inherent shortcomings of the AIMD and CMD. This study also discusses some limitations of MLP for modeling glasses.

Large-scale atomistic study of plasticity in amorphous gallium oxide with a machine-learning potential

Jiahui Zhang; Junlei Zhao; Jesper Byggmästar; Erkka J. Frankberg; Antti Kuronen. arXiv preprint (2024). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/2404.17353v1

Abstract: Compared to the widely investigated crystalline polymorphs of gallium oxide (Ga2O3), knowledge about its amorphous state is still limited. With the help of a machine-learning interatomic potential, we conducted large-scale atomistic simulations to investigate the glass transition and mechanical behavior of amorphous Ga2O3 (a-Ga2O3). During the quenching simulations, amorphization of gallium oxide melt is observed at ultrahigh cooling rates, including a distinct glass transition. The final densities at room temperature have up to 4% variance compared to experiments. The glass transition temperature is evaluated to range from 1234 K to 1348 K at different cooling rates. Structural analysis of the amorphous structure shows evident similarities in structural properties between a-Ga2O3 and amorphous alumina (a-Al2O3), such as radial distribution function, coordination distribution, and bond angle distribution. An amorphous gallium oxide structure that contains approximately one million atoms is prepared for the tension simulation. A highly plastic behavior is observed at room temperature in the tension simulations, comparable to amorphous alumina. With quantitative characterization methods, we show that a-Ga2O3 can possibly has a higher nucleation rate of localized plastic strain events compared to a-Al2O3, which can increase the material's resistance to shear banding formation during deformation.

Validation Workflow for Machine Learning Interatomic Potentials for Complex Ceramics

Kimia Ghaffari; Salil Bavdekar; Douglas E. Spearot; Ghatu Subhash. arXiv preprint (2024). Cited by 0. Found by: arxiv. DOI: 10.1016/j.commatsci.2024.112983

Abstract: The number of published Machine Learning Interatomic Potentials (MLIPs) has increased significantly in recent years. These new data-driven potential energy approximations often lack the physics-based foundations that inform many traditionally-developed interatomic potentials and hence require robust validation methods for their applicability, accuracy, computational efficiency, and transferability to the intended applications. This work presents a sequential, three-stage workflow for MLIP validation: (i) preliminary validation, (ii) static property prediction, and (iii) dynamic property prediction. This material-agnostic procedure is demonstrated in a tutorial approach for the development of a robust MLIP for boron carbide (B4C), a widely employed, structurally complex ceramic that undergoes a deleterious deformation mechanism called "amorphization" under high-pressure loading. It is shown that the resulting B4C MLIP offers a more accurate prediction of properties compared to the available empirical potential.

Atomistic Insights into Cu/amorphous-Ta$_x$N Interfacial Adhesion via Machine Learning Interatomic Potentials: Effects of Stoichiometry and Interface Construction

Jeong Min Choi; Jaehoon Kim; Ji-Hwan Lee; Won-Joon Son; Seungwu Han. arXiv preprint (2025). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/2509.20662v2

Abstract: Accurate understanding and control of interfacial adhesion between Cu and Ta$_x$N diffusion barriers are essential for ensuring the mechanical reliability and integrity of Cu interconnect systems in semiconductor devices. Amorphous tantalum nitride (a-Ta$_x$N) barriers are particularly attractive due to their superior barrier performance, attributed to the absence of grain boundaries. However, a systematic atomistic investigation of how varying Ta stoichiometries influences adhesion strength at Cu/a-Ta$_x$N interfaces remains lacking, hindering a comprehensive understanding of interface optimization strategies. In this study, we employ machine learning interatomic potentials (MLIPs) to perform steered molecular dynamics (SMD) simulations of Cu/a-Ta$_x$N interfaces. We simultaneously evaluate three distinct interface construction approaches–static relaxation, high-temperature annealing, and simulated Cu deposition–to comprehensively investigate their influence on adhesion strength across varying Ta compositions ($x=1, 2, 4$). Peak force and work of adhesion values from SMD simulations quantitatively characterize interface strength, while atomic stress and strain analyses elucidate detailed deformation behavior, highlighting the critical role of interfacial morphologies. Additionally, we explore the atomistic mechanisms underlying cohesive failure, revealing how targeted incorporation of Ta atoms into Cu layers enhances the cohesive strength of the interface. This study demonstrates how MLIP-driven simulations can elucidate atomic-scale relationships between interface morphology and adhesion behavior, providing insights that can guide future atomistic engineering strategies toward enhancing intrinsic barrier adhesion, potentially enabling liner-free interconnect technologies.

Machine Learning Potential Powered Insights into the Mechanical Stability of Amorphous Li-Si Alloys

Zixiong Wei; Nongnuch Artrith. arXiv preprint (2024). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/2402.08834v

Abstract: Understanding the mechanical properties of solid-state materials at the atomic scale is crucial for developing novel materials. For example, amorphous LiSi alloys are attractive anode materials for solid-state Li-ion batteries but face mechanical instabilities due to significant volume variations with changing Li content. A fundamental grasp of the mechanical behavior in such systems is essential to address their poor mechanical integrity. Experimental methods offer insufficient information to elaborate on dynamic mechanical degradation mechanisms at the atomic scale, and computationally demanding first-principles methods, like DFT, struggle to access the system sizes needed for modeling mechanical phenomena. Machine learning potentials (MLPs) can overcome the computational constraints of traditional DFT-based simulations, enabling large-scale, accurate, and efficient simulations. Here, we provide a concise tutorial on developing and applying MLPs to investigate mechanical properties in materials systems, ranging from bulk to nanoparticles, using Li-Si alloys as an example. Trained on a comprehensive dataset (~45,000 DFT structures) with the aenet package accelerated by PyTorch, a robust MLP is constructed to reproduce results consistent with previous experimental observations. We demonstrate applying the MLP to realistic structures to visualize the deformation mechanism and determine the origin of mechanical instabilities caused by fracturing. This work aims to establish MLP-based simulations as a tool to understand the atomic-scale mechanical behavior in different materials systems.

Is the Future of Materials Amorphous? Challenges and Opportunities in Simulations of Amorphous Materials

Ata Madanchi; Emna Azek; Karim Zongo; Laurent K. Béland; Normand Mousseau; Lena Simine. arXiv preprint (2024). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/2410.05035v2

Abstract: Amorphous solids form an enormous and underutilized class of materials. In order to drive the discovery of new useful amorphous materials further we need to achieve a closer convergence between computational and experimental methods. In this review, we highlight some of the important gaps between computational simulations and experiments, discuss popular state-of-the-art computational techniques such as the Activation Relaxation Technique nouveau (ARTn) and Reverse Monte Carlo (RMC), and introduce more recent advances: machine learning interatomic potentials (MLIPs) and generative machine learning for simulations of amorphous matter, e.g., the Morphological Autoregressive Protocol (MAP). Examples are drawn from the amorphous silicon and silica literature as well as from molecular glasses. Our outlook stresses the need for new computational methods to extend the time- and length- scales accessible through numerical simulations.

Machine-learning-driven modelling of amorphous and polycrystalline BaZrS$_{3}$

Laura-Bianca Paşca; Yuanbin Liu; Andy S. Anker; Ludmilla Steier; Volker L. Deringer. arXiv preprint (2025). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/2506.01517v1

Abstract: The chalcogenide perovskite material BaZrS$_{3}$ is of growing interest for emerging thin-film photovoltaics. Here we show how machine-learning-driven modelling can be used to describe the material's amorphous precursor as well as polycrystalline structures with complex grain boundaries. Using a bespoke machine-learned interatomic potential (MLIP) model for BaZrS$_{3}$, we study the atomic-scale structure of the amorphous phase, quantify grain-boundary formation energies, and create realistic-scale polycrystalline structural models which can be compared to experimental data. Beyond BaZrS$_{3}$, our work exemplifies the increasingly central role of MLIPs in materials chemistry and marks a step towards realistic device-scale simulations of materials that are gaining momentum in the fields of photovoltaics and photocatalysis.

Resolving Structural Avalanches in Amorphous Carbon with Arclength Continuation

Fraser Birks; Ibrahim Ghanem; Lars Pastewka; James Kermode; Maciej Buze. arXiv preprint (2026). Cited by 0. Found by: arxiv. DOI: 10.1103/6n5m-rxc1

Abstract: Plastic deformation in amorphous solids is carried by localized shear transformations that self-organize into avalanches. In amorphous carbon modeled with a machine-learned interatomic potential, we find that the energetics and organization of these avalanches can be resolved by systematically following the underlying energy landscape. With a pseudo-arclength numerical continuation framework, we decompose avalanches into constituent shear transformations and determine their strain-dependent energetics. Our analysis shows that, prior to onset, avalanches have a latent structure that consists of well-separated local minima. We further demonstrate that arclength continuation yields an event driven framework for following avalanche dynamics, eliminating time-step effects on statistical avalanche properties such as distributions of stress drops.

On phase separation and crystallization of Ge-rich GeSbTe alloys from atomistic simulations with a machine learning interatomic potential

Omar Abou El Kheir; Dario Baratella; Marco Bernasconi. arXiv preprint (2026). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/2604.13843v1

Abstract: We developed a machine learning interatomic potential (MLIP) for Ge-rich GeSbTe alloys of interest for applications in phase change memories embedded in microcontrollers. The MLIP was generated by fitting with a neural network method a large database of energies and forces computed within density functional theory of elemental, binary, stoichiometric and non-stoichiometric ternary alloys in the Ge-Sb-Te phase diagram. The MLIP is demonstrated to be highly transferable to large regions of the phase diagram around the compositions included in the dataset. The MLIP is then exploited to simulate the crystallization with phase separation of three Ge-rich alloys on the Ge-Sb$_2$Te$_3$ and Ge- Ge$_2$Sb$_2$Te$_5$ tie-lines that correspond to the set process of the memory cell. The transformation on the ns time scale and at 600 K, comparable to the operation conditions of the memory, yields crystalline cubic GeTe slightly Sb-doped and amorphous GeSb and Ge. These metastable phases differ from the thermodynamically stable products and form due to kinetics effects on the short time span of the set operation in phase change memories.

Thermal Conductivity and Temperature-Induced Band Gap Renormalization in Crystalline and Amorphous Ga$_2$O$_3$

Rustam Arabov; Jiaxuan Li; Xiaotong Chen; Nikita Rybin; Alexander Shapeev. arXiv preprint (2026). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/2603.29484v2

Abstract: The lattice thermal conductivity (LTC) and electron-phonon interactions in crystalline and amorphous gallium oxide are herein determined by coupling a machine-learned interatomic potential, namely the moment tensor potential (MTP) model, to first-principles calculations. Crystalline $ $-Ga$_2O_3$ exhibits a substantial band gap renormalization (BGR) of $\sim$0.45 eV at 700 K, with $\sim$0.2 eV caused by zero-point BGR. The computed temperature dependence of BGR induced by classical nuclear motion in $ $-Ga$_2O_3$ is stronger than that in amorphous Ga$_2$O$_3$, with the difference in BGR reaching $\sim$0.18 eV at 900 K. Thermal transport calculations reveal that the LTC of amorphous Ga$_2$O$_3$ remains near 0.9 W$\cdot$ m^{-1}$\cdot$K^{-1} for temperatures between 300 K and 700 K, which is approximately an order of magnitude lower than that of crystalline $ $-Ga$_2O_3$. Overall, the presented framework provides a computationally tractable and reliable route for predicting properties of semiconductors (both crystalline and amorphous) under operating conditions relevant to microelectronics and optoelectronics.

From oligomers to entangled polymers: How to train a transferable machine learning interatomic potential

Mirko Fischer; Andreas Heuer. arXiv preprint (2026). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/2608.01162v1

Abstract: Over the past decade, Machine Learning Interatomic Potentials (MLIPs) have emerged as a powerful technique for performing molecular dynamics (MD) simulations with nearly ab initio accuracy. Alongside the development of new descriptors and advanced machine learning architectures, sophisticated procedures for the generation of diverse and accurate reference datasets have been established. To date, research has focused primarily on MLIPs for crystalline or amorphous inorganic and small molecular systems; however, large macromolecules such as polymers remain underrepresented in the literature, despite beeing an important class of materials. In this work, we investigate several aspects of developing MLIPs for polymers, utilizing polyethylene as a representative, yet simple model system. First, we compare various local atomic descriptors, identifying the Atomic Cluster Expansion (ACE) as the most effective for this application. Second, we implement and automatized active learning scheme to efficiently generate diverse training data and demonstrate that ACE potentials fitted on small oligomers are transferable to larger polymers. Given that the accurate reproduction of the density depends critically on a correct description of intermolecular interactions, which are far more complex to learn than intramolecular interactions, we carefully evaluate the performance of the ACE potentials with respect to non-bonded interactions. By utilizing the computationally efficient OPLS-AA force field as a ground truth reference, we are able to perform a direct comparison of nanosecond-scale MD trajectories resulting from the ACE and reference potential. We find that the ACE potential accurately reproduces key thermodynamic, structural and dynamical properties.

ATLAS: A Foundation Neural Sampler for Amorphous Materials

Mouyang Cheng; Denis Blessing; Botao Yu; Gerhard Neumann; Mingda Li; Carles Domingo-Enrich; Yuanqi Du. arXiv preprint (2026). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/2607.19198v1

Abstract: Amorphous materials exhibit exceptional mechanical and functional properties, yet their rugged energy landscapes are notoriously difficult to sample. Below the glass-transition temperature, conventional molecular dynamics and Monte Carlo become inefficient because equilibration relies on rare barrier-crossing events, while data-driven generative models are constrained by scarce and biased reference ensembles. Here, we introduce ATLAS, an efficient sampler that learns a diffusion process to generate Boltzmann-distributed amorphous structures directly from a target energy function. Parameterized by an equivariant graph neural network, ATLAS generalizes across system size, temperature, and composition. By exploiting the time reversal of the diffusion process, it enables efficient estimation of thermodynamic quantities and steering toward target observables. In two-dimensional Kob-Andersen systems, ATLAS reproduces parallel tempering Markov chain Monte Carlo structural distributions, free energies and entropies, achieving below 0.2% free energy error in the low-temperature glass regime with over 500-fold fewer energy evaluations. In Cu-Zr and Cr-Co-Ni metallic glasses, ATLAS recovers experimentally observed short-range-order trends and steers structures toward prescribed order parameters and optimized bulk moduli. Moreover, composition-amortized pretraining outperforms composition-specific training from scratch, reduces inverse-design costs by several hundred-fold, and enables sampling with expensive universal machine learning interatomic potentials. Coupled to a large language model agent, ATLAS searches an eight-element space for high-entropy metallic glasses balancing stiffness and ductility, identifying a converged Pareto frontier within 480 oracle evaluations. Together, these results establish ATLAS as a foundation model for sampling, steering and designing amorphous materials.

Polymorphic crystallites model for monolayer amorphous materials

Le-Ye Zhu; Xi Zhang; Yun-Peng Wang; Jieheng Shi; Junwei Zhang; Shixuan Du; Yu-Yang Zhang. arXiv preprint (2026). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/2605.01881v1

Abstract: Modeling the atomic structure of amorphous materials has long been a critical challenge in materials science. Recent advances in monolayer amorphous materials enable direct observation of their atomic structures, paving the way for a better understanding of their atomic-scale models. Here, we investigate amorphous multielement monolayers using machine learning potential from first-principles total energies via energy-driven kinetic Monte Carlo based active-learning framework. A polymorphic crystallite model is proposed to describe the atomic configuration of monolayer amorphous boron nitride, as it consists of coexisting crystallite of $o-B_2N_2$ and $o-B_4N_4$ structural motifs. Generality of the polymorphic crystallite model is further validated in two other multielement monolayer amorphous systems. Monolayer amorphous LiCl shows coexisting hexagonal and tetragonal crystallites, while monolayer amorphous BCN contains a combination of graphene-like, h-BN-like, and borophene-like crystallites. These findings expand the classical picture of amorphous structure models and offer new insight into the microscopic structure of amorphous materials.

A Unified Deep Neural Network Potential Capable of Predicting Thermal Conductivity of Silicon in Different Phases

Ruiyang Li; Eungkyu Lee; Tengfei Luo. arXiv preprint (2019). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/1912.05044v1

Abstract: Molecular dynamics simulations have been extensively used to predict thermal properties, but simulating different phases with similar precision using a unified force field is often difficult, due to the lack of accurate and transferrable interatomistic potential fields. As a result, this issue has become a major barrier to predicting the phase change of materials and their transport properties with atomistic-level modeling techniques. Recently, machine learning based algorithms have emerged as promising tools to develop accurate potentials for molecular dynamics simulations. In this work, we approach the problem of predicting the thermal conductivity of silicon in different phases by performing molecular dynamics simulations with a deep neural network potential. This neural network potential is trained with ab-initio data of silicon in the crystalline, liquid and amorphous phases. The accuracy of our potential is first validated through reproducing the atomistic structures during the phase transition, where other empirical potentials usually fail. The thermal conductivity of different phases is then calculated, showing a good agreement with the experimental results and ab-initio calculation results. Our work shows that a unified neural network-based potential can be a promising tool for studying phase change and thermal transport of materials with high accuracy.

Atomistic insights into hydrogen migration in IGZO from machine-learning interatomic potential: linking atomic diffusion to device performance

Hyunsung Cho; Minseok Moon; Jaehoon Kim; Eunkyung Koh; Hyeon-Deuk Kim; Rokyeon Kim; Gyehyun Park; Seungwu Han; Youngho Kang. arXiv preprint (2025). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/2508.19674v1

Abstract: Understanding hydrogen diffusion is critical for improving the reliability and performance of oxide thin-film transistors (TFTs), where hydrogen plays a key role in carrier modulation and bias instability. In this work, we investigate hydrogen diffusion in amorphous IGZO (a-IGZO) and c-axis aligned crystalline IGZO (CAAC-IGZO) using machine learning interatomic potential molecular dynamics (MLIP-MD) simulations. We construct accurate phase-specific MLIPs by fine-tuning SevenNet-0, a universal pretrained MLIP, and validate the models against a comprehensive dataset covering hydrogen-related configurations and diffusion environments. Hydrogen diffusivity is evaluated over 650–1700 K, revealing enhanced mobility above 750 K in a-IGZO due to the glassy matrix, while diffusion at lower temperatures is constrained by the rigid network. Arrhenius extrapolation of the diffusivity indicates that hydrogen in a-IGZO can reach the channel/insulator interface within 10^{4} seconds at 300–400 K, likely contributing to negative bias stress-induced device degradation. Trajectory analysis reveals that long-range diffusion in a-IGZO is enabled by a combination of hydrogen hopping and flipping mechanisms. In CAAC-IGZO, hydrogen exhibits high in-plane diffusivity but severely restricted out-of-plane transport due to a high energy barrier along the c-axis. This limited vertical diffusion in CAAC-IGZO suggests minimal impact on bias instability. This work bridges the atomic-level hydrogen transport mechanism and device-level performance in oxide TFTs by leveraging large-scale MLIP-MD simulations.

Thermal Conductivity Modeling using Machine Learning Potentials: Application to Crystalline and Amorphous Silicon

Xin Qian; Shenyou Peng; Xiaobo Li; Yujie Wei; Ronggui Yang. arXiv preprint (2019). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/1907.09088v1

Abstract: First-principles based modeling on phonon dynamics and transport using density functional theory and Boltzmann transport equation has proven powerful in predicting thermal conductivity of crystalline materials, but it remains unfeasible for modeling complex crystals and disordered solids due to the prohibitive computational cost to capture the disordered structure, especially when the quasiparticle "phonon" model breaks down. Recently, machine-learning regression algorithms show great promises for building high-accuracy potential fields for atomistic modeling with length and time scales far beyond those achievable by first-principles calculations. In this work, using both crystalline and amorphous silicon as examples, we develop machine learning based potential fields for predicting thermal conductivity. The machine learning based interatomic potential is derived from density functional theory calculations by stochastically sampling the potential energy surface in the configurational space. The thermal conductivities of both amorphous and crystalline silicon are then calculated using equilibrium molecular dynamics, which agree well with experimental measurements. This work documents the procedure for training the machine-learning based potentials for modeling thermal conductivity, and demonstrates that machine-learning based potential can be a promising tool for modeling thermal conductivity of both crystalline and amorphous materials with strong disorder.

Amorphous materials as a frontier challenge for universal interatomic potentials

Natascia L. Fragapane; Volker L. Deringer. arXiv preprint (2026). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/2607.11384v1

Abstract: Pre-trained or 'foundational' machine-learned interatomic potentials (MLIPs) are now widely used in materials modelling. However, early pre-trained models and benchmarks have largely focused on ordered, crystalline structures, and their transferability to non-crystalline solids remains unclear. Here, we show that the amorphous state is indeed a central challenge for future universal MLIPs, based on a systematic evaluation of current mainstream models in this domain. We introduce a benchmarking framework built on a curated reference dataset of canonical amorphous systems, as well as validation for structures and properties. Our study identifies limitations in the transferability of many current pre-trained models and investigates fine-tuning strategies tailored to disordered phases. Together, our results can facilitate future applications of MLIPs in the fast-growing field of amorphous functional materials, and they provide guidance for designing next-generation training datasets and transferable atomistic models.

Efficient training of machine learning potentials for metallic glasses: CuZrAl validation

Antoni Wadowski; Anshul D. S. Parmar; Filip Kaśkosz; Jesper Byggmästar; Jan S. Wróbel; Mikko J. Alava; Silvia Bonfanti. arXiv preprint (2024). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/2501.00589v2

Abstract: Interatomic potentials are key to uncovering microscopic structure-property relationships, essential for multiscale simulations and high-throughput experiments. For metallic glasses, their disordered atomic structure makes the development of potentials particularly challenging, resulting in the scarcity of chemistry-specific parametrizations for this important class of materials. We address this gap by introducing an efficient methodology to design machine learning interatomic potentials (MLIPs), benchmarked on the CuZrAl system. Using a Lennard-Jones surrogate model, swap-Monte Carlo sampling, and single-point Density Functional Theory (DFT) corrections, we capture amorphous structures spanning 14 decades of supercooling. These representative configurations, competing with the experimental time scale, enable robust model training across diverse states, while minimizing the need for extensive DFT datasets. The resulting MLIP matches the experimental data and predictions of the classical embedded atom method (EAM) for structural, dynamical, energetic, and mechanical properties. This approach offers a scalable path to develop accurate MLIPs for complex metallic glasses, including emerging multi-component and high-entropy systems.

Constitutive relations for plasticity of amorphous carbon

Richard Jana; Julian von Lautz; S. Mostafa Khosrownejad; W. Beck Andrews; Michael Moseler; Lars Pastewka. J. Phys. Mater. 3, 035005 (2020) (2020). Cited by 0. Found by: arxiv. DOI: 10.1088/2515-7639/ab953c

Abstract: We deform representative volume elements of amorphous carbon obtained from melt-quenches in molecular dynamics calculations using bond-order and machine learning interatomic potentials. A Drucker-

Prager law with a zero-pressure flow stress of 41.2~GPa and an internal friction coefficient of 0.39 describes the deviatoric stress during flow as a function of pressure. We identify the mean coordination number as the order parameter describing this flow surface. However, a description of the dynamical relaxation of the quenched samples towards steady-state flow requires an additional order parameter. We suggest an intrinsic strain of the samples as a possible order parameter and present equations for its evolution. Our results provide insights into rehybridization and pressure dependence of friction between coated surfaces as well as routes towards the description of amorphous carbon in macroscale models of deformation.

Crystal-like thermal transport in amorphous carbon

Jaeyun Moon; Zhiting Tian. arXiv preprint (2024). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/2405.07298v1

Abstract: Thermal transport properties of amorphous carbon has attracted increasing attention due to its extreme thermal properties: It has been reported to have among the highest thermal conductivity for bulk amorphous solids up to $\sim$ 37 Wm^{-1}K^{-1}, comparable to crystalline sapphire ($ $-Al\textsubscript{2}O\textsubscript{3}). Further, large density dependence in thermal conductivity demonstrates a potential for largely tunable thermal conductivity. However, mechanism behind the high thermal conductivity and its large density dependence remains elusive due to many variables at play. In this work, we perform large-scale ($\sim$ 10^{5} atoms) molecular dynamics simulations utilizing a machine learning potential based on neural networks. Through spectral decomposition of thermal conductivity which enables a quantum correction to classical heat capacity, we find that propagating vibrational excitations govern thermal transport in amorphous carbon ($\sim$ 100 \% of thermal conductivity) in sharp contrast to the conventional wisdom that diffusive vibrational excitations dominate thermal transport in amorphous solids. Instead, this remarkable behavior resembles thermal transport in simple crystals. Moreover, our temperature dependent spectral diffusivity and velocity current correlation analyses reveal that the density dependent thermal conductivity originates from anharmonicity sensitive propagating excitations. Our work suggests a novel insight and design principle into developing mechanically hard, thermally conductive amorphous solids.

Structural and elastic properties of amorphous carbon from simulated quenching at low rates

Richard Jana; Daniele Savio; Volker L. Deringer; Lars Pastewka. Modelling Simul. Mater. Sci. Eng. 27, 085009 (2019) (2019). Cited by 0. Found by: arxiv. DOI: 10.1088/1361-651X/ab45da

Abstract: We generate representative structural models of amorphous carbon (a-C) from constant-volume quenching from the liquid with subsequent relaxation of internal stresses in molecular dynamics simulations using empirical and machine-learning interatomic potentials. By varying volume and quench rate we generate structures with a range of density and amorphous morphologies. We find that all a-C samples show a universal relationship between hybridization, bulk modulus and density despite having distinct cohesive energies. Differences in cohesive energy are traced back to slight changes in the distribution of bond-angles that will likely affect thermal stability of these structures.

A Gaussian Approximation Potential for Amorphous Si:H

Davis Unruh; Reza Vatan Meidanshahi; Stephen M. Goodnick; Gábor Csányi; Gergely T. Zimányi. arXiv preprint (2021). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/2106.02946v3

Abstract: Hydrogenation of amorphous silicon (a-Si:H) is critical for reducing defect densities, passivating mid-gap states and surfaces, and improving photoconductivity in silicon-based electro-optical devices. Modelling the atomic scale structure of this material is critical to understanding these processes, which in turn is needed to describe c-Si/a-Si:H heterjunctions that are at the heart of the modern solar cells with world record efficiency. Density functional theory (DFT) studies achieve the required high accuracy but are limited to moderate system sizes a hundred atoms or so by their high computational cost. Simulations of amorphous materials in particular have been hindered by this high cost because large structural models are required to capture the medium range order that is characteristic of such materials. Empirical potential models are much faster, but their accuracy is not sufficient to correctly describe the frustrated local structure. Data driven, "machine learned" interatomic potentials have broken this impasse, and have been highly successful in describing a variety of amorphous materials in their elemental phase. Here we extend the Gaussian approximation potential (GAP) for silicon by incorporating the interaction with hydrogen, thereby significantly improving the degree of realism with which amorphous silicon can be modelled. We show that our Si:H GAP enables the simulation of hydrogenated silicon with an accuracy very close to DFT, but with computational expense and run times reduced by several orders of magnitude for large structures. We demonstrate the capabilities of the Si:H GAP by creating models of hydrogenated liquid and amorphous silicon, and showing that their energies, forces and stresses are in excellent agreement with DFT results, and their structure as captured by bond and angle distributions, with both DFT and experiments.

On the origin of in-gap states in amorphous Ge$_2$Sb$_2$Te$_5$

Omar Abou El Kheir; Marco Bernasconi. arXiv preprint (2026). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/2602.15446v1

Abstract: The localized states in the band gap of amorphous phase change alloys like Ge$_2$Sb$_2$Te$_5$ control the electrical conduction via the Poole-Frenkel mechanism. Understanding the origin of in-gap states and their evolution in time during aging of the glass is therefore important for the control of the resistance drift in phase change memory devices. Here, we use a machine learning interatomic potential to generate several models of Ge$_2$Sb$_2$Te$_5$ whose electronic structure is then analyzed within density functional theory with a hybrid functional. A detailed statistical analysis of the structural motifs on which the in-gap states are localized, reveals that the vast majority of in-gap states involve wrong bonds (homopolar or Ge-Sb bonds) often accompanied by Ge in tetrahedral configurations or overcoordinated Ge and Sb atoms. Metadynamics simulations mimicking glass aging support the picture that structural relaxations lead to the depletion of in-gap states and then to an increase of resistance. The simulations thus provide important insights for the mitigation of the resistance drift in phase change memory devices.

A Systematic Approach to Generating Accurate Neural Network Potentials: the Case of Carbon

Yusuf Shaidu; Emine Kucukbenli; Ruggero Lot; Franco Pellegrini; Efthimios Kaxiras; Stefano de Gironcoli. arXiv preprint (2020). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/2011.04604v1

Abstract: Availability of affordable and widely applicable interatomic potentials is the key needed to unlock the riches of modern materials modelling. Artificial neural network based approaches for generating potentials are promising; however neural network training requires large amounts of data, sampled adequately from an often unknown potential energy surface. Here we propose a self-consistent approach that is based on crystal structure prediction formalism and is guided by unsupervised data analysis, to construct an accurate, inexpensive and transferable artificial neural network potential. Using this approach, we construct an interatomic potential for Carbon and demonstrate its ability to reproduce first principles results on elastic and vibrational properties for diamond, graphite and graphene, as well as energy ordering and structural properties of a wide range of crystalline and amorphous phases.

Modulation of structural short-range order due to chemical patterning in multi-component amorphous interfacial complexions

Esther C. Hessong; Zhengyu Zhang; Tianjiao Lei; Mingjie Xu; Toshihiro Aoki; Timothy J. Rupert. arXiv preprint (2025). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/2509.06166v3

Abstract: Amorphous interfacial complexions have been shown to restrict grain growth and improve damage tolerance in nanocrystalline alloys, with increased chemical complexity stabilizing the complexions themselves. Here, we investigate local chemical composition and structural short-range order in Cu-rich, multi-component nanocrystalline alloys to understand how dopants self-organize within these amorphous complexions and how local structure is altered. High resolution scanning transmission electron microscopy and elemental analysis are used to study both grain boundaries and interphase boundaries, with chemical partitioning observed for both. Notably, the amorphous-crystalline transition region is observed to be enriched in certain dopant species and depleted of others as compared to the interior of the amorphous complexions. This chemical patterning can be explained in terms of the elemental preference for ordered or disordered grain boundary environments. As only a qualitative measure of structural short-range order can be obtained with nanobeam electron diffraction for these specimens, atomistic simulations with a custom-built machine learning interatomic potential are then used to probe how dopant patterning affects local structural state. Increased grain boundary chemical complexity is found to result in a more disordered complexion structure, with segregation to the amorphous-crystalline transition regions driving changes in local structure that are sensitive to dopant ratios. As a whole, the intimate connection between local chemistry and order in amorphous interfacial complexions is demonstrated, opening the door for microstructural engineering within the amorphous complexions themselves.

Medium-range structural order in amorphous arsenic

Yuanbin Liu; Yuxing Zhou; Richard Ademuwagun; Luc Walterbos; Janine George; Stephen R. Elliott; Volker L. Deringer. arXiv preprint (2025). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/2509.02484v1

Abstract: Medium-range order (MRO) is a key structural feature of amorphous materials, but its origin and nature remain elusive. Here, we reveal the MRO in amorphous arsenic (a-As) using advanced atomistic simulations, based on machine-learned potentials derived using automated workflows. Our simulations accurately reproduce the experimental structure factor of a-As, especially the first sharp diffraction peak (FSDP), which is a signature of MRO. We compare and contrast the structure of a-As with that of its lighter homologue, red amorphous phosphorus (a-P), identifying the dihedral-angle distribution as a key factor differentiating the MRO in both. The pressure-dependent structural behaviors of a-As and a-P differ as well, which we link to the interplay of ring topology and structural entropy. We finally show that the origin of the FSDP is closely correlated with the size and spatial distribution of voids in the amorphous networks. Our work provides fundamental insights into MRO in an amorphous elemental system, and more widely it illustrates the usefulness of automation for machine-learning-driven atomistic simulations.

From Bulk to Surface: Structure and Dynamics of Amorphous Alumina from Deep Potential Molecular Dynamics

Zheng Yu; Jiayan Xu; Abhirup Patra; Sharan Shetty; Detlef Hohl; Roberto Car. arXiv preprint (2026). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/2605.05361v1

Abstract: Understanding the atomic-scale structure and dynamics of amorphous oxide surfaces is essential for interpreting their chemical reactivity, mechanical stability, and interfacial behavior, yet direct experimental characterization remains challenging. We employ Deep Potential (DP) molecular dynamics to generate large-scale, ab initio-quality models of amorphous Al$_2$O$_3$ bulk glasses and melt-quenched free surfaces, enabling a quantitative analysis of both structure and relaxation dynamics with statistical confidence inaccessible to direct ab initio simulation. The trained DP model reproduces experimental liquid and glass structure, captures the cooling-rate dependence of the bulk glass transition, and corrects systematic biases in the polyhedral populations predicted by widely used classical force fields. At the free surface, mass density recovers to bulk values over ~10 $\unicode{x212B}$, while local coordination requires a slightly wider subsurface region to fully converge. The outermost layer is oxygen-enriched, exhibits altered polyhedral connectivity with contracted Al-O bonds, and hosts a broad population of under-coordinated motifs (notably AlO$_3$ and OAl$_2$) whose abundances are governed by glass stability. These reactive Lewis acid and Br$\unicode{x00F8}$nsted base sites are locally paired in a manner consistent with bond-valence compensation, yet remain spatially dispersed rather than aggregating into extended clusters. Despite this pronounced structural heterogeneity, the surface relaxes on the same timescale as the bulk and exhibits a comparable glass transition temperature, suggesting that the disordered surface is kinetically stable once formed. Together, these results establish a molecular-level picture of amorphous alumina surfaces and demonstrate the capability of machine-learned potentials to resolve structure-property relationships in disordered oxide interfaces.

Recent Advances in Metallic Glasses

Silvia Bonfanti; Ralf Busch; Jesper Byggmästar; Jeppe C. Dyre; Jürgen Eckert; Spencer Fajardo; Michael L. Falk; Isabella Gallino; Jamie J. Kruzic; Jiayin Lu; Giulio Monaco; Misaki Ozawa; Anshul D. S. Parmar; Chris H. Rycroft; Srikanth Sastry. arXiv preprint (2025). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/2512.16590v1

Abstract: This paper reviews recent advances in the field of metallic glasses, focusing on the development of novel experimental techniques and in silico models. We discuss progress in experimental characterization, additive manufacturing, multiscale modeling approaches, and the growing role of machine learning in understanding and designing these complex materials. On the experimental side, we highlight measurements of thermophysical properties of supercooled liquids via fast chip calorimetry and enhancements in mechanical properties through rejuvenation treatments. This work underscores the crucial role of short-range order and medium-range order in controlling metallic glass mechanical properties. Recent progress in structural probes allows in situ observations of deformation mechanisms, positioning the field well to further advance our understanding of mechanical properties. Additive manufacturing of metallic glasses is discussed as one encouraging new manufacturing route for metallic glasses. We examine laser powder-bed fusion process physics and the central trade-off between amorphicity and densification, including heat affected zone devitrification and defects formation, together with emerging mitigation strategies and applications. On the theoretical and simulation side, we review advances in nanoscale, mesoscale, and continuum modeling of metallic glasses that have led to promising approaches

by which multiscale schemes can incorporate data sourced from atomic-scale simulations. These efforts have helped to elucidate the connection between the glass structure and mechanical and rheological responses. We also cover the development of machine learning interatomic potentials for metallic glasses, along with machine learning driven prediction of glass forming ability and inverse design methods. Finally, challenges and directions for future research are presented and discussed.

Bridging classical and quantum interpretation of chemical state analysis by XPS/HAXPES to resolve short-range order in amorphous alumina films

Simon Gramatte; Xing Wang; Michael Alejandro Hernández Bertrán; Claudia Cancellieri; Giovanni Pizzi; Deborah Prezzi; Iurii Timrov; Olivier Politano; Ivo Utke; Lars P. H. Jeurgens; Vladyslav Turlo. arXiv preprint (2024). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/2408.08255v2

Abstract: Probing the local structure and chemistry of wide-bandgap amorphous oxide thin films remains challenging due to the limitations of lab-based spectroscopy. This work integrates X-ray photoelectron spectroscopy (XPS), hard X-ray photoemission spectroscopy (HAXPES), molecular dynamics simulations using machine-learning interatomic potentials, density-functional theory (DFT) calculations, and classical electrostatic modeling of final-state core-ionization effects in Al atoms to uncover the structure and chemistry of amorphous alumina polymorphs made with atomic layer deposition (ALD). DFT calculations using the Delta Kohn-Sham method supported the interpretation of final-state effects and validated electrostatic model assumptions. Shifts in the measured Auger parameter were interpreted as extra-atomic relaxation energies, revealing sensitivity to the local coordination environment. Structural disorder and thermal fluctuations were found to govern the distribution of extra-atomic relaxation energies, suggesting that cryo-XPS can isolate and reveal intrinsic structural building blocks of amorphous oxides. Simulated heating and annealing demonstrated that Auger parameter shifts can serve as indicators of phase decomposition in H-supersaturated ALD amorphous alumina. These findings provide a pathway for comprehensive interpretation and predictive modeling of XPS spectra in amorphous wide-bandgap oxides.

Machine-Learning-Guided Insights into Solid-Electrolyte Interphase Conductivity: Are Amorphous Lithium Fluorophosphates the Key?

Peichen Zhong; Kristin A. Persson. arXiv preprint (2025). Cited by 0. Found by: arxiv. DOI: 10.1021/acsenergylett.5c03526

Abstract: Despite decades of study, the identity of the dominant \ce{Li+}-conducting phase within the inorganic SEI of Li-ion batteries remains unresolved. While the mosaic model describes LiF/\ce{Li2O}/\ce{Li2CO3} nanocrystallites within a disordered matrix, these crystalline phases inherently offer limited ionic conductivity. Growing evidence suggests that interfaces, grain boundaries, and amorphous phases may instead host the primary fast-ion pathways. Using diffusion-based generative structure prediction and machine-learning interatomic potentials (MLIPs), we investigate lithium difluorophosphate (\ce{LiPO2F2}), a key mixed-anion decomposition product of phosphorus- and fluorine-containing electrolytes. We identify a stable crystalline polymorph and demonstrate that the amorphous counterpart is conductive, with projected room-temperature $ \approx 0.18$ mS cm^{-1} and $E_\mathrm{a} \approx 0.40$ eV. This enhancement stems from structural disorder flattening the Li site-energy landscape and a low formation energy for Li-interstitial defects, which supplies additional mobile carriers. We propose amorphous mixed-anion Li–P–O–F phases as a promising conducting medium in the SEI, offering a specific target for engineering improved battery interfaces.

Atomic insight into Li$^+$ ion transport in amorphous electrolytes Li$_x$AlO$_y$Cl$_{3+x-2y}$ (0.5 $\leq$ x $\leq$ 1.5, 0.25 $\leq$ y $\leq$ 0.75)

Yang Qifan; Xu Jing; Fu Xiao; Lian Jingchen; Wang Liqi; Gong Xuhe; Xiao Ruijuan; Li Hong. arXiv preprint (2024). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/2409.16712v1

Abstract: The recent study of viscoelastic amorphous oxychloride electrolytes has opened up a new field of research for solid-state electrolytes. In this work, we chose Li-Al-O-Cl system containing disordered structures with varying O/Cl ratio and Li$^+$ content to study their structural characteristics and ion transport mechanism using ab-initio molecular dynamics (AIMD) simulation and machine learning interatomic potential based molecular dynamics (MLIP-based MD) simulation. It is found that O-doping results in the presence of a skeleton of Al-chains formed by AlOCl tetrahedra and the increase of glass forming ability, causing Cl atoms' rotation around centered-Al within the tetrahedron thus facilitating the motion of Li$^+$ ions. However, further increase of O/Cl ratio decreases the number of rotating Cl atoms, weakening the transport of Li$^+$.

So increasing glass forming ability without reducing Cl content or by methods through controlling synthesis conditions, are useful to promote Li$^+$ ion conducting of oxychloride electrolytes.

Turing Pattern Engineering Enables Kinetically Ultrastable yet Ductile Metallic Glasses

Huanrong Liu; Qingan Li; Shan Zhang; Rui Su; Yunjiang Wang; Pengfei Guan. arXiv preprint (2025). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/2512.20196v1

Abstract: Enhancing the kinetic stability of glasses often necessitates deepening thermodynamic stability, which typically compromises ductility due to increased structural rigidity. Decoupling these properties remains a critical challenge for functional applications. Here, we demonstrate that pattern engineering in metallic glasses (MGs) enables unprecedented kinetic ultrastability while retaining thermodynamic metastability and intrinsic plasticity. Through atomistic simulations guided by machine-learning interatomic potentials and replica-exchange molecular dynamics, we reveal that clustering oxygen contents, driven by reaction-diffusion-coupled pattern dynamics, act as localized pinning sites. These motifs drastically slow structural relaxation, yielding kinetic stability comparable to crystal-like ultrastable glasses while retaining an energetic as-cast state. Remarkably, the thermodynamically metastable state preserves heterogeneous atomic mobility, allowing strain delocalization under mechanical stress. By tailoring oxygen modulation via geometric patterning, we achieve an approximately 200 K increase in the onset temperature of the glass transition (Tonset) while maintaining fracture toughness akin to conventional MGs. This work establishes a paradigm of kinetic stabilization without thermodynamic compromise, offering a roadmap to additively manufacture bulk amorphous materials with combined hyperstability and plasticity.

Realistic atomistic structure of amorphous silicon from machine-learning-driven molecular dynamics

Volker L. Deringer; Noam Bernstein; Albert P. Bartók; Matthew J. Cliffe; Rachel N. Kerber; Lauren E. Marbella; Clare P. Grey; Stephen R. Elliott; Gábor Csányi. arXiv preprint (2018). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/1803.02802v1

Abstract: Amorphous silicon (a-Si) is a widely studied non-crystalline material, and yet the subtle details of its atomistic structure are still unclear. Here, we show that accurate structural models of a-Si can be obtained by harnessing the power of machine-learning algorithms to create interatomic potentials. Our best a-Si network is obtained by cooling from the melt in molecular-dynamics simulations, at a rate of 10^{11} K/s (that is, on the 10 ns timescale). This structure shows a defect concentration of below 2% and agrees with experiments regarding excess energies, diffraction data, as well as ^{29}Si solid-state NMR chemical shifts. We show that this level of quality is impossible to achieve with faster quench simulations. We then generate a 4,096-atom system which correctly reproduces the magnitude of the first sharp diffraction peak (FSDP) in the structure factor, achieving the closest agreement with experiments to date. Our study demonstrates the broader impact of machine-learning interatomic potentials for elucidating accurate structures and properties of amorphous functional materials.

Density dependence of thermal conductivity in nanoporous and amorphous carbon with machine-learned molecular dynamics

Yanzhou Wang; Zheyong Fan; Ping Qian; Miguel A. Caro; Tapio Ala-Nissila. arXiv preprint (2024). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/2408.12390v2

Abstract: Disordered forms of carbon are an important class of materials for applications such as thermal management. However, a comprehensive theoretical understanding of the structural dependence of thermal transport and the underlying microscopic mechanisms is lacking. Here we study the structure-dependent thermal conductivity of disordered carbon by employing molecular dynamics (MD) simulations driven by a machine-learned interatomic potential based on the efficient neuroevolution potential approach. Using large-scale MD simulations, we generate realistic nanoporous carbon (NP-C) samples with density varying from 0.3 to 1.5 g cm^{-3} dominated by sp^2 motifs, and amorphous carbon (a-C) samples with density varying from 1.5 to 3.5 g cm^{-3} exhibiting mixed sp^2 and sp^3 motifs. Structural properties including short- and medium-range order are characterized by atomic coordination, pair correlation function, angular distribution function and structure factor. Using the homogeneous nonequilibrium MD method and the associated quantum-statistical correction scheme, we predict a linear and a superlinear density dependence of thermal conductivity for NP-C and a-C, respectively, in good agreement with relevant experiments. The distinct density dependences are attributed to the different impacts of the sp^2 and sp^3 motifs on the spectral heat capacity, vibrational

mean free paths and group velocity. We additionally highlight the significant role of structural order in regulating the thermal conductivity of disordered carbon.

The ab initio amorphous materials database: Empowering machine learning to decode diffusivity

Hui Zheng; Eric Sivonxay; Max Gallant; Ziyao Luo; Matthew McDermott; Patrick Huck; Kristin A. Persson. arXiv preprint (2024). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/2402.00177v1

Abstract: Amorphous materials exhibit unique properties that make them suitable for various applications in science and technology, ranging from optical and electronic devices and solid-state batteries to protective coatings. However, data-driven exploration and design of amorphous materials is hampered by the absence of a comprehensive database covering a broad chemical space. In this work, we present the largest computed amorphous materials database to date, generated from systematic and accurate \textit{ab initio} molecular dynamics (AIMD) calculations. We also show how the database can be used in simple machine-learning models to connect properties to composition and structure, here specifically targeting ionic conductivity. These models predict the Li-ion diffusivity with speed and accuracy, offering a cost-effective alternative to expensive density functional theory (DFT) calculations. Furthermore, the process of computational quenching amorphous materials provides a unique sampling of out-of-equilibrium structures, energies, and force landscape, and we anticipate that the corresponding trajectories will inform future work in universal machine learning potentials, impacting design beyond that of non-crystalline materials.

Structural properties of amorphous Na$_3$OCl electrolyte by first-principles and machine learning molecular dynamics

T. -L. Pham; M. Guerboub; S. D. Wansi Wendj; A. Bouzid; C. Tugène; M. Boero; C. Massobrio; Y. -H. Shin; G. Ori. arXiv preprint (2024). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/2404.11442v1

Abstract: Solid-state electrolytes mark a significant leap forward in the field of electrochemical energy storage, offering improved safety and efficiency compared to conventional liquid electrolytes. Among these, antiperovskite electrolytes, particularly those based on Li and Na, have emerged as promising candidates due to their superior ionic conductivity and straightforward synthesis processes. This study focuses on the amorphous phase of antiperovskite Na$_3$OCl, assessing its structural properties through a combination of first-principles molecular dynamics (FPMD) and machine learning interatomic potential (MLIP) simulations. Our comprehensive analysis spans models ranging from 135 to 3645 atoms, allowing for a detailed examination of X-ray and neutron structure factors, total and partial pair correlation functions, coordination numbers, and structural unit distributions. We demonstrate the minimal, albeit partially present, size effects on these structural features and validate the accuracy of the MLIP model in reproducing the intricate details of the amorphous Na$_3$OCl structure described at the FPMD level.

Enhanced ionic conductivity through crystallization of glass-Li$_3$PS$_4$ by machine learning molecular dynamics simulations

Koji Shimizu; Parth Bahuguna; Shigeo Mori; Akitoshi Hayashi; Satoshi Watanabe. arXiv preprint (2023). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/2312.06963v1

Abstract: Understanding the atomistic mechanism of ion conduction in solid electrolytes is critical for the advancement of all-solid-state batteries. Glass-ceramics, which undergo crystallization from a glass state, frequently exhibit unique properties including enhanced ionic conductivities compared to both the original crystalline and glass forms. Despite these distinctive features, specific details regarding the behavior of ion conduction in glass-ceramics, particularly concerning conduction pathways, remain elusive. In this study, we demonstrate the crystallization process of glass-Li$_3$PS$_4$ through molecular dynamics simulations employing machine learning interatomic potentials constructed from first principles calculation data. Our analyses of Li conduction using the obtained partially crystallized structures reveal that the diffusion barriers of Li decrease as the crystallinity in Li$_3$PS$_4$ glass-ceramics increases. Furthermore, Li displacements predominantly occur in the precipitated crystalline portion, suggesting that percolation conduction plays a significant role in enhanced Li conduction. These findings provide valuable insights for the future utilization of glass-ceramic materials.

Machine learning potential for predicting thermal conductivity of -phase and amorphous Tantalum Nitride

Zhicheng Zong; Yangjun Qin; Jiahong Zhan; Haisheng Fang; Nuo Yang. arXiv preprint (2025). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/2508.03297v1

Abstract: Tantalum nitride (TaN) has attracted considerable attention due to its unique electronic and thermal properties, high thermal conductivity, and applications in electronic components. However, for the -phase of TaN, significant discrepancies exist between previous experimental measurements and theoretical predictions. In this study, deep potential models for TaN in both the -phase and amorphous phase were developed and employed in molecular dynamics simulations to investigate the thermal conductivities of bulk and nanofilms. The simulation results were compared with reported experimental and theoretical results, and the mechanism for differences were discussed. This study provides insights into the thermal transport mechanisms of TaN, offering guidance for its application in advanced electronic and thermal management devices.

Optimization and Validation of a Deep Learning CuZr Atomistic Potential: Robust Applications for Crystalline and Amorphous Phases with near-DFT Accuracy

Christopher M. Andolina; Philip Williamson; Wissam A. Saidi. J. Chem. Phys. 152, 154701 (2020) (2020). Cited by 0. Found by: arxiv. DOI: 10.1063/5.0005347

Abstract: We show that a deep-learning neural network potential (DP) based on density functional theory (DFT) calculations can well describe Cu-Zr materials, an example of a binary alloy system that can coexist in several ordered intermetallics and as an amorphous phase. The complex phase diagram for Cu-Zr makes it a challenging system for traditional atomistic force-fields that fail to describe well the different properties and phases. Instead, we show that a DP approach using a large database with ~300k configurations can render results generally on par with DFT. The training set includes configurations of pristine and bulk elementary metals and intermetallics in the liquid and solid phases in addition to slab and amorphous configurations. The DP model was validated by comparing bulk properties such as lattice constants, elastic constants, bulk moduli, phonon spectra, surface energies to DFT values for identical structures. Further, we contrast the DP results with values obtained using well-established two embedded atom method potentials. Overall, our DP potential provides near DFT accuracy for the different Cu-Zr phases but with a fraction of its computational cost, thus enabling accurate computations of realistic atomistic models especially for the amorphous phase.

Mechanisms of alkali ionic transport in amorphous oxyhalides solid state conductors

Luca Binci; KyuJung Jun; Bowen Deng; Gerbrand Ceder. arXiv preprint (2026). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/2601.06384v1

Abstract: Amorphous oxyhalides have attracted significant attention due to their relatively high ionic conductivity (\$>\$1 mS cm\$^{-1}\$), excellent chemical stability, mechanical softness, and facile synthesis routes via standard solid-state reactions. These materials exhibit an ionic conductivity that is almost independent of the underlying chemistry, in stark contrast to what occurs in crystalline conductors. In this work, we employ an accurately fine-tuned machine learning interatomic potential to construct large-scale molecular dynamics trajectories encompassing hundreds of nanoseconds to obtain statistically converged transport properties. We find that the amorphous state consists of chain fragments of metal-anion tetrahedra of various lenght. By analyzing the residence time of alkali cations migrating around tetrahedrally-coordinated trivalent metal ions, we find that oxygen anions on the metal-anion tetrahedra limit alkali diffusion. By computing the full Einstein expression of the ionic conductivity, we demonstrate that the alkali transference number of these materials is strongly influenced by distinct-particles correlations, while at the same time they are characterized by an alkali Haven ratio close to one, implying that ionic transport is largely dictated by uncorrelated self-diffusion. Finally, by extending this analysis to chemical compositions \$AMX_{2.5}\textsf{O}_{0.75}\$, spanning different alkaline (\$A\$ = Li, Na, K), metallic (\$M\$ = Al, Ga, In), and halogen (\$X\$ = Cl, Br, I) species, we clarify why the diffusion properties of these materials remain largely insensitive to variations in atomic chemistry.

Neural-Network Assisted Study of Nitrogen Atom Dynamics on Amorphous Solid Water – II. Diffusion

Viktor Zaverkin; Germán Molpeceres; Johannes Kästner. arXiv preprint (2021). Cited by 0. Found by: arxiv. DOI: 10.1093/mnras/stab3631

Abstract: The diffusion of atoms and radicals on interstellar dust grains is a fundamental ingredient for predicting accurate molecular abundances in astronomical environments. Quantitative values of diffusivity and diffusion barriers usually rely heavily on empirical rules. In this paper, we compute the diffusion coefficients of adsorbed nitrogen atoms by combining machine-learned interatomic potentials, metadynamics, and kinetic Monte Carlo simulations. With this approach, we obtain a diffusion coefficient of nitrogen atoms on the surface of amorphous solid water of merely $(3.5 \pm 1.1)10^{-34}$cm^{2}s^{-1} at 10 K for a bare ice surface. Thus, we find that nitrogen, as a paradigmatic case for light and weakly bound adsorbates, is unable to diffuse on bare amorphous solid water at 10 K. Surface coverage has a strong effect on the diffusion coefficient by modulating its value over 9–12 orders of magnitude at 10 K and enables diffusion for specific conditions. In addition, we have found that atom tunneling has a negligible effect. Average diffusion barriers of the potential energy surface (2.56 kJ mol^{-1}) differ strongly from the effective diffusion barrier obtained from the diffusion coefficient for a bare surface (6.06 kJ mol^{-1}) and are, thus, inappropriate for diffusion modeling. Our findings suggest that the thermal diffusion of N on water ice is a process that is highly dependent on the physical conditions of the ice.

Machine Learning Assisted Modeling of Amorphous TiO$_2$-Doped GeO$_2$ for Advanced LIGO Mirror Coatings

Jun Jiang; Rui Zhang; Kiran Prasai; Riccardo Bassiri; James N. Fry; Martin M. Fejer; Hai-Ping Cheng. arXiv preprint (2025). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/2503.21701v1

Abstract: The mechanical loss angle of amorphous TiO$_2$-doped GeO$_2$ can be lower than 10^{-4}, making it a candidate for Laser Interferometer Gravitational-wave Observatory (LIGO) mirror coatings. Amorphous oxides have complex atomic structures that are influenced by various factors, including doping concentration, preparation, and thermal history, resulting in different mass densities and physical properties. Modeling at atomistic level enables capturing these effects by generating atomic structure models according to experimental conditions. In order to obtain reliable and physical amorphous models at an affordable cost, we develop classical and machine-learning potentials (MLP) to speed up simulations. First-principles calculations are used to train and validate MLP as well as validating structure models. To better reproduce properties such as elastic modulus, radial distribution function (RDF) and the variations in mass density of doped amorphous oxides, density functional theory (DFT) calculations are used to optimize the final models. We find that the mass densities of amorphous systems are correlated with the total void volume. The experimental mass density matches the models with the most symmetric potential energy wells under volume change. The elastic response of the metal-oxygen network is also studied. The 27\% TiO$_2$ doped GeO$_2$ system shows the least number of large atom-atom distance changes, while for 44\% TiO$_2$ doped GeO$_2$, a majority of Ti-O distances are significantly changed. In response to strains, the metal-oxygen network at low mass densities prefers to adjust bond angles, while at high mass densities, the adjustment is mainly done by changing atom-atom distance.

Fast Isotropic Li-Ion Diffusion in Zeolitic Imidazolate Framework Glass Electrolytes for Batteries

Yong Li; Tao Du; Timothée Jamin; Zhencai Li; Kasper Tolborg; Yuanzheng Yue; Morten M. Smedskjaer. arXiv preprint (2026). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/2608.06902v1

Abstract: All-solid-state lithium batteries require solid electrolytes that combine rapid room-temperature ion transport with mechanical robustness and interfacial compatibility. Zeolitic imidazolate framework (ZIF) glasses, with ZIFs being a sub-set of metal-organic frameworks, offer an attractive yet relatively underexplored platform because they combine an grainboundary-free and amorphous topology with chemically tunable frameworks. Here, we reveal that structural disorder unlocks fast and isotropic lithium diffusion in ZIF glasses. This is realized by using a machine learning interatomic potential to simulate Li+ transport in crystalline and glassy ZIF-4 and ZIF-62. Structural disorder reduces the activation energy for Li+ migration from ~0.35 eV to 0.16 eV and increases the extrapolated room-temperature diffusion coefficient by more than one order of magnitude for ZIF-4 and nearly sevenfold for ZIF-62. Analyses of non-Gaussian dynamics and van Hove correlation functions reveal that Li+ diffusion in crystalline ZIFs occurs via rare, dynamically heterogeneous hopping events among well-defined cages, whereas Li+ diffusion in glassy ZIFs is more homogeneous, continuous, and Fickian-like, benefiting from a wide distribution of coordination geometries and migration barriers. Li+ diffusion in crystalline ZIFs is strongly anisotropic, reflecting that ordered orientations of imidazolate and benzimidazolate rings impose distinct energy barriers along different crystallographic directions. Upon vitrification, these ring orientations become randomized, and hence, the diffusion of Li+ becomes isotropic or near-isotropic. These findings imply that well-designed metal-organic framework glasses are a promising candidate as high-performance solid-state electrolytes.

kALDo 2.0: Scalable Thermal Transport from First Principles and Machine Learning Potentials

Giuseppe Barbalinardo; Zekun Chen; Dylan Folkner; Bohan Li; Nicholas W. Lundgren; Nathaniel Troup; Alfredo Fiorentino; Davide Donadio. arXiv preprint (2026). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/2602.2372

Abstract: We introduce kALDo2.0, an open-source Python package for computing vibrational, elastic, and thermal transport properties of solids from first principles and machine-learned interatomic potentials. Building on the anharmonic lattice dynamics (ALD) framework, kALDo2.0 provides efficient CPU and GPU-accelerated implementations of the Boltzmann transport equation (BTE) for crystals and the quasi-harmonic Green-Kubo (QHGK) method. QHGK extends thermal transport predictions beyond crystals to disordered materials, including glasses, alloys, and complex nanostructures. kALDo2.0 introduces native integration with modern machine-learned potentials (MLPs), enabling thermal transport workflows that combine the accuracy of first-principles methods with the scalability of classical force fields. It also features comprehensive support for temperature-dependent effective potentials workflows, flexible storage backends for large-scale calculations, and advanced quantification of anharmonicity. The software seamlessly interfaces with electronic structure codes (Quantum ESPRESSO, VASP), molecular dynamics packages (LAMMPS), and MLPs (ACE, NEP, MACE, MatterSim, Orb), enabling thermal transport studies from 0 K to finite temperatures. kALDo2.0 implements multiple BTE solution strategies and essential physical corrections, including isotopic scattering and non-analytical terms for polar materials. A modular Python architecture with lazy evaluation and multiple storage formats (ASCII, NumPy, HDF5) enables simulations of systems containing up to tens of thousands of atoms. This paper describes the theoretical framework, implementation details, software architecture, and validation examples demonstrating kALDo2.0's capabilities for studying complex materials, including halide perovskites with strong anharmonicity and polar oxides requiring long-range electrostatic corrections.

Neural-Network Assisted Study of Nitrogen Atom Dynamics on Amorphous Solid Water. I. Adsorption & Desorption

Germán Molpeceres; Viktor Zaverkin; Johannes Kästner. arXiv preprint (2020). Cited by 0. Found by: arxiv. DOI: 10.1093/mnras/staa2891

Abstract: Dynamics of adsorption and desorption of (4S)-N on amorphous solid water are analyzed using molecular dynamics simulations. The underlying potential energy surface was provided by machine-learned interatomic potentials. Binding energies confirm the latest available theoretical and experimental results. The nitrogen sticking coefficient is close to unity at dust temperatures of 10 K but decreases at higher temperatures. We estimate a desorption time scale of 1 s at 28 K. The estimated time scale allows chemical processes mediated by diffusion to happen before desorption, even at higher temperatures. We found that the energy dissipation process after a sticking event happens on the picosecond timescale at dust temperatures of 10 K, even for high energies of the incoming adsorbate. Our approach allows the simulation of large systems for reasonable time scales at an affordable computational cost and ab-initio accuracy. Moreover, it is generally applicable for the study of adsorption dynamics of interstellar radicals on dust surfaces.

Simulations of High Temperature Decomposition of Metal-Organic Frameworks to form Amorphous Catalysts

Connor W. Edwards; Oliver M. Linder-Patton; Jack D. Evans. arXiv preprint (2026). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/2601.16459v1

Abstract: Metal-organic framework (MOF) derived materials formed through high temperature processes show great potential as catalysts. However, understanding of structure-property relationships between the initial MOF and the resulting MOF-derived catalyst is limited because the amorphous nature of the catalyst challenges standard structural characterization methods. Neural network approaches that learn interatomic potentials from density functional theory offer a promising solution. We simulated the pyrolysis of UiO-66, UiO-67 and MIP-206 using both foundational and fine-tuned machine learned interatomic potentials (MLIPs). To mimic experimental conditions, an atmosphere of CO2 and H2 was introduced and the structures were doped with 20 wt% copper to probe the effect of copper on the structural evolution of MOFs. These simulations provide atomistic insights into gas evolution, metal nanoparticle formation, and linker decomposition that were compared to available experimental data. Overall, this work demonstrates the potential of MLIPs to accurately model high temperature MOF dynamics under experimentally relevant conditions and guide the design of new catalytic materials.

Thermal conductivity of glasses above the plateau: first-principles theory and applications

Michele Simoncelli; Francesco Mauri; Nicola Marzari. arXiv preprint (2022). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/2209.11201v1

Abstract: Predicting the thermal conductivity of glasses from first principles has hitherto been a prohibitively complex problem. In fact, past works have highlighted challenges in achieving computational convergence with respect to length and/or time scales using either the established Allen-Feldman or Green-Kubo formulations, endorsing the concept that atomistic models containing thousands of atoms – thus beyond the capabilities of first-principles calculations – are needed to describe the thermal conductivity of glasses. In addition, these established formulations either neglect anharmonicity (Allen-Feldman) or miss the Bose-Einstein statistics of atomic vibrations (Green Kubo), thus leaving open the question on the relevance of these effects. Here, we present a first-principles formulation to address the thermal conductivity of glasses above the plateau, which can account comprehensively for the effects of structural disorder, anharmonicity, and quantum Bose-Einstein statistics. The protocol combines the Wigner formulation of thermal transport with convergence-acceleration techniques, and is validated in vitreous silica using both first-principles calculations and a quantum-accurate machine-learned interatomic potential. We show that models of vitreous silica containing less than 200 atoms can already reproduce the thermal conductivity in the macroscopic limit and that anharmonicity negligibly affects heat transport in vitreous silica. We discuss the microscopic quantities that determine the trend of the conductivity at high temperature, highlighting the agreement of the calculations with experiments in the temperature range above the plateau where radiative effects remain negligible ($50<T<450$ K).

Simulating alternating bias assisted annealing of amorphous oxide tunnel junctions

Alexander C. Tyner; Alexander V. Balatsky. arXiv preprint (2025). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/2512.18420v1

Abstract: Amorphous oxide tunneling barriers, primarily formed from aluminum, represent one of the most widely adopted platforms for superconducting quantum bits (qubits). To overcome challenges associated with defects and sample variance among the tunneling barriers, the methodology of alternating bias assisted annealing (ABAA) was introduced in Pappas et. al[1]. The process of applying alternating bias to the barrier and subsequently aging before use was shown to reduce defects in the barrier. Namely, defects that give rise to two-level systems, coupling to the qubit and expediting decoherence. In this work we replicate an expedited ABAA process through a combination of ab-initio molecular dynamics and machine-learned potentials, illuminating how ABAA effects the energy landscape of the barrier.

Origin of negative thermal expansion and pressure induced amorphization in zirconium tungstate from machine-learning potential

Ri He; Hongyu Wu; Yi Lu; Zhicheng Zhong. arXiv preprint (2022). Cited by 0. Found by: arxiv. DOI: 10.1103/PhysRevB.106.174101

Abstract: Understanding various macroscopic pressure-volume-temperature properties of materials on the atomistic level has always been an ambition for physicists and material scientists. Particularly, some materials such as zirconium tungstate (ZrW2O8), exhibit multiple exotic properties including negative thermal expansion (NTE) and pressure-induced amorphization (PIA). Here, using machine-learning based deep potential, we trace both of the phenomena in ZrW2O8 back to a common atomistic origin, where the nonbridging O atoms play a critical role. We demonstrate that the nonbridging O atoms confer great flexibility to vibration of polyhedrons, and kinetically drive volume shrinking on heating, or NTE. In addition, beyond a certain critical pressure, we find that the migration of nonbridging O atoms leads to additional bond formation that lowers the potential energy, suggesting that the PIA is a potential-driven first-order phase transition. Most importantly, we identify a second critical pressure beyond which the amorphous phase of ZrW2O8 undergoes a hidden phase transition from a reversible phase to an irreversible one.

Harnessing Structural Disorder: Unraveling Hydrogen Evolution in Monolayer Amorphous Carbon via First-Principles Simulations and Machine-Learned Potentials

Sreehari M S; Ashutosh Krishna Amaram; Raghavan Ranganathan. arXiv preprint (2026). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/2605.07670v1

Abstract: Disorder and defective coordination in the catalytic plane are crucial for enhancing the Hydrogen Evolution Reaction (HER) on two-dimensional catalysts. Amorphous materials are disordered, making them catalytically adaptive for many reactions. In this work, the HER capabilities of Monolayer Amorphous Carbon (MAC) were studied in comparison with crystalline carbon derivatives, such as pristine graphene (GE) and graphyne derivatives. MAC generated from melt-quench simulations revealed a diverse framework of predominantly sp2 and sp3 carbons with numerous 5-, 6-, and 7-membered rings. Density Functional Theory (DFT) calculations investigated free-energy variations in hydrogen adsorption for each material. According to Sabatier's principle, optimum activity is achieved when the Gibbs free energy (Delta GH) change approaches zero. Crystalline carbon materials possess limited active sites, with beta-graphyne showing the best Delta GH value of +0.34 eV. The adsorption study for MAC was conducted in 30 distinct local environments, where core structural properties were analyzed against varying radii. Calculations showed a Delta GH distribution for MAC ranging from -0.02 eV to +1.35 eV. To evaluate activity across the entire MAC surface, a MACE MLIP foundation model was finetuned, achieving optimal energy and force fitting of 1.67 meV/atom and 29.15 meV/A, respectively. The MLIP predicted Delta GH values from -0.91 eV to +1.70 eV, with approximately 15% of sites exhibiting values below +0.25 eV. Feature analysis revealed that 7-membered rings, curvature, and ripple height enhance HER activity. Our findings suggest that, with careful optimization of local features, MAC can be tuned to compete with noble metal catalysts.

Thermal transport of glasses via machine learning driven simulations

Paolo Pegolo; Federico Grasselli. arXiv preprint (2024). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/2402.06479v

Abstract: Accessing the thermal transport properties of glasses is a major issue for the design of production strategies of glass industry, as well as for the plethora of applications and devices where glasses are employed. From the computational standpoint, the chemical and morphological complexity of glasses calls for atomistic simulations where the interatomic potentials are able to capture the variety of local environments, composition, and (dis)order that typically characterize glassy phases. Machine-learning potentials (MLPs) are emerging as a valid alternative to computationally expensive ab initio simulations, inevitably run on very small samples which cannot account for disorder at different scales, as well as to empirical force fields, fast but often reliable only in a narrow portion of the thermodynamic and composition phase diagrams. In this article, we make the point on the use of MLPs to compute the thermal conductivity of glasses, through a review of recent theoretical and computational tools and a series of numerical applications on vitreous silica and vitreous silicon, both pure and intercalated with lithium.

Machine learning Hamiltonian enables scalable and accurate defect calculations: The case of oxygen vacancies in amorphous SiO$_2$

Zhenxing Dai; Zhong Yang; Mingjue Ni; Menglin Huang; Hongjun Xiang; Xin-Gao Gong; Shiyou Chen. arXiv preprint (2026). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/2604.07197v1

Abstract: Point defects critically influence the properties of materials and devices, yet density functional theory (DFT) remains computationally demanding for defect supercell calculations. Machine learning interatomic potentials (MLIPs) offer high efficiency but require extensive datasets. MLIPs trained only on defect configurations in small supercells exhibit systematic energy errors in larger supercells, demonstrating limited transferability. Here, we present a machine learning Hamiltonian (MLH) model-based method for calculating total energies and atomic forces in defect supercells with linear-scaling computational cost, enabling efficient structural relaxation and accurate formation energy predictions. We take oxygen vacancies in amorphous SiO$_2$ as an example and train the MLH model on defect configurations in 95-atom supercells, with the training data derived from 120 self-consistent field calculations and 12 structural relaxations. The MLH model enables efficient structural relaxations for host (defect-free) and defect systems in larger supercells, avoiding the systematic energy errors observed in MLIPs. The cancellation of energy errors between host and defect systems yields accurate formation energy predictions, with deviations from DFT below 50 meV. The proposed method holds significant potential for defect simulations in complex materials.

Synthesizability, hardness, and stacking order in multicomponent transition metal carbides from machine-learned potentials

Xin Liu; Anirudh Raju Natarajan. arXiv preprint (2026). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/2605.2948

Abstract: Multicomponent transition metal carbides are promising for extreme-environment applications, but identifying compositions that are both synthesizable and hard remains challenging. We fine-tune the MACE machine-learned interatomic potential on approximately 28,000 density functional theory calculations spanning the composition space of groups 4-6 transition metals and carbon to predict the thermodynamic stability and elastic properties of multicomponent carbides. The fine-tuned model achieves formation energy errors of ∼

10 meV/atom for thermodynamically relevant structures with only 20% of the training data. We screen over 1500 equiatomic compositions across rocksalt, hexagonal, and hcp prototypes, combining free energy models with elasticity-based hardness surrogates. Synthesizability predictions at 1500°C agree well with experimental reports for both single-phase and multiphase carbides. The group number of the constituent metals governs both stability and hardness. Free energy contributions from short-range order are small, typically a few meV/atom, indicating that a perfectly disordered solid solution provides a reasonable approximation for high-throughput screening. For compositions mixing group 4/5 and group 6 metals, we identify a new family of stacking-ordered phases with formation energies well below those of disordered rocksalt or hexagonal structures. DFT calculations corroborate these predictions and suggest that stacking-ordered phases should be experimentally accessible in multicomponent carbides. This study provides a framework for screening synthesizable multicomponent materials with target properties, identifies promising carbide compositions across the full nine-component space, and reveals a new class of stacking-ordered carbides accessible only in multicomponent compositions.

Efficient and accurate simulation of vitrification in multi-component metallic liquids with neural-network potentials

Rui Su; Jieyi Yu; Pengfei Guan; Weihua Wang. arXiv preprint (2022). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/2211.03350v3

Abstract: Constructing accurate interatomic potential and overcoming the exponential growth of structural equilibration time are challenges to the atomistic investigations of the composition-dependent structure and dynamics during the vitrification process of deeply supercooled multi-component metallic liquids. In this work, we describe a state-of-the-art strategy to address these challenges simultaneously. In the case of the representative Zr-Cu-Al system, in combination with a general algorithm for generating the neural-network potentials (NNP) of multi-component metallic glasses effectively and accurately, we propose a highly efficient atom-swapping hybrid Monte Carlo (SHMC) algorithm for accelerating the thermodynamic equilibration of deeply supercooled liquids. Extensive calculations demonstrate that the newly developed NNP faithfully reproduces the phase stabilities and structural characteristics obtained from the ab initio calculations and experiments. In the combined NNP-SHMC algorithm, the structure equilibration time in the deeply supercooled temperatures is accelerated by at least five orders of magnitudes, and the quenched glassy samples exhibit comparable stability to those prepared in the laboratory. Our results pave the way for the next-generation studies of the vitrification process and, thereby the composition-dependent glass-forming ability and physical properties of multi-component metallic glasses.

Atomic cluster expansion potential for the Si-H system

Louise A. M. Rosset; Volker L. Deringer. arXiv preprint (2025). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/2510.15633v1

Abstract: The silicon-hydrogen system is of key interest for solar-cell devices, including both crystalline and amorphous modifications. Elemental amorphous Si is now well understood, but the atomic-scale effects of hydrogenating the silicon matrix remain to be fully explored. Here, we present a machine-learned interatomic potential model based on the atomic cluster expansion (ACE) framework that can describe a wide range of Si-H phases, from crystalline and amorphous bulk structures to surfaces and molecules. We perform numerical and physical validation across a range of hydrogen concentrations and compare our results to experimental findings. Our work constitutes an advancement toward the exploration of large structural models of a-Si:H at realistic device scales.

Mechanisms of temperature-dependent thermal transport in amorphous silica from machine-learning molecular dynamics

Ting Liang; Penghua Ying; Ke Xu; Zhenqiang Ye; Chao Ling; Zheyong Fan; Jianbin Xu. Physical Review B 108, 184203 (2023) (2023). Cited by 0. Found by: arxiv. DOI: 10.1103/PhysRevB.108.184203

Abstract: Amorphous silica (a-SiO$_2$) is a foundational disordered material for which the thermal transport properties are important for various applications. To accurately model the interatomic interactions in classical molecular dynamics (MD) simulations of thermal transport in a-SiO$_2$, we herein develop an accurate yet highly efficient machine-learned potential model that allowed us to generate a-SiO$_2$ samples closely resembling experimentally produced ones. Using the homogeneous nonequilibrium MD method and a proper quantum-statistical correction to the classical MD results, quantitative agreement with experiments is achieved for the thermal conductivities of bulk and 190 nm-thick a-SiO$_2$ films over a wide range of temperatures. To interrogate the thermal vibrations at different temperatures, we calculated the current correlation functions corresponding to the transverse acoustic (TA) and longitudinal acoustic (LA) collective vibrations. The results reveal that below the Ioffe-Regel crossover frequency, phonons as well-defined excitations, remain applicable in a-SiO$_2$ and play a predominant role at low temperatures, resulting in a temperature-dependent increase in thermal conductivity. In the high-temperature region, more phonons are excited, accompanied by a more intense liquid-like diffusion event. We attribute the temperature-independent thermal conductivity in the high-temperature range of a-SiO$_2$ to the collaborative involvement of excited phonon scattering and liquid-like diffusion in heat conduction. These findings provide physical insights into the thermal transport of a-SiO$_2$ and are expected to be applied to a vast range of amorphous materials.

Atomic-scale factors that control the rate capability of nanostructured amorphous Si for high-energy-density batteries

Nongnuch Artrith; Alexander Urban; Yan Wang; Gerbrand Ceder. arXiv preprint (2019). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/1901.09272v1

Abstract: Nanostructured Si is the most promising high-capacity anode material to substantially increase the energy density of Li-ion batteries. Among the remaining challenges is its low rate capability as compared to conventional materials. To understand better what controls the diffusion of Li in the amorphous Li-Si alloy, we use a novel machine-learning potential trained on more than 40,000 ab-initio calculations and nanosecond-scale molecular dynamics simulations, to visualize for the first time the delithiation of entire LiSi nanoparticles. Our results show that the Si host is not static but undergoes a dynamic rearrangement from isolated atoms, to chains, and clusters, with the Li diffusion strongly governed by this Si rearrangement. We find that the Li diffusivity is highest when Si segregates into clusters, so that Li diffusion proceeds via hopping between the Si clusters. The average size of Si clusters and the concentration range over which Si clustering occurs can thus function as design criteria for the development of rate-improved anodes based on modified Si.

CO Adsorption Sites on Interstellar Water Ices Explored with Machine Learning Potentials. Binding energy distributions and snowline

Giulia M. Bovolenta; Germán Molpeceres; Kenji Furuya; Johannes Kästner; Stefan Vogt-Geisse. A&A 703, A172 (2025) (2025). Cited by 0. Found by: arxiv. DOI: 10.1051/0004-6361/202555836

Abstract: Context. Carbon monoxide (CO) is arguably the most important molecule for interstellar organic chemistry. Its binding to amorphous solid water (ASW) ice regulates both diffusion and desorption processes. Accurately characterizing the CO binding energy (BE) is essential for realistic astrochemical modeling. Aims. We aim to derive a statistically robust and physically accurate distribution of CO BEs on ASW surfaces, and to evaluate its implications for laboratory temperature-programmed desorption experiments and interstellar chemistry, with a focus on protoplanetary disks. Methods. We trained a machine-learned potential (MLP) on 8321 density functional theory (DFT) energies and gradients of CO interacting with differently-sized water clusters (22-60 water molecules). The DFT method was selected after extensive benchmark. With this potential we built realistic non-porous and porous ASW surfaces, and computed a BE distribution. We used symmetry-adapted perturbation theory to rationalize the interaction of CO on the different binding sites. Results. We find that both ASW morphologies yield similar Gaussian-like BE distributions with mean values near 900 K. However, the nature of the binding interactions is rather different and critically depends on surface roughness and dangling-OH bonds. Simulated TPD curves reproduce experimental trends across several coverage regimes. From an astrochemical point of view, the application of the full BE distribution has a dramatic influence on the CO distribution in protoplanetary disks, leading to a broader CO snowline region, improving predictions of CO gas-ice partitioning, and suggesting an equally broader distribution of organics in these objects.

AI-Aided Mapping of the Structure-Composition-Conductivity Relationships of Glass-Ceramic Lithium Thiophosphate Electrolytes

Haoyue Guo; Qian Wang; Alexander Urban; Nongnuch Artrith. arXiv preprint (2022). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/2201.11203v1

Abstract: Lithium thiophosphates (LPS) with the composition (Li$_2$S)$_x$(P$_2$S$_5$)$_{1-x}$ are among the most promising prospective electrolyte materials for solid-state batteries (SSBs), owing to their superionic conductivity at room temperature ($>10^{-3}$ S cm^{-1}), soft mechanical properties, and low grain boundary resistance. Several glass-ceramic (gc) LPS with different compositions and good Li conductivity have been previously reported, but the relationship between composition, atomic structure, stability, and Li conductivity remains unclear due to the challenges in characterizing non-crystalline phases in experiments or simulations. Here, we mapped the LPS phase diagram by combining first principles and artificial intelligence

(AI) methods, integrating density functional theory, artificial neural network potentials, genetic-algorithm sampling, and ab initio molecular dynamics simulations. By means of an unsupervised structure-similarity analysis, the glassy/ceramic phases were correlated with the local structural motifs in the known LPS crystal structures, showing that the energetically most favorable Li environment varies with the composition. Based on the discovered trends in the LPS phase diagram, we propose a candidate solid-state electrolyte composition, (Li$_{2}$S)$_{x}$(P$_{2}$S$_{5}$)$_{1-x}$ ($x\sim{}0.725$), that exhibits high ionic conductivity ($>10^{-2}$ S cm^{-1}) in our simulations, thereby demonstrating a general design strategy for amorphous or glassy/ceramic solid electrolytes with enhanced conductivity and stability.

Nine-element machine-learned interatomic potentials for multiphase refractory alloys

Jesper Byggmästar; Tiago Lopes; Zheyong Fan; Tapio Ala-Nissila. arXiv preprint (2026). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/2603.04147v1

Abstract: New refractory alloys are being continuously designed and characterised for applications requiring good high-temperature mechanical properties and stability. Computational design from atomistic simulations is limited by interatomic potentials missing key elements, being too inaccurate, or computationally too slow for large-scale simulations. Here we present development of a refractory alloy database and two computationally efficient and general-purpose machine-learned potentials (tabGAP and NEP). We also design a cross-sampling strategy for effective sampling of training data using predictions from two potentials with completely different underlying architecture. The potentials support arbitrary alloy compositions of elements in groups four to six in the periodic table (Ti, Zr, Hf, V, Nb, Ta, Cr, Mo, W). The database is diverse yet multitargeted to enable simulations of refractory metals and alloys across different pure-metal, solid-solution, intermetallic, and glassy phases. We demonstrate the usefulness of the potentials by reproducing known pressure-, temperature-, and solute-induced phase transitions, grain boundary segregation, and simulations of radiation damage in the WTaCrVHf metallic glass.

Reaction dynamics on amorphous solid water surfaces using interatomic machine learned potentials. Microscopic energy partition revealed from the P + H -> PH reaction

Germán Molpeceres; Viktor Zaverkin; Kenji Furuya; Yuri Aikawa; Johannes Kästner. A&A 673, A51 (2023) (2023). Cited by 0. Found by: arxiv. DOI: 10.1051/0004-6361/202346073

Abstract: Energy redistribution after a chemical reaction is one of the few mechanisms to explain the diffusion and desorption of molecules which require more energy than the thermal energy available in quiescent molecular clouds (10 K). This energy distribution can be important in phosphorous hydrides, elusive yet fundamental molecules for interstellar prebiotic chemistry. We studied the reaction dynamics of the \ce{P + H -> PH} reaction on amorphous solid water, a reaction of astrophysical interest, using \emph{ab-initio} molecular dynamics with atomic forces evaluated by a neural network interatomic potential. We found that the exact nature of the initial phosphorous binding sites is less relevant for the energy dissipation process because the nascent PH molecule rapidly migrates to sites with higher binding energy after the reaction. Non-thermal diffusion and desorption-after-reaction were observed and occurred early in the dynamics, essentially decoupled from the dissipation of the chemical reaction energy. From an extensive sampling of reactions on sites, we constrained the average dissipated reaction energy within the simulation time (50 ps) to be between 50 and 70 %. Most importantly, the fraction of translational energy acquired by the formed molecule was found to be mostly between 1 and 5 %. Including these values, specifically for the test cases of 2% and 5% of translational energy conversion, in astrochemical models, reveals very low gas-phase abundances of PH$_{x}$ molecules and reflects that considering binding energy distributions is paramount for correctly merging microscopic and macroscopic modelling of non-thermal surface astrochemical processes. Finally, we found that PD molecules dissipate more of the reaction energy. This effect can be relevant for the deuterium fractionation and preferential distillation of molecules in the interstellar medium.

Experiment-driven atomistic materials modeling: A case study combining X-ray photoelectron spectroscopy and machine learning potentials to infer the structure of oxygen-rich amorphous carbon

Tigany Zarrouk; Rina Ibragimova; Albert P. Bartók; Miguel A. Caro. J. Am. Chem. Soc. 146, 14645 (2024) (2024). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/2402.03219v2

Abstract: An important yet challenging aspect of atomistic materials modeling is reconciling experimental and computational results. Conventional approaches involve generating numerous configurations through molecular dynamics or Monte Carlo structure optimization and selecting the one with the closest match to experiment.

However, this inefficient process is not guaranteed to succeed. We introduce a general method to combine atomistic machine learning (ML) with experimental observables that produces atomistic structures compatible with experiment by design. We use this approach in combination with grand-canonical Monte Carlo within a modified Hamiltonian formalism, to generate configurations that agree with experimental data and are chemically sound (low in energy). We apply our approach to understand the atomistic structure of oxygenated amorphous carbon (a-CO$_{x}$), an intriguing carbon-based material, to answer the question of how much oxygen can be added to carbon before it fully decomposes into CO and CO$_2$. Utilizing an ML-based X-ray photoelectron spectroscopy (XPS) model trained from GW and density functional theory (DFT) data, in conjunction with an ML interatomic potential, we identify a-CO$_{x}$ structures compliant with experimental XPS predictions that are also energetically favorable with respect to DFT. Employing a network analysis, we accurately deconvolve the XPS spectrum into motif contributions, both revealing the inaccuracies inherent to experimental XPS interpretation and granting us atomistic insight into the structure of a-CO$_{x}$. This method generalizes to multiple experimental observables and allows for the elucidation of the atomistic structure of materials directly from experimental data, thereby enabling experiment-driven materials modeling with a degree of realism previously out of reach.

When More Data Hurts: Optimizing Data Coverage While Mitigating Diversity Induced Underfitting in an Ultra-Fast Machine-Learned Potential

Jason B. Gibson; Tesia D. Janicki; Ajinkya C. Hire; Chris Bishop; J. Matthew D. Lane; Richard G. Hennig. arXiv preprint (2024). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/2409.07610v1

Abstract: Machine-learned interatomic potentials (MLIPs) are becoming an essential tool in materials modeling. However, optimizing the generation of training data used to parameterize the MLIPs remains a significant challenge. This is because MLIPs can fail when encountering local enviroments too different from those present in the training data. The difficulty of determining \textit{a priori} the environments that will be encountered during molecular dynamics (MD) simulation necessitates diverse, high-quality training data. This study investigates how training data diversity affects the performance of MLIPs using the Ultra-Fast Force Field (UF^3) to model amorphous silicon nitride. We employ expert and autonomously generated data to create the training data and fit four force-field variants to subsets of the data. Our findings reveal a critical balance in training data diversity: insufficient diversity hinders generalization, while excessive diversity can exceed the MLIP's learning capacity, reducing simulation accuracy. Specifically, we found that the UF^3 variant trained on a subset of the training data, in which nitrogen-rich structures were removed, offered vastly better prediction and simulation accuracy than any other variant. By comparing these UF^3 variants, we highlight the nuanced requirements for creating accurate MLIPs, emphasizing the importance of application-specific training data to achieve optimal performance in modeling complex material behaviors.

Large Scale Structure Prediction of Near-Stoichiometric Magnesium Oxide Based on a Machine-Learned Interatomic Potential: Novel Crystalline Phases and Oxygen-Vacancy Ordering

Hossein Tahmasbi; Stefan Goedecker; S. Alireza Ghasemi. Phys. Rev. Materials 5, 083806 (2021) (2021). Cited by 0. Found by: arxiv. DOI: 10.1103/PhysRevMaterials.5.083806

Abstract: Using a fast and accurate neural network potential we are able to systematically explore the energy landscape of large unit cells of bulk magnesium oxide with the minima hopping method. The potential is trained with a focus on the near-stoichiometric compositions, in particular on suboxides, i.e., Mg$_x$O$_{1-x}$ with $0.50<x<0.60$. Our extensive exploration demonstrates that for bulk stoichiometric compounds, there are several new low-energy rocksalt-like structures in which Mg atoms are octahedrally six–coordinated and form trigonal prismatic motifs with different stacking sequences. Furthermore, we find a dense spectrum of novel non-stoichiometric crystal phases of Mg$_x$O$_{1-x}$ for each composition of x. These structures are mostly similar to the rock salt structure with octahedral coordination and five–coordinated Mg atoms. Due to the removal of one oxygen atom, the energy landscape becomes more glass-like with oxygen-vacancy type structures that all lie very close to each other energetically. For the same number of magnesium and oxygen atoms our oxygen-deficient structures are lower in energy if the vacancies are aligned along lines or planes than rock salt structures with randomly distributed oxygen vacancies. We also found the putative global minima configurations for each composition of the non-stoichiometric suboxide structures. These structures are predominantly composed of (111) slabs of the rock salt structure which are terminated with Mg atoms at the top and bottom, and are stacked in different sequences along the z-direction. Like other Magnéli-type phases, these structures have properties that differ considerably from their stoichiometric counterparts such as low lattice thermal conductivity and high electrical conductivity.

Extracting Atomic Environments for Machine Learning Interatomic Potentials

Jared C. Stimac; Fei Zhou; Kyle Bushick; Bo Lei; Sebastien Hamel; Amit Samanta; Vincenzo Lordi. arXiv preprint (2026). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/2607.26018v2

Abstract: In order to appropriately capture large-scale material features and emergent phenomena via atomistic simulations, such as Molecular Dynamics (MD), the system scale can range up to hundreds of millions of atoms. However, the force-field models that drive those simulations are generally trained with Density Functional Theory (DFT) reference data, limited to relatively small configurations on the order of 100s or 1000s of atoms. To compute DFT forces on atoms in regions of interest, for example for active-learning or on-the-fly training of interatomic potentials, one needs to extract a small set of atoms from the larger simulation box, and typically work with periodic boundary conditions for DFT. However, methods to select the shape and size of this extracted set of atoms, as well as to generate a potentially necessary passivating envelope, have not been systematically analyzed. In this work, we benchmark several techniques, including a generative diffusion-based artificial intelligence (AI) approach, for extracting atomic environments from large, bulk configurations and embedding them into smaller configurations suitable for DFT calculations with periodic boundary conditions. We test with a diverse set of material systems, which includes amorphous $\mathrm{SiO_2}$, Ta with screw dislocations, and molten C. We demonstrated a notably simple procedure, a method we refer to as deletions, yields superior performance over an array of alternative extraction methods.

Particle Swarm Based Hyper-Parameter Optimization for Machine Learned Interatomic Potentials

Suresh Kondati Natarajan; Miguel A. Caro. arXiv preprint (2020). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/2101.00049v1

Abstract: Modeling non-empirical and highly flexible interatomic potential energy surfaces (PES) using machine learning (ML) approaches is becoming popular in molecular and materials research. Training an ML-PES is typically performed in two stages: feature extraction and structure-property relationship modeling. The feature extraction stage transforms atomic positions into a symmetry-invariant mathematical representation. This representation can be fine-tuned by adjusting on a set of so-called "hyper-parameters" (HPs). Subsequently, an ML algorithm such as neural networks or Gaussian process regression (GPR) is used to model the structure-PES relationship based on another set of HPs. Choosing optimal values for the two sets of HPs is critical to ensure the high quality of the resulting ML-PES model. In this paper, we explore HP optimization strategies tailored for ML-PES generation using a custom-coded parallel particle swarm optimizer (available freely at https://github.com/suresh0807/PPSO.git). We employ the smooth overlap of atomic positions (SOAP) descriptor in combination with GPR-based Gaussian approximation potentials (GAP) and optimize HPs for four distinct systems: a toy C dimer, amorphous carbon, $ $-Fe, and small organic molecules (QM9 dataset). We propose a two-step optimization strategy in which the HPs related to the feature extraction stage are optimized first, followed by the optimization of the HPs in the training stage. This strategy is computationally more efficient than optimizing all HPs at the same time by means of significantly reducing the number of ML models needed to be trained to obtain the optimal HPs. This approach can be trivially extended to other combinations of descriptor and ML algorithm and brings us another step closer to fully automated ML-PES generation.

An Accurate and Transferable Machine Learning Potential for Carbon

Patrick Rowe; Volker L Deringer; Piero Gasparotto; Gábor Csányi; Angelos Michaelides. arXiv preprint (2020). Cited by 0. Found by: arxiv. DOI: 10.1063/5.0005084

Abstract: We present an accurate machine learning (ML) model for atomistic simulations of carbon, constructed using the Gaussian approximation potential (GAP) methodology. The potential, named GAP-20, describes the properties of the bulk crystalline and amorphous phases, crystal surfaces and defect structures with an accuracy approaching that of direct ab initio simulation, but at a significantly reduced cost. We combine structural databases for amorphous carbon and graphene, which we extend substantially by adding suitable configurations, for example, for defects in graphene and other nanostructures. The final potential is fitted to reference data computed using the optB88-vdW density functional theory (DFT) functional. Dispersion interactions, which are crucial to describe multilayer carbonaceous materials, are therefore implicitly included. We additionally account for long-range dispersion interactions using a semianalytical two-body term and show that an improved model can be obtained through an optimisation of the many-body smooth overlap of atomic positions (SOAP) descriptor. We rigorously test the potential on lattice parameters, bond lengths, formation energies and phonon dispersions of numerous carbon allotropes. We compare the formation energies of an extensive set of defect structures, surfaces and surface reconstructions to DFT reference calculations. The present work demonstrates

the ability to combine, in the same ML model, the previously attained flexibility required for amorphous carbon [Phys. Rev. B, 95, 094203, (2017)] with the high numerical accuracy necessary for crystalline graphene [Phys. Rev. B, 97, 054303, (2018)], thereby providing an interatomic potential that will be applicable to a wide range of applications concerning diverse forms of bulk and nanostructured carbon.

Structure-property relations of silicon oxycarbides studied using a machine learning interatomic potential

Niklas Leimeroth; Jochen Rohrer; Karsten Albe. arXiv preprint (2024). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/2403.10154v1

Abstract: Silicon oxycarbides show outstanding versatility due to their highly tunable composition and microstructure. Consequently, a key challenge is a thorough knowledge of structure-property relations in the system. In this work, we fit an atomic cluster expansion potential to a set of actively learned DFT training data spanning a wide configurational space. We demonstrate the ability of the potential to produce realistic amorphous structures and rationalize the formation of different morphologies of the turbostratic free carbon phase. Finally, we relate the materials stiffness to its composition and microstructure, finding a delicate dependence on Si-C bonds that contradicts commonly assumed relations to the free carbon phase.

SiC-TGAP: A machine learning interatomic potential for radiation damage simulations in 3C-SiC

Ali Hamedani; Andrea E. Sand. arXiv preprint (2025). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/2510.06966v1

Abstract: Silicon carbide (SiC) has long been a subject of study for its application in harsh environments. Existing empirical interatomic potentials for 3C-SiC show significant discrepancies in predicting the properties that are crucial in describing the evolution of defects generated in collision cascades. We present a Gaussian approximation potential model for 3C-SiC (TGAP) trained by two-body and the turboSOAP many-body descriptors. The dataset covers crystalline, liquid and amorphous phases. To accurately capture defect dynamics, twenty-one defect types have been included in the dataset. TGAP captures the experimentally observed decomposition of carbon atoms in the liquid phase at atmospheric pressure, while also accurately reproducing the radial distribution function of the high-temperature homogeneous liquid phase across a range of densities. Moreover, it predicts the melting point in very good agreement with density functional theory and experiments. The potential is equipped with the Nordlund-Lehtola-Hobler repulsive potential to capture the high repulsion of recoils in the collision cascades. TGAP provides an accurate tool for atomistic simulation of radiation damage in cubic SiC.

Liquid anomalies and Fragility of Supercooled Antimony

Flavio Giuliani; Francesco Guidarelli Mattioli; Yuhan Chen; Daniele Dragoni; Marco Bernasconi; John Russo; Lilia Boeri; Riccardo Mazzarello. arXiv preprint (2025). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/2510.25920

Abstract: Phase-change materials (PCMs) based on group IV, V, and VI elements, such as Ge, Sb, and Te, exhibit distinctive liquid-state features, including thermodynamic anomalies and unusual dynamical properties, which are believed to play a key role in their fast and reversible crystallization behavior. Antimony (Sb), a monoatomic PCM with ultrafast switching capabilities, stands out as the only elemental member of this group for which the properties of the liquid and supercooled states have so far remained unknown. In this work, we use large-scale molecular dynamics simulations with a neural network potential trained on first-principles data to investigate the liquid, supercooled, and amorphous phases of Sb across a broad pressure-temperature range. We uncover clear signatures of anomalous behavior, including a density maximum and non-monotonic thermodynamic response functions, and introduce a novel octahedral order parameter that captures the structural evolution of the liquid. Moreover, extrapolation of the viscosity to the glass transition, based on configurational and excess entropies, indicates that Sb is a highly fragile material. Our results present a compelling new case for the connection between the liquid-state properties of phase-change materials and their unique ability to combine high amorphous-phase stability with ultrafast crystallization.

Unveiling hole-facilitated amorphisation in pressure-induced phase transformation of silicon

Tong Zhao; Shulin Zhong; Yuxin Sun; Defan Wu; Chunyi Zhang; Rui Shi; Hao Chen; Zhenyi Ni; Xiaodong Pi; Xiangyang Ma; Yunhao Lu; Deren Yang. arXiv preprint (2024). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/2412.04732v1

Abstract: Pressure-induced phase transformation occurs during silicon (Si) wafering processes. \b{eta}-tin (Si-II) phase is formed at high pressures, followed by the transformation to Si-XII, Si-III or/and amorphous Si (-Si) phases during the subsequent decompression. While the imposed pressure and its release rate are known to dictate the phase transformation of Si, the effect of charge carriers are ignored. Here, we experimentally unveil that the increased hole concentration facilitates the amorphization in the pressure-induced phase transformation of Si. The underlying mechanism is elucidated by the theoretical calculations based on machine-learning interatomic potentials. The hole-facilitated amorphization is also experimentally confirmed to occur in the indented Ge, GaAs or SiC. We discover that hole concentration is another determining factor for the pressure-induced phase transformations of the industrially important semiconductors.

Critical role of phase-dependent properties in modeling photothermal sintering of LiCoO2 cathodes

Yang Hu; Benoit Sklénard; Wouter Vels; Yaroslav E. Romanyuk; Vladyslav Turlo. arXiv preprint (2026). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/2604.21842v2

Abstract: Photothermal (photonic) sintering crystallizes as-deposited amorphous LiCoO2 (LCO) cathodes for solid-state thin-film batteries using millisecond, surface-localized heating. However, process design often relies on 1D models with phase-averaged, temperature-independent properties, which can mispredict peak temperatures and thermal damage margins. Here we develop a multiscale, data-driven framework that provides phase- and grain size-resolved thermophysical inputs for stoichiometric LCO. We train an Allegro neural network potential with near-ab initio accuracy, enabling Green-Kubo calculations of thermal conductivity for crystalline and amorphous phases. The low, weakly density-dependent conductivity of amorphous LCO motivates its use as an effective intergranular phase in a thin-interface model that reproduces observed grain-size-dependent thermal transport. Combined with measured wavelength-resolved optical properties in 1D multiphysics simulations, we show amorphous LCO absorbs more strongly and reaches higher peak temperatures than crystalline LCO; thus crystalline, constant-property models systematically overestimate safe operating windows.

Unifying the description of hydrocarbons and hydrogenated carbon materials with a chemically reactive machine learning interatomic potential

Rina Ibragimova; Mikhail S. Kuklin; Tigany Zarrouk; Miguel A. Caro. arXiv preprint (2024). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/2409.08194v1

Abstract: We present a general-purpose machine learning (ML) interatomic potential for carbon and hydrogen which is capable of simulating various materials and molecules composed of these elements. This ML interatomic potential is trained using the Gaussian approximation potential (GAP) framework and an extensive dataset of C-H configurations obtained from density functional theory. The dataset is constructed through iterative training and structure-search techniques that generate a broad range of configurations to comprehensively sample the potential energy surface. Furthermore, the dataset is supplemented with relevant bulk, molecular, and high-pressure structures. Finally, long-range van der Waals interactions are added as a locally parametrized model. The accuracy and generality of the potential are validated through the analysis of different simulations under a wide range of conditions, including weak interactions, high temperature, and high pressure. We show that our CH GAP model describes different problems such as the formation of simple and complex alkanes, aromatic hydrocarbons, hydrogenated amorphous carbon (a-C:H), and CH systems at extreme conditions, while retaining good accuracy for pure carbon materials. We use this model to generate hydrocarbons of different sizes and complexity without prior knowledge of organic chemistry rules, and to highlight intrinsic limitations to the simultaneous description on intra and intermolecular interactions within a single computational framework. Our general-purpose ML interatomic potential has the capability to significantly advance research in the field of H-containing carbon materials and compounds, particularly in the areas where longer dynamics, reactivity and large-scale effects may be important.

Unconventional crystallization pathway bypassing the intermediate cubic phase in phase-change superlattices

Bai-Qian Wang; Nian-Ke Chen; Yao-Jie Wang; Jia Sun; Yu-Ting Huang; Ming Xu; Shengbai Zhang; Xian-Bin Li. arXiv preprint (2026). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/2606.03309v1

Abstract: The Ge-Sb-Te (GST) superlattice phase-change material is a promising candidate for overcoming the high power-consumption of phase-change memory (PCM). However, the working mechanism of the superlattice PCM remains controversial. Partial amorphization, which is currently considered the most plausible mechanism, remains hotly debated: how does the partially amorphized GST recrystallize into its superlattice phase

instead of the conventionally expected cubic phase? Here, we address this issue using large-scale molecular dynamics simulations enabled by a machine-learning interatomic potential. Starting from a partially melted GST superlattice, we demonstrate that the residual crystalline regions serve as nuclei, enabling the amorphous GST to recrystallize directly into the superlattice phase without passing through the intermediate cubic phase. Moreover, the recrystallized phase is not an ideal superlattice, but rather a structurally ordered and chemically disordered defective superlattice characterized by anti-site defects and stacking faults. The defective superlattice region is also more susceptible to melting than the defect-free superlattice, and thereby can act as the active region of the PCM device. These results help to clarify the longstanding debates concerning the mechanism of superlattice-based PCM.

Modeling phase separation in polymer-derived silicon carbonitride ceramics through extended machine learning molecular dynamics

Fabien Mortier; Sylvian Cadars; Olivier Masson; Mauro Boero; Guido Ori; Yun Wang; Samuel Bernard; Assil Bouzid. arXiv preprint (2026). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/2605.20358v2

Abstract: Polymer-derived ceramics combine the thermal stability of ceramics with the versatile properties of carbon domains, but modeling their atomic-scale evolution during processing remains elusive due to the limitations of traditional computational methods. To address this issue, here we develop and apply a machine learning interatomic potential for silicon carbonitride-based (Si-C-N-H) systems, trained on a diversified database of over 9000 configurations – including amorphous models, high-temperature states, surfaces, and crystal structure predictions – to capture the full complexity of these materials. This potential enables large-scale molecular dynamics simulations of 8000-atom systems revealing the atomic-scale evolution of the polymer-derived ceramic during thermal treatment. A key result of this work is the occurrence of a phase separation where carbon domains progressively nucleate from the amorphous SiCN matrix during thermal processing, forming distinct graphene-like sheets while preserving the integrity of the ceramic network. The resulting models reproduce the experimental atomic pair distribution functions with exceptional fidelity, validating our approach and providing microscopic explanations for the material unique combination of ceramic and graphitic properties. In this process, defective 5- and/or 7-member carbon rings, mediate the transformation to stable 6-member aromatic structures. These findings offer new atomic-scale insights into the thermal stability and structural transformation pathways of polymer-derived ceramics, while our methodology opens avenues for studying complex amorphous systems with first-principles accuracy at experimentally relevant scales.

Understanding phase transitions of $ $-quartz under dynamic compression conditions by machine-learning driven atomistic simulations

Linus C. Erhard; Christoph Otzen; Jochen Rohrer; Clemens Prescher; Karsten Albe. npj Comput Mater 11, 58 (2025) (2024). Cited by 0. Found by: arxiv. DOI: 10.1038/s41524-025-01542-4

Abstract: Characteristic shock effects in silica serve as a key indicator of historical impacts at geological sites. Despite this geological significance, atomistic details of structural transformations under high pressure and shock compression remain poorly understood. This ambiguity is evidenced by conflicting experimental observations of both amorphization and crystallization transitions. Utilizing a newly developed machine-learning interatomic potential, we examine the response of $ $-quartz to shock compression with a peak pressure of 60 GPa over nano-second timescales. We initially observe amorphization before recrystallization into a d-NiAs-structured silica with disorder on the silicon sublattice, accompanied by the formation of domains with partial order of silicon. Investigating a variety of strain conditions enables us to identify the non-hydrostatic stress and strain states that allow the direct diffusionless formation of rosiaite-structured silica.

Simulation of the crystallization kinetics of Ge$_2$Sb$_2$Te$_5$ nanoconfined in superlattice geometries for phase change memories

Debdipto Acharya; Omar Abou El Kheir; Simone Marcorini; Marco Bernasconi. arXiv preprint (2025). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/2501.18370v1

Abstract: Phase change materials are the most promising candidates for the realization of artificial synapsis for neuromorphic computing. Different resistance levels corresponding to analogic values of the synapsis conductance can be achieved by modulating the size of an amorphous region embedded in its crystalline matrix. Recently, it has been proposed that a superlattice made of alternating layers of the phase change compound Sb$_2$Te$_3$ and of the TiTe$_2$ confining material allows for a better control of multiple intermediate resistance states and for a lower drift with time of the electrical resistance of the amorphous phase. In this work, we consider to substitute Sb$_2$Te$_3$ with the Ge$_2$Sb$_2$Te$_5$ prototypical phase change compound

that should feature better data retention. By exploiting molecular dynamics simulations with a machine learning interatomic potential, we have investigated the crystallization kinetics of Ge$_2$Sb$_2$Te$_5$ nanoconfined in geometries mimicking Ge$_2$Sb$_2$Te$_5$/TiTe$_2$ superlattices. It turns out that nanoconfinement induces a slight reduction in the crystal growth velocities with respect to the bulk, but also an enhancement of the nucleation rate due to heterogeneous nucleation. The results support the idea of investigating Ge$_2$Sb$_2$Te$_5$/TiTe$_2$ superlattices for applications in neuromorphic devices with improved data retention. The effect on the crystallization kinetics of the addition of van der Waals interaction to the interatomic potential is also discussed.

Local Structure Dictates Ionic Transport and Mechanical Properties in Glassy Solid Electrolytes for Lithium Batteries

Yong Li; Tao Du; Rasmus Christensen; Timothée Jamin; Zhencai Li; Qi Zhang; Xiaoyi Xu; Kasper Tolborg; Yuanzheng Yue; Morten M. Smedskjaer. arXiv preprint (2026). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/2608.06895v1

Abstract: Electrolytes composed of sulfide and halide glasses are promising candidates for all-solid-state lithium batteries owing to their processability, lack of grain boundaries, and relatively high ionic conductivity. Nevertheless, their ionic conductivity and mechanical properties are still not satisfying for the real-world applications. Significant advances in solid electrolytes require a thorough understanding of their microstructures. Here, we reveal the connections among structure, ionic transport properties, and mechanical stability in a series of glassy solid electrolytes by employing molecular dynamics simulations based on a machine learning interatomic potential. Specifically, we explore how the interplay between B-S and P-S networks in glassy Li-S-P-B-I (LSPBI) governs ionic conductivity and deformation behavior. The introduction of P2S5 into a B2S3-based glass induces a critical structural transformation, through which both ionic conductivity and mechanical nano-ductility can be enhanced. For a moderate P2S5 content, incorporated PS4 units depolymerize the rigid boron framework, creating percolative diffusion pathways for fast ionic transport. Concurrently, the flexible P-S-P configurations enable energy dissipation through bond bending, leading to the brittle-to-ductile transition. However, excessive P2S5 increases the fraction of polyphosphates (e.g., P2S6 and P2S7), thereby polymerizing the structural network and ultimately impeding Li+ mobility. Our work thus provides atomistic principles for engineering glass electrolytes with balanced ionic conductivity and mechanical robustness.

Machine learning exploration of binding energy distributions of H2O at astrochemically relevant dust grain surfaces

Anant Vaishnav; Niels M. Mikkelsen; Mie Andersen. arXiv preprint (2026). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/2602.11050v1

Abstract: Binding energies (BEs) of adsorbates on interstellar dust grains critically control adsorption, desorption, diffusion, and surface reactivity, and therefore strongly influence astrochemical models of star- and planet-forming regions. While recent computational studies increasingly report full distributions of BEs rather than single representative values, these distributions are typically derived for either bare grain surfaces or thick water-ice mantles. In this work, we bridge these regimes by systematically investigating the BE distributions of water on partially and fully ice-covered dust grain surfaces. We employ machine-learning interatomic potentials (MLIPs) based on graph neural networks to model water adsorption on graphene and on the Mg-terminated (010) surface of forsterite, representing carbonaceous and silicate grains, respectively. The models enable extensive sampling of adsorption sites on water clusters, monolayers, and bilayers generated under both crystalline (thermally processed) and amorphous (low-temperature) growth conditions. At submonolayer coverage, the chemical nature of the underlying grain strongly affects both ice morphology and binding energies, with Mg-O interactions on silicate surfaces producing particularly deep binding sites. From monolayer coverage onward, adsorption on both substrates is dominated by hydrogen bonding within the ice, reducing the influence of the grain material. Across all coverages, amorphous ice structures systematically shift the BE distributions toward stronger binding compared to crystalline ice, introducing highly stable defect and pocket sites. These results demonstrate that BE distributions in the submonolayer to few-layer ice regime are broad and highly surface dependent, and they provide physically motivated input for next-generation astrochemical models incorporating surface heterogeneity.

Atomic cluster expansion for quantum-accurate large-scale simulations of carbon

Minaam Qamar; Matous Mrovec; Yury Lysogorskiy; Anton Bochkarev; Ralf Drautz. arXiv preprint (2022). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/2210.09161v4

Abstract: We present an atomic cluster expansion (ACE) for carbon that improves over available classical and machine learning potentials. The ACE is parameterized from an exhaustive set of important carbon structures at extended volume and energy range, computed using density functional theory (DFT). Rigorous validation reveals that ACE predicts accurately a broad range of properties of both crystalline and amorphous carbon phases while being several orders of magnitude more computationally efficient than available machine learning models. We demonstrate the predictive power of ACE on three distinct applications, brittle crack propagation in diamond, evolution of amorphous carbon structures at different densities and quench rates and nucleation and growth of fullerene clusters under high pressure and temperature conditions.

A high-efficiency neuroevolution potential for tobermorite and calcium silicate hydrate systems with ab initio accuracy

Xiao Xu; Shijie Wang; Haifeng Qin; Zhiqiang Zhao; Zheyong Fan; Zhuhua Zhang; Hang Yin. arXiv preprint (2025). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/2505.18993v1

Abstract: Tobermorite and Calcium Silicate Hydrate (C-S-H) systems are indispensable cement materials but still lack a satisfactory interatomic potential with both high accuracy and high computational efficiency for better understanding their mechanical performance. Here, we develop a Neuroevolution Machine Learning Potential (NEP) with Ziegler-Biersack-Littmark hybrid framework for tobermorite and C-S-H systems, which conveys unprecedented efficiency in molecular dynamics simulations with substantially reduced training datasets. Our NEP model achieves prediction accuracy comparable to DFT calculations using just around 300 training structures, significantly fewer than other existing machine learning potentials trained for tobermorite. Critically, the GPU-accelerated NEP computations enable scalable simulations of large tobermorite systems, reaching several thousand atoms per GPU card with high efficiency. We demonstrate the NEP's versatility by accurately predicting mechanical properties, phonon density of states, and thermal conductivity of tobermorite. Furthermore, we extend the NEP application to large-scale simulations of amorphous C-S-H, highlighting its potential for comprehensive analysis of structural and mechanical behaviors under various realistic conditions.

How Realistic are Idealized Copper Surfaces? A Machine Learning Study of Rough Copper-Water Interfaces

Linus C. Erhard; Johannes Schörghuber; Aleix Comas-Vives; Georg K. H. Madsen. arXiv preprint (2025). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/2509.17833v1

Abstract: Copper is a highly promising catalyst for the electrochemical CO$_2$ reduction reaction (CO2RR) since it is the only pure metal that can form highly added-value products such as ethylene and ethanol. Since the CO2RR takes place in aqueous solution, the detailed atomic structure of the water-copper interface is essential for unraveling the key reaction mechanisms. In this study, we investigate copper-water interfaces exhibiting nanometer-scale roughnesses. We introduce two molecular dynamics protocols to create rough copper surfaces, which are subsequently brought into contact with water. From these interfaces, we sample additional training configurations from machine-learning-interatomic-potential-driven molecular dynamics simulations containing hundreds of thousands of atoms. An active learning workflow is developed to identify regions with high spatially resolved uncertainty and convert them into DFT-feasible cells through a modified amorphous matrix embedding approach. Finally, we analyze the local environments at the interface using unsupervised machine-learning techniques. Unique environments emerge on the rough copper surfaces absent from model systems, including stacking-fault-induced configurations and undercoordinated corner atoms. Notably, corner atoms consistently feature chemisorbed water molecules in our simulations, indicating their potential importance in catalytic processes.

Softening of Vibrational Modes and Anharmonicity Induced Thermal Conductivity Reduction in a-Si:H at High Temperatures

Zhuo Chen; Yuejin Yuan; Yanzhou Wang; Penghua Ying; Shouhang Li; Cheng Shao; Wenyang Ding; Gang Zhang; Meng An. arXiv preprint (2025). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/2502.05584v2

Abstract: Hydrogenated amorphous silicon (a-Si:H) has garnered considerable attention in the semiconductor industry, particularly for its use in solar cells and passivation layers for high performance silicon solar cells, owing to its exceptional photoelectric properties and scalable manufacturing processes. A comprehensive understanding of thermal transport mechanism in a-Si:H is essential for optimizing thermal management and ensuring the reliable operation of these devices. In this study, we developed a neuroevolution machine learning potential based on first-principles calculations of energy, forces, and virial, which enables accurate modeling of interatomic interactions in both a-Si:H and a-Si systems. Using the homogeneous nonequilibrium molecular

dynamics (HNEMD) method, we systematically investigated the thermal conductivity of a-Si:H and a-Si across a temperature range of 300-1000 K and hydrogen concentrations ranging from 6 to 12 at%. Our simulation results found that thermal conductivity of a-Si:H with 12 at% hydrogen was significantly reduced by 12% compared to that of a-Si at 300 K. We analyzed the spectral thermal conductivity, vibrational density of states and lifetimes of vibrational modes, and revealed the softening of vibrational modes and anharmonicity effects contribute to the reduction of thermal conductivity as temperature and hydrogen concentration increase. Furthermore, the influence of hydrogen concentration and temperature on diffuson and propagon contribution to thermal conductivity of a-Si:H was revealed. This study provides valuable insights for developing thermal management strategies in silicon-based semiconducting devices and advances the understanding of thermal transport in amorphous systems.

How to Train a Shallow Ensemble

Moritz Schäfer; Matthias Kellner; Johannes Kästner; Michele Ceriotti. arXiv preprint (2026). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/2602.15747v1

Abstract: Shallow ensembles provide a convenient strategy for uncertainty quantification in machine learning interatomic potentials, that is computationally efficient because the different ensemble members share a large part of the model weights. In this work, we systematically investigate training strategies for shallow ensembles to balance calibration performance with computational cost. We first demonstrate that explicit optimization of a negative log-likelihood (NLL) loss improves calibration with respect to approaches based on ensembles of randomly initialized models, or on a last-layer Laplace approximation. However, models trained solely on energy objectives yield miscalibrated force estimates. We show that explicitly modeling force uncertainties via an NLL objective is essential for reliable calibration, though it typically incurs a significant computational overhead. To address this, we validate an efficient protocol: full-model fine-tuning of a shallow ensemble originally trained with a probabilistic energy loss, or one sampled from the Laplace posterior. This approach results in negligible reduction in calibration quality compared to training from scratch, while reducing training time by up to 96%. We evaluate this protocol across a diverse range of materials, including amorphous carbon, ionic liquids (BMIM), liquid water (H$_2$O), barium titanate (BaTiO$_3$), and a model tetrapeptide (Ac-Ala3-NHMe), establishing practical guidelines for reliable uncertainty quantification in atomistic machine learning.

Atomistic Structure Generation and Neural-Network Screening of Hard Carbons to Identify High-Capacity Sodium Storage

Harry Mclean; Aiden Daniel Emery; Theodore Thomas Walton; Ned Thaddeus Taylor; Steven Paul Hepplestone. arXiv preprint (2026). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/2608.17716v1

Abstract: Hard carbons are established anodes for lithium-ion batteries and leading candidates for sodium-ion batteries, yet their electrochemical performance is governed by a heterogeneous network of graphitic domains, defects, and nanopores that conventional atomistic methods cannot model at the required length scales. We combine universal machine-learned interatomic potentials with the RAFFLE structure-generation framework to construct 13,096 realistic hard carbon models containing up to 4,378 atoms, matching experimentally measured densities, porosities, and sp^2/sp^3 bonding fractions. Explicit sodium intercalation of representative structures reproduces the characteristic sloping-to-plateau voltage profiles, revealing that capacity increases with decreasing carbon density and increasing porosity. To screen the full library, we train a lightweight neural-network surrogate that predicts capacity directly from the host using frozen universal-potential descriptors augmented by geometric void features. The surrogate identifies high-capacity candidates exceeding 800 mAh g^{-1}, which are validated by full intercalation calculations. This scalable framework links hard carbon microstructure to sodium-storage performance and provides atomistic design principles for high-capacity anodes. More broadly, the workflow enables systematic exploration of synthesis-dependent amorphous microstructures, including precursor chemistry, pyrolysis, heteroatom doping, and pore engineering, providing a route toward atomistically informed hard carbon design.

Nuclear Quantum Effects in the Acetylene:Ammonia Plastic Co-crystal

Atul C. Thakur; Richard C. Remsing. arXiv preprint (2023). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/2310.00

Abstract: Organic molecular solids can exhibit rich phase diagrams. In addition to structurally unique phases, translational and rotational degrees of freedom can melt at different state points, giving rise to partially disordered solid phases. The structural and dynamic disorder in these materials can have a significant impact on the physical properties of the organic solid, necessitating a thorough understanding of disorder at the atomic scale. When these disordered phases form at low temperatures, especially in crystals with light nuclei, the

prediction of materials properties can be complicated by the importance of nuclear quantum effects. As an example, we investigate nuclear quantum effects on the structure and dynamics of the orientationally-disordered, translationally-ordered plastic phase of the acetylene:ammonia (1:1) co-crystal that is expected to exist on the surface of Saturn's moon Titan. Titan's low surface temperature (~90 K) suggests that the quantum mechanical behavior of nuclei may be important in this and other molecular solids in these environments. By using neural network potentials combined with ring polymer molecular dynamics simulations, we show that nuclear quantum effects increase orientational disorder and rotational dynamics within the acetylene:ammonia (1:1) co-crystal by weakening hydrogen bonds. Our results suggest that nuclear quantum effects are important to accurately model molecular solids and their physical properties in low temperature environments.

The transformation mechanisms among cuboctahedra, Ino's decahedra and icosahedra structures of magic-size gold nanoclusters

Ehsan Rahmatizad Khajehpasha; Mohammad Ismaeil Safa; Nasrin Eyvazi; Marco Krummenacher; Stefan Goedecker. arXiv preprint (2026). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/2601.10434v1

Abstract: Gold nanoclusters possess multiple competing structural motifs with small energy differences, enabling structural coexistence and interconversion. Using a high-accuracy machine learned potential trained on some 20'000 density functional theory reference data points, we investigate transformation pathways connecting both high-symmetry and amorphous cuboctahedra, Ino's decahedra and icosahedra for Au55, Au147, Au309 and Au561 nanoclusters. Our saddle point searches reveal that high-symmetry transformations from cuboctahedra and Ino's decahedra to icosahedra proceed through a single barrier and represent soft-mode-driven jitterbug-type and slip-dislocation motions. In addition, we identify lower-barrier asymmetric transformation pathways that drive the system into disordered, Jahn-Teller-stabilized amorphous icosahedra. Minima Hopping sampling further uncovers, in this context, many such low-symmetry minima. Some of the newly identified global minima for Au309 and Au561 have energies that are up to 2.8 eV lower than the previously reported global minima. Hence, both the shapes and the transformation pathways studied in previous investigations are not the physically relevant ones. In contrast to the previously studied pathways, our transformation pathways give reasonable transformation times that are in rough agreement with experiments.

Advances in modeling complex materials: The rise of neuroevolution potentials

Penghua Ying; Cheng Qian; Rui Zhao; Yanzhou Wang; Feng Ding; Shunda Chen; Zheyong Fan. Chem. Phys. Rev. 6, 011310 (2025) (2025). Cited by 0. Found by: arxiv. DOI: 10.1063/5.0259061

Abstract: Interatomic potentials are essential for driving molecular dynamics (MD) simulations, directly impacting the reliability of predictions regarding the physical and chemical properties of materials. In recent years, machine-learned potentials (MLPs), trained against first-principles calculations, have become a new paradigm in materials modeling as they provide a desirable balance between accuracy and computational cost. The neuroevolution potential (NEP) approach, implemented in the open-source GPUMD software, has emerged as a promising machine-learned potential, exhibiting impressive accuracy and exceptional computational efficiency. This review provides a comprehensive discussion on the methodological and practical aspects of the NEP approach, along with a detailed comparison with other representative state-of-the-art MLP approaches in terms of training accuracy, property prediction, and computational efficiency. We also demonstrate the application of the NEP approach to perform accurate and efficient MD simulations, addressing complex challenges that traditional force fields typically can not tackle. Key examples include structural properties of liquid and amorphous materials, chemical order in complex alloy systems, phase transitions, surface reconstruction, material growth, primary radiation damage, fracture in two-dimensional materials, nanoscale tribology, and mechanical behavior of compositionally complex alloys under various mechanical loadings. This review concludes with a summary and perspectives on future extensions to further advance this rapidly evolving field.

NEP89: Universal neuroevolution potential for inorganic and organic materials across 89 elements

Ting Liang; Ke Xu; Eric Lindgren; Zherui Chen; Rui Zhao; Jiahui Liu; Esmée Berger; Benrui Tang; Bohan Zhang; Yanzhou Wang; Keke Song; Penghua Ying; Nan Xu; Haikuan Dong; Shunda Chen; Paul Erhart; Zheyong Fan; Tapio Ala-Nissila; Jianbin Xu. Nature Computational Science (2026) (2025). Cited by 0. Found by: arxiv. DOI: 10.1038/s43588-026-01009-6

Abstract: While machine-learned interatomic potentials offer near-quantum-mechanical accuracy for atomistic simulations, many are material-specific or computationally intensive, limiting their broader use. Here, we introduce NEP89, a foundation model based on neuroevolution potential architecture, delivering near-empirical-potential speed and high accuracy across 89 elements. A compact yet comprehensive training dataset covering inorganic and organic materials was curated through descriptor-space subsampling and iterative refinement across multiple datasets. NEP89 achieves competitive accuracy compared to representative foundation models while being three to four orders of magnitude more computationally efficient, enabling previously impractical large-scale atomistic simulations of inorganic and organic systems. In addition to its out-of-the-box applicability to diverse scenarios, including million-atom-scale compression of compositionally complex alloys, ion diffusion in solid-state electrolytes and water, rocksalt dissolution, methane combustion, and protein-ligand dynamics, NEP89 also supports fine-tuning for rapid adaptation to user-specific applications, such as mechanical, thermal, structural, and spectral properties of two-dimensional materials, metallic glasses, and organic crystals.

Reconstructions and Dynamics of $ $-Lithium Thiophosphate Surfaces

Hanna Türk; Davide Tisi; Michele Ceriotti. PRX Energy 4(3), 033010 (2025) (2025). Cited by 0. Found by: arxiv. DOI: 10.1103/5hf9-hlj6

Abstract: Lithium thiophosphate (LPS) is a promising solid electrolyte for next-generation lithium-ion batteries due to its superior energy storage, high ionic conductivity, and low-flammability components. Despite its potential, the high reactivity of LPS with common contaminants such as atmospheric water, preparation solvents, and electrode materials poses significant challenges for commercialization. The lack of understanding regarding the structure, morphology, and chemical behavior of LPS's surface slows down the search for solutions to these issues. Here, we utilize a machine learning interatomic potential to achieve a fundamental, atomistic understanding of the mechanical and chemical properties of the $ $-Li$_3PS_4$ surfaces. Employing molecular dynamics simulations, we identify relevant surface complexions formed by surface reconstructions, determine their surface energies and compute the Wulff shape of $ $-LPS. The most stable complexions exhibit properties distinctly different from the bulk, including amorphization, increased density, decreased conductivity and large deformation of the structure building blocks. We demonstrate that these surfaces are not static, but undergo significant dynamical activity which is clearly identified by an analysis featuring a time-averaged structural descriptor. Finally, we examine the changes of the electronic structure induced by the surface complexions, which provides us with details on changes in surface reactivity and active sites, underlining the importance to investigate surface complexions under realistic conditions.

Single-model uncertainty quantification in neural network potentials does not consistently outperform model ensembles

Aik Rui Tan; Shingo Urata; Samuel Goldman; Johannes C. B. Dietschreit; Rafael Gómez-Bombarelli. arXiv preprint (2023). Cited by 0. Found by: arxiv. DOI: 10.1038/s41524-023-01180-8

Abstract: Neural networks (NNs) often assign high confidence to their predictions, even for points far out-of-distribution, making uncertainty quantification (UQ) a challenge. When they are employed to model interatomic potentials in materials systems, this problem leads to unphysical structures that disrupt simulations, or to biased statistics and dynamics that do not reflect the true physics. Differentiable UQ techniques can find new informative data and drive active learning loops for robust potentials. However, a variety of UQ techniques, including newly developed ones, exist for atomistic simulations and there are no clear guidelines for which are most effective or suitable for a given case. In this work, we examine multiple UQ schemes for improving the robustness of NN interatomic potentials (NNIPs) through active learning. In particular, we compare incumbent ensemble-based methods against strategies that use single, deterministic NNs: mean-variance estimation, deep evidential regression, and Gaussian mixture models. We explore three datasets ranging from in-domain interpolative learning to more extrapolative out-of-domain generalization challenges: rMD17, ammonia inversion, and bulk silica glass. Performance is measured across multiple metrics relating model error to uncertainty. Our experiments show that none of the methods consistently outperformed each other across the various metrics. Ensembling remained better at generalization and for NNIP robustness; MVE only proved effective for in-domain interpolation, while GMM was better out-of-domain; and evidential regression, despite its promise, was not the preferable alternative in any of the cases. More broadly, cost-effective, single deterministic models cannot yet consistently match or outperform ensembling for uncertainty quantification in NNIPs.

Revisit Many-body Interaction Heat Current and Thermal Conductivity Calculation in Moment Tensor Potential/LAMMPS Interface

Siu Ting Tai; Chen Wang; Ruihuan Cheng; Yue Chen. arXiv preprint (2024). Cited by 0. Found by: arxiv. DOI: 10.1021/acs.jctc.4c01659

Abstract: The definition of heat current operator for systems for non-pairwise additive interactions and its impact on related lattice thermal conductivity ($ _{L}$) via molecular dynamics simulation (MD) are ambiguous and controversial when migrating from conventional empirical potential models to machine learning potential (MLP) models. Empirical model descriptions are often limited to three- to four-body interaction while a sophisticated representation of the many-body physics could be resembled in MLPs. Herein, we study and compare the significance of many-body interaction to the heat current computation in one of the most popular MLP models, the Moment Tensor Potential (MTP). Non-equilibrium MD simulations and equilibrium MD simulations among four different materials, $PbTe$, amorphous $Sc_{0.2}Sb_{2}Te_{3}$, graphene, and BAs, were performed. We found inconsistency between the simulation thermostat and its implemented heat current operator in our non-equilibrium MD results which violate law of energy conservation and suggest a need for revision. We revisit the virial stress tensor expression within the calculator and identified the lack of a generalised many-body heat current description in it. We uncover the influence of the modified heat current formula that could alter the $ _{L}$ results 29% to 64% using the equilibrium MD computational approach. Our work demonstrates the importance of a many-body description during thermal analysis in MD simulations when MLPs are in concern. This work sheds light on a better understanding of the relationship between interatomic interaction and its heat transport mechanism.

Million-atom simulation of the set process in phase change memories at the real device scale

Omar Abou El Kheir; Marco Bernasconi. arXiv preprint (2025). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/2501.07384v1

Abstract: Phase change materials are exploited in several enabling technologies such as storage class memories, neuromorphic devices and memories embedded in microcontrollers. A key functional property for these applications is the fast crystal nucleation and growth in the supercool liquid phase. Over the last decade, atomistic simulations based on density functional theory (DFT) have provided crucial insights on the early stage of this process. These simulations are, however, restricted to a few hundred atoms for at most a few ns. More recently, the scope of the DFT simulations have been greatly extended by leveraging on machine learning techniques. In this paper, we show that the exploitation of a recently devised neural network potential for the prototypical phase change compound Ge$_2$Sb$_2$Te$_5$, allows simulating the crystallization process in a multimillion atom model at the length and time scales of the real memory devices. The simulations provide a vivid atomistic picture of the subtle interplay between crystal nucleation and crystal growth from the crystal/amorphous rim. Moreover, the simulations have allowed quantifying the distribution of point defects controlling electronic transport, in a very large crystallite grown at the real conditions of the set process of the device.

Cluster expansion constructed over Jacobi-Legendre polynomials for accurate force fields

Michelangelo Domina; Urvesh Patil; Matteo Cobelli; Stefano Sanvito. arXiv preprint (2022). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/2208.10292v3

Abstract: We introduce a compact cluster expansion method, constructed over Jacobi and Legendre polynomials, to generate highly accurate and flexible machine-learning force fields. The constituent many-body contributions are separated, interpretable and adaptable to replicate the physical knowledge of the system. In fact, the flexibility introduced by the use of the Jacobi polynomials allows us to impose, in a natural way, constrains and symmetries to the cluster expansion. This has the effect of reducing the number of parameters needed for the fit and of enforcing desired behaviours of the potential. For instance, we show that our Jacobi-Legendre cluster expansion can be designed to generate potentials with a repulsive tail at short inter-atomic distances, without the need of imposing any external function. Our method is here continuously compared with available machine-learning potential schemes, such as the atomic cluster expansion and potentials built over the bispectrum. As an example we construct a Jacobi-Legendre potential for carbon, by training a slim and accurate model capable of describing crystalline graphite and diamond, as well as liquid and amorphous elemental carbon.

Nucleation mechanism for the direct graphite-to-diamond phase transition

Rustam Z. Khaliullin; Hagai Eshet; Thomas D. Kuhne; Jorg Behler; Michele Parrinello. arXiv preprint (2011). Cited by 0. Found by: arxiv. DOI: 10.1038/nmat3078

Abstract: Graphite and diamond have comparable free energies, yet forming diamond from graphite is far from easy. In the absence of a catalyst, pressures that are significantly higher than the equilibrium coexistence pressures are required to induce the graphite-to-diamond transition. Furthermore, the formation of the metastable hexagonal polymorph of diamond instead of the more stable cubic diamond is favored at lower temperatures. The concerted mechanism suggested in previous theoretical studies cannot explain these phenomena. Using an ab initio quality neural-network potential we performed a large-scale study of the graphite-to-diamond transition assuming that it occurs via nucleation. The nucleation mechanism accounts for the observed phenomenology and reveals its microscopic origins. We demonstrated that the large lattice distortions that accompany the formation of the diamond nuclei inhibit the phase transition at low pressure and direct it towards the hexagonal diamond phase at higher pressure. The nucleation mechanism proposed in this work is an important step towards a better understanding of structural transformations in a wide range of complex systems such as amorphous carbon and carbon nanomaterials.

Structure and solidification of the (Fe0.75B0.15Si0.1)100-xTax (x=0-2) melts: experiment and machine learning

I. V. Sterkhova; L. V. Kamaeva; V. I. Ladyanov; N. M. Chtchelkatchev. arXiv preprint (2022). Cited by 0. Found by: arxiv. DOI: 10.1016/j.jpcs.2022.111143

Abstract: Fe-B-Si system is a matrix for synthesis of new functional materials with exceptional magnetic and mechanical properties. Progress in this area is associated with the search for optimal doping conditions. This theoretical and experimental study is aimed to address the influence of Ta alloying on the structure of undercooled (Fe0.75B0.15Si0.1)100-xTax (x=0-2) melts, their undercoolability and the processes of structure formation during solidification. Small concentration of Ta complicates standard ab initio and machine learning investigations. We developed a technique for fast and stable training of machine learning interatomic potential (MLIP) in this case and uncovered the structure of the undercooled melts. Molecular dynamic simulations with MLIP showed that at Ta concentration of 1 at.% there is a sharp change in the chemical short-range ordering in the melt associated with a change in the interaction of Ta atoms. This effect leads to a restructuring of the cluster formation in the system. At the same time, our experimental investigation shows that melts with a Ta content of 1 at.% have the greatest tendency to undercoolability. Alloying with Ta promotes the formation of primary crystals of Fe2B, and at a concentration of more than 1.5 at.% Ta, also of FeTaB. Herewith, near 1 at.% Ta, the crystallization of the melt proceeds nontrivially: with the formation of two intermediate metastable phases Fe3B and Fe2Ta Laves phase. Also, the highest tendency to amorphization under conditions of quick quenching is exhibited by a melt with a Ta concentration of 1 at.%. The results not only provide understanding of optimal alloying of Fe-B-Si materials but also promote a machine learning method for numerical design of metallic alloys with a small dopant concentration.

An experimentally validated end-to-end framework for operando modeling of intrinsically complex metallosilicates

Jong Hyun Jung; Tom Schächtel; Yongliang Ou; Selina Itzigehl; Marc Högler; Niels Hansen; Johanna R. Bruckner; Blazej Grabowski. arXiv preprint (2025). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/2512.02309v2

Abstract: Structurally and chemically complex materials such as amorphous metallosilicates underpin major catalytic and separation technologies, yet their intrinsic complexity challenges reliable atomistic modeling under realistic conditions. Consequently, simulations that connect composition to material properties remain largely inaccessible for these materials. Here, we enable quantitative operando atomistic modeling of intrinsically complex materials through an experimentally validated end-to-end computational framework. The approach combines separation of simulation domains, lightweight machine-learning potentials trained on high-fidelity data, and large-scale de novo in silico synthesis that mimics experimental procedures. We apply the framework to realistic mesoporous SiO$_2$(Al$_2$O$_3$)$\_{x/2}$ (0 $\leq x \leq$ 0.4) and validate the results experimentally. Simulations quantitatively reproduce multiple experimental observables, including bulk densities, pair distribution functions, infrared spectra, and hydroxyl densities. Beyond prediction, the framework enables analysis of acid sites and vibrations for catalytic and adsorption processes. By integrating simulation and experiment within a unified workflow, we advance the realism and reliability of atomistic modeling for intrinsically complex materials.

Accelerated Discovery of Crystalline Materials with Record Ultralow Lattice Thermal Conductivity via a Universal Descriptor

Xingchen Shen; Jiongzhi Zheng; Michael Marek Koza; Petr Levinsky; Jiri Hejtmanek; Philippe Boullay; Bernard Raveau; Jinghui Wang; Jun Li; Pierric Lemoine; Christophe Candolfi; Emmanuel Guilmeau. arXiv preprint (2025). Cited by 0. Found by: arxiv. DOI: 10.1038/s41467-025-67333-z

Abstract: Ultralow glass-like lattice thermal conductivity in crystalline materials is crucial for enhancing energy conversion efficiency in thermoelectrics and thermal insulators. We introduce a universal descriptor for thermal conductivity that relies only on the atomic number in the primitive cell and the sound velocity, enabling fast and scalable materials screening. Coupled with high-throughput workflows and universal machine learning

potentials, we identify the candidate materials with ultralow thermal conductivity from over 25, 000 materials. We further validate this approach by experimentally confirming record-low thermal conductivity values of 0.15-0.16 W/m/K from 170 to 400 K in the halide metal CsAg2I3. Combining inelastic neutron scattering with first-principles calculations, we attribute the ultralow thermal conductivity to the intrinsically small sound velocity, strong anharmonicity, and structural complexity. Our work illustrates how a universal descriptor, combined with high-throughput screening, machine-learning potential and experiment, enables the efficient discovery of materials with ultralow thermal conductivity.

Cluster Models for Next-Generation, Machine-Learning-Based Energy Functions for Molecular Simulations

JingChun Wang; Meenu Upadhyay; Eric D. Boittier; Kham Lek Chaton; Valerii Andreichev; Mike Devereux; Shimoni Patel; Sena Aydin; Kai Töpfer; Markus Meuwly. arXiv preprint (2025). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/2509.11639v1

Abstract: Energy functions for pure and heterogenous systems are one of the backbones for molecular simulation of condensed phase systems. With the advent of machine learned potential energy surfaces (ML-PESs) a new era has started. Statistical models allow the representation of reference data from electronic structure calculations for chemical systems of almost arbitrary complexity at unprecedented detail and accuracy. Here, kernel- and neural network-based approaches for intramolecular degrees of freedom are combined with distributed charge models for long range electrostatics to describe the interaction energies of condensed phase systems. The main focus is on illustrative examples ranging from pure liquids (dichloromethane, water) to chemically and structurally heterogeneous systems (eutectic liquids, CO on amorphous solid water), reactions (Menshutkin), and spectroscopy (triatomic probes for protein dynamics). For all examples, small to medium-sized clusters are used to represent and improve the total interaction energy compared with reference quantum chemical calculations. Although remarkable accuracy can be achieved for some systems (chemical accuracy for dichloromethane and water), it is clear that more realistic models are required for van der Waals contributions and improved water models need to be used for more quantitative simulations of heterogeneous chemical and biological systems.

Tuning the lattice thermal conductivity in van-der-Waals structures through rotational (dis)ordering

Fredrik Eriksson; Erik Fransson; Christopher Linderälv; Zheyong Fan; Paul Erhart. ACS Nano 17, 25565 (2023) (2023). Cited by 0. Found by: arxiv. DOI: 10.1021/acsnano.3c09717

Abstract: It has recently been demonstrated that MoS2 with irregular interlayer rotations can achieve an extreme anisotropy in the lattice thermal conductivity (LTC), which is for example of interest for applications in waste heat management in integrated circuits. Here, we show by atomic scale simulations based on machine-learned potentials that this principle extends to other two-dimensional materials including C and BN. In all three materials introducing rotational disorder drives the through-plane LTC to the glass limit, while the in-plane LTC remains almost unchanged compared to the ideal bulk materials. We demonstrate that the ultralow through-plane LTC is connected to the collapse of their transverse acoustic modes in the through-plane direction. Furthermore, we find that the twist angle in periodic moiré structures representing rotational order provides an efficient means for tuning the through-plane LTC that operates for all chemistries considered here. The minimal through-plane LTC is obtained for angles between 1 and 4 degree depending on the material, with the biggest effect in MoS2. The angular dependence is correlated with the degree of stacking disorder in the materials, which in turn is connected to the slip surface. This provides a simple descriptor for predicting the optimal conditions at which the LTC is expected to become minimal.

Sub-micrometer phonon mean free paths in metal-organic frameworks revealed by machine-learning molecular dynamics simulations

Penghua Ying; Ting Liang; Ke Xu; Jin Zhang; Jianbin Xu; Zheng Zhong; Zheyong Fan. ACS Appl.Mater.Interfaces, 2023,15, 36412 (2023). Cited by 0. Found by: arxiv. DOI: 10.1021/acsami.3c07770

Abstract: Metal-organic frameworks (MOFs) are a family of materials that have high porosity and structural tunability and hold great potential in various applications, many of which requiring a proper understanding of the thermal transport properties. Molecular dynamics (MD) simulations play an important role in characterizing the thermal transport properties of various materials. However, due to the complexity of the structures, it is difficult to construct accurate empirical interatomic potentials for reliable MD simulations of MOFs. To this end, we develop a set of accurate yet highly efficient machine-learned potentials for three typical MOFs, including MOF-5, HKUST-1, and ZIF-8, using the neuroevolution potential approach as implemented in the GPUMD package, and perform extensive MD simulations to study thermal transport in the three MOFs. Although the lattice thermal conductivity (LTC) values of the three MOFs are all predicted to be smaller than 1 $\rm{W/(m\ K)}$ at room temperature, the phonon mean free paths (MFPs) are found to reach the sub-micrometer scale in the low-frequency region. As a consequence, the apparent LTC only converges to the diffusive limit for micrometer single crystals, which means that the LTC is heavily reduced in nanocrystalline MOFs. The sub-micrometer phonon MFPs are also found to be correlated with a moderate temperature dependence of LTC between those in typical crystalline and amorphous materials. Both the large phonon MFPs and the moderate temperature dependence of LTC fundamentally change our understanding of thermal transport in MOFs.

Unconventional mechanical and thermal behaviors of MOF CALF-20

Dong Fan; Supriyo Naskar; Guillaume Maurin. Nature Communication 15, 3251 (2024) (2023). Cited by 0. Found by: arxiv. DOI: 10.1038/s41467-024-47695-6

Abstract: CALF-20 was recently identified as a novel benchmark sorbent for CO$_2$ capture at the industrial scale, however comprehensive atomistic insight into its mechanical/thermal properties under working conditions is still lacking. In this study, we developed a general-purpose machine-learned potential (MLP) for the CALF-20 MOF framework that predicts the thermodynamic and mechanical properties of the structure at finite temperatures within first-principles accuracy. Interestingly, CALF-20 was demonstrated to exhibit both negative area compression and negative thermal expansion. Most strikingly, upon application of the tensile strain along the [001] direction, CALF-20 was shown to display a distinct two-step elastic deformation behavior, unlike typical MOFs that undergo plastic deformation after elasticity. Furthermore, this MOF was shown to exhibit a spectacular fracture strain of up to 27% along the [001] direction at room temperature comparable to that of MOF glasses. These abnormal thermal and mechanical properties make CALF-20 as attractive material for flexible and stretchable electronics and sensors.

Heat transport in superionic materials via machine-learned molecular dynamics

Wenjiang Zhou; Benrui Tang; Zheyong Fan; Federico Grasselli; Stefano Baroni; Bai Song. arXiv preprint (2025). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/2512.04718v2

Abstract: Precise modeling and understanding of heat transport in the superionic phase are of great interest. Although simulations combining Green-Kubo (GK) molecular dynamics with machine-learned potentials (MLPs) stand as a promising approach, substantial challenges remain due to the crucial impact of atomic diffusion. Here, we first show that the thermal conductivity ($ $) of superionic materials calculated via conventional GK integral of the energy flux varies notably with the MLP model. Subsequently, we highlight that reliable, model-independent $ $ values can be obtained by applying Onsager's reciprocal relations to correctly capture the coupled heat and mass transport. Remarkably, an anomalously invariant $ $ can be observed over a wide temperature range, distinct from the characteristic trends in traditional crystals and glasses. In addition, we illustrate that conventional $ $ decompositions into kinetic, potential, and cross terms suffer from ambiguities in the physical interpretation, despite their mathematical rigor. Finally, we propose a criterion for the necessity of the Onsager correction and reveal the underlying mechanism as a competition between thermally and chemically driven ion fluxes.

Machine learning for predicting ultralow thermal conductivity and high ZT in complex thermoelectric materials

Yuzhou Hao; Yuting Zuo; Jiongzhi Zheng; Wenjie Hou; Hong Gu; Xiaoying Wang; Xuejie Li; Jun Sun; Xiangdong Ding; Zhibin Gao. arXiv preprint (2024). Cited by 0. Found by: arxiv. DOI: 10.1021/acsami.4c09043

Abstract: Efficient and precise calculations of thermal transport properties and figure of merit, alongside a deep comprehension of thermal transport mechanisms, are essential for the practical utilization of advanced thermoelectric materials. In this study, we explore the microscopic processes governing thermal transport in the distinguished crystalline material Tl$_9$SbTe$_6$ by integrating a unified thermal transport theory with machine learning-assisted self-consistent phonon calculations. Leveraging machine learning potentials, we expedite the analysis of phonon energy shifts, higher-order scattering mechanisms, and thermal conductivity arising from various contributing factors like population and coherence channels. Our finding unveils an exceptionally low thermal conductivity of 0.31 W m^{-1} K^{-1} at room temperature, a result that closely correlates with experimental observations. Notably, we observe that the off-diagonal terms of heat flux operators play a significant role in shaping the overall lattice thermal conductivity of Tl$_9$SbTe$_6$, where the ultralow thermal conductivity resembles that of glass due to limited group velocities. Furthermore, we achieve a maximum ZT value of 3.17 in the c-axis orientation for \textit{p}-type Tl$_9$SbTe$_6$ at 600 K, and an optimal ZT value of 2.26 in the a-axis and b-axis direction for \textit{n}-type Tl$_9$SbTe$_6$ at 500

K. The crystalline Tl_9SbTe_6 not only showcases remarkable thermal insulation but also demonstrates impressive electrical properties owing to the dual-degeneracy phenomenon within its valence band. These results not only elucidate the underlying reasons for the exceptional thermoelectric performance of Tl_9SbTe_6 but also suggest potential avenues for further experimental exploration.

Incipient ionic conductors: Ion-constrained lattices achieving superionic-like thermal conductivity by extreme anharmonicity

Yongheng Li; Chunqiu Lu; Bin Wei; Cong Lu; Xingang Jiang; Daisuke Ishikawa; Taishun Manjo; Caofeng Pan; Alfred Q. R. Baron; Jiawang Hong. arXiv preprint (2025). Cited by 0. Found by: arxiv. DOI: 10.1002/adma.202513381

Abstract: Phonon liquid-like thermal conduction in the solid state enables superionic conductors to serve as efficient thermoelectric device candidates. While liquid-like motion of ions effectively suppresses thermal conductivity ( ), their high mobility concurrently triggers material degradation due to undesirable ion migration and consequent metal deposition, making it still a challenge to balancing low and high stability. Here, we report a superionic-like thermal transport alongside restricted long-range ion migration in $CsCu_2I_3$ with incipient ionic conduction, using synchrotron X-ray diffraction, inelastic X-ray scattering, and machine-learning potential-based simulations. We reveal that the Cu ions exhibit confined migration between CuI_4 tetrahedra at high temperatures, displaying extreme anharmonicity of dominated phonons beyond conventional rattling and comparable to that in superionic conductorsl. Consequently, a glass-like (~0.3 W m^{-1} K^{-1} at 300 K) following the relationship of $\sim T^{0.17}$, was achieved along the x-direction, where Cu ion migration is three oders of magnitude lower than in superionic conductors. These results highlight the advantage of incipient ionic conductors in simultaneously maintaining both low and high stability, elucidate the thermal transport mechanism via ion migration constraints, and pave an effective pathway toward ultralow thermal conductivity in ionic conductors.

Understanding correlations in BaZrO3:Structure and dynamics on the nano-scale

Erik Fransson; Petter Rosander; Paul Erhart; Göran Wahnström. Chemistry of Materials 36, 514 (2024) (2023). Cited by 0. Found by: arxiv. DOI: 10.1021/acs.chemmater.3c02548

Abstract: Barium zirconate BaZrO3 is one of few perovskites that is claimed to retain an average cubic structure down to 0K at ambient pressure, while being energetically very close to a tetragonal phase obtained by condensation of a soft phonon mode at the R-point. Previous studies suggest, however, that the local structure of BaZrO3 may change at low temperature forming nanodomains or a glass-like phase. Here, we investigate the global and local structure of BaZrO3 as a function of temperature and pressure via molecular dynamics simulations using a machine-learned potential with near density functional theory (DFT) accuracy. We show that the softening of the octahedral tilt mode at the R-point gives rise to weak diffuse superlattice reflections at low temperatures and ambient pressure, which are also observed experimentally. However, we do not observe any static nanodomains but rather soft dynamic fluctuations of the ZrO6 octahedra with a correlation length of 2 to 3nm over time-scales of about 1ps. This soft dynamic behaviour is the precursor of a phase transition and explains the emergence of weak superlattice peaks in measurements. On the other hand, when increasing the pressure at 300K we find a phase transition from the cubic to the tetragonal phase at around 16GPa, also in agreement with experimental studies.

Block QC (439 unique)

Hyperuniform states of matter

Salvatore Torquato. Physics Reports (2018). Cited by 485. Found by: openalex. DOI: 10.1016/j.physrep.2018.03.001

Localization and delocalization of light in photonic moiré lattices

Peng Wang; Yuanlin Zheng; Xianfeng Chen; Changming Huang; Yaroslav V. Kartashov; Lluis Torner; Vladimir V. Konotop; Fangwei Ye. Nature (2019). Cited by 482. Found by: openalex. DOI: 10.1038/s41586-019-1851-6

Delocalization of a disordered bosonic system by repulsive interactions

B. Deissler; Matteo Zaccanti; G. Roati; Chiara D'Errico; M. Fattori; M. Modugno; Giovanni Carlo Modugno; M. Inguscio. Nature Physics (2010). Cited by 273. Found by: openalex. DOI: 10.1038/nphys1635

Topological triple phase transition in non-Hermitian Floquet quasicrystals

Sebastian Weidemann; Mark Kremer; Stefano Longhi; Alexander Szameit. Nature (2022). Cited by 225. Found by: openalex. DOI: 10.1038/s41586-021-04253-0

Abstract: Abstract Phase transitions connect different states of matter and are often concomitant with the spontaneous breaking of symmetries. An important category of phase transitions is mobility transitions, among which is the well known Anderson localization1, where increasing the randomness induces a metal–insulator transition. The introduction of topology in condensed-matter physics2–4 lead to the discovery of topological phase transitions and materials as topological insulators5. Phase transitions in the symmetry of non-Hermitian systems describe the transition to on-average conserved energy6 and new topological phases7–9. Bulk conductivity, topology and non-Hermitian symmetry breaking seemingly emerge from different physics and, thus, may appear as separable phenomena. However, in non-Hermitian quasicrystals, such transitions can be mutually interlinked by forming a triple phase transition10. Here we report the experimental observation of a triple phase transition, where changing a single parameter simultaneously gives rise to a localization (metal–insulator), a topological and parity–time symmetry-breaking (energy) phase transition. The physics is manifested in a temporally driven (Floquet) dissipative quasicrystal. We implement our ideas via photonic quantum walks in coupled optical fibre loops11. Our study highlights the intertwinement of topology, symmetry breaking and mobility phase transitions in non-Hermitian quasicrystalline synthetic matter. Our results may be applied in phase-change devices, in which the bulk and edge transport and the energy or particle exchange with the environment can be predicted and controlled.

Disorder-Enhanced Transport in Photonic Quasicrystals

Liad Levi; Mikael C. Rechtsman; Barak Freedman; Tal Schwartz; Ofer Manela; Mordechai Segev. Science (2011). Cited by 205. Found by: openalex. DOI: 10.1126/science.1202977

Abstract: Quasicrystals are aperiodic structures with rotational symmetries forbidden to conventional periodic crystals; examples of quasicrystals can be found in aluminum alloys, polymers, and even ancient Islamic art. Here, we present direct experimental observation of disorder-enhanced wave transport in quasicrystals, which contrasts directly with the characteristic suppression of transport by disorder. Our experiments are carried out in photonic quasicrystals, where we find that increasing disorder leads to enhanced expansion of the beam propagating through the medium. By further increasing the disorder, we observe that the beam progresses through a regime of diffusive-like transport until it finally transitions to Anderson localization and the suppression of transport. We study this fundamental phenomenon and elucidate its origins by relating it to the basic properties of quasicrystalline media in the presence of disorder.

Generalized Aubry-André self-duality and mobility edges in non-Hermitian quasiperiodic lattices

Tong Liu; Hao Guo; Yong Pu; Stefano Longhi. Physical review. B./Physical review. B (2020). Cited by 148. Found by: openalex. DOI: 10.1103/physrevb.102.024205

Abstract: We demonstrate the existence of generalized Aubry-Andr\'e self-duality in a class of non-Hermitian quasiperiodic lattices with complex potentials. From the self-duality relations, the analytical expression of mobility edges is derived. Compared to Hermitian systems, mobility edges in non-Hermitian ones not only separate localized from extended states but also indicate the coexistence of complex and real eigenenergies, making possible a topological characterization of mobility edges. An experimental scheme, based on optical pulse propagation in synthetic photonic mesh lattices, is suggested to implement a non-Hermitian quasicrystal displaying mobility edges.

Topological Phase Transitions and Mobility Edges in Non-Hermitian Quasicrystals

Quan Lin; Tianyu Li; Lei Xiao; Kunkun Wang; Wei Yi; Peng Xue. Physical Review Letters (2022). Cited by 136. Found by: openalex. DOI: 10.1103/physrevlett.129.113601

Abstract: Non-Hermiticity significantly enriches the properties of topological models, leading to exotic features such as the non-Hermitian skin effects and non-Bloch bulk-boundary correspondence that have no counterparts in Hermitian settings. Its impact is particularly illustrating in non-Hermitian quasicrystals where the interplay between non-Hermiticity and quasiperiodicity results in the concurrence of the delocalization-localization transition, the parity-time (PT)-symmetry breaking, and the onset of the non-Hermitian skin effects. Here, we experimentally simulate non-Hermitian quasicrystals using photonic quantum walks. Using dynamic observables, we demonstrate that the system can transit from a delocalized, PT-symmetry broken phase that features non-Hermitian skin effects, to a localized, PT-symmetry unbroken phase with no non-Hermitian skin effects. The measured critical point is consistent with the theoretical prediction through a spectral winding number, confirming the topological origin of the phase transition. More interestingly, we also provide the first experimental evidence for mobility edges which are induced by non-Hermiticity. Our Letter opens the avenue of investigating the interplay of non-Hermiticity, quasiperiodicity, and spectral topology in open quantum systems.

Localization and delocalization of light in photonic moiré lattices

Peng Wang; Yuanlin Zheng; Xianfeng Chen; Changming Huang; Yaroslav V. Kartashov; Lluis Torner; V. V. Konotop; Fangwei Ye. LA Referencia (Red Federada de Repositorios Institucionales de Publicaciones Científicas) (2020). Cited by 121. Found by: openalex. URL: https://openalex.org/W3102201054

Abstract: Abstract Moiré lattices consist of two superimposed identical periodic structures with a relative rotation angle. Moiré lattices have several applications in everyday life, including artistic design, the textile industry, architecture, image processing, metrology and interferometry. For scientific studies, they have been produced using coupled graphene–hexagonal boron nitride monolayers1, 2, graphene–graphene layers3, 4 and graphene quasicrystals on a silicon carbide surface5. The recent surge of interest in moiré lattices arises from the possibility of exploring many salient physical phenomena in such systems; examples include commensurable–incommensurable transitions and topological defects2, the emergence of insulating states owing to band flattening3, 6, unconventional superconductivity4 controlled by the rotation angle7, 8, the quantum Hall effect9, the realization of non-Abelian gauge potentials10 and the appearance of quasicrystals at special rotation angles11. A fundamental question that remains unexplored concerns the evolution of waves in the potentials defined by moiré lattices. Here we experimentally create two-dimensional photonic moiré lattices, which—unlike their material counterparts—have readily controllable parameters and symmetry, allowing us to explore transitions between structures with fundamentally different geometries (periodic, general aperiodic and quasicrystal). We observe localization of light in deterministic linear lattices that is based on flat-band physics6, in contrast to previous schemes based on light diffusion in optical quasicrystals12, where disorder is required13 for the onset of Anderson localization14 (that is, wave localization in random media). Using commensurable and incommensurable moiré patterns, we experimentally demonstrate the two-dimensional localization–delocalization transition of light. Moiré lattices may feature an almost arbitrary geometry that is consistent with the crystallographic symmetry groups of the sublattices, and therefore afford a powerful tool for controlling the properties of light patterns and exploring the physics of periodic–aperiodic phase transitions and two-dimensional wavepacket phenomena relevant to several areas of science, including optics, acoustics, condensed matter and atomic physics.

Bose-Einstein condensates in optical quasicrystal lattices

Laurent Sanchez-Palencia; L. Santos. Physical Review A (2005). Cited by 117. Found by: openalex. DOI: 10.1103/physreva.72.053607

Abstract: We analyze the physics of Bose-Einstein condensates confined in two dimensional (2D) quasiperiodic optical lattices, which offer an intermediate situation between ordered and disordered systems. First, we analyze the time-of-flight interference pattern that reveals quasiperiodic long-range order. Second, we demonstrate localization effects associated with quasidisorder as well as quasiperiodic Bloch oscillations associated with the extended nature of the wave function of a Bose-Einstein condensate in an optical quasicrystal. In addition, we discuss in detail the crossover between diffusive and localized regimes when the quasiperiodic potential is switched on, as well as the effects of interactions.

Observing Localization in a 2D Quasicrystalline Optical Lattice

Matteo Sbroscia; Konrad Viebahn; Edward Carter; Jr-Chiun Yu; Alexander L. Gaunt; Ulrich Schneider. Physical Review Letters (2020). Cited by 105. Found by: openalex. DOI: 10.1103/physrevlett.125.200604

Abstract: Quasicrystals are long-range ordered but not periodic, representing an interesting middle ground between order and disorder. We experimentally and numerically study the localization transition in the ground state of noninteracting and weakly interacting bosons in an eightfold symmetric quasicrystalline optical lattice. In contrast to typically used real space in situ techniques, we probe the system in momentum space by recording matter wave diffraction patterns. Shallow lattices lead to extended states whereas we observe a localization transition at a critical lattice depth of V_{0} 1.78(2)E_{rec} for the noninteracting system. Our measurements and Gross-Pitaevskii simulations demonstrate that in interacting systems the transition is shifted to deeper lattices, as expected from superfluid order counteracting localization. Quasiperiodic potentials, lacking conventional rare regions, provide the ideal testing ground to realize many-body localization in 2D.

Power dependent soliton location and stability in complex photonic structures

Yannis Kominis; Kyriakos Hizanidis. Optics Express (2008). Cited by 73. Found by: openalex. DOI: 10.1364/oe.16.012124

Abstract: The presence of spatial inhomogeneity in a nonlinear medium results in the breaking of the translational invariance of the underlying propagation equation. As a result traveling wave soliton solutions do not exist in general for such systems, while stationary solitons are located in fixed positions with respect to the inhomogeneous spatial structure. In simple photonic structures with monochromatic modulation of the linear refractive index, soliton position and stability do not depend on the characteristics of the soliton such as power, width and propagation constant. In this work, we show that for more complex photonic structures where either one of the refractive indices (linear or nonlinear) is modulated by more than one wavenumbers, or both of them are modulated, soliton position and stability depends strongly on its characteristics. The latter results in additional functionality related to soliton discrimination in such structures. The respective power (or width/propagation constant) dependent bifurcations are studied in terms of a Melnikov-type theory. The latter is used for the determination of the specific positions, with respect to the spatial structure, where solitons can be located. A wide variety of cases are studied, including solitons in periodic and quasiperiodic lattices where both the linear and the nonlinear refractive index are spatially modulated. The investigation of a wide variety of inhomogeneities provides physical insight for the design of a spatial structure and the control of the position and stability of a localized wave.

Localized modes in photonic quasicrystals with Penrose-type lattice

Andrea Villa; Stéfan Enoch; G. Tayeb; Filippo Capolino; V. Pierro; Vincenzo Galdi. Optics Express (2006). Cited by 60. Found by: openalex. DOI: 10.1364/oe.14.010021

Abstract: We investigate the properties of the resonant modes that occur in the transparency bands of two-dimensional finite-size Penrose-type photonic quasicrystals made of dielectric cylindrical rods. These modes stem from the natural local arrangements of the quasicrystal structure rather than, as originally thought, from fabrication-related imperfections. Examples of local density of states and field maps are shown for different wavelengths. Calculations of local density of states show that these modes mainly originate from the interactions between a limited numbers of rods.

Octonacci photonic quasicrystals

E.R. Brandão; Carlos H. Costa; M.S. Vasconcelos; D.H.A.L. Anselmo; V.D. Mello. Optical Materials (2015). Cited by 56. Found by: openalex. DOI: 10.1016/j.optmat.2015.04.051

Observation of localization of light in linear photonic quasicrystals with diverse rotational symmetries

Peng Wang; Qidong Fu; V. V. Konotop; Yaroslav V. Kartashov; Fangwei Ye. Nature Photonics (2024). Cited by 54. Found by: openalex. DOI: 10.1038/s41566-023-01350-6

Localized photonic band edge modes and orbital angular momenta of light in a golden-angle spiral

Seng Fatt Liew; Heeso Noh; Jacob Trevino; Luca Dal Negro; Hui Cao. Optics Express (2011). Cited by 47. Found by: openalex. DOI: 10.1364/oe.19.023631

Abstract: We present a numerical study on photonic bandgap and band edge modes in the golden-angle spiral array of air cylinders in dielectric media. Despite the lack of long-range translational and rotational order, there is a large PBG for the TE polarized light. Due to spatial inhomogeneity in the air hole spacing, the band edge modes are spatially localized by Bragg scattering from the parastichies in the spiral structure. They have discrete angular momenta that originate from different families of the parastichies whose numbers correspond to the Fibonacci numbers. The unique structural characteristics of the golden-angle spiral lead to distinctive features of the band edge modes that are absent in both photonic crystals and quasicrystals.

Photonic quasicrystal single-cell cavity mode

Sun-Kyung Kim; Jeehye Lee; Se-Heon Kim; In-Kag Hwang; Yong-Hee Lee; Sung-Bock Kim. Applied Physics Letters (2005). Cited by 45. Found by: openalex. openalex_2yr: 3.881 (2026). DOI: 10.1063/1.1852716

Abstract: We propose and realize the photonic quasicrystal (PQC) single-cell resonator based on a InP-InGaAsP freestanding slab. A well-defined hexapole-like localized state following a C6 symmetry is identified from the PQC single-cell resonator. In this type of hexapole mode, the electromagnetic energy is strongly concentrated on the dielectric region, in contrast to that in a triangular lattice photonic crystal. By tailoring the structural parameters, the hexapole mode shows a maximum theoretical quality factor of 20000. The PQC single-cell resonator lases in a single hexapole mode with a threshold of 0.6mW at room temperature.

Intrinsic photonic wave localization in a three-dimensional icosahedral quasicrystal

Seung-Yeol Jeon; Hyungho Kwon; Kahyun Hur. Nature Physics (2017). Cited by 43. Found by: openalex. DOI: 10.1038/nphys4002

Mode localization and band-gap formation in defect-free photonic quasicrystals

Khaled Mnaymneh; Robert C. Gauthier. Optics Express (2007). Cited by 42. Found by: openalex. DOI: 10.1364/oe.15.005089

Abstract: Defects in photonic crystals are local regions in which the translational symmetry is broken. The same definition can be applied to photonic quasicrystals except in this case the symmetry is the 2pi/n rotational symmetry, where n is the rotational fold number. In this context, if no such defects are present, the structure is called "defect-free". Even though photonic quasicrystal patterns can be defect-free, localized modes can still exist in such structures. These modes resemble those of a central potential that suggests that localization in photonic quasicrystals are actually "extended" modes of the rotational symmetry. A possible connection is suggested between these localized modes and short-range dependence of the photonic band gap (PBG). Such a connection implies a tight-binding description of PBG formation of photonic quasicrystals - making them more similar to electronic semiconductors than regular photonic crystals.

Experimental Observation of Intrinsic Light Localization in Photonic Icosahedral Quasicrystals

Artem D. Sinelnik; Ivan I. Shishkin; Xiaochang Yu; K. B. Samusev; Pavel A. Belov; M. F. Limonov; Pavel Ginzburg; Mikhail V. Rybin. Advanced Optical Materials (2020). Cited by 39. Found by: openalex. openalex_2yr: 5.548 (2026). DOI: 10.1002/adom.202001170

Abstract: Abstract One of the most intriguing problems of light transport in solids is the localization that has been observed in various disordered photonic structures. The light localization in defect-free icosahedral quasicrystals has recently been predicted theoretically without experimental verification. Herein, the fabrication of submicron-size dielectric icosahedral quasicrystals is reported and the results of detailed studies of the photonic properties of these structures are demonstrated. The first direct experimental observation of intrinsic light localization in defect-free quasicrystals is presented. This result is obtained in time-resolved measurements at different laser wavelengths in the visible. The localization is linked with the aperiodicity of the icosahedral structure, which leads to uncompensated scattering of light from an individual structural element over the entire sphere, providing multiple scattering inside the sample and, as a result, the intrinsic localization of light.

Observation of Topological Corner State Arrays in Photonic Quasicrystals

Aoqian Shi; Yiwei Peng; Jiapei Jiang; Yuchen Peng; Peng Peng; Jianzhi Chen; Hongsheng Chen; Shuangchun Wen; Xiao Lin; Fei Gao; Jianjun Liu. Laser & Photonics Review (2024). Cited by 35. Found by: openalex. DOI: 10.1002/lpor.202300956

Abstract: Abstract Recently, the studies of topological corner states (TCSs) have been extended from crystals to quasicrystals, which are referred to as higher-order topological quasicrystalline insulators (HOTQIs). However, the TCSs of complete quasi-periodic structure in photonic systems have yet to be demonstrated. Moreover, there is only one TCS in each corner region in higher-order topological insulators (HOTIs). Increasing the number of TCS is expected to increase the application potential of TCSs. In this work, HOTQIs in photonic systems are experimentally observed. It is found that HOTQIs possess TCS arrays, and each TCS array contains several TCSs. Furthermore, the universal theoretical framework of the multimer analysis method is improved, and the difference in the average charge density is proposed as a real-space topological index. These results will open

up new ideas for investigating highly integrated, multi-region localized TCSs and are expected to provide new ways to explore topological phenomena and the applications of photonic quasicrystals.

Dephasing-Induced Mobility Edges in Quasicrystals

Stefano Longhi. Physical Review Letters (2024). Cited by 34. Found by: openalex. DOI: 10.1103/physrevlett.132.236301

Abstract: Mobility edges (ME), separating Anderson-localized states from extended states, are known to arise in the single-particle energy spectrum of certain one-dimensional lattices with aperiodic order. Dephasing and decoherence effects are widely acknowledged to spoil Anderson localization and to enhance transport, suggesting that ME and localization are unlikely to be observable in the presence of dephasing. Here it is shown that, contrary to such a wisdom, ME can be created by pure dephasing effects in quasicrystals in which all states are delocalized under coherent dynamics. Since the lifetimes of localized states induced by dephasing effects can be extremely long, rather counterintuitively decoherence can enhance localization of excitation in the lattice. The results are illustrated by considering photonic quantum walks in synthetic mesh lattices.

Enhanced Soliton Transport in Quasiperiodic Lattices with Introduced Aperiodicity

Andrey A. Sukhorukov. Physical Review Letters (2006). Cited by 33. Found by: openalex. DOI: 10.1103/physrevlett.96.113902

Abstract: We study linear transmission and nonlinear soliton transport through quasiperiodic structures, where the lattice profiles are described by multiple modulation frequencies. We show that resonant scattering at mixed-frequency resonances limits transmission efficiency of localized wave packets, leading to radiation and possible trapping of solitons. We obtain an explicit analytical expression for optimal quasiperiodic lattice profiles, where additional aperiodic modulations suppress mixed-frequency resonances, resulting in dramatic enhancement of soliton mobility. Our results can be applied to the design of photonic waveguide structures, and arrays of magnetic micro-traps for atomic Bose-Einstein condensates.

Exact mobility edges and topological phase transition in two-dimensional non-Hermitian quasicrystals

Zhihao Xu; X. C. Xia; Shu Chen. Science China Physics Mechanics and Astronomy (2021). Cited by 31. Found by: openalex. DOI: 10.1007/s11433-021-1802-4

Quasiperiodic photonic crystal fiber [Invited]

Exian Liu; Jianjun Liu. Chinese Optics Letters (2023). Cited by 30. Found by: openalex. openalex_2yr: 4.254 (2026). DOI: 10.3788/col202321.060603

Abstract: In the fields of light manipulation and localization, quasiperiodic photonic crystals, or photonic quasicrystals (PQs), are causing an upsurge in research because of their rotational symmetry and long-range orientation of transverse lattice arrays, as they lack translational symmetry. It allows for the optimization of well-established light propagation properties and has introduced new guiding features. Therefore, as a class, quasiperiodic photonic crystal fibers, or photonic quasicrystal fibers (PQFs), are considered to add flexibility and richness to the optical properties of fibers and are expected to offer significant potential applications to optical fiber fields. In this review, the fundamental concept, working mechanisms, and invention history of PQFs are explained. Recent progress in optical property improvement and its novel applications in fields such as dispersion control, polarization-maintenance, supercontinuum generation, orbital angular momentum transmission, plasmon-based sensors and filters, and high nonlinearity and topological mode transmission, are then reviewed in detail. Bandgap-type air-guiding PQFs supporting low attenuation propagation and regulation of photonic density states of quasiperiodic cladding and in which light guidance is achieved by coherent Bragg scattering are also summarized. Finally, current challenges encountered in the guiding mechanisms and practical preparation techniques, as well as the prospects and research trends of PQFs, are also presented.

Luminescence properties of a Fibonacci photonic quasicrystal

V. Passias; N. V. Valappil; Z. Shi; L. Deych; A. A. Lisyansky; V. M. Menon. Optics Express (2009). Cited by 28. Found by: openalex. DOI: 10.1364/oe.17.006636

Abstract: An active one-dimensional Fibonacci photonic quasi-crystal is realized via spin coating. Luminescence properties of an organic dye embedded in the quasi-crystal are studied experimentally and compared to theoretical simulations. The luminescence occurs via the pseudo-bandedge mode and follows the dispersion properties of the Fibonacci crystal. Time resolved luminescence measurement of the active structure shows faster spontaneous emission rate, indicating the effect of the large photon densities available at the bandedge due to the presence of critically localized states. The experimental results are in good agreement with the theoretical calculations for steady-state luminescence spectra.

Non-Hermitian Delocalization in a Two-Dimensional Photonic Quasicrystal

Zhaoyang Zhang; Shun Liang; I. Septembre; Jiawei Yu; Yongping Huang; Maochang Liu; Yanpeng Zhang; Min Xiao; G. Malpuech; D. D. Solnyshkov. Physical Review Letters (2024). Cited by 27. Found by: openalex. DOI: 10.1103/physrevlett.132.263801

Abstract: Theoretical and experimental studies suggest that both Hermitian and non-Hermitian quasicrystals show localization due to the fractal spectrum and to the transition to diffusive bands via exceptional points, respectively. Here, we present an experimental study of a dodecagonal photonic quasicrystal based on electromagnetically induced transparency in a Rb vapor cell. First, we observe the suppression of the wave packet expansion in the Hermitian case. We then discover a new regime, where increasing the non-Hermiticity leads to delocalization, demonstrating that the behavior in non-Hermitian quasicrystals is richer than previously thought.

Simultaneous large band gaps and localization of electromagnetic and elastic waves in defect-free quasicrystals

Tianbao Yu; Zhong Wang; Wenxing Liu; Tongbiao Wang; Nian-Hua Liu; Qinghua Liao. Optics Express (2016). Cited by 27. Found by: openalex. DOI: 10.1364/oe.24.007951

Abstract: We report numerically large and complete photonic and phononic band gaps that simultaneously exist in eight-fold phoxonic quasicrystals (PhXQCs). PhXQCs can possess simultaneous photonic and phononic band gaps over a wide range of geometric parameters. Abundant localized modes can be achieved in defect-free PhXQCs for all photonic and phononic polarizations. These defect-free localized modes exhibit multiform spatial distributions and can confine simultaneously electromagnetic and elastic waves in a large area, thereby providing rich selectivity and enlarging the interaction space of optical and elastic waves. The simulated results based on finite element method show that quasiperiodic structures formed of both solid rods in air and holes in solid materials can simultaneously confine and tailor electromagnetic and elastic waves; these structures showed advantages over the periodic counterparts.

Parity-Induced Thermalization Gap in Disordered Ring Lattices

Yao Wang; Jun Gao; Xiao-Ling Pang; Zhi-Qiang Jiao; Hao Tang; Yuan Chen; Lu-Feng Qiao; Zhen-Wei Gao; Jian-Peng Dou; Ai-Lin Yang; Xian-Min Jin. Physical Review Letters (2019). Cited by 26. Found by: openalex. DOI: 10.1103/physrevlett.122.013903

Abstract: The gaps separating two different states widely exist in various physical systems: from the electrons in periodic lattices to the analogs in photonic, phononic, plasmonic systems, and even quasicrystals. Recently, a thermalization gap, an inaccessible range of photon statistics, was proposed for light in disordered structures [Nat. Phys. 11, 930 (2015)NPAHAX1745-247310.1038/nphys3482], which is intrinsically induced by the disorder-immune chiral symmetry and can be reflected by the photon statistics. The lattice topology was further identified as a decisive role in determining the photon statistics when the chiral symmetry is satisfied. Being very distinct from one-dimensional lattices, the photon statistics in ring lattices are dictated by its parity, i.e., odd or even sited. Here, we for the first time experimentally observe a parity-induced thermalization gap in strongly disordered ring photonic structures. In a limited scale, though the light tends to be localized, we are still able to find clear evidence of the parity-dependent disorder-immune chiral symmetry and the resulting thermalization gap by measuring photon statistics, while strong disorder-induced Anderson localization overwhelms such a phenomenon in larger-scale structures. Our results shed new light on the relation among symmetry, disorder, and localization, and may inspire new resources and artificial devices for information processing and quantum control on a photonic chip.

Photonic band gaps and localization in two-dimensional metallic quasicrystals

Mehmet Bayındır; Ertugrul Cubukcu; İrfan Bulu; Ekmel Özbay. Europhysics Letters (EPL) (2001). Cited by 25. Found by: openalex. openalex_2yr: 2.066 (2026). DOI: 10.1209/epl/i2001-00485-9

Abstract: We report on experimental observation of photonic stop bands in two-dimensional metallic Penrose lattice. We investigated the defect characteristics, and observed strongly localized cavity modes below the plasma frequency. The absence of the translational symmetry allowed to change the defect frequencies and localization properties of the defect modes. Propagation of photons along highly localized coupled-cavity modes was experimentally demonstrated in quasiperiodic metallic structures.

Dielectric refractive index dependence of the focusing properties of a dielectric-cylinder-type decagonal photonic quasicrystal flat lens and its photon localization

Jianjun Liu; Exian Liu; Zhigang Fan; Xiong Zhang. Applied Physics Express (2015). Cited by 23. Found by: openalex. openalex_2yr: 3.957 (2026). DOI: 10.7567/apex.8.112003

Abstract: Dielectric refractive index dependence of the focusing properties of a dielectric-cylinder-type decagonal photonic quasicrystal flat lens and its photon localization, Liu, Jianjun, Liu, Exian, Fan, Zhigang, Zhang, Xiong

Defect-free localized modes and coupled-resonator acoustic waveguides constructed in two-dimensional phononic quasicrystals

Mingda Zhang; Wei Zhong; Xiangdong Zhang. Journal of Applied Physics (2012). Cited by 22. Found by: openalex. DOI: 10.1063/1.4721372

Abstract: The localization of acoustic waves in defect-free two-dimensional quasiperiodic phononic crystals has been investigated by using an exact multiple-scattering method. Different types of localized states have been found in eightfold, tenfold, and twelvefold two-dimensional phononic quasicrystals without introducing any defect. The defect-free coupled-resonator acoustic waveguides, based on these localized modes, have been designed, and the resonant transmission properties of acoustic waves in the waveguides have been demonstrated by numerical simulations. It is anticipated that our findings can be potentially applied to phononic devices.

Reentrant delocalization transition in one-dimensional photonic quasicrystals

Sachin Vaidya; Christina Jörg; Kyle Linn; M.J. Goh; Mikael C. Rechtsman. Physical Review Research (2023). Cited by 21. Found by: openalex. DOI: 10.1103/physrevresearch.5.033170

Abstract: Waves propagating in certain one-dimensional quasiperiodic lattices are known to exhibit a sharp localization transition. We theoretically predict and experimentally observe that the localization of light in one-dimensional photonic quasicrystals is followed by a second delocalization transition for some states on increasing quasiperiodic modulation strength—an example of a reentrant transition. We further propose that this phenomenon can be qualitatively captured by a dimerized tight-binding model with long-range couplings. This work furthers our understanding of localization physics in complex systems and the impact of quasiperiodicity on light transport in photonic devices.

Light Localization in Local Isomorphism Classes of Quasicrystals

Chaney Lin; Paul J. Steinhardt; Salvatore Torquato. Physical Review Letters (2018). Cited by 20. Found by: openalex. DOI: 10.1103/physrevlett.120.247401

Abstract: We study a continuum of photonic quasicrystal heterostructures derived from local isomorphism (LI) classes of pentagonal quasicrystal tilings. These tilings are obtained by direct projection from a five-dimensional hypercubic lattice. We demonstrate that, with the sole exception of the Penrose LI class, all other LI classes result in degenerate, effectively localized states, with precisely predictable and tunable properties (frequencies, frequency splittings, and densities). We show that localization and tunability are related to a mathematical property of the pattern known as ”restorability,” i.e., whether the tiling can be uniquely specified given only a set of rules that fix all allowed clusters smaller than some bound.

Effects of graphene on light transmission spectra in Dodecanacci photonic quasicrystals

E.F. Silva; M.S. Vasconcelos; Carlos H. Costa; D.H.A.L. Anselmo; V.D. Mello. Optical Materials (2019). Cited by 20. Found by: openalex. DOI: 10.1016/j.optmat.2019.109450

Simultaneous localization of photons and phonons within the transparency bands of LiNbO_3 phoxonic quasicrystals

Zhong Wang; Wenxing Liu; Tianbao Yu; Tongbiao Wang; Haoming Li; Nian-Hua Liu; Qinghua Liao. Optics Express (2016). Cited by 20. Found by: openalex. DOI: 10.1364/oe.24.023353

Abstract: PhXQCs provide a good candidate to enhance phononic-photonic interaction and show considerable advantage over the periodic counterparts.

Spin waves in planar quasicrystal of Penrose tiling

Justyna Rychły-Gruszecka; Szymon Mieszczak; Jarosław W. Kłos. Journal of Magnetism and Magnetic Materials (2017). Cited by 20. Found by: openalex. DOI: 10.1016/j.jmmm.2017.03.029

Non-Hermitian topological mobility edges and transport in photonic quantum walks

Stefano Longhi. Optics Letters (2022). Cited by 18. Found by: openalex. DOI: 10.1364/ol.460484

Abstract: In non-Hermitian quasicrystals, mobility edges (ME) separating localized and extended states in the complex energy plane can arise as a result of non-Hermitian terms in the Hamiltonian. Such ME are of topological nature, i.e., the energies of localized and extended states exhibit distinct topological structures in the complex energy plane. However, depending on the origin of non-Hermiticity, i.e., asymmetry of hopping amplitudes or complexification of the incommensurate potential phase, different winding numbers are introduced, corresponding to different transport features in the bulk of the lattice: while ballistic transport is allowed in the former case, pseudo-dynamical localization is observed in the latter case. The results are illustrated by considering non-Hermitian photonic quantum walks in synthetic mesh lattices.

Exciton-polaritonic quasicrystalline and aperiodic structures

Alexander N. Poddubny; Laura Pilozzi; M. M. Voronov; E. L. Ivchenko. Physical Review B (2009). Cited by 18. Found by: openalex. DOI: 10.1103/physrevb.80.115314

Abstract: We have theoretically studied propagation of exciton-polaritons in deterministic aperiodic multiple-quantum-well structures, particularly, in the Fibonacci and Thue-Morse chains. The attention is concentrated on the structures tuned to the resonant Bragg condition with two-dimensional quantum-well exciton. Depending on the number of wells, the super-radiant either photonic-quasicrystal regimes are realized in these aperiodic structures. For moderate values of the exciton nonradiative damping rate $\ensuremath{\Gamma}$, the developed theory based on the two-wave approximation allows one to perceive and describe analytically the exact transfer-matrix computations for transmittance and reflectance spectra in the whole frequency range except for a narrow region near the exciton resonance ${\ensuremath{\omega}}_{0}$. In this region the optical spectra and the exciton-polariton dispersion demonstrate scaling invariance and self-similarity which can be interpreted in terms of the “band-edge” cycle of the trace map, in the case of Fibonacci structures, and in terms of zero reflection frequencies, in the case of Thue-Morse structures. With decreasing $\ensuremath{\Gamma}$, in the whole allowed polariton band the two-wave approximation stops to be valid, and a transition occurs from Bloch-like to localized states, with modes closer to ${\ensuremath{\omega}}_{0}$ becoming localized first.

Light Localisation and Lasing: Random and Quasi-random Photonic Structures

Mher Ghulinyan; Lorenzo Pavesi. Medical Entomology and Zoology (2014). Cited by 17. Found by: openalex. URL: https://openalex.org/W641369541

Abstract: List of contributors Preface 1. Light propagation and emission in complex photonic media W. L. Vos, A. Lagendijk and A. P. Mosk 2. Transport of localized waves via modes and channels A. Genack and Z. Shi 3. Modes structure and interaction in random lasers M. Leonetti and C. Lopez 4. Ordered and disordered light transport in couple microring resonators S. Mookherjea 5. One-dimensional photonic quasicrystals M. Ghulinyan 6. 2D pseudo-random and deterministic aperiodic lasers H. Cao, H. Noh and L. Dal Negro 7. 3D photonic quasicrystal and deterministic aperiodic structures A. Ledermann, M. Renner and G. von Freymann 8. Cavity quantum electrodynamics with three-dimensional photonic bandgap crystals W. L. Vos and L. A. Woldering References Index.

Anderson localization and Brewster anomalies in photonic disordered quasiperiodic lattices

E. Reyes-Gómez; Alexys Bruno-Alfonso; S. B. Cavalcanti; L. E. Oliveira. Physical Review E (2011). Cited by 15. Found by: openalex. DOI: 10.1103/physreve.84.036604

Abstract: A comprehensive study of the properties of light propagation through one-dimensional photonic disordered quasiperiodic superlattices, composed of alternating layers with random thicknesses of air and a dispersive metamaterial, is theoretically performed. The superlattices consist of the successive stacking of N quasiperiodic Fibonacci or Thue-Morse heterostructures. The width of the slabs in the photonic superlattice may randomly fluctuate around its mean value, which introduces a structural disorder into the system. It is assumed that the left-handed layers have a Drude-type dispersive response for both the dielectric permittivity and magnetic permeability, and Maxwell's equations are solved for oblique incidence by using the transfer-matrix formalism. The influence of both quasiperiodicity and structural disorder on the localization length and Brewster anomalies are thoroughly discussed.

Bose-Einstein condensates in quasiperiodic lattices: Bosonic Josephson junction, self-trapping, and multimode dynamics

Henrique C. Prates; Dmitry A. Zezyulin; V. V. Konotop. Physical Review Research (2022). Cited by 15. Found by: openalex. DOI: 10.1103/physrevresearch.4.033219

Abstract: A Bose-Einstein condensate loaded in a one-dimensional bichromatic optical lattice with constituent sublattices having incommensurate periods is considered. Using the rational approximations for the incommensurate periods, we show that below the mobility edge the localized states are distributed nearly homogeneously in the space and explore the versatility of such potentials. We show that superposition of symmetric and antisymmetric localized states can be used to simulate various physical dynamical regimes, known to occur in double-well and multiwell traps. As examples, we obtain an alternative realization of a bosonic Josephson junction, whose coherent oscillations display beatings or switching in the weakly nonlinear regime, and describe self-trapping and four-mode dynamics, mimicking coherent oscillations and self-trapping in four-well potentials. These phenomena can be observed for different pairs of modes, which are localized due to the interference rather than due to a confining trap. The results obtained using few-mode approximations are compared with the direct numerical simulations of the one-dimensional Gross-Pitaevskii equation. The localized states and the related dynamics are found to persist for long times even in the repulsive condensates. We also described bifurcations of the families of nonlinear modes, the symmetry breaking, and stable minigap solitons.

Fundamental and vortex gap solitons in quasiperiodic photonic lattices

Changming Huang; Liangwei Dong; Hanying Deng; Xiao Zhang; Penghui Gao. Optics Letters (2021). Cited by 15. Found by: openalex. DOI: 10.1364/ol.443051

Abstract: We address the existence and stability of fundamental, single-charged vortex, and double-charged vortex gap solitons in two-dimensional quasiperiodic photonic lattices imprinted in a Kerr-type medium. Fundamental and vortex gap solitons can bifurcate from linear localized states or their combination supported by quasiperiodic lattices for both defocusing and focusing nonlinearities. We find that the three types of solitons mentioned above are stable in the entire existence domain for defocusing nonlinearities, and that they can also be stable at a lower power level for focusing nonlinearities. At higher power, unstable solitons are characterized by a ring-shaped symmetry-breaking distribution, and the unique spot profile formed is repeatedly observed with changes in propagation distance.

Stationary and traveling solitons via local dissipation in Bose-Einstein condensates in ring optical lattices

Russell Campbell; Gian-Luca Oppo. Physical Review A (2016). Cited by 14. Found by: openalex. DOI: 10.1103/physreva.94.043626

Abstract: A model of a Bose-Einstein condensate in a ring optical lattice with atomic dissipations applied at a stationary or at a moving location on the ring is presented. The localized dissipation is shown to generate and stabilize both stationary and traveling lattice solitons. Among many localized solutions, we have generated spatially stationary quasiperiodic lattice solitons and a family of traveling lattice solitons with two intensity peaks per potential well with no counterpart in the discrete case. Collisions between traveling and stationary lattice solitons as well as between two traveling lattice solitons display a critical dependence from the lattice depth. Stable counterpropagating solitons in ring lattices can find applications in gyroscope interferometers

with ultracold gases.

A study of optical reflectance and localization modes of 1-D Fibonacci photonic quasicrystals using different graded dielectric materials

Bipin K. Singh; Praveen C. Pandey. Journal of Modern Optics (2014). Cited by 13. Found by: openalex. DOI: 10.1080/09500340.2014.914587

Abstract: In this paper, we present an analytical study on the reflection properties of light through one-dimensional (1-D) quasi-periodic multilayer structures. The considered structures are as follows: F7, F8, F9, (F2)10, (F3)10 and some combinations such as: [(F2)10 (F7) (F2)10], [(F2)10 (F8) (F2)10], [(F3)10 (F7) (F3)10], [(F3)10 (F8) (F3)10], [(F2)10(F3)10], [(F2)10 (F7) (F3)10] and [(F2)10 (F8) (F3)10], where (Fj)n represents n period of the Fibonacci sequence of jth generation. These multilayer structures are considered of two types of layers. One type of layer is considered of graded material like normal, linear or exponential graded material, and the second type of layer is considered of constant refractive index material. Transfer matrix method is utilized to calculate the reflection spectra and localization modes of such structures in the frequency range 150–450 THz. This work would provide the basis of understanding of the effect of graded materials on the reflection and localization modes in Fibonacci photonic quasicrystal structures and obtained spectra can be used in the recognition of grading of materials. The considered heterostructures provide the broad reflection band and some localization modes in the calculated region.

Localized modes in defect-free two-dimensional circular photonic crystals

Wei Zhong; Xiangdong Zhang. Physical Review A (2010). Cited by 13. Found by: openalex. DOI: 10.1103/physreva.81.013805

Abstract: The localization of electromagnetic waves in defect-free circular photonic crystals (CPCs) is investigated using a multiple-scattering method. It is shown that electromagnetic waves of certain freqencies are localized in some special regions inside a perfect CPC with a high order of rotational symmetry. Localized modes with a high \$Q\$ factor, greater than \${10}^{6}\$, are obtained. In particular, some unique localized modes such as circular modes and localized modes for both polarized waves are found in the present system, which are completely different from those in defect-free photonic quasicrystals and periodic structures. Their physical properties are also analyzed.

Study of Optical Propagation in Hybrid Periodic/Quasiregular Structures Based on Porous Silicon

Jose Escorcia-GarcÃa; M.E. Mora-Ramos. PIERS Online (2009). Cited by 13. Found by: openalex. DOI: 10.2529/piers080906010703

Abstract: In this work we report the propagation of the light waves within Bragg-Fibonacci (BM-FN) hybrid structures and the appearance of photonic bandgaps in the re∞ectance spec- trum of these systems, which are studied theoretically with the use of oblique incidence transfer matrix technique. In all cases, the discussion is made regarding the possible application of these structures as photonic quasicrystals. For the simulation we have adopted the typical parame- ters of a Porous-Silicon dielectric multilayer. It is found that a hybrid conflguration of the type BM-FN-BM is a promising candidate for resonant microcavities with strong mode localization.

Acousto-optic interactions for terahertz waves using phoxonic quasicrystals

Zhong Wang; Tianbao Yu; Tongbiao Wang; Wenxing Liu; Nian-Hua Liu; Qinghua Liao. Journal of Physics D Applied Physics (2018). Cited by 12. Found by: openalex. DOI: 10.1088/1361-6463/aaa98c

Abstract: Abstract Phoxonic quasicrystals possess abundant localized modes and low-index-contrast bandgaps, thus showing much advantage over periodic counterparts. We calculated the modulation of photonic modes for terahertz waves by phononic modes in defect-free phoxonic quasicrystals (FDQs) and quasicrystals having center point defects (PDQs). Both the moving interface (MI) and photoelastic effects are considered in acousto-optic coupling. Terahertz modes can be efficiently modified by acoustic modes due to the simultaneous confinement of electromagnetic and elastic fields both inside and outside the bandgaps. The acousto-optic coupling strength depends on the selection of the acoustic and terahertz defect modes. The predominant coupling mechanism is determined by the overlapping between the terahertz and acoustic modes in PDQs, while the MI effect becomes predominant in FDQs. Our results are important for terahertz wave modulation with ultra-small footprint size.

Resonant transmission of electromagnetic waves through two-dimensional photonic quasicrystals

Yair Neve-Oz; Therese Pollok; Sven Burger; M. Golosovsky; Dan Davidov. Journal of Applied Physics (2010). Cited by 10. Found by: openalex. DOI: 10.1063/1.3329542

Abstract: We present numerical simulations of electromagnetic millimeter-wave propagation in a two-dimensional lattice of dielectric rods arranged in a tenfold Penrose tiling. We find (i) isotropic photonic band gap as expected for quasicrystals and (ii) localized states. We demonstrate that the high frequency edge of the second band gap is characterized by a very small refractive index (fast light). We study the transmission of electromagnetic waves in the frequency range corresponding to fast light and demonstrate that it is related to tunneling through localized states. We use the fast light phenomenon to design a focusing device—a planoconcave lens.

Pseudospectral Renormalization Method for Solitons in Quasicrystal Lattice with the Cubic-Quintic Nonlinearity

Nalan Antar. Journal of Applied Mathematics (2014). Cited by 10. Found by: openalex. DOI: 10.1155/2014/848153

Abstract: We modified the spectral renormalization method as the pseudospectral renormalization method in order to find the localized solutions. The pseudospectral renormalization method can be applied to a large class of problems including different homogeneities. Using this computational method, we demonstrate the existence of two different solitons in optical media described by the self-focusing cubic and the self-defocusing quintic nonlinear Schrödinger equation with quasicrystal lattice. It is shown that there are two different lattice solitons corresponding to the first and the second renormalization factors for the self-focusing cubic and the self-defocusing quintic model. However, the self-focusing quintic nonlinearity without optical lattice does not support two different solitons. We showed that the lattice solitons corresponding to the first and the second renormalization factors have the same powers and amplitudes. We also demonstrate that quintic nonlinearity supports bistable solitons by adding the optical lattice such as a quasicrystal lattice. The linear and nonlinear stabilities of these solitons are investigated using direct simulation of the nonlinear Schrödinger equation with the cubic-quintic nonlinearity and its linearized equation.

Robust Anderson transition in non-Hermitian photonic quasicrystals

Stefano Longhi. Optics Letters (2024). Cited by 9. Found by: openalex. DOI: 10.1364/ol.517182

Abstract: Anderson localization, i.e., the suppression of diffusion in lattices with a random or incommensurate disorder, is a fragile interference phenomenon that is spoiled out in the presence of dephasing effects or a fluctuating disorder. As a consequence, Anderson localization-delocalization phase transitions observed in Hermitian systems, such as in one-dimensional quasicrystals when the amplitude of the incommensurate potential is increased above a threshold, are washed out when dephasing effects are included. Here we consider localization-delocalization spectral phase transitions occurring in non-Hermitian (NH) quasicrystals with local incommensurate gain and loss and show that, contrary to the Hermitian case, the non-Hermitian phase transition is robust against dephasing effects. The results are illustrated by considering synthetic quasicrystals in photonic mesh lattices.

The square Thue–Morse tiling for photonic application

Luigi Moretti; Vito Mocella. The Philosophical Magazine A Journal of Theoretical Experimental and Applied Physics (2008). Cited by 9. Found by: openalex. DOI: 10.1080/14786430802047076

Abstract: The photonic properties of a two-dimensional (2D) photonic aperiodic crystal based on the Thue–Morse (ThMo) substitutional sequence were investigated theoretically. In contrast to traditional photonic quasicrystals based on the Penrose tiling, these structures were obtained by removing the lattice points from a square arrangement, following the inflation rules emerging from the ThMo sequence. The resulting structure does not exhibit the typical translational symmetry of photonic crystals. In particular, it is well known that the ThMo sequence has a singular continuous Fourier transform. This property was transferred directly on the 2D ThMo photonic aperiodic crystal represented by an array of pillars in air. The electromagnetic field distribution can be described as a quasi-localized state, with characteristics lying between the localized states, corresponding to the defect state in a photonic crystal, and the Bloch states, as in the case of the eigenmode in a photonic crystal. The photonic bandgap formation was explored as a function of pillar radius. Furthermore, a preliminary investigation of the defect behaviour in square ThMo tiling was carried out. The electric field in the defect state was revealed to be strictly localized in the defect pillar. These structures provide interesting properties, which could be used to design novel optical devices.

Topological wave phenomena in photonic time quasicrystals

Xiang Ni; Shixiong Yin; Huanan Li; Andrea Alù. Physical review. B./Physical review. B (2025). Cited by 8. Found by: openalex. DOI: 10.1103/physrevb.111.125421

Abstract: Time interfaces consist of abrupt and spatially uniform changes in the optical properties of a medium. Their periodic occurrence forms photonic time crystals, which offer opportunities to tailor classical and quantum light-matter interactions. Here we explore one-dimensional photonic time quasicrystals, formed when time interfaces occur in a quasiperiodic fashion featuring long-range order. We unveil the emergence of topological phases and Hofstadter butterfly spectra in these systems, and demonstrate that their temporal response emulates the localization of topological edge states, enabling the temporal analog of topological Thouless pumping. Our findings open avenues for topological photonics leveraging time as a synthetic dimension, with functionalities that go beyond their spatial counterparts.

Tunable defect modes through the (YBCO-Yttria) based on Octonacci photonic quasicrystals

Youssef Trabelsi; Francis Segovia-Chaves; Naim Ben Ali. Results in Physics (2022). Cited by 8. Found by: openalex. DOI: 10.1016/j.rinp.2022.106176

Abstract: The transmission spectra by a one-dimensional (1D) inhomogeneous stratified medias organized following the Octonacci sequence are investigated. The transfer-matrix method and the Frequency-dependent dispersion formula according to the two-fluid Gorter–Casimir theory are deployed. The proposed hetero-structures are made up of Yttrium oxide, known as yttria, Y2O3 (A) and superconductor (B): yttrium–barium–copper oxide (YBa2Cu3O7), resulting in the (BTO/Y2O3)N/YBCO/(Y2O3/BTO)N multilayered nanostructure array. We report the localization of modes through the multilayered stacks built according to the inflation rule of quasi-periodic Octonacci sequence. Localized defect modes were obtained for specific generation of Octonacci structure and for both TE and TM modes. The optical properties exhibiting different regions with zero transmission called photonic bandgaps (PBGs) that can be tuned by Octonacci order and temperature of Bulk superconductor. The performance of defect modes that allows the propagation of optical signals and the interchannels are evaluated by tailoring the superconductor temperature, the Octonacci order, the direction of the incident wave, and the deformation degree (parameter that control the thickness of superconductor lattices). The proposed system exhibits similar localized resonators operating at cryogenic temperature environments. We verify that bandgaps modifications are influenced by the deformation strain applied along whole thicknesses slab.

Spectral properties of two coupled Fibonacci chains

Anouar Moustaj; Malte Röntgen; Christian V. Morfonios; Peter Schmelcher; C. Morais Smith. New Journal of Physics (2023). Cited by 8. Found by: openalex. DOI: 10.1088/1367-2630/acf0e0

Abstract: Abstract The Fibonacci chain, i.e. a tight-binding model where couplings and/or on-site potentials can take only two different values distributed according to the Fibonacci word, is a classical example of a one-dimensional quasicrystal. With its many intriguing properties, such as a fractal eigenvalue spectrum, the Fibonacci chain offers a rich platform to investigate many of the effects that occur in three-dimensional quasicrystals. In this work, we study the eigenvalues and eigenstates of two identical Fibonacci chains coupled to each other in different ways. We find that this setup allows for a rich variety of effects. Depending on the coupling scheme used, the resulting system (i) possesses an eigenvalue spectrum featuring a richer hierarchical structure compared to the spectrum of a single Fibonacci chain, (ii) shows a coexistence of Bloch and critical eigenstates, or (iii) possesses a large number of degenerate eigenstates, each of which is perfectly localized on only four sites of the system. If additionally, the system is infinitely extended, the macroscopic number of perfectly localized eigenstates induces a perfectly flat quasi band. Especially the second case is interesting from an application perspective, since eigenstates that are of Bloch or of critical character feature largely different transport properties. At the same time, the proposed setup allows for an experimental realization, e.g. with evanescently coupled waveguides, electric circuits, or by patterning an anti-lattice with adatoms on a metallic substrate.

Interface states in two-dimensional quasicrystals with broken inversion symmetry

Danilo Beli; Matheus I. N. Rosa; Luca Lomazzi; Carlos De Marqui; Massimo Ruzzene. Physical Review Applied (2025). Cited by 7. Found by: openalex. DOI: 10.1103/physrevapplied.23.024039

Abstract: Topological metamaterials exhibit remarkable functionality in robust energy localization and waveguiding. Passive systems mostly use periodic structures with nontrivial topological features in their band structures. While quasicrystals could significantly expand this design space by introducing other symmetry orders, their lack of periodicity adds challenges. This work demonstrates the existence of interface states induced by broken inversion symmetries in two-dimensional quasicrystals: Mass dimerization breaks inversion symmetry in a tenfold-symmetric quasicrystal lattice, producing two fivefold sublattices to create domain-wall interfaces for diffractionless waveguiding.

Radiative heat transfer mediated by topological phonon polaritons in a family of quasiperiodic nanoparticle chains

Boxiang Wang; Changying Zhao. International Journal of Heat and Mass Transfer (2023). Cited by 7. Found by: openalex. DOI: 10.1016/j.ijheatmasstransfer.2023.124163

Diffraction and pseudospectra in non-Hermitian quasiperiodic lattices

Ananya Ghatak; Dimitrios Kaltsas; Manas Kulkarni; Konstantinos G. Makris. Physical review. E (2024). Cited by 6. Found by: openalex. DOI: 10.1103/physreve.110.064228

Abstract: Wave dynamics in disordered open media is an intriguing topic and has lately attracted a lot of attention in non-Hermitian physics, especially in photonics. In fact, spatial distributions of gain and loss elements are physically possible in the context of integrated photonic waveguide arrays. In these type of lattices, counterintuitive quantized jumps along the propagation direction appear in the strong disorder limit (where all eigenstates are localized), and they have also been recently experimentally observed. We systematically study the non-Hermitian quasiperiodic Aubry-André-Harper model with on-site gain and loss distribution, with an emphasis on the spectral sensitivity based on pseudospectra analysis. Moreover, diffraction dynamics and the quantized jumps as well as the effect of saturable nonlinearity are investigated in detail. In this paper, we reveal the intricate relation between the nonlinearity and non-hermiticity.

Enhanced Light Scattering Using a Two-Dimensional Quasicrystal-Decorated 3D-Printed Nature-Inspired Bio-photonic Architecture

Partha Kumbhakar; Ashim Pramanik; Shashank Mishra; Raphael M. Tromer; Krishanu Biswas; Arup Dasgupta; Douglas S. Galvão; Chandra Sekhar Tiwary. The Journal of Physical Chemistry C (2023). Cited by 6. Found by: openalex. DOI: 10.1021/acs.jpcc.3c00513

Abstract: A number of strategies have been exploited so far to trap photons inside living cells to obtain high-contrast imaging. Also, launching light inside biological materials is technically challenging. Using photon confinement in a three-dimensional (3D)-printed biomimetic architecture in the presence of a localized surface plasmon resonance (LSPR) promoter can overcome some of these issues. This work compares optical confinement in natural and 3D-printed photonic architectures, namely, fish scale, in the presence of atomically thin Al 70 Co 10 Fe 5 Ni 10 Cu 5 quasicrystals (QCs). Due to their wideband LSPR response, the QCs work as photon scattering hotspots. The architecture acts as an additive source of excitation for the two-dimensional (2D) QCs via total internal reflection (TIR). The computational analysis describes the surface plasmon-based scattering property of 2D QCs. The 3D-printed fish scale's image contrast with the 2D Al 70 Co 10 Fe 5 Ni 10 Cu 5 QC has been compared with other 2D materials (graphene, h-BN, and MoS 2) and outperforms them. The present study conceptually presents a new approach for obtaining high-quality imaging of biological imaging, even using high-energy photons.

Stability of quasicrystalline ultracold fermions to dipolar interactions

Paolo Molignini. Physical Review Research (2025). Cited by 6. Found by: openalex. DOI: 10.1103/szdc-61nl

Abstract: Quasiperiodic potentials can be used to interpolate between localization and delocalization in one dimension. With the rise of optical platforms engineering dipolar interactions, a key question is the stability of quasicrystalline phases under these long-range interactions. In this work, we study repulsive ultracold dipolar fermions in a quasiperiodic optical lattice to characterize the behavior of interacting quasicrystals. We simulate the full time evolution of the typical experimental protocols used to probe quasicrystalline order and localization properties. We extract experimentally measurable dynamical observables and correlation functions to characterize the three phases observed in the noninteracting setting: localized, intermediate, and extended. We then study the stability of such phases to repulsive dipolar interactions. We find that dipolar interactions can completely alter the shape of the phase diagram by stabilizing the intermediate phase, mostly at the expense

of the extended phase. Moreover, in the strongly interacting regime, a resonance-like behavior characterized by density oscillations appears. Remarkably, strong dipolar repulsions can also localize particles even in the absence of quasiperiodicity if the primary lattice is sufficiently deep. Our work shows that dipolar interactions in a quasiperiodic potential can give rise to a complex, tunable coexistence of localized and extended quantum states.

One-dimensional photonic quasicrystals

Mher Ghulinyan. Cambridge University Press eBooks (2014). Cited by 5. Found by: openalex. openalex_2yr: 0.7 (2026). DOI: 10.1017/cbo9781139839501.006

Abstract: Introduction Self-organization is one of the most extraordinary tools in Nature which assembles its elementary building blocks, atoms and molecules, in the form of solid substances. Crystals, in this sense, represent the class of a vast number of solid materials in which atoms (molecules) are brought together in a perfect, spatially periodic manner. A macroscopic crystal, therefore, is made by repeating its smallest microscopic period – the crystal unit cell – infinitely in space (Fig. 5.1(a)). Therefore, when translating the crystal by an integer number of unit cell size, the crystal coincides with itself. This property is in the basis of modern solid-state physics and crystallography and is known as the discrete translational symmetry . Crystals are thus ordered materials in which atoms show both short- (within a couple of nearest neighbors) and long-range order. Because of the periodic lattice potential, the energy states within a crystal are extended, and the electrons can diffuse freely through it. At the opposite extreme of crystalline order there are the disordered or amorphous solid materials, in which the short-range order may still exist, while the long-range order is completely missing (Fig. 5.1(c)). In these kind of solids, e.g. in heavily doped semiconductors or glasses, periodicity is lacking and the atomic potential varies in a random manner. In certain cases, when the degree of randomness is large enough, electron wavefunctions localize exponentially near individual atomic sites and the diffusion of charge carriers vanishes (Anderson localization) [11]. With the advances in X-ray diffraction tools at the beginning of the twentieth century, scientists learned to image the signatures of long-range order of crystals by mapping the symmetries of their reciprocal space lattices. The crystal symmetries for all possible arrangements of perfect periodicity are classified in 230 space groups. Contrary to this, the diffraction form, a disordered solid, shows cloud-like concentric rings indicating diffraction from a totally random atomic lattice and, consequently, a lack of any symmetry or, equivalently, any preferential direction in the solid.

Observation of Topological Corner State Arrays in Photonic Quasicrystals

Aoqian Shi; Yiwei Peng; Jiapei Jiang; Yuchen Peng; Peng Peng; Jianzhi Chen; Hongsheng Chen; Shuangchun Wen; Xiao Lin; Fei Gao; Jianjun Liu. arXiv (Cornell University) (2022). Cited by 5. Found by: openalex. DOI: 10.48550/arxiv.2209.05751

Abstract: Recently, the studies of topological corner states (TCSs) are extended from crystals to quasicrystals, which are referred to as higher-order topological quasicrystalline insulators (HOTQIs). However, the TCSs of complete quasi-periodic structure in photonic systems have yet to be demonstrated. Moreover, there is only one TCS in each corner region in higher-order topological insulators (HOTIs). Increasing the number of TCS is expected to increase application potential of TCSs. In this work, HOTQIs in photonic systems are experimentally observed. It is found that HOTQIs possess TCS arrays, and each TCS array contains several TCSs. Furthermore, the universal theoretical framework of the multimer analysis method is improved, and the difference in the average charge density is proposed as a real-space topological index. These results will open up new ideas for investigating highly integrated, multi-region localized TCSs and are expected to provide new ways to explore topological phenomena and the applications of photonic quasicrystals.

Observation of Light Localization at the Edges of Quasicrystal Waveguide Arrays

Sergey K. Ivanov; S. A. Zhuravitskii; N. S. Kostyuchenko; Nikolay N. Skryabin; I. V. Dyakonov; A. A. Kalinkin; S. P. Kulik; V. O. Kompanets; . . ; Sergey Alyatkin; A. Ferrando; Yaroslav V. Kartashov; Victor N. Zadkov. Physical Review Letters (2025). Cited by 4. Found by: openalex. DOI: 10.1103/physrevlett.134.113803

Abstract: Quasicrystals are unique systems that, unlike periodic structures, lack translational symmetry but exhibit long-range order dramatically enriching the system properties. While evolution of light in the bulk of photonic quasicrystals is well studied, experimental evidences of light localization near the edge of truncated photonic quasicrystal structures are practically absent. In this Letter, we observe both linear and nonlinear localization of light at the edges of radially cropped quasicrystal waveguide arrays, forming an aperiodic Penrose tiling. Our theoretical analysis reveals that for certain truncation radii, the system exhibits linear eigenstates localized at the edge of the truncated array, whereas for other radii, this localization does not occur, highlighting the significant influence of truncation on edge light localization. Using single-waveguide excitations, we experimentally confirm the presence of localized states in Penrose arrays inscribed by a femtosecond laser and investigate the effects of nonlinearity on these states. Our theoretical findings identify a family of edge solitons, and experimentally, we observe a transition from linear localized states to edge solitons as the power of the input pulse increases. Our results represent the first experimental demonstration of localization phenomena induced by the selective truncation of quasiperiodic photonic systems.

Optical properties of 1D photonic time quasicrystals

Anny Araújo; Carlos Costa; S. Azevedo; G. M. Viswanathan; Claudionor Bezerra. Journal of the Optical Society of America B (2025). Cited by 4. Found by: openalex. DOI: 10.1364/josab.542001

Abstract: Photonic time crystals are materials in which the refractive index varies periodically and suddenly in time. This material exhibits unusual properties, such as momentum bands separated by gaps. Because of the sudden change in the optical properties, the light propagating in these materials experiences time refraction and time reflection, analogous to refraction and reflection in conventional photonic crystals. Interference between time-refracted and time-reflected waves gives rise to Floquet–Bloch states and dispersion bands, which are momentum gapped. In this paper, we employ a transfer-matrix treatment to study the propagation of light waves in photonic time quasicrystals composed of two alternating building slabs, A and B , constructed according to Fibonacci, Thue–Morse, and Double-Period sequences. We present numerical results for the photonic band structure in terms of the dimensionless wave-vector k ¯= k / k 0 and the normalized dimensionless Bloch's angular frequency Ω T / . Our numerical results show the bandgap behavior as a function of the ratio between refractive indices n BA = n B / n A and layer thicknesses t BA = t B / t A . We also studied the localization and self-similar behavior of the band structures, whose fractality can be described by a power law. We show that the power law scaling index , which can be identified as being a diffusion constant of the spectra, presents a non-monotonic dependence on n BA and t BA . Finally, plots of the transmission/reflection spectra, also calculated via the transfer-matrix method, in terms of the dimensionless wave-vector k ¯= k / k 0 , are shown for different values of refractive indices n BA and layer thicknesses t BA .

Symmetry-Induced Light Confinement in a Photonic Quasicrystal-Based Mirrorless Cavity

Gianluigi Zito; Giulia Rusciano; Antonio Sasso; Sergio De Nicola. Crystals (2016). Cited by 4. Found by: openalex. openalex_2yr: 3.465 (2026). DOI: 10.3390/cryst6090111

Abstract: We numerically investigate the electromagnetic field localization in a two-dimensional photonic quasicrystal generated with a holographic tiling. We demonstrate that light confinement can be induced into an air mirrorless cavity by the inherent symmetry of the spatial distribution of the dielectric scatterers forming the side walls of the open cavity. Furthermore, the propagation direction can be controlled by suitable designs of the structure. This opens up new avenues for designing photonic materials and devices.

Resonant modes of 12-fold symmetric defect free photonic quasicrystal

Minfeng Chen; Yunjing Li; Yuh-Jen Cheng; Yai-Chung Chang; Chun-Yen Chang. Optics Express (2014). Cited by 4. Found by: openalex. DOI: 10.1364/oe.22.002007

Abstract: This work investigates the resonant modes of a 12-fold symmetric defect free photonic quasicrystal (PQC) nanorod array using finite difference time domain (FDTD) simulation. Localized modes can exist in PQC without introducing defects due to the lack of translational symmetry. The resonant modes of the unit cell PQC and the one time expanded PQC from unit cell are systematically examined. The resonant spectrum is that of a single rod modified by the interaction among PQC nanorods. The mode confinement is contributed by guided resonance and destructive interference scattering. The self-scaling similarity of resonant spectrum and mode profile are also investigated.

Tunability in the terahertz range of a one-dimensional photonic quasicrystal containing an InSb semiconductor

Francis Segovia-Chaves; Hussein A. Elsayed; Ahmed Mehaney. Optik (2021). Cited by 4. Found by: openalex. DOI: 10.1016/j.ijleo.2021.167675

Diffraction propagation of light wave and image distortionless transmission in ten-fold symmetric photonic quasicrystal lattices

Wentao Jin; Meng Song; Wen-Cheng Hu; Yan Xue; Yuanmei Gao. Optics Communications (2021). Cited by 4. Found by: openalex. DOI: 10.1016/j.optcom.2021.127329

Properties of defect modes in two-dimensional organic octagonal quasicrystals at low dielectric contrast

Yuanyuan Cai; Zhi Wang; Xiao Chen; Changwei Li; Xiaoqing Wang; Shuai Feng; Yiquan Wang. Optics Communications (2018). Cited by 4. Found by: openalex. DOI: 10.1016/j.optcom.2018.06.010

Isotropic photonic bandgap in Penrose textured metallic mircocavity

J. Y. Zhang; Hoi Lam Tam; W.H. Wong; Edwin Yue-Bun Pun; Jian-Bai Xia; Kok-Wai Cheah. Solid State Communications (2006). Cited by 4. Found by: openalex. DOI: 10.1016/j.ssc.2006.02.035

Anomalous optical Anderson localization in mixed one dimensional photonic quasicrystals

Cunding Liu; Mingdong Kong; Bincheng Li. Optics Express (2015). Cited by 3. Found by: openalex. DOI: 10.1364/oe.23.0a1297

Abstract: Anomalous optical Anderson localization (AOAL) in mixed one-dimensional (1D) photonic quasicrystals with matching impedance is obtained when the average refractive index of left- and right-handed layers is approximately zero. The transport properties of ordinary and anomalous localization are investigated and compared. The difficulties in expressing analytically the scaling factors of the mixed photonic quasicrystals are illustrated from Hamiltonian-map analysis. An approach based on transfer matrix method is proposed to simulate the localization behavior. From the simulation, it is found that the narrow distribution of the phase shift is responsible for AOAL in the mixed photonic structures. The scaling factors of AOAL decrease with the broadening of the phase shift distribution. The maximum phase shifts of the mixed photonic structures determine the lower boundary of the anomalous localization.

Interaction of fast electron beam with photonic quasicrystals

Wei Zhong; Jinying Xu; Xiangdong Zhang. Optics Express (2009). Cited by 3. Found by: openalex. DOI: 10.1364/oe.17.013270

Abstract: The interactions of two-dimensional 10-fold and 12-fold photonic quasicrystals with a moving electron beam have been studied by using the multiple-scattering method. The electron energy loss spectroscopy (EELS) and three-dimensional local density of states in these systems have been calculated. Some three-dimensional localized states in these defect-free two-dimensional systems have been found. It has been demonstrated that these localized states can be explored by means of the EELS.

Programmable 2D photonic quasicrystals with real-time reconfigurability

Peisheng Sun; Xuesong Hu; Ruonan Li; Jingjing Xue; Haitao Zhou; Linhua Ye; Yun Zhang; Li-Gang Wang; Junxiang Zhang. Optics Letters (2025). Cited by 3. Found by: openalex. DOI: 10.1364/ol.568662

Abstract: We demonstrate a programmable approach to generate 2D photonic quasicrystals (PQCs) with real-time reconfigurability. By encoding complex amplitude onto the spatial light modulator (SLM), we directly emulate tunable PQC lattices, where the engineered diffractive patterns define the PQC structure. This approach enables real-time switching (60 Hz) between diverse PQC lattices with up to 26-fold rotational symmetry, tailorable orbital angular momentum (OAM), and distortion control. Moreover, parameter optimization can further customize the PQC structures to meet additional demands. This SLM-based method enables scalable PQCs with experimentally validated reliability. It offers direct applications in localization, topological photonics, and parallel optical trapping, providing a versatile platform for emerging optical and quantum applications.

Shear based gap control in 2D photonic quasicrystals of dielectric cylinders

Ángel Andueza; Joaquín Sevilla; Jesús Pérez-Conde; Kang Wang. Optics Express (2021). Cited by 3. Found by: openalex. DOI: 10.1364/oe.427235

Abstract: 2D dielectric photonic quasicrystals can be designed to show isotropic band gaps. In this work we study a quasiperiodic lattice made of silicon dielectric cylinders (= 12) arranged as periodic unit cell based on a decagonal approximant of a quasiperiodic Penrose lattice. We analyze the bulk properties of the resulting lattice as well as the bright states excited in the gap, which correspond to localized resonances of the electromagnetic field in specific cylinder clusters of the lattice. Then we introduce a controlled shear deformation which breaks the decagonal symmetry and evaluate the width reduction of the gap together with the evolution of the resonances, for all shear values compatible with physical constraints (cylinder contact). The gap width reduction reaches 18.5% while different states change their frequency in several ways. Realistic analysis of the actual transmission of the electromagnetic radiation, often missing in the literature, has been performed for a finite "slice" of the proposed quasicrystals structure. Two calculation procedures based on MIT Photonic Bands (MPB) and Finite Integration Technique (FIT) are used for the bulk and the finite structures showing an excellent agreement between them.

Properties of microcavities in two-dimensional photonic quasicrystals with octagonal rotational symmetry

D. M. Beggs; M. A. Kaliteevski; R. A. Abram. Journal of Modern Optics (2007). Cited by 3. Found by: openalex. DOI: 10.1080/09500340601102441

Abstract: It is now well established that two-dimensional structures with dielectric variations based on quasicrystalline tilings are able to support photonic band gaps. Here, we have investigated the properties of the localized defect modes with frequencies within the photonic band gap for a certain kind of two-dimensional octagonal photonic quasicrystal with lattice vacancies breaking the quasi-periodic symmetry. The eigenfrequencies of such localized modes are given as a function of filling fractions for four distinct microcavity designs, and the electromagnetic field profiles of the localized modes are explored. The calculations of the eigenfrequencies and electromagnetic field profiles were performed using a supercell approximation in a plane wave method.

Simultaneous localization of photons and phonons in defect-free dodecagonal phoxonic quasicrystals

Bihang Xu; Zhong Wang; Yixiang Tan; Tianbao Yu. Modern Physics Letters B (2018). Cited by 3. Found by: openalex. DOI: 10.1142/s0217984918500963

Abstract: In dodecagonal phoxonic quasicrytals (PhXQCs) with a very high rotational symmetry, we demonstrate numerically large phoxonic band gaps (PhXBGs, the coexistence of photonic and phononic band gaps). By computing the existence and dependence of PhXBGs on the choice of radius of holes, we find that PhXQCs can possess simultaneous photonic and phononic band gaps over a rather wide range of geometric parameters. Furthermore, localized modes of THz photons and tens of MHz phonons may exist inside and outside band gaps in defect-free PhXQCs. The electromagnetic and elastic field can be confined simultaneously around the quasicrytals center and decay in a length scale of several basic cells. As a kind of quasiperiodic structures, 12-fold PhXQCs provide a good candidate for simultaneously tailoring electromagnetic and elastic waves. Moreover, these structures exhibit some interesting characteristics due to the very high symmetry.

Interface states in two-dimensional quasicrystals with broken inversion symmetry

Danilo Beli; Matheus I. N. Rosa; Luca Lomazzi; Carlos De Marqui; Massimo Ruzzene. arXiv (Cornell University) (2023). Cited by 3. Found by: openalex. DOI: 10.48550/arxiv.2304.09628

Abstract: We investigate the existence of interface states induced by broken inversion symmetries in two-dimensional quasicrystal lattices. We introduce a 10-fold rotationally symmetric quasicrystal lattice whose inversion symmetry is broken through a mass dimerization that produces two 5-fold symmetric sub-lattices. By considering resonator scatterers attached to an elastic plate, we illustrate the emergence of bands of interface states that accompany a band inversion of the quasicrystal spectrum as a function of the dimerization parameter. These bands are filled by modes which are localized along domain-wall interfaces separating regions of opposite inversion symmetry. These features draw parallels to the dynamic behavior of topological interface states in the context of the valley Hall effect, which has been so far limited to periodic lattices. We numerically and experimentally demonstrate wave-guiding in a quasicrystal lattice featuring a zig-zag interface with sharp turns of 36 degrees, which goes beyond the limitation of 60 degrees associated with 6-fold symmetric (i.e., honeycomb) periodic lattices. Our results provide new opportunities for symmetry-based quasicrystalline topological waveguides that do not require time-reversal symmetry breaking, and that allow for higher freedom in the design of their waveguiding trajectories by leveraging higher-order rotational symmetries.

Observation of Topological Corner State Arrays in Photonic Quasicrystals (Laser Photonics Rev. 18(7)/2024)

Aoqian Shi; Yiwei Peng; Jiapei Jiang; Yuchen Peng; Peng Peng; Jianzhi Chen; Hongsheng Chen; Shuangchun Wen; Xiao Lin; Fei Gao; Jianjun Liu. Laser & Photonics Review (2024). Cited by 2. Found by: openalex. DOI: 10.1002/lpor.202470041

Abstract: Topological Corner States in Photonic Quasicrystals Higher-order topological quasicrystalline insulators (HOTQIs) in photonic systems are theoretically and experimentally observed by Fei Gao, Jianjun Liu, and co-workers (see article number 2300956). It is found that HOTQIs possess topological corner state (TCS) arrays, with each TCS array containing several TCSs. This work introduces new ideas for investigating highly integrated, multi-region localized TCSs and is expected to provide novel approaches for exploring topological phenomena and the applications of photonic quasicrystals.

Negative refraction and localized states of a classical wave in high-symmetry quasicrystals

Xiangdong Zhang; Wei Zhong; Zhifang Feng; Yiquan Wang; Zhi-Yuan Li; Daozhong Zhang. The Philosophical Magazine A Journal of Theoretical Experimental and Applied Physics (2010). Cited by 2. Found by: openalex. DOI: 10.1080/14786435.2010.512576

Abstract: Recently, negative refraction of electromagnetic waves in periodic photonic crystals has been demonstrated experimentally and sub-wavelength images observed. A theoretical and experimental investigation is reported of the electromagnetic wave transport in high-symmetry photonic quasicrystals (QCs). It is shown that negative refraction can appear in these transparent quasicrystalline photonic structures. It is interesting that highly symmetric two-dimensional photonic QCs possess a universal feature for non-near-field focus of two kinds of polarized waves (S wave and P wave). That is, the non-near-field focus for two kinds of polarized waves can be realized by using flat lenses, which consist of some high-symmetric two-dimensional photonic QCs with the same structures and parameters. In addition, some two-dimensional and three-dimensional localized states in defect-free photonic QCs have been found. It is evident that these unusual localized states can be explored by means of electron energy loss spectroscopy.

Light wave propagation and diffraction inhibition in three-dimensional photonic quasicrystal lattices

Meng Song; Jin Wentao; Shaochun Fu; Yuanmei Gao. Results in Physics (2023). Cited by 2. Found by: openalex. DOI: 10.1016/j.rinp.2023.106884

Abstract: We have investigated the linear propagation of light wave in three-dimensional photonic quasicrystal lattices by numerical simulation. The diffraction characteristics of the light waves in the longitudinal non-modulated, in-phase modulated and out-of-phase modulated quasicrystal lattices have been compared and discussed. We have demonstrated the diffraction inhibition of light wave in three-dimensional photonic quasicrystals. Linear transverse light localization has been realized in the longitudinal out-of-phase modulated three-dimensional quasicrystal lattices by adjusting the structural parameters appropriately. Furthermore, the results also show that the longitudinal out-of-phase modulated photonic quasicrystals are easier to produce diffraction inhibition than the longitudinal out-of-phase modulated periodic lattices under low transverse refractive index contrast. Distortionless dot image transmission has numerically demonstrated in out-of-phase modulated photonic quasicrystals array. It provides a new way for distortion-free image transmission, which has promising applications in optical information transmission and all-optical signal processing.

Probing Structural Correlated Disorder with Photonic Transport

Ying-Yue Yang; Yi-Jun Chang; Yong-Heng Lu; Ka-Di Zhu; Xian-Min Jin. ACS Photonics (2024). Cited by 2. Found by: openalex. openalex_2yr: 5.567 (2026). DOI: 10.1021/acsphotonics.3c01385

Abstract: Wave spreading has been widely studied in various disordered materials including quasiperiodic, amorphous, and completely random materials based on the spatial order. However, there is no clear observable mechanism connecting the underlying correlated disorder to wave transport. Here, we experimentally probe correlated disorder with different characteristic length scales in disordered materials by directly observing the photonic transport on a chip. The density of states of corresponding aperiodic structures shows different degrees of singularity, suggesting diverse behaviors of photons. By analyzing localization and transport properties of photons, we demonstrate that compared to the nearly ballistic transport in the quasiperiodic lattice with long-range order, photon behavior is partly destroyed from ballistic transport in amorphous lattices with short-range

order while totally deviating from ballistic spreading in completely random lattices without spatial order. The spreading coefficient related to the photon variance represents an evident difference for correlated disorder with different characteristic length scales. The resolution of diverse transport behaviors is sufficient to distinguish different disorder types, suggesting an eligible identifier for disorder materials.

Topological states in Penrose-square photonic crystals

Qichen Zhang; Jianzhi Chen; Dongyang Liu; Jianjun Liu. Journal of the Optical Society of America A (2024). Cited by 2. Found by: openalex. DOI: 10.1364/josaa.520606

Abstract: Topological edge states (TESs) and topological corner states (TCSs) in photonic crystals (PCs) provide an effective way to control the propagation and localization of light. The topological performance of integrated photonic devices can be improved by introducing the basic structural unit of photonic quasicrystals (PQCs) into PCs. However, the previous works arranged the basic structural unit of Stampfli-type and 12-fold Penrose-type photonic quasicrystals into triangular lattices, which have a complex structure and allow light to only propagate around 60° or 120° corners, limiting their applications. In this paper, a Penrose-square PC is proposed, which realizes both TESs and TCSs, and light successfully propagates around 90° corners. This work may reduce the difficulties encountered in the preparation of topological photonic crystals (TPCs) structured by arranging the basic structural units of PQCs periodically. It also provides a new, to the best of our knowledge, platform for studying TPCs and new ideas for improving the performance of integrated photonic devices.

About optical localization in photonic quasicrystals

Faris Mohammed; Alexander Quandt. Optical and Quantum Electronics (2016). Cited by 1. Found by: openalex. DOI: 10.1007/s11082-016-0648-1

Three- and four-material photonic quasicrystal

Chittaranjan Nayak; Jitendra K. Behera; Rajesh Moharana; Leonid Kotov. Optical Engineering (2025). Cited by 1. Found by: openalex. DOI: 10.1117/1.oe.64.7.077101

Abstract: Photonic quasicrystals (PQCs) are a unique class of materials characterized by aperiodic yet ordered structures, blending the periodicity of PQCs with the randomness of disordered media. These materials exhibit photonic bandgaps (PBGs) and localized modes, enabling advanced optical control. Although traditional photonic quasicrystals have been extensively studied, the exploration of multi-component photonic quasicrystals, such as tribonacci PQCs (TPQCs) and quadribonacci PQCs (QPQCs) structures, remains uncharted. We investigate the optical properties of TPQCs and QPQCs composed of three and four materials, respectively, using the transfer matrix method. Our findings reveal that TPQCs and QPQCs exhibit PBGs at longer wavelengths, influenced by material configuration and composition. In addition, the PBGs' position, width, and intensity are highly sensitive to the refractive index and arrangement of the constituent materials. This research provides new insights into the design of advanced photonic devices based on multi-component PQCs, highlighting their potential for enhanced optical performance and functionality.

Mode localization in defect-free GaN-based bottom-up photonic quasicrystal lasers

Tzeng-Tsong Wu; Chih-Cheng Chen; Hao Chen; Tien-Chang Lu; Shing-Chung Wang; Cheng-Huang Kuo. (2012). Cited by 1. Found by: openalex. DOI: 10.1109/islc.2012.6348389

Abstract: GaN-based bottom-up photonic quasicrystal (PQC) lasers were realized and characterized. Photonic quasicrystal nanopillars were realized by nanoimprint technique and selective area growth. Highly localized mode was identified for the first time in GaN-based PQC lasers.

Fast Light and Focusing in 2D Photonic Quasicrystals

Yair Neve-Oz; Therese Pollok; Sven Burger; M. Golosovsky; Dan Davidov. PIERS Online (2009). Cited by 1. Found by: openalex. DOI: 10.2529/piers081223080021

Abstract: We present numerical studies of the electromagnetic wave propagation in a meta- material built from dielectric rods. The rods were arranged in a quasicrystalline lattice based on 10-fold Penrose tiling. We find wide isotropic band gaps and localized states, as expected in a quasicrystalline lattice. Our most important finding is the presence of the band gap edge states that are characterized by a very small refractive index (fast light). We use this phenomenon to design a focusing device | a plano-concave lens.

Emission enhancement in hybrid Tamm plasmon/photonic quasicrystal structure

Konstantin M. Morozov; . . ; Aleksei V. Belonovskii; Elizaveta I. Girshova. SN Applied Sciences (2019). Cited by 1. Found by: openalex. DOI: 10.1007/s42452-019-1375-6

Formation of nonlinear modes in one-dimensional quasiperiodic lattices with a mobility edge

Dmitry A. Zezyulin; G. L. Alfimov. Physical Review A (2024). Cited by 1. Found by: openalex. DOI: 10.1103/physreva.110.063304

Abstract: We investigate the formation of steady states in one-dimensional Bose-Einstein condensates of repulsively interacting ultracold atoms loaded into a quasiperiodic potential created by two incommensurate periodic lattices. We study the transformations between linear and nonlinear modes and describe the general patterns that govern the birth of nonlinear modes emerging in spectral gaps near band edges. We show that nonlinear modes in a symmetric potential undergo both symmetry-breaking pitchfork bifurcations and saddle-node bifurcations, mimicking the prototypical behaviors of symmetric and asymmetric double-well potentials. The properties of the nonlinear modes differ for bifurcations occurring below and above the mobility edge. In the generic case, when the quasiperiodic potential consists of two incommensurate lattices with a nonzero phase shift between them, the formation of localized modes in the spectral gaps occurs through a cascade of saddle-node bifurcations. Because of the analogy between the Gross-Pitaevskii equation and the nonlinear Schr\"odinger equation, our results can also be applied to optical modes guided by quasiperiodic photonic lattices.

Coupling photonic quasicrystals with a one-dimensional critical high-temperature superconducting cavity

Francis Segovia-Chaves; Youssef Trabelsi; T.A. Taha. Optik (2023). Cited by 1. Found by: openalex. DOI: 10.1016/j.ijleo.2023.170847

Novel hybrid organic/inorganic 2D photonic quasicrystals with 8-fold and 12-fold diffraction symmetries

Lucia Petti; Massimo Rippa; Rossella Capasso; Marco Zanella; Liberato Manna; Jun Zhou; Weijie Song; Pasquale Mormile. Proceedings of SPIE, the International Society for Optical Engineering/Proceedings of SPIE (2012). Cited by 1. Found by: openalex. DOI: 10.1117/12.923183

Abstract: Quasiperiodic crystals (QCs) are a new class of materials that exhibit long-range aperiodic translational order and high rotational symmetries. Unlike periodically arranged photonic crystals (or photonic band-gaps), PQCs possess unique light localization and transport properties related to their complex, multifractal energy spectra. Advances in 2D photonic structures are expected in the introduction of non linear and/or active functionality into a 2D PQC. One-dimensional semiconductor nanostructures are likewise promising materials both in fundamental research and in practical applications. CdSe/CdS rods present the appealing characteristics of strong and tunable light emission from green to red, are highly fluorescent and show linearly polarized emission. These characteristics open the way to a new class of hybrid devices based on polymers and colloidal NRs in which the unique optical properties of the inorganic moiety are combined with the processability of the host matrix to develop new high performing optical devices such as organic light-emitting diodes, ultra-low threshold lasers and non-linear devices. One of the challenges of these applications is the incorporation of inorganic nanoparticles into organic polymer matrices, since this is usually accompanied by phase separation, aggregation of nanoparticles, loss of transparency and luminescence quenching due to exciton energy transfer. In this paper two-dimensional (2D) hybrid PQCs which consist of air rods in a nanocomposite prepared by incorporating CdSe/CdS core/shell nanorods (NR) in a polymer are proposed and experimentally demonstrated. Scanning electron microscopy and far field diffraction are used to characterize the experimental structures.

Quantum Diffusion in a Photonic Fibonacci Chain: From Localization to Ballistic Dynamics

Jiankun Zhu; Yao Qin; Yuxiang Guo; Jizhou Wu; Sheng-Jun Yang; Yucheng Wang; Jingyun Fan. Physical Review Letters (2026). Cited by 1. Found by: openalex. DOI: 10.1103/tjm4-gjx8

Abstract: Quantum transport remains a central yet experimentally challenging problem in condensed matter and quantum physics. Here we report the first complete experimental characterization of the full spectrum of quantum transport behaviors in a one-dimensional Fibonacci chain-the paradigmatic quasicrystalline model-spanning localization, subdiffusion, normal and superdiffusion, and ballistic transport. Using a tunable photonic quantum-walk platform, these regimes are unambiguously resolved through their distinct power-law scalings of the mean square displacement and smooth autocorrelation function, together with pronounced oscillatory dynamical structures. These signatures arise from the intrinsic multifractal spectra and hyperuniform order of the Fibonacci quasicrystal, long predicted but never before experimentally resolved. Beyond mapping a comprehensive transport regime diagram, our highly controllable platform provides a powerful and versatile framework for exploring quasiperiodicity, multifractal criticality, and emergent quantum transport phenomena.

Vortex solitons in disclination quasicrystals

Hua Zhong; Yaroslav Kartashov; Yongdong Li; Fangwei Ye; Y Zhang. Reports on Progress in Physics (2026). Cited by 1. Found by: openalex. DOI: 10.1088/1361-6633/ae73b5

Abstract: Quasicrystals are ubiquitous in nature. They represent unique aperiodic materials featuring long-range order that occupy an intermediate niche between exactly periodic and disordered materials. These properties are reflected in the unusual evolutionary dynamics of excitations and unique localization properties of linear eigenmodes supported by quasicrystals. In particular, being the structures characterized by discrete rotational symmetryC of order , photonic quasicrystals can support the stable propagation of linear vortex-carrying light beams and vortex solitons. However, the impact of discrete rotational symmetry of quasicrystals on the properties of vortex light states has not been investigated so far, as only the systems constructed using the simplest Penrose tiling or corresponding optically induced Penrose lattices are considered in this context. Here, we propose a broad class of quasicrystals with global topological defects-disclinations-introduced into their structure that allows to produce novel quasicrystalline structures with any desired order of discrete rotational symmetry from the basic Penrose structure. Such global topological deformation substantially enriches the linear spectrum of quasicrystals, allowing them to support new types of linear vortex states and bifurcating from them families of stable thresholdless vortex solitons with unusual intensity and phase distributions. We find two different classes of stable vortex solitons comprising in-phase or out-of-phase pairs of closely located bright spots, with total intensity distribution reflecting a particular discrete rotational symmetry of the quasicrystal with disclination. Remarkably, even low-charge vortex solitons can be stable in quasicrystals with disclinations, whereas stability intervals for them broaden with the decrease of the discrete rotational symmetryC of quasicrystals. Our results extend the theory of localization in quasicrystals to structures with global topological deformation, highlighting emerging prospects for the robust transmission of power or information arising in these systems.

A tribute to Sydney G. Davison

Aart W. Kleyn. Progress in Surface Science (2003). Cited by 1. Found by: openalex. DOI: 10.1016/j.progsurf.2003.09.003

Stability of quasicrystalline ultracold fermions to dipolar interactions

Paolo Molignini. arXiv (Cornell University) (2024). Cited by 1. Found by: openalex. DOI: 10.48550/arxiv.2403.04830

Abstract: Quasiperiodic potentials can be used to interpolate between localization and delocalization in one dimension. With the rise of optical platforms engineering dipolar interactions, a key question is the stability of quasicrystalline phases under these long-range interactions. In this work, we study repulsive ultracold dipolar fermions in a quasiperiodic optical lattice to characterize the behavior of interacting quasicrystals. We simulate the full time evolution of the typical experimental protocols used to probe quasicrystalline order and localization properties. We extract experimentally measurable dynamical observables and correlation functions to characterize the three phases observed in the noninteracting setting: localized, intermediate, and extended. We then study the stability of such phases to repulsive dipolar interactions. We find that dipolar interactions can completely alter the shape of the phase diagram by stabilizing the intermediate phase, mostly at the expense of the extended phase. Moreover, in the strongly interacting regime, a resonance-like behavior characterized by density oscillations appears. Remarkably, strong dipolar repulsions can also localize particles even in the absence of quasiperiodicity if the primary lattice is sufficiently deep. Our work shows that dipolar interactions in a quasiperiodic potential can give rise to a complex, tuneable coexistence of localized and extended quantum states.

Engineering Aperiodic Spiral Order in Nanophotonics: Fundamentals and Device Applications

Luca Dal Negro; Nate Lawrence; Jacob Trevino. (2015). Cited by 1. Found by: openalex. DOI: 10.1201/b19365-5

Abstract: In this chapter, we present our work on the engineering of aperiodic spiral order for nanophotonic device applications. After introducing the guiding ideas behind aperiodic nanophotonics, we discuss the distinctive structural and optical properties of arrays of metal- dielectric nanoparticles with Vogel spiral geometry. This fascinating class of optical nanostructures offers unprecedented opportunities to manipulate light scattering and resonant phenomena over broad frequency spectra. In particular, photonic Vogel spirals support large photonic bandgaps (PBGs) as well as a high density of radially localized optical modes that cannot be found in regular photonic crystals or quasicrystal structures, and are ideally suited to boost polarization-insensitive light-matter coupling on planar substrates. In addition, optical Vogel spirals feature distinctive scattering resonances carrying orbital angular momentum (OAM) of light with a rich azimuthal spectrum, uniquely tailored by their distinctive aperiodic geometry. The direct generation and manipulation of structured light on optical chips using nanophotonic Vogel spirals provide exciting opportunities for engineering applications to broadband optical sources, secure communication, photovoltaics, and optical sensing.

Colloquium: Synthetic quantum matter in non-standard geometries

Tobias Graß; Dario Bercioux; Utso Bhattacharya; Maciej Lewenstein; Hai Son Nguyen; Christof Weitenberg. arXiv (Cornell University) (2024). Cited by 1. Found by: openalex. DOI: 10.48550/arxiv.2407.06105

Abstract: Quantum simulation is making a significant impact on scientific research. The prevailing tendency of the field is to build quantum simulators that get closer to real-world systems of interest, in particular electronic materials. However, progress in the microscopic design also provides an opportunity for an orthogonal research direction: building quantum many-body systems beyond real-world limitations. This colloquium takes this perspective: Concentrating on synthetic quantum matter in non-standard lattice geometries, such as fractal lattices or quasicrystals, higher-dimensional or curved spaces, it aims at providing a fresh introduction to the field of quantum simulation aligned with recent trends across various quantum simulation platforms, including atomic, photonic, and electronic devices. We also shine light on the novel phenomena which arise from these geometries: Condensed matter physicists may appreciate the variety of different localization properties as well as novel topological phases which are offered by such exotic quantum simulators. But also in the search of quantum models for gravity and cosmology, quantum simulators of curved spaces can provide a useful experimental tool.

Localized resonant states and transmission in a two- dimensional photonic quasicrystal

Yair Neve-Oz; Therese Pollok; Sven Burger; M. Golosovsky; Dan Davidov. International Conference on Applied Mathematics (2011). Cited by 0. Found by: openalex. URL: https://openalex.org/W135497521

Abstract: We report numerical studies of the electromagnetic wave transmission through a two-dimensional photonic quasicrystal (PQC) based on a rhombic Penrose tiling. We find a frequency range where the wavelength in the array is especially large. The direction of propagating waves emerging from such an array can be controlled by adjusting the array's shape. Moreover, we show that due to the quasiperiodic structure, high transmission can be achieved by tunneling through the chain of localized resonant states.

Photonic band gaps and localization in two-dimensional metallic quasicrystals

Enrique Maciá. (2016). Cited by 0. Found by: openalex. URL: https://openalex.org/W2560550406

Multiple -wavelength Fabry-Perot filters with localized modes in symmetric photonics quasicrystals

Liu Cunding; Mingdong Kong; Bincheng Li. (2016). Cited by 0. Found by: openalex. DOI: 10.1364/oic.2016.fa.9

Abstract: Optical filters at different wavelengths are constructed from effective band gaps of one dimensional Fibonacci quasicrystals. The construction condition, performances and possible applications of the filters are discussed.

Finite-difference methods used to model photonic wave localization in 3D quasicrystals

Aditi Risbud. MRS Bulletin (2017). Cited by 0. Found by: openalex. DOI: 10.1557/mrs.2017.39

Diffraction and pseudospectra in non-Hermitian quasiperiodic lattices

Ananya Ghatak; Dimitrios Kaltsas; Manas Kulkarni; Konstantinos G. Makris. arXiv (Cornell University) (2024). Cited by 0. Found by: openalex. DOI: 10.48550/arxiv.2410.09185

Abstract: Wave dynamics in disordered open media is an intriguing topic, and has lately attracted a lot of attention in non-Hermitian physics, especially in photonics. In fact, spatial distributions of gain and loss elements are physically possible in the context of integrated photonic waveguide arrays. In particular, in these type of lattices, counter-intuitive quantized jumps along the propagation direction appear in the strong disorder limit (where all eigenstates are localized) and they have also been recently experimentally observed. We systematically study the non-Hermitian quasiperiodic Aubry-André-Harper model with on-site gain and loss distribution (NHAAH), with an emphasis on the spectral sensitivity based on pseudospectra analysis. Moreover, diffraction dynamics and the quantized jumps, as well as, the effect of saturable nonlinearity, are investigated in detail. Our study reveals the intricate relation between the nonlinearity and non-Hermiticity.

Reentrant delocalization transition in one-dimensional photonic quasicrystals

Sachin Vaidya; Christina Jörg; Kyle Linn; Megan Goh; Mikael C. Rechtsman. arXiv (Cornell University) (2022). Cited by 0. Found by: openalex. DOI: 10.48550/arxiv.2211.06047

Abstract: Waves propagating in certain one-dimensional quasiperiodic lattices are known to exhibit a sharp localization transition. We theoretically predict and experimentally observe that the localization of light in one-dimensional photonic quasicrystals may be followed by a second delocalization transition for some states on increasing quasiperiodic modulation strength - an example of a reentrant transition. We further propose that this phenomenon can be qualitatively captured by a dimerized tight-binding model with long-range couplings.

Appearance of Mobility Edge in Self-Dual Quasiperiodic Lattices

Gang Wang; Nianbei Li; Tsuneyoshi Nakayama. arXiv (Cornell University) (2013). Cited by 0. Found by: openalex. DOI: 10.48550/arxiv.1312.0844

Abstract: Within the framework of the Aubry-Andre model, one kind of self-dual quasiperiodic lattice, it is known that a sharp transition occurs from \emph{all} eigenstates being extended to \emph{all} being localized. The common perception for this type of quasiperiodic lattice is that the self-duality excludes the appearance of the mobility edge separating localized from extended states. In this work, we propose a multi-chromatic quasiperiodic lattice model retaining the self-duality identical to the Aubry-Andre model, and demonstrate numerically the occurrence of the localization-delocalization transition with definite mobility edges. This contrasts with the Aubry-Andre model. As a result, the diffusion of wave packet exhibits a transition from ballistic to diffusive motion, and back to ballistic motion. We point out that experimental realizations of the predicted transition can be accessed with light waves in photonic lattices and matter waves in optical lattices.

Non-Hermitian delocalization in a 2D photonic quasicrystal

Zhaoyang Zhang; Shun Liang; I. Septembre; Jiawei Yu; Yongping Huang; Maochang Liu; Yanpeng Zhang; Min Xiao; G. Malpuech; D. D. Solnyshkov. arXiv (Cornell University) (2023). Cited by 0. Found by: openalex. DOI: 10.48550/arxiv.2312.09322

Abstract: Quasicrystals show long-range order, but lack translational symmetry. So far, theoretical and experimental studies suggest that both Hermitian and non-Hermitian quasicrystals show localized eigenstates. This localization is due to the fractal structure of the spectrum in the Hermitian case and to the transition to diffusive bands via exceptional points in the non-Hermitian case. Here, we present an experimental study of a dodecagonal (12-fold) photonic quasicrystal based on electromagnetically-induced transparency in a Rb vapor cell. The transition to a quasicrystal is obtained by superposing two honeycomb lattices at 30° with a continuous tuning of their amplitudes. Non-Hermiticity is controlled independently. We study the spatial expansion of a probe wavepacket. In the Hermitian case, the wavepacket expansion is suppressed when the amplitude of the second lattice is increased (quasicrystal localization). We find a new regime, where increasing the non-Hermitian potential in the quasicrystal enhances spatial expansion, with the C_{12} symmetry becoming visible in the wavepacket structure. This real-space expansion is due to a k-space localization on specific quasicrystal modes. Our results show that the non-Hermitian quasicrystal behavior is richer than previously thought. The localization properties of the quasicrystals can be used for beam tailoring in photonics, but are also important in other fields.

Vortex and corner solitons in Stampfli-tiling dodecagonal quasiperiodic lattices

Boquan Ren; Yongli Qu; Milivoj R. Belić; Yongdong Li; Yiqi Zhang. Chaos Solitons & Fractals (2025). Cited by 0. Found by: openalex. openalex_2yr: 5.898 (2026). DOI: 10.1016/j.chaos.2025.117285

Localization in photonic crystals

Faris Siedahmend Mohammed Osman. University of the Witwatersrand, Johannesburg Institutional Repository on DSpace (University of the Witwatersrand, Johannesburg) (2017). Cited by 0. Found by: openalex. URL: https://openalex.org/W2794744957

Abstract: This thesis is an accumulation of the work and that was carried out and published as two articles \nand two book chapters. Throughout the thesis, we develop and present theoretical as well as \nnumerical model to extend the existing techniques to study the optical properties of photonic \ncrystals, plasmonic photonic crystals and photonic quasicrystals. \nWe start with a background review, where we cover the theoretical aspects of light–matter \ninteraction. That is followed by a review of the physics of photonic crystals. In that chapter, we \ndiscuss the different properties of photonic crystals, plasmonic photonic crystals as well as the \ntopic of localization. We then delve into the numerical aspects of the subject. We provide a \nreview on the frequency domain method and the finite–differences–time–domain methods which \nthey are both used in the work to perform different types of simulations. \nThe frequency domain method is, then, extended to enable the numerical analysis of the optical \nproperties in plasmonic photonic crystals. We use first order perturbation theory to study the \neffect of surface plasmon polaritons on the photonic band structure of plasmonic photonic crystals. \nWe developed a simple numerical tool that extends the standard frequency domain methods to \ncompute the photonic band structure of plasmonic photonic crystals. \nWe then employ the two stage cut and project scheme to generate a dodecagonal two–dimensional \nquasiperiodic structure. The finite-differences-time–domain method is applied to simulate the \npropagation of electromagnetic modes in the system. We compute the transmission coefficients \nas well as the inverse participation ratio for a quasicrystal consisting of dielectric cylindrical rods. \nThe analysis has shown that crystal has critical states. Furthermore, we apply the frequency \ndomain method to quantify the localized modes in the vicinity of defects in a two–dimensional \nphotonic crystal. We compute the intensity of those modes in the surroundings of the defects sites \nto identify their nature. Finally, we use the finite–differences–time–domain method to provide a \nsecond example of a quasicrystalline structure, where the states are localized.

Optical properties of 1D-photonic time quasicrystals

Claudionor Bezerra; Anny Araújo; S. Azevedo; Carlos da Silva Costa; G. M. Viswanathan. (2024). Cited by 0. Found by: openalex. DOI: 10.1364/opticaopen.26973478

Abstract: Photonic time crystals are materials in which the refractive index varies periodically and suddenly in time. This material exhibits unusual properties such as momentum bands separated by gaps. Because of the sudden change in the optical properties, the light propagating in these materials experiences time-refraction and time-reflection, analogous to refraction and reflection in conventional photonic crystals. Interference between time-refracted and time-reflected waves gives rise to Floquet-Bloch states and dispersion bands, which are momentum gapped. In this paper, we employ a transfer-matrix treatment to study the propagation of light waves in photonic time quasicrystals composed of two alternating building slabs, \$A\$ and \$B\$, constructed according to Fibonacci, Thue-Morse and Double-Period sequences. We present numerical results for the photonic band structure in terms of the dimensionless wave-vector \$\bar{k} = k / k_{0}\$ and the normalized dimensionless Bloch's angular frequency \$\Omega T/\pi\$. Our numerical results show the band-gap behavior as a function of the ratio between refraction indices \$n_{BA} = n_{B}/n_{A}\$ and layer thicknesses \$t_{BA} = t_{B}/t_{A}\$. We also studied the localization and self-similar behavior of the band structures, whose fractality can be described by a power law. We show that the power law scaling index \$\delta\$, which can be identified as being a diffusion constant of the spectra, presents a non-monotonic dependence on \$n_{BA}\$ and \$t_{BA}\$. Finally, plots of the transmission/reflection spectra, also calculated via transfer-matrix method, in terms of the dimensionless wave-vector \$\bar{k} = k / k_{0}\$, are shown for different values of refraction indices \$n_{BA}\$ and layer thicknesses \$t_{BA}\$.

Optical properties of 1D-photonic time quasicrystals

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Reentrant delocalization transition in one-dimensional photonic quasicrystals

Kyle Linn; Sachin Vaidya; Christina Jörg; Megan Goh; Mikael C. Rechtsman. (2023). Cited by 0. Found by: openalex. DOI: 10.1364/cleo_fs.2023.ftu4d.8

Abstract: We theoretically predict and experimentally observe that the localization of light in one-dimensional photonic quasicrystals is followed by a second delocalization transition on increasing quasiperiodic modulation strength - an example of a reentrant transition.

Robust Anderson transition in non-Hermitian photonic quasicrystals

Stefano Longhi. (2024). Cited by 0. Found by: openalex. DOI: 10.1364/opticaopen.25134767

Abstract: Anderson localization, i.e. the suppression of diffusion in lattices with random or incommensurate disorder, is a fragile interference phenomenon which is spoiled out in the presence of dephasing effects or fluctuating disorder. As a consequence, Anderson localization-delocalization phase transitions observed in Hermitian systems, such as in one-dimensional quasicrystals when the amplitude of the incommensurate potential is increased above a threshold, are washed out when dephasing effects are included. Here we consider localization-delocalization spectral phase transitions occurring in non-Hermitian quasicrystals with local incommensurate gain and loss, and show that, contrary to the Hermitian case, the non-Hermitian phase transition is robust against dephasing effects. The results are illustrated by considering synthetic quasicrystals in photonic mesh lattices.

Robust Anderson transition in non-Hermitian photonic quasicrystals

Stefano Longhi. (2024). Cited by 0. Found by: openalex. DOI: 10.1364/opticaopen.25134767.v1

Abstract: Anderson localization, i.e. the suppression of diffusion in lattices with random or incommensurate disorder, is a fragile interference phenomenon which is spoiled out in the presence of dephasing effects or fluctuating disorder. As a consequence, Anderson localization-delocalization phase transitions observed in Hermitian systems, such as in one-dimensional quasicrystals when the amplitude of the incommensurate potential is increased above a threshold, are washed out when dephasing effects are included. Here we consider localization-delocalization spectral phase transitions occurring in non-Hermitian quasicrystals with local incommensurate gain and loss, and show that, contrary to the Hermitian case, the non-Hermitian phase transition is robust against dephasing effects. The results are illustrated by considering synthetic quasicrystals in photonic mesh lattices.

Photonic Crystal/Quasicrystal Nanolasers and Spontaneous Emission Control

Kengo Nozaki; Toshihiko Baba. The Review of Laser Engineering (2006). Cited by 0. Found by: openalex. DOI: 10.2184/lsj.34.756

Abstract: We demonstrate the laser operation by an ultrasmall mode in photonic crystal (PC) and quasiperiodic photonic crystal (QPC) nanocavities fabricated into GaInAsP slab. In the former, the smallest modal volume of

0.019 m3 =0.15 (/n) 3 is achieved by using a PC point-shift nanocavity. In the latter, as well as the nanocavity mode lasing, unique localization of Bragg modes arising from fractal lattices of the QPC is observed. These nanocavities are effective for the enhancement of the spontaneous emission rate (so-called Purcell effect). The time-resolved measurement shows 28 times faster spontaneous emission decay in the point-shift nanocavity.

Formation of nonlinear modes in one-dimensional quasiperiodic lattices with a mobility edge

Dmitry A. Zezyulin; G. L. Alfimov. arXiv (Cornell University) (2024). Cited by 0. Found by: openalex. DOI: 10.48550/arxiv.2411.13936

Abstract: We investigate the formation of steady states in one-dimensional Bose-Einstein condensates of repulsively interacting ultracold atoms loaded into a quasiperiodic potential created by two incommensurate periodic lattices. We study the transformations between linear and nonlinear modes and describe the general patterns that govern the birth of nonlinear modes emerging in spectral gaps near band edges. We show that nonlinear modes in a symmetric potential undergo both symmetry-breaking pitchfork bifurcations and saddle-node bifurcations, mimicking the prototypical behaviors of symmetric and asymmetric double-well potentials. The properties of the nonlinear modes differ for bifurcations occurring below and above the mobility edge. In the generic case, when the quasiperiodic potential consists of two incommensurate lattices with a nonzero phase shift between them, the formation of localized modes in the spectral gaps occurs through a cascade of saddle-node bifurcations. Because of the analogy between the Gross-Pitaevskii equation and the nonlinear Schrödinger equation, our results can also be applied to optical modes guided by quasiperiodic photonic lattices.

Linear light localization in two-dimensional fractional diffraction quasicrystals

Zeyun Shi; Lu Qin; Pengfei Li; Yuan Zhou; Xiaoqian Zhao; Guanghui Wang. Optics Letters (2025). Cited by 0. Found by: openalex. DOI: 10.1364/ol.566565

Abstract: We numerically investigate linear light localization in two-dimensional photonic quasicrystals incorporating fractional diffraction. Our findings reveal that localization arises when the potential depth exceeds a critical threshold, which diminishes rapidly with increasing order of discrete rotational symmetry. Notably, fractional diffraction modifies these thresholds, enabling localization in significantly shallower potentials compared to conventional systems-particularly under lower diffraction orders (where fractional effects are most pronounced). The resulting localized modes exhibit remarkable robustness, remaining insensitive to the initial position of the input beam. These results establish fractional-diffraction photonic quasicrystals as a versatile, tunable platform for precise manipulation of wave localization phenomena.

Analytical mobility edge in nonreciprocal quasiperiodic lattices with next-nearest-neighbor hopping

Wenmin Wang; Xiaosen Yang; Xianqi Tong. arXiv (Cornell University) (2026). Cited by 0. Found by: openalex. DOI: 10.48550/arxiv.2607.10996

Abstract: We investigate localization transitions and spectral topology in a one-dimensional non-Hermitian generalization of the Aubry-André model in which both the nearest-neighbor and the next-nearest-neighbor hopping amplitudes are nonreciprocal. By extending the Fermi-surface point-matching method to nonreciprocal hopping, we derive a closed-form expression for the energy-dependent mobility edge in which the two nonreciprocity parameters are absorbed into exponentially renormalized effective hopping amplitudes. The mobility edge forms a single parabola in the energy–potential plane: nearest-neighbor nonreciprocity rigidly shifts the localization boundary toward stronger potentials, whereas next-nearest-neighbor nonreciprocity reduces the curvature of the boundary and thereby broadens the energy window in which extended and localized states coexist. Exact diagonalization confirms the analytical boundary for purely nearest-neighbor, purely next-nearest-neighbor, and combined nonreciprocity, and recovers the known Hermitian mobility edge in the reciprocal limit. We further analyze the spectral topology under periodic boundary conditions and show that the spectral winding numbers evaluated at base energies near the two band edges directly bracket the mixed phase: the winding number at the lower band edge drops when the mobility edge enters the spectrum and the first localized states appear, while the winding number at the upper band edge drops when the last extended states localize, delineating the full potential-strength window over which extended and localized states coexist. These results provide a compact analytical framework that connects energy-dependent localization, spectral topology, and nonreciprocity in quasiperiodic lattices, and they are directly testable in photonic, atomic, and electrical-circuit platforms.

Spectral Distribution of one-dimensional Photonic Quasicrystals: The Role of Irrational Numbers

Hui Quan; Wei Si; Kai Jiang. arXiv (Cornell University) (2026). Cited by 0. Found by: openalex. URL: https://openalex.org/W7124117353

Abstract: In this paper, we construct a one-dimensional photonic quasicrystal by combining two incommensurate spatial harmonics, where the ratio of their periods is the irrational number . We evaluate the photonic quasicrystal accurately by a generalized spectral method that embeds the quasiperiodic structure into a higher-dimensional periodic system. We study the spectral distribution of one-dimensional photonic quasicrystals and find some interesting phenomena. As the computational resolution N increases, there are more eigenvalues within finite frequency bandwidths, and the maximum localization always occurs at spectral gap edges for states near index N + 1. By varying within the range of (0,1), we present a butterfly-shaped spectral structure with abundant band gaps. We find that the spectral structure factor Q (defined as I_{mg}/N, where I_{mg} is the maximum gap index) exhibits different linear patterns as changes: Q = 1 - when < c, while Q = when > c, where c \approx 0.424 is the transition point. This linear relationship holds robustly in the strong quasiperiodic regime (away from 0 or 1) and is independent of the specific type of irrational number used. The relationship disappears (weak quasiperiodic regime) near = 0 or = 1. It demonstrates that the intrinsic spectral properties of one-dimensional photonic quasicrystals are fundamentally governed by the magnitude of the irrational parameter .

Super-ballistic diffusion induced by nonlinear interactions in a one-dimensional quasiperiodic lattice

J. K. Dong; Long-Hua Gu; Luchen Zhang; Zhi Li; Dan-Wei Zhang. Physics Letters A (2024). Cited by 0. Found by: openalex. DOI: 10.1016/j.physleta.2024.129549

Analytical mobility edge in nonreciprocal quasiperiodic lattices with next-nearest-neighbor hopping

Wenmin Wang; Xiaosen Yang; Xianqi Tong. arXiv (Cornell University) (2026). Cited by 0. Found by: openalex. URL: https://openalex.org/W7168433559

Abstract: We investigate localization transitions and spectral topology in a one-dimensional non-Hermitian generalization of the Aubry-André model in which both the nearest-neighbor and the next-nearest-neighbor hopping amplitudes are nonreciprocal. By extending the Fermi-surface point-matching method to nonreciprocal hopping, we derive a closed-form expression for the energy-dependent mobility edge in which the two nonreciprocity parameters are absorbed into exponentially renormalized effective hopping amplitudes. The mobility edge forms a single parabola in the energy–potential plane: nearest-neighbor nonreciprocity rigidly shifts the localization boundary toward stronger potentials, whereas next-nearest-neighbor nonreciprocity reduces the curvature of the boundary and thereby broadens the energy window in which extended and localized states coexist. Exact diagonalization confirms the analytical boundary for purely nearest-neighbor, purely next-nearest-neighbor, and combined nonreciprocity, and recovers the known Hermitian mobility edge in the reciprocal limit. We further analyze the spectral topology under periodic boundary conditions and show that the spectral winding numbers evaluated at base energies near the two band edges directly bracket the mixed phase: the winding number at the lower band edge drops when the mobility edge enters the spectrum and the first localized states appear, while the winding number at the upper band edge drops when the last extended states localize, delineating the full potential-strength window over which extended and localized states coexist. These results provide a compact analytical framework that connects energy-dependent localization, spectral topology, and nonreciprocity in quasiperiodic lattices, and they are directly testable in photonic, atomic, and electrical-circuit platforms.

Spectral Distribution of one-dimensional Photonic Quasicrystals: The Role of Irrational Numbers

Hui Quan; Wei Si; Kai Jiang. arXiv (Cornell University) (2026). Cited by 0. Found by: openalex. DOI: 10.48550/arxiv.2601.06482

Abstract: In this paper, we construct a one-dimensional photonic quasicrystal by combining two incommensurate spatial harmonics, where the ratio of their periods is the irrational number . We evaluate the photonic quasicrystal accurately by a generalized spectral method that embeds the quasiperiodic structure into a higher-dimensional periodic system. We study the spectral distribution of one-dimensional photonic quasicrystals and find some interesting phenomena. As the computational resolution N increases, there are more eigenvalues within finite frequency bandwidths, and the maximum localization always occurs at spectral gap edges for states near index N + 1. By varying within the range of (0,1), we present a butterfly-shaped spectral structure

with abundant band gaps. We find that the spectral structure factor Q (defined as I_{mg}/N, where I_{mg} is the maximum gap index) exhibits different linear patterns as changes: Q = 1 - when < c, while Q = when > c, where c \approx 0.424 is the transition point. This linear relationship holds robustly in the strong quasiperiodic regime (away from 0 or 1) and is independent of the specific type of irrational number used. The relationship disappears (weak quasiperiodic regime) near = 0 or = 1. It demonstrates that the intrinsic spectral properties of one-dimensional photonic quasicrystals are fundamentally governed by the magnitude of the irrational parameter .

Time-Domain Interferometric Measurement of Photonic Band Structure in a One-Dimensional Quasicrystal

Toshiaki Hattori; Noriaki Tsurumachi; Sakae Kawato; Hiroki Nakatsuka. Ultrafast Phenomena (1994). Cited by 0. Found by: openalex. DOI: 10.1364/up.1994.md.14

Abstract: Although ultrashort optical pulses are used for the time-domain study of transmission of pulses through dispersive media, interferometric measurements using white light with an ultrashort correlation time can very often be much more powerful and useful [1,2]. Here, we present an application of Fourier-transform interferometry to the study of photonic band structure in a one-dimensional quasiperiodic structure. Photonic band formation in periodical dielectric structures and localization of light in random structures have been a subject of great interest in recent years [3]. Studies on photonic quasiperiodical structures will extend this new area of research furthermore. Our sample of one-dimensional quasiperiodic system was a photonic Fibonacci lattice, which was proposed by Kohmoto et al. [4]. Fractal behavior of wave functions and the energy spectrum in it has been theoretically predicted.

Families of localized modes of Bose-Einstein condensates enabled by incommensurate optical lattice and photon-atom interactions

Pedro S. Gil; V. V. Konotop. Physical Review Research (2026). Cited by 0. Found by: openalex. DOI: 10.1103/5hcq-tfjz

Abstract: We consider a Bose-Einstein condensate (BEC) loaded into a one-dimensional optical cavity under the combined action of an external potential and atom-cavity coupling with mutually incommensurate periods. Such configuration enables the localization of matter waves even in the absence of two-body interactions. We study families of localized modes within the mean-field approximation for red and blue detunings from atomic and cavity resonances in relatively shallow quasiperiodic lattices, beyond the validity of the tight-binding approximation. The parameter regimes supporting localization of atomic wave packets are identified. The system exhibits two types of bistability manifested as distinct photon numbers under otherwise identical conditions. One type arises from the coexistence of multiple families of localized modes, typical of conservative nonlinear systems, while the other stems from the multivalued dependence of the families on system parameters, characteristic of systems exhibiting hysteresis. BEC in a cavity may also display pseudodegeneracy, understood as the existence of two distinct atomic-density distributions corresponding to the same atomic and photon numbers (although different chemical potentials). The stability of the localized modes is analyzed. It is shown that, owing to the strong impact of long-range interactions on stability, a two-localized-mode configuration can operate as an logic gate.

Families of localized modes of Bose-Einstein condensates enabled by incommensurate optical lattice and photon-atom interactions

Pedro S. Gil; V. V. Konotop. arXiv (Cornell University) (2026). Cited by 0. Found by: openalex. DOI: 10.48550/arxiv.2602.17780

Abstract: We consider a Bose-Einstein condensate (BEC) loaded into a one-dimensional optical cavity under the combined action of an external potential and atom-cavity coupling with mutually incommensurate periods. Such configuration enables the localization of matter waves even in the absence of two-body interactions. We study families of localized modes within the mean-field approximation for red and blue detunings from atomic and cavity resonances in relatively shallow quasiperiodic lattices, beyond the validity of the tight-binding approximation. The parameter regimes supporting localization of atomic wave packets are identified. The system exhibits two types of bistability manifested as distinct photon numbers under otherwise identical conditions. One type arises from the coexistence of multiple families of localized modes, typical of conservative nonlinear systems, while the other stems from the multivalued dependence of the families on system parameters, characteristic of systems exhibiting hysteresis. BEC in a cavity may also display pseudodegeneracy, understood as the existence of two distinct atomic-density distributions corresponding to the same atomic and photon numbers

(although different chemical potentials). The stability of the localized modes is analyzed. It is shown that, owing to the strong impact of long-range interactions on stability, a two-localized-mode configuration can operate as an XOR logic gate.

Micro-cylinder mode in photonic quasicrystal observed by near-field optical microscopy

Zheyu Fang; Tao Dai; Bei Zhang; Xing Zhu. Proceedings of SPIE, the International Society for Optical Engineering/Proceedings of SPIE (2007). Cited by 0. Found by: openalex. DOI: 10.1117/12.767590

Abstract: We propose and realize a micro-cylinder mode in photonic quasicrystal (PQC) on the top of an edge-emitting GaN light emitting diodes (LEDs) with 30 m indium tin oxides (ITO) covered and 8 m GaN exposed broad-stripe. A well-defined hexagon-like localized state following a C6v symmetry is identified from the PQC micro-cylinder arrays. Enhancement factor of the surface light extraction from 12-fold PQC patterns on electrical current injected GaN-based light emitters is 2.4 times higher than non-patterned regions, which is observed by scanning near-field optical microscopy (SNOM). According to the near-field optical images, micro-cylinders can be considered as nano-optics sources because the electromagnetic energy is strongly concentrated due to the wave guide modes and the beaming effect of the surface microstructures.

Optical QuasiCrystals -Properties andDynamics

Demetrios N. Christodoulides; Jason W. Fleischer. (2005). Cited by 0. Found by: openalex. URL: https://openalex.org/W218

Abstract: Wepresent thefirst observation ofwavedynamics in2DPenrose-Tile quasi-periodic optical lattices, including experiments onlinear discrete diffraction fromvarious lattice sites, nonlinear (self- focusing) localization, andexperiments onnonlinearly interacting quasi-periodic lattices. ©2005 Optical Society ofAmerica OCIScodes: (190.5330) Photorefractive nonlinear optics; (090.7330) Volume holographic gratings; (999.9999) Photonic crystals

Photonic band gap effect and localization in two-dimensional Penrose lattice

Mehmet Bayındır; Ertugrul Cubukcu; İrfan Bulu; Ekmel Özbay. (2002). Cited by 0. Found by: openalex. DOI: 10.1109/qels.2001.961940

Abstract: Summary form only given. Photonic crystals are artificial periodic structures in which the refractive index modulation gives rise to stop bands for electromagnetic waves (EM) within a certain frequency range in all directions. Recently it was recognized that the photonic gaps can exist in two-dimensional (2D) octagonal quasicrystals. A quasiperiodic system is characterized by a lack of long range periodic translational order. But the quasiperiodic system has long-range band orientational order, so that it can be considered as an intermediate between periodic and random systems.In this work, we report on observation of the photonic band gap effect in a 2D Penrose quasicrystal consisting of dielectric rods.

Photonic band gap effect and localization in two-dimensional Penrose lattice

Mehmet Bayındır; Ertugrul Cubukcu; İrfan Bulu; Ekmel Özbay. (2001). Cited by 0. Found by: openalex. DOI: 10.1109/cleo.2001.948207

Abstract: Summary form only given. Photonic crystals are artificial periodic structures in which the refractive index modulation gives rise to stop bands for electromagnetic waves (EM) within a certain frequency range in all directions. Recently it was recognized that the photonic gaps can exist in two-dimensional (2D) octagonal quasicrystals. A quasiperiodic system is characterized by a lack of long range periodic translational order. But the quasiperiodic system has long-range band orientational order, so that it can be considered as an intermediate between periodic and random systems. In this work, we report on observation of the photonic band gap effect in a 2D Penrose quasicrystal consisting of dielectric rods. Defect characteristics of various inequivalent sites of the crystal were investigated. The Penrose tiles are composed of fat and skinny rhombic unit cells and fill the 2D plane nonperiodically.

Critical states and anomalous wave transport in an aperiodic polariton monotile

Valtýr Kári Daníelsson; Helgi Sigurðsson. arXiv (Cornell University) (2026). Cited by 0. Found by: openalex. DOI: 10.48550/arxiv.2605.29023

Abstract: Recently "the Hat" monotile was introduced into the family of aperiodic tilings and quasicrystals boasting physical properties lying at the boundary of ordered and disordered systems. Here we study the two-dimensional wave transport, transverse localization and scaling properties of the quantum modes in a Monotile quasilattice. Our system is based on reconfigurable optical lattices for cavity-polaritons which provide flexible means to study wavepacket dynamics, strong nonlinear phenomena, and power-driven condensation in this new type of an aperiodic tiling. We confirm the existence of localized and critical states in the Monotile through direct diagonalization of the Schrödinger equation. Scaling analysis on the moments of the wavefunction distribution reveals anomalous transport regimes of super-diffusive and near sub-diffusive polariton transport associated with the fractal structure of the Monotile Hilbert space. We propose a strategy using resonantly excited polariton fluids to verify our findings.

Critical states and anomalous wave transport in an aperiodic polariton monotile

Valtýr Kári Daníelsson; Helgi Sigurðsson. arXiv (Cornell University) (2026). Cited by 0. Found by: openalex. URL: https://openalex.org/W7162894222

Abstract: Recently "the Hat" monotile was introduced into the family of aperiodic tilings and quasicrystals boasting physical properties lying at the boundary of ordered and disordered systems. Here we study the two-dimensional wave transport, transverse localization and scaling properties of the quantum modes in a Monotile quasilattice. Our system is based on reconfigurable optical lattices for cavity-polaritons which provide flexible means to study wavepacket dynamics, strong nonlinear phenomena, and power-driven condensation in this new type of an aperiodic tiling. We confirm the existence of localized and critical states in the Monotile through direct diagonalization of the Schrödinger equation. Scaling analysis on the moments of the wavefunction distribution reveals anomalous transport regimes of super-diffusive and near sub-diffusive polariton transport associated with the fractal structure of the Monotile Hilbert space. We propose a strategy using resonantly excited polariton fluids to verify our findings.

Dephasing-induced mobility edges in quasicrystals

Stefano Longhi. arXiv (Cornell University) (2024). Cited by 0. Found by: openalex. DOI: 10.48550/arxiv.2405.07198

Abstract: Mobility edges (ME), separating Anderson-localized states from extended states, are known to arise in the single-particle energy spectrum of certain one-dimensional lattices with aperiodic order. Dephasing and decoherence effects are widely acknowledged to spoil Anderson localization and to enhance transport, suggesting that ME and localization are unlikely to be observable in the presence of dephasing. Here it is shown that, contrary to such a wisdom, ME can be created by pure dephasing effects in quasicrystals in which all states are delocalized under coherent dynamics. Since the lifetimes of localized states induced by dephasing effects can be extremely long, rather counter-intuitively decoherence can enhance localization of excitation in the lattice. The results are illustrated by considering photonic quantum walks in synthetic mesh lattices.

Study of Localization-Delocalization Transition of Light in Photonic Moiré Lattices Fabricated with Saturable Nonlinear Materials

Trong Dat Ngo; Chinh Cuong Duong; Luong Thien Nguyen; Thanh Luan Nguyen; Đức Anh Nguyễn; Nguyen Viet Hung. Diffusion and defect data, solid state data. Part B, Solid state phenomena/Solid state phenomena (2024). Cited by 0. Found by: openalex. openalex_2yr: 0.68 (2026). DOI: 10.4028/p-ttxk2p

Abstract: Recently, Moiré lattices have received much attention from physicists and materials scientists. These structures have opened the door to the exploration of numerous physical phenomena such as superconductivity, the commensurate-incommensurate transition, the appearance of quasicrystals at special rotation angles, or the two-dimensional localization-delocalization transition of light in linear systems. In this study, we propose photonic Moiré lattices induced by saturable nonlinear materials. After performing numerical simulations, it is observed that there exists a transition between delocalized and localized formation of laser beams under different geometrical conditions, commensurate and incommensurate lattices. The results suggest that Moiré lattices with their compactness and tunability would be utilized to control the light patterns in integrated optical devices.

Non-diffusion transport in decoherent non-Hermitian quasicrystals

Yudong Ren; Rui Zhao; Kangpeng Ye; Lu Zhang; Hongsheng Chen; Haoran Xue; Yihao Yang. arXiv (Cornell University) (2025). Cited by 0. Found by: openalex. DOI: 10.48550/arxiv.2505.04205

Abstract: Disorder and coherence jointly govern wave transport in complex media. In Hermitian systems, a long-established paradigm since Anderson's work holds that disorder-induced localization relies on phase-coherent interference, and that the loss of coherence inevitably suppresses localization and restores featureless diffusive transport at long times. Whether this intuition remains valid in non-Hermitian systems, where transport can be governed by dissipation rather than interference, has remained largely open. Here we theoretically and experimentally demonstrate that this paradigm fundamentally breaks down in decoherent non-Hermitian quasicrystals. Using a programmable photonic lattice with independently engineered dissipation and fully programmable dephasing, we access regimes spanning from fully coherent to fully incoherent dynamics. While decoherence washes out localization and enforces structureless diffusion in Hermitian lattices, we find that decoherent non-Hermitian quasicrystals retain nontrivial, non-diffusive transport structures even in the incoherent limit. These include dissipation-induced localization, diffusion-localization transitions, and decoherence-induced mobility edges, phenomena with no counterparts in Hermitian disordered systems. We develop a unified theoretical framework that captures the ensemble-averaged dynamics across the entire coherence landscape, continuously connecting coherent and incoherent regimes, and reveals how dissipation and decoherence cooperate to shape transport. Our results establish decoherent non-Hermitian lattices as a distinct class of transport systems, in which dissipation and incoherence generate structured, non-diffusive phases, beyond the conventional Anderson picture.

Simulating non-Hermitian quasicrystals with single-photon quantum walks

Quan Lin; Tianyu Li; Lei Xiao; Kunkun Wang; Wei Yi; Peng Xue. arXiv (Cornell University) (2021). Cited by 0. Found by: openalex. DOI: 10.48550/arxiv.2112.15024

Abstract: Non-Hermiticity significantly enriches the properties of topological models, leading to exotic features such as the non-Hermitian skin effects and non-Bloch bulk-boundary correspondence that have no counterparts in Hermitian settings. Its impact is particularly illustrating in non-Hermitian quasicrystals where the interplay between non-Hermiticity and quasiperiodicity results in the concurrence of the delocalization-localization transition, the parity-time (PT)-symmetry breaking, and the onset of the non-Hermitian skin effects. Here we experimentally simulate non-Hermitian quasicrystals using photonic quantum walks. Using dynamic observables, we demonstrate that the system can transit from a delocalized, PT-symmetry broken phase that features non-Hermitian skin effects, to a localized, PT-symmetry unbroken phase with no non-Hermitian skin effects. The measured critical point is consistent with the theoretical prediction through a spectral winding number, confirming the topological origin of the phase transition. Our work opens the avenue of investigating the interplay of non-Hermiticity, quasiperiodicity, and spectral topology in open quantum systems.

Floquet Harper-Hofstadter Butterflies and non-Hermitian phase transition in quasicrystals

Mark Kremer; Sebastian Weidemann; Stefano Longhi; MartinWimmer; Ulf Peschel; Alexander Szameit. Conference on Lasers and Electro-Optics (2021). Cited by 0. Found by: openalex. openalex_2yr: 0 (2026). DOI: 10.1364/cleo_qels.2021.fw2m.2

Abstract: The non-Hermitian Aubry-André-Harper model exhibits a non-Hermitian phase transition that coincides with a localization transition. We provide a high-resolution measurement of the Butterfly spectrum and propose experiments to measure those transitions, using a photonic lattice.

Advances in classical and quantum wave dynamics on quasiperiodic lattices : a dissertation submitted for the degree of Doctor of Philosophy in Physics, Centre for Theoretical Chemistry and Physics, New Zealand Institute for Advanced Study, Massey University, Albany, New Zealand

Carlo Danieli. Massey Research Online (Massey University) (2016). Cited by 0. Found by: openalex. URL: https://openalex.org/W2698686299

Abstract: Lattices and discrete networks are cornerstones of a number of scientific subjects. In condensed \nmatter, optical lattices allowed the experimental realization of several theoretically \npredicted phenomena. Indeed, these structures constitute ideal benchmarks for light and \nwave propagation experiments involving interacting particles, such as clouds of ultra-cold \natoms that Bose-Einstein condensate. Moreover, they allow experimental design of particular \nlattice topologies, as well as the implementation of several classes of spatial perturbations. \nFor example, Anderson localization being observed for the first time in atomic \nBose-Einstein condensate experiments and Aubry-André localization discovered with light \npropagating through networks of optical waveguide. \nThis thesis considers different types of lattices in the presence of quasiperiodic modulations, \nmainly the celebrated Aubry-André potential. Particular attention will be given to \nspectral properties of models, localization features of eigenmodes and the transition from \ndelocalized

(metallic) eigenstates to localized (insulating) ones within the energy spectrum. \nWe additionally discuss the relation between the model's properties and the dynamics of \nparticles hopping along the lattice. \nAfter introducing the linear discrete Schrödinger equation, we first discuss the spectral \nproperties of the Aubry-André model. We then study the transition between metallic \nand insulating regimes of a class of quasiperiodic potentials constructed as an iterative \nsuperposition of periodic potentials with increasing spatial period. Next, we discuss the \nAubry-André perturbation of flat-band topologies, their energy-dependent transition (mobility \nedge), which can be expressed in analytical forms in case of specific onsite energy \ncorrelations, highlighting existence of zeroes, singularities and divergences. We then discuss \ntwo cases of driven one-dimensional lattices, namely an Aubry-André chain with a weak \ntime-space periodic driving and an Anderson chain with a quasiperiodic multi-frequency \ndriving. We show anaytically and numerically how drivings can lift the respective localization \nand generate delocalization by design. Finally we discuss the problem of the possible \ngeneration of correlated metallic states of two interacting particles problem in one dimensional \nAubry-André chains, under a coherent drive of the interaction.

Exploring Light-Induced Phases of 2D Materials in a Modulated 1D Quasicrystal

Yifei Bai; Anna R. Dardia; Toshihiko Shimasaki; David Weld. Physical Review X (2026). Cited by 0. Found by: openalex. DOI: 10.1103/37vk-qz71

Abstract: Light-induced quantum phases offer the potential for simple and powerful tuning of material properties. For example, simply illuminating 2D materials in the integer quantum Hall regime with polarized light is predicted to drive quantum phase transitions. Such phenomena are largely beyond the current frontier of solid state experiments due to technical limitations on laser intensity and material purity. However, the Harper-Hofstadter mapping which relates a two-dimensional integer quantum Hall system to a 1D quasicrystal enables the same polarization-dependent light-induced phase transitions to be observed using a quantum gas in a driven quasiperiodic optical lattice. We report results of such an experiment. We observe an interlaced phase diagram of localization-delocalization phase transitions as a function of drive polarization and amplitude. Elliptically polarized driving can stabilize an extended critical phase featuring multifractal wave functions; we observe signatures of this phenomenon in anomalous polarization-dependent subdiffusive transport. In this regime, increasing the strength of the quasiperiodic potential can enhance rather than suppress transport. These experiments demonstrate a simple method for synthesizing exotic multifractal states and exploring light-induced quantum phases across different dimensionalities.

Exploring light-induced phases of 2D materials in a modulated 1D quasicrystal

Yifei Bai; Anna R. Dardia; Toshihiko Shimasaki; David Weld. arXiv (Cornell University) (2025). Cited by 0. Found by: openalex. DOI: 10.48550/arxiv.2506.11984

Abstract: Light-induced quantum phases offer the potential for simple and powerful tuning of material properties. For example, simply illuminating 2D materials in the integer quantum Hall regime with polarized light is predicted to drive quantum phase transitions. Such phenomena are largely beyond the current frontier of solid state experiments due to technical limitations on laser intensity and material purity. However, the Harper-Hofstadter mapping which relates a two-dimensional integer quantum Hall system to a 1D quasicrystal enables the same polarization-dependent light-induced phase transitions to be observed using a quantum gas in a driven quasiperiodic optical lattice. We report results of such an experiment. We observe an interlaced phase diagram of localization-delocalization phase transitions as a function of drive polarization and amplitude. Elliptically polarized driving can stabilize an extended critical phase featuring multifractal wavefunctions; we observe signatures of this phenomenon in anomalous polarization-dependent subdiffusive transport. In this regime, increasing the strength of the quasiperiodic potential can enhance rather than suppress transport. These experiments demonstrate a simple method for synthesizing exotic multifractal states and exploring light-induced quantum phases across different dimensionalities.

Multicomponent Photonic Crystals with Inhomogeneous Scatterers

Alexander B. Khanikaev; Mikhail V. Rybin; M. F. Limonov. Series in optics and optoelectronics (2012). Cited by 0. Found by: openalex. DOI: 10.1201/b12175-11

Abstract: The concept of quasicrystal as a nonperiodic structure with perfect long-ranged order was brought in solid-state physics by Levine and Steinhardt.1 At present, it has become clear that, in addition to crystalline and amorphous materials, there exists a third form of solids that unexpectedly lls the gap between the two well-de ned condensed-matter states. Moreover, this intermediate class called aperiodic deterministic structures

includes the famous Fibonacci sequence ABAAB . . . and other quasicrystals that can be described by a projection onto the n-dimensional (nD) space with n = 1, 2, or 3 of an mD periodic lattice with dimensionality m > n. Examples of aperiodic structures different from quasicrystals are Thue-Morse and period-doubling sequences. Discovery of quasicrystals and other deterministic aperiodic structures initiated new elds of research in photonics. The studies of aperiodic long-range-ordered systems were extended to optics in the work by Kohmoto et al.,2 where a 1D quasicrystal constructed of dielectric layers forming the Fibonacci sequence was proposed. Since then, photonic quasicrystals and other arti cial long-rangeordered aperiodic objects have aroused an increasing interest in optical spectroscopy of solids.3-5 In this chapter, we rst de ne the quasicrystals and present their structure factors. Then we consider light propagation in aperiodic photonic structures and pay particular attention CONTENTS 2.6.1 De nition and Structure Factor .... 132 2.6.1.1 One-Dimensional Quasicrystals .... 132 2.6.1.2 Fibonacci Structures .... 133 2.6.1.3 Structure Factor.. 134 2.6.2 Two-Wave Approximation .... 137 2.6.3 Re ection and Transmission from Fibonacci Structures .... 138 2.6.3.1 Nonresonant Fibonacci Binary Chains .... 139 2.6.3.2 Fibonacci QW Structures .... 140 2.6.4 Scaling Features and Localization of Excitonic Polaritons....................................... 143 2.6.5 Variety of Aperiodic Long-Range-Order Photonic Structures .... 146 2.6.5.1 Non-Fibonacci Aperiodic Sequences .... 146 2.6.5.2 Two-Dimensional Structures .... 147 2.6.5.3 Three-Dimensional Structures .... 148 2.6.6 Summary .... 148 References .... 149 to application of the two-wave approximation (TWA). The latter allows one to interpret the optical spectra of aperiodic structures in terms of the periodic objects and underline the speci c features arising as a result of the nonperiodicity. To illustrate, we analyze not only the binary Fibonacci optical superlattices built of two constituent layers A and B but also the recently proposed arti cial objects, namely, Fibonacci multiple quantum-well (QW) structures. An important point is that the regimes where TWA is invalid demonstrate the properties of optical spectra that are forbidden for periodic structures: (i) the localization characteristic for disordered systems and (ii) scaling and self-similarity that are absent in both conventional crystals and disordered materials. In the nal part of this chapter, we brie y enumerate other proposed and studied photonic aperiodic long-range-ordered photonic structures-1D, 2D, and 3D.

Time-resolved light propoagation at the band-edge states of 1D Fibonacci quasicrystals

Luca Dal Negro; Claudio J. Otón; Z. Gaburro; Lorenzo Pavesi; Philip J. M. Johnson; Ad Lagendijk; Diederik S. Wiersma. MRS Proceedings (2002). Cited by 0. Found by: openalex. DOI: 10.1557/proc-722-l6.9

Exploring Aperiodic Order in Photonic Time Crystals

Marino Coppolaro; Massimo Moccia; Giuseppe Castaldi; Vincenzo Galdi. arXiv (Cornell University) (2025). Cited by 0. Found by: openalex. DOI: 10.48550/arxiv.2505.20039

Abstract: We present a theoretical framework for analyzing aperiodically ordered photonic time quasicrystals (PTQCs), which are the temporal analogs of spatial photonic quasicrystals. Using a general two-symbol substitutional sequence to model temporal modulations, we extend the trace and anti-trace map formalism used for spatial photonic quasicrystals to the temporal domain. Focusing on the Thue-Morse sequence as a representative example, we examine the band structure and wave-transport properties, discussing their physical origins and highlighting both similarities and key differences with conventional periodic photonic time crystals. Furthermore, we investigate the peculiar features of PTQCs, such as multiscale spectral response and localization effects. Our findings provide valuable insights into the complex interplay between aperiodic order and wave dynamics in time-varying media, highlighting its potential to enable the development of advanced photonic devices.

Strong light confinement of tunable resonances in low symmetric quasicrystal through orientational variations

Döne Yılmaz; Mediha Tutgun; Hamza Kurt. (2018). Cited by 0. Found by: openalex. DOI: 10.1117/12.2307331

Abstract: We propose octagonal quasi-crystal designs providing effective light confinement for different resonance frequencies through the structural modification with the utilization of low-symmetric photonic unit cells. The effect of rotational symmetry reduction on the cavity resonance appearing in the corresponding photonic bandgap of each structure has been investigated. Relatively small dielectric cylinders have been additionally located at discrete angular positions with particular distances from the center of the each core cylinder and

the noteworthy resonance peaks have been observed to emerge in the bandgaps. Rotational symmetry of the proposed structures is to be modified by varying the angular displacement of the smaller quasi-crystalline rods with the angle in terms of the x-axis of the small rod. The successful demonstration of tunable resonance modes has been achieved numerically and experimentally for the first time by tailoring the positional parameters and reducing the crystalline symmetry. Strongly localized modes in the proposed quasi-crystals have great potential for various slow light applications along with other technologies such as sensors, lasers and memory units.

Topological States in 1D Photonic Quasi-crystals

M.S. Vasconcelos; E.L. Albuquerque. MRS Proceedings (2014). Cited by 0. Found by: openalex. DOI: 10.1557/opl.2014.839

Exploration of localization physics with atomic, molecular, and optical platforms

Jun Gao; Hang Li; Val Zwiller; Ali W. Elshaari. Journal of Physics Condensed Matter (2026). Cited by 0. Found by: openalex. DOI: 10.1088/1361-648x/ae6aee

Abstract: Anderson localization (AL)-a fundamental wave phenomenon resulting from coherent interference in disordered potentials-has attracted substantial attention in quantum optics and ultracold atomic physics due to its universal nature and controllable experimental realizations. In this review, we survey recent advances in the study of localization physics, focusing primarily on AL, quasi-periodic modulation, and the emergence of mobility edges in optical and atomic systems. After briefly introducing the theoretical origins and historical context, we discuss the theoretical frameworks underlying localization phenomena, highlighting key models such as the Anderson tight-binding model and the Aubry-André quasiperiodic lattice. The core emphasis of this review lies in experimental developments: we comprehensively describe how ultracold atomic gases and optical platforms-including disordered waveguides and fiber loop systems-have enabled precise observation and characterization of localization transitions and mobility edges. Experimental methodologies, diagnostics, and challenges unique to each system are thoroughly addressed. Finally, we conclude by outlining promising future directions, such as many-body localization, non-Hermitian Anderson physics, and synthetic-dimension systems, emphasizing how the interplay between theory and experimental innovation continues to drive the field forward.

Photonic Band Structure of Fibonacci Superlattices with Metamaterials. Part 1: The Algebraic Method of Calculation for Lossless Layers

Karol Tarnowski; W. Salejda. Photonics Letters of Poland (2009). Cited by 0. Found by: openalex. DOI: 10.4302/plp.2009.3.09

Abstract: We study the properties of the photonic band structure (PBS) of infinite and binary Fibonacci superlattices (FS) containing metamaterial lossless layers. The transfer matrix method (TMM) and periodic boundary conditions are applied. The algebraic method of PBS calculations, based on the Bloch theorem, is described and used. The dispersion relations, for both type of light polarization, characterizing peculiarities of PBS are calculated numerically and presented. Fractal properties of a photonic band structure and the existence of so called zero-n gaps are established. Full Text: PDF References E. Macia,"The role of aperiodic order in science and technology", Rep. Prog. Phys. 69, 397 (2006) [CrossRef] W. Steurer, D. Suter-Widmer,"Photonic and phononic quasicrystals", J. Phys. D: Appl. Phys. 40, R229 (2007) [CrossRef] S. A. Ramakrishna, T. M. Grzegorczyk, "Physics and Applications of Negative Refractive Index Materials", SPIE Press and CRC Press 2009 J.E. Lugo et al,"Multiband negative refraction in one-dimensional photonic crystals", Optics Express, 17, 3042 (2009) [CrossRef] H. Zhang, X. Chen, Y. Li, Y. Fu, N. Yuan,"The Bragg gap vanishing phenomena in one-dimensional photonic crystals", Optics Express, 17, 7800 (2009) [CrossRef] M. de Dios-Leyva, J.C. Drake-Perez,"Zero-width band gap associated with the nŻ =0 condition in photonic crystals containing left-handed materials", Phys. Rev. E, 79, 036608 (2009) [CrossRef] M. de Dios-Leyva, O.E. Gonzales-Vasquez,"Band structure and associated electromagnetic fields in one-dimensional photonic crystals with left-handed materials", Phys. Rev. B, 77,125102 (2008) [CrossRef] R. Srivastava, K.B. Thapa, S. Pati, S.P. Ojha,"Negative refraction in 1D photonic crystals", Sol. State Comm., 147, 157 (2008) [CrossRef] Y. Fang, S. He,"Transparent structure consisting of metamaterial layers and matching layers", Phys. Rev. A, 78, 023813 (2008) [CrossRef] A. Bruno-Alfonso, E. Reyes-Gomez, S.B. Cavalcanti, L.E. Oliveira,"Band edge states of the ?n?=0 gap of Fibonacci photonic lattices", Phys. Rev. A, 78, 035801 (2008) [CrossRef] Eds. C. M. Krowne, Y. Zhang, "Physics of Negative Refraction and Negative Index Materials", Springer 2007 Y.-T Fang, J. Zhou, E.Y.B. Pun,"High-Q filters based on one-dimensional photonic crystals using epsilon-negative materials", Appl. Phys. B, 86, 587 (2007) [CrossRef] X. Hu, Z. Liu, Q. Gong,"A multichannel filter in a photonic crystal heterostructure containing single-negative materials", J. Opt. A: Pure Appl. Opt., 9, 877 (2007) [CrossRef] F. F. de

Medeiros, E. L. Albuquerque, M. S. Vasconcelos,"Transmission spectra in photonic band-gap Fibonacci nanostructures", Surface Science, 601, 4492 (2007) [CrossRef] K. Tarnowski, W. Salejda, M. H. Tyc,"Propagation of polarized light through optical nanosuperlattices", Optica Applicata, 37, 387, 2007 M.H. Tyc, W. Salejda, A. Klauzer-Kruszyna, K. Tarnowski,"Photonic band structure of one-dimensional aperiodic superlattices composed of negative refraction metamaterials", SPIE Proceedings Series, 2007, 6581, 658112 [CrossRef] M.H. Tyc, W. Salejda, A. Klauzer-Kruszyna, K. Tarnowski,"Propagation of polarized light through superlattices composed of left- and right-handed materials", SPIE Proceedings Series, 2007, 6581, 658113 [CrossRef] P. Yeh, "Optical Waves in Layered Media", Wiley-Interscience 2005 H.X. Da, C. Xu, Z.Y. Li,"Omnidirectional reflection from one-dimensional quasi-periodic photonic crystal containing left-handed material", Phys. Lett. A, 345, 459 (2005) [CrossRef] W. Salejda, A. Klauzer-Kruszyna, M. H. Tyc, K. Tarnowski,"One-dimensional photonic quasicrystals: application to Bragg reflectors", SPIE Proceedings Series, 2005, 5950, 59501Q [CrossRef] W. Salejda, A. Klauzer-Kruszyna, M. H. Tyc, K. Tarnowski,"Electromagnetic wave propagation through aperiodic superlattices composed of left- and right-handed materials", SPIE Proceedings Series, 2005, 5955, 595514 [CrossRef] J. Li, D. Zhao, Z. Liu,"Zero n - photonic band gap in a quasiperiodic stacking of positive and negative refractive index materials", Phys. Lett. A, 332, 461 (2004) [CrossRef] A. Klauzer- Kruszyna, W. Salejda, M.H. Tyc,"Polarized Light Transmission through Generalized Fibonacci Multilayers. I. Dynamical maps approach", Optik, 115, 257 (2004) A. Klauzer- Kruszyna, W. Salejda, M.H. Tyc,"Polarized Light Transmission through Generalized Fibonacci Multilayers. II. Numerical Results", Optik, 115, 267 (2004) M. Kohomoto, B. Sutherland, K. Iguchi,"Localization of optics: Quasiperiodic media", Phys. Rev. Lett., 58, 2436 (1987) [CrossRef] J. Li, L. Zhou, C.T. Chan, P. Shang,"Photonic Band Gap from a Stack of Positive and Negative Index Materials", Phys. Rev. Lett., 90, 083901 (2003) [CrossRef]

Quasicrystalline Structures for Electromagnetic Wave Manipulation

Mikhail V. Rybin. (2024). Cited by 0. Found by: openalex. DOI: 10.1109/piers62282.2024.10618664

Abstract: Quasicrystal-based photonic structures are rigorously ordered but aperiodic systems and the theoretical description of their properties is relatively poor developed yet. The lack of translation symmetry enables many wave phenomena that are prohibited in periodic systems. Quasicrystal-based structures give more degrees of freedom for designing photonic systems that demonstrate novel effects. On the other hand, their rigorous order still limits the degree of freedom to a reasonable value. The aim of this article is to review some recent wave effects observed in quasicrystal-based structures including the intrinsic light localization, complete photonic bandgap formation in low-refractive index systems and frequency-angular selective absorption of electromagnetic radiation.

(no authors). Chinese Optics Letters (2023). Cited by 0. Found by: openalex. openalex_2yr: 4.254 (2026). DOI: 10.3788/col/2023/21/6

Abstract: In the fields of light manipulation and localization, quasiperiodic photonic crystals, or photonic quasicrystals (PQs), are causing an upsurge in research because of their rotational symmetry and long-range orientation of transverse lattice arrays, as they lack translational symmetry.It allows for the optimization of well-established light propagation properties and has introduced new guiding features.Therefore, as a class, quasiperiodic photonic crystal fibers, or photonic quasicrystal fibers (PQFs), are considered to add flexibility and richness to the optical properties of fibers and are expected to offer significant potential applications to optical fiber fields.In this review, the fundamental concept, working mechanisms, and invention history of PQFs are explained.Recent progress in optical property improvement and its novel applications in fields such as dispersion control, polarization-maintenance, supercontinuum generation, orbital angular momentum transmission, plasmon-based sensors and filters, and high nonlinearity and topological mode transmission, are then reviewed in detail.Bandgap-type air-guiding PQFs supporting low attenuation propagation and regulation of photonic density states of quasiperiodic cladding and in which light guidance is achieved by coherent Bragg scattering are also summarized.Finally, current challenges encountered in the guiding mechanisms and practical preparation techniques, as well as the prospects and research trends of PQFs, are also presented.

Boundary modes in quasiperiodic elastic structures

Matheus I. N. Rosa; Raj Kumar Pal; José Roberto de França Arruda; Massimo Ruzzene. (2018). Cited by 0. Found by: openalex. DOI: 10.1117/12.2300887

Abstract: Topological metamaterials are a new class of materials that support topological modes such as edge modes and interface modes, which are commonly immune to scattering and imperfections. This novelty has been the subject of extensive research in many branches of physics such as electronics, photonics, phononics, and acoustics. The nontrivial topological properties related to the presence of topological modes are tipically found in periodic media. However, it was recently demonstrated that structures called quasicrystals may also exhibit nontrivial topological behavior attributed to dimensions higher than that of the quasicrystal. While quasiperiodicity has received a lot of attention in the fields of crystallography and photonics, research into quasiperiodic elastic structures has been scarce. In this paper, we show how the concepts of quasiperiodicity may be applied to the design of topological mechanical metamaterials. We start by investigating the boundary modes present in quasiperiodic 1D phononic lattices. These modes have the interesting property of being localized at either one of the two different boundaries depending on the value of an additional parameter, which is remnant of the higher dimension. A smooth variation of this parameter in either time or a spatial dimension can lead to a robust transfer of energy between two sites of the structure. We present an idealized mechanical system composed by an array of coupled rods that may be used as a platform for realizing this kind of robust transfer of energy. These are preliminary investigations into a entirely new class of structures which may lead to novel engineering applications.

Spectral Properties of Two Coupled Fibonacci Chains

Anouar Moustaj; Malte Röntgen; Christian V. Morfonios; Peter Schmelcher; C. Morais Smith. arXiv (Cornell University) (2022). Cited by 0. Found by: openalex. DOI: 10.48550/arxiv.2208.05178

Abstract: The Fibonacci chain, i.e., a tight-binding model where couplings and/or on-site potentials can take only two different values distributed according to the Fibonacci word, is a classical example of a one-dimensional quasicrystal. With its many intriguing properties, such as a fractal eigenvalue spectrum, the Fibonacci chain offers a rich platform to investigate many of the effects that occur in three-dimensional quasicrystals. In this work, we study the eigenvalues and eigenstates of two identical Fibonacci chains coupled to each other in different ways. We find that this setup allows for a rich variety of effects. Depending on the coupling scheme used, the resulting system (i) possesses an eigenvalue spectrum featuring a richer hierarchical structure compared to the spectrum of a single Fibonacci chain, (ii) shows a coexistence of Bloch and critical eigenstates, or (iii) possesses a large number of degenerate eigenstates, each of which is perfectly localized on only four sites of the system. If additionally, the system is infinitely extended, the macroscopic number of perfectly localized eigenstates induces a perfectly flat quasi band. Especially the second case is interesting from an application perspective, since eigenstates that are of Bloch or of critical character feature largely different transport properties. At the same time, the proposed setup allows for an experimental realization, e.g., with evanescently coupled waveguides, electric circuits, or by patterning an anti-lattice with adatoms on a metallic substrate.

Aubry-André-Harper momentum-state chain in curved spacetime

Yiyi MAO; Hanning DAI. Acta Physica Sinica (2024). Cited by 0. Found by: openalex. openalex_2yr: 0 (2026). DOI: 10.7498/aps.74.20241592

Abstract: Anderson localization is a profound phenomenon in condensed matter physics, representing a fundamental transition in eigenstates, which is triggered off by disorder. The one-dimensional Aubry-André-Harper (AAH) model, an iconic quasiperiodic lattice model, is one of the simplest models that demonstrate the Anderson localization transition. Recently, with the growth of interest in quantum lattice models in curved spacetime (CST), the AAH model in CST has been proposed to explore the interplay between Anderson localization and CST physics. Several CST lattice models have been realized in optical waveguide systems to date, but there are still significant challenges to the experimental preparation and measurement of states, primarily due to the difficulty in dynamically modulating the lattices in such systems. In this work, we propose an experimental scheme using a momentum-state lattice (MSL) in an ultracold atom system to realize the AAH model in CST and study the Anderson localization in this context. Due to the individually controllable coupling between adjacent momentum states in each pair, the coupling amplitude in the MSL can be encoded as a power-law position-dependent \begin{document}$J_n \propto n^{\sigma}$\end{document} , which is conducive to the effective simulation of CST. The numerical calculation results of the MSL Hamiltonian show that the phase separation appears in a 34-site AAH chain in CST, where wave packet dynamics exhibit the localized behavior on one side of the critical site and the extended behavior on the other side. The critical site of phase separation is identified by extracting the turning points of the evolving fractal dimension and wave packet width from the evolution simulations. Furthermore, by modulating the spacetime curvature parameter , we propose a method of preparing the eigenstates of the AAH chain in CST, and perform numerical simulations in the MSL. By calculating the fractal dimension of eigenstates prepared using the aforementioned method, we analyze the localization properties of eigenstates under various quasiperiodic modulation phases, confirming the coexistence of localized phase, swing phase, and extended phase in the energy spectrum. Unlike traditional localized and

extended phases, eigenstates in the swing phase of the AAH model in CST exhibit different localization properties under different modulation phases, indicating the existence of a swing mobility edge. Our results provide a feasible experimental method for studying Anderson localization in CST and presents a new platform for realizing quantum lattice models in curved spacetime.

Efficient local classification of parity-based material topology

Stephan Wong; Ichitaro Yamazaki; Chris Siefert; Iain Duff; Terry A. Loring; Alexander Cerjan. arXiv (Cornell University) (2026). Cited by 0. Found by: openalex. DOI: 10.48550/arxiv.2601.13598

Abstract: Although the classification of crystalline materials can be generally handled by momentum-space-based approaches, topological classification of aperiodic materials remains an outstanding challenge, as the absence of translational symmetry renders such conventional approaches inapplicable. Here, we present a numerically efficient real-space framework for classifying parity-based $\mathbb{Z}\_2$ topology in aperiodic systems based on the spectral localizer framework and the direct computation of the sign of a Pfaffian associated with a large sparse skew-symmetric matrix. Unlike projector-based or momentum-space-based approaches, our method does not rely on translational symmetry, spectral gaps in the Hamiltonian's bulk, or gapped auxiliary operators such as spin projections, and instead provides a local, energy-resolved topological invariant accompanied by an intrinsic measure of topological protection. A central contribution of this work is the development of a scalable sparse factorization algorithm that enables the reliable determination of the Pfaffian's sign for large sparse matrices, making the approach practical to realistic physical materials. We apply this framework to identify the quantum spin Hall effect in quasicrystalline class AII systems, including gapless heterostructures, and to diagnose fragile topology in a large $C\_2 \mathcal{T}$-symmetric photonic quasicrystal. Overall, our results demonstrate that the spectral localizer, combined with efficient sparse numerical methods, provides a unified and robust tool for diagnosing parity-based topological phases in aperiodic electronic, photonic, and acoustic materials where conventional band-theoretic indexes are inapplicable.

Efficient local classification of parity-based material topology

Stephan Wong; Ichitaro Yamazaki; Chris Siefert; Iain Duff; Terry A. Loring; Alexander Cerjan. arXiv (Cornell University) (2026). Cited by 0. Found by: openalex. URL: https://openalex.org/W7125351703

Abstract: Although the classification of crystalline materials can be generally handled by momentum-space-based approaches, topological classification of aperiodic materials remains an outstanding challenge, as the absence of translational symmetry renders such conventional approaches inapplicable. Here, we present a numerically efficient real-space framework for classifying parity-based $\mathbb{Z}\_2$ topology in aperiodic systems based on the spectral localizer framework and the direct computation of the sign of a Pfaffian associated with a large sparse skew-symmetric matrix. Unlike projector-based or momentum-space-based approaches, our method does not rely on translational symmetry, spectral gaps in the Hamiltonian's bulk, or gapped auxiliary operators such as spin projections, and instead provides a local, energy-resolved topological invariant accompanied by an intrinsic measure of topological protection. A central contribution of this work is the development of a scalable sparse factorization algorithm that enables the reliable determination of the Pfaffian's sign for large sparse matrices, making the approach practical to realistic physical materials. We apply this framework to identify the quantum spin Hall effect in quasicrystalline class AII systems, including gapless heterostructures, and to diagnose fragile topology in a large $C\_2 \mathcal{T}$-symmetric photonic quasicrystal. Overall, our results demonstrate that the spectral localizer, combined with efficient sparse numerical methods, provides a unified and robust tool for diagnosing parity-based topological phases in aperiodic electronic, photonic, and acoustic materials where conventional band-theoretic indexes are inapplicable.

Colloidal monolayers on quasiperiodic laser fields

Jules Mikhael. OPUS Publication Server of the University of Stuttgart (University of Stuttgart) (2010). Cited by 0. Found by: openalex. DOI: 10.18419/opus-4924

Abstract: Quasicrystals are somewhat paradoxical structures which exhibit many amazing properties distinguishing them from ordinary crystals. Although the atoms are not localized at periodic positions, quasicrystals posses perfect long-range order. Until the early 1980s it was unanimously established that ordered matter is always periodic. Accordingly, the rotational symmetry in real space was thought to be limited to n=2,3,4 and 6. However more than a hundred complex metal alloys, for instance the discretely diffracting icosahedral AlPdMn or decagonal AlNiCo, have defied these crystallographic rules and self-organized into quasicrystals. Although the majority of the identified quasicrystals are complex metal alloys synthesized in the laboratory, recent experimental results proved that quasiperiodic order is not limited to metals. Matter also organizes itself aperiodically at larger length scales where thermal fluctuations play an important role. Recent experiments have shown that quasiperiodic order is also oberved in soft matter systems, such as micellars, polymers, and binary nanoparticles. Quasicrystals show many interesting properties which are quite different from that of periodic crystals. Accordingly, they are considered as materials with high technological potential e.g. as surface coatings, thermal barriers, catalysts or photonic materials. Quasicrystalline structures have been theoretically predicted also in systems with a single type of particles. Nevertheless, experimentally their spontaneous formation has been only observed in binary, ternary or even more complex alloys. Accordingly, their surfaces exhibit a high degree of structural and chemical complexity and show intriguing properties. In order to understand the origin of those characteristics it would be helpful to disentangle structural and chemical aspects which can be achieved by growing single-element monolayers to quasicrystalline surfaces. Apart from understanding how quasicrystalline properties can be transferred to such monolayers, this approach might allow fabrication of materials with novel properties. First heteroepitatic growth experiments on decagonal and icosahedral surfaces indeed demonstrate the formation of Pb, Bi and Sb monolayers with a high degree of quasicrystalline order as determined by low-energy electron diffraction and elastic helium atom scattering experiments. Compared to reciprocal space studies, only recently atomically resolved scanning tunneling microscopy investigations of the adsorbate morphology became possible. Even then, however, it is difficult to relate the structure of the adsorbate to that of the underlying substrate. In that respect, the study of the phase behaviour of colloidal particles interacting with quasiperiodic laser fields can throw new light on fundamental problems of broad interest in the physics of quasicrystals and in condensed matter physics. In fact colloidal systems are meanwhile established as excellent model for atomic systems and colloidal physics have demonstrated that such systems can give answers to many basic physics questions. Depending on the pair-interaction and the concentration, colloidal systems show analogues of all the states of atomic systems: gas, liquid and solid states. The mesoscopic size (nm-µm), the time scales (ms-s) and the tunability of the pair interaction in colloidal systems make them a convenient model system for experimental and theoretical studies. As a consequence, real space analysis by means of video microscopy allows tracking the trajectories of the individual particles and makes the time evolution of the system accessible in detail. Such information is inaccessible in systems investigated by diffraction experiments, as the scattering information is available only averaged over the scattering area. Because in a colloidal system there is direct access to real space information, the strength and nature of the different interactions, the origins of the complex phase behavior could be in different examples identified. In conclusion, the study of the rich phase behavior of colloidal suspensions provides ideal conditions for experimental and theoretical studies. In this Thesis, we report on a real-space investigation of the phase behaviour of charged colloidal monolayers interacting with quasicrystalline decagonal or tetradecagonal substrates created by interfering five or seven laser beams. Different starting configurations, such as dense fluid and triangular crystals with different densities, are prepared. At low intensities and high particle densities, the electrostatic colloidal repulsion dominates over the colloid-substrate interaction and the crystalline structure remains mainly intact. As expected, at very high intensities the colloid-substrate interaction dominates and a quasiperiodic ordering is observed. Interestingly, at intermediate intensities we observe the alignment of crystalline domains along the 5 directions of the quasicrystalline substrate. This is in agreement with observations of Xenon atoms adsorbed on the ten-fold decagonal Al-Ni-Co surface and numerical simulations of weakly adsorbed atomic systems. Intermediate phases are observed for colloid-substrate interactions strong enough to produce defects in the crystal. These defects adapt the form of rows of quadratic tiles. Surprisingly, for specific particle densities (at which the colloid-substrate interaction is minimized) we identify a novel pseudomorphic ordering. This intermediate phase which exhibits likewise crystalline and quasicrystalline structural properties can be described by an Archimedean-like tiling consisting of alternating rows of quadratic and triangular tiles. The calculated diffraction pattern of this phase is in agreement with recent observations of copper adsorbed on icosahedral AlPdMn surfaces. Interestingly, we also observe the formation of the same phase on tetradecagonal substrates also at densities for which the potential energy of the colloidal system is minimized. Although the structure can also be described by rows of triangles and rows of squares, a closer analysis reveals substantial differences. Here, large domains with almost periodic ordering are found. We show that this behavior is closely related to the low density of highly symmetric local motifs in the substrate potential. In the second part of this Thesis the conditions under which quasicrystals form are investigated. Currently, it is not clear why most quasicrystals hold 5- or 10-fold symmetry but no single example with 7 or 9-fold symmetry has ever been observed. Since the properties of quasicrystals are strongly connected to their atomic structure, a better understanding of their growth mechanisms is of great importance. In contrast to crystals which are periodic in all three dimensions, quasiperiodicity is always (except for icosahedral quasicrystals) restricted to two dimensions. Accordingly, three-dimensional quasicrystals are comprised of a periodic stacking of quasiperiodic layers and any hurdle in the formation of quasiperiodic order within a single layer will eventually prohibit their growth along the periodic direction. In this Thesis, we also report on geometrical constraints which impede the formation of quasicrystals with certain symmetries in a colloidal model system. This is achieved by subjecting a colloidal monolayer to N=5- and 7-beam quasiperiodic potential landscapes. Our results clearly demonstrate that quasicrystalline order is much easier established for $N = 5$ compared to $N = 7$. With increasing laser intensity we observe that the colloids first adopt quasiperiodic order at local areas which then laterally grow until an extended quasicrystalline layer forms. As nucleation sites where quasiperiodicity originates, we identify highly symmetric motifs in the laser pattern. We find that their density strongly varies with n and surprisingly is smallest exactly for those quasicrystalline symmetries which have never been observed in atomic systems. Since such high symmetry motifs also exist in atomic quasicrystals where they act as preferential adsorption sites, this suggests that it is indeed the deficiency of such motifs which accounts for the absence of e.g. materials with 7-fold symmetry. In addition to the fundamental aspects, we report in this Thesis on the fabrication of large colloidal quasiperiodic layers incorporated in a polymer hydrogel matrix. Because quasicrystals have higher point group symmetry than ordinary crystals, micrometer-scale quasicrystalline materials are expected to exhibit large and isotropic photonic bandgaps in the visible range. In our case, the quasiperiodic symmetries are induced using extended light fields. The reported gelled colloidal quasicrystals are unique in that they have large sizes as well as good optical uniformity. With laser diffraction the in situ variable length scale of such materials is demonstrated. In conclusion, we have studied the phase behavior of charged colloidal particles interacting with quasiperiodic laser fields. We showed that novel pseudomorphic growth can lead to the formation of a phase which exhibits likewise crystalline and quasicrystalline structural properties. We also performed unconventional measurements in order to understand why the formation of quasicrystals is limited to specific rotational symmetries. We have found that geometrical hurdles play a crucial role in the proliferation of quasiperiodicity and that such hurdles can hindered or even prohibited the formation of e.g. 7- or 9-fold symmetry. And finally, we have shown that the combination of extended light fields and hydrogel matrices leads to the formation of large quasiperiodically ordered colloidal materials.

Topological critical states and anomalous electronic transmittance in one dimensional quasicrystals

Junmo Jeon; SungBin Lee. Phys. Rev. Research 3, 013168 (2021) (2020). Cited by 0. Found by: arxiv. DOI: 10.1103/PhysRevResearch.3.013168

Abstract: Due to the absence of periodic length scale, electronic states and their topological properties in quasicrystals have been barely understood. Here, we focus on one dimensional quasicrystal and reveal that their electronic critical states are topologically robust. Based on tiling space cohomology, we exemplify the case of one dimensional aperiodic tilings especially Fibonacci quasicrystal and prove the existence of topological critical states at zero energy. Furthermore, we also show exotic electronic transmittance behavior near such topological critical states. Within the perturbative regime, we discuss lack of translational symmetries and presence of topological critical states lead to unconventional scaling behavior in transmittance. Considering both analytic analysis and numerics, electronic transmittance is computed in cases where the system is placed in air or is connected by semi-infinite periodic leads. Finally, we also discuss generalization of our analysis to other quasicrystals. Our findings open a new class of topological quantum states which solely exist in quasicrystals due to exotic tiling patterns in the absence of periodic length scale, and their anomalous electronic transport properties applicable to many experiments.

Non-Fermi-Liquid Behavior in Metallic Quasicrystals with Local Magnetic Moments

Eric C. Andrade; Anuradha Jagannathan; Eduardo Miranda; Matthias Vojta; Vladimir Dobrosavljević. Phys. Rev. Lett. 115, 036403 (2015) (2014). Cited by 0. Found by: arxiv. DOI: 10.1103/PhysRevLett.115.036403

Abstract: Motivated by the intrinsic non-Fermi-liquid behavior observed in the heavy-fermion quasicrystal $\mbox{Au}_{51}\mbox{Al}_{34}\mbox{Yb}_{15}$, we study the low-temperature behavior of dilute magnetic impurities placed in metallic quasicrystals. We find that a large fraction of the magnetic moments are not quenched down to very low temperatures T, leading to a power-law distribution of Kondo temperatures $P(T_{\mbox{K}})\sim T_{\mbox{K}}^{ -1}$, with a nonuniversal exponent $ $, in a remarkable similarity to the Kondo-disorder scenario found in disordered heavy-fermion metals. For $ <1$, the resulting singular $P(T_{\mbox{K}})$ induces non-Fermi-liquid behavior with diverging thermodynamic responses as $T\rightarrow0$.

Emergent Localization in Dodecagonal Bilayer Quasicrystals

Moon Jip Park; Hee Seung Kim; SungBin Lee. Phys. Rev. B 99, 245401 (2019) (2018). Cited by 0. Found by: arxiv. DOI: 10.1103/PhysRevB.99.245401

Abstract: A new type of long-range ordering in the absence of translational symmetry gives rise to drastic revolution of our common knowledge in condensed matter physics. Quasicrystal, as such unconventional system, became a plethora to test our insights and to find exotic states of matter. In particular, electronic properties

in quasicrystal have gotten lots of attention along with their experimental realization and controllability in twisted bilayer systems. In this work, we study how quasicrystalline order in bilayer systems can induce unique localization of electrons without any extrinsic disorders. We focus on dodecagonal quasicrystal that has been demonstrated in twisted bilayer graphene system in recent experiments. In the presence of small gap, we show the localization generically occurs due to non-periodic nature of quasicrystal, which is evidenced by the inverse participation ratio and the energy level statistics. We understand the origin of such localization by approximating the dodecagonal quasicrystals as an impurity scattering problem.

Quantum critical state in a magnetic quasicrystal

Kazuhiko Deguchi; Shuya Matsukawa; Noriaki K. Sato; Taisuke Hattori; Kenji Ishida; Hiroyuki Takakura; Tsutomu Ishimasa. Nature Materials (Advance Online Publication 7 October 2012) (2012). Cited by 0. Found by: arxiv. DOI: 10.1038/Nmat3432

Abstract: Quasicrystals are metallic alloys that possess long-range, aperiodic structures with diffraction symmetries forbidden to conventional crystals. Since the discovery of quasicrystals by Schechtman et al. at 1984 (ref. 1), there has been considerable progress in resolving their geometric structure. For example, it is well known that the golden ratio of mathematics and art occurs over and over again in their crystal structure. However, the characteristic properties of the electronic states - whether they are extended as in periodic crystals or localized as in amorphous materials - are still unresolved. Here we report the first observation of quantum (T = 0) critical phenomena of the Au-Al-Yb quasicrystal - the magnetic susceptibility and the electronic specific heat coefficient arising from strongly correlated 4f electrons of the Yb atoms diverge as T -> 0. Furthermore, we observe that this quantum critical phenomenon is robust against hydrostatic pressure. By contrast, there is no such divergence in a crystalline approximant, a phase whose composition is close to that of the quasicrystal and whose unit cell has atomic decorations (that is, icosahedral clusters of atoms) that look like the quasicrystal. These results clearly indicate that the quantum criticality is associated with the unique electronic state of the quasicrystal, that is, a spatially confined critical state. Finally we discuss the possibility that there is a general law underlying the conventional crystals and the quasicrystals.

Quantum-geometric origin of superfluid weight in quasicrystals with critical states

Kazuma Saito; Ryo Okugawa; Yusuke Kato; Takami Tohyama. arXiv preprint (2026). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/2606.09989v1

Abstract: A distinctive feature of many quasiperiodic systems is the presence of critical states that are neither extended nor exponentially localized. We investigate the geometric effect on the superfluid weight in quasiperiodic systems with critical states at zero temperature. We employ both real-space and momentum-space approaches to superfluid weight in quasicrystals, which allows us to separate the conventional and quantum geometric contributions. We find that the superfluid weight is dominated by the geometric contribution in quasiperiodic systems with critical states. This finding reveals a fundamental interplay between superconductivity and critical states in quasicrystals.

Quantum metric and localization in a quasicrystal

Quentin Marsal; Patric Holmvall; Annica M. Black-Schaffer. arXiv preprint (2025). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/2506.15575v3

Abstract: We use the quantum metric to understand the properties of quasicrystals, represented by the one-dimensional (1D) Fibonacci chain. We show that the quantum metric can relate the localization properties of the eigenstates to the self-similarity of both the chain and its energy spectrum. In particular, the quantum metric incorporates information about distances between the local symmetry centers of each eigenstate, making it much more sensitive to the localization properties of quasicrystals than other measures of localization, such as the inverse participation ratio. Importantly, we further find that a complete description of localization requires us to, in addition, introduce a new phasonic component to the quantum metric, along with a similarly mixed phason-position Chern number. Using this, we show that the sum of both position and phasonic components of the quantum metric is lower-bounded by the gap labels associated with each energy gap of the Fibonacci chain, which stem from the Chern number. This establishes a direct link through the quantum geometry between spatial localization and fractal energy spectrum of quasicrystals. Taken together, quantum geometry provides a unifying, yet accessible, understanding of quasicrystals, rooted in their self-similarity and with intriguing consequences also for many-body physics.

Critical analysis of the re-entrant localization transition in a one-dimensional dimerized quasiperiodic lattice

Shilpi Roy; Sourav Chattopadhyay; Tapan Mishra; Saurabh Basu. arXiv preprint (2021). Cited by 0. Found by: arxiv. DOI: 10.1103/PhysRevB.105.214203

Abstract: A re-entrant localization transition has been predicted recently in a one-dimensional quasiperiodic lattice with dimerized hopping between the nearest-neighbour sites (Phys. Rev. Lett. {\bf 126} 106803 (2021)) \cite{PhysRevLett.126.106803}. It has been shown that the interplay between the hopping dimerization and a staggered quasi-periodic disorder manifests two localization transitions through two intermediate phases resulting in four critical points as a function of the quasiperiodic potential. In this paper, we study the phenomenon of the re-entrant localization transition by examining the spectral properties of the states. By performing a systematic finite-size scaling analysis for a fixed value of the hopping dimerization, we obtain accurate critical disorder strengths for different transitions and the associated critical exponents. Moreover, through a multifractal analysis, we study the critical nature of the states across the localization transitions by computing the mass exponents and the corresponding fractal dimensions of the states. Further, we complement the critical nature of the states by computing the Hausdorff dimensions.

Localization and topological transitions in non-Hermitian quasiperiodic lattices

Ling-Zhi Tang; Guo-Qing Zhang; Ling-Feng Zhang; Dan-Wei Zhang. Phys. Rev. A 103, 033325 (2021) (2021). Cited by 0. Found by: arxiv. DOI: 10.1103/PhysRevA.103.033325

Abstract: We investigate the localization and topological transitions in a one-dimensional (interacting) non-Hermitian quasiperiodic lattice, which is described by a generalized Aubry-André-Harper model with irrational modulations in the off-diagonal hopping and on-site potential and with non-Hermiticities from the nonreciprocal hopping and complex potential phase. For noninteracting cases, we reveal that the nonreciprocal hopping (the complex potential phase) can enlarge the delocalization (localization) region in the phase diagrams spanned by two quasiperiodical modulation strengths. We show that the localization transition are always accompanied by a topological phase transition characterized the winding numbers of eigenenergies in three different non-Hermitian cases. Moreover, we find that a real-complex eigenenergy transition in the energy spectrum coincides with (occurs before) these two phase transitions in the nonreciprocal (complex potential) case, while the real-complex transition is absent under the coexistence of the two non-Hermiticities. For interacting spinless fermions, we demonstrate that the extended phase and the many-body localized phase can be identified by the entanglement entropy of eigenstates and the level statistics of complex eigenenergies. By making the critical scaling analysis, we further show that the many-body localization transition coincides with the real-complex transition and occurs before the topological transition in the nonreciprocal case, which are absent in the complex phase case.

Non-Abelian generalization of non-Hermitian quasicrystal: PT-symmetry breaking, localization, entanglement and topological transitions

Longwen Zhou. Phys. Rev. B 108, 014202 (2023) (2023). Cited by 0. Found by: arxiv. DOI: 10.1103/PhysRevB.108.014202

Abstract: Non-Hermitian quasicrystal forms a unique class of matter with symmetry-breaking, localization and topological transitions induced by gain and loss or nonreciprocal effects. In this work, we introduce a non-Abelian generalization of the non-Hermitian quasicrystal, in which the interplay between non-Hermitian effects and non-Abelian quasiperiodic potentials create mobility edges and rich transitions among extended, critical and localized phases. These generic features are demonstrated by investigating three non-Abelian variants of the non-Hermitian Aubry-André-Harper model. A unified characterization is given to their spectrum, localization, entanglement and topological properties. Our findings thus add new members to the family of non-Hermitian quasicrystal and uncover unique physics that can be triggered by non-Abelian effects in non-Hermitian systems.

New Classes of Quasicrystals and Marginal Critical States

Nobuhisa Fujita; Komajiro Niizeki. Phys. Rev. Lett. 85, 4924 (2000) (2000). Cited by 0. Found by: arxiv. DOI: 10.1103/PhysRevLett.85.4924

Abstract: One-dimensional quasilattices, namely, the geometrical objects to represent quasicrystals, are classified into mutual local-derivability (MLD) classes. Besides the familiar class, there exist an infinite number of new MLD classes, and different MLD classes are distinguished by the inflation rules of their representatives. It has been found that electronic properties of a new MLD class are characterized by the presence of marginal

critical states, which are considered to be nearly localized states.

Localization and spectral structure in two-dimensional quasicrystal potentials

Zhaoxuan Zhu; Shengjie Yu; Dean Johnstone; Laurent Sanchez-Palencia. Phys. Rev. A 109, 013314 (2024) (2023). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/2307.09527v2

Abstract: Quasicrystals, a fascinating class of materials with long-range but nonperiodic order, have revolutionized our understanding of solid-state physics due to their unique properties at the crossroads of long-range-ordered and disordered systems. Since their discovery, they continue to spark broad interest for their structural and electronic properties. The quantum simulation of quasicrystals in synthetic quantum matter systems offers a unique playground to investigate these systems with unprecedented control parameters. Here, we investigate the localization properties and spectral structure of quantum particles in 2D quasicrystalline optical potentials. While states are generally localized at low energy and extended at high energy, we find alternating localized and critical states at intermediate energies. Moreover, we identify a complex succession of gaps in the energy spectrum. We show that the most prominent gap arises from strongly localized ring states, with the gap width determined by the energy splitting between states with different quantized winding numbers. In addition, we find that these gaps are stable for quasicrystals with different rotational symmetries and potential depths, provided that other localized states do not enter the gap generated by the ring states. Our findings shed light on the unique properties of quantum quasicrystals and have implications for their many-body counterparts.

Localization transitions in non-Hermitian quasiperiodic lattice

Aruna Prasad Acharya; Sanjoy Datta. arXiv preprint (2023). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/2306.0

Abstract: The delocalization-localization (DL) transition in non-Hermitian systems exhibits intriguing features distinct from their Hermitian counterparts. In this study, we investigate the DL transition in a generalized non-Hermitian lattice with asymmetric hopping and complex quasi-periodic potential. Irrespective of the boundary conditions, the lattice undergoes a DL transition at a critical strength of the quasiperiodic potential with identical modulation of its real and complex parts. For periodic boundary conditions (PBC), we obtained an analytical expression that accurately predicts this critical point. Our numerical results indicate that the critical point remains the same with the open boundary condition (OBC) as well. Interestingly, we observe that a difference in the modulation of the real and the complex part of potential leads to a mixed phase that appears between the delocalized and the localized phases. Intriguingly, within the mixed state region, we observed a coexistence of skin modes and localized states in the case of OBC, while in the case of PBC, a mixed phase is created by a coexistence of delocalized and localized states. We mapped out the phase diagrams for different scenarios offering valuable insights into the role of different parameters in a wide class of non-Hermitian quasiperiodic lattices.

Field-Tunable Valley Coupling and Localization in a Dodecagonal Semiconductor Quasicrystal

Zhida Liu; Qiang Gao; Yanxing Li; Xiaohui Liu; Fan Zhang; Dong Seob Kim; Yue Ni; Miles Mackenzie; Hamza Abudayyeh; Kenji Watanabe; Takashi Taniguchi; Chih-Kang Shih; Eslam Khalaf; Xiaoqin Li. arXiv preprint (2024). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/2408.02176v1

Abstract: Quasicrystals are characterized by atomic arrangements possessing long-range order without periodicity. Van der Waals (vdW) bilayers provide a unique opportunity to controllably vary atomic alignment between two layers from a periodic moiré crystal to an aperiodic quasicrystal. Here, we reveal a remarkable consequence of the unique atomic arrangement in a dodecagonal WSe2 quasicrystal: the K and Q valleys in separate layers are brought arbitrarily close in momentum space via higher-order Umklapp scatterings. A modest perpendicular electric field is sufficient to induce strong interlayer K-Q hybridization, manifested as a new hybrid excitonic doublet. Concurrently, we observe the disappearance of the trion resonance and attribute it to quasicrystal potential driven localization. Our findings highlight the remarkable attribute of incommensurate systems to bring any pair of momenta into close proximity, thereby introducing a novel aspect to valley engineering.

Localization of Electronic Wave Functions on Quasiperiodic Lattices

Thomas Rieth; Uwe Grimm; Michael Schreiber. Proceedings of the 6th International Conference on Quasicrystals, Eds. S. Takeuchi and T. Fujiwara (World Scientific, Singapore, 1998), pp. 639-642 (1998). Cited by 0. Found by: arxiv. DOI: 10.1088/0953-8984/10/4/008

Abstract: We study electronic eigenstates on quasiperiodic lattices using a tight-binding Hamiltonian in the vertex model. In particular, the two-dimensional Penrose tiling and the three-dimensional icosahedral Ammann-Kramer tiling are considered. Our main interest concerns the decay form and the self-similarity of the electronic wave functions, which we compute numerically for periodic approximants of the perfect quasiperiodic structure. In order to investigate the suggested power-law localization of states, we calculate their participation numbers and structural entropy. We also perform a multifractal analysis of the eigenstates by standard box-counting methods. Our results indicate a rather different behaviour of the two- and the three-dimensional systems. Whereas the eigenstates on the Penrose tiling typically show power-law localization, this was not observed for the icosahedral tiling.

Appearance of Mobility Edge in Self-Dual Quasiperiodic Lattices

Gang Wang; Nianbei Li; Tsuneyoshi Nakayama. arXiv preprint (2013). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/1312.0844v1

Abstract: Within the framework of the Aubry-Andre model, one kind of self-dual quasiperiodic lattice, it is known that a sharp transition occurs from \emph{all} eigenstates being extended to \emph{all} being localized. The common perception for this type of quasiperiodic lattice is that the self-duality excludes the appearance of the mobility edge separating localized from extended states. In this work, we propose a multi-chromatic quasiperiodic lattice model retaining the self-duality identical to the Aubry-Andre model, and demonstrate numerically the occurrence of the localization-delocalization transition with definite mobility edges. This contrasts with the Aubry-Andre model. As a result, the diffusion of wave packet exhibits a transition from ballistic to diffusive motion, and back to ballistic motion. We point out that experimental realizations of the predicted transition can be accessed with light waves in photonic lattices and matter waves in optical lattices.

Spin-wave localization on phasonic defects in one-dimensional magnonic quasicrystal

Szymon Mieszczak; Maciej Krawczyk; Jarosław W. Kłos. arXiv preprint (2022). Cited by 0. Found by: arxiv. DOI: 10.1103/PhysRevB.106.064430

Abstract: We report on the evolution of the spin-wave spectrum under structural disorder introduced intentionally into one-dimensional magnonic quasicrystal. We study theoretically a system composed of ferromagnetic strips arranged in a Fibonacci sequence. We considered several stages of disorder in the form of phasonic defects, where different rearrangements of strips are introduced. By transition from the quasiperiodic order towards disorder, we show a gradual degradation of spin-waves fractal spectra and closing of the frequency gaps. In particular, the phasonic defects lead to the disappearance of the van Hove singularities at the frequency gap edges by moving modes into the frequency gaps and appearing new modes inside the frequency gaps. These modes disperse and eventually can close the gap, with increasing disorder levels. The work reveals how the the presence of disorder modifies the intrinsic spin wave localization existing in undefected magnonic quasicrystals. The paper contributes to the knowledge of magnonic Fibonacci quasicrystals and opens the way to study of the phasonic defects in two-dimensional magnonic quasicrystals.

Localization transitions of correlated particles in nonreciprocal quasicrystals

Lei Wang; Juan Kang; Ni Liu; Chaohua Wu; Gang Chen. arXiv preprint (2024). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/2412.18755v1

Abstract: The interplay among interaction, non-Hermiticity, and disorder opens a new avenue for engineering novel phase transitions. We here study the spectral and localization features of two interacting bosons in one-dimensional nonreciprocal quasicrystals. Specifically, by considering a quasiperiodic Hubbard lattice with nonreciprocal hoppings, we show that the interaction can lead to a mobility edge, which arises from the fact that the bound states display a much lower threshold for spectral and extended-localized transitions than scattering states. The localization transition of bound or scattering states is accompanied by a complex-real spectrum transition. Moreover, while the two-particle localized states are robust to the boundary conditions, the two-particle extended states turn into skin modes under open boundary condition. We also show the correlated dynamics to characterize these localization transitions. Finally, we reveal that the bound states can form mobility edge on their own by introducing a dimerized nonreciprocal quasicrystal. Our paper may pave the way for the study of non-Hermitian few-body physics.

Critical states and anomalous mobility edges in two-dimensional diagonal quasicrystals

Callum W. Duncan. Phys. Rev. B 109, 014210 (2024) (2023). Cited by 0. Found by: arxiv. DOI: 10.1103/PhysRevB.109.014210

Abstract: We study the single-particle properties of two-dimensional quasicrystals where the underlying geometry of the tight-binding lattice is crystalline but the on-site potential is quasicrystalline. We will focus on the 2D generalised Aubry-André model which has a varying form to its quasiperiodic potential, through a deformation parameter and varied irrational periods of cosine terms, which allows a continuous family of on-site quasicrystalline models to be studied. We show that the 2D generalised Aubry-André model exhibits single-particle mobility edges between extended and localised states and a localisation transition in a similar manner to the prior studied one-dimensional limit. However, we find that such models in two dimensions are dominated across large parameter regions by critical states. The presence of critical states results in anomalous mobility edges between both extended and critical and localised and critical states in the single-particle spectrum, even when there is no mobility edge between extended and localised states present. Due to this, these models exhibit anomalous diffusion of initially localised states across the majority of parameter regions, including deep in the normally localised regime. The presence of critical states in large parameter regimes and throughout the spectrum will have consequences for the many-body properties of quasicrystals, including the formation of the Bose glass and the potential to host a many-body localised phase.

Anomalous localization and multifractality in a kicked quasicrystal

Toshihiko Shimasaki; Max Prichard; H. Esat Kondakci; Jared Pagett; Yifei Bai; Peter Dotti; Alec Cao; Tsung-Cheng Lu; Tarun Grover; David M. Weld. arXiv preprint (2022). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/2203.09442v2

Abstract: Multifractal states offer a "third way" for quantum matter, neither fully localized nor ergodic, exhibiting singular continuous spectra, self-similar wavefunctions, and transport and entanglement scaling exponents intermediate between extended and localized states. While multifractality in equilibrium systems generally requires fine-tuning to a critical point, externally driven quantum matter can exhibit multifractal states with no equilibrium counterpart. We report the experimental observation of multifractal matter and anomalous localization in a kicked Aubry-André-Harper quasicrystal. Our cold-atom realization of this previously-unexplored model is enabled by apodized Floquet engineering techniques which expand the accessible phase diagram by five orders of magnitude. This kicked quantum quasicrystal exhibits a rich phase diagram including not only fully localized and fully delocalized phases but also an extended region comprising an intricate nested pattern of localized, delocalized, and multifractal states. Mapping transport properties throughout the phase diagram, we observe disorder-driven re-entrant delocalization and sub-ballistic transport, and present a theoretical explanation of these phenomena based on eigenstate multifractality. These results open up the exploration of new states of matter characterized by an intricate interplay of fractal structure and quantum dynamics.

Fractal Spectrum of the Aubry-Andre Model

Ang-Kun Wu. arXiv preprint (2021). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/2109.07062v2

Abstract: The Aubry-Andre model is a one-dimensional lattice model for quasicrystals with localized and delocalized phases. At the localization transition point, the system displays fractal spectrum, which relates to the Hofstadter butterfly. In this work, we uncover the exact self-similarity structures in the energy spectrum. We separate the fractal structures into two parts: the fractal filling positions of gaps and the scaling of gap sizes. We show that the fractal fillings emerge for a certain type of incommensurate periodicity regardless of potential strength. However, the power-law scaling of gap sizes is characteristic for general incommensurability at the critical point of the model.

Quantum Antiferromagnetism in Quasicrystals

Stefan Wessel; Anuradha Jagannathan; Stephan Haas. Phys. Rev. Lett. 90, 177205 (2003). (2002). Cited by 0. Found by: arxiv. DOI: 10.1103/PhysRevLett.90.177205

Abstract: The antiferromagnetic Heisenberg model is studied on a two-dimensional bipartite quasiperiodic lattice. The distribution of local staggered magnetic moments is determined on finite square approximants with up to 1393 sites, using the Stochastic Series Expansion Quantum Monte Carlo method. A non-trivial inhomogeneous ground state is found. For a given local coordination number, the values of the magnetic moments are spread out, reflecting the fact that no two sites in a quasicrystal are identical. A hierarchical

structure in the values of the moments is observed which arises from the self-similarity of the quasiperiodic lattice. Furthermore, the computed spin structure factor shows antiferromagnetic modulations that can be measured in neutron scattering and nuclear magnetic resonance experiments. This generic model is a first step towards understanding magnetic quasicrystals such as the recently discovered Zn-Mg-Ho icosahedral structure.

Boundary-dependent Self-dualities, Winding Numbers and Asymmetrical Localization in non-Hermitian Quasicrystals

Xiaoming Cai. Phys. Rev. B 103, 014201 (2021) (2020). Cited by 0. Found by: arxiv. DOI: 10.1103/PhysRevB.103.014201

Abstract: We study a non-Hermitian Aubry-André-Harper model with both nonreciprocal hoppings and complex quasiperiodical potentials, which is a typical non-Hermitian quasicrystal. We introduce boundary-dependent self-dualities in this model and obtain analytical results to describe its Asymmetrical Anderson localization and topological phase transitions. We find that the Anderson localization is not necessarily in accordance with the topological phase transitions, which are characteristics of localization of states and topology of energy spectrum respectively. Furthermore, in the localized phase, single-particle states are asymmetrically localized due to non-Hermitian skin effect and have energy-independent localization lengths. We also discuss possible experimental detections of our results in electric circuits.

Engineering mobility in quasiperiodic lattices with exact mobility edges

Zhenbo Wang; Yu Zhang; Li Wang; Shu Chen. Phys. Rev. B 108, 174202 (2023) (2023). Cited by 0. Found by: arxiv. DOI: 10.1103/PhysRevB.108.174202

Abstract: We investigate the effect of an additional modulation parameter $ $ on the mobility properties of quasiperiodic lattices described by a generalized Ganeshan-Pixley-Das Sarma model with two on site modulation parameters. For the case with bounded quasiperiodic potential, we unveil the existence of self-duality relation, independent of $ $. By applying Avila's global theory, we analytically derive Lyapunov exponents in the whole parameter space, which enables us to determine mobility edges or anomalous mobility edges exactly. Our analytical results indicate that the mobility edge equation is described by two curves and their intersection with the spectrum gives the true mobility edge. Tuning the strength parameter $ $ can change the spectrum of the quasiperiodic lattice, and thus engineers the mobility of quasi-periodic systems, giving rise to completely extended, partially localized, and completely localized regions. For the case with unbounded quasiperiodic potential, we also obtain the analytical expression of the anomalous mobility edge, which separates localized states from critical states. By increasing the strength parameter $ $, we find that the critical states can be destroyed gradually and finally vanishes.

Extended-localized transition in diffusive quasicrystals

Zhoufei Liu; Pei-Chao Cao; Ying Li; Jiping Huang. Phys. Rev. Appl. 21, 064035 (2024) (2022). Cited by 0. Found by: arxiv. DOI: 10.1103/PhysRevApplied.21.064035

Abstract: Compared to periodic systems, quasicrystals without translational invariance exhibit unexpected localization properties. The extended-localized transition in quasicrystals has been observed in both quantum and classical wave systems. However, its manifestation in diffusion systems, which serve as novel platforms for exploring phases of matter in condensed matter physics, remains unexplored. Here, we present the implementation of the extended-localized transition in a diffusive quasicrystal based on the coupled ring chain structure. By modulating the thermal conductivities of rings, we obtain the diffusive one-dimensional Aubry-André-Harper (AAH) model, which exhibits an extended-localized transition. Thanks to the ring-shaped chain, we clearly demonstrate the extended-localized transition under the uniform excitation through temperature field simulations. For the localized state, the temperature field clearly demonstrates a multiple localization centers phenomenon, which has no counterpart in wave systems. We also quantitatively investigate the temperature evolution and size effect of this transition. Furthermore, the local excitation has been adopted to demonstrate the temperature field for both the extended and localized states. Besides, we implement the non-Hermitian diffusive AAH model by rotating rings, whose temperature field shows a moving multiple localization centers phenomenon in the localized phase. Finally, we give the experimental suggestions for the diffusive AAH model and propose a potential application named as double-trace distributed generator. Our results can facilitate the design of flexible thermal devices and efficient heat management.

Interplay of non-Hermitian skin effects and Anderson localization in non-reciprocal quasiperiodic lattices

Hui Jiang; Li-Jun Lang; Chao Yang; Shi-Liang Zhu; Shu Chen. Phys. Rev. B 100, 054301 (2019) (2019). Cited by 0. Found by: arxiv. DOI: 10.1103/PhysRevB.100.054301

Abstract: Non-Hermiticity from non-reciprocal hoppings has been shown recently to demonstrate the non-Hermitian skin effect (NHSE) under open boundary conditions (OBCs). Here we study the interplay of this effect and the Anderson localization in a \textit{non-reciprocal} quasiperiodic lattice, dubbed non-reciprocal Aubry-André model, and a \textit{rescaled} transition point is exactly proved. The non-reciprocity can induce not only the NHSE, but also the asymmetry in localized states with two Lyapunov exponents for both sides. Meanwhile, this transition is also topological, characterized by a winding number associated with the complex eigenenergies under periodic boundary conditions (PBCs), establishing a \textit{bulk-bulk} correspondence. This interplay can be realized by an elaborately designed electronic circuit with only linear passive RLC devices instead of elusive non-reciprocal ones, where the transport of a continuous wave undergoes a transition between insulating and amplifying. This initiative scheme can be immediately applied in experiments to other non-reciprocal models, and will definitely inspires the study of interplay of NHSEs and more other quantum/topological phenomena.

Composite-State Localization Beyond the External Landscape in Non-Hermitian Quasicrystals

Tian Zhou; Xue-Bin Wang; Zhongmin Yang; Tao Liu. arXiv preprint (2026). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/2608.08105v1

Abstract: A composite excitation need not inherit the localization behavior of its constituents. We show that an interacting non-Hermitian quasiperiodic ladder realizes a controllable and reversible localization inversion between composite and unbound excitations, where internal configuration, rather than only the external potential, becomes a control parameter for localization. Opposite complex potentials on the two legs cancel at first order for a same-rung pair but act directly on separated particles, allowing extended composite states to persist while the unpaired sector becomes localized. A strong-coupling theory identifies the composite state as an emergent weakly modulated non-Hermitian quasicrystal generated by virtual unpaired configurations. Breaking the potential antisymmetry restores a direct modulation of the composite band and reverses the localization hierarchy. Engineering configuration-space pathways further stabilizes an extended composite band embedded within a localized continuum, the inverse of the conventional bound-state-in-the-continuum scenario. Our results establish internal configuration as a reversible control parameter for localization.

Non-Hermitian delocalization in a 2D photonic quasicrystal

Zhaoyang Zhang; Shun Liang; Ismael Septembre; Jiawei Yu; Yongping Huang; Maochang Liu; Yanpeng Zhang; Min Xiao; Guillaume Malpuech; Dmitry Solnyshkov. arXiv preprint (2023). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/2312.09322v1

Abstract: Quasicrystals show long-range order, but lack translational symmetry. So far, theoretical and experimental studies suggest that both Hermitian and non-Hermitian quasicrystals show localized eigenstates. This localization is due to the fractal structure of the spectrum in the Hermitian case and to the transition to diffusive bands via exceptional points in the non-Hermitian case. Here, we present an experimental study of a dodecagonal (12-fold) photonic quasicrystal based on electromagnetically-induced transparency in a Rb vapor cell. The transition to a quasicrystal is obtained by superposing two honeycomb lattices at 30° with a continuous tuning of their amplitudes. Non-Hermiticity is controlled independently. We study the spatial expansion of a probe wavepacket. In the Hermitian case, the wavepacket expansion is suppressed when the amplitude of the second lattice is increased (quasicrystal localization). We find a new regime, where increasing the non-Hermitian potential in the quasicrystal enhances spatial expansion, with the C_{12} symmetry becoming visible in the wavepacket structure. This real-space expansion is due to a k-space localization on specific quasicrystal modes. Our results show that the non-Hermitian quasicrystal behavior is richer than previously thought. The localization properties of the quasicrystals can be used for beam tailoring in photonics, but are also important in other fields.

Nonlinear topological Toda quasicrystal

Motohiko Ezawa. J. Phys. Soc. Jpn. 91, 084703 (2022) (2022). Cited by 0. Found by: arxiv. DOI: 10.7566/JPSJ.91.084703

Abstract: Topological edge states are known to emerge in certain quasicrystals. We investigate a topological quasicrystal in the presence of nonlinearity by generalizing the Toda lattice to include modulated periodic hoppings, where the period is taken irrational to the original lattice. It is found that topological edge states in a quasicrystal survive against nonlinearity based on the quench dynamics. It is also found that an extended-localization transition is induced by the quasicrystal hopping modulation. The present model is experimentally realizable by a transmission line with variable capacitance diodes, where the inductance is modulated.

Theory of Localized States in Quasiperiodic Lattices

Jin-Rong Chen; Xin-Yu Guo; Shi-Ping Ding; Tian-Le Wu; Miao Liang; Jin-Hua Gao; X. C. Xie. arXiv preprint (2025). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/2509.05950v1

Abstract: The physics of localized states in quasiperiodic lattices has been extensively studied for decades, but still lacks an comprehensive theoretical framework. Recently, we developed a incommensurate energy band (IEB) theory, which extends the concept of energy bands to quasiperiodic systems lacking translational symmetry, thereby achieving a breakthrough in elucidating extended states. Here, we demonstrate that, due to the inherent duality between momentum and real space, the IEB theory also offers a comprehensive framework for elucidating localized states. Specifically, via a so-called spiral (module) mapping, the energy spectrum of localized states can be represented as a function defined on a compact circular manifold-akin to the Brillouin zone-whose form resembles conventional energy bands. These localized state energy bands (LSEBs) fully characterize all the properties of the localized states. Moreover, we show that quasiperiodic systems with mobility edges exhibit a unique hybrid band structure: the IEB for extended states (momentum space) and LSEB for localized states (real space), separated by mobility edges. Our theory thus establishes a comprehensive framework for analyzing the localized states in quasiperiodic lattices.

Emergent extended states in an unbounded quasiperiodic lattice

Jia-Ming Zhang; Shan-Zhong Li; Shi-Liang Zhu; Zhi Li. arXiv preprint (2025). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/2502.13503v1

Abstract: Previous studies have established that quasiperiodic lattice models with unbounded potentials can exhibit localized and multifractal states, yet preclude the existence of extended states. In this work, we introduce a quasiperiodic system that incorporates both unbounded potentials and unbounded hopping amplitudes, where extended states emerge as a direct consequence of the unbounded hopping terms overcoming the localization constraints imposed by the unbounded potential, thereby facilitating enhanced particle transport. By using Avila's global theory, we derive analytical expressions for the phase boundaries, with exact results aligning closely with numerical simulations.Intriguingly, we uncover a hidden self-duality in the proposed model by establishing a mapping to the Aubry-André model, revealing a profound structural connection between these systems.

Nonmonotonic crossover and scaling behaviors in a disordered 1D quasicrystal

Anuradha Jagannathan; Piyush Jeena; Marco Tarzia. Phys. Rev. B 99, 054203 (2019) (2018). Cited by 0. Found by: arxiv. DOI: 10.1103/PhysRevB.99.054203

Abstract: We consider a noninteracting disordered 1D quasicrystal in the weak disorder regime. We show that the critical states of the pure model approach strong localization in strikingly different ways, depending on their renormalization properties. A finite size scaling analysis of the inverse participation ratios of states (IPR) of the quasicrystal shows that they are described by several kinds of scaling functions. While most states show a progressively increasing IPR as a function of the scaling variable, other states exhibit a nonmonotonic 're-entrant' behavior wherein the IPR first decreases, and passes through a minimum, before increasing. This surprising behavior is explained in the framework of perturbation renormalization group treatment, where wavefunctions can be computed analytically as a function of the hopping amplitude ratio and the disorder, however it is not specific to this model. Our results should help to clarify results of recent studies of localization due to random and quasiperiodic potentials.

Growing Perfect Decagonal Quasicrystals by Local Rules

Hyeong-Chai Jeong. Phys. Rev. Lett. 98, 135501 (2007) (2007). Cited by 0. Found by: arxiv. DOI: 10.1103/PhysRevLett.98.135501

Abstract: A local growth algorithm for a decagonal quasicrystal is presented. We show that a perfect Penrose tiling (PPT) layer can be grown on a decapod tiling layer by a three dimensional (3D) local rule growth. Once a PPT layer begins to form on the upper layer, successive 2D PPT layers can be added on top resulting in a perfect decagonal quasicrystalline structure in bulk with a point defect only on the bottom surface layer. Our growth rule shows that an ideal quasicrystal structure can be constructed by a local growth algorithm in 3D, contrary to the necessity of non-local information for a 2D PPT growth.

Renormalization-Group Theory of 1D quasiperiodic lattice models with commensurate approximants

Miguel Gonçalves; Bruno Amorim; Eduardo V. Castro; Pedro Ribeiro. Phys. Rev. B 108, L100201 (2023) (2022). Cited by 0. Found by: arxiv. DOI: 10.1103/PhysRevB.108.L100201

Abstract: We develop a renormalization group (RG) description of the localization properties of onedimensional (1D) quasiperiodic lattice models. The RG flow is induced by increasing the unit cell of subsequent commensurate approximants. Phases of quasiperiodic systems are characterized by RG fixed points associated with renormalized single-band models. We identify fixed-points that include many previously reported exactly solvable quasiperiodic models. By classifying relevant and irrelevant perturbations, we show that phase boundaries of more generic models can be determined with exponential accuracy in the approximant's unit cell size, and in some cases analytically. Our findings provide a unified understanding of widely different classes of 1D quasiperiodic systems.

Dephasing-induced distinct mobility edges in a dimerized off-diagonal quasicrystal

Ming-Jie Tao; Yi-Ting Wang; Jing Li; Hongsheng Hou; Xiang-Ping Jiang; Lei Pan. arXiv preprint (2026). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/2608.17937v1

Abstract: Anderson localization and the mobility edge (ME) have been extensively studied in isolated aperiodic systems. Conventional theory suggests that dephasing and decoherence should disrupt localization and facilitate transport. In this work, we investigate localization behaviors in a dimerized off-diagonal Aubry-Andre-Harper (AAH) quasicrystal subject to on-site pure dephasing. In the strong-dephasing limit, we apply adiabatic elimination within the Lindblad master equation framework to derive an effective classical Markov transition matrix that governs the dissipative relaxation dynamics. Counterintuitively, we demonstrate that pure dephasing can induce distinct MEs, including both conventional MEs separating extended and localized states and anomalous MEs separating multifractal critical states from localized states, even when all eigenstates of the original closed coherent system are delocalized or multifractal. Using fractal dimension finite-size scaling, wave-packet spreading dynamics, and energy spectrum statistics, we numerically verify the coexistence of fully extended, multifractal critical, and localized regions within the relaxation spectrum of the dissipative system, and construct the global dissipative phase diagram. These findings reveal that dephasing can see as a powerful mechanism for controlling localization transitions, thus enhancing our understanding of dissipative quasicrystal systems.

Spatially Localized Quasicrystals

P. Subramanian; A. J. Archer; E. Knobloch; A. M. Rucklidge. New J. Phys. 20 (2018) 122002 (2017). Cited by 0. Found by: arxiv. DOI: 10.1088/1367-2630/aaf3bd

Abstract: We investigate quasicrystal-forming soft matter using a two-scale phase field crystal model. At state points near thermodynamic coexistence between bulk quasicrystals and the liquid phase, we find multiple metastable spatially localized quasicrystals embedded in a background of liquid. In three dimensions we obtain spatially localized icosahedral quasicrystals. In two dimensions, we compute several families of spatially localized quasicrystals with dodecagonal structure and investigate their properties as a function of the system parameters. In both cases the localized quasicrystals are metastable, and so correspond to energetically locally favored structures. The presence of such structures is expected to crucially affect the dynamics of the crystallization process.

Floquet engineering of topological localization transitions and mobility edges in one-dimensional non-Hermitian quasicrystals

Longwen Zhou. Phys. Rev. Research 3, 033184 (2021) (2021). Cited by 0. Found by: arxiv. DOI: 10.1103/PhysRevResearch.3.033184

Abstract: Time-periodic driving fields could endow a system with peculiar topological and transport features. In this work, we find dynamically controlled localization transitions and mobility edges in non-Hermitian quasicrystals via shaking the lattice periodically. The driving force dresses the hopping amplitudes between lattice sites, yielding alternate transitions between localized, mobility edge and extended non-Hermitian quasicrystalline phases. We apply our Floquet engineering approach to five representative models of non-Hermitian quasicrystals, obtain the conditions of photon-assisted localization transitions and mobility edges, and find the expressions of Lyapunov exponents for some models. We further introduce topological winding numbers of Floquet quasienergies to distinguish non-Hermitian quasicrystalline phases with different localization nature. Our discovery thus extend the study of quasicrystals to non-Hermitian Floquet systems, and provide an efficient way of modulating the topological and transport properties of these unique phases.

Dimerization induced mobility edges and multiple reentrant localization transitions in non-Hermitian quasicrystals

Wenqian Han; Longwen Zhou. Phys. Rev. B 105, 054204 (2022) (2021). Cited by 0. Found by: arxiv. DOI: 10.1103/PhysRevB.105.054204

Abstract: Non-Hermitian effects could create rich dynamical and topological phase structures. In this work, we show that the collaboration between lattice dimerization and non-Hermiticity could generally bring about mobility edges and multiple localization transitions in one-dimensional quasicrystals. Non-Hermitian extensions of the Aubry-André-Harper (AAH) model with staggered onsite potential and dimerized hopping amplitudes are introduced to demonstrate our results. Reentrant localization transitions due to the interplay between quasiperiodic gain/loss and lattice dimerization are found. Quantized winding numbers are further adopted as topological invariants to characterize transitions among phases with distinct spectrum and transport nature. Our study thus enriches the family of non-Hermitian quasicrystals by incorporating effects of lattice dimerization, and offering a convenient way to modulate localization transitions and mobility edges in non-Hermitian systems.

First-order Quantum Phase Transitions and Localization in the 2D Haldane Model with Non-Hermitian Quasicrystal Boundaries

Xianqi Tong; Su-Peng Kou. arXiv preprint (2023). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/2309.09173v1

Abstract: The non-Hermitian extension of quasicrystals (QC) are highly tunable system for exploring novel material phases. While extended-localized phase transitions have been observed in one dimension, quantum phase transition in higher dimensions and various system sizes remain unexplored. Here, we show the discovery of a new critical phase and imaginary zeros induced first-order quantum phase transition within the two-dimensional (2D) Haldane model with a quasicrystal potential on the upper boundary. Initially, we illustrate a phase diagram that evolves with the amplitude and phase of the quasiperiodic potential, which is divided into three distinct phases by two critical boundaries: phase (I) with extended wave functions, PT-restore phase (II) with localized wave functions, and a critical phase (III) with multifunctional wave functions. To describe the wavefunctions in these distinct phases, we introduce a low-energy approximation theory and an effective two-chain model. Additionally, we uncover a first-order structural phase transition induced (FOSPT) by imaginary zeros. As we increase the size of the potential boundary, we observe the critical phase splitting into regions in proportion to the growing number of potential zeros. Importantly, these observations are consistent with groundstate fidelity and energy gap calculations. Our research enhances the comprehension of phase diagrams associated with high-dimensional quasicrystal potentials, offering valuable contributions to the exploration of unique phases and quantum phase transition.

How Do Quasicrystals Grow?

Aaron S. Keys; Sharon C. Glotzer. Phys. Rev. Lett. 99, pp. 235503 (2007) (2007). Cited by 0. Found by: arxiv. DOI: 10.1103/PhysRevLett.99.235503

Abstract: Using molecular simulations, we show that the aperiodic growth of quasicrystals is controlled by the ability of the growing quasicrystal ‘nucleus’ to incorporate kinetically trapped atoms into the solid phase with minimal rearrangement. In the system under investigation, which forms a dodecagonal quasicrystal, we show that this process occurs through the assimilation of stable icosahedral clusters by the growing quasicrystal. Our results demonstrate how local atomic interactions give rise to the long-range aperiodicity of quasicrystals.

One-dimensional L{é}vy Quasicrystal

Pallabi Chatterjee; Ranjan Modak. J. Phys.: Condens. Matter 35 505602 (2023) (2022). Cited by 0. Found by: arxiv. DOI: 10.1088/1361-648X/acf9d4

Abstract: Space-fractional quantum mechanics (SFQM) is a generalization of the standard quantum mechanics when the Brownian trajectories in Feynman path integrals are replaced by L{é}vy flights. We introduce L{é}vy quasicrystal by discretizing the space-fractional Schr$\ddot{\text{o}}$dinger equation using the Gr$\ddot{\text{u}}$nwald-Letnikov derivatives and adding on-site quasiperiodic potential. The discretized version of the usual Schr$\ddot{\text{o}}$dinger equation maps to the Aubry-Andr{é} Hamiltonian, which supports localization-delocalization transition even in one dimension. We find the similarities between L{é}vy quasicrystal and the Aubry-Andr{é} (AA) model with power-law hopping and show that the L{é}vy quasicrystal supports a delocalization-localization transition as one tunes the quasiperiodic potential strength and shows the coexistence of localized and delocalized states separated by mobility edge. Hence, a possible realization of SFQM in optical experiments should be a new experimental platform to test the predictions of AA models in the presence of power-law hopping.

Re-entrant topological phase transition in a non-Hermitian quasiperiodic lattice

Ashirbad Padhan; Soumya Ranjan Padhi; Tapan Mishra. arXiv preprint (2023). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/2306.11084v1

Abstract: We predict a re-entrant topological transition in a one dimensional non-Hermitian quasiperiodic lattice. By considering a non-Hermitian generalized Aubry-André-Harper (AAH) model with quasiperiodic potential, we show that the system first undergoes a transition from the delocalized phase to the localized phase and then to the delocalized phase as a function of the hermiticity breaking parameter. This re-entrant delocalization-localization-delocalization transition in turn results in a re-entrant topological transition identified by associating the phases with spectral winding numbers. Moreover, we find that these two transitions occur through intermediate phases hosting both extended and localized states having real and imaginary energies, respectively. We find that these phases also possess non-trivial winding numbers which are different from that of the localized phase.

A patchy-particle 3-dimensional octagonal quasicrystal

Akie Kowaguchi; Savan Mehta; Jonathan P. K. Doye; Eva G. Noya. J. Chem. Phys. 163, 244904 (2025) (2024). Cited by 0. Found by: arxiv. DOI: 10.1063/5.0292922

Abstract: We devise an ideal 3-dimensional octagonal quasicrystal that is based upon the 2-dimensional Ammann-Beenker tiling and that is potentially suitable for realization with patchy particles. Based on an analysis of its local environments we design a binary system of 8- and 5-patch particles that in simulations assembles into a 3-dimensional octagonal quasicrystal. The local structure is subtly different from the original ideal quasicrystal possessing a narrower coordination-number distribution; in fact, the 8-patch particles are not needed and a one-component system of the 5-patch particles assembles into an essentially identical octagonal quasicrystal. We also consider a one-component system of the 8-patch particles; this assembles into a cluster with a number of crystalline domains, but which, because of the coherent boundaries between the crystallites, has approximate eight-fold order. We envisage that these systems could be realized using DNA origami or protein design.

Perpendicular space accounting of localized states in a quasicrystal

Murod Mirzhalilov; M. Ö. Oktel. Phys. Rev. B 102, 064213 (2020) (2020). Cited by 0. Found by: arxiv. DOI: 10.1103/PhysRevB.102.064213

Abstract: Quasicrystals can be described as projections of sections of higher dimensional periodic lattices into real space. The image of the lattice points in the projected out dimensions, called the perpendicular space, carries valuable information about the local structure of the real space lattice. In this paper, we show that perpendicular space projection can be used to analyze the elementary excitations of a quasicrystal. In particular, we consider the vertex tight binding model on the two dimensional Penrose lattice and investigate the properties of strictly localized states using perpendicular space images. Our method reproduces the previously reported frequencies for the six types of localized states in this model. We also calculate the overlaps between different localized states and show that the number of type-5 and type-6 localized states which are independent from the four other types is a factor of golden ratio $ =(1+\sqrt{5})/2$ higher than previously reported values. Two orientations

of the same type-5 or type-6 which are supported around the same site are shown to be linearly dependent with the addition of other types. We also show through exhaustion of all lattice sites in perpendicular space that any point in the Penrose lattice is either in the support of at least one localized state or is forbidden by local geometry to host a strictly localized state.

Purely local growth of a quasicrystal

Thomas Fernique; Ilya Galanov. arXiv preprint (2022). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/2201.00789v

Abstract: Self-assembly is the process in which the components of a system, whether molecules, polymers, or macroscopic particles, are organized into ordered structures as a result of local interactions between the components themselves, without exterior guidance. In this paper, we speak about the self-assembly of aperiodic tilings. Aperiodic tilings serve as a mathematical model for quasicrystals - crystals that do not have any translational symmetry. Because of the specific atomic arrangement of these crystals, the question of how they grow remains open. In this paper, we state the theorem regarding purely local and deterministic growth of Golden-Octagonal tilings. Showing, contrary to the popular belief, that local growth of aperiodic tilings is possible.

Mott transition and correlation effects on strictly localized states in an octagonal quasicrystal

Efe Yelesti; Onur Erten; M. O. Oktel. Phys. Rev. B 112, 144204 (2025) (2025). Cited by 0. Found by: arxiv. DOI: 10.1103/hxhj-7tvt

Abstract: Flat-band systems have attracted significant attention as platforms for studying strongly correlated electron physics, where the dominance of electron-electron interactions over kinetic energy gives rise to a variety of emergent phenomena. Quasicrystals are compelling systems for studying these phenomena as they host degenerate strictly localized states at zero energy due to perfect destructive interference patterns. In this study, we use the slave-rotor mean-field approach to investigate the effects of electron interactions within the Hubbard model on the Ammann-Beenker quasicrystal. The phase diagram characterizing metallic and Mott insulator regions indicates a first-order phase transition. Our analysis shows that the local coordination number affects the local quasiparticle weight, displaying varying metallicity across the sites. Furthermore, we focus on the strictly localized states that arise in the non-interacting limit. We find that interactions and deviation from particle-hole symmetry induce spectral splitting, broadening, and partial delocalization of the localized states, depending on the local environment. In particular, certain localized states with higher coordination numbers remain more robust compared to others. Our results highlight the critical role of local geometry in shaping correlation effects in flat-band quasicrystals.

Phononic Topological States in 1D Quasicrystals

J. R. M. Silva; M. S. Vasconcelos; D. H. A. L. Anselmo; V. D. Mello. arXiv preprint (2019). Cited by 0. Found by: arxiv. DOI: 10.1088/1361-648X/ab312a

Abstract: We theoretically analyze the spectrum of phonons of a one-dimensional quasiperiodic lattice. We simulate the quasicrystal from the classic system of spring-bound atoms with a force constant modulated by the Aubry-André model, so that its value is slightly different in each site of the lattice. From the equations of motion, we obtained the equivalent phonon spectrum of the Hofstadter butterfly, characterizing a multifractal. In this spectrum, we obtained the extended, critical and localized regimes, and we observed that the multifractal characteristic is sensitive to the number of atoms and the $ $ parameter of our model. We also verified the presence of border states for phonons, where some modes in the system boundaries present vibrations. Through the measurement of localization of the individual displacements in each site, we verify the presence of a phase transition through the Inverse Participation Rate (IPR) for $ = 1.0 $, where the system changes from extended to localized.

Emergent multi-loop nested point gap in a non-Hermitian quasiperiodic lattice

Yi-Qi Zheng; Shan-Zhong Li; Zhi Li. Phys. Rev. B 111, 104204 (2025) (2024). Cited by 0. Found by: arxiv. DOI: 10.1103/PhysRevB.111.104204

Abstract: We propose a geometric series modulated non-Hermitian quasiperiodic lattice model, and explore its localization and topological properties. The results show that with the ever-increasing summation terms of the geometric series, multiple mobility edges and non-Hermitian point gaps with high winding number can be induced in the system. The point gap spectrum of the system has a multi-loop nested structure in the complex

plane, resulting in a high winding number. In addition, we analyze the limit case of summation of infinite terms. The results show that the mobility edges merge together as only one mobility edge when summation terms are pushed to the limit. Meanwhile, the corresponding point gaps are merged into a ring with winding number equal to one. Through Avila's global theory, we give an analytical expression for mobility edges in the limit of infinite summation, reconfirming that mobility edges and point gaps do merge and will result in a winding number that is indeed equal to one.

Spin susceptibility in quasicrystal superconductor with spin-orbit interactions

Yasuhiro Tada. arXiv preprint (2025). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/2512.15093v1

Abstract: We study impacts of spin-orbit interactions on the spin susceptibility in quasicrystal superconductors, motivated by the anomolous superconducting properties in the van der Waals quasicrystal Ta$_{1.6}$Te under magnetic fields. We consider the Penrose tiling model with s-wave pairing as a representative system and include anisotropic spin-orbit interactions allowed for the quasicrystal structure. It is shown that the spin-momentum locking locally takes place in the quasicrystal, and electron motion and spin directions are tightly connected at each spatial position in the real space. As a result, the spin susceptibility is enhanced for both in-plane and out-of-plane magnetic fields in the presence of a Rashba-type spin-orbit interaction. For an Ising-type spin-orbit interaction, the spin susceptibility only for in-plane magnetic fields is increased, while the out-of-plane spin susceptibility is almost unchanged. When there are both kinds of the spin-orbit interactions, temperature dependence of the spin susceptibility for any field directions is suppressed.

Localization of Pairs in One-Dimensional Quasicrystals with Power-Law Hopping

G. A. Domínguez-Castro; R. Paredes. arXiv preprint (2022). Cited by 0. Found by: arxiv. DOI: 10.1103/PhysRevB.106.134208

Abstract: Pair localization in one-dimensional quasicrystals with nearest-neighbor hopping is independent of whether short-range interactions are repulsive or attractive. We numerically demonstrate that this symmetry is broken when the hopping follows a power law $1/r^ $. In particular, for repulsively bound states, we find that the critical quasiperiodicity that signals the transition to localization is always bounded by the standard Aubry-André critical point, whereas attractively bound dimers get localized at larger quasiperiodic modulations when the range of the hopping increases. Extensive numerical calculations establish the contrasting nature of the pair energy gap for repulsive and attractive interactions, as well as the behavior of the algebraic localization of the pairs as a function of quasiperiodicity, interaction strength, and power-law hops. The results here discussed are of direct relevance to the study of the quantum dynamics of systems with power-law couplings.

Hybrid Quasicrystals, Transport and Localization in Products of Minimal Sets

Tulio O. Carvalho; Cesar R. de Oliveira. arXiv preprint (2007). Cited by 0. Found by: arxiv. DOI: 10.1007/s10955-007-9299-8

Abstract: We consider convex combinations of finite-valued almost periodic sequences (mainly substitution sequences) and put them as potentials of one-dimensional tight-binding models. We prove that these sequences are almost periodic. We call such combinations {\em hybrid quasicrystals} and these studies are related to the minimality, under the shift on both coordinates, of the product space of the respective (minimal) hulls. We observe a rich variety of behaviors on the quantum dynamical transport ranging from localization to transport.

Localization, $\mathcal{PT}$-Symmetry Breaking and Topological Transitions in non-Hermitian Quasicrystals

Aruna Prasad Acharya; Aditi Chakrabarty; Deepak Kumar Sahu; Sanjoy Datta. Phys. Rev. B 105, 014202 (2022) (2021). Cited by 0. Found by: arxiv. DOI: 10.1103/PhysRevB.105.014202

Abstract: According to the topological band theory of a Hermitian system, the different electronic phases are classified in terms of topological invariants, wherein the transition between the two phases characterized by a different topological invariant is the primary signature of a topological phase transition. Recently, it has been argued that the delocalization-localization transition in a quasicrystal, described by the non-Hermitian $\mathcal{PT}$-symmetric extension of the Aubry-André-Harper (AAH) Hamiltonian can also be identified as a topological phase transition. Interestingly, the $\mathcal{PT}$-symmetry also breaks down at the same critical point. However, in this article, we have shown that the delocalization-localization transition and the $\mathcal{PT}$-symmetry breaking are not connected to a topological phase transition. To demonstrate this, we have studied the non-Hermitian $\mathcal{PT}$-symmetric AAH Hamiltonian in the presence of Rashba Spin-Orbit (RSO) coupling. We have obtained an analytical expression of the topological transition point and compared it with the numerically obtained critical points. We have found that, except in some special cases, the critical point and the topological transition point are not the same. In fact, the delocalization-localization transition takes place earlier than the topological transition whenever they do not coincide.

Localized States in Local Isomorphism Classes of Pentagonal Quasicrystals

M. Ö. Oktel. Phys. Rev. B 106, 024201 (2022) (2022). Cited by 0. Found by: arxiv. DOI: 10.1103/PhysRevB.106.024201

Abstract: A family of pentagonal quasicrystals can be defined by projecting a section of the five-dimensional cubic lattice to two dimensions. A single parameter, the sum of intercepts $Γ=\sum_j _j$, describes this family. Each value of $0\le Γ\le \frac{1}{2}$ defines a unique local isomorphism class for these quasicrystals, with $Γ=0$ giving the Penrose lattice. Except for a few special values of $Γ$, these lattices lack simple inflation-deflation rules making it hard to count how frequently a given local configuration is repeated. We consider the vertex-tight-binding model on these quasicrystals and investigate the strictly localized states (LS) for all values of $Γ$. We count the frequency of localized states both by numerical exact diagonalization on lattices of $~10^5$ sites and by identifying localized state types and calculating their perpendicular space images. While the imbalance between the number of sites of the sublattices of the bipartite quasicrystal grows monotonically with $Γ$, we find that the localized state fraction first decreases and then increases as the distance from the Penrose lattice grows. The highest LS fraction of $~10.17\%$ is attained at $Γ=0.5$ while the minimum is $~4.5\%$ at $Γ\simeq 0.12$. The LS on the even sublattice are generally concentrated near sites with high symmetry, while the LS on the odd sublattice are more uniformly distributed. The odd sublattice has a higher LS fraction, having almost three times the LS frequency of the even sublattice at $Γ=0.5$. We identify 20 LS types on the even sublattice, and their total frequency agrees well with the numerical exact diagonalization result for all values of $Γ$. For the odd sublattice, we identify 45 LS types. However, their total frequency remains significantly below the numerical calculation, indicating the possibility of more independent LS types.

Exploring Metallic-Insulating Transition and Thermodynamic Applications of Fibonacci Quasicrystals

He-Guang Xu; Shujie Cheng. arXiv preprint (2024). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/2410.17550v1

Abstract: Extended and critical states are two common phenomena in Fibonacci quasicrystals. In this paper, we first reveal the difference between the extended phase and the critical phase in the extended-critical Fibonacci quasicrystal from the perspectives of quantum transport and Wigner distribution. The transport conductance indicates that the extended-critical transition resembles a metallic-insulating transition. Moreover, the Wigner distributions show that the Wigner distribution of the extended wave function is localized in the momentum direction of the phase space, while that of the critical wave function is sub-extended in the momentum direction of the phase space. Based on the results of entanglement entropy, the extended-critical transition is a thermodynamic phase transition because it is accompanied by decreasing entropy. We engineer a quantum heat cycle engine with the extended-critical quasicrystal as the working medium, and find that there are rich working modes in the engine, such as quantum accelerator, quantum heater and quantum heat engine. Importantly, the extended quasicrystals are more conducive to the realization of quantum heat engines, while the critical quasicrystals are more conducive to the realization of quantum heaters. Our work is an important step toward exploring the rich thermodynamic applications of Fibonacci quasicrystals.

Self-induced Bose glass phase in quantum cluster quasicrystals

Matheus Grossklags; Matteo Ciardi; Vinicius Zampronio; Fabio Cinti; Alejandro Mendoza-Coto. Results in Physics, 65, 107991, (2024) (2023). Cited by 0. Found by: arxiv. DOI: 10.1016/j.rinp.2024.107991

Abstract: We study the emergence of Bose glass phases in self sustained bosonic quasicrystals induced by a pair interaction between particles of Lifshitz-Petrich type. By using a mean field variational method designed in momentum space as well as Gross-Pitaevskii simulations we determine the phase diagram of the model. The study of the local and global superfluid fraction allows the identification of supersolid, super quasicrystal, Bose glass and insulating phases. The Bose glass phase emerges as a quasicrystal phase in which the global superfluidity is essentially zero, while the local superfluidity remains finite in certain ring structures of the quasicrystalline pattern. Furthermore, we perform continuous space Path Integral Monte Carlo simulations for a case in which the interaction between particles stabilizes a quasicrystal phase. Our results show that as the

strength of the interaction between particles is increased the system undergoes a sequence of states consistent with the super quasicrystal, Bose glass, and quasicrystal insulator thermodynamic phases.

Dislocations in quasicrystals and their interaction with cluster-like obstacles

R. Mikulla; P. Gumbsch; H. -R. Trebin. arXiv preprint (1997). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/cond-mat/9708102v2

Abstract: A dislocation moving through a quasicrystal is leaving in its wake a fault denoted phason wall. For a two-dimensional model quasicrystal the disregistry energy of this phason wall is studied to determine possible Burgers vectors of the quasicrystalline structure. Contrary to periodic crystals, the disregistry energy is an average quantity with large fluctuations on the atomic scale. Therefore the dislocation core structure and mobility cannot be linked to this quantity e.g. by a Peierls-Nabarro model. Atomistic simulations show that dislocation motion is controlled by local obstacles inherent to the atomic structure of the quasicrystal.

Computational Self-Assembly of a Six-Fold Chiral Quasicrystal

Nydia Roxana Varela-Rosales; Michael Engel. arXiv preprint (2024). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/2408.01984v2

Abstract: Quasicrystals are unique materials characterized by long-range order without periodicity. They are observed in systems such as metallic alloys, soft matter, and particle simulations. Unlike periodic crystals, which are invariant under real-space symmetry operations, quasicrystals possess symmetry that requires description by a space group in reciprocal space. In this study, we report the self-assembly of a six-fold chiral quasicrystal using molecular dynamics simulations of a two-dimensional particle system. The particles interact via the Lennard-Jones-Gauss pair potential and are subjected to a periodic substrate potential. We confirm the presence of chiral symmetry through diffraction patterns and order parameters, revealing unique local motifs in both real and reciprocal space. The quasicrystal's properties, including the tiling structure and symmetry and the extent of diffuse scattering, are strongly influenced by substrate potential depth and temperature. Our results provide insights into the mechanisms of chiral quasicrystal formation and the role and potential of external fields in tailoring quasicrystal structures.

Controlling light with nonlinear quasicrystal metasurfaces

Yutao Tang; Junhong Deng; King Fai Li; Mingke Jin; Jack Ng; Guixin Li. arXiv preprint (2019). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/1901.10138v1

Abstract: Metasurface, a kind of two-dimensional structured medium, represents a novel platform to manipulate the propagation of light at subwavelength scale. In linear optical regime, many interesting topics such as planar metalens, metasurface optical holography and so on have been widely investigated. Recently, metasurfaces go into nonlinear optical regime. While it is recognized that the local symmetry of the meta-atoms plays vital roles, its relationship with global symmetry of the nonlinear metasurfaces remains elusive. According to the Penrose tiling and the newly proposed hexagonal quasicrystalline tiling, here we designed and fabricated the nonlinear optical quasicrystal metasurfaces based on the geometric phase controlled plasmonic meta-atoms with local rotational symmetry. The second harmonic waves will be determined by both the tiling schemes of quasicrystal metasurfaces and the local symmetry of meta-atoms they consist of. The proposed concept opens new routes for designing nonlinear metasurface crystals with desired optical functionalities.

Intrinsic vortex pinning in superconducting quasicrystals

Yuki Nagai. Phys. Rev. B 106, 064506 (2022) (2021). Cited by 0. Found by: arxiv. DOI: 10.1103/PhysRevB.106.064506

Abstract: We numerically show that a vortex pinning occurs in superconducting quasicrystal without impurities and defects. This vortex pinning is intrinsic since the superconducting order parameter in quasicrystal is always inhomogeneous due to the lack of the translational symmetry. We propose that experiments influenced by vortex pinning effects can detect the atomic-scale inhomogeneous superconducting order parameter in quasicrystals. We develop a numerical method to solve the Bogoliubov-de Gennes equations and gap equations in large systems, which is based on the localized-Krylov subspace and a sparse modeling technique. Two two-dimensional quasicrystals, the Penrose and Amman-Beenker tiling, are considered.

Driving-induced multiple ${\cal PT}$-symmetry breaking transitions and reentrant localization transitions in non-Hermitian Floquet quasicrystals

Longwen Zhou; Wenqian Han. Phys. Rev. B 106, 054307 (2022) (2022). Cited by 0. Found by: arxiv. DOI: 10.1103/PhysRevB.106.054307

Abstract: The cooperation between time-periodic driving fields and non-Hermitian effects could endow systems with distinctive spectral and transport properties. In this work, we uncover an intriguing class of non-Hermitian Floquet matter in one-dimensional quasicrystals, which is characterized by the emergence of multiple driving-induced ${\cal PT}$-symmetry breaking/restoration, mobility edges, and reentrant localization transitions. These findings are demonstrated by investigating the spectra, level statistics, inverse participation ratios and wavepacket dynamics of a periodically quenched nonreciprocal Harper model. Our results not only unveil the richness of localization phenomena in driven non-Hermitian quasicrystals, but also highlight the advantage of Floquet approach in generating unique types of nonequilibrium phases in open systems.

Hyperuniform electron distributions controlled by electron interactions in quasicrystal

Shiro Sakai; Ryotaro Arita; Tomi Ohtsuki. Phys. Rev. B 105, 054202 (2022) (2021). Cited by 0. Found by: arxiv. DOI: 10.1103/PhysRevB.105.054202

Abstract: We study how the electron-electron interactions influence the charge distributions in the metallic state of quasicrystals. As a simple theoretical model, we introduce an extended Hubbard model on the Penrose lattice, and numerically solve the model (up to ~1.4 million sites) within the Hartree-Fock approximation. Because each site on the quasiperiodic lattice has a different local geometry, the Coulomb interaction, in particular the intersite one, works in a site-dependent way, leading to a nontrivial redistribution of the charge. The resultant charge distribution patterns are not multifractal but characterized by hyperuniformity, which offers a measure to distinguish various inhomogeneous but ordered distributions. We clarify how the electron interactions alter the order metric of the hyperuniformity, revealing that the intersite interaction considerably affects the hyperuniformity in particular on the electron-rich side.

Stripe order in quasicrystals

Rafael M. P. Teixeira; Eric C. Andrade. Eur. Phys. J. B 98, 188 (2025) (2025). Cited by 0. Found by: arxiv. DOI: 10.1140/epjb/s10051-025-01040-y

Abstract: We explore the emergence of magnetic order in geometrically frustrated quasiperiodic systems, focusing on the interplay between local tile symmetry and frustration-induced constraints. In particular, we study the J_1-J_2 Ising model on the two-dimensional Ammann–Beenker quasicrystal. Through large-scale Monte Carlo simulations and general arguments, we map the phase diagram of the model. For small J_2, a Néel phase appears, whereas a stripe phase is stable for dominant antiferromagnetic J_2, despite the system's lack of periodicity. Although long-range stripe order emerges below a critical temperature, unlike in random systems, it is softened by the nucleation of competing stripe domains pinned at specific quasiperiodic sites. This behavior reveals a unique mechanism of symmetry breaking in quasiperiodic lattices, where geometric frustration and local environment effects compete to determine the magnetic ground state. Our results show how long-range order adapts to non-periodic structures, with implications for understanding nematic phases and other broken-symmetry states in quasicrystals.

Topological Anderson insulator phase in a quasicrystal lattice

Rui Chen; Dong-Hui Xu; Bin Zhou. Phys. Rev. B 100, 115311 (2019) (2019). Cited by 0. Found by: arxiv. DOI: 10.1103/PhysRevB.100.115311

Abstract: Motivated by the recent experimental realization of the topological Anderson insulator and research interest on the topological quasicrystal lattices, we investigate the effects of disorder on topological properties of a two-dimensional Penrose-type quasicrystal lattice that supports the quantum spin Hall insulator (QSHI) and normal insulator (NI) phases in the clean limit. It is shown that the helical edge state of the QSHI phase is robust against weak disorder. Most saliently, it is found that disorder can induce a phase transition from NI to QSHI phase in the quasicrystal system. The numerical results based on a two-terminal device show that a quantized conductance plateau can arise inside the energy gap of NI phase for moderate Anderson disorder strength. Further, it is confirmed that the local current distributions of the disorder-induced quantized conductance plateau are located on the two edges of sample. Finally, we identify this disorder-induced phase as topological Anderson insulator phase by computing the disorder-averaged spin Bott index.

Anomalous Localization and Mobility Edges in Non-Hermitian Quasicrystals with Disordered Imaginary Gauge Fields

Guolin Nan; Zhijian Li; Feng Mei; Zhihao Xu. arXiv preprint (2026). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/2601.14754v2

Abstract: Localization in non-Hermitian quasicrystals can differ fundamentally from its Hermitian counterpart when non-reciprocity is spatially disordered. Here we study a one-dimensional non-Hermitian Aubry-André-Harper chain with a Bernoulli imaginary gauge field and quasiperiodic onsite modulation. In the nearest-neighbor limit, we identify an anomalous transition from a fully erratic non-Hermitian skin effect (ENHSE) phase to a fully localized phase. Although the fractal dimension vanishes in both regimes, the Lyapunov exponent and the fluctuation of the eigenstate center of mass sharply distinguish them. For generic finite-size realizations, this transition is further accompanied by a complex-to-real spectral change under periodic boundary conditions and a change of spectral winding from nontrivial to trivial. With weak next-nearest-neighbor hopping, we uncover an anomalous mobility edge at the same location as in the Hermitian generalized Aubry-André-Harper model, but separating Anderson-localized states from ENHSE-type macroscopic-accumulation states rather than extended states. We further show that this anomalous localization structure is reflected in spectral winding and wave-packet dynamics: single realizations exhibit winding-dependent drift, winding-resolved averaging preserves opposite directional responses, and full disorder averaging largely restores Hermitian-like transport. Our results establish practical diagnostics of anomalous localization and mobility edges in non-Hermitian quasicrystals.

Bose-Einstein Condensation and quasicrystals

Moorad Alexanian; Vanik E. Mkrtchian. Armenian Journal of Physics, 2021, vol. 14, issue 2, pp. 128-131 (2021). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/2107.02901v1

Abstract: We consider interacting Bose particles in an external local potential. It is shown that large class of external quasicrystal potentials cannot sustain any type of Bose-Einstein condensates. Accordingly, at spatial dimensions $D\leq 2$ in such quasicrystal potentials a supersolid is not possible via Bose-Einstein condensates at finite temperatures. The latter also hold true for the two-dimensional Fibonacci tiling. However, supersolids do arise at $D\leq 2$ via Bose-Einstein condensates from infinitely long-range, nonlocal interparticle potentials.

Modelling Quasicrystal Growth

Uwe Grimm; Dieter Joseph. Quasicrystals - An Introduction to Structure, Physical Properties and Applications, edited by J.-B. Suck, M. Schreiber and P. H"aussler (Springer, Berlin, 2002) pp. 199-218 (1999). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/cond-mat/9903074v1

Abstract: Understanding the growth of quasicrystals poses a challenging problem, not the least because the quasiperiodic order present in idealized mathematical models of quasicrystals prohibit simple local growth algorithms. This can only be circumvented by allowing for some degree of disorder, which of course is always present in real quasicrystalline samples. In this review, we give an overview of the present state of theoretical research, addressing the problems, the different approaches and the results obtained so far.

Quantum Cluster Quasicrystals

Guido Pupillo; Primoz Ziherl; Fabio Cinti. Phys. Rev. B 101, 134522 (2020) (2019). Cited by 0. Found by: arxiv. DOI: 10.1103/PhysRevB.101.134522

Abstract: Quasicrystals remain among the most intriguing materials in physics and chemistry. Their structure results in many unusual properties including anomalously low friction as well as poor electrical and thermal conductivity but it also supports superconductivity, which shows that quantum effects in quasicrystals can be quite unique. Here we theoretically study superfluidity in a model quantum cluster quasicrystal. Using path-integral Monte Carlo simulations, we explore a 2D ensemble of bosons with the Lifshitz-Petrich-Gaussian pair potential, finding that moderate quantum fluctuations do not destroy the dodecagonal quasicrystalline order. This quasicrystal is characterized by a small yet finite superfluidity, demonstrating that particle clustering combined with the local cogwheel structure can underpin superfluidity even in the almost classical regime. This type of distributed superfluidity may also be expected in certain open crystalline lattices. Large quantum fluctuations are shown to induce transitions to cluster solids, supersolids and superfluids, which we characterize fully quantum-mechanically.

Nonlinearity-Induced Thouless Pumping in Quasiperiodic Lattices

Xiao-Xiao Hu; Dun Zhao; Hong-Gang Luo. arXiv preprint (2026). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/2604.00639v1

Abstract: Nonlinear Thouless pumping has been established in periodic lattices; its counterpart in quasiperiodic lattices remains unexplored. Here, we show a nonlinear topological pumping of gap solitons in quasiperiodic lattices where the local nonlinear self-consistent potentials lead to a lattice potential reconstruction; as a result, an emergent topological structure induced by this local reconstruction governs the dynamics of the gap solitons. This enables solitons to adiabatically occupy a single topological band, realizing quasi-quantized Thouless pumping. In addition, the intrinsic lattice perturbations disrupt this band occupation, which drives solitons into a non-quantized drifting regime. However, even in this regime, we also find that the soliton transport is constrained by the topological properties of a critical rational approximant. Tuning nonlinearity or lattice scaling reveals a controllable switching among topological pumping, drifting, and localization. Our work uncovers a mechanism for nonlinearity-induced topological behavior in complex lattice potentials.

Strictly localized states in the octagonal Ammann-Beenker quasicrystal

M. Ö. Oktel. Phys. Rev. B 104, 014204 (2021) (2021). Cited by 0. Found by: arxiv. DOI: 10.1103/PhysRevB.104.014204

Abstract: Ammann-Beenker lattice is a two-dimensional quasicrystal with eight-fold symmetry, which can be described as a projection of a cut from a four-dimensional simple cubic lattice. We consider the vertex tight-binding model on this lattice and investigate the strictly localized states at the center of the spectrum. We use a numerical method based on the generation of finite lattices around a given perpendicular space point and QR decomposition of the Hamiltonian to count the strictly localized states. We apply this method to count the frequency of localized states in lattices of up to 100 000 sites. We obtain an orthogonal set of compact localized states by diagonalizing the position operator projected onto the manifold spanned by the zero energy states. We identify twenty localized state types and calculate their exact frequencies through their perpendicular space images. Unlike the Penrose lattice, all the localized state types are eight-fold symmetric around an eight edge vertex, and all vertex types can support localized states. The total frequency of these twenty types gives a lower bound of $f_{LS}=30796-21776\sqrt{2} \simeq 0.08547$ for the fraction of strictly localized states in the spectrum. This value is in agreement with the numerical calculation and very close to the recently conjectured exact fraction of localized states $f_{Ex}=3/2-\sqrt{2}\simeq 0.08579$.

Family of self-dual quasicrystals with critical Phases

Wenzhi Wang; Wei Yi; Tianyu Li. W. Wang, W. Yi and T. Li, Phys. Rev. B 112, L140201 (2025) (2025). Cited by 0. Found by: arxiv. DOI: 10.1103/yntg-6yh2

Abstract: We propose a general framework for constructing self-dual one-dimensional quasiperiodic lattice models with arbitrary-range hoppings and multifractal behaviors. Our framework generates a broad spectrum of one dimensional quasicrystals, ranging from the off-diagonal Aubry-André-Harper models on one end, to those featuring long-range hoppings with varied quasiperiodic modulations on another. Focusing on models with off-diagonal quasiperiodic hoppings with power-law decay, we exploit the fact that, when the self-dual condition is satisfied, the system must be in the critical state with multifractal properties. This enables the engineering of models with competing extended, critical, and localized phases, with richly structured mobility edges separating them. As an outstanding example, we show that a limiting case of our family of self-dual quasicrystals can be implemented using Rydberg-atom arrays. Our work offers a systematic route toward critical phases from self-duality considerations, and would facilitate the experimental simulation of these exotic states.

Length scale formation in the Landau levels of quasicrystals

Junmo Jeon; Moon Jip Park; SungBin Lee. arXiv preprint (2021). Cited by 0. Found by: arxiv. DOI: 10.1103/PhysRevB.105.045146

Abstract: Exotic tiling patterns of quasicrystals have motivated extensive studies of quantum phenomena such as critical states and phasons. Nevertheless, the systematic understanding of the Landau levels of quasicrystals in the presence of the magnetic field has not been established yet. One of the main obstacles is the complication of the quasiperiodic tilings without periodic length scales, thus it has been thought that the system cannot possess any universal features of the Landau levels. In this paper, contrary to these assertions, we develop a generic theory of the Landau levels for quasicrystals. Focusing on the two dimensional quasicrystals with

rotational symmetries, we highlight that quasiperiodic tilings induce anomalous Landau levels where electrons are localized near the rotational symmetry centers. Interestingly, the localization length of these Landau levels has a universal dependence on n for quasicrystals with n-fold rotational symmetry. Furthermore, macroscopically degenerate zero energy Landau levels are present due to the chiral symmetry of the rhombic tilings. In this case, each Landau level forms an independent island where electrons are trapped at given fields, but with field control, the interference between the islands gives rise to an abrupt change in the local density of states. Our work provide a general scheme to understand the electron localization behavior of the Landau levels in quasicrystals.

Asymmetric transfer matrix analysis of Lyapunov exponents in one-dimensional non-reciprocal quasicrystals

Shan-Zhong Li; Enhong Cheng; Shi-Liang Zhu; Zhi Li. Phys. Rev. B 110, 134203 (2024) (2024). Cited by 0. Found by: arxiv. DOI: 10.1103/PhysRevB.110.134203

Abstract: The Lyapunov exponent, serving as an indicator of the localized state, is commonly utilized to identify localization transitions in disordered systems. In non-Hermitian quasicrystals, the non-Hermitian effect induced by non-reciprocal hopping can lead to the manifestation of two distinct Lyapunov exponents on opposite sides of the localization center. Building on this observation, we here introduce a comprehensive approach for examining the localization characteristics and mobility edges of non-reciprocal quasicrystals, referred to as asymmetric transfer matrix analysis. We demonstrate the application of this method to three specific scenarios: the non-reciprocal Aubry-André model, the non-reciprocal off-diagonal Aubry-André model, and the non-reciprocal mosaic quasicrystals. This work may contribute valuable insights to the investigation of non-Hermitian quasicrystal and disordered systems.

The Fibonacci quasicrystal: case study of hidden dimensions and multifractality

Anuradha Jagannathan. arXiv preprint (2020). Cited by 0. Found by: arxiv. DOI: 10.1103/RevModPhys.93.045001

Abstract: The distinctive electronic properties of quasicrystals stem from their long range structural order, with invariance under rotations and under discrete scale change, but without translational invariance. d-dimensional quasicrystals can be described in terms of lattices of higher dimension $D>d$, and many of their properties can be simply derived from analyses that take into account the extra "hidden" dimensions. In particular, as recent theoretical and experimental studies have shown, quasicrystals can have topological properties inherited from the parent crystals. These properties are discussed here for the simplest of quasicrystals, the 1D Fibonacci chain. The Fibonacci noninteracting tight-binding Hamiltonians are characterized by multifractality of spectrum and states, which is manifested in many of its physical properties, notably in transport. Perturbations due to disorder and re-entrance phenomena are described, along with the crossover to strong Anderson localization. Perturbations due to boundary conditions also give information on the spatial and topological electronic properties, as is shown for the superconducting proximity effect. Related models including phonon and mixed Fibonacci models are discussed, as well as generalizations to other quasiperiodic chains and higher dimensional extensions. Interacting quasiperiodic systems and the case for many body localization are briefly discussed. Some experimental realizations of the 1D quasicrystal and their potential applications are described.

Universal self-assembly of one-component three-dimensional dodecagonal quasicrystals

Roman Ryltsev; Nikolay Chtchelkatchev. arXiv preprint (2017). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/1705.01517v1

Abstract: Using molecular dynamics simulations, we study computational self-assembly of one-component three-dimensional dodecagonal (12-fold) quasicrystals in systems with two-length-scale potentials. Existing criteria for three-dimensional quasicrystal formation are quite complicated and rather inconvenient for particle simulations. So to localize numerically the quasicrystal phase, one should usually simulate over a wide range of system parameters. We show how to universally localize the parameters values at which dodecagonal quasicrystal order may appear for a given particle system. For that purpose, we use a criterion recently proposed for predicting decagonal quasicrystal formation in one-component two-length-scale systems. The criterion is based on two dimensionless effective parameters describing the fluid structure which are extracted from radial distribution function. The proposed method allows reducing the time spent for searching the parameters favoring certain solid structure for a given system. We show that the method works well for dodecagonal quasicrystals; this results is verified on four systems with different potentials: Dzugutov potential, oscillating potential which mimics metal interactions, repulsive shoulder potential describing effective interaction for core/shell model of colloids and embedded-atom model potential for aluminum. Our results suggest that mechanism of dodecagonal

quasicrystal formation is universal for both metallic and soft-matter systems and it is based on competition between interparticle scales.

Growth modes of quasicrystals

C. V. Achim; M. Schmiedeberg; H. Löwen. Phys. Rev. Lett. 112, 255501 (2014) (2014). Cited by 0. Found by: arxiv. DOI: 10.1103/PhysRevLett.112.255501

Abstract: The growth of quasicrystals, i.e., aperiodic structures with long-range order, seeded from the melt is investigated using a dynamical phase field crystal model. Depending on the thermodynamic conditions, two different growth modes are detected, namely defect-free growth of the stable quasicrystal and a mode dominated by phasonic flips which are incorporated as local defects into the grown structure such that random tiling-like ordering emerges. The latter growth mode is unique to quasicrystals and can be verified in experiments on one-component mesoscopic systems.

Non-Hermitian quasicrystal in dimerized lattices

Longwen Zhou; Wenqian Han. arXiv preprint (2021). Cited by 0. Found by: arxiv. DOI: 10.1088/1674-1056/ac1efc

Abstract: Non-Hermitian quasicrystals possess PT and metal-insulator transitions induced by gain and loss or nonreciprocal effects. In this work, we uncover the nature of localization transitions in a generalized Aubry-Andre-Harper model with dimerized hopping amplitudes and complex onsite potential. By investigating the spectrum, adjacent gap ratios and inverse participation ratios, we find an extended phase, a localized phase and a mobility edge phase, which are originated from the interplay between hopping dimerizations and non-Hermitian onsite potential. The lower and upper bounds of the mobility edge are further characterized by a pair of topological winding numbers, which undergo quantized jumps at the boundaries between different phases. Our discoveries thus unveil the richness of topological and transport phenomena in dimerized non-Hermitian quasicrystals.

On the characterisation of a hitherto unreported icosahedral quasicrystal phase in additively manufactured aluminium alloy AA7075

Shravan K. Kairy; Oumaima Gharbi; Juan Nicklaus; Derui Jiang; Christopher R. Hutchinson; Nick Birbilis. arXiv preprint (2018). Cited by 0. Found by: arxiv. DOI: 10.1007/s11661-018-5025-1

Abstract: Aluminium alloy AA7075 (Al-Zn-Mg-Cu) specimens were prepared using selective laser melting, also known as powder bed fusion additive manufacturing. In the as-manufactured state, which represents a locally rapidly solidified condition, the prevalence of a previously unreported icosahedral quasicrystal with 5-fold symmetry was observed. The icosahedral quasicrystal, which has been termed nu-phase, was comprised of Zn, Cu and Mg.

Conventional superconductivity in quasicrystals

Ronaldo N. Araújo; Eric C. Andrade. Phys. Rev. B 100, 014510 (2019) (2019). Cited by 0. Found by: arxiv. DOI: 10.1103/PhysRevB.100.014510

Abstract: Motivated by a recent experimental observation of superconductivity in the Al-Zn-Mg quasicrystal, we study the low-temperature behavior of electrons moving in the quasiperiodic potential of the Ammann-Beenker tiling in the presence of a local attraction. We employ the Bogoliubov-de Gennes approach for approximants of different sizes and determine the local pairing amplitude Δ_{i} as well its spatial average, Δ_{0}, the superconducting order parameter. Due to the lack of periodicity of the octagonal tiling, the resulting superconducting state is inhomogeneous, but we find no evidence of the superconductivity islands, as observed in disordered systems, with $\Delta_{i}\rightarrow0$ at T_{c} for all sites. In the weak-coupling regime, we find that the superconducting order parameter depends appreciably on the approximant size only if the Fermi energy sits at a pseudogap in the noninteracting density of states, with Δ_{0} decreasing as the system size increases. These results are in line with the experimental observations for the Al-Zn-Mg quasicrystal, and they suggest that, despite their electronic structure, quasicrystals are prone to display conventional BCS-like superconductivity.

Reentrant delocalization transition in one-dimensional photonic quasicrystals

Sachin Vaidya; Christina Jörg; Kyle Linn; Megan Goh; Mikael C. Rechtsman. arXiv preprint (2022). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/2211.06047v1

Abstract: Waves propagating in certain one-dimensional quasiperiodic lattices are known to exhibit a sharp localization transition. We theoretically predict and experimentally observe that the localization of light in one-dimensional photonic quasicrystals may be followed by a second delocalization transition for some states on increasing quasiperiodic modulation strength - an example of a reentrant transition. We further propose that this phenomenon can be qualitatively captured by a dimerized tight-binding model with long-range couplings.

A Database of 2,500 Quasicrystal Cells

Tony Robbin; George Francis; Kurt Baumann. arXiv preprint (2018). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/1805.11457v2

Abstract: Here is a database of quasicrystal cells computed by the deBruijn Grand Dual Method. The database is in a form that can be converted and read by a variety of geometry programs. Proof of the accuracy of the computations is given by the consistency of the two values of the volumes of the cells. How the deBruijn algorithm works, and the possible use of the algorithm for modeling non-local phenomena is also discussed.

Non-Hermitian butterfly spectra in a family of quasiperiodic lattices

Li Wang; Zhenbo Wang; Shu Chen. Phys. Rev. B 110, L060201 (2024) (2024). Cited by 0. Found by: arxiv. DOI: 10.1103/PhysRevB.110.L060201

Abstract: We propose a family of exactly solvable quasiperiodic lattice models with analytical complex mobility edges, which can incorporate mosaic modulations as a straightforward generalization. By sweeping a potential tuning parameter $ $, we demonstrate a kind of interesting butterfly-like spectra in complex energy plane, which depicts energy-dependent extended-localized transitions sharing a common exact non-Hermitian mobility edge. Applying Avila's global theory, we are able to analytically calculate the Lyapunov exponents and determine the mobility edges exactly. For the minimal model without mosaic modulation, a compactly analytic formula for the complex mobility edges is obtained, which, together with analytical estimation of the range of complex energy spectrum, gives the true mobility edge. The non-Hermitian mobility edge in complex energy plane is further verified by numerical calculations of fractal dimension and spatial distribution of wave functions. Tuning parameters of non-Hermitian potentials, we also investigate the variations of the non-Hermitian mobility edges and the corresponding butterfly spectra, which exhibit richness of spectrum structures.

Noncoplanar ferrimagnetism and local crystalline-electric-field anisotropy in the quasicrystal approximant Au$_{70}$Si$_{17}$Tb$_{13}$

Takanobu Hiroto; Taku J Sato; Huibo Cao; Takafumi Hawai; Tetsuya Yokoo; Shinichi Itoh; Ryuji Tamura. arXiv preprint (2019). Cited by 0. Found by: arxiv. DOI: 10.1088/1361-648X/ab997d

Abstract: Neutron scattering experiments have been performed to elucidate magnetic properties of the quasicrystal approximant Au$_{70}$Si$_{17}$Tb$_{13}$, consisting of icosahedral spin clusters in a body-centered-cubic lattice. Bulk magnetic measurements performed on the single crystalline sample unambiguously confirm long-range ordering at $T_{\rm C} = 11.6 \pm 1$ K. In contrast to the simple ferromagnetic response in the bulk measurements, single crystal neutron diffraction confirms a formation of intriguing non-collinear and non-coplanar magnetic order. The magnetic moment direction was found to be nearly tangential to the icosahedral cluster surface in the local mirror plane, which is quite similar to that recently found in the antiferromagnetic quasicrystal approximant Au$_{72}$Al$_{14}$Tb$_{14}$. Inelastic neutron scattering on the powdered sample exhibits a very broad peak centered at $\hbar \simeq 4$ meV. The observed inelastic spectrum was explained by the crystalline-electric-field model taking account of the chemical disorder at the fractional Au/Si sites. The resulting averaged anisotropy axis for the crystalline-electric-field ground state is consistent with the ordered moment direction determined in the magnetic structure analysis, confirming that the non-coplanar magnetic order is stabilized by the local uniaxial anisotropy.

Topological Hofstadter Insulators in a Two-Dimensional Quasicrystal

Duc-Thanh Tran; Alexandre Dauphin; Nathan Goldman; Pierre Gaspard. Phys. Rev. B 91, 085125 (2015) (2014). Cited by 0. Found by: arxiv. DOI: 10.1103/PhysRevB.91.085125

Abstract: We investigate the properties of a two-dimensional quasicrystal in the presence of a uniform magnetic field. In this configuration, the density of states (DOS) displays a Hofstadter butterfly-like structure when it is represented as a function of the magnetic flux per tile. We show that the low-DOS regions of the energy spectrum are associated with chiral edge states, in direct analogy with the Chern insulators realized with periodic lattices. We establish the topological nature of the edge states by computing the topological Chern number associated with the bulk of the quasicrystal. This topological characterization of the non-periodic lattice is achieved through a local (real-space) topological marker. This work opens a route for the exploration of topological insulating materials in a wide range of non-periodic lattice systems, including photonic crystals and cold atoms in optical lattices.

Interface states in two-dimensional quasicrystals with broken inversion symmetry

Danilo Beli; Matheus I. N. Rosa; Luca Lomazzi; Carlos De Marqui; Massimo Ruzzene. arXiv preprint (2023). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/2304.09628v1

Abstract: We investigate the existence of interface states induced by broken inversion symmetries in two-dimensional quasicrystal lattices. We introduce a 10-fold rotationally symmetric quasicrystal lattice whose inversion symmetry is broken through a mass dimerization that produces two 5-fold symmetric sub-lattices. By considering resonator scatterers attached to an elastic plate, we illustrate the emergence of bands of interface states that accompany a band inversion of the quasicrystal spectrum as a function of the dimerization parameter. These bands are filled by modes which are localized along domain-wall interfaces separating regions of opposite inversion symmetry. These features draw parallels to the dynamic behavior of topological interface states in the context of the valley Hall effect, which has been so far limited to periodic lattices. We numerically and experimentally demonstrate wave-guiding in a quasicrystal lattice featuring a zig-zag interface with sharp turns of 36 degrees, which goes beyond the limitation of 60 degrees associated with 6-fold symmetric (i.e., honeycomb) periodic lattices. Our results provide new opportunities for symmetry-based quasicrystalline topological waveguides that do not require time-reversal symmetry breaking, and that allow for higher freedom in the design of their waveguiding trajectories by leveraging higher-order rotational symmetries.

Reversible phasonic control of a quantum phase transition in a quasicrystal

Toshihiko Shimasaki; Yifei Bai; H. Esat Kondakci; Peter Dotti; Jared E. Pagett; Anna R. Dardia; Max Prichard; André Eckardt; David M. Weld. arXiv preprint (2023). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/2312.00976v1

Abstract: Periodic driving can tune the quasistatic properties of quantum matter. A well-known example is the dynamical modification of tunneling by an oscillating electric field. Here we show experimentally that driving the phasonic degree of freedom of a cold-atom quasicrystal can continuously tune the effective quasi-disorder strength, reversibly toggling a localization-delocalization quantum phase transition. Measurements agree with fit-parameter-free theoretical predictions, and illuminate a fundamental connection between Aubry-André localization in one dimension and dynamic localization in the associated two-dimensional Harper-Hofstadter model. These results open up new experimental possibilities for dynamical coherent control of quantum phase transitions.

Interaction tuned pattern-selective superconductivity: Application to the dodecagonal quasicrystal

Junmo Jeon; SungBin Lee. arXiv preprint (2025). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/2503.14593v2

Abstract: Quasicrystals exhibit superconductivity under the unique interplay of long-range order and strong inhomogeneity, distinguishing them from both crystalline and amorphous systems. Understanding how this structural complexity affects superconducting states and phase transitions remains an important open question. Here, we unveil anomalous superconductivity in a dodecagonal quasicrystal using the attractive Hubbard model within the Bogoliubov-de Gennes framework. We show that both the gap structure and critical temperature depend on the local pattern due to inhomogeneous charge distribution and kinetic terms. This leads to unconventional phase transitions between mixed phases where superconducting and normal metal regions coexist, even without external fields, which we term pattern-selective superconductivity. Furthermore, superconductivity can be anomalously suppressed even under stronger attractive interactions due to the alignment of the Fermi level with a spectral gap in the fragmented Hartree-shifted spectrum. These findings, not observed in conventional crystals, amorphous systems, or previous studies of quasicrystal superconductivity, highlight the distinct role of quasicrystalline order.

The Aubry-Andre Anderson model: Magnetic impurities coupled to a fractal spectrum

Ang-Kun Wu; Daniel Bauernfeind; Xiaodong Cao; Sarang Gopalakrishnan; Kevin Ingersent; J. H. Pixley. arXiv preprint (2022). Cited by 0. Found by: arxiv. DOI: 10.1103/PhysRevB.106.165123

Abstract: The Anderson model for a magnetic impurity in a one-dimensional quasicrystal is studied using the numerical renormalization group (NRG). The main focus is elucidating the physics at the critical point of the Aubry-Andre (AA) Hamiltonian, which exhibits a fractal spectrum with multifractal wave functions, leading to an AA Anderson (AAA) impurity model with an energy-dependent hybridization function defined through the multifractal local density of states at the impurity site. We first study a class of Anderson impurity models with uniform fractal hybridization functions that the NRG can solve to arbitrarily low temperatures. Below a Kondo scale T_K, these models approach a fractal strong-coupling fixed point where impurity thermodynamic properties oscillate with $\log_b T$ about negative average values determined by the fractal dimension of the spectrum. The fractal dimension also enters into a power-law dependence of T_K on the Kondo exchange coupling J_K. To treat the AAA model, we combine the NRG with the kernel polynomial method (KPM) to form an efficient approach that can treat hosts without translational symmetry down to a temperature scale set by the KPM expansion order. The aforementioned fractal strong-coupling fixed point is reached by the critical AAA model in a simplified treatment that neglects the wave-function contribution to the hybridization. The temperature-averaged properties are those expected for the numerically determined fractal dimension of 0.5. At the AA critical point, impurity thermodynamic properties become negative and oscillatory. Under sample-averaging, the mean and median Kondo temperatures exhibit power-law dependences on J_K with exponents characteristic of different fractal dimensions. We attribute these signatures to the impurity probing a distribution of fractal strong-coupling fixed points with decreasing temperature.

Energy levels and their correlations in quasicrystals

A. Jagannathan; F. Piechon. arXiv preprint (2006). Cited by 0. Found by: arxiv. DOI: 10.1080/14786430701196990

Abstract: Quasicrystals can be considered, from the point of view of their electronic properties, as being intermediate between metals and insulators. For example, experiments show that quasicrystalline alloys such as AlCuFe or AlPdMn have conductivities far smaller than those of the metals that these alloys are composed from. Wave functions in a quasicrystal are typically intermediate in character between the extended states of a crystal and the exponentially localized states in the insulating phase, and this is also reflected in the energy spectrum and the density of states. In the theoretical studies we consider in this review, the quasicrystals are described by a pure hopping tight binding model on simple tilings. We focus on spectral properties, which we compare with those of other complex systems, in particular, the Anderson model of a disordered metal.

Correlation-induced phase transitions and mobility edges in an interacting non-Hermitian quasicrystal

Tian Qian; Yongjian Gu; Longwen Zhou. Phys. Rev. B 109, 054204 (2024) (2023). Cited by 0. Found by: arxiv. DOI: 10.1103/PhysRevB.109.054204

Abstract: Non-Hermitian quasicrystal constitutes a unique class of disordered open system with PT-symmetry breaking, localization and topological triple phase transitions. In this work, we uncover the effect of quantum correlation on phase transitions and entanglement dynamics in non-Hermitian quasicrystals. Focusing on two interacting bosons in a Bose-Hubbard lattice with quasiperiodically modulated gain and loss, we find that the onsite interaction between bosons could drag the PT and localization transition thresholds towards weaker disorder regions compared with the noninteracting case. Moreover, the interaction facilitates the expansion of the critical point of a triple phase transition in the noninteracting system into a critical phase with mobility edges, whose domain could be flexibly controlled by tuning the interaction strength. Systematic analyses of the spectrum, inverse participation ratio, topological winding number, wavepacket dynamics and entanglement entropy lead to consistent predictions about the correlation-driven phases and transitions in our system. Our findings pave the way for further studies of the interplay between disorder and interaction in non-Hermitian quantum matter.

Wigner distribution, Wigner entropy and Quantum Refrigerator of a One-Dimensional Off-diagonal Quasicrystal

Shan Suo; Ao Zhou; Yanting Chen; Shujie Cheng; Gao Xianlong. arXiv preprint (2026). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/2601.17691v1

Abstract: We investigate an off-diagonal quasicrystal featuring simultaneous off-diagonal and diagonal quasiperiodic modulations. By analyzing the fractal dimension, we map out the delocalization-localization phase diagram. We demonstrate that delocalized and localized states can be distinguished via the Wigner distribution, while extended, critical, and localized phases are separated using the Wigner entropy. Furthermore, we explore the quantum thermodynamic properties, revealing that localized states facilitate the emergence of a quantum heater mode, alongside the appearance of a refrigerator mode. These findings enhance our understanding of localization phenomena and expand the thermodynamic applications of quasiperiodic systems.

Localization transition, spectrum structure and winding numbers for one-dimensional non-Hermitian quasicrystals

Yanxia Liu; Qi Zhou; Shu Chen. Phys. Rev. B 104, 024201 (2021) (2020). Cited by 0. Found by: arxiv. DOI: 10.1103/PhysRevB.104.024201

Abstract: By analyzing the Lyapunov exponent (LE), we develop a rigorous, fundamental scheme for the study of general non-Hermitian quasicrystals with both complex phase factor and non-reciprocal hopping. Specially, the localization-delocalization transition point, $\mathcal{PT}$-symmetry-breaking point and the winding number transition points are determined by LEs of its dual Hermitian model. The analysis was based on Avila's global theory, and we found that winding number is directly related to the acceleration, the slope of the LE, while quantization of acceleration is the crucial ingredient of Avila's global theory. This result applies as well to the models with higher winding, not only the simplest Aubry-André model. As typical examples, we obtain the analytical phase boundaries of localization transition for non-Hermitian Aubry-André model in the whole parameter space, and the complete phase diagram is straightforwardly determined. For the non-Hermitian Soukoulis-Economou model, a high winding model, we show how the phase boundaries of localization transition and winding number transitions relate to the LEs of its dual Hermitian model. Moreover, we discover an intriguing feature of robust spectrum, i.e., the spectrum keeps invariant when one changes the complex phase parameter h or non-reciprocal parameter g in the region of $h<|h_c|$ or $g<|g_c|$ if the system is in the extended or localized state, respectively.

Bloch-like Electronic Wave Functions in Two-Dimensional Quasicrystals

Shahar Even-Dar Mandel; Ron Lifshitz. arXiv preprint (2008). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/0808.3659v1

Abstract: Electrons in quasicrystals generically possess critical wave functions that are neither exponentially-localized nor extended, but rather decay algebraically in space. Nevertheless, motivated by recent calculations on the square and cubic Fibonacci quasicrystals, we investigate whether it is possible to obtain extended wave functions expressed as linear combinations of degenerate, or nearly-degenerate, critical eigenfunctions. We find that not only is this possible, but that the wave functions that emerge are Bloch-like, exhibiting the quasiperiodic long-range order of the underlying quasicrystal. We discuss the significance of this result for the study of electronic properties of real quasicrystals.

Quasicrystals in pattern formation, Part I: Local existence and basic properties

Ian Melbourne; Jens Rademacher; Bob Rink; Sergey Zelik. arXiv preprint (2024). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/2410.19967v2

Abstract: In this paper, we propose a general mechanism for the existence of quasicrystals in spatially extended systems (partial differential equations with Euclidean symmetry). We argue that the existence of quasicrystals with higher order rotational symmetry, icosahedral symmetry, etc, is a natural and universal consequence of spontaneous symmetry breaking, bypassing technical issues such as Diophantine properties and hard implicit function theorems. The diffraction diagrams associated with these quasicrystal solutions are not Delone sets, so strictly speaking they do not conform to the definition of a "mathematical quasicrystal". But they do appear to capture very well the features of the diffraction diagrams of quasicrystals observed in nature. For the Swift-Hohenberg equation, we obtain more detailed information, including that the ℓ^2 norm of the diffraction diagram grows like the square root of the bifurcation parameter.

Transition pathways connecting crystals and quasicrystals

Jianyuan Yin; Kai Jiang; An-Chang Shi; Pingwen Zhang; Lei Zhang. arXiv preprint (2020). Cited by 0. Found by: arxiv. DOI: 10.1073/pnas.2106230118

Abstract: Due to structural incommensurability, the emergence of a quasicrystal from a crystalline phase represents a challenge to computational physics. Here the nucleation of quasicrystals is investigated by using an efficient computational method applied to a Landau free-energy functional. Specifically, transition pathways connecting different local minima of the Lifshitz-Petrich model are obtained by using the high-index saddle dynamics. Saddle points on these paths are identified as the critical nuclei of the 6-fold crystals and 12-fold quasicrystals. The results reveal that phase transitions between the crystalline and quasicrystalline phases could follow two possible pathways, corresponding to a one-stage phase transition and a two-stage phase transition involving a metastable lamellar quasicrystalline state, respectively.

Computation and visualization of photonic quasicrystal spectra via Blochs theorem

Alejandro W. Rodriguez; Alexander P. McCauley; Yehuda Avniel; Steven G. Johnson. Physical Review B, 77 104201, 2008 (2007). Cited by 0. Found by: arxiv. DOI: 10.1103/PhysRevB.77.104201

Abstract: Previous methods for determining photonic quasicrystal (PQC) spectra have relied on the use of large supercells to compute the eigenfrequencies and/or local density of states (LDOS). In this manuscript, we present a method by which the energy spectrum and the eigenstates of a PQC can be obtained by solving Maxwells equations in higher dimensions for any PQC defined by the standard cut-and-project construction, to which a generalization of Blochs theorem applies. In addition, we demonstrate how one can compute band structures with defect states in the higher-dimensional superspace with no additional computational cost. As a proof of concept, these general ideas are demonstrated for the simple case of one-dimensional quasicrystals, which can also be solved by simple transfer-matrix techniques.

Atomistic mechanisms of dynamics in a two-dimensional dodecagonal quasicrystal

Kun Zhao; Matteo Baggioli; Wen-Sheng Xu; Jack F. Douglas; Yun-Jiang Wang. J. Chem. Phys. 162, 234503 (2025) (2024). Cited by 0. Found by: arxiv. DOI: 10.1063/5.0270291

Abstract: Quasicrystals have been observed in a variety of materials ranging from metal alloys to block copolymers. However, their structural and dynamical properties cannot be readily described in terms of conventional solid-state models of liquids and solids. We may expect the dynamics of this specific class of quasicrystalline materials to be more like glass-forming liquids in the sense of exhibiting large fluctuations in the local mobility ("dynamic heterogeneity") and non-Arrhenius temperature dependence of relaxation and diffusion. In this work, we investigate a model dodecagonal quasicrystal material in two dimensions (2D) using molecular dynamics (MD) simulations, with a focus on heterogeneous dynamics and non-Arrhenius relaxation and diffusion. As observed in glass-forming liquids and heated crystals, we observe a two-stage relaxation dynamics in the self-intermediate scattering function $F_s(k,t)$ of our quasicrystal material. It involves a fast $ $-relaxation and $ $ relaxation process having a highly temperature dependent relaxation time whose activation energy varies in concert with the extent of string-like collective motion, a phenomenon recognized to occur in glass-forming liquids at low temperatures and crystalline materials at elevated temperatures. After examining the dynamics of our dodecagonal quasicrystalline material in great detail, we conclude that the dynamics of these materials more closely resembles observations on metallic glass-forming liquids than crystalline materials.

Magnons in a Quasicrystal: Propagation, Localization and Extinction of Spin Waves in Fibonacci Structures

Filip Lisiecki; Justyna Rychły; Piotr Kuświk; Hubert Głowiński; Jarosław W. Kłos; Felix Groß; Iuliia Bykova; Markus Weigand; Mateusz Zelent; Eberhard Goering; Gisela Schütz; Maciej Krawczyk; Feliks Stobiecki; Janusz Dubowik; Joachim Gräfe. Phys. Rev. Applied 11, 054061 (2019) (2018). Cited by 0. Found by: arxiv. DOI: 10.1103/PhysRevApplied.11.054061

Abstract: Magnonic quasicrystals exceed the possibilities of spin wave (SW) manipulation offered by regular magnonic crystals, because of their more complex SW spectra with fractal characteristics. Here, we report the direct x-ray microscopic observation of propagating SWs in a magnonic quasicrystal, consisting of dipolarly coupled permalloy nanowires arranged in a one-dimensional Fibonacci sequence. SWs from the first and second band as well as evanescent waves from the band gap between them are imaged. Moreover, additional miniband gaps in the spectrum are demonstrated, directly indicating an influence of the quasiperiodicity of the system. The experimental results are interpreted using numerical calculations and we deduce a simple model to estimate the frequency position of the magnonic gaps in quasiperiodic structures. The demonstrated features of SW spectra in one-dimensional magnonic quasicrystals allows utilizing this class of metamaterials for magnonics and makes them an ideal basis for future applications.

Al-Cu-Fe alloys: the relationship between the quasicrystal and its melt

L. V. Kamaeva; R. E. Ryltsev; V. I. Lad'yanov; N. M. Chtchelkatchev. arXiv preprint (2019). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/1908.03931v1

Abstract: Understanding the mechanisms which relate properties of liquid and solid phases is crucial for fabricating new advanced solid materials, such as glasses, quasicrystals and high-entropy alloys. Here we address this issue for quasicrystal-forming Al-Cu-Fe alloys which can serve as a model for studying microscopic mechanisms of quasicrystal formation. We study experimentally two structural-sensitive properties of the liquid – viscosity and undercoolability – and compare results with \textit{ab initio} investigations of short-range order (SRO). We observe that SRO in Al-Cu-Fe melts is polytetrahedral and mainly presented by distorted Kasper polyhedra. However, topologically perfect icosahedra are almost absent an even stoichiometry of icosahedral quasicrystal phase that suggests the topological structure of local polyhedra does not survive upon melting. It is shown that the main features of interatomic interaction in Al-Cu-Fe system, extracted from radial distribution function and bong-angle distribution function, are the same for both liquid and solid states. In particular, the system demonstrates pronounced repulsion between Fe and Cu as well as strong chemical interaction between Fe and Al, which are almost concentration-independent. We argue that SRO and structural-sensitive properties of a melt may serve as useful indicators of solid phase formation. In particular, in the concentration region corresponding to the composition of the icosahedral phase, a change in the chemical short-range order is observed, which leads to minima on the viscosity and udercoolability isotherms and has a noticeable effect on the initial stage of solidification.

Existence of quasicrystals and universal stable sampling and interpolation in LCA groups

E. Agora; J. Antezana; C. Cabrelli; B. Matei. arXiv preprint (2016). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/1611.05804v2

Abstract: We characterize all the locally compact abelian (LCA) groups that contain quasicrystals (a class of model sets). Moreover, we describe all possible quasicrystals in the group constructing an appropriate lattice associated with the cut and project scheme that produces it. On the other hand, if an LCA group G admits a simple quasicrystal, we prove that recent results of Meyer and Matei for the case of the n-dimensional Euclidean space can be extended to G. More precisely, we prove that simple quasicrystals are universal sets of stable sampling and universal sets of stable interpolation in generalized Paley-Wiener spaces.

How Quasicrystals Remember: Hierarchical Memory Under Cyclic Shear

Edwin A. Bedolla-Montiel; Marjolein Dijkstra. arXiv preprint (2026). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/2607.12674v1

Abstract: Quasicrystals occupy a unique middle ground between periodically ordered crystals and disordered glasses, making them an ideal platform for examining the interplay between disorder and the emergence of mechanical memory. Using athermal quasistatic shear simulations, we show that two-dimensional dodecagonal quasicrystals encode and recover memory under cyclic driving. Above the yielding transition, the response becomes irreversible, characterized by persistent shear bands and locally transformed regions. Below yielding, cyclic shear with varying amplitudes produces a hierarchy of nested hysteresis loops in the stress-strain response characteristic of loop-return point memory. By resolving the underlying reversible plastic events, we reveal localized phason-like tile rearrangements as the elementary switching units and identify tile-switch hysterons responsible for memory in the quasicrystal. Such a microscopic identification of the fundamental switching units is considerably more challenging, and often impossible, in amorphous solids. Despite their structural diversity, these rearrangements share a compact core, sharp bistability, and an Eshelby-compatible elastic far field. In contrast, a periodic approximant of the quasicrystal lacks both the structural disorder and the bistable tile-switch rearrangements required for cyclic-shear memory, linking phason degrees of freedom to bistable hysterons.

Electronic properties of ternary quasicrystals in one dimension

Nobuhisa Fujita; Komajiro Niizeki. Phys. Rev. B 64 (2001) 144207. (2001). Cited by 0. Found by: arxiv. DOI: 10.1103/PhysRevB.64.144207

Abstract: The one-electron properties of a certain class of one-dimensional ternary quasicrystals are investigated. In particular, we show in detail the presence of a special kind of critical states called marginal critical states in these QCs. By the use of a real-space renormalization-group method, it is shown that the scaling properties of marginal critical states are characterized by stretched exponentials. These states are virtually localized, so that

their presence may make a QC less conductive.

One-Dimensional Quasicrystals from Incommensurate Charge Order

Felix Flicker; Jasper van Wezel. Phys. Rev. Lett. 115, 236401 (2015) (2015). Cited by 0. Found by: arxiv. DOI: 10.1103/PhysRevLett.115.236401

Abstract: Artificial quasicrystals are nowadays routinely manufactured, yet only two naturally occurring examples are known. We present a class of systems with the potential to be realized both artificially and in nature, in which the lowest energy state is a one-dimensional quasicrystal. These systems are based on incommensurately charge-ordered materials, in which the quasicrystalline phase competes with the formation of a regular array of discommensurations as a way of interpolating between incommensurate charge order at high temperatures and commensurate order at low temperatures. The nonlocal correlations characteristic of the quasicrystalline state emerge from a free-energy contribution localized in reciprocal space. We present a theoretical phase diagram showing that the required material properties for the appearance of such a ground state allow for one-dimensional quasicrystals to form in real materials. The result is a potentially wide class of one-dimensional quasicrystals.

Magnetic ground state of a prototype quasicrystal approximant: a candidate for octahedral spin ice physics

Leonie Woodland; Farid Labib; Dmitry Khalyavin; Pascal Manuel; Fabio Orlandi; Hubertus Luetkens; Ryuji Tamura. arXiv preprint (2026). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/2605.26044v1

Abstract: Magnetic ordering in quasicrystals has recently emerged as a fertile ground for discovering unconventional magnetic states beyond the framework of periodic crystals. However, elucidating the microscopic origin of such states remains challenging due to the intrinsic aperiodicity of quasicrystals. Here, we address this issue by investigating the prototypical Tsai-type quasicrystal approximant Cd6Tb, which preserves the essential local geometry and connectivity of icosahedral quasicrystals while allowing detailed structural and magnetic characterization due to its translational periodicity. Using neutron diffraction measurements, we find a noncoplanar multi-k magnetic ground state composed of Ising-like Tb moments arranged on a network of corner-sharing octahedra, the ingredients required to host octahedral spin-ice physics. Remarkably, only one third of the Tb moments develop long-range magnetic order, whereas the remaining moments display strongly reduced static order accompanied by persistent spin dynamics on microsecond timescales, as evidenced by muon spin rotation. This coexistence of ordered and fluctuating moments constitutes a potential realization of magnetic fragmentation - a key prediction of octahedral spin-ice physics - in a quasicrystal-related material.

Valence Change Driven by Constituent Element Substitution in the Mixed-Valence Quasicrystal and Approximant Au-Al-Yb

Shuya Matsukawa; Katsumasa Tanaka; Mika Nakayama; Kazuhiko Deguchi; Keiichiro Imura; Hiroyuki Takakura; Shiro Kashimoto; Tsutomu Ishimasa; Noriaki K. Sato. J. Phys. Soc. Jpn. 83 (2014) 034705 (5 Pages) (2014). Cited by 0. Found by: arxiv. DOI: 10.7566/JPSJ.83.034705

Abstract: Quantum criticality has been considered to be specific to crystalline materials such as heavy fermions. Very recently, however, the Tsai-type quasicrystal Au51Al34Yb15 has been reported to show unusual quantum critical behavior. To obtain a deeper understanding of this new material, we have searched for other Tsai-type cluster materials. Here, we report that the metal alloys Au44Ga41Yb15 and Ag47Ga38Yb15 are members of the 1/1 approximant to the Tsai-type quasicrystal and that both possess no localized magnetic moment. We suggest that the Au-Al-Yb system is located near the border of the divalent and trivalent states of the Yb ion; we also discuss a possible origin of the disappearance of magnetism, associated with the valence change, by the substitution of the constituent elements.

Quantum Entanglement of Non-Hermitian Quasicrystals

Li-Mei Chen; Yao Zhou; Shuai A. Chen; Peng Ye. Phys. Rev. B 105, L121115 (2022) (2021). Cited by 0. Found by: arxiv. DOI: 10.1103/PhysRevB.105.L121115

Abstract: As a hallmark of pure quantum effect, quantum entanglement has provided unconventional routes to condensed matter systems. Here, from the perspective of quantum entanglement, we disclose exotic quantum physics in non-Hermitian quasicrystals. We present a class of experimentally realizable models for non-Hermitian quasicrystal chains, in which asymmetric hopping and complex potential coexist. We diagnose global phase diagram by means of entanglement from both real-space and momentum-space partition. By measuring

entanglement entropy, we numerically determine the metal-insulator transition point. We combine real-space and momentum-space entanglement spectra to complementarily characterize the delocalization phase and the localization phase. Inspired by entanglement spectrum, we further analytically prove that a duality exists between the two phase regions. The transition point is self-dual and exact, further validating the numerical result from diagonalizing non-Hermitian matrices. Finally, we identify mobility edge by means of entanglement.

Nature of Protected Zero Energy States in Penrose Quasicrystals

Ezra Day-Roberts; Rafael M. Fernandes; Alex Kamenev. Phys. Rev. B 102, 064210 (2020) (2020). Cited by 0. Found by: arxiv. DOI: 10.1103/PhysRevB.102.064210

Abstract: The electronic spectrum of the Penrose rhombus quasicrystal exhibits a macroscopic fraction of exactly degenerate zero energy states. In contrast to other bipartite quasicrystals, such as the kite-and-dart one, these zero energy states cannot be attributed to a global mismatch $Δn$ between the number of sites in the two sublattices that form the quasicrystal. Here, we argue that these zero energy states are instead related to a local mismatch $Δn(\bf r)$. Although $Δn(\bf r)$ averages to zero, its staggered average over self-organized domains gives the correct number of zero energy states. Physically, the local mismatch is related to a hidden structure of nested self-similar domains that support the zero energy states. This allows us to develop a real space renormalization-group scheme, which yields the scaling law for the fraction of zero energy states, Z, versus size of their support domain, N, as $Z\propto N^{- }$ with $ =1-\ln 2/\ln(1+) \approx 0.2798$ (where $ $ is the golden ratio). It also reproduces the known total fraction of the zero energy states, $81-50 \approx 0.0983$. We also show that the exact degeneracy of these states is protected against a wide variety of local perturbations, such as irregular or random hopping amplitudes, magnetic field, random dilution of the lattice, etc. We attribute this robustness to the hidden domain structure and speculate about its underlying topological origin.

Emergence of multiple localization transitions in a one-dimensional quasiperiodic lattice

Ashirbad Padhan; Mrinal Kanti Giri; Suman Mondal; Tapan Mishra. Phys. Rev. B 105, L220201 (2022) (2021). Cited by 0. Found by: arxiv. DOI: 10.1103/PhysRevB.105.L220201

Abstract: Low dimensional quasiperiodic systems exhibit localization transitions by turning all quantum states localized after a critical quasidisorder. While certain systems with modified or constrained quasiperiodic potential undergo multiple localization transitions in one dimension, we predict an emergence of multiple localization transitions without directly imposing any constraints on the quasiperiodic potential. By considering a one-dimensional system described by the Aubry-Andŕe (AA) model, we show that an additional staggered onsite potential can drive the system through a series of localization transitions as a function of the staggered potential. Interestingly, we find that the number of localization transitions strongly depends on the strength of the quasiperiodic potential. Moreover, we obtain the signatures of these localization transitions in the expansion dynamics and propose an experimental scheme for their detection in the quantum gas experiment.

Origin of Quantum Criticality in Yb-Al-Au Approximant Crystal and Quasicrystal

Shinji Watanabe; Kazumasa Miyake. J. Phys. Soc. Jpn. 85 (2016) 063703 (2016). Cited by 0. Found by: arxiv. DOI: 10.7566/JPSJ.85.063703

Abstract: To get insight into the mechanism of emergence of unconventional quantum criticality observed in quasicrystal Yb$_{15}$Al$_{34}$Au$_{51}$, the approximant crystal Yb$_{14}$Al$_{35}$Au$_{51}$ is analyzed theoretically. By constructing a minimal model for the approximant crystal, the heavy quasiparticle band is shown to emerge near the Fermi level because of strong correlation of 4f electrons at Yb. We find that charge-transfer mode between 4f electron at Yb on the 3rd shell and 3p electron at Al on the 4th shell in Tsai-type cluster is considerably enhanced with almost flat momentum dependence. The mode-coupling theory shows that magnetic as well as valence susceptibility exhibits $ \sim T^{-0.5}$ for zero-field limit and is expressed as a single scaling function of the ratio of temperature to magnetic field T/B over four decades even in the approximant crystal when some condition is satisfied by varying parameters, e.g., by applying pressure. The key origin is clarified to be due to the strong locality of the critical Yb-valence fluctuation and small Brillouin zone reflecting the large unit cell, giving rise to the extremely-small characteristic energy scale. This also gives a natural explanation for the quantum criticality in the quasicrystal corresponding to the infinite limit of the unit-cell size.

Fractal Surface States in Three-Dimensional Topological Quasicrystals

Zhu-Guang Chen; Cunzhong Lou; Kaige Hu; Lih-King Lim. Phys. Rev. Research 6, 043054 (2024) (2024). Cited by 0. Found by: arxiv. DOI: 10.1103/PhysRevResearch.6.043054

Abstract: We study topological states of matter in quasicrystals, which do not rely on crystalline orders. In the absence of a bandstructure description and spin-orbit coupling, we show that a three-dimensional quasicrystal can nevertheless form a topological insulator. It relies on a combination of noncrystallographic rotational symmetry of quasicrystals and electronic orbital space symmetry, which is the quasicrystalline counterpart of topological crystalline insulator. The resulting topological state obeys a non-trivial twisted bulk-boundary correspondence and lacks a good metallic surface. The topological surface states, localized on the top and bottom planes respecting the quasicrystalline symmetry, exhibit a new kind of multifractality with probability density concentrates mostly on high symmetry patches. They form a near-degenerate manifold of 'immobile' states whose number scales proportionally with the macroscopic sample size. This can open the door to a novel platform for topological surface physics distinct from the crystalline counterpart.

Host-atom-driven transformation of a honeycomb oxide into a dodecagonal quasicrystal

Martin Haller; Julia Hewelt; V. Y. M. Rajesh Chirala; Loi Vinh Tran; Ankur Bhide; Muriel Wegner; Stefan Förster; Wolf Widdra. Phys. Rev. Lett. 136, 156201 (2026) (2025). Cited by 0. Found by: arxiv. DOI: 10.1103/ws7j-tvty

Abstract: Dodecagonal oxide quasicrystals (OQCs) have so far been limited to a few elemental systems, with no general formation mechanism established. Here, we demonstrate a versatile approach to OQC formation via a host-atom-induced transformation of a metal-oxide honeycomb (HC) network. Adsorption of Ba, Sr, or Eu onto the HC layer triggers its reorganization into a dodecagonal tiling, as revealed by low-energy electron diffraction and scanning tunneling microscopy. Full conversion occurs when 73% of the honeycomb rings are occupied. Kelvin probe and UV photoelectron spectroscopy show a linear decrease in work function with increasing host coverage, followed by a sharp increase upon quasicrystal formation due to reduced host dipoles. This transformation mechanism enables the fabrication of structurally precise OQCs, including a new Eu-Ti-O phase that extends the field to lanthanide quasicrystals, forming a 2D grid of localized magnetic moments. The method offers a general route to explore lattice-matched substrates for epitaxial growth and may be adapted to other 2D honeycomb materials - such as graphene, hexagonal ice, and silica - paving the way for engineered aperiodic systems beyond transition metal oxides.

Closing of gaps and gap labeling and passage from molecular states to critical states in a 2D quasicrystal

Anuradha Jagannathan. arXiv preprint (2023). Cited by 0. Found by: arxiv. DOI: 10.1103/PhysRevB.108.115109

Abstract: The single electron spectrum and wavefunctions in quasicrystals continue to be a fascinating problem, with few known solutions, especially in two and higher dimensions. This paper investigates the energy spectra and gap structures in tight-binding models on a quasiperiodic tiling in two dimensions. By varying a continuous parameter, we follow the evolution of the band structure from the discrete molecular or atomic states, to the multifractal states well-known from previous studies. We propose a scheme for labeling gaps in finite approximants. It is equivalent in the limit of infinite systems to the one presented by Kellendonk and Putnam based on the algebraic structure of this quasiperiodic system.

Ground states of a family of frustrated spin models for quasicrystals and their approximants

Anuradha Jagannathan. arXiv preprint (2025). Cited by 0. Found by: arxiv. DOI: 10.1103/ht6g-5dwl

Abstract: Many new families of quasicrystal-forming magnetic alloys have been synthesized and studied in recent years. For small changes of composition, the alloys can go from quasiperiodic to periodic (approximant crystals) while conserving most of the local atomic environments. Experiments show that many of the periodic approximants order at low temperatures, with clear signatures of ferromagnetic or antiferromagnetic transitions, and also in some cases undergo non-equilibrium spin glass transitions. In contrast, the quasicrystals are mostly found to be spin glasses. Systematically studying these alloys could help elucidate the role played by quasiperiodicity in (de)stabilizing long range magnetic order. In this work, we study cluster spin models with the aim of understanding the mechanisms behind various types of long range magnetic ordering in approximants and quasicrystals. These models embody key features of real systems, and to some extent are analytically tractable, both for periodic and quasiperiodic cases. For the quasicrystal, we describe two novel magnetic phases with quasiperiodic ordering. Our results should serve to motivate further studies with detailed numerical explorations of this family of models.

Phonon transmittance of one dimensional quasicrystals

Junmo Jeon; SungBin Lee. arXiv preprint (2020). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/2009.12378v1

Abstract: In quasicrystals, special tiling patterns could give rise to unique physical phenomena such as critical states distinct from periodic systems. In this paper, we study how quasi-periodicity in aperiodic systems results in anomalous phonon modes, especially focusing on thermal transmittance in one-dimensional quasicrystals. Unlike periodic or compeletely random systems, we classify certain quasicrystals could host critical phonon modes whose transport properties are topologically protected based on their pattern equivariant cohomology group of supertilings. Starting from discussing general rule to find such critical phonon modes, we discuss classification of topologically distinct thermal transmittance in quasiperiodic systems. To be more specific, we exemplify (decorated) metallic-mean tilings and Cantor tiling, and derive universal features for resonant and decaying phonon modes as a function of quasi-periodic strength. Our study paves a new way to understand thermal transmittance of quasi-periodic systems based on the topological classification and offers quasicrystals as strong candidates to control drastic phonon modes.

Site-selective correlations in interacting ”flat-band” quasicrystals

Yuxi Zhang; Richard T. Scalettar; Rafael M. Fernandes. arXiv preprint (2024). Cited by 0. Found by: arxiv. DOI: 10.1103/77l3-krmj

Abstract: Model lattices such as the kagome and Lieb lattices have been widely investigated to elucidate the properties of interacting flat-band systems. While a quasicrystal does not have proper bands, the non-interacting density of states of several of them displays the typical signature of a flat band pinned at the Fermi level: a delta-function zero-energy peak. Here, we employ quantum Monte Carlo simulations to determine the effect of onsite repulsion on these quasicrystals. While global properties such as the antiferromagnetic structure factor and the specific heat behave similarly as in the case of periodic lattices undergoing a Mott transition, the behavior of the local density of states depends on the coordination number of the site. In particular, sites with the smallest coordination number, which give the dominant spectral-weight contribution to the zero-energy peak, are the ones most strongly impacted by the interaction. Besides establishing site-selective correlations in quasicrystals, our work also points to the importance of the real-space structure of flat bands in interacting systems.

Density Distribution in Soft Matter Crystals and Quasicrystals

Priya Subramanian; Daniel J. Ratliff; Alastair M. Rucklidge; Andrew J. Archer. Phys. Rev. Lett. 126, 218003 (2021) (2020). Cited by 0. Found by: arxiv. DOI: 10.1103/PhysRevLett.126.218003

Abstract: The density distribution in solids is often represented as a sum of Gaussian peaks (or similar functions) centred on lattice sites or via a Fourier sum. Here, we argue that representing instead the $\mathit{logarithm}$ of the density distribution via a Fourier sum is better. We show that truncating such a representation after only a few terms can be highly accurate for soft matter crystals. For quasicrystals, this sum does not truncate so easily, nonetheless, representing the density profile in this way is still of great use, enabling us to calculate the phase diagram for a 3-dimensional quasicrystal forming system using an accurate non-local density functional theory.

Quantum state transfer and maximal entanglement between distant qubits using a minimal quasicrystal pump

Arnob Kumar Ghosh; Rubén Seoane Souto; Vahid Azimi-Mousolou; Annica M. Black-Schaffer; Patric Holmvall. Phys. Rev. B 112, 205427 (2025) (2025). Cited by 0. Found by: arxiv. DOI: 10.1103/8zys-w2v4

Abstract: Coherent quantum state transfer over macroscopic distances between non-neighboring elements in quantum circuits is a crucial component to increase connectivity and simplify quantum information processing. To facilitate such transfers, an efficient and easily controllable quantum pump would be highly beneficial. In this work, we demonstrate such a quantum pump based on a one-dimensional quasicrystal Fibonacci chain~(FC). In particular, we utilize the unique properties of quasicrystals to pump the edge-localized winding states between the two distant ends of the chain by only minimal manipulation of the FC at its end points. We establish the necessary conditions for successful state transfer within a fully time-dependent picture and also demonstrate

robustness of the transfer protocol against disorder. We then couple external qubits to each end of the FC and establish highly adaptable functionality as a quantum bus with both on-demand switching of the qubit states and generation of maximally entangled Bell states between the qubits. Thanks to the minimal control parameters, the setup is well-suited for implementation across diverse experimental platforms, thus establishing quasicrystals as an efficient platform for versatile quantum information processing.

Scale-free localization versus Anderson localization in unidirectional quasiperiodic lattices

Yu Zhang; Luhong Su; Shu Chen. Phys. Rev. B 111, L140201(2025) (2025). Cited by 0. Found by: arxiv. DOI: 10.1103/PhysRevB.111.L140201

Abstract: Scale-free localization emerging in non-Hermitian physics has recently garnered significant attention. In this work, we explore the interplay between scale-free localization and Anderson localization by investigating a unidirectional quasiperiodic model with generalized boundary conditions. We derive analytical expressions of Lyapunov exponent from the bulk equations. Together with the boundary equation, we can determine properties of eigenstates and spectrum and establish their exact relationships with the quasiperiodic potential strength and boundary parameter. While eigenstates exhibit scale-free localization in the weak disorder regime, they become localized in the strong disorder regime. The scale-free and Anderson localized states satisfy the boundary equation in distinct ways, leading to different localization properties and scaling behaviors. Generalizing our framework, we design a model with exact energy edges separating the scale-free and Anderson localized states via the mosaic modulation of quasiperiodic potentials. Our models can be realized experimentally in electric circuits.

Quantum interferences in quasicrystals

Stephan Roche. arXiv preprint (1999). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/cond-mat/9907476v2

Abstract: Contributions of quantum interference effects occuring in quasicrystals are emphasized. First conversely to metallic systems, quasiperiodic ones are shown to enclose original alterations of their conductive properties while downgrading long range order. Besides, origins of localization mechanisms are outlined within the context of the original metal-insulator transition (MIT) found in these materials.

Cut-and-Project Density Functional Theory for Quasicrystals

Gavin N. Nop; Jonathan D. H. Smith; Thomas Koschny; Durga Paudyal. arXiv preprint (2026). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/2603.14590v1

Abstract: Cut-and-project from a symmetric structure in a higher-dimensional space is a standard method for describing the structure of a large class of quasicrystals. By means of a novel localization procedure, we now show how local physical interactions within these quasicrystals are also accurately described by cut-and-project, from corresponding physical interactions in the higher-dimensional space. A density functional theory (DFT++) formulation allows the cut-and-project method to handle the Schroedinger equation for interactions in quasicrystals. The theory is both rigorous and computationally tractable. The resulting ab initio approach specifies quasicrystalline quantum states, in contrast to previous approaches which only worked with crystalline approximants of the quasi-periodic structures.

Fractalized magnon transport on the quasicrystal with enhanced stability

Junmo Jeon; Se Kwon Kim; SungBin Lee. arXiv preprint (2022). Cited by 0. Found by: arxiv. DOI: 10.1103/PhysRevB.106.134431

Abstract: Magnonics has been receiving significant attention in magnetism and spintronics because of its premise for devices using spin current carried by magnons, quanta of spin-wave excitations of a macroscopically ordered magnetic media. Although magnonics has clear energy-wise advantage over conventional electronics due to the absence of the Joule heating, the inherent magnon-magnon interactions give rise to finite lifetime of the magnons which has been hampering the efficient realizations of magnonic devices. To promote magnonics, it is imperative to identify the delocalized magnon modes that are minimally affected by magnon-magnon interactions and thus possess a long lifetime and use them to achieve efficient magnon transport. Here, we suggest that quasicrystals may offer the solution to this problem via the critical magnon modes that are neither extended nor localized. We find that the critical magnon exhibits fractal characteristics that are absent in conventional magnon modes in regular solids such as a unique power-law scaling and a self-similar distribution of distances showing perfect

magnon transmission. Moreover, the critical magnons have longer lifetimes compared to the extended ones in a periodic system, by suppressing the magnon-magnon interaction decay rate. Such enhancement of the magnon stability originates from the presence of the quasi-periodicity and intermediate localization behavior of the critical magnons. Thus, we offer the utility of quasicrystals and their critical spin wave functions in magnonics as unique fractal transport characteristics and enhanced stability.

Common quantum phase transition in quasicrystals and heavy-fermion metals

V. R. Shaginyan; A. Z. Msezane; K. G. Popov; G. S. Japaridze; V. A. Khodel. Phys. Rev. B 87, 245122 (2013) (2013). Cited by 0. Found by: arxiv. DOI: 10.1103/PhysRevB.87.245122

Abstract: Extraordinary new materials named quasicrystals and characterized by noncrystallographic rotational symmetry and quasiperiodic translational properties have attracted scrutiny. Study of quasicrystals may shed light on the most basic notions related to the quantum critical state observed in heavy-fermion metals. We show that the electronic system of some quasicrystals is located at the fermion condensation quantum phase transition without tuning. In that case the quasicrystals possess the quantum critical state with the non-Fermi liquid behavior which in magnetic fields transforms into the Landau Fermi-liquid one. Remarkably, the quantum critical state is robust despite the strong disorder experienced by the electrons. We also demonstrate for the first time that quasicrystals exhibit the typical scaling behavior of their thermodynamic properties such as the magnetic susceptibility, and belong to the famous family of heavy-fermion metals. Our calculated thermodynamic properties are in good agreement with recent experimental observations.

High-harmonic spectroscopy of mobility edges in one-dimensional quasicrystals

H. K. Avetissian; B. R. Avchyan; A. Brown; G. F. Mkrtchian. arXiv preprint (2025). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/2509.06647v2

Abstract: Quasicrystals occupy a unique position between periodic and disordered systems, where localization phenomena such as Anderson transitions and mobility edges can emerge even in the absence of disorder. This distinctive behavior motivates the development of robust, all-optical diagnostic tools capable of probing the structural, topological, and dynamical properties of such systems. In this work, focusing on generalized Aubry-André-Harper models and on an incommensurate potential in the continuum limit, we demonstrate that high-harmonic generation phenomenon serves as a powerful probe of localization transitions and mobility edges in quasicrystals. We introduce a new parameter–dipole mobility–which captures the impact of intraband dipole transitions and enables classification of nonlinear optical regimes, where excitation and high-harmonic generation yield can differ by orders of magnitude. We show that the cutoff frequency of harmonics is strongly influenced by the position of the mobility edge, providing a robust and experimentally accessible signature of localization transitions in quasicrystals.

Anomalous electronic conductance in quasicrystals

Stephan Roche. arXiv preprint (1999). Cited by 0. Found by: arxiv. DOI: 10.1103/PhysRevB.61.6048

Abstract: Generic quantum interference effects occuring in 1D-quasicrystals are reviewed with emphasis on the joint effect of phason disorder on electronic localization and propagation modes. In close conjunction with properties of real materials, the contributions of quantum interferences in several regimes close to the metal insulator transition are outlined.

The Aubry-André model as the hobbyhorse for understanding localization phenomenon

G. A. Domínguez-Castro; R. Paredes. arXiv preprint (2018). Cited by 0. Found by: arxiv. DOI: 10.1088/1361-6404/ab1670

Abstract: We present a thorough pedagogical analysis of the single particle localization phenomenon in a quasiperiodic lattice in one dimension. Description of disorder in the lattice is represented by the Aubry-André model. Characterization of localization is performed through the analysis of both, stationary and dynamical properties. The stationary properties investigated are the inverse participation ratio (IPR), the normalized participation ratio (NPR) and the energy spectrum as a function of the disorder strength. As expected, the distinctive Hofstadter pattern is found. Two dynamical quantities allow discerning the localization phenomenon, being the spreading of an initially localized state and the evolution of population imbalance in even and odd sites across the lattice.

Tensor network method for real-space topology in quasicrystal Chern mosaics

Tiago V. C. Antão; Yitao Sun; Adolfo O. Fumega; Jose L. Lado. Phys. Rev. Lett. 136, 156601 (2026) (2025). Cited by 0. Found by: arxiv. DOI: 10.1103/hhdf-xpwg

Abstract: Computing topological invariants in two-dimensional quasicrystals and super-moire matter is a remarkable open challenge, due to the absence of translational symmetry and the colossal number of sites inherent to these systems. Here, we establish a method to compute local topological invariants of exceptionally large systems using tensor networks, enabling the computation of invariants for Hamiltonians with hundreds of millions of sites, several orders of magnitude above the capabilities of conventional methodologies. Our approach leverages a tensor-network representation of the density matrix using a Chebyshev tensor network algorithm, enabling large-scale calculations of topological markers in quasicrystalline and moire systems. We demonstrate our methodology with two-dimensional quasicrystals featuring C_8 and C_{10} rotational symmetries and mosaics of Chern phases. Our work establishes a powerful method to compute topological phases in exceptionally large-scale topological systems, providing the required tool to rationalize generic supe-moire and quasicrystalline topological matter.

Interlayer Decoupling in 30° Twisted Bilayer Graphene Quasicrystal

Bing Deng; Binbin Wang; Ning Li; Rongtan Li; Yani Wang; Jilin Tang; Qiang Fu; Zhen Tian; Peng Gao; Jiamin Xue; Hailin Peng. ACS Nano (2020) (2020). Cited by 0. Found by: arxiv. openalex_2yr: 16.894 (2026). DOI: 10.1021/acsnano.9b07091

Abstract: Stacking order has strong influence on the coupling between the two layers of twisted bilayer graphene (BLG), which in turn determines its physical properties. Here, we report the investigation of the interlayer coupling of the epitaxially grown single-crystal 30° twisted BLG on Cu(111) at the atomic scale. The stacking order and morphology of BLG is controlled by a rationally designed two-step growth process, that is, the thermodynamically controlled nucleation and kinetically controlled growth. The crystal structure of the 30°-twisted bilayer graphene (30°-tBLG) is determined to have the quasicrystal like symmetry. The electronic properties and interlayer coupling of the 30°-tBLG is investigated using scanning tunneling microscopy (STM) and spectroscopy (STS). The energy-dependent local density of states (DOS) with in-situ electrostatic doping shows that the electronic states in two graphene layers are decoupled near the Dirac point. A linear dispersion originated from the constituent graphene monolayers is discovered with doubled degeneracy. This study contributes to controlled growth of twist-angle-defined BLG, and provides insights of the electronic properties and interlayer coupling in this intriguing system.

Spin waves in one-dimensional bi-component quasicrystals

J. Rychły; J. W. Kłos; M. Mruczkiewicz; M. Krawczyk. Phys. Rev. B 92, 054414 (2015) (2014). Cited by 0. Found by: arxiv. DOI: 10.1103/PhysRevB.92.054414

Abstract: We studied finite Fibonacci sequence of Co and Py stripes aligned side-by-side and in direct contact with each other. Calculations based on continuous model including exchange and dipole interactions were performed for structures feasible for fabrication and characterization of main properties of magnonic quasicrystals. We have shown the fractal structure of the magnonic spectrum with a number of magnonics gaps of different width. Also localization of spin waves in quasicrystals and existence of surface spin waves in finite quaiscrystal structure is demonstrated.

Equilibrium configurations for generalized Frenkel-Kontorova models on quasicrystals

Rodrigo Treviño. arXiv preprint (2017). Cited by 0. Found by: arxiv. DOI: 10.1007/s00220-019-03557-7

Abstract: I study classes of generalized Frenkel-Kontorova models whose potentials are given by almost-periodic functions which are closely related to aperiodic Delone sets of finite local complexity. Since such Delone sets serve as good models for quasicrytals, this setup presents Frenkel-Kontorova models for the type of aperiodic crystals which have been discovered since Shechtman's discovery of quasicrystals. Here I consider models with configurations $u:\mathbb{Z}^r \rightarrow \mathbb{R}^d$, where d is the dimension of the quasicrystal, for any r and d. The almost-periodic functions used for potentials are called pattern-equivariant and I show that if the interactions of the model satisfies a mild C^2 requirement, and if the potential satisfies a mild non-degeneracy assumption, then there exist equilibrium configurations of any prescribed rotation rotation number/vector/plane. The assumptions are general enough to satisfy the classical Frenkel-Kontorova models and its multidimensional analogues. The proof uses the idea of the anti-integrable limit.

Localization, multifractality, and many-body localization in periodically kicked quasiperiodic lattices

Yu Zhang; Bozhen Zhou; Haiping Hu; Shu Chen. Phys. Rev. B 106, 054312 (2022) (2022). Cited by 0. Found by: arxiv. DOI: 10.1103/PhysRevB.106.054312

Abstract: We study the combined effect of quasiperiodic disorder, driven and interaction in the periodically kicked Aubry-André model. In the non-interacting limit, by analyzing the quasienergy spectrum statistics, we verify the existence of a dynamical localization transition in the high-frequency region, whereas the spectrum statistics becomes intricate in the low-frequency region due to the emergence of the extended/localized-to-multifractal edges in the quasienergy spectrum, which separate the multifractal states from the extended (localized) states. When the interaction is introduced, we find the periodically kicked incommensurate potential can lead to a transition from ergodic to many-body-localization phase in the high-frequency region. However, the many-body localization phase vanishes in the low-frequency region even for strong quasiperiodic disorder. Our studies demonstrate that the periodically kicked Aubry-André model displays rich dynamical phenomena and the driving frequency plays an important role in the formation of many-body localization in addition to the disorder strength.

A parametric study of the lensing properties of dodecagonal photonic quasicrystals

E. Di Gennaro; D. Morello; C. Miletto; S. Savo; A. Andreone; G. Castaldi; V. Galdi; V. Pierro. arXiv preprint (2007). Cited by 0. Found by: arxiv. DOI: 10.1016/j.photonics.2007.12.001

Abstract: We present a study of the lensing properties of two-dimensional (2-D) photonic quasicrystal (PQC) slabs made of dielectric cylinders arranged according to a 12-fold-symmetric square-triangle aperiodic tiling. Our full-wave numerical analysis confirms the results recently emerged in the technical literature and, in particular, the possibility of achieving focusing effects within several frequency regions. However, contrary to the original interpretation, such focusing effects turn out to be critically associated to local symmetry points in the PQC slab, and strongly dependent on its thickness and termination. Nevertheless, our study reveals the presence of some peculiar properties, like the ability to focus the light even for slabs with a reduced lateral width, or beaming effects, which render PQC slabs potentially interesting and worth of deeper investigation. Key words: Photonic quasicrystals; negative refraction; superlensing.

Exact anomalous mobility edges in one-dimensional non-Hermitian quasicrystals

Xiang-Ping Jiang; Weilei Zeng; Yayun Hu; Lei Pan. arXiv preprint (2024). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/2409.03591v1

Abstract: Recent research has made significant progress in understanding localization transitions and mobility edges (MEs) that separate extended and localized states in non-Hermitian (NH) quasicrystals. Here we focus on studying critical states and anomalous MEs, which identify the boundaries between critical and localized states within two distinct NH quasiperiodic models. Specifically, the first model is a quasiperiodic mosaic lattice with both nonreciprocal hopping term and on-site potential. In contrast, the second model features an unbounded quasiperiodic on-site potential and nonreciprocal hopping. Using Avila's global theory, we analytically derive the Lyapunov exponent and exact anomalous MEs. To confirm the emergence of the robust critical states in both models, we conduct a numerical multifractal analysis of the wave functions and spectrum analysis of level spacing. Furthermore, we investigate the transition between real and complex spectra and the topological origins of the anomalous MEs. Our results may shed light on exploring the critical states and anomalous MEs in NH quasiperiodic systems.

Identification and properties of topological states in the bulk of quasicrystals

Frode Balling-Ansø; Jeppe Lykke Krogh; Ella Elisabeth Lassen; Anne E. B. Nielsen. Phys. Rev. B 113, 075134 (2026) (2025). Cited by 0. Found by: arxiv. DOI: 10.1103/56vs-zwr8

Abstract: In contrast to the usual bulk-boundary correspondence, topological states localized within the bulk of the system have been numerically identified in quasicrystalline structures, termed bulk localized transport (BLT) states. These states exhibit properties different from edge states, one example being that the number of BLT states scales with system size, while the number of edge states scales with system perimeter. Here, we define a procedure to identify BLT states, which is based on the physically motivated crosshair marker and robustness analyses. Applying the procedure to the Hofstadter model on the Ammann-Beenker tiling, we find that the BLT states appear mainly for magnetic fluxes within a specific interval. While edge states appear at

low densities of states, we find that BLT states can appear at many different densities of states. Many of the BLT states are found to have real-space localization that follows geometric patterns characteristic of the given quasicrystal. Furthermore, BLT states can appear both isolated and in groups within the energy spectrum which could imply greater robustness for the states within such groups. The spatial localization of the states within a certain group can change depending on the Fermi energy.

Relationship between Structure and Dynamics of an Icosahedral Quasicrystal using Unsupervised Machine Learning

Edwin A. Bedolla-Montiel; Susana Marín-Aguilar; Marjolein Dijkstra. arXiv preprint (2025). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/2507.15731v1

Abstract: We present a comprehensive study of the structure, formation, and dynamics of a one-component model system that self-assembles into an icosahedral quasicrystal (IQC). Using molecular dynamics simulations combined with unsupervised machine learning techniques, we identify and characterize the unique structural motifs of IQCs, including icosahedral and dodecahedral arrangements, and quantify the evolution of local environments during the IQC formation process. Our analysis reveals that the formation of the IQC is driven by the emergence of distinct local clusters that serve as precursors to the fully developed quasicrystalline phase. Additionally, we examine the dynamics of the system across a range of temperatures, identifying transitions from vibrationally restricted motion to activated diffusion, and uncovering signatures of dynamic heterogeneity inherent to the quasicrystalline state. To directly connect structure and dynamics, we use a machine-learning-based order parameter to quantify the presence of distinct local environments across temperatures. We find that regions with high structural order, as captured by specific machine-learned classes, correlate with suppressed self-diffusion and minimal dynamical heterogeneity, consistent with phason-like motion within the IQC. In contrast, regions with lower structural order exhibit enhanced collective motion and increased dynamical heterogeneity. These results establish a quantitative framework for understanding the coupling between structural organization and dynamical processes in quasicrystals, providing new insights into the mechanisms governing IQC stability and dynamics.

Quantum bridge states and sub-dimensional transports in quasicrystals

Junmo Jeon; SungBin Lee. arXiv preprint (2022). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/2205.15343v1

Abstract: Exotic tiling patterns of quasicrystals have gotten a lot of attention for unique quantum phenomena such as critical state and multifractality. In this regard, finding new quasi-periodic tiling patterns and the relevant quantum states is one of the main interests in quasicrystals. Here, we focus on new types of quasicrystals described by coexisting phason flips and discuss their quantum transports with distinct local tiling patterns. It turns out that such new tilings are uniquely descended from the translation of high-dimensional lattices and their projection, in particular, a four-dimensional hypercubic lattice. Such unconventional tiling gives rise to sub-dimensionally confined states forming a bridge, so-called quantum bridge states. It turns out that electrons in such states could be strongly localized by applying a magnetic field at a particular time but are transported to the sites which share the same local patterns along the bridge with time evolution. The speed of transport is controlled via magnetic field strength, and one can even expect electron transportation between multiple bridges where their numbers are determined by the translation vectors of a four-dimensional hypercubic lattice. Our work paves a way to discover anomalous quantum states and their transports uniquely present in quasicrystals with new tiling structures.

1D quasicrystals and topological markers

Joseph Sykes; Ryan Barnett. arXiv preprint (2022). Cited by 0. Found by: arxiv. DOI: 10.1088/2633-4356/ac75a6

Abstract: Local topological markers are effective tools for determining the topological properties of both homogeneous and inhomogeneous systems. The Chern marker is an established topological marker that has previously been shown to effectively reveal the topological properties of 2D systems. In an earlier work, the present authors have developed a marker that can be applied to 1D time-dependent systems which can be used to explore their topological properties, like charge pumping under the presence of disorder. In this paper, we show how to alter the 1D marker so that it can be applied to quasiperiodic and aperiodic systems. We then verify its effectiveness against different quasicrystal Hamiltonians, some which have been addressed in previous studies using existing methods, and others which possess topological structures that have been largely unexplored. We also demonstrate that the altered 1D marker can be productively applied to systems that are fully aperiodic.

Fragile magnetic order in metallic quasicrystals

Ronaldo N. Araújo; Carlo C. Bellinati; Eric C. Andrade. Phys. Rev. B 110, 165131 (2024) (2023). Cited by 0. Found by: arxiv. DOI: 10.1103/PhysRevB.110.165131

Abstract: Inspired by recent experimental studies of local magnetic moments interacting with a metallic quasicrystal, we study the low-temperature fate of spins placed in two-dimensional tilings. In the diluted local moment limit, we calculate the spin relaxation rate $1/T_{1}$, as measured by electron spin resonance, and show that it displays a marked dependence on the system size N and the filing n of the electronic bath. For a finite concentration of spins, we integrate out the conduction electrons and generate an effective magnetic coupling between the local moments, which we treat as Ising spins. Despite the strongly frustrating nature of the magnetic couplings and the lack of periodicity in the problem, we find long-range orders for finite N in our large-scale Monte Carlo simulations. However, the resulting magnetically state is fragile, as clusters of essentially free spins fluctuate down to very low temperatures.

Quantum Fluctuations and Excitations in Antiferromagnetic Quasicrystals

Stefan Wessel; Igor Milat. Phys. Rev. B 71, 104427 (2005). (2004). Cited by 0. Found by: arxiv. DOI: 10.1103/PhysRevB.71.104427

Abstract: We study the effects of quantum fluctuations and the excitation spectrum for the antiferromagnetic Heisenberg model on a two-dimensional quasicrystal, by numerically solving linear spin-wave theory on finite approximants of the octagonal tiling. Previous quantum Monte Carlo results for the distribution of local staggered magnetic moments and the static spin structure factor are reproduced well within this approximate scheme. Furthermore, the magnetic excitation spectrum consists of magnon-like low-energy modes, as well as dispersionless high-energy states of multifractal nature. The dynamical spin structure factor, accessible to inelastic neutron scattering, exhibits linear-soft modes at low energies, self-similar structures with bifurcations emerging at intermediate energies, and flat bands in high-energy regions. We find that the distribution of local staggered moments stemming from the inhomogeneity of the quasiperiodic structure leads to a characteristic energy spread in the local dynamical spin susceptibility, implying distinct nuclear magnetic resonance spectra, specific for different local environments.

Exploring light-induced phases of 2D materials in a modulated 1D quasicrystal

Yifei Bai; Anna R. Dardia; Toshihiko Shimasaki; David M. Weld. arXiv preprint (2025). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/2506.11984v3

Abstract: Light-induced quantum phases offer the potential for simple and powerful tuning of material properties. For example, simply illuminating 2D materials in the integer quantum Hall regime with polarized light is predicted to drive quantum phase transitions. Such phenomena are largely beyond the current frontier of solid state experiments due to technical limitations on laser intensity and material purity. However, the Harper-Hofstadter mapping which relates a two-dimensional integer quantum Hall system to a 1D quasicrystal enables the same polarization-dependent light-induced phase transitions to be observed using a quantum gas in a driven quasiperiodic optical lattice. We report results of such an experiment. We observe an interlaced phase diagram of localization-delocalization phase transitions as a function of drive polarization and amplitude. Elliptically polarized driving can stabilize an extended critical phase featuring multifractal wavefunctions; we observe signatures of this phenomenon in anomalous polarization-dependent subdiffusive transport. In this regime, increasing the strength of the quasiperiodic potential can enhance rather than suppress transport. These experiments demonstrate a simple method for synthesizing exotic multifractal states and exploring light-induced quantum phases across different dimensionalities.

Emergent Nonperturbative Universal Floquet Localization

Soumadip Pakrashi; Atanu Rajak; Sambuddha Sanyal. arXiv preprint (2026). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/2601.09793v1

Abstract: We show that a robust, nonperturbative localization plateau emerges in periodically driven quasiperiodic lattices, independent of the static localization properties and drive protocol. Using exact Floquet dynamics, Floquet perturbation theory, and optimal-order van Vleck analysis, we identify a fine-tuned amplitude-to-frequency ratio where all Floquet states become localized despite dense resonances. The van Vleck expansion achieves superasymptotic accuracy up to an optimal orde; it ultimately breaks down due to resonant hybridization at a weak quasiperiodic potential, revealing that the observed localization is nonperturbative.

Localization transition in weakly-interacting Bose superfluids in one-dimensional quasiperdiodic lattices

Samuel Lellouch; Laurent Sanchez-Palencia. Physical Review A, American Physical Society, 2014, 90, pp.061602(R) (2014). Cited by 0. Found by: arxiv. DOI: 10.1103/PhysRevA.90.061602

Abstract: We study the localization of collective pair excitations in weakly-interacting Bose superfluids in one-dimensional quasiperiodic lattices. The localization diagram is first determined numerically. For intermediate interaction and quasiperiodic amplitude we find a sharp localization transition, with extended low-energy states and localized high-energy states. We then develop an analytical treatment, which allows us to quantitatively map the localization transition into that of an effective multiharmonic quasiperiodic system.

Twisted Multilayer Graphene: Superperiodicity and quasicrystals

Pedro Alcázar Guerrero. arXiv preprint (2026). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/2607.25411v1

Abstract: This thesis investigates how superperiodicity, quasiperiodicity, and disorder shape electronic and spin transport in graphene-based systems, with an emphasis on experimentally relevant length scales and realistic atomistic modeling. Using large-scale real-space quantum-transport methods, it first establishes controlled transport fingerprints that distinguish conventional Bloch propagation in periodic structures from the anomalous dynamics induced by quasiperiodic modulations. Building on this framework, the thesis analyzes magic-angle twisted bilayer graphene and shows that, within a finite disorder window where flat-band features remain robust, moderate Anderson disorder can counterintuitively enhance the mean free path. This disorder-induced delocalization is further linked to changes in the quantum metric extracted from optical conductivity, revealing a direct connection between transport, electronic geometry, and the real-space extent of the underlying states. The study then turns to graphene quasicrystal approximants and hybrid multilayer stacks, identifying sub-ballistic transport and self-similar localization patterns as signatures of quasicrystalline order, while also demonstrating their strong fragility against disorder and interlayer proximity effects. Finally, the thesis addresses spin transport in suspended monolayer graphene, showing that atomic-scale corrugations generate short-range fluctuating Rashba fields that can limit spin lifetimes to the nanosecond range even when charge transport remains close to ballistic. Taken together, these results provide a unified picture of how geometry, disorder, and structural complexity govern transport phenomena in twisted and corrugated graphene systems.

Gapless superconductivity and its real-space topology in quasicrystals

Kazuma Saito; Masahiro Hori; Ryo Okugawa; K. Tanaka; Takami Tohyama. Physical Review Research 7, L022077 (2025) (2024). Cited by 0. Found by: arxiv. DOI: 10.1103/nb7l-f44r

Abstract: We study superconductivity in Ammann-Beenker quasicrystals under magnetic field. By assuming an intrinsic s-wave pairing interaction and solving for mean-field equations self-consistently, we find gapless superconductivity in the quasicrystals at and near half filling. We show that gapless superconductivity originates in broken translational symmetry and confined states unique to the quasicrystals. When Rashba spin-orbit coupling is present, the quasicrystalline gapless superconductor can be topologically nontrivial and characterized by a nonzero pseudospectrum invariant given by a spectral localizer. The gapless topological superconducting phase exhibits edge states with near-zero energy. These findings suggest that quasicrystals can be a unique platform for realizing gapless superconductivity with nontrivial topology.

Robust flat bands in twisted trilayer graphene quasicrystals

Chen-Yue Hao; Zhen Zhan; Pierre A. Pantaleón; Jia-Qi He; Ya-Xin Zhao; Kenji Watanabe; Takashi Taniguchi; Francisco Guinea; Lin He. arXiv preprint (2024). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/2401.09010v1

Abstract: Moiré structures formed by twisting three layers of graphene with two independent twist angles present an ideal platform for studying correlated quantum phenomena, as an infinite set of angle pairs is predicted to exhibit flat bands. Moreover, the two mutually incommensurate moiré patterns among the twisted trilayer graphene (TTG) can form highly tunable moiré quasicrystals. This enables us to extend correlated physics in periodic moiré crystals to quasiperiodic systems. However, direct local characterization of the structure of the moiré quasicrystals and of the resulting flat bands are still lacking, which is crucial to fundamental understanding and control of the correlated moiré physics. Here, we demonstrate the existence of flat bands in a series of TTGs with various twist angle pairs and show that the TTGs with different magic angle pairs are strikingly dissimilar in their atomic and electronic structures. The lattice relaxation and the interference between moiré patterns are highly dependent on the twist angles. Our direct spatial mappings, supported by

theoretical calculations, reveal that the localization of the flat bands exhibits distinct symmetries in different regions of the moiré quasicrystals.

Non-diffusion transport in decoherent non-Hermitian quasicrystals

Yudong Ren; Rui Zhao; Kangpeng Ye; Lu Zhang; Hongsheng Chen; Haoran Xue; Yihao Yang. arXiv preprint (2025). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/2505.04205v2

Abstract: Disorder and coherence jointly govern wave transport in complex media. In Hermitian systems, a long-established paradigm since Anderson's work holds that disorder-induced localization relies on phase-coherent interference, and that the loss of coherence inevitably suppresses localization and restores featureless diffusive transport at long times. Whether this intuition remains valid in non-Hermitian systems, where transport can be governed by dissipation rather than interference, has remained largely open. Here we theoretically and experimentally demonstrate that this paradigm fundamentally breaks down in decoherent non-Hermitian quasicrystals. Using a programmable photonic lattice with independently engineered dissipation and fully programmable dephasing, we access regimes spanning from fully coherent to fully incoherent dynamics. While decoherence washes out localization and enforces structureless diffusion in Hermitian lattices, we find that decoherent non-Hermitian quasicrystals retain nontrivial, non-diffusive transport structures even in the incoherent limit. These include dissipation-induced localization, diffusion-localization transitions, and decoherence-induced mobility edges, phenomena with no counterparts in Hermitian disordered systems. We develop a unified theoretical framework that captures the ensemble-averaged dynamics across the entire coherence landscape, continuously connecting coherent and incoherent regimes, and reveals how dissipation and decoherence cooperate to shape transport. Our results establish decoherent non-Hermitian lattices as a distinct class of transport systems, in which dissipation and incoherence generate structured, non-diffusive phases, beyond the conventional Anderson picture.

Consistent Simulation of Fibonacci Anyon Braiding within a Qubit Quasicrystal Inflation Code

Marcelo M. Amaral. arXiv preprint (2025). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/2506.21643v1

Abstract: The simulation of non-Abelian anyon braiding is a critical step towards fault-tolerant quantum computation. We introduce a framework for this task based on a one-dimensional Quasicrystal Inflation Code (QIC). The code is defined by a local Hamiltonian whose ground state manifold enforces Fibonacci tiling constraints and possesses the correct Fibonacci degeneracy. We derive the corresponding braid operators and demonstrate that, while they are formally non-local, they possess an exact, local 3-qubit structure. This allows us to distill their action into a single, physically realizable 8×8 gate, which we term the B_{gate}. We prove through comparative compilation for systems with different numbers of qubits that constructing circuits by composing these local gates is dramatically more scalable than compiling the equivalent global unitary, showing a greater than tenfold reduction in circuit depth. To validate the framework, we successfully executed a braiding algorithm for the Jones polynomial, a topological invariant of knots, on an IBM Quantum processor, quantifying the signal degradation due to hardware noise. Finally, we numerically confirm for systems up to 17 qubits that our construction rigorously satisfies the required Temperley-Lieb and braid group relations while preserving the code subspace. This work establishes a validated and physically grounded pathway for the scalable simulation of anyonic braiding on programmable quantum systems.

Holographic Foliations: Self-Similar Quasicrystals from Hyperbolic Honeycombs

Latham Boyle; Justin Kulp. Phys. Rev. D 111, 046001 (2025) (2024). Cited by 0. Found by: arxiv. DOI: 10.1103/PhysRevD.111.046001

Abstract: Discrete geometries in hyperbolic space are of longstanding interest in pure mathematics and have come to recent attention in holography, quantum information, and condensed matter physics. Working at a purely geometric level, we describe how any regular tessellation of ($d+1$)-dimensional hyperbolic space naturally admits a d-dimensional boundary geometry with self-similar "quasicrystalline" properties. In particular, the boundary geometry is described by a local, invertible, self-similar substitution tiling, that discretizes conformal geometry. We greatly refine an earlier description of these local substitution rules that appear in the 1D/2D example and use the refinement to give the first extension to higher dimensional bulks; including a detailed account for all regular 3D hyperbolic tessellations. We comment on global issues, including the reconstruction of bulk geometries from boundary data, and introduce the notion of a "holographic foliation": a foliation by a stack of self-similar quasicrystals, where the full geometry of the bulk (and of the foliation itself) is encoded in any single leaf in a local invertible way. In the $\{3,5,3\}$ tessellation of 3D hyperbolic space by regular icosahedra, we find a 2D boundary quasicrystal admitting points of 5-fold symmetry which is not the Penrose

tiling, and record and comment on a related conjecture of William Thurston. We end with a large list of open questions for future analytic and numerical studies.

Evidence of local effects in anomalous refraction and focusing properties of dodecagonal photonic quasicrystals

Emiliano Di Gennaro; Carlo Miletto; Salvatore Savo; Antonello Andreone; Davide Morello; Vincenzo Galdi; Giuseppe Castaldi; Vincenzo Pierro. arXiv preprint (2008). Cited by 0. Found by: arxiv. DOI: 10.1103/PhysRevB.77.193104

Abstract: We present the key results from a comprehensive study of the refraction and focusing properties of a two-dimensional dodecagonal photonic 'quasicrystal'' (PQC), carried out via both full-wave numerical simulations and microwave measurements on a slab made of alumina rods inserted in a parallel-plate waveguide. We observe anomalous refraction and focusing in several frequency regions, confirming some recently published results. However, our interpretation, based on numerical and experimental evidence, differs substantially from the one in terms of 'effective negative refractive-index" that was originally proposed. Instead, our study highlights the critical role played by short-range interactions associated with local order and symmetry.

Plasmonic quasicrystals for designable ultra broadband transmission enhancement and second harmonic generation

Sachin Kasture; Ajith P R; V J Yallapragada; Raj Patil; Nikesh V. V.; Gajendra Mulay; Achanta Venu Gopal. arXiv preprint (2013). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/1309.3286v1

Abstract: Quasi-crystals are intriguing as they exhibit rotational symmetry and long range ordering but lack translational symmetry. 2-dimensional metal-dielectric patterns are interesting to make use of surface plasmon polariton (SPP) mediated local field enhancement and for near dispersionless SPP modes. In plasmonic crystals, the orientation and periodicity of the pattern dictate the polarization response and the discrete plasmon resonances while the interfaces define the plasmon dispersion. However, unique properties of plasmonic quasicrystals lead to polarization independence, designable k-space and broadband transmission enhancement due to SPP mediation. These are useful in many applications like energy harvesting, nonlinear optics and quantum plasmonics. We demonstrate design and fabrication of large area quasicrystal air hole patterns of pi/5 symmetry in metal film in which broadband, launch angle and polarization independent transmission enhancement as well as broadband second harmonic generation are observed. Designable transmission response, other symmetries and tilings are possible.

Which wavenumbers determine the thermodynamic stability of soft matter quasicrystals?

D. J. Ratliff; A. J. Archer; P. Subramanian; A. M. Rucklidge. Phys. Rev. Lett. 123, 148004 (2019) (2019). Cited by 0. Found by: arxiv. DOI: 10.1103/PhysRevLett.123.148004

Abstract: For soft matter to form quasicrystals an important ingredient is to have two characteristic lengthscales in the interparticle interactions. To be more precise, for stable quasicrystals, periodic modulations of the local density distribution with two particular wavenumbers should be favored, and the ratio of these wavenumbers should be close to certain special values. So, for simple models, the answer to the title question is that only these two ingredients are needed. However, for more realistic models, where in principle all wavenumbers can be involved, other wavenumbers are also important, specifically those of the second and higher reciprocal lattice vectors. We identify features in the particle pair interaction potentials which can suppress or encourage density modes with wavenumbers associated with one of the regular crystalline orderings that compete with quasicrystals, enabling either the enhancement or suppression of quasicrystals in a generic class of systems.

Space-Time Quasicrystal Structures and Inflationary and Late Time Evolution Dynamics in Accelerating Cosmology

Sergiu I. Vacaru. Clas. Quant. Grav. 35 (2018) no. 24, 245009 (2018). Cited by 0. Found by: arxiv. DOI: 10.1088/1361-6382/aaec93

Abstract: We construct new classes of cosmological solutions in modified and Einstein gravity theories encoding space-time quasicrystal, STQC, configurations modelled by nonlinear self-organized and pattern forming quasiperiodic structures. Such solutions are defined by generic off-diagonal locally anisotropic and inhomogeneous metrics depending via generating and integration functions on all spacetime coordinates. There are defined

nonholonomic variables and conditions for the generating/integration functions and sources for effective descriptions, or approximations, as "quasi" Friedmann-Lamaître-Robertson-Walker (FLRW) metrics. Such (off-) diagonal STQC-FLRW configurations contain memory on nonlinear classical and/or quantum interactions and may describe new acceleration cosmology scenarios. For special time-periodic conditions on nonlinear gravitational and matter field interactions, we can model at cosmological scales certain analogous of time crystal like structures originally postulated by Frank Wilczek in condensed matter physics. We speculate how STQC quasi-FLRW configurations could explain modern cosmology data and provide viable descriptions for the inflation and structure formation in our Universe. Finally, it is discussed systematically and critically how a unified description of inflation with dark energy era can be explained by (modified) cosmological STQC-scenarios.

Quasiperiodic pairing in graphene quasicrystals

Rasoul Ghadimi; Bohm-Jung Yang. Nano Lett. 2025, 25, 5, 1808 1815 (2024). Cited by 0. Found by: arxiv. DOI: 10.1021/acs.nanolett.4c04386

Abstract: We investigate the superconducting instabilities of twisted bilayer graphene quasicrystals (TBGQC) obtained by stacking two monolayer graphene sheets with a 30° relative twisting. The electronic energy spectrum of TBGQC contains periodic energy ranges (PER) and quasiperiodic energy ranges (QER), where the underlying local density of states (LDOS) exhibits periodic and quasiperiodic distribution, respectively. We found that superconductivity in the PER is a simple superposition of two monolayer superconductors. This is because, particularly near the charge neutrality point of TBGQC, the two layers are weekly coupled, leading to pairing instabilities with uniform distribution in real space. On the other hand, within QER, the inhomogeneous distribution of the LDOS enhances the superconducting instability with a non-uniform distribution of pairing amplitudes, leading to quasiperiodic superconductivity. Our study can qualitatively explain the superconductivity in recently discovered moiré quasicrystals, which show superconductivity in its QER.

Electronic states of a disordered 2D quasiperiodic tiling: from critical states to Anderson localization

Anuradha Jagannathan; Marco Tarzia. arXiv preprint (2022). Cited by 0. Found by: arxiv. DOI: 10.1103/PhysRevB.107.054206

Abstract: We consider critical eigenstates in a two dimensional quasicrystal and their evolution as a function of disorder. By exact diagonalization of finite size systems we show that the evolution of properties of a typical wave-function is non-monotonic. That is, disorder leads to states delocalizing, until a certain crossover disorder strength is attained, after which they start to localize. Although this non-monotonic behavior is only present in finite-size systems and vanishes in the thermodynamic limit, the crossover disorder strength decreases logarithmically slowly with system size, and is quite large even for very large approximants. The non-monotonic evolution of spatial properties of eigenstates can be observed in the anomalous dimensions of the wave-function amplitudes, in their multifractal spectra, and in their dynamical properties. We compute the two-point correlation functions of wave-function amplitudes and show that these follow power laws in distance and energy, consistent with the idea that wave-functions retain their multifractal structure on a scale which depends on disorder strength. Dynamical properties are studied as a function of disorder. We find that the diffusion exponents do not reflect the non-monotonic wave-function evolution. Instead, they are essentially independent of disorder until disorder increases beyond the crossover value, after which they decrease rapidly, until the strong localization regime is reached. The differences between our results and earlier studies on geometrically disordered "phason-flip" models lead us to propose that the two models are in different universality classes. We conclude by discussing some implications of our results for transport and a proposal for a Mott hopping mechanism between power law localized wave-functions, in moderately disordered quasicrystals.

Binary icosahedral quasicrystals of hard spheres in spherical confinement

Da Wang; Tonnishtha Dasgupta; Ernest B. van der Wee; Daniele Zanaga; Thomas Altantzis; Yaoting Wu; Gabriele M. Coli; Christopher B. Murray; Sara Bals; Marjolein Dijkstra; Alfons van Blaaderen. arXiv preprint (2019). Cited by 0. Found by: arxiv. DOI: 10.1038/s41567-020-1003-9

Abstract: The influence of geometry on the local and global packing of particles is important to many fundamental and applied research themes such as the structure and stability of liquids, crystals and glasses. Here, we show by experiments and simulations that a binary mixture of hard-sphere-like particles crystallizing into the MgZn2 Laves phase in bulk, spontaneously forms 3D icosahedral quasicrystals in slowly drying droplets. Moreover, the local symmetry of 70-80% of the particles changes to that of the MgCu2 Laves phase. Both of these findings are significant for photonic applications. If the stoichiometry deviates from that of the Laves phase, our experiments show that the crystallization of MgZn2 is hardly affected by the spherical confinement. Our simulations show that the quasicrystals nucleate away from the spherical boundary and grow along five-fold symmetric structures. Our findings not only open the way for particle-level studies of nucleation and growth of 3D quasicrystals, but also of binary crystallization.

Topological metal-insulator transitions in one-dimensional non-Hermitian quasicrystals: beyond PT-symmetry

Guangjie Zhang; Bing Shao; Longwen Zhou; Jiangbin Gong; Weiwei Zhu. arXiv preprint (2026). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/2605.25072v1

Abstract: One-dimensional non-Hermitian quasicrystals with parity and time-reversal (PT) symmetry can simultaneously exhibit localization-delocalization transition, topological phase transition, and PT-symmetry-breaking transition. This motivates this work to investigate how the absence of PT symmetry impacts topological metal-insulator transitions in non-Hermitian quasicrystals. We propose a non-Hermitian quasiperiodic model that generally does not preserve PT symmetry and demonstrate that, in most parameter regions, such a system supports triple phase transitions that encompass localization, topology, and degeneracy-breaking. The system may also exhibit a particular type of localization-delocalization transition analogous to the Hermitian case, namely, without activating topological phase transitions or degeneracy-breaking transitions. Our work extends the topological metal-insulator transitions previously studied in PT-symmetric systems to a more general class of non-Hermitian setting, and further reveals that non-Hermitian systems can host distinct types of localization behavior.

Asymptotic estimates of large gaps between directions in certain planar quasicrystals

Gustav Hammarhjelm; Andreas Strömbergsson; Shucheng Yu. arXiv preprint (2024). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/2409.19514v2

Abstract: For quasicrystals of cut-and-project type in $\mathbb{R}^d$, it was proved by Marklof and Strömbergsson that the limit local statistical properties of the directions to the points in the set are described by certain $\operatorname{SL}_d(\mathbb{R})$-invariant point processes. In the present paper we make a detailed study of the tail asymptotics of the limiting gap statistics of the directions, for certain specific classes of planar quasicrystals.

Phasons and the Plastic Deformation of Quasicrystals

Maurice Kleman. Eur. phys. J. B 31 (2003) 315-325 (2002). Cited by 0. Found by: arxiv. DOI: 10.1140/epjb/e2003-00037-3

Abstract: The plastic deformation of a quasicrystal (QC) is ruled by singularities of its 'phonon' strain field and of its 'phason' strain field. In the framework of the topological theory of defects, and the QC being defined as an irrational subset of a high dimensional crystal, both types of defects appear as distinct components of the same entity, a 'disvection'. Each of them can also be described in classical terms, within a detailed analysis of the Volterra process:(a)- the 'phonon' singularity breaks some symmetry of translation, represented by a Burgers vector projected from the high dimensional crystal onto the physical space; it is akin to an ordinary perfect dislocation, (b)- the 'phason' singularities are dislocation dipoles whose Burgers vectors are of a special type; they break not only a particular symmetry of translation but also, in the QCs' jargon, the 'class of local isomorphism' of the QC. In fact, such dipoles, if they open up into loops, bound stacking faults; a phason singularity is thus an imperfect dislocation. It is suggested that the interplay of perfect and imperfect dislocations is at the origin of the peculiar characters of the plastic deformation of QCs, namely the brittle-ductile transition followed by a stage of work softening; in particular the brittle-ductile transition could be related to a cooperative transition of the Kosterlitz-Thouless type which affects the dipoles and turn them into (imperfect) dislocation loops.

Starobinsky Inflation and Dark Energy and Dark Matter Effects from Quasicrystal Like Spacetime Structures

R. Aschheim; L. Bubuianu; Fang Fang; Klee Irwin; V. Ruchin; S. Vacaru. Annals of Physics 394 (2018) 120-138 (2016). Cited by 0. Found by: arxiv. DOI: 10.1016/j.aop.2018.04.033

Abstract: The goal of this work on mathematical cosmology and geometric methods in modified gravity theories, MGTs, is to investigate Starobinsky-like inflation scenarios determined by gravitational and scalar field configurations mimicking quasicrystal, QC, like structures. Such spacetime aperiodic QCs are different from

those discovered and studied in solid state physics but described by similar geometric methods. We prove that an inhomogeneous and locally anisotropic gravitational and matter field effective QC mixed continuous and discrete "aether" can be modeled by exact cosmological solutions in MGTs and Einstein gravity. The coefficients of corresponding generic off-diagonal metrics and generalized connections depend (in general) on all spacetime coordinates via generating and integration functions and certain smooth and discrete parameters. Imposing additional nonholonomic constraints, prescribing symmetries for generating functions and solving the boundary conditions for integration functions and constants, we can model various nontrivial torsion QC structures or extract cosmological Levi–Civita configurations with diagonal metrics reproducing de Sitter (inflationary) like and other types homogeneous inflation and acceleration phases. Finally, we speculate how various dark energy and dark matter effects can be modeled by off-diagonal interactions and deformations of a nontrivial QC like gravitational vacuum structure and analogous scalar matter fields.

Raman study of lattice dynamics in quasicrystals

Yu. S. Ponosov; N. I. Shchegolikhina; A. F. Prekul. arXiv preprint (2009). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/0908.4535v1

Abstract: We present the first Raman investigation of icosahedral quasicrystals. Broad structured bands in the energy range up to ~ 500 $cm^{-1} have been observed in a series of AlFeCu, AlPdMn and AlPdRe systems. Original information on the vibrational density of states g(omega) was obtained for AlPdRe; for AlFeCu and AlPdMn estimated g(omega) shows a good agreement with the previous neutron results, but demonstrates finer structure. Strong increase in the parameter of electron-vibrational coupling for the low-energy vibrations and its correlation with changes in electronic conductivity has been observed in the series from AlFeCu to AlPdRe. This suggests the increase of the degree of localization for these vibrational excitations and involved electronic states.

Localization Transitions in a Half-Filled Helical Aubry-André Model

Taylan Yildiz; B. Tanatar; Balázs Hetényi. arXiv preprint (2026). Cited by 0. Found by: arxiv. DOI: 10.1103/c5wk-3ym5

Abstract: We study localization in a one-dimensional quasiperiodic lattice obtained by extending the Aubry-André model with an additional Nth-neighbor hopping term of strength J_{N}. This long-range tunneling couples successive windings of an effective helical chain and introduces a second control parameter beyond the quasiperiodic potential strength $Δ$. Working with noninteracting fermions (typically at half filling), we diagnose the delocalization-localization transition using extensions of the modern theory of polarization. Specifically, we compute the polarization amplitudes of the many-body Slater-determinant ground state and construct a geometric Binder cumulant from polarization amplitudes. The critical potential where the localization transition happens is extracted from the sign change (zero crossing) of the geometric Binder cumulant. We map critical potential as a function of J_N and the helical range N, finding that stronger helical hopping generally stabilizes the extended phase (shifting critical potential upward), while the N-dependence can display pronounced commensurability-induced spikes. We further compare the geometric Binder cumulant with the Fermi gap, which remains near zero at small values of potential and opens in the same parameter regime where the geometric Binder cumulant departs from extended phase. Finally, to take a controlled thermodynamic limit along Fibonacci system sizes, we employ a Zeckendorf-shift construction that fixes the many-body sector consistently as system size goes to infinity.

Reentrant Localization Transitions in a Topological Anderson Insulator: A Study of a Generalized Su-Schrieffer-Heeger Quasicrystal

Zhanpeng Lu; Yunbo Zhang; Zhihao Xu. Front. Phys. 20, 024204 (2025) (2023). Cited by 0. Found by: arxiv. DOI: 10.15302/frontphys.2025.024204

Abstract: We study the topology and localization properties of a generalized Su-Schrieffer-Heeger (SSH) model with a quasi-periodic modulated hopping. It is found that the interplay of off-diagonal quasi-periodic modulations can induce topological Anderson insulator (TAI) phases and reentrant topological Anderson insulator (RTAI), and the topological phase boundaries can be uncovered by the divergence of the localization length of the zero-energy mode. In contrast to the conventional case that the TAI regime emerges in a finite range with the increase of disorder, the TAI and RTAI are robust against arbitrary modulation amplitude for our system. Furthermore, we find that the TAI and RTAI can induce the emergence of reentrant localization transitions. Such an interesting connection between the reentrant localization transition and the TAI/RTAI can be

detected from the wave-packet dynamics in cold atom systems by adopting the technique of momentum-lattice engineering.

Anderson Localization and Swing Mobility Edge in Curved Spacetime

Shan-Zhong Li; Xue-Jia Yu; Shi-Liang Zhu; Zhi Li. Phys. Rev. B 108, 094209 (2023) (2022). Cited by 0. Found by: arxiv. DOI: 10.1103/PhysRevB.108.094209

Abstract: We construct a quasiperiodic lattice model in curved spacetime to explore the crossover concerning both condensed matter and curved spacetime physics. We study the related Anderson localization and find that the model has a clear boundary of localized-extended phase separation, which leads to a swing mobility edge, i.e., the coexistence of localized, swing and sub-extended phases. The swing mobility edge, first reported here, features the phase-dependent eigenstate, that is, the eigenstate swing between the extended and localized state for differnt phase parameter of the quasiperiodic potential. Furthermore, A novel self-consistent segmentation method is developed to calculate the analytical expression of the critical point of phase separation, and the rich phase diagram is obtained by calculating the fractal dimension and scaling index in multifractal analysis.

Generic non-Hermitian mobility edges in a class of duality-breaking quasicrystals

Xiang-Ping Jiang; Mingdi Xu; Lei Pan. Results in Physics 70 (2025) 108146 (2024). Cited by 0. Found by: arxiv. DOI: 10.1016/j.rinp.2025.108146

Abstract: We provide approximate solutions for the mobility edge (ME) that demarcates localized and extended states within a specific class of one-dimensional non-Hermitian (NH) quasicrystals. These NH quasicrystals exhibit a combination of nonreciprocal hopping terms and complex quasiperiodic on-site potentials. Our analytical approach is substantiated by rigorous numerical calculations, demonstrating significant accuracy. Furthermore, our ansatz closely agrees with the established limiting cases of the NH Aubry-Andr{é}-Harper (AAH) and Ganeshan-Pixley-Das Sarma (GPD) models, which have exact results, thereby enhancing its credibility. Additionally, we have examined their dynamic properties and discovered distinct behaviors in different regimes. Our research provides a practical methodology for estimating the position of MEs in a category of NH quasicrystals that break duality.

Quantum transport in quasicrystals and complex metallic alloys

Didier Mayou; Guy Trambly De Laissardière. Quasicrystals, Elsevier, Amsterdam (Ed.) (2008) 209-265 (2007). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/cond-mat/0701639v1

Abstract: The semi-classical Bloch-Boltzmann theory is at the heart of our understanding of conduction in solids, ranging from metals to semi-conductors. Physical systems that are beyond the range of applicability of this theory are thus of fundamental interest. This is the case of disordered systems which present quantum interferences in the diffusive regime, i.e. Anderson localization effects. It appears that in quasicrystals and related complex metallic alloys another type of breakdown of the semi-classical Bloch-Boltzmann theory operates. This type of quantum transport is related to the specific propagation mode of electrons in these systems. We develop a theory of quantum transport that applies to a normal ballistic law but also to these specific diffusion laws. As we show phenomenological models based on this theory describe correctly the experimental transport properties. Ab-initio calculations performed on approximants confirm also the validity of this anomalous quantum diffusion scheme. Although the present chapter focuses on electrons in quasicrystals and complex metallic alloys, the concept that are developed here can be useful for phonons in these systems. There is also a deep analogy between the type of quantum transport described here and the conduction properties of other systems where charge carriers are also slow, such as some heavy fermions or polaronic systems.

Hybrid skin-topological effect induced by eight-site cells and arbitrary adjustment of the localization of topological edge states

Jianzhi Chen; Aoqian Shi; Yuchen Peng; Peng Peng; Jianjun Liu. Chinese Physics Letters 41, 037103 (2024) Chinese Physics Letters 41, 037103 (2024) (2024). Cited by 0. Found by: arxiv. DOI: 10.1088/0256-307X/41/3/037103

Abstract: The hybrid skin-topological effect (HSTE) in non-Hermitian systems exhibits both the skin effect and topological protection, offering a novel mechanism for the localization of topological edge states (TESs) in electrons, circuits, and photons. However, it remains unclear whether the HSTE can be realized in quasicrystals, and the unique structure of quasicrystals with multi-site cells may provide novel localization phenomena for

TESs induced by the HSTE. We propose an eight-site cell in two-dimensional quasicrystals and realize the HSTE with eight-site nonreciprocal intracell hoppings. Furthermore, we can arbitrarily adjust the eigenfield distributions of the TESs and discover domain walls associated with effective dissipation and their correlation with localization. We present a new scheme to precisely adjust the energy distribution in non-Hermitian quasicrystals with arbitrary polygonal outer boundaries.

Su-Schrieffer-Heeger quasicrystal: Topology, localization, and mobility edge

D. A. Miranda; T. V. C. Antão; N. M. R Peres. Phys. Rev. B 109, 195427, Published 21 May 2024 (2024). Cited by 0. Found by: arxiv. DOI: 10.1103/PhysRevB.109.195427

Abstract: In this paper we discussed the topological transition between trivial and nontrivial phases of a quasi-periodic (Aubry-André like) mechanical Su-Schrieffer-Heeger (SSH) model. We find that there exists a nontrivial boundary separating the two topological phases and an analytical expression for this boundary is found. We discuss the localization of the vibrational modes using the calculation of the inverse participation ratio (IPR) and access the localization nature of the states of the system. We find three different regimes: extended, localized, and critical, depending on the intensity of the Aubry-André spring. We further study the energy dependent mobility edge (ME) separating localized from extended eigenstates and find its analytical expression for both commensurate and incommensurate modulation wavelengths, thus enlarging the library of models possessing analytical expressions for the ME. Our results extend previous results for the theory of fermionic topological insulators and localization theory in quantum matter to the classical realm.

The scales of disorder in perfect quasicrystals

Alan Rodrigo Mendoza Sosa; Atahualpa S. Kraemer; Michael Schmiedeberg; Erdal C. Oğuz. arXiv preprint (2026). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/2607.09274v1

Abstract: The classical dichotomy between crystalline order and amorphous disorder is increasingly challenged by novel states that lack conventional crystalline symmetries while retaining crystal-like properties. Quasicrystals occupy a distinctive position within this expanding framework by possessing long-range order without translational periodicity, thereby permitting arbitrary \$N\$-fold rotational symmetry. Paradoxically, far from their unique symmetry center, high-symmetry quasicrystals closely resemble disordered patterns, raising the question of how deterministic order can be detected. Here we show that increasing rotational symmetry progressively suppresses local statistical signatures of quasiperiodicity, while preserving its underlying exact long-range order. This order is thus concealed below an emergent crossover length that grows linearly with \$N\$. Therefore, as \$N \rightarrow \infty\$, the disorder-like regime expands without bound, defining a symmetry-controlled geometric critical point at which deterministic order and randomness become statistically indistinguishable over any finite observation window. For finite \$N\$, however, quasiperiodic order becomes detectable beyond this crossover, revealing a second emergent length scale that we identify as the size of a \textit{statistical unit cell} – finite patches over which statistical properties recur despite the absence of conventional translational periodicity. In one dimension, the statistical-unit-cell size coincides with the crossover length, whereas in two dimensions it grows as \$N^2\$, remaining smaller than the size of typical approximants and establishing a hierarchy of emergent length scales. Together, the disorder-to-order crossover and statistical unit cells provide a quantitative framework connecting crystals, quasicrystals, and amorphous matter, showing how apparent disorder can emerge from purely deterministic geometry.

Entropic crystallization of geometrically frustrated magnets on 1/1 approximant Tsai-type quasicrystal

Oscar Novat; Ludovic D. C. Jaubert; Masafumi Udagawa. arXiv preprint (2026). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/2604.02180v1

Abstract: We have studied the antiferromagnetic Ising model on the icosahedral bcc lattice, as a model system of 1/1 approximant Tsai-type quasicrystals. We addressed thermal equilibrium properties of this system with Markov-chain Monte Carlo simulation supplemented with the parallel tempering technique to accelerate the relaxation dynamics. As a result, we found a second-order phase transition takes place to the magnetic ordered phase with \${\mathbb Z_3}\times {\mathbb Z_2}\$ symmetry breaking. Despite the ordering, the low-temperature phase keeps macroscopic degeneracy as identified by finite residual entropy, \$\mathcal{S}\sim0.1767/{\rm spin}\$. Remarkably, the existence of residual entropy turns out to play a major role in the formation of magnetic order. Generation of domain wall is suppressed, as it reduces the residual entropy locally stored in icosahedra, beyond the gain of configurational entropy due to domain wall patterns. Magnetic

order arises out of this competition as entropic crystallization, which manifest universal mechanism of strongly frustrated systems with large geometrical units.

Unveiling exotic magnetic phase diagram of a non-Heisenberg quasicrystal approximant

Farid Labib; Kazuhiro Nawa; Shintaro Suzuki; Hung-Cheng Wu; Asuka Ishikawa; Kazuki Inagaki; Takenori Fujii; Katsuki Kinjo; Taku J. Sato; Ryuji Tamura. arXiv preprint (2023). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/2310.14291v1

Abstract: A magnetic phase diagram of the non-Heisenberg Tsai-type 1/1 Au-Ga-Tb approximant crystal (AC) has been established across a wide electron-per-atom (e/a) range via magnetization and powder neutron diffraction measurements. The diagram revealed exotic ferromagnetic (FM) and antiferromagnetic (AFM) orders that originate from the unique local spin icosahedron common to icosahedral quasicrystals (iQCs) and ACs; The noncoplanar whirling AFM order is stabilized as the ground state at the e/a of 1.72 or less whereas a noncoplanar whirling FM order was found at the larger e/a of 1.80, with magnetic moments tangential to the Tb icosahedron in both cases. Moreover, the FM/AFM phase selection rule was unveiled in terms of the nearest neighbour (J1) and next nearest neighbour (J2) interactions by numerical calculations on a non-Heisenberg single icosahedron. The present findings will pave the way for understanding the intriguing magnetic orders of not only non-Heisenberg FM/AFM ACs but also non-Heisenberg FM/AFM iQCs, the latter of which are yet to be discovered.

Non-Fermi-Liquid Behaviors in Weakly Interacting Two-Dimensional Quasicrystals

Junmo Jeon; Shiro Sakai. arXiv preprint (2026). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/2607.18394v1

Abstract: Non-Fermi-liquid (NFL) behavior is usually associated with strong electronic correlations, quantum criticality, or singular fluctuations that invalidate the quasiparticle picture. Quasicrystals are known to provide a fundamentally different route to such physics: their lack of translational symmetry gives rise to critical single-particle states and fragmented spectra even in the absence of electron-electron interactions. Here, we numerically study the NFL behaviors in quasicrystals with weak interactions. Using Penrose and Ammann–Beenker tilings as representative quasiperiodic systems, we analyze thermodynamic, magnetic, and transport properties within a weak-coupling perturbative framework. The Sommerfeld coefficient and magnetic susceptibility show anomalous temperature dependences governed by the singular energy dependence of the density of states near the Fermi level. The Wilson ratio reveals that low-temperature NFL behavior in quasicrystals is accompanied by enhanced spin fluctuations over a broad range of filling fractions, while its magnitude and temperature dependence vary sensitively with filling fraction. The self-energy shows anomalous scaling and a finite residual scattering rate, indicating the breakdown of quasiparticles. Our site-resolved analysis reveals a pronounced enhancement of the residual scattering rate in regions with a high local density of states. These results serve as a guide to analyze experimental data for quasicrystals and to identify the microscopic mechanism of the NFL behaviors.

Existence results in the linear dynamics of quasicrystals with phason diffusion and non-linear gyroscopic effects

Luca Bisconti; Paolo Maria Mariano. arXiv preprint (2015). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/1511.06

Abstract: Quasicrystals are characterized by quasi-periodic arrangements of atoms. The description of their mechanics involves deformation and a (so called phason) vector field accounting at macroscopic scale of local phase changes, due to atomic flips necessary to match quasi-periodicity under the action of the external environment. Here we discuss the mechanics of quasicrystals, commenting the shift from its initial formulation, as standard elasticity in a space with dimension twice the ambient one, to a more elaborated setting avoiding physical inconveniences of the original proposal. In the new setting we tackle two problems. First we discuss the linear dynamics of quasicrystals including a phason diffusion. We prove existence of weak solutions and their uniqueness under rather general boundary and initial conditions. We then consider phason rotational inertia, non-linearly coupled with the curl of the macroscopic velocity, and prove once again existence of weak solutions to the pertinent balance equations.

Striking Similarities in Dynamics and Vibrations of 2D Quasicrystals and Supercooled Liquids

Edwin A. Bedolla-Montiel; Marjolein Dijkstra. arXiv preprint (2025). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/2508.18856v2

Abstract: We investigate the interplay between structure and dynamics in two structurally distinct two-dimensional systems: a dodecagonal quasicrystal (DDQC) and a supercooled binary liquid. Using molecular dynamics simulations, we uncover striking dynamical similarities despite their fundamentally different structural organizations. Both systems exhibit pronounced dynamic heterogeneities, as evidenced by the cage-trapping plateaus in the mean-squared displacement and the pronounced peaks in the non-Gaussian parameter. In both cases, we observe a strong correlation between local structural order and dynamic propensity, indicating similar structure-dynamics relationships, albeit driven by distinct microscopic mechanisms. Despite these parallels, their vibrational properties diverge: the DDQC exhibits multiple peaks linked to phason dynamics, while the supercooled liquid displays a characteristic boson peak. Analysis of vibrational eigenmodes shows that both systems exhibit extended modes at low frequencies. At high frequencies, however, the DDQC maintains a higher density of topological defects, reflecting its quasi-long-range order. Finally, we contextualize these findings by comparing both systems to a square crystal. While the dynamics appears similar across all three systems, the vibrational and topological features clearly distinguish the DDQC and glass from the crystalline state. These results underscore a surprising universality in dynamical behavior across structurally diverse systems and provide new insights into how structural organization shapes motion in soft-matter systems.

Cholesteric shells: two-dimensional blue fog and finite quasicrystals

Livio Nicola Carenza; Giuseppe Gonnella; Davide Marenduzzo; Giuseppe Negro; Enzo Orlandini. Physical Review Letters 2022 (2021). Cited by 0. Found by: arxiv. DOI: 10.1103/PhysRevLett.128.027801

Abstract: We study the phase behaviour of a quasi-two dimensional cholesteric liquid crystal shell. We characterise the topological phases arising close to the isotropic-cholesteric transition, and show that they differ in a fundamental way from those observed on a flat geometry. For spherical shells, we discover two types of quasi-two dimensional topological phases: finite quasicrystals and amorphous structures, both made up by mixtures of polygonal tessellations of half-skyrmions. These structures generically emerge instead of regular double twist lattices because of geometric frustration, which disallows a regular hexagonal tiling of curved space. For toroidal shells, the variations in the local curvature of the surface stabilises heterogeneous phases where cholesteric patterns coexist with hexagonal lattices of half-skyrmions. Quasicrystals, amorphous and heterogeneous structures could be sought experimentally by self assembling cholesteric shells on the surface of emulsion droplets.

Mapping of strained graphene into one-dimensional Hamiltonians: quasicrystals and modulated crystals

Gerardo G. Naumis; Pedro Roman-Taboada. arXiv preprint (2014). Cited by 0. Found by: arxiv. DOI: 10.1103/PhysRevB.89.241404

Abstract: The electronic properties of graphene under any arbitrary uniaxial strain field are obtained by an exact mapping of the corresponding tight-binding Hamiltonian into an effective one-dimensional modulated chain. For a periodic modulation, the system displays a rich behavior, including quasicrystals and modulated crystals with a complex spectrum, gaps at the Fermi energy and interesting localization properties. These features are explained by the incommensurate or commensurate nature of the potential, which leads to a dense filling of the reciprocal space in the former case. Thus, the usual perturbation theory approach breaks down in some cases, as is proved by analyzing a special momentum that uncouples the model into dimers.

The fate of Wannier-Stark localization and skin effect in periodically driven non-Hermitian quasiperiodic lattices

Aditi Chakrabarty; Sanjoy Datta. arXiv preprint (2024). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/2412.11740

Abstract: The eigenstates of one-dimensional Hermitian and non-Hermitian tight-binding systems (in the presence/absence of quasiperiodic potential) and an external electric field undergo complete localization with equally spaced eigenenergies, known as the Wannier-Stark (WS) localization. In this work, we demonstrate that when the electric field is slowly modulated with time, new non-trivial phases with multiple mobility edges emerge in place of WS localized phase, which persists up to a certain strength of the non-Hermiticity. On the other hand, for a large driving frequency, we retrieve the usual sharp delocalization-localization transition to the usual (no WS) localized phase, similar to the static non-Hermitian Aubry-André-Harper type without any electric field. This vanishing of WS localization can be attributed solely to the time-periodic drive and occurs irrespective of the non-Hermiticity. Interestingly, under the open boundary condition (OBC), we find that contrary to the undriven systems where an external electric field destroys the SE completely, the SE appears in certain regime of the parameter space when the electric field is temporally driven. This appearance of SE is closely related to the absence of extended unitarity. In addition, in the presence of the drive, the skin states are found to be

multifractal, contrary to its usual nature in such non-Hermitian systems. An in-depth understanding about the behavior of the states in the driven system is established from the long-time dynamics of an initial excitation.

Information Theoretic Signatures of Localization and Mobility Edges in Quasiperiodic Systems

Arpita Goswami. arXiv preprint (2026). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/2603.27175v1

Abstract: We investigate localization transitions and mobility edge phenomena in one-dimensional quasiperiodic lattice models using an information theoretic framework based on the Tsallis entropy of single particle eigenstates.We employ the Tsallis entropy as a continuous, normalized functional of wavefunction amplitudes, where the entropic index q provides a tunable sensitivity to different regions of the probability distribution, enhancing the contribution of localized peaks ($q>1$) or extended components ($q<1$). Building on this framework, we introduce an entropy-gradient susceptibility defined from the energy dependence of the Tsallis entropy, which probes variations in eigenstate structure across the spectrum. We show that this quantity clearly distinguishes global localization transitions from mobility edge physics. In the Aubry Andre model, where all eigenstates undergo a uniform transition, the entropy varies smoothly, resulting in a broad crossover in the susceptibility. In contrast, in systems hosting mobility edges, including a quasiperiodically modulated Su Schrieffer Heeger chain and the generalized Aubry Andre model, the coexistence of localized and extended states produces sharp spectral variations, leading to a pronounced and system size independent peak. The qualitative behavior persists over a broad range of the entropic parameter q, with systematic variations reflecting its role as a tunable probe of spectral structure. Our results establish an information theoretic approach that leverages the continuous q dependence of Tsallis entropy to construct a derivative based measure of spectral heterogeneity, providing a complementary and physically transparent diagnostic of mobility edge phenomena beyond conventional state resolved measures.

Quasilattice-conserved optimization of the atomic structure of decagonal Al-Co-Ni quasicrystals

Xiao-Tian Li; Xiao-Bao Yang; Yu-Jun Zhao. arXiv preprint (2014). Cited by 0. Found by: arxiv. DOI: 10.1088/0256-307X/32/3/036102

Abstract: The detailed atomic structure of quasicrystals has been an open question for decades. Here, we present a quasilattice-conserved optimization method (quasiOPT), with particular quasiperiodic boundary conditions. As the atomic coordinates described by basic cells and quasilattices, we are able to maintain the self-similarity characteristics of qusicrystals with the atomic structure of the boundary region updated timely following the relaxing region. Exemplified with the study of decagonal Al-Co-Ni (d-Al-Co-Ni), we propose a more stable atomic structure model based on Penrose quasilattice and our quasiOPT simulations. In particular, "rectangle-triangle" rules are suggested for the local atomic structures of d-Al-Co-Ni quasicrystals.

Quasiperiodic Skin Criticality in an Exactly Solvable Non-Hermitian Quasicrystal

Zhangyuan Chen; Muhammad Idrees; Ying Yang; Xianqi Tong; Xiaosen Yang. Phys. Rev. B 113, 235423(2026) (2026). Cited by 0. Found by: arxiv. DOI: 10.1103/wlsw-zyq9

Abstract: Critical states in quasiperiodic systems defy the conventional dichotomy between extended and localized states. In this work, we demonstrate that non-Hermiticity fundamentally reshapes this paradigm by giving rise to an exactly solvable quasiperiodic critical phase with no energy selectivity. We introduce a non-Hermitian quasiperiodic lattice based on a modulated Hatano-Nelson model and uncover a new universality class of quasiperiodic skin criticality, in which all eigenstates share an identical multifractal spatial structure. Through a nonunitary gauge transformation, the system is mapped onto a disorder-free lattice, enabling exact analytical solutions for the full spectrum and eigenstates. As a consequence, the inverse participation ratio is strictly energy-independent and controlled solely by a global phase. We further show that this criticality persists in multiband lattices, establishing a general and analytically controlled framework for non-Hermitian quasiperiodic critical phenomena.

Localization characteristics of two-dimensional quasicrystals consisting of metal nanoparticles

Jian-Wen Dong; Kin Hung Fung; C. T. Chan; He-Zhou Wang. Phys. Rev. B 80, 155118 (2009) (2009). Cited by 0. Found by: arxiv. DOI: 10.1103/PhysRevB.80.155118

Abstract: Using the eigen-decomposition method, we investigated the plasmonic modes in a two-dimensional quasicrystalline array of metal nanoparticles. Various properties of the plasmonic modes, such as their symmetry, radiation loss and spatial localization are studied. Some collective plasmonic modes are found to be leaky with

out-of-plane radiation loss, but some modes can have very high fidelity. To facilitate the analysis on the complex behavior of the collective plasmonic modes, we considered a trajectory map in a three dimension parameter space defined by the frequency of the mode, the participation ratio of the mode that gives the spatial distribution, and the eigenvalue that is related to the inverse lifetime of the mode. With the help of the trajectory map method, we found that there are modes that are localized in some special ways and the plasmonic mode with highest spatial localization and highly fidelity is found to be a type of anti-phase ring mode. In general, different localized modes have very different radial decay behaviors. There is no special relationship between the fidelity of the modes and their spatial localization.

Symmetry-breaking dynamics of the finite-size Lipkin-Meshkov-Glick model near ground state

Yi Huang; Tongcang Li; Zhang-qi Yin. Phys. Rev. A 97, 012115 (2018) (2017). Cited by 0. Found by: arxiv. DOI: 10.1103/PhysRevA.97.012115

Abstract: We study the dynamics of the Lipkin-Meshkov-Glick (LMG) model with finite number of spins. In the thermodynamic limit, the ground state of the LMG model with isotropic Hamiltonian in broken phase breaks to a mean-field ground state with a certain direction. However, when the spins number N is finite, the exact ground state is always unique and is not given by a classical mean-field ground state. Here we prove that for N is large but finite, through a tiny external perturbation, a localized state which is close to a mean-field ground state can be prepared, which mimics a spontaneous symmetry breaking (SSB). Besides, we find the localized in-plane spin polarization oscillates with two different frequencies $\sim O(1/N)$, and the lifetime of the localized state is long enough to exhibit this oscillation. We numerically test the analytical results and find that they agree with each other very well. Finally, we link the phenomena to quantum time crystals and quasicrystals.

Entanglement phase transitions in non-Hermitian quasicrystals

Longwen Zhou. Phys. Rev. B 109, 024204 (2024) (2023). Cited by 0. Found by: arxiv. DOI: 10.1103/PhysRevB.109.024204

Abstract: The scaling law of entanglement entropy could undergo qualitative changes during the nonunitary evolution of a quantum many-body system. In this work, we uncover such entanglement phase transitions in one-dimensional non-Hermitian quasicrystals (NHQCs). We identify two types of entanglement transitions with different scaling laws and critical behaviors due to the interplay between non-Hermitian effects and quasiperiodic potentials. The first type represents a typical volume-law to area-law transition, which happens together with a PT-symmetry breaking and a localization transition. The second type features an abnormal log-law to area-law transition, which is mediated by a critical phase with a volume-law scaling in the steady-state entanglement entropy. These entangling phases and transitions are demonstrated in two representative models of NHQCs. Our results thus advanced the study of entanglement transitions in non-Hermitian disordered systems and further disclosed the rich entanglement patterns in NHQCs.

Antiferromagnetism in two-dimensional quasicrystals

Anuradha Jagannathan; Attila Szallas. arXiv preprint (2008). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/0810.

Abstract: Quasiperiodic structures possess long range positional order, but are freed of constraints imposed by translational invariance. For spins interacting via Heisenberg couplings, one may expect therefore to find novel magnetic configurations in such structures. We have studied magnetic properties for simple two dimensional models, as a first step towards understanding experimentally studied magnetic quasicrystals such as the Zn-Mg-R(rare earth) compounds. We analyse properties of the antiferromagnetic ground state and magnon excitation modes for bipartite tilings such as the octagonal and Penrose tilings. In the absence of frustration, one has an inhomogeneous Neel-ordered ground state, with local quantum fluctuations of the local staggered magnetic order parameter. We study the spin wave spectrum and wavefunctions of such antiferromagnets within the linear spin wave approximation. Some results for spin-spin correlations in these systems are discussed.

First-principles evidence for conventional superconductivity in a quasicrystal approximant

Pedro N. Ferreira; Roman Lucrezi; Sangmin Lee; Lucy Nathwani; Matthew Julian; Rohit P. Prasankumar; Warren E. Pickett; Chris J. Pickard; Philip Kim; Christoph Heil. arXiv preprint (2025). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/2511.09224v2

Abstract: Quasicrystals (QCs) host long-range order without translational symmetry, a regime in which the very foundations of BCS theory are not straightforwardly applicable, yet experiments on QCs and their approximant crystals (ACs) point to conventional, s-wave, electron-phonon coupled superconductivity. Here we test the predictive power of the electron-phonon framework in a representative decagonal AC from first principles. Using state-of-the-art \textit{ab initio} methods, we compute the superconducting properties of the recently discovered AC Al$_{13}$Os$_4$ and quantitatively reproduce its bulk T_c. This constitutes, to our knowledge, the first \textit{ab initio} determination of T_c for an AC and establishes that the electron-phonon framework is predictive in these systems as well. Using the generalized quasichemical approximation for alloy modeling in the decagonal Al–Os family, we predict tunable superconductivity in Al$_{13}$Os$_{4-x}$Re$_x$ and Al$_{13}$Os$_{4-x}$Ir$_x$; in particular, Al$_{13}$Re$_4$ is dynamically stable and estimated to have a T_c about 30% above Al$_{13}$Os$_4$. Finally, we discuss the role of ACs as high-fidelity proxies for their parent QCs. Although long-range quasiperiodicity may introduce subtle electronic features, our findings indicate that the key ingredients for superconductivity are already encoded in the local structural motifs preserved by the AC. This places the Al–Os and Al–Re families among the most promising candidates for the highest-T_c quasicrystalline superconductivity.

Quasicrystalline 30° twisted bilayer graphene: Fractal patterns and electronic localization properties

Kevin J. U. Vidarte; Caio Lewenkopf. Frontiers in Carbon 3, 1496179 (2024) (2024). Cited by 0. Found by: arxiv. DOI: 10.3389/frcrb.2024.1496179

Abstract: The recently synthesized 30° twisted bilayer graphene (30°-TBG) systems are unique quasicrystal systems possessing dodecagonal symmetry with graphene's relativistic properties. We employ a real-space numerical atomistic framework that respects both the dodecagonal rotational symmetry and the massless Dirac nature of the electrons to describe the local density of states of the system. The approach we employ is very efficiency for systems with very large unit cells and does not rely on periodic boundary conditions. These features allow us to address a broad class of multilayer two-dimensional crystal with incommensurate configurations, particularly TBGs. Our results reveal that the 30°-TBG electronic spectrum consist of extended states together with a set of localized wave functions. The localized states exhibit fractal patterns consistent with the quasicrystal tiling.

Bulk Localised Transport States in Infinite and Finite Quasicrystals via Magnetic Aperiodicity

Dean Johnstone; Matthew J. Colbrook; Anne E. B. Nielsen; Patrik Öhberg; Callum W. Duncan. Phys. Rev. B 106, 045149 (2022) (2021). Cited by 0. Found by: arxiv. DOI: 10.1103/PhysRevB.106.045149

Abstract: Robust edge transport can occur when particles in crystalline lattices interact with an external magnetic field. This system is well described by Bloch's theorem, with the spectrum being composed of bands of bulk states and in-gap edge states. When the confining lattice geometry is altered to be quasicrystaline, then Bloch's theorem breaks down. However, we still expect to observe the basic characteristics of bulk states and current carrying edge states. Here, we show that for quasicrystals in magnetic fields, there is also a third option; the bulk localised transport states. These states share the in-gap nature of the well-known edge states and can support transport along them, but they are fully contained within the bulk of the system, with no support along the edge. We consider both finite and infinite systems, using rigorous error controlled computational techniques that are not prone to finite-size effects. The bulk localised transport states are preserved for infinite systems, in stark contrast to the normal edge states. This allows for transport to be observed in infinite systems, without any perturbations, defects, or boundaries being introduced. We confirm the in-gap topological nature of the bulk localised transport states for finite and infinite systems by computing common topological measures; namely the Bott index and local Chern marker. The bulk localised transport states form due to a magnetic aperiodicity arising from the interplay of length scales between the magnetic field and quasiperiodic lattice. Bulk localised transport could have interesting applications similar to those of the edge states on the boundary, but that could now take advantage of the larger bulk of the lattice. The infinite size techniques introduced here, especially the calculation of topological measures, could also be widely applied to other crystalline, quasicrystalline, and disordered models.

Correlation-enhanced Friedel oscillations in amorphous alloys and quasicrystals

Johann Kroha. J. Noncryst. Solids, 250-252, 865 (1999) (1998). Cited by 0. Found by: arxiv. DOI: 10.1016/S0022-3093(99)00194-5

Abstract: We show that quantum correlations induced by electron-electron interactions in the presence of random impurity scattering can play an important role in the thermal stabilization of amorphous Hume-Rothery systems: When there is strong backscattering off local, concentrical ion clusters, the static electron density response $ (0,q)$ acquires a powerlaw divergence at $q=2k_F$ even at elevated temperature. This leads to an enhancement as well as to a systematical phase shift of the Friedel oscillations, both consistent with experiments. The possible importance of this effect in icosahedral quasicrystals is discussed.

Monte Carlo study on critical exponents of the classical Heisenberg model in ferromagnetic icosahedral quasicrystal

Shinji Watanabe; Tsunetomo Yamada; Hiroyuki Takakura; Nobuhisa Fujita. Phys. Rev. Research 7 (2025) 043113 (2025). Cited by 0. Found by: arxiv. DOI: 10.1103/4zb8-2zjq

Abstract: Quasicrystals (QCs) lack three-dimensional periodicity of atomic arrangement but possess long-range structural order, which are distinct from periodic crystals and random systems. Here, we show how the ferromagnetic (FM) order arises in the icosahedral QC (i-QC) on the basis of the Monte Carlo simulation of the Heisenberg model on the Yb lattice of Cd$_{5.7}$Yb composed of regular icosahedrons. By finite-size scaling of the Monte Carlo data, we identified the critical exponents of the magnetization, magnetic susceptibility, and spin correlation length, $ =0.508(30)$, $ =1.361(59)$, and $ =0.792(17)$, respectively. We confirmed that our data satisfy the hyperscaling relation and estimated the other critical exponents $ =-0.376(51)$, $ =3.68(23)$, and $ =0.282(65)$. These results show a new universality class inherent in the i-QC, which is different from those in periodic magnets and spin glasses. In the i-QC, each Yb site at vertices of the regular icosahedrons is classified into 8 classes with respect to the coordination numbers of the nearest-neighbor and next-nearest-neighbor bonds. We revealed the FM-transition mechanism by showing that the difference in the local environment of each site is governed by cooperative evolution of spin correlations upon cooling, giving rise to the critical phenomena.

Catalytic mechanism of LENR in quasicrystals based on localized anharmonic vibrations and phasons

Volodymyr Dubinko; Denis Laptev; Klee Irwin. arXiv preprint (2016). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/1609.06625v1

Abstract: Quasicrystals (QCs) are a novel form of matter, which are neither crystalline nor amorphous. Among many surprising properties of QCs is their high catalytic activity. We propose a mechanism explaining this peculiarity based on unusual dynamics of atoms at special sites in QCs, namely, localized anharmonic vibrations (LAVs) and phasons. In the former case, one deals with a large amplitude (~ fractions of an angstrom) time-periodic oscillations of a small group of atoms around their stable positions in the lattice, known also as discrete breathers, which can be excited in regular crystals as well as in QCs. On the other hand, phasons are a specific property of QCs, which are represented by very large amplitude (~angstrom) oscillations of atoms be-tween two quasi-stable positions determined by the geometry of a QC. Large amplitude atomic motion in LAVs and phasons results in time-periodic driving of adjacent potential wells occupied by hydrogen ions (protons or deuterons) in case of hydrogenated QCs. This driving may result in the increase of amplitude and energy of zero-point vibrations (ZPV). Based on that, we demonstrate a drastic increase of the D-D or D-H fusion rate with increasing number of modulation periods evaluated in the framework of Schwinger model, which takes into account suppression of the Coulomb barrier due to lattice vibrations. In this context, we present numerical solution of Schrodinger equation for a particle in a non-stationary double well potential, which is driven time-periodically imitating the action of a LAV or phason. We show that the rate of tunneling of the particle through the potential barrier separating the wells is enhanced drastically by the driving, and it increases strongly with increasing amplitude of the driving. These results support the concept of nuclear catalysis in QCs that can take place at special sites provided by their inherent topology.

A general approach to the exact localized transition points of 1D mosaic disorder models

Yanxia Liu. arXiv preprint (2022). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/2208.02762v2

Abstract: In this paper, we present a general correspondence between the mosaic and non-mosaic models, which can be used to obtain the exact solution for the mosaic ones. This relation holds not only for the quasicrystal models, but also for the Anderson models. Despite the different localization properties of the specific models, this relationship shares a unified form. Applying our method to the mosaic Anderson models, we find that there is a discrete set of extended states. At last, we also give the general analytical mobility edge for the mosaic slowly varying potential models and the mosaic Ganeshan-Pixley-Das Sarma models.

Optical reflectivity as a simple diagnostic method for testing structural quality of icosahedral quasicrystals

Valerie Brien; Anne Dauscher; F. Machizaud. Journal of Applied Physics, American Institute of Physics, 2006, 100 (4), pp.043503 (2020). Cited by 0. Found by: arxiv. DOI: 10.1063/1.2234558

Abstract: Optical reflectivity as a simple diagnostic method for testing structural quality of icosahedral quasicrystals 2 The optical reflectivity of Al-based and Ti-based quasicrystalline and approximant samples were investigated versus the quality of their structural morphology using optical reflectometry, X-ray diffraction and transmission electron microscopy. The different structural morphologies were obtained using three different preparation processes : sintering, pulsed laser deposition and reactive cathodic magnetron sputtering. The work demonstrates that the canonical behaviour of icosahedral state in specular reflectivity is extremely sensitive to different and very fine aspects of the microstructure : sizes of grains smaller than 50 nm, slight local diffuse disorder and shifts away from the icosahedral crystallographic structure (approximants). The work explains why the optical properties of the same kind of quasicrystals found in literature sometimes reveal a different behaviour from one author to another. The study then confirms the work of some authors and definitely shows that the canonical behaviour of icosahedral state in specular reflectivity over the 30000-50000 cm-1 domain is characterized by a decreasing function made of steps. It also shows that this behaviour can be interpreted thanks to the cluster hierarchy of the model of Janot.

Magnetic properties of icosahedral quasicrystals and their cubic approximants in the Cd-Mg-RE (RE = Gd, Tb, Dy, Ho, Er, and Tm) systems

Farid Labib; Daisuke Okuyama; Nobuhisa Fujita; Tsunetomo Yamada; Satoshi Ohhashi; Taku J. Sato; An-Pang Tsai. arXiv preprint (2020). Cited by 0. Found by: arxiv. DOI: 10.1088/1361-648X/ab9343

Abstract: A systematic investigation has been performed to elucidate effects of Rare-Earth (RE) type and local atomic configuration on magnetic properties of icosahedral quasicrystal (iQC) and their cubic approximants (2/1 and 1/1 ACs) in the ternary Cd-Mg-RE (RE = Gd, Tb, Dy, Ho, Er, and Tm) systems. At low temperatures, iQC and 2/1 ACs exhibit spin-glass-like freezing for RE = Gd, Tb, Dy, and Ho, while Er and Tm systems do not show freezing behaviour down to the base temperature ~ 2 K. The 1/1 ACs exhibit either spin-glass-like freezing or antiferromagnetic (AFM) ordering depending on their constituent Mg content. The Tf values show increasing trend from iQC to 2/1 and 1/1 ACs. In contrast, the absolute values of Weiss temperature for iQCs are larger than those in 2/1 and 1/1 ACs, indicating that the total AFM interactions between the neighboring spins are larger in aperiodic, rather than periodic systems. Competing spin interactions originating from the long-range Ruderman-Kittel-Kasuya-Yoshida mechanism along with chemical disorder of Cd/Mg ions presumably account for the observed spin-glass-like behavior in Cd-Mg-RE iQCs and ACs.

Delone measures of finite local complexity and applications to spectral theory of one-dimensional continuum models of quasicrystals

Steffen Klassert; Daniel Lenz; Peter Stollmann. arXiv preprint (2010). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/1003.3574v1

Abstract: We study measures on the real line and present various versions of what it means for such a measure to take only finitely many values. We then study perturbations of the Laplacian by such measures. Using Kotani-Remling theory, we show that the resulting operators have empty absolutely continuous spectrum if the measures are not periodic. When combined with Gordon type arguments this allows us to prove purely singular continuous spectrum for some continuum models of quasicrystals.

Self-duality of One-dimensional Quasicrystals with Spin-Orbit Interaction

Deepak Kumar Sahu; Aruna Prasad Acharya; Debajyoti Choudhuri; Sanjoy Datta. Phys. Rev. B 104, 054202 (2021) (2021). Cited by 0. Found by: arxiv. DOI: 10.1103/PhysRevB.104.054202

Abstract: Non-interacting spinless electrons in one-dimensional quasicrystals, described by the Aubry-André-Harper (AAH) Hamiltonian with nearest neighbour hopping, undergoes metal to insulator transition (MIT) at a critical strength of the quasi-periodic potential. This transition is related to the self-duality of the AAH Hamiltonian. Interestingly, at the critical point, which is also known as the self-dual point, all the single particle wave functions are multifractal or non-ergodic in nature, while they are ergodic and delocalized (localized) below (above) the critical point. In this work, we have studied the one dimensional quasi-periodic AAH Hamiltonian in the presence of spin-orbit (SO) coupling of Rashba type, which introduces an additional spin conserving complex

hopping and a spin-flip hopping. We have found that, although the self-dual nature of AAH Hamiltonian remains unaltered, the self-dual point gets affected significantly. Moreover, the effect of the complex and spin-flip hoppings are identical in nature. We have extended the idea of Kohn's localization tensor calculations for quasi-particles and detected the critical point very accurately. These calculations are followed by detailed multifractal analysis along with the computation of inverse participation ratio and von Neumann entropy, which clearly demonstrate that the quasi-particle eigenstates are indeed multifractal and non-ergodic at the critical point. Finally, we mapped out the phase diagram in the parameter space of quasi-periodic potential and SO coupling strength.

Real-space Formalism for the Euler Class and Fragile Topology in Quasicrystals and Amorphous Lattices

Dexin Li; Citian Wang; Huaqing Huang. SciPost Phys. 17, 086 (2024) (2023). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/2311.01557v1

Abstract: We propose a real-space formalism of the topological Euler class, which characterizes the fragile topology of two-dimensional systems with real wave functions. This real-space description is characterized by local Euler markers whose macroscopic average coincides with the Euler number, and it applies equally well to periodic and open boundary conditions for both crystals and noncrystalline systems. We validate this by diagnosing topological phase transitions in clean and disordered crystalline systems with the reality endowed by the space-time inversion symmetry $\mathcal{I}_{ST}$. Furthermore, we demonstrated the topological Euler phases in quasicrystals and even in amorphous lattices lacking any spatial symmetries. Our work not only provides a local characterization of the fragile topology but also significantly extends its territory beyond $\mathcal{I}_{ST}$-symmetric crystalline materials.

Logarithmically Correlated Landscapes and Localization in Non-Hermitian Quasicrystals

Xianqi Tong; Qifeng Ding; Xiaosen Yang. arXiv preprint (2026). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/2608.11700v1

Abstract: We study a one-dimensional Hatano-Nelson ring whose nonreciprocal hopping is quasiperiodically modulated through zero. A gauge transformation maps every eigenstate onto a single spatial envelope, whose logarithm becomes a deterministic, logarithmically correlated field once the hopping vanishes along the quasiperiodic orbit. We derive an exact Fourier representation and prove that the landscape variance grows logarithmically with system size, with a stiffness set by the modulation power and the arithmetic of the incommensurate frequency. The extended phase terminates at an algebraic boundary given by Jensen's formula. Inside the singular regime, localization requires the stiffness to exceed a critical threshold: above it, the wavefunction weight concentrates on the few highest landscape peaks and the state is localized; below it, the weight spreads over too many competing peaks and the state is a critical multifractal. The extreme-value statistics are anomalous: peak gaps grow as a power of the logarithm of rank, with an exponent that encodes the continued-fraction type of the frequency, distinct from the Anderson, Aubry-André, and random-gauge universality classes. The stiffness adds across channels in multiband lattices, and all signatures survive percent-level component disorder, placing the mechanism within reach of nonreciprocal topolectrical circuits.

Long-range Magnetic Order in Models for Rare Earth Quasicrystals

Stefanie Thiem; J. T. Chalker. Phys. Rev. B 92, 224409 (2015) (2015). Cited by 0. Found by: arxiv. DOI: 10.1103/PhysRevB.92.224409

Abstract: We take a two-step theoretical approach to study magnetism of rare earth quasicrystals by considering Ising spins on quasiperiodic tilings, coupled via RKKY interactions. First, we compute RKKY interactions from a tight-binding Hamiltonian defined on the two-dimensional quasiperiodic tilings. We find that the magnetic interactions are frustrated and strongly dependent on the local environment. This results in the formation of clusters with strong bonds at certain patterns of the tilings that repeat quasiperiodically. Second, we examine the statistical mechanics of Ising spins with these RKKY interactions, using extensive Monte Carlo simulations. Although models that have frustrated interactions and lack translational invariance might be expected to display spin glass behaviour, we show that the spin system has a phase transition to low-temperature states with long-range quasiperiodic magnetic order. Additionally, we find that in some of the systems spin clusters can fluctuate much below the ordering temperature.

Exact mobility edges, $\mathcal{PT}$-symmetry breaking and skin effect in one-dimensional non-Hermitian quasicrystals

Yanxia Liu; Yucheng Wang; Xiong-Jun Liu; Qi Zhou; Shu Chen. Phys. Rev. B 103, 014203 (2021) (2020). Cited by 0. Found by: arxiv. DOI: 10.1103/PhysRevB.103.014203

Abstract: We propose a general analytic method to study the localization transition in one-dimensional quasicrystals with parity-time ($\mathcal{PT}$) symmetry, described by complex quasiperiodic mosaic lattice models. By applying Avila's global theory of quasiperiodic Schrödinger operators, we obtain exact mobility edges and prove that the mobility edge is identical to the boundary of $\mathcal{PT}$-symmetry breaking, which also proves the existence of correspondence between extended (localized) states and $\mathcal{PT}$-symmetry ($\mathcal{PT}$-symmetry-broken) states. Furthermore, we generalize the models to more general cases with non-reciprocal hopping, which breaks $\mathcal{PT}$ symmetry and generally induces skin effect, and obtain a general and analytical expression of mobility edges. While the localized states are not sensitive to the boundary conditions, the extended states become skin states when the periodic boundary condition is changed to open boundary condition. This indicates that the skin states and localized states can coexist with their boundary determined by the mobility edges.

Anomalous electronic transport in Quasicrystals and related Complex Metallic Alloys

Guy Trambly de Laissardière; Didier Mayou. C. R. Physique 15 (2014) 70-81 (2013). Cited by 0. Found by: arxiv. DOI: 10.1016/j.crhy.2013.09.010

Abstract: We analyze the transport properties in approximants of quasicrystals alpha-AlMnSi, 1/1-AlCuFe and for the complex metallic phase lambda-AlMn. These phases presents strong analogies in their local atomic structures and are related to existing quasicrystalline phases. Experimentally they present unusual transport properties with low conductivities and a mix of metallic-like and insulating-like characteristics. We compute the band structure and the quantum diffusion in the perfect structure without disorder and introduce simple approximations that allow to treat the effect of disorder. Our results demonstrate that the standard Bloch-Boltzmann theory is not applicable to these intermetallic phases. Indeed their dispersion relation are flat indicating small band velocities and corrections to quantum diffusion that are not taken into account in the semi-classical Bloch-Boltzmann scheme become dominant. We call this regime the small velocity regime. A simple Relaxation Time Approximation to treat the effect of disorder allows us to reproduce the main experimental facts on conductivity qualitatively and even quantitatively.

Macroscopically degenerate localized zero-energy states of quasicrystalline bilayer systems in strong coupling limit

Hyunsoo Ha; Bohm-Jung Yang. Phys. Rev. B 104, 165112 (2021) (2021). Cited by 0. Found by: arxiv. DOI: 10.1103/PhysRevB.104.165112

Abstract: When two identical two-dimensional (2D) periodic lattices are stacked in parallel after rotating one layer by a certain angle relative to the other layer, the resulting bilayer system can lose lattice periodicity completely and become a 2D quasicrystal. Twisted bilayer graphene with 30-degree rotation is a representative example. We show that such quasicrystalline bilayer systems generally develop macroscopically degenerate localized zero-energy states (ZESs) in strong coupling limit where the interlayer couplings are overwhelmingly larger than the intralayer couplings. The emergent chiral symmetry in strong coupling limit and aperiodicity of bilayer quasicrystals guarantee the existence of the ZESs. The macroscopically degenerate ZESs are analogous to the flat bands of periodic systems, in that both are composed of localized eigenstates, which give divergent density of states. For monolayers, we consider the triangular, square, and honeycomb lattices, comprised of homogenous tiling of three possible planar regular polygons: the equilateral triangle, square, and regular hexagon. We construct a compact theoretical framework, which we call the quasiband model, that describes the low energy properties of bilayer quasicrystals and counts the number of ZESs using a subset of Bloch states of monolayers. We also propose a simple geometric scheme in real space which can show the spatial localization of ZESs and count their number. Our work clearly demonstrates that bilayer quasicrystals in strong coupling limit are an ideal playground to study the intriguing interplay of flat band physics and the aperiodicity of quasicrystals.

Quantum transport in a one-dimensional quasicrystal with mobility edges

Yan Xing; Lu Qi; Xuedong Zhao; Zhe Lü; Shutian Liu; Shou Zhang; Hong-Fu Wang. Physical Review A 105, 032443 (2022) (2022). Cited by 0. Found by: arxiv. DOI: 10.1103/PhysRevA.105.032443

Abstract: Quantum transport in a one-dimensional (1D) quasiperiodic lattice with mobility edges is explored. We first investigate the adiabatic pumping between left and right edge modes by resorting to two edge-bulk-edge channels and demonstrate that the success or failure of the adiabatic pumping depends on whether the corresponding bulk subchannel undergoes a localization-delocalization transition. Compared with the paradigmatic Aubry-André (AA) model, the introduction of mobility edges triggers an opposite outcome for successful pumping in the two channels, showing a discrepancy of critical condition, and facilitates the robustness of the adiabatic pumping against quasidisorder. We also consider the transfer between excitations at both boundaries of the lattice and an anomalous phenomenon characterized by the enhanced quasidisorder contributing to the excitation transfer is found. Furthermore, there exists a parametric regime where a nonreciprocal effect emerges in the presence of mobility edges, which leads to a unidirectional transport for the excitation transfer and enables potential applications in the engineering of quantum diodes.

Super Hamiltonian in superspace for incommensurate superlattices and quasicrystals

Manuel Valiente; Callum W. Duncan; Nikolaj T. Zinner. arXiv preprint (2019). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/1908.03214v1

Abstract: Infinite quasiperiodic arrangements in space, such as quasicrystals, are typically described as projections of higher-dimensional periodic lattices onto the physical dimension. The concept of a reference higher-dimensional space, called a superspace, has proved useful in relation to quasiperiodic systems. Although some quantum-mechanical systems in quasiperiodic media have been shown to admit quasiperiodic states, any sort of general Hamiltonian formalism in superspace is lacking to this date. Here, we show how to extend generic quantum-mechanical Hamiltonians to higher dimensions in such a way that eigenstates of the original Hamiltonian are obtained as projections of the Hamiltonian in superspace, which we call the super Hamiltonian. We apply the super Hamiltonian formalism to a simple, yet realistic one-dimensional quantum particle in a quasiperiodic potential without the tight-binding approximation, and obtain continuously labelled eigenstates of the system corresponding to a continuous spectrum. All states corresponding to the continuum are quasiperiodic. We also obtain the Green's functions for continuum states in closed form and, from them, the density of states and local density of states, and scattering states off defects and impurities. The closed form of this one-dimensional Green's function is equally valid for any continuum state in any one-dimensional single-particle quantum system admitting continuous spectrum. With the basis set we use, which is periodic in superspace, and therefore quasiperiodic in physical space, we find that Anderson-localised states are also quasiperiodic if distributional solutions are admitted, but circumvent this difficulty by generalising the superspace method to open boundary conditions (cont'd).

Hierarchical Structures of Quantum Geometric Spectrum in Quasicrystals: A Renormalization-Group Study

Jundi Wang; Yuxiao Chen; Huaqing Huang. Physical Review Letters (2026) (2025). Cited by 0. Found by: arxiv. DOI: 10.1103/v4ky-6v45

Abstract: Quantum geometry, characterized by the quantum metric and Berry curvature, is a powerful framework for understanding diverse physical phenomena in quantum materials, but its behavior in non-periodic systems remains largely uncharted. Here, we uncover a universal mechanism for the divergent enhancement of the quantum metric in one-dimensional quasiperiodic systems, governed by the interplay of wavefunction criticality and spectral fractality. Using the paradigmatic Fibonacci chain, we demonstrate that the quantum metric displays a hierarchical scaling structure that mirrors the fractal organization of the energy spectrum. A real-space renormalization-group analysis yields an analytic power-law scaling, $\mathcal{G} \propto (ΔE)^{-k}$, between the quantum metric $\mathcal{G}$ and spectral gap $ΔE$, with the exponent k dictated by the system's self-similarity. This scaling persists in the critical Aubry-André-Harper model but disappears in both its localized and extended phases, confirming its universality across different quasiperiodic paradigms and its unique link to criticality. Our results show that the quantum metric provides a sensitive geometric indicator of quasiperiodic criticality, and highlight quasicrystals as promising platforms for realizing unconventional giant quantum geometric effects beyond the limits of periodic crystals.

Topological delocalization transitions and mobility edges in the nonreciprocal Maryland model

Longwen Zhou; Yongjian Gu. J. Phys.: Condens. Matter 34, 115402 (2022) (2021). Cited by 0. Found by: arxiv. DOI: 10.1088/1361-648X/ac4530

Abstract: Non-Hermitian effects could trigger spectrum, localization and topological phase transitions in quasiperiodic lattices. We propose a non-Hermitian extension of the Maryland model, which forms a paradigm

in the study of localization and quantum chaos by introducing asymmetry to its hopping amplitudes. The resulting nonreciprocal Maryland model is found to possess a real-to-complex spectrum transition at a finite amount of hopping asymmetry, through which it changes from a localized phase to a mobility edge phase. Explicit expressions of the complex energy dispersions, phase boundaries and mobility edges are found. A topological winding number is further introduced to characterize the transition between different phases. Our work introduces a unique type of non-Hermitian quasicrystal, which admits exactly obtainable phase diagrams, mobility edges, and holding no extended phases at finite nonreciprocity in the thermodynamic limit.

Exact non-Hermitian mobility edges in one-dimensional quasicrystal lattice with exponentially decaying hopping and its dual lattice

Yanxia Liu; Yongjian Wang; Zuohuan Zheng; Shu Chen. Phys. Rev. B 103, 134208 (2021) (2020). Cited by 0. Found by: arxiv. DOI: 10.1103/PhysRevB.103.134208

Abstract: We analytically determine the non-Hermitian mobility edges of a one-dimensional quasiperiodic lattice model with exponential decaying hopping and complex potentials as well as its dual model, which is just a non-Hermitian generalization of the Ganeshan-Pixley-Das Sarma model with nonreciprocal nearest-neighboring hopping. The presence of non-Hermitian term destroys the self-duality symmetry and thus prevents us exploring the localization-delocalization point through looking for self-dual points. Nevertheless, by applying Avila's global theory, the Lyapunov exponent of the Ganeshan-Pixley-Das Sarma model can be exactly derived, which enables us to get an analytical expression of mobility edge of the non-Hermitian dual model. Consequently, the mobility edge of the original model is obtained by using the dual transformation, which creates exact mappings between the spectra and wavefunctions of these two models.

Spectral and Diffusive Properties of Silver-Mean Quasicrystals in 1,2, and 3 Dimensions

V. Z. Cerovski; M. Schreiber; U. Grimm. Phys. Rev. B 72, 054203 (2005) (2004). Cited by 0. Found by: arxiv. DOI: 10.1103/PhysRevB.72.054203

Abstract: Spectral properties and anomalous diffusion in the silver-mean (octonacci) quasicrystals in d=1,2,3 are investigated using numerical simulations of the return probability C(t) and the width of the wave packet w(t) for various values of the hopping strength v. In all dimensions we find C(t)\sim t^{- }, with results suggesting a crossover from <1 to =1 when v is varied in d=2,3, which is compatible with the change of the spectral measure from singular continuous to absolute continuous; and we find w(t)\sim t^ with 0< (v)<1 corresponding to anomalous diffusion. Results strongly suggest that (v) is independent of d. The scaling of the inverse participation ratio suggests that states remain delocalized even for very small hopping amplitude v. A study of the dynamics of initially localized wavepackets in large three-dimensional quasiperiodic structures furthermore reveals that wavepackets composed of eigenstates from an interval around the band edge diffuse faster than those composed of eigenstates from an interval of the band-center states: while the former diffuse anomalously, the latter appear to diffuse slower than any power law.

Decoding flat bands from compact localized states

Yuge Chen; Juntao Huang; Kun Jiang; Jiangping Hu. arXiv preprint (2022). Cited by 0. Found by: arxiv. DOI: 10.1016/j.scib.2023.11.032

Abstract: The flat band system is an ideal quantum platform to investigate the kaleidoscope created by the electron-electron correlation effects. The central ingredient of realizing a flat band is to find its compact localized states. In this work, we develop a systematic way to generate the compact localized states by designing destructive interference pattern from 1-dimensional chains. A variety of 2-dimensional new flat band systems are constructed with this method. Furthermore, we show that the method can be extended to generate the compact localized states in multi-orbital systems by carefully designing the block hopping scheme, as well as in quasicrystal and disorder systems.

Magnetic and transport properties of i-R-Cd icosahedral quasicrystals (R = Y, Gd-Tm)

Tai Kong; Sergey L. Bud'ko; Anton Jesche; John McArthur; Andreas Kreyssig; Alan I. Goldman; Paul C. Canfield. arXiv preprint (2014). Cited by 0. Found by: arxiv. DOI: 10.1103/PhysRevB.90.014424

Abstract: We present a detailed characterization of the recently discovered i-R-Cd (R = Y, Gd-Tm) binary quasicrystals by means of x-ray diffraction, temperature-dependent dc and ac magnetization, temperature-dependent resistance and temperature-dependent specific heat measurements. Structurally, the broadening of x-ray diffraction peaks found for i-R-Cd is dominated by frozen-in phason strain, which is essentially independent of R. i-Y-Cd is weakly diamagnetic and manifests a temperature-independent susceptibility. i-Gd-Cd can be characterized as a spin-glass below 4.6 K via dc magnetization cusp, a third order non-linear magnetic susceptibility peak, a frequency-dependent freezing temperature and a broad maximum in the specific heat. i-R-Cd (R = Ho-Tm) is similar to i-Gd-Cd in terms of features observed in thermodynamic measurements. i-Tb-Cd and i-Dy-Cd do not show a clear cusp in their zero-field-cooled dc magnetization data, but instead show a more rounded, broad local maximum. The resistivity for i-R-Cd is of order 300 $ Ω$ cm and weakly temperature-dependent. The characteristic freezing temperatures for i-R-Cd (R = Gd-Tm) deviate from the de Gennes scaling, in a manner consistent with crystal electric field splitting induced local moment anisotropy.

Mobility Crossover in Two-Dimensional Berry Crystals

Zixuan Chai; Si-Yuan Chen; Chenzheng Yu; Anton M. Graf; Joonas Keski-Rahkonen; Eric J. Heller. arXiv preprint (2024). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/2412.07117v3

Abstract: A Berry crystal is a random superposition of N plane waves of equal amplitude and fixed wavevector magnitude, propagating in different directions. Using numerical simulations of wavepacket dynamics, spectral analysis based on autocorrelation functions, and scaling of Inverse Participation Ratio, the nature of eigenstates across the energy spectrum of a two-dimensional Berry crystal is characterized. It exhibits Anderson localization and critical (extended but non-ergodic) states, reminiscent of quasicrystals, which sit in the middle ground between periodic and disordered systems and can host critical states. However, in contrast to quasicrystals that display sharp mobility edges separating extended and localized phases, the Berry crystal exhibits an extended regimes of critical states. We name this a "mobility crossover". At weak potential strength, low-energy states are extended while higher-energy states near the backscattering momentum are critical. As the potential strength increases to become comparable with the recoil energy, these critical states evolve into localized states, yielding a transition from extended non-ergodic to localized behavior near the backscattering momentum. An estimate for the boundaries of ergodic extended regimes is given by the mapping onto an effective Anderson model on a Bethe Lattice. The results shed light on the relation between backscattering and Anderson localization in continuous two-dimensional aperiodic systems.

Exact complex mobility edges and flagellate spectra for non-Hermitian quasicrystals with exponential hoppings

Li Wang; Jiaqi Liu; Zhenbo Wang; Shu Chen. Phys. Rev. B 110, 144205 (2024) (2024). Cited by 0. Found by: arxiv. DOI: 10.1103/PhysRevB.110.144205

Abstract: We propose a class of general non-Hermitian quasiperiodic lattice models with exponential hoppings and analytically determine the genuine complex mobility edges by solving its dual counterpart exactly utilizing Avila's global theory. Our analytical formula unveils that the complex mobility edges usually form a loop structure in the complex energy plane. By shifting the eigenenergy a constant t, the complex mobility edges of the family of models with different hopping parameter t can be described by a unified formula, formally independent of t. By scanning the hopping parameter, we demonstrate the existence of a type of intriguing flagellate-like spectra in complex energy plane, in which the localized states and extended states are well separated by the complex mobility edges. Our result provides a firm ground for understanding the complex mobility edges in non-Hermitian quasiperiodic lattices.

Non-Hermitian mobility edges in one-dimensional quasicrystals with parity-time symmetry

Yanxia Liu; Xiang-Ping Jiang; Junpeng Cao; Shu Chen. Phys. Rev. B 101, 174205 (2020) (2020). Cited by 0. Found by: arxiv. DOI: 10.1103/PhysRevB.101.174205

Abstract: We investigate localization-delocalization transition in one-dimensional non-Hermitian quasiperiodic lattices with exponential short-range hopping, which possess parity-time ($\mathcal{PT}$) symmetry. The localization transition induced by the non-Hermitian quasiperiodic potential is found to occur at the $\mathcal{PT}$-symmetry-breaking point. Our results also demonstrate the existence of energy dependent mobility edges, which separate the extended states from localized states and are only associated with the real part of eigen-energies. The level statistics and Loschmidt echo dynamics are also studied.

The local atomic quasicrystal structure of the icosahedral Mg25Y11Zn64 alloy

S. Bruehne; E. Uhrig; C. Gross; W. Assmus; A. S. Masadeh; S. J. L. Billinge. J. Phys.: Condens. Matter 17 (2005) 1561-1572 (2004). Cited by 0. Found by: arxiv. DOI: 10.1088/0953-8984/17/10/011

Abstract: A local and medium range atomic structure model for the face centred icosahedral (fci) Mg25Y11Zn64 alloy has been established in a sphere of r = 27 A. The model was refined by least squares techniques using the atomic pair distribution (PDF) function obtained from synchrotron powder diffraction. Three hierarchies of the atomic arrangement can be found: (i) five types of local coordination polyhedra for the single atoms, four of which are of Frank-Kasper type. In turn, they (ii) form a three-shell (Bergman) cluster containing 104 atoms, which is condensed sharing its outer shell with its neighbouring clusters and (iii) a cluster connecting scheme corresponding to a three-dimensional tiling leaving space for few glue atoms. Inside adjacent clusters, Y8-cubes are tilted with respect to each other and thus allow for overall icosahedral symmetry. It is shown that the title compound is essentially isomorphic to its holmium analogue. Therefore fci-Mg-Y-Zn can be seen as the representative structure type for the other rare earth analogues fci-Mg-Zn-RE (RE = Dy, Er, Ho, Tb) reported in the literature.

Classifying extended, localized and critical states in quasiperiodic lattices via unsupervised learning

Bohan Zheng; Siyu Zhu; Xingping Zhou; Tong Liu. arXiv preprint (2024). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/2410.15061v1

Abstract: Classification of quantum phases is one of the most important areas of research in condensed matter physics. In this work, we obtain the phase diagram of one-dimensional quasiperiodic models via unsupervised learning. Firstly, we choose two advanced unsupervised learning algorithms, Density-Based Spatial Clustering of Applications with Noise (DBSCAN) and Ordering Points To Identify the Clustering Structure (OPTICS), to explore the distinct phases of Aubry-André-Harper model and quasiperiodic p-wave model. The unsupervised learning results match well with traditional numerical diagonalization. Finally, we compare the similarity of different algorithms and find that the highest similarity between the results of unsupervised learning algorithms and those of traditional algorithms has exceeded 98\%. Our work sheds light on applications of unsupervised learning for phase classification.

Strictly Localized States on the Socolar Dodecagonal Lattice

M. Akif Keskiner; M. Ö Oktel. Phys. Rev. B 106, 064207 (2022) (2022). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/2207.05552v2

Abstract: Socolar dodecagonal lattice is a quasicrystal closely related to the better-known Ammann-Beenker and Penrose lattices. The cut and project method generates this twelve-fold rotationally symmetric lattice from the six-dimensional simple cubic lattice. We consider the vertex tight-binding model on this lattice and use the acceptance domains of the vertices in perpendicular space to count the frequency of strictly localized states. We numerically find that these states span $f_{\mathrm{Num}}\simeq 7.61$ \% of the Hilbert space. We give 18 independent localized state types and calculate their frequencies. These localized state types provide a lower bound of $f_{\mathrm{LS}} =\frac{10919-6304\sqrt{3}}{2} \simeq 0.075854$, accounting for more than 99% of the zero-energy manifold. Numerical evidence points to larger localized state types with smaller frequencies, similar to the Ammann-Beenker lattice. On the other hand, we find sites forbidden by local connectivity to host localized states. Forbidden sites do not exist for the Ammann-Beenker lattice but are common in the Penrose lattice. We find a lower bound of $f_{\mathrm{Forbid}}\simeq 0.038955$ for the frequency of forbidden sites. Finally, all the localized state types we find can be chosen to have constant density and alternating signs over their support, another feature shared with the Ammann-Beenker lattice.

Fate of Majorana zero modes, critical states and non-conventional real-complex transition in non-Hermitian quasiperiodic lattices

Tong Liu; Shujie Cheng; Hao Guo; Gao Xianlong. Phys. Rev. B 103, 104203 (2021) (2020). Cited by 0. Found by: arxiv. DOI: 10.1103/PhysRevB.103.104203

Abstract: We study a one-dimensional p-wave superconductor subject to non-Hermitian quasiperiodic potentials. Although the existence of the non-Hermiticity, the Majorana zero mode is still robust against the disorder perturbation. The analytic topological phase boundary is verified by calculating the energy gap closing point and the topological invariant. Furthermore, we investigate the localized properties of this model, revealing that

the topological phase transition is accompanied with the Anderson localization phase transition, and a wide critical phase emerges with amplitude increments of the non-Hermitian quasiperiodic potentials. Finally, we numerically uncover a non-conventional real-complex transition of the energy spectrum, which is different from the conventional $\mathcal{PT}$ symmetric transition.

Two-body interaction induced phase transitions and intermediate phases in nonreciprocal non-Hermitian quasicrystals

Yalun Zhang; Longwen Zhou. Phys. Rev. B 111, 064204 (2025) (2024). Cited by 0. Found by: arxiv. DOI: 10.1103/PhysRevB.111.064204

Abstract: Non-Hermitian phenomena, such as exceptional points, non-Hermitian skin effects, and topologically nontrivial phases have attracted continued attention. In this work, we reveal how interactions and nonreciprocal hopping could collectively influence the behavior of two interacting bosons on quasiperiodic lattices. Focusing on the Bose-Hubbard model with Aubry-André-Harper quasiperiodic modulations and hopping asymmetry, we discover that interactions could enlarge the localization transition point of the noninteracting system into an intermediate mobility edge phase, in which localized doublons formed by bosonic pairs can coexist with delocalized states. Under the open boundary condition, the bosonic doublons could further show non-Hermitian skin effects, realizing doublon condensation at the edges, and their direction of skin-localization can be flexibly tuned by the hopping parameters. A framework is developed to characterize the spectral, localization, and topological transitions accompanying these phenomena. Our work advances the understanding of localization and topological phases in non-Hermitian systems, particularly in relation to multiparticle interactions.

Conduction in quasi-periodic and quasi-random lattices: Fibonacci, Riemann, and Anderson models

Vipin Kerala Varma; Sebastiano Pilati; Vladimir E. Kravtsov. Phys. Rev. B 94, 214204 (2016) (2016). Cited by 0. Found by: arxiv. DOI: 10.1103/PhysRevB.94.214204

Abstract: We study the ground state conduction properties of noninteracting electrons in aperiodic but non-random one-dimensional models with chiral symmetry, and make comparisons against Anderson models with non-deterministic disorder. The first model we consider is the Fibonacci lattice, which is a paradigmatic model of quasicrystals; the second is the Riemann lattice, which we define inspired by Dyson's proposal on the possible connection between the Riemann hypothesis and a suitably defined quasicrystal. Our analysis is based on Kohn's many-particle localization tensor defined within the modern theory of the insulating state. In the Fibonacci quasicrystal, where all single-particle eigenstates are critical (i.e., intermediate between ergodic and localized), the noninteracting electron gas is found to be a conductor at most electron densities, including the half-filled case; however, at various specific fillings $ $, including the values $ = 1/g^n$, where g is the golden ratio and n is any integer, the gas turns into an insulator due to spectral gaps. Metallic behaviour is found at half-filling in the Riemann lattice as well; however, in contrast to the Fibonacci quasicrystal, the Riemann lattice is generically an insulator due to single-particle eigenstate localization, likely at all other fillings. Its behaviour turns out to be alike that of the off-diagonal Anderson model, albeit with different system-size scaling of the band-centre anomalies. The advantages of analysing the Kohn's localization tensor instead of other measures of localization familiar from the theory of Anderson insulators (such as the participation ratio or the Lyapunov exponent) are highlighted.

Lattice Bounded Distance Equivalence for 1D Delone Sets with Finite Local Complexity

Petr Ambrož; Zuzana Masáková; Edita Pelantová. arXiv preprint (2020). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/2010.03884v1

Abstract: Spectra of suitably chosen Pisot-Vijayaraghavan numbers represent non-trivial examples of self-similar Delone point sets of finite local complexity, indispensable in quasicrystal modeling. For the case of quadratic Pisot units we characterize, dependingly on digits in the corresponding numeration systems, the spectra which are bounded distance to an average lattice. Our method stems in interpretation of the spectra in the frame of the cut-and-project method. Such structures are coded by an infinite word over a finite alphabet which enables us to exploit combinatorial notions such as balancedness, substitutions and the spectrum of associated incidence matrices.

Interlaced wire medium with quasicrystal lattice

Eugene A. Koreshin; Mikhail V. Rybin. arXiv preprint (2022). Cited by 0. Found by: arxiv. DOI: 10.1103/PhysRevB.105.024307

Abstract: We propose a design of interlaced wire medium with quasicrystalline lattice based on five-fold rotation symmetry Penrose tiling. The transport properties of this structure are studied. We distinguish two transport regimes, namely, propagation regime related to the low-frequency interval and localization regime in the high-frequency interval. While the former is observed in structures both with and without translation symmetry property, the latter is exclusive for aperiodic structures only. We show that the localization regime is promising for many applications including engineering of effective multi-channel devices for telecommunication and imaging systems.

Local growth of icosahedral quasicrystalline tilings

Connor Hann; Joshua E. S. Socolar; Paul J. Steinhardt. Phys. Rev. B 94, 014113 (2016) (2016). Cited by 0. Found by: arxiv. DOI: 10.1103/PhysRevB.94.014113

Abstract: Icosahedral quasicrystals (IQCs) with extremely high degrees of translational order have been produced in the laboratory and found in naturally occurring minerals, yet questions remain about how IQCs form. In particular, the fundamental question of how locally determined additions to a growing cluster can lead to the intricate long-range correlations in IQCs remains open. In answer to this question, we have developed an algorithm that is capable of producing a perfectly ordered IQC, yet relies exclusively on local rules for sequential, face-to-face addition of tiles to a cluster. When the algorithm is seeded with a special type of cluster containing a defect, we find that growth is forced to infinity with high probability and that the resultant IQC has a vanishing density of defects. The geometric features underlying this algorithm can inform analyses of experimental systems and numerical models that generate highly ordered quasicrystals.

Mobility rings in a non-Hermitian non-Abelian quasiperiodic lattice

Rui-Jie Chen; Guo-Qing Zhang; Zhi Li; Dan-Wei Zhang. Phys. Rev. A 112, 013320 (2025) (2025). Cited by 0. Found by: arxiv. DOI: 10.1103/zfrn-53lz

Abstract: We study localization and topological properties in spin-1/2 non-reciprocal Aubry-André chain with SU(2) non-Abelian artificial gauge fields. The results reveal that, different from the Abelian case, mobility rings, will emerge in the non-Abelian case accompanied by the non-Hermitian topological phase transition. As the non-Hermitian extension of mobility edges, such mobility rings separate Anderson localized eigenstates from extended eigenstates in the complex energy plane under the periodic boundary condition. Based on the topological properties, we obtain the exact expression of the mobility rings. Furthermore, the corresponding indicators such as inverse participation rate, normalized participation ratio, winding number, non-Hermitian spectral structures and wave functions are numerically studied. The numerical results are in good agreement with the analytical expression, which confirms the emergence of mobility rings.

Anyon Quasilocalization in a Quasicrystalline Toric Code

Soumya Sur; Mohammad Saad; Adhip Agarwala. arXiv preprint (2025). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/2511.17144v2

Abstract: An exactly solvable model of a quantum spin liquid on a quasicrystal, akin to Kitaev's honeycomb model, was introduced in Kim \textit{et al.}, \href{https://doi.org/10.1103/PhysRevB.110.214438}{\text{Phys. Rev. B} \textbf{110}, 214438 (2024)}. It was shown that in contrast to the translationally invariant models, such a spin liquid stabilizes a gapped ground state with a finite irrational flux density. In this work, we analyze the strong bond-anisotropic limit of the model and demonstrate that the aperiodic lattice geometry naturally generates a hierarchy of exponentially separated coupling constants in the resulting toric code Hamiltonian. Furthermore, a perturbative magnetic field leads to anomalous localization properties where an anyonic excitation sequentially delocalizes over subsets of sites forming equipotential contours in the quasicrystal. In addition, certain background flux configurations, together with the underlying geometry, give rise to strictly localized eigenstates that remain decoupled from the rest of the spectrum. Using numerical studies, we uncover the key mechanisms responsible for this unconventional localization behavior. Our study highlights that topologically ordered phases, in the presence of geometrical constraints can lead to highly anomalous localization properties of fractionalized charges.

Breakdown of the correspondence between the real-complex and delocalization-localization transitions in non-Hermitian quasicrystals

Wen Chen; Shujie Cheng; Ji Lin; Reza Asgari; Gao Xianlong. arXiv preprint (2022). Cited by 0. Found by: arxiv. DOI: 10.1103/PhysRevB.106.144208

Abstract: The correspondence between the real-complex transition in energy and delocalization-localization transition is well-established in a class of Aubry-Andr'e-Harper model with exponential non-Hermitian on-site potentials. In this paper, we study a generalized Aubry-Andr'e model with off-diagonal modulation and non-Hermitian on-site potential. We find that, when there exists an incommensurate off-diagonal modulation, the correspondence breaks down, although the extended phase is maintained in a wide parameter range of the strengths of the on-site potential and the off-diagonal hoppings. An additional intermediate phase with a non-Hermitian mobility edge emerges when the off-diagonal hoppings become commensurate. This phase is characterized by the real and complex sections of the energy spectrum corresponding to the extended and localized states. In this case, the aforementioned correspondence reappears due to the recovery of the PT-symmetry.

N-independent Localized Krylov Bogoliubov-de Gennes Method: Ultra-fast Numerical Approach to Large-scale Inhomogeneous Superconductors

Yuki Nagai. J. Phys. Soc. Jpn. 89, 074703 (2020) (2020). Cited by 0. Found by: arxiv. DOI: 10.7566/JPSJ.89.074703

Abstract: We propose the ultra-fast numerical approach to large-scale inhomogeneous superconductors, which we call the Localized Krylov Bogoliubov-de Gennes method (LK-BdG). In the LK-BdG method, the computational complexity of the local Green's function, which is used to calculate the local density of states and the mean-fields, does not depend on the system size N. The calculation cost of self-consistent calculations is ${\cal O}(N)$, which enables us to open a new avenue for treating extremely large systems with millions of lattice sites. To show the power of the LK-BdG method, we demonstrate a self-consistent calculation on the 143806-site Penrose quasicrystal lattice with a vortex and a calculation on 1016064-site two-dimensional nearest-neighbor square-lattice tight-binding model with many vortices. We also demonstrate that it takes less than 30 seconds with one CPU core to calculate the local density of states with whole energy range in 100-millions-site tight-binding model.

Discrete quasiperiodic sets with predefined local structure

Nicolae Cotfas. Journal of Geometry and Physics 56 (2006) 2415-2428 (2004). Cited by 0. Found by: arxiv. DOI: 10.1016/j.geomphys.2005.12.008

Abstract: Model sets play a fundamental role in structure analysis of quasicrystals. The diffraction diagram of a quasicrystal admits as symmetry group a finite group G, and there is a G-cluster C (union of orbits of G) such that the quasicrystal can be regarded as a quasiperiodic packing of interpenetrating copies of C. We present an algorithm which leads from any G-cluster C directly to a multi-component model set Q such that the arithmetic neighbours of any point x belonging to Q are distributed on the sites of the translated copy x+C of C. Our mathematical algorithm may be useful in quasicrystal physics.

Growth of a One Dimensional Quasiperiodic Covering with Locally Determined Decorations

Hyeong-Chai Jeong. arXiv preprint (2008). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/0801.3509v1

Abstract: A growth mechanism for a perfect one-dimensional (1D) quasiperiodic structure is presented with a local covering rule. We use rectangular tiles with two different types of string decorations. The string position in a tile is allowed to move when the tile is attached to an existing patch. By adjusting the position properly with local information, we show that a growth of perfect quasiperiodic structure is possible. This observation may provide new insight into how quasicrystals grow with perfect quasiperiodic order.

Local 3D real space atomic structure of the simple icosahedral Ho11Mg15Zn74 quasicrystal from PDF data

S. Bruehne; E. Uhrig; C. Gross; W. Assmus. Cryst. Res. Technol. 38(12) (2003) 1023-1036. (2003). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/cond-mat/0308161v1

Abstract: We present a new complementary strategy to quasicrystalline structure determination: The local atomic structure of simple icosahedral (si) Ho11Mg15Zn74 [a(6D)=5.144(3)A] in a sphere of up to r=17A was

refined using the atomic pair distribution function (PDF) from in-house X-ray powder diffraction data (MoKa1, Qmax=13.5A-1; R=20.4%). The basic building block is a 105-atom Bergman-Cluster {Ho8Mg12Zn85}. Its center is occupied by a Zn atom - in contrast to a void in face centred icosahedral (fci) Ho9Mg26Zn65. The center is then surrounded by another 12 Zn atoms, forming an icosahedron (1st shell). The 2nd shell is made up of 8 Ho atoms arranged on the vertices of a cube which in turn is completed to a pentagon dodecahedron by 12 Mg atoms, the dodecahedron then being capped by 12 Zn atoms. The 3rd shell is a distorted soccer ball of 60 Zn atoms, reflecting the higher Zn content of the si phase compared to the fci phase. In our model, 7% of all atoms are situated in between the clusters. The model corresponds to a hypothetical 1/1-approximant of the icosahedral (i) phase. The local coordinations of the single atoms are of a much distorted Frank-Kasper type and call to mind those present in 0/1-Mg2Zn11.

Mobility edge and intermediate phase in one-dimensional incommensurate lattice potentials

Xiao Li; S. Das Sarma. Phys. Rev. B 101, 064203 (2020) (2018). Cited by 0. Found by: arxiv. DOI: 10.1103/PhysRevB.101.064203

Abstract: We study theoretically the localization properties of two distinct one-dimensional quasiperiodic lattice models with a single-particle mobility edge (SPME) separating extended and localized states in the energy spectrum. The first one is the familiar Soukoulis-Economou trichromatic potential model with two incommensurate potentials, and the second is a system consisting of two coupled 1D Aubry-Andre chains each containing one incommensurate potential. We show that as a function of the Hamiltonian model parameters, both models have a wide single-particle intermediate phase (SPIP), defined as the regime where localized and extended single-particle states coexist in the spectrum, leading to a behavior intermediate between purely extended or purely localized when the system is dynamically quenched from a generic initial state. Our results thus suggest that both systems could serve as interesting experimental platforms for studying the interplay between localized and extended states, and may provide insight into the role of the coupling of small baths to localized systems. We also calculate the Lyapunov (or localization) exponent for several incommensurate 1D models exhibiting SPME, finding that such localization critical exponents for quasiperiodic potential induced localization are nonuniversal and depend on the microscopic details of the Hamiltonian.

Multifractal-enriched mobility edges and emergent quantum phases in Rydberg atomic arrays

Shan-Zhong Li; Yi-Cai Zhang; Yucheng Wang; Shanchao Zhang; Shi-Liang Zhu; Zhi Li. Sci. China-Phys. Mech. Astron. 69, 217212 (2026) (2025). Cited by 0. Found by: arxiv. DOI: 10.1007/s11433-025-2774-2

Abstract: Anderson localization describes disorder-induced phase transitions, distinguishing between localized and extended states. In quasiperiodic systems, a third multifractal state emerges, characterized by unique energy and wave functions. However, the corresponding multifractal-enriched mobility edges and three-state-coexisting quantum phases have yet to be experimentally detected. In this work, we propose exactly-solvable one-dimensional quasiperiodic lattice models that simultaneously host three-state-coexisting quantum phases, with their phase boundaries analytically derived via Avila's global theorem. Furthermore, we propose experimental protocols via Rydberg atom arrays to realize these states. Notably, we demonstrate a spectroscopic technique capable of measuring inverse participation ratios across real-space and dual-space domains, enabling simultaneous characterization of localized, extended, and multifractal quantum phases in systems with up to tens of qubits. Our work opens new avenues for the experimental exploration of Anderson localization and multifractal states in artificial quantum systems.

Local Rules for Computable Planar Tilings

Thomas Fernique; Mathieu Sablik. EPTCS 90, 2012, pp. 133-141 (2012). Cited by 0. Found by: arxiv. DOI: 10.4204/EPTCS.90.11

Abstract: Aperiodic tilings are non-periodic tilings characterized by local constraints. They play a key role in the proof of the undecidability of the domino problem (1964) and naturally model quasicrystals (discovered in 1982). A central question is to characterize, among a class of non-periodic tilings, the aperiodic ones. In this paper, we answer this question for the well-studied class of non-periodic tilings obtained by digitizing irrational vector spaces. Namely, we prove that such tilings are aperiodic if and only if the digitized vector spaces are computable.

Local spectroscopy of moiré-induced electronic structure in gate-tunable twisted bilayer graphene

Dillon Wong; Yang Wang; Jeil Jung; Sergio Pezzini; Ashley M. DaSilva; Hsin-Zon Tsai; Han Sae Jung; Ramin Khajeh; Youngkyou Kim; Juwon Lee; Salman Kahn; Sajjad Tollabimazraehno; Haider Rasool; Kenji Watanabe; Takashi Taniguchi; Alex Zettl; Shaffique Adam; Allan H. MacDonald; Michael F. Crommie. Phys. Rev. B 92, 155409 (2015) (2015). Cited by 0. Found by: arxiv. DOI: 10.1103/PhysRevB.92.155409

Abstract: Twisted bilayer graphene (tBLG) forms a quasicrystal whose structural and electronic properties depend on the angle of rotation between its layers. Here we present a scanning tunneling microscopy study of gate-tunable tBLG devices supported by atomically-smooth and chemically inert hexagonal boron nitride (BN). The high quality of these tBLG devices allows identification of coexisting moiré patterns and moiré super-superlattices produced by graphene-graphene and graphene-BN interlayer interactions. Furthermore, we examine additional tBLG spectroscopic features in the local density of states beyond the first van Hove singularity. Our experimental data is explained by a theory of moiré bands that incorporates ab initio calculations and confirms the strongly non-perturbative character of tBLG interlayer coupling in the small twist-angle regime.

Topological zero-modes of the spectral localizer of trivial metals

Selma Franca; Adolfo G. Grushin. Phys. Rev. B 109, 195107 (2024) (2023). Cited by 0. Found by: arxiv. DOI: 10.1103/PhysRevB.109.195107

Abstract: Topological insulators are described by topological invariants that can be computed by integrals over momentum space, but also as traces over local, real-space topological markers. These markers are useful to detect topological insulating phases in disordered crystals, quasicrystals and amorphous systems. Among these markers, only the spectral localizer operator can be used to distinguish topological metals, that show zero-modes of the localizer spectrum. However, it remains unclear whether trivial metals also display zero-modes, and if their localizer spectrum is distinguishable from topological ones. Here, we show that trivial metals generically display zero-modes of the localizer spectrum. The localizer zero-modes are determined by the zero-mode solutions of a Dirac equation with a varying mass parameter. We use this observation, valid in any dimension, to determine the difference between the localizer spectrum of trivial and topological metals, and conjecture the spectrum of the localizer for fractional quantum Hall edges. Because the localizer is a local, real-space operator, it may be used as a tool to differentiate between non-crystalline topological and trivial metals, and characterize strongly correlated systems, for which local topological markers are scarce.

Hidden dualities in 1D quasiperiodic lattice models

Miguel Gonçalves; Bruno Amorim; Eduardo V. Castro; Pedro Ribeiro. SciPost Phys. 13, 046 (2022) (2021). Cited by 0. Found by: arxiv. DOI: 10.21468/SciPostPhys.13.3.046

Abstract: We find that quasiperiodicity-induced transitions between extended and localized phases in generic 1D systems are associated with hidden dualities that generalize the well-known duality of the Aubry-André model. These spectral and eigenstate dualities are locally defined near the transition and can, in many cases, be explicitly constructed by considering relatively small commensurate approximants. The construction relies on auxiliary 2D Fermi surfaces obtained as functions of the phase-twisting boundary conditions and of the phase-shifting real-space structure. We show that, around the critical point of the limiting quasiperiodic system, the auxiliary Fermi surface of a high-enough-order approximant converges to a universal form. This allows us to devise a highly-accurate method to obtain mobility edges and duality transformations for generic 1D quasiperiodic systems through their commensurate approximants. To illustrate the power of this approach, we consider several previously studied systems, including generalized Aubry-André models and coupled Moiré chains. Our findings bring a new perspective to examine quasiperiodicity-induced extended-to-localized transitions in 1D, provide a working criterion for the appearance of mobility edges, and an explicit way to understand the properties of eigenstates close to and at the transition.

An efficient saddle search method for ordered phase transitions involving translational invariance

Gang Cui; Kai Jiang; Tiejun Zhou. arXiv preprint (2023). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/2310.0710

Abstract: In this work, we propose an efficient nullspace-preserving saddle search (NPSS) method for a class of phase transitions involving translational invariance, where the critical states are often degenerate. The NPSS method includes two stages, escaping from the basin and searching for the index-1 generalized saddle point. The NPSS method climbs upward from the generalized local minimum in segments to overcome the challenges of degeneracy. In each segment, an effective ascent direction is ensured by keeping this direction orthogonal to the nullspace of the initial state in this segment. This method can escape the basin quickly and converge to the transition states. We apply the NPSS method to the phase transitions between crystals, and between crystal and quasicrystal, based on the Landau-Brazovskii and Lifshitz-Petrich free energy functionals. Numerical results show a good performance of the NPSS method.

Invariable mobility edge in a quasiperiodic lattice

Tong Liu; Yufei Zhu; Shujie Cheng; Feng Li; Hao Guo; Yong Pu. arXiv preprint (2021). Cited by 0. Found by: arxiv. DOI: 10.1088/1674-1056/ac140e

Abstract: In this paper, we study a one-dimensional tight-binding model with tunable incommensurate potentials. Through the analysis of the inverse participation rate, we uncover that the wave functions corresponding to the energies of the system exhibit different properties. There exists a critical energy under which the wave functions corresponding to all energies are extended. On the contrary, the wave functions corresponding to all energies above the critical energy are localized. However, we are surprised to find that the critical energy is a constant independent of the potentials. We use the self-dual relation to solve the critical energy, namely the mobility edge, and then we verify the analytical results again by analyzing the spatial distributions of the wave functions. Finally, we give a brief discussion on the possible experimental observation of the invariable mobility edge in the system of ultracold atoms in optical lattices.

Interplay of Quasiperiodic Criticality and the Non-Hermitian Skin Effect

Zhangyuan Chen; Xianqi Tong; Xiaosen Yang. arXiv preprint (2026). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/2607.08294v1

Abstract: Quasiperiodic lattices can host critical eigenstates, whereas nonreciprocal hopping in non-Hermitian lattices can induce non-Hermitian skin effect. In this work, we investigate localization phenomena in a Hatano–Nelson model with quasiperiodically modulated hopping amplitudes, where nonreciprocity arises from unequal modulation strengths of the right and left hoppings. Using a non-unitary gauge transformation, we map the non-Hermitian system into a Hermitian quasiperiodic system and obtain an exact analytical expression for the Lyapunov exponent in the thermodynamic limit. Under periodic boundary conditions, inverse participation ratios and finite-size scaling analysis are used to identify the quasiperiodic critical regimes. The comparison shows that parameter regimes hosting quasiperiodic critical states under periodic boundary conditions can exhibit the non-Hermitian skin effect under open boundary conditions. Furthermore, the non-Hermitian skin effect associated with quasiperiodic critical regimes is also observed in representative long-range hopping models and multiband extensions. Our results provide an analytically controlled perspective on how quasiperiodicity, modulated nonreciprocity, and boundary conditions jointly shape the non-Hermitian skin effect in critical regimes.

Computing with almost periodic functions

R. V. Moody; M. Nesterenko; J. Patera. arXiv preprint (2008). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/0808.1814v1

Abstract: The paper develops a method for discrete computational Fourier analysis of functions defined on quasicrystals and other almost periodic sets. A key point is to build the analysis around the emerging theory of quasicrystals and diffraction in the setting on local hulls and dynamical systems. Numerically computed approximations arising in this way are built out of the Fourier module of the quasicrystal in question, and approximate their target functions uniformly on the entire infinite space. The methods are entirely group theoretical, being based on finite groups and their duals, and they are practical and computable. Examples of functions based on the standard Fibonacci quasicrystal serve to illustrate the method (which is applicable to all quasicrystals modeled on the cut and project formalism).

Quantum criticality and Kibble-Zurek scaling in the Aubry-André-Stark model

En-Wen Liang; Ling-Zhi Tang; Dan-Wei Zhang. Phys. Rev. B 110, 024207 (2024) (2024). Cited by 0. Found by: arxiv. DOI: 10.1103/PhysRevB.110.024207

Abstract: We explore quantum criticality and Kibble-Zurek scaling (KZS) in the Aubry-Andre-Stark (AAS) model, where the Stark field of strength ε is added onto the one-dimensional quasiperiodic lattice. We perform scaling analysis and numerical calculations of the localization length, inverse participation ratio (IPR), and energy gap between the ground and first excited states to characterize critical properties of the delocalization-localization transition. Remarkably, our scaling analysis shows that, near the critical point, the localization length $ $ scales with ε as $ \propto\varepsilon^{- }$ with $ \approx0.3$ a new critical exponent for the AAS model, which is different from the counterparts for both the pure Aubry-Andre (AA) model and Stark model. The IPR $\mathcal{I}$ scales as $\mathcal{I}\propto\varepsilon^{s}$ with the critical exponent $s\approx0.098$, which is also different from both two pure models. The energy gap $ΔE$ scales as $ΔE\propto \varepsilon^{ z}$ with the same critical exponent $z\approx2.374$ as that for the pure AA model. We further reveal hybrid scaling functions in the overlap between the critical regions of the Anderson and Stark localizations. Moreover, we investigate the driven dynamics of the localization transitions in the AAS model. By linearly changing the Stark (quasiperiodic) potential, we calculate the evolution of the localization length and the IPR, and study their dependence on the driving rate. We find that the driven dynamics from the ground state is well described by the KZS with the critical exponents obtained from the static scaling analysis. When both the Stark and quasiperiodic potentials are relevant, the KZS form includes the two scaling variables. This work extends our understanding of critical phenomena on localization transitions and generalizes the application of the KZS to hybrid models.

Single-particle and many-body analyses of a quasiperiodic integrable system after a quench

Kai He; Lea F. Santos; Tod M. Wright; Marcos Rigol. Phys. Rev. A 87, 063637 (2013) (2013). Cited by 0. Found by: arxiv. DOI: 10.1103/PhysRevA.87.063637

Abstract: In general, isolated integrable quantum systems have been found to relax to an apparent equilibrium state in which the expectation values of few-body observables are described by the generalized Gibbs ensemble. However, recent work has shown that relaxation to such a generalized statistical ensemble can be precluded by localization in a quasiperiodic lattice system. Here we undertake complementary single-particle and many-body analyses of noninteracting spinless fermions and hard-core bosons within the Aubry-Andre model to gain insight into this phenomenon. Our investigations span both the localized and delocalized regimes of the quasiperiodic system, as well as the critical point separating the two. Considering first the case of spinless fermions, we study the dynamics of the momentum distribution function and characterize the effects of real-space and momentum-space localization on the relevant single-particle wave functions and correlation functions. We show that although some observables do not relax in the delocalized and localized regimes, the observables that do relax in these regimes do so in a manner consistent with a recently proposed Gaussian equilibration scenario, whereas relaxation at the critical point has a more exotic character. We also construct various statistical ensembles from the many-body eigenstates of the fermionic and bosonic Hamiltonians and study the effect of localization on their properties.

Coexistence of topological Anderson insulator and multifractal critical phase in a non-Hermitian quasicrystal

Qi-Bo Zeng; Rong Lü. Phys. Rev. B 113, 224203 (2026) (2026). Cited by 0. Found by: arxiv. DOI: 10.1103/n9rf-zk9t

Abstract: The interplay of topology, disorder, and non-Hermiticity gives rise to phenomena beyond the conventional classification of quantum phases. We propose a one-dimensional non-Hermitian Su-Schrieffer-Heeger model with quasiperiodically modulated nonreciprocal intracell hopping. We show that quasiperiodic modulation can substantially enhance the topological regime and, remarkably, induce a non-Hermitian topological Anderson insulator (TAI) phase. Beyond the topological transition, increasing nonreciprocity drives a cascade of localization transitions in which all bulk eigenstates evolve from extended to multifractal critical and ultimately to localized states. Strikingly, the extended-to-critical transition coincides exactly with a real-complex spectral transition. We establish complete phase diagrams and derive exact analytical boundaries for both topological and localization transitions, uncovering an unanticipated coexistence of TAI and multifractal critical phases. Finally, we propose a feasible implementation in topolectrical circuits. Our results reveal a new paradigm for studying the cooperative effects of topology, quasiperiodicity, and non-Hermiticity.

Quasicrystalline Spin Liquid

Sunghoon Kim; Mohammad Saad; Dan Mao; Adhip Agarwala; Debanjan Chowdhury. Phys. Rev. B 110, 214438, 2024 (2024). Cited by 0. Found by: arxiv. DOI: 10.1103/PhysRevB.110.214438

Abstract: The interplay of electronic interactions and frustration in crystalline systems leads to a panoply of correlated phases, including exotic Mott insulators with non-trivial patterns of entanglement. Disorder introduces additional quantum interference effects that can drive localization phenomena. Quasicrystals, which are neither disordered nor perfectly crystalline, are interesting playgrounds for studying the effects of interaction,

frustration, and quantum interference. Here we consider a solvable example of a quantum spin liquid on a tri-coordinated quasicrystal. We extend Kitaev's original construction for the spin model to our quasicrystalline setting and perform a large scale flux-sampling to find the ground-state configuration in terms of the emergent majorana fermions and flux excitations. This reveals a fully gapped and time-reversal symmetric quantum spin liquid, regardless of the exchange anisotropies, accompanied by a tendency towards non-trivial (de-)localization at the edge and the bulk. The advent of moiré materials and a variety of quantum simulators provide a new platform to bring phases of quasicrystalline quantum matter to life in a controlled fashion.

Disordered, Quasicrystalline and Crystalline Phases of Densely Packed Tetrahedra

Amir Haji-Akbari; Michael Engel; Aaron S. Keys; Xiaoyu Zheng; Rolfe G. Petschek; Peter Palffy-Muhoray; Sharon C. Glotzer. Nature 462, 773–777 (2009) (2010). Cited by 0. Found by: arxiv. DOI: 10.1038/nature08641

Abstract: All hard, convex shapes are conjectured by Ulam to pack more densely than spheres, which have a maximum packing fraction of = /\sqrt18 ~ 0.7405. For many shapes, simple lattice packings easily surpass this packing fraction. For regular tetrahedra, this conjecture was shown to be true only very recently; an ordered arrangement was obtained via geometric construction with = 0.7786, which was subsequently compressed numerically to = 0.7820. Here we show that tetrahedra pack much better than this, and in a completely unexpected way. Following a conceptually different approach, using thermodynamic computer simulations that allow the system to evolve naturally towards high-density states, we observe that a fluid of hard tetrahedra undergoes a first-order phase transition to a dodecagonal quasicrystal, which can be compressed to a packing fraction of = 0.8324. By compressing a crystalline approximant of the quasicrystal, the highest packing fraction we obtain is = 0.8503. If quasicrystal formation is suppressed, the system remains disordered, jams, and compresses to = 0.7858. Jamming and crystallization are both preceded by an entropy-driven transition from a simple fluid of independent tetrahedra to a complex fluid characterized by tetrahedra arranged in densely packed local motifs that form a percolating network at the transition. The quasicrystal that we report represents the first example of a quasicrystal formed from hard or non-spherical particles. Our results demonstrate that particle shape and entropy can produce highly complex, ordered structures.

Photoinduced Phase Transition in Two-Band model on Penrose Tiling

Ken Inayoshi; Yuta Murakami; Akihisa Koga. J. Phys.: Conf. Ser. 2164, 012050 (2022) (2021). Cited by 0. Found by: arxiv. DOI: 10.1088/1742-6596/2164/1/012050

Abstract: We study the effects of the photo irradiation on the band insulating state in the two-band Hubbard model on the Penrose tiling. Examining the time- and site-dependent physical quantities, we find that the excitionic state is dynamically induced with site-dependent order parameters. It is also clarified that, in the excitonic state induced by the photo irradiation, local oscillatory behavior appears in the electron number as well as in the order parameter, which should be characteristic of the quasiperiodic lattice.

Quasicrystalline electronic states in 30° rotated twisted bilayer graphene

Pilkyung Moon; Mikito Koshino; Young-Woo Son. Phys. Rev. B 99, 165430 (2019) (2019). Cited by 0. Found by: arxiv. DOI: 10.1103/PhysRevB.99.165430

Abstract: The recently realized bilayer graphene system with a twist angle of 30° offers a new type of quasicrystal which unites the dodecagonal quasicrystalline nature and graphene's relativistic properties. Here, we introduce a concise theoretical framework that fully respects both the dodecagonal rotational symmetry and the massless Dirac nature, to describe the electronic states of the system. We find that the electronic spectrum consists of resonant states labeled by 12-fold quantized angular momentum, together with the extended relativistic states. The resulting quasi-band structure is composed of the nearly flat bands with spiky peaks in the density of states, where the wave functions exhibit characteristic patterns which fit to the fractal inflations of the quasicrystal tiling. We also demonstrate that the 12-fold resonant states appear as spatially-localized states in a finite-size geometry, which is another hallmark of quasicrystal. The theoretical method introduced here is applicable to a broad class of "extrinsic quasicrystals" composed of a pair of two-dimensional crystals overlaid on top of the other with incommensurate configurations.

Observation of subdiffusion of a disordered interacting system

E. Lucioni; B. Deissler; L. Tanzi; G. Roati; M. Modugno; M. Zaccanti; M. Larcher; F. Dalfovo; M. Inguscio; G. Modugno. Phys. Rev. Lett. 106, 230403 (2011) (2010). Cited by 0. Found by: arxiv. DOI: 10.1103/Phys-

RevLett.106.230403

Abstract: We study the transport dynamics of matter-waves in the presence of disorder and nonlinearity. An atomic Bose-Einstein condensate that is localized in a quasiperiodic lattice in the absence of atom-atom interaction shows instead a slow expansion with a subdiffusive behavior when a controlled repulsive interaction is added. The measured features of the subdiffusion are compared to numerical simulations and a heuristic model. The observations confirm the nature of subdiffusion as interaction-assisted hopping between localized states and highlight a role of the spatial correlation of the disorder.

Correlation function of weakly interacting bosons in a disordered lattice

B. Deissler; E. Lucioni; M. Modugno; G. Roati; L. Tanzi; M. Zaccanti; M. Inguscio; G. Modugno. New J. Phys. 13 (2011) 023020 (2010). Cited by 0. Found by: arxiv. DOI: 10.1088/1367-2630/13/2/023020

Abstract: One of the most important issues in disordered systems is the interplay of the disorder and repulsive interactions. Several recent experimental advances on this topic have been made with ultracold atoms, in particular the observation of Anderson localization, and the realization of the disordered Bose-Hubbard model. There are however still questions as to how to differentiate the complex insulating phases resulting from this interplay, and how to measure the size of the superfluid fragments that these phases entail. It has been suggested that the correlation function of such a system can give new insights, but so far little experimental investigation has been performed. Here, we show the first experimental analysis of the correlation function for a weakly interacting, bosonic system in a quasiperiodic lattice. We observe an increase in the correlation length as well as a change in shape of the correlation function in the delocalization crossover from Anderson glass to coherent, extended state. In between, the experiment indicates the formation of progressively larger coherent fragments, consistent with a fragmented BEC, or Bose glass.

Theory of spin Bott index for quantum spin Hall states in non-periodic systems

Huaqing Huang; Feng Liu. Phys. Rev. B 98, 125130 (2018) (2018). Cited by 0. Found by: arxiv. DOI: 10.1103/PhysRevB.98.125130

Abstract: This is a joint publication with the Letter by H. Huang and F. Liu [Phys. Rev. Lett. 121, 126401 (2018)]. In this work, we propose the spin Bott index to identify the quantum spin Hall (QSH) state in both crystalline and non-periodic systems. The applicability of the spin Bott index is confirmed by analyzing the periodic and disorder Kane-Mele models. As an example of non-periodic systems, we systematically investigate the QSH effect in a Penrose-type quasicrystal lattice (QL). We characterized the nontrivial electronic topology of the QL by directly calculating the spin Bott index. In addition, the topological edge states, the localization of wavefunctions and quantized transport signatures are also studied in detail.

Time-quasiperiodic topological superconductors with Majorana Multiplexing

Yang Peng; Gil Refael. Phys. Rev. B 98, 220509 (2018) (2018). Cited by 0. Found by: arxiv. DOI: 10.1103/PhysRevB.98.220509

Abstract: Time-quasiperiodic Majoranas are generalizations of Floquet Majoranas in time-quasiperiodic superconducting systems. We show that in a system driven at d mutually irrational frequencies, there are up to 2^d types of such Majoranas, coexisting despite spatial overlap and lack of time-translational invariance. Although the quasienergy spectrum is dense in such systems, the time-quasiperiodic Majoranas can be stable and robust against resonances due to localization in the periodic-drives induced synthetic dimensions. This is demonstrated in a time-quasiperiodic Kitaev chain driven at two frequencies. We further relate the existence of multiple Majoranas in a time-quasiperiodic system to the time quasicrystal phase introduced recently. These time-quasiperiodic Majoranas open a new possibility for braiding which will be pursued in the future.

Hubbard Models for Quasicrystalline Potentials

Emmanuel Gottlob; Ulrich Schneider. Phys. Rev. B 107, 144202 (2023) (2022). Cited by 0. Found by: arxiv. DOI: 10.1103/PhysRevB.107.144202

Abstract: Quasicrystals are long-range ordered, yet not periodic, and thereby present a fascinating challenge for condensed matter physics, as one cannot resort to the usual toolbox based on Bloch's theorem. Here, we present a numerical method for constructing the Hubbard Hamiltonian of non-periodic potentials without making use of Bloch's theorem and apply it to the case of an eightfold rotationally symmetric 2D optical quasicrystal that was recently realized using cold atoms. We construct maximally localised Wannier functions and use

them to extract on-site energies, tunneling amplitudes, and interaction energies. In addition, we introduce a configuration-space representation, where sites are ordered in terms of shape and local environment, that leads to a compact description of the infinite-size quasicrystal in which all Hamiltonian parameters can be expressed as smooth functions. This configuration-space picture allows one to construct arbitrarily large tight-binding graphs for numerical many-body calculations and enables new analytic arguments on the topological structure and many-body physics of these models, for instance the conclusion that this quasicrystal will host unit-filling Mott insulators in the thermodynamic limit.

Affine Dihedral Subgroups of Higher Dimensional Cubic Lattices $\mathbb{Z}^n$ and Quasicrystallography

Nazife Ozdes Koca; Mehmet Koca; Ramazan Koc; Amira Al-Maqbali. Sultan Qaboos University Journal For Science 2025 : Vol. 30: Iss. 2, 102-116 (2023). Cited by 0. Found by: arxiv. DOI: 10.53539/2414-536X.1405

Abstract: Quasicrystals described as the projections of higher dimensional cubic lattices, and the particular affine extensions of the dihedral group $I_2(h)$ of order $2h$, $h=2n$ being the Coxeter number, as a subgroup of affine B_n offers a different perspective to h-fold symmetric quasicrystallography. Affine $I_2(h)$ is constructed as the subgroup of the affine B_n, the symmetry of the cubic lattice $\mathbb{Z}^n$. The infinite discrete group with local dihedral symmetry of order $2h$ operates on the concentric h-gons obtained by projecting the Voronoi cell of the cubic lattice with 2^n vertices onto the Coxeter plane. Voronoi cells tile the space facet to facet, consequently, leading to the tilings of the Coxeter plane with some overlaps of the rhombic tiles. It is noted that the projected Voronoi cell is the overlap of h copy of the h-gons tiled with some rhombi and rotated by the angle $2 /h$. After a general discussion on the lattice $\mathbb{Z}^n$ with the affine symmetry $\tilde B_n$ and its affine dihedral subgroup $\tilde I_2(h)$ its projection onto the Coxeter plane has been worked out with some examples. The cubic lattices with affine symmetry $\tilde B_n$ $(n=1,2,3,4,5)$ have been presented and shown that the projection of the lattice B_3 leads to the hexagonal lattice, the projection of the lattice B_4 describes the Ammann-Beenker quasicrystal lattice with 8-fold local symmetry and the projection of the lattice B_5 describes a quasicrystal structure with local 10-fold symmetry with thick and thin rhombi. It is then straight forward to show that the projections of the cubic lattices with even higher dimensions onto the Coxeter plane may lead to the quasicrystal structures with 12-fold, 18-fold symmetries and so on.

Magnetic states induced by electron-electron interactions in a plane quasiperiodic tiling

A. Jagannathan; H. J. Schulz. arXiv preprint (1996). Cited by 0. Found by: arxiv. DOI: 10.1103/PhysRevB.55.8045

Abstract: We consider the Hubbard model for electrons in a two-dimensional quasiperiodic tiling using the Hartree–Fock approximation. Numerical solutions are obtained for the first three square approximants of the perfect octagonal tiling. At half-filling, the magnetic state is antiferromagnetic. We calculate the distributions of local magnetizations and their dependence on the local environments as U (Hubbard repulsion strength) is varied. The inflation symmetry of quasicrystals results in a corresponding inflation symmetry of the magnetic configurations when passing from one tiling to the next in the progression towards the infinite quasicrystal.

Amplitude-dependent edge states and discrete breathers in nonlinear modulated phononic lattices

Matheus I. N. Rosa; Michael J. Leamy; Massimo Ruzzene. arXiv preprint (2022). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/2201.05526v2

Abstract: We investigate the spectral properties of one-dimensional spatially modulated nonlinear phononic lattices, and their evolution as a function of amplitude. In the linear regime, the stiffness modulations define a family of periodic and quasiperiodic lattices whose bandgaps host topological edge states localized at the boundaries of finite domains. With cubic nonlinearities, we show that edge states whose eigenvalue branch remains within the gap as amplitude increases remain localized, and therefore appear to be robust with respect to amplitude. In contrast, edge states whose corresponding branch approaches the bulk bands experience delocalization transitions. These transitions are predicted through continuation studies on the linear eigenmodes as a function of amplitude, and are confirmed by direct time domain simulations on finite lattices. Through our predictions, we also observe a series of amplitude-induced localization transitions as the bulk modes detach from the nonlinear bulk bands and become discrete breathers that are localized in one or more regions of the domain. Remarkably, the predicted transitions are independent of the size of the finite lattice, and exist for both periodic and quasiperiodic lattices. These results highlight the co-existence of topological edge states and discrete breathers in nonlinear modulated lattices. Their interplay may be exploited for amplitude-induced

eigenstate transitions, for the assessment of the robustness of localized states, and as a strategy to induce discrete breathers through amplitude tuning.

Superconducting properties of Fibonacci chains with enhanced superconducting pairing at the boundaries

Quanyong Zhu; Guo-Qiao Zha; A. A. Shanenko; Yajiang Chen. Physical Review B 112, 134503 (2025) (2024). Cited by 0. Found by: arxiv. DOI: 10.1103/j8tj-82ty

Abstract: Recently, the superconducting properties of Fibonacci quasicrystals have attracted considerable attention. By numerically solving the self-consistent Bogoliubov-de Gennes equations for an $s-$wave superconducting Fibonacci chain, we find that the system exhibits universal end superconductivity, where the pair potential at the chain ends can persist at higher temperatures compared to the bulk critical temperature (T_{cb}) of the condensate in the chain center. Furthermore, our study reveals two distinct critical temperatures at the left (T_{cL}) and right (T_{cR}) ends of the chain. This complex behavior arises from the competition between topological bound states and critical states, a characteristic of quasicrystals. With the chosen parameters, the maximal enhancement of T_{cR} reaches up to 66% relative to T_{cb}, while T_{cL} can increase by up to 31%. Our study sheds light on the phenomenon of end superconductivity in Fibonacci quasicrystals, pointing to alternative pathways for increasing the superconducting critical temperature.

Chemical localization

V. F. Gantmakher. UFN 172(11), 1283 (2002) [English translation in Physics-Uspekhi 45(11), (2002)] (2002). Cited by 0. Found by: arxiv. DOI: 10.1070/PU2002v045n11ABEH001246

Abstract: Analysis of experimental data shows that the metal–insulator transition is possible in materials composed of atoms of only metallic elements. Such a transition may occur in spite of the high concentration of valence electrons. It requires stable atomic configurations to act as deep potential traps absorbing dozens of valence electrons. This means in essence that bulk metallic space transforms into an assembly of identical quantum dots. Depending on the parameters, such a material either does contain delocalized electrons (metal) or does not contain such electrons (insulator). The degree of disorder is one of these parameters. Two types of substances with such properties are discussed: liquid binary alloys with both components being metallic, and thermodynamically stable quasicrystals.

Sub-diffusive electronic states in octagonal tiling

G. Trambly de Laissardière; C. Oguey; D. Mayou. Journal of Physics: Conf. Series 809 (2017) 012020 (2016). Cited by 0. Found by: arxiv. DOI: 10.1088/1742-6596/809/1/012020

Abstract: We study the quantum diffusion of charge carriers in octagonal tilings. Our numerical results show a power law decay of the wave-packet spreading, $L(t) \propto t^ $, characteristic of critical states in quasicrystals at large time t. For many energies states are sub-diffusive, i.e. $ < 0.5$, and thus conductivity increases when the amount of defects (static defects and/or temperature) increases.

Phase transitions and bunching of correlated particles in a non-Hermitian quasicrystal

Stefano Longhi. Phys. Rev. B 108, 075121 (2023) (2023). Cited by 0. Found by: arxiv. DOI: 10.1103/PhysRevB.108.075121

Abstract: Non-interacting particles in non-Hermitian quasi crystals display localization-delocalization and spectral phase transitions in complex energy plane, that can be characterized by point-gap topology. Here we investigate the spectral and dynamical features of two interacting particles in a non-Hermitian quasi crystal, described by an effective Hubbard model in an incommensurate sinusoidal potential with a complex phase, and unravel some intriguing effects without any Hermitian counterpart. Owing to the effective decrease of correlated hopping introduced by particle interaction, doublon states, i.e. bound particle states, display a much lower threshold for spectral and localization-delocalization transitions than single-particle states, leading to the emergence of mobility edges. Remarkably, since doublons display longer lifetimes, two particles initially placed in distant sites tend to bunch and stick together, forming a doublon state in the long time limit of evolution, a phenomenon that can be dubbed {\em non-Hermitian particle bunching}.

Higher-dimensional Hofstadter butterfly on Penrose lattice

Rasoul Ghadimi; Takanori Sugimoto; Takami Tohyama. arXiv preprint (2022). Cited by 0. Found by: arxiv. DOI: 10.1103/PhysRevB.106.L201113

Abstract: Quasicrystal is now open to search for novel topological phenomena enhanced by its peculiar structure characterized by an irrational number and high-dimensional primitive vectors. Here we extend the concept of a topological insulator with an emerging staggered local magnetic flux (i.e., without external fields), similar to the Haldane's honeycomb model, to the Penrose lattice as a quasicrystal. The Penrose lattice consists of two different tiles, where the ratio of the numbers of tiles corresponds to an irrational number. Contrary to periodic lattices, the periodicity of energy spectrum with respect to the magnetic flux no longer exists reflecting the irrational number in the Penrose lattice. Calculating the Bott index as a topological invariant, we find topological phases appearing in a fractal energy spectrum like the Hofstadter butterfly. More intriguingly, by folding the one-dimensional aperiodic magnetic flux into a two-dimensional periodic flux space, the fractal structure of energy spectrum is extended to higher dimension, whose section corresponds to the Hofstadter butterfly.

Lyapunov exponent for pure and random Fibonacci chains

M. T. Velhinho; I. R. Pimentel. Phys. Rev. B 61, 1043 (2000) (2000). Cited by 0. Found by: arxiv. DOI: 10.1103/PhysRevB.61.1043

Abstract: We study the Lyapunov exponent for electron and phonon excitations, in pure and random Fibonacci quasicrystal chains, using an exact real space renormalization group method, which allows the calculation of the Lyapunov exponent as a function of the energy. It is shown that the Lyapunov exponent on a pure Fibonacci chain has a self-similar structure, characterized by a scaling index that is independent of the energy for the electron excitations, "diagonal" or "off-diagonal" quasiperiodic, but is a function of the energy for the phonon excitations. This scaling behavior implies the vanishing of the Lyapunov exponent for the states on the spectrum, and hence the absence of localization on the Fibonacci chain, for the various excitations considered. It is also shown that disordered Fibonacci chains, with random tiling that introduces phason flips at certain sites on the chain, exhibit the same Lyapunov exponent as the pure Fibonacci chain, and hence this type of disorder is irrelevant, either in the case of electron or phonon excitations.

Exploring Aperiodic Order in Photonic Time Crystals

Marino Coppolaro; Massimo Moccia; Giuseppe Castaldi; Vincenzo Galdi. arXiv preprint (2025). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/2505.20039v1

Abstract: We present a theoretical framework for analyzing aperiodically ordered photonic time quasicrystals (PTQCs), which are the temporal analogs of spatial photonic quasicrystals. Using a general two-symbol substitutional sequence to model temporal modulations, we extend the trace and anti-trace map formalism used for spatial photonic quasicrystals to the temporal domain. Focusing on the Thue-Morse sequence as a representative example, we examine the band structure and wave-transport properties, discussing their physical origins and highlighting both similarities and key differences with conventional periodic photonic time crystals. Furthermore, we investigate the peculiar features of PTQCs, such as multiscale spectral response and localization effects. Our findings provide valuable insights into the complex interplay between aperiodic order and wave dynamics in time-varying media, highlighting its potential to enable the development of advanced photonic devices.

Ground state of a two dimensional quasiperiodic antiferromagnet

A. Jagannathan. arXiv preprint (2004). Cited by 0. Found by: arxiv. DOI: 10.1103/PhysRevB.71.115101

Abstract: We consider the spin 1/2 Heisenberg antiferromagnet on a two dimensional quasiperiodic tiling. The T=0 state is long range ordered, with an inhomogeneous distribution of the staggered moment. An approximate real space renormalization scheme first developed for the square lattice is generalized and used to obtain information on the ground state energy as well as the distribution of local correlations in the quasicrystal.

On the Spectra of Sieved Schrödinger Operators

Jake Fillman; Alexandro Luna. arXiv preprint (2025). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/2505.08763v1

Abstract: We give a family of examples of discrete Schrödinger operators whose spectral dimension is not invariant under sieving. The examples are produced from the Fibonacci Hamiltonian, which is one of the main models of a one-dimensional quasicrystal. We also give a family of examples in which the local Hausdorff dimension tends to zero in some parts of the spectrum as the sieving parameter is sent to infinity.

Complete Structural Determination of Mesostructural Dodecagonal Quasicrystalline Particles

Xueliang Zhang; Xi Wang; Nobuhisa Fujita; Osamu Terasaki; Lu Han. arXiv preprint (2026). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/2608.02142v1

Abstract: Quasicrystals have revolutionized our understanding of order in solids by demonstrating exotic structural and physicochemical properties with diverse potential applications. Despite the development of various theoretical models and experimental techniques to describe quasicrystal structures, the precise determination of local three dimensional (3D) arrangements of constituent atoms, or of secondary building units such as clusters or micelles, remains elusive. This challenge is particularly acute in self assembled soft matter quasicrystalline systems, where the complex assembly of molecular groups introduces additional defects and structural modulations. Herein, we report the first complete structural determination of self-assembled mesostructural dodecagonal quasicrystalline particles. Employing advanced electron tomography, combined with dedicated structural tracing and processing workflows, the 3D coordinates of all nodal sites were extracted. This approach reveals that the actual structure deviates from the conventionally assumed tetrahedral close packing geometry, exhibiting diverse coordination environments and displacive fluctuations. We identified and quantified rotational intergrowths arising from node exchange, as well as various defects and disorder, with these features discernible only through 3D analysis. Additionally, we propose a simplified two-layer stacking of isomorphic hexagonal model to form dodecagonal quasicrystal. This work advances our understanding of soft-matter dodecagonal quasicrystals and paves the way for detailed structural elucidation of self-assembled systems.

Antiferromagnetic order in the Hubbard Model on the Penrose Lattice

Akihisa Koga; Hirokazu Tsunetsugu. Phys. Rev. B 96, 214402 (2017) (2017). Cited by 0. Found by: arxiv. DOI: 10.1103/PhysRevB.96.214402

Abstract: We study an antiferromagnetic order in the ground state of the half-filled Hubbard model on the Penrose lattice and investigate the effects of quasiperiodic lattice structure. In the limit of infinitesimal Coulomb repulsion $U \rightarrow +0$, the staggered magnetizations persist to be finite, and their values are determined by confined states, which are strictly localized with thermodynamics degeneracy. The magnetizations exhibit an exotic spatial pattern, and have the same sign in each of cluster regions, the size of which ranges from 31 sites to infinity. With increasing U, they continuously evolve to those of the corresponding spin model in the $U=\infty$ limit. In both limits of U, local magnetizations exhibit a fairly intricate spatial pattern that reflects the quasiperiodic structure, but the pattern differs between the two limits. We have analyzed this pattern change by a mode analysis by the singular value decomposition method for the fractal-like magnetization pattern projected into the perpendicular space.

Mott transition for a Lieb-Liniger gas in a shallow quasiperiodic potential: Delocalization induced by disorder

Hepeng Yao; Luca Tanzi; Laurent Sanchez-Palencia; Thierry Giamarchi; Giovanni Modugno; Chiara D'Errico. Phys. Rev. Lett. 133, 123401 (2024) (2024). Cited by 0. Found by: arxiv. DOI: 10.1103/PhysRevLett.133.123401

Abstract: Disorder or quasi-disorder is known to favor the localization in many-body Bose systems. Here in contrast, we demonstrate an anomalous delocalization effect induced by incommensurability in quasiperiodic lattices. Loading ultracold atoms in two shallow periodic lattices with equal amplitude and either equal or incommensurate spatial periods, we show the onset of a Mott transition not only in the periodic case but also in the quasiperiodic case. Upon increase of the incommensurate component of the potential we find that the Mott insulator turns into a delocalized superfluid. Our experimental results agree with quantum Monte Carlo calculations, showing anomalous delocalization induced by the interplay between the commensuration and interaction.

Understanding shape entropy through local dense packing

Greg van Anders; Daphne Klotsa; N. Khalid Ahmed; Michael Engel; Sharon C. Glotzer. PNAS 111, E4812-E4821 (2014) (2013). Cited by 0. Found by: arxiv. DOI: 10.1073/pnas.1418159111

Abstract: Entropy drives the phase behavior of colloids ranging from dense suspensions of hard spheres or rods to dilute suspensions of hard spheres and depletants. Entropic ordering of anisotropic shapes into complex crystals, liquid crystals, and even quasicrystals has been demonstrated recently in computer simulations and experiments. The ordering of shapes appears to arise from the emergence of directional entropic forces (DEFs) that align neighboring particles, but these forces have been neither rigorously defined nor quantified in generic systems. Here, we show quantitatively that shape drives the phase behavior of systems of anisotropic particles upon crowding through DEFs. We define DEFs in generic systems, and compute them for several hard particle systems. We show that they are on the order of a few kT at the onset of ordering, placing DEFs on par with traditional depletion, van der Waals, and other intrinsic interactions. In experimental systems with these other interactions, we provide direct quantitative evidence that entropic effects of shape also contribute to self-assembly. We use DEFs to draw a distinction between self-assembly and packing behavior. We show that the mechanism that generates directional entropic forces is the maximization of entropy by optimizing local particle packing. We show that this mechanism occurs in a wide class of systems, and we treat, in a unified way, the entropy-driven phase behavior of arbitrary shapes incorporating the well-known works of Kirkwood, Onsager, and Asakura and Oosawa.

Approximating Metal-Insulator Transitions

C. Danieli; K. Rayanov; B. Pavlov; G. Martin; S. Flach. arXiv preprint (2014). Cited by 0. Found by: arxiv. DOI: 10.1142/S0217979215500368

Abstract: We consider quantum wave propagation in one-dimensional quasiperiodic lattices. We propose an iterative construction of quasiperiodic potentials from sequences of potentials with increasing spatial period. At each finite iteration step the eigenstates reflect the properties of the limiting quasiperiodic potential properties up to a controlled maximum system size. We then observe approximate metal-insulator transitions (MIT) at the finite iteration steps. We also report evidence on mobility edges which are at variance to the celebrated Aubry-Andre model. The dynamics near the MIT shows a critical slowing down of the ballistic group velocity in the metallic phase similar to the divergence of the localization length in the insulating phase.

Engineering bands of extended electronic states in a class of topologically disordered and quasiperiodic lattices

Biplab Pal; Arunava Chakrabarti. Physics Letters A, Volume 378, Issue 37, Page 2782 (2014) (2014). Cited by 0. Found by: arxiv. DOI: 10.1016/j.physleta.2014.07.034

Abstract: We show that a discrete tight-binding model representing either a random or a quasiperiodic array of bonds, can have the entire energy spectrum or a substantial part of it absolutely continuous, populated by extended eigenfunctions only, when atomic sites are coupled to the lattice locally, or non-locally from one side. The event can be fine-tuned by controlling only the host-adatom coupling in one case, while in two other cases cited here an additional external magnetic field is necessary. The delocalization of electronic states for the group of systems presented here is sensitive to a subtle correlation between the numerical values of the Hamiltonian parameters - a fact that is not common in the conventional cases of Anderson localization. Our results are analytically exact, and supported by numerical evaluation of the density of states and electronic transmission coefficient.

Combined energy – diffraction data refinement of decagonal AlNiCo

M. Mihalkovič; C. L. Henley; M. Widom. arXiv preprint (2003). Cited by 0. Found by: arxiv. DOI: 10.1016/j.jnoncrysol.2003.11.034

Abstract: We incorporate realistic pair potential energies directly into a non-linear least-square fit of diffraction data to quantitatively compare structure models with experiment for the Ni-rich d(AlNiCo) quasicrystal. The initial structure models are derived from a few {\it a priori} assumptions (gross features of the Patterson function) and the pair potentials. In place of the common hyperspace approach to the structure refinement of quasicrystals, we use a real-space tile decoration scheme, which does not rely on strict quasiperiodicity, and makes it easy to enforce sensible local arrangements of the atoms. Inclusion of the energies provides information complementary to the diffraction data and protects the fit procedure from converging on spurious solutions. The method pinpoints sites which are likely to break the symmetry of their local environment.

Electronic and optical properties of strained graphene and other strained 2D materials: a review

Gerardo G. Naumis; Salvador Barraza-Lopez; Maurice Oliva-Leyva; Humberto Terrones. Rep. Prog. Phys. 80 096501 (2017) (2016). Cited by 0. Found by: arxiv. DOI: 10.1088/1361-6633/aa74ef

Abstract: This review presents the state of the art in strain and ripple-induced effects on the electronic and optical properties of graphene. It starts by providing the crystallographic description of mechanical deformations, as well as the diffraction pattern for different kinds of representative deformation fields. Then, the focus turns to the unique elastic properties of graphene, and to how strain is produced. Thereafter, various theoretical approaches used to study the electronic properties of strained graphene are examined, discussing the advantages of each. These approaches provide a platform to describe exotic properties, such as a fractal spectrum related with quasicrystals, a mixed Dirac-Schrödinger behavior, emergent gravity, topological insulator states, in molecular graphene and other 2D discrete lattices. The physical consequences of strain on the optical properties are reviewed next, with a focus on the Raman spectrum. At the same time, recent advances to tune the optical conductivity of graphene by strain engineering are given, which open new paths in device applications. Finally, a brief review of strain effects in multilayered graphene and other promising 2D materials like silicene and materials based on other group-IV elements, phosphorene, dichalcogenide- and monochalcogenide-monolayers is presented, with a brief discussion of interplays among strain, thermal effects, and illumination in the latter material family.

Fate of Majorana zero modes by a modified real-space-Pfaffian method and mobility edges in a one-dimensional quasiperiodic lattice

Shujie Cheng; Yufei Zhu; Gao Xianlong; Tong Liu. arXiv preprint (2021). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/2102.00737v2

Abstract: We aim to study a one-dimensional p-wave superconductor with quasiperiodic on-site potentials. A modified real-space-Pfaffian method is applied to calculate the topological invariants. We confirm that the Majorana zero mode is protected by the nontrivial topology the topological phase transition is accompanied by the energy gap closing and reopening. In addition, we numerically find that there are mobility edges which originate from the competition between the extended p-wave pairing and the localized quasi-disorder. We qualitatively analyze the influence of superconducting pairing parameters and on-site potential strength on the mobility edge. In general, our work enriches the research on the p-wave superconducting models with quasiperiodic potentials.

Stochastic Flips on Dimer Tilings

Thomas Fernique; Damien Regnault. arXiv preprint (2011). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/1111.72

Abstract: This paper introduces a Markov process inspired by the problem of quasicrystal growth. It acts over dimer tilings of the triangular grid by randomly performing local transformations, called {\em flips}, which do not increase the number of identical adjacent tiles (this number can be thought as the tiling energy). Fixed-points of such a process play the role of quasicrystals. We are here interested in the worst-case expected number of flips to converge towards a fixed-point. Numerical experiments suggest a bound quadratic in the number n of tiles of the tiling. We prove a O(n^2.5) upper bound and discuss the gap between this bound and the previous one. We also briefly discuss the average-case.

Cleaved surface of i-AlPdMn quasicrystals: Influence of the local temperature elevation at the crack tip on the fracture surface roughness

Laurent Ponson; Daniel Bonamy; Luc Barbier. Physical Review B 74 (20/11/2006) 184205 (2006). Cited by 0. Found by: arxiv. DOI: 10.1103/PhysRevB.74.184205

Abstract: Roughness of i-AlPdMn cleaved surfaces are presently analysed. From the atomic scale to 2-3 nm, they are shown to exhibit scaling properties hiding the cluster (0.45 nm) aperiodic structure. These properties are quantitatively similar to those observed on various disordered materials, albeit on other ranges of length scales. These properties are interpreted as the signature of damage mechanisms occurring within a 2-3 nm wide zone at the crack tip. The size of this process zone finds its origin in the local temperature elevation at the crack tip. For the very first time, this effect is reported to be responsible for a transition from a perfectly brittle behavior to a nanoductile one.

Resonant metallic states in driven quasiperiodic lattices

L. Morales-Molina; E. Doerner; C. Danieli; S. Flach. arXiv preprint (2014). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/1405.0765v1

Abstract: We consider a quasiperiodic Aubry-Andre (AA) model and add a weak time-periodic and spatially quasiperiodic perturbation. The undriven AA model is chosen to be well in the insulating regime. The spatial quasiperiodic perturbation extends the model into two dimensions in reciprocal space. For a spatial resonance which reduces the reciprocal space dynamics to an effective one-dimensional two-leg ladder case, the ac perturbation resonantly couples certain groups of localized eigenstates of the undriven AA model and turns them into extended metallic ones. Slight detuning of the spatial and temporal frequencies off resonance returns these states into localized ones. We analyze the details of the resonant metallic eigenstates using Floquet representations. In particular, we find that their size grows linearly with the system size. Initial wave packets overlap with resonant metallic eigenstates and lead to ballistic spreading.

On the spectrum of 1D quantum Ising quasicrystal

W. N. Yessen. arXiv preprint (2011). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/1110.6894v6

Abstract: We consider one dimensional quantum Ising spin-1/2 chains with two-valued nearest neighbor couplings arranged in a quasi-periodic sequence, with uniform, transverse magnetic field. By employing the Jordan-Wigner transformation of the spin operators to spinless fermions, the energy spectrum can be computed exactly on a finite lattice. By employing the transfer matrix technique and investigating the dynamics of the corresponding trace map, we show that in the thermodynamic limit the energy spectrum is a Cantor set of zero Lebesgue measure. Moreover, we show that local Hausdorff dimension is continuous and nonconstant over the spectrum. This forms a rigorous counterpart of numerous numerical studies.

Emergent periodic and quasiperiodic lattices on surfaces of synthetic Hall tori and synthetic Hall cylinders

Yangqian Yan; Shao-Liang Zhang; Sayan Choudhury; Qi Zhou. Phys. Rev. Lett. 123, 260405 (2019) (2018). Cited by 0. Found by: arxiv. DOI: 10.1103/PhysRevLett.123.260405

Abstract: Synthetic spaces allow physicists to bypass constraints imposed by certain physical laws in experiments. Here, we show that a synthetic torus, which consists of a ring trap in the real space and internal states of ultracold atoms cyclically coupled by Laguerre-Gaussian Raman beams, could be threaded by a net effective magnetic flux through its surface—an impossible mission in the real space. Such synthetic Hall torus gives rise to a periodic lattice in the real dimension, in which the periodicity of density modulation of atoms fractionalizes that of the Hamiltonian. Correspondingly, the energy spectrum is featured by multiple bands grouping into clusters with nonsymmorphic symmetry protected band crossings in each cluster, leading to braidings of wavepackets in Bloch oscillations. Our scheme allows physicists to glue two synthetic Hall tori such that localization may emerge in a quasicrystalline lattice. If the Laguerre-Gaussian Raman beams and ring traps were replaced by linear Raman beams and ordinary traps, a synthetic Hall cylinder could be realized and deliver many of the aforementioned phenomena.

Observation of an aperiodic polariton monotile

Sergey Alyatkin; Yaroslav V. Kartashov; Kirill Sitnik; Philipp Grigoryev; Pavlos G. Lagoudakis. arXiv preprint (2026). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/2605.13206v1

Abstract: A plethora of unconventional localization phenomena and fractal features of linear spectrum observed in quasiperiodic structures have been accompanied by a long-standing quest for the geometrical elements and structures that permit tilings of the plane, but only in a non-periodic manner. Until 2024, it was believed that such quasiperiodic structures, or quasicrystals, could only be composed of at least two different tiles. Surprisingly, a newly discovered class of quasicrystals requires only one elementary monotile. However, its physical realization and study of propagating coherent excitations in this novel setting remained elusive. Here we optically sculpt aperiodic quasicrystals composed of ”einstein” monotiles in an inorganic microcavity and observe nontrivial relative phases of the exciton-polariton condensates nonresonantly excited at the vertices of each monotile. Utilizing energy-resolved tomography in momentum-space, we reveal the formation of distinct Bragg peaks with six-fold symmetry and Dirac-like spectral fingerprints, intrinsic to the underlying graphene-like structure, while interferometric phase reconstruction shows a nontrivial synchronization pattern distinct from both periodic triangular lattices and Penrose quasicrystals. Our work demonstrates that monotiles can be converted into a programmable driven-dissipative artificial material, where long-range coherence coexists with enforced geometric aperiodicity, producing synchronization and spectral responses distinct from both periodic and conventional quasicrystalline tilings.

Unraveling Multifractality and Mobility Edges in Quasiperiodic Aubry-André-Harper Chains through High-Harmonic Generation

Marlena Dziurawiec; Jessica O. de Almeida; Mohit Lal Bera; Marcin Płodzień; Maciej M. Maśka; Maciej Lewenstein; Tobias Grass; Utso Bhattacharya. arXiv preprint (2023). Cited by 0. Found by: arxiv. DOI: 10.1103/PhysRevB.110.014209

Abstract: Quasicrystals are fascinating and important because of their unconventional atomic arrangements, which challenge traditional notions of crystalline structures. Unlike regular crystals, they lack translational symmetry and generate unique mechanical, thermal, and electrical properties, holding promise for numerous applications. In order to probe the electronic properties of quasicrystals, tools beyond linear response transport measurements are needed, since all spectral regions can be affected by the non-periodic geometry. Here we show that high-harmonic spectroscopy offers an advanced avenue for this goal. Focusing on the quasiperiodic 1D Aubry-André-Harper (AAH) model, we leverage high-harmonic spectroscopy to delve into their intricate characteristics: By carefully analyzing emitted harmonic intensities, we extract the multifractal spectrum – an essential indicator of the spatial distribution of electronic states in quasicrystals. Additionally, we address the detection of mobility edges, vital energy thresholds that demarcate localized and extended eigenstates within generalized AAH models. The precise identification of these mobility edges sheds light on the metal-insulator transition and the behavior of electronic states near these boundaries. Merging high-harmonic spectroscopy with the AAH model provides a powerful framework for understanding the interplay between localization and extended states in quasicrystals for an extremely wide energy range not captured within linear response studies, thereby offering valuable insights for guiding future experimental investigations.

Phase Diagram of Hard Tetrahedra

Amir Haji-Akbari; Michael Engel; Sharon C. Glotzer. J. Chem. Phys. 135, 194101 (2011) (2011). Cited by 0. Found by: arxiv. DOI: 10.1063/1.3651370

Abstract: Advancements in the synthesis of faceted nanoparticles and colloids have spurred interest in the phase behavior of polyhedral shapes. Regular tetrahedra have attracted particular attention because they prefer local symmetries that are incompatible with periodicity. Two dense phases of regular tetrahedra have been reported recently. The densest known tetrahedron packing is achieved in a crystal of triangular bipyramids (dimers) with packing density 4000/4671=85.63%. In simulation a dodecagonal quasicrystal is observed; its approximant, with periodic tiling (3.4.3^2.4), can be compressed to a packing fraction of 85.03%. Here, we show that the quasicrystal approximant is more stable than the dimer crystal for packing densities below 84% using Monte Carlo computer simulations and free energy calculations. To carry out the free energy calculations, we use a variation of the Frenkel-Ladd method for anisotropic shapes and thermodynamic integration. The enhanced stability of the approximant can be attributed to a network substructure, which maximizes the free volume (and hence the 'wiggle room') available to the particles and facilitates correlated motion of particles, which further contributes to entropy and leads to diffusion for packing densities below 65%. The existence of a solid-solid transition between structurally distinct phases not related by symmetry breaking – the approximant and the dimer crystal– is unusual for hard particle systems.

Superconductivity on a Quasiperiodic Lattice: Extended-to-Localized Crossover of Cooper Pairs

Shiro Sakai; Nayuta Takemori; Akihisa Koga; Ryotaro Arita. Phys. Rev. B 95, 024509 (2017) (2016). Cited by 0. Found by: arxiv. DOI: 10.1103/PhysRevB.95.024509

Abstract: We study a possible superconductivity in quasiperiodic systems, by portraying the issue within the attractive Hubbard model on a Penrose lattice. Applying a real-space dynamical mean-field theory to the model consisting of 4181 sites, we find a superconducting phase at low temperatures. Reflecting the nonperiodicity of the Penrose lattice, the superconducting state exhibits an inhomogeneity. According to the type of the inhomogeneity, the superconducting phase is categorized into three different regions which cross over each other. Among them, the weak-coupling region exhibits spatially extended Cooper pairs, which are nevertheless distinct from the conventional pairing of two electrons with opposite momenta.

Subdiffusion of nonlinear waves in quasiperiodic potentials

Marco Larcher; Tetyana V. Laptyeva; Joshua D. Bodyfelt; Franco Dalfovo; Michele Modugno; Sergej Flach. New Journal of Physics 14 (2012) 103036 (2012). Cited by 0. Found by: arxiv. DOI: 10.1088/1367-2630/14/10/103036

Abstract: We study the spatio-temporal evolution of wave packets in one-dimensional quasiperiodic lattices which localize linear waves. Nonlinearity (related to two-body interactions) has destructive effect on localization, as recently observed for interacting atomic condensates [Phys. Rev. Lett. 106, 230403 (2011)]. We extend the analysis of the characteristics of the subdiffusive dynamics to large temporal and spatial scales. Our results for the second moment m_2 consistently reveal an asymptotic $m_2 \sim t^{1/3}$ and intermediate $m_2 \sim t^{1/2}$ laws. At variance to purely random systems [Europhys. Lett. 91, 30001 (2010)] the fractal gap structure of the linear wave spectrum strongly favors intermediate self-trapping events. Our findings give a new dimension to the theory of wave packet spreading in localizing environments.

Diffraction and pseudospectra in non-Hermitian quasiperiodic lattices

Ananya Ghatak; Dimitrios H. Kaltsas; Manas Kulkarni; Konstantinos G. Makris. arXiv preprint (2024). Cited by 0. Found by: arxiv. URL: http://arxiv.org/abs/2410.09185v1

Abstract: Wave dynamics in disordered open media is an intriguing topic, and has lately attracted a lot of attention in non-Hermitian physics, especially in photonics. In fact, spatial distributions of gain and loss elements are physically possible in the context of integrated photonic waveguide arrays. In particular, in these type of lattices, counter-intuitive quantized jumps along the propagation direction appear in the strong disorder limit (where all eigenstates are localized) and they have also been recently experimentally observed. We systematically study the non-Hermitian quasiperiodic Aubry-André-Harper model with on-site gain and loss distribution (NHAAH), with an emphasis on the spectral sensitivity based on pseudospectra analysis. Moreover, diffraction dynamics and the quantized jumps, as well as, the effect of saturable nonlinearity, are investigated in detail. Our study reveals the intricate relation between the nonlinearity and non-Hermiticity.

Discrete quasiperiodic sets with predefined covering cluster

Nicolae Cotfas. Philosophical Magazine 86 (2006) 895-900 (2005). Cited by 0. Found by: arxiv. DOI: 10.1080/14786430500263413

Abstract: Some of the most remarkable tilings and discrete quasiperiodic sets used in quasicrystal physics can be obtained by using strip projection method in a superspace of dimension four, five or six, and the projection of a unit hypercube as a window of selection. We present some mathematical results which allow one to use this very elegant method in superspaces of dimension much higher, and to generate discrete quasiperiodic sets with a more complicated local structure by starting from the corresponding covering cluster. Hundreds of points of these sets can be obtained in only a few minutes by using our computer programs.

Localization control born of intertwined quasiperiodicity and non-Hermiticity

Jeon, Junmo; Lee, SungBin. SciPost Phys.Core (2022). Cited by 0. Found by: inspire. DOI: 10.21468/SciPostPhysCore.6.4.077

Topological Pumping over a Photonic Fibonacci Quasicrystal

Verbin, Mor; Zilberberg, Oded; Lahini, Yoav; Kraus, Yaacov E.; Silberberg, Yaron. Phys.Rev.B (2014). Cited by 0. Found by: inspire. DOI: 10.1103/PhysRevB.91.064201

Anderson localization in Bose-Einstein condensates

Modugno, Giovanni. Rept.Prog.Phys. (2010). Cited by 0. Found by: inspire. DOI: 10.1088/0034-4885/73/10/102401

Support Vector Machine Kernels as Quantum Propagators

Kuo, Nan-Hong; Wong, Renata. INSPIRE record (2025). Cited by 0. Found by: inspire. URL: https://inspirehep.net/api/lite

Anomalous localization in a kicked quasicrystal

Shimasaki, Toshihiko; Prichard, Max; Kondakci, H. Esat; Pagett, Jared E.; Bai, Yifei; Dotti, Peter; Cao, Alec; Dardia, Anna R.; Lu, Tsung-Cheng; Grover, Tarun; Weld, David M.. Nature Phys. (2022). Cited by 0. Found by: inspire. DOI: 10.1038/s41567-023-02329-4

Colloquium: Synthetic quantum matter in nonstandard geometries

Grass, Tobias; Bercioux, Dario; Bhattacharya, Utso; Lewenstein, Maciej; Nguyen, Hai Son; Weitenberg, Christof. Rev.Mod.Phys. (2024). Cited by 0. Found by: inspire. DOI: 10.1103/RevModPhys.97.011001

Observing the two-dimensional Bose glass in an optical quasicrystal

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Phasonic Spectroscopy of a Quantum Gas in a Quasicrystalline Lattice

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Non-Hermitian physics

Ashida, Yuto; Gong, Zongping; Ueda, Masahito. Adv.Phys. (2020). Cited by 0. Found by: inspire. DOI: 10.1080/00018732.2021.1876991

Cavity QED with Quantum Gases: New Paradigms in Many-Body Physics

Mivehvar, Farokh; Piazza, Francesco; Donner, Tobias; Ritsch, Helmut. Adv.Phys. (2021). Cited by 0. Found by: inspire. DOI: 10.1080/00018732.2021.1969727

Source overlap

Source	Retrieved	Found only here	Filtered venues
openalex	656	599	37
arxiv	613	567	0
inspire	17	10	0

'Found only here' counts unique records no other source returned – a measure of each database's marginal contribution.

Distributions

Publication year

Year	Records
1994	1
1996	1
1997	1
1998	2
1999	3
2000	2
2001	3
2002	5
2003	3
2004	5
2005	4
2006	6
2007	9
2008	7

Year	Records
2009	8
2010	10
2011	9
2012	9
2013	9
2014	23
2015	25
2016	64
2017	56
2018	84
2019	99
2020	89
2021	79
2022	87
2023	94
2024	152
2025	154
2026	123

Top venues

Venue	Records	OpenAlex 2-yr mean citedness
arXiv preprint	354	
arXiv (Cornell University)	57	
Advanced Energy Materials	34	18.63 (2026)
Advanced Materials	23	22.086 (2026)
Advanced Functional Materials	19	14.975 (2026)
ACS Applied Materials & Interfaces	19	8.382 (2026)
The Journal of Physical Chemistry C	16	
Nature Energy	15	
ACS Energy Letters	15	16.712 (2026)
The Journal of Chemical Physics	13	
Journal of the American Chemical Society	12	
Nano Energy	12	
Journal of Materials Chemistry A	11	
Optics Express	11	
Science	10	

Most frequent authors

Author	Records
Michaël Grätzel	21
Marco Bernasconi	19
Shaik M. Zakeeruddin	15
Zheyong Fan	14
Nam-Gyu Park	13
Omar Abou El Kheir	13
Shu Chen	11
Volker L. Deringer	10
Stefano Longhi	10
Un-Gi Jong	10
Chol-Jun Yu	10
Longwen Zhou	10
Anders Hagfeldt	9
Michael D. McGehee	9
Henry J. Snaith	9

Journal metrics

Metric: OpenAlex 2-yr mean citedness (openalex_2yr), from lit/journals/metrics.json (journals.py). Values are kept per year; the evolution table shows every year on file for venues that appear in this record set.

Venues in this set by OpenAlex 2-yr mean citedness

Venue	Records	OpenAlex 2-yr mean citedness	Year	Q
Chemical Reviews	1	53.886	2026	
Chemical Society Reviews	1	42.455	2026	
Energy & Environmental Science	1	30.475	2026	
Advanced Materials	23	22.086	2026	
Advanced Energy Materials	34	18.63	2026	
ACS Nano	3	16.894	2026	
ACS Energy Letters	15	16.712	2026	
Angewandte Chemie International Edition	6	15.006	2026	
Advanced Functional Materials	19	14.975	2026	

Venue	Records	OpenAlex 2-yr mean citedness	Year	Q
Energy Storage Materials	1	13.488	2026	
Cement and Concrete Research	1	13.05	2026	
Chem	2	11.7	2026	
Applied Physics Reviews	1	11.309	2026	
Chemical Engineering Journal	3	10.953	2026	
Acta Materialia	3	10.171	2026	
ACS Materials Letters	1	9.392	2026	
Artificial Intelligence Chemistry	1	9.034	2026	
Advanced Science	6	8.903	2026	
Current Opinion in Chemical Engineering	1	8.446	2026	
ACS Applied Materials & Interfaces	19	8.382	2026	
EcoMat	1	7.557	2026	
Chemistry of Materials	3	6.387	2026	
Cell Reports Physical Science	1	5.933	2026	
Chaos Solitons & Fractals	1	5.898	2026	
Ceramics International	1	5.869	2026	

Filtered non-curated venues

Venue	Records removed
Figshare	15
Zenodo	11
SSRN Electronic Journal	9
Open MIND	1
Preprints.org	1

Count history

Per-block totals across archived runs (counts_history.csv); drift shows how the indexes – or your queries – changed.

Block	20260815T095908	20260828T091142	20260828T100120	20260828T160426
PSC	-	-	14,620	4,744
MLP	-	239	838	376
NOV	-	0	0	0
QC	-	-	978	585

Per-source raw results (before deduplication)

MLP_arxiv (138 records)

Title	Authors	Year	Venue	Cited	DOI
Accelerating Amorphous Alloy Discovery: Data-Driven Property Prediction via General-Purpose Machine Learning Interatomic Potential	Xuhe Gong; Hengbo Zhao; Xiao Fu; Jingchen Lian; Qifan Yang; Ran Li; Ruijuan Xiao; Tao Zhang; Hong Li	2025	arXiv preprint	0	(link)
Unveiling defect motifs in amorphous GeSe using machine learning interatomic potentials	Minseok Moon; Seungwoo Hwang; Jaesun Kim; Yutack Park; Changho Hong; Seungwu Han	2025	arXiv preprint	0	(link)
Li-P-S Electrolyte Materials as a Benchmark for Machine-Learned Interatomic Potentials	Natascia L. Fragapane; Volker L. Deringer	2025	arXiv preprint	0	(link)
Persistent homology-based descriptor for machine-learning potential of amorphous structures	Emi Minamitani; Ippei Obayashi; Koji Shimizu; Satoshi Watanabe	2022	arXiv preprint	0	(link)
Investigating Ionic Diffusivity in Amorphous Solid Electrolytes using Machine Learned Interatomic Potentials	Aqshat Seth; Rutvij Pankaj Kulkarni; Gopalakrishnan Sai Gautam	2024	arXiv preprint	0	10.1021/acsmaterialsau.4c00117
Unraveling the Crystallization Kinetics of the Ge$_2$Sb$_2$Te$_5$ Phase Change Compound with a Machine-Learned Interatomic Potential	Omar Abou El Kheir; Luigi Bonati; Michele Parrinello; Marco Bernasconi	2023	npj Comput Mater 10, 33 (2024)	0	10.1038/s41524-024-01217-6
Unveiling the crystallization kinetics in Ge-rich Ge$_x$Te alloys by large scale simulations with a machine-learned interatomic potential	Dario Baratella; Omar Abou El Kheir; Marco Bernasconi	2024	arXiv preprint	0	(link)

Title	Authors	Year	Venue	Cited	DOI
Melt-Quench Failures and Practical Solutions for Universal Machine-Learning Interatomic Potentials in Amorphous Structure Generation	Shuwei Li; Yuqi An; Xingyu Guo; Wenqiang Yang; Zhenbin Wang	2026	arXiv preprint	0	(link)
Comparison of intermediate-range order in GeO$_2$ glass: molecular dynamics using machine-learning interatomic potential vs.\ reverse Monte Carlo fitting to experimental data	Kenta Matsutani; Shusuke Kasamatsu; Takeshi Usuki	2024	arXiv preprint	0	(link)
Machine-learned potential for amorphous Indium-Tin-Oxide alloys	Shuaiyang Guo; Yuan Wang; Wei Zhang	2026	arXiv preprint	0	(link)
Revealing Phonon Bridge Effect for Amorphous vs Crystalline Metal-Silicide Layers at Si/Ti Interfaces by a Machine Learning Potential	Mayur Singh; Lokanath Patra; Chengyang Zhang; Greg MacDougall; Suman Datta; David Cahill; Satish Kumar	2025	arXiv preprint	0	10.1021/acsaelm.5c02592
Modeling Short-Range and Three-Membered Ring Structures in Lithium Borosilicate Glasses using Machine Learning Potential	Shingo Urata	2022	arXiv preprint	0	(link)
Large-scale atomistic study of plasticity in amorphous gallium oxide with a machine-learning potential	Jiahui Zhang; Junlei Zhao; Jesper Byggmästar; Erkka J. Frankberg; Antti Kuronen	2024	arXiv preprint	0	(link)
Viscosity, breakdown of Stokes-Einstein relation and dynamical heterogeneity in supercooled liquid Ge$_2$Sb$_2$Te$_5$ from simulations with a neural network potential	Simone Marcorini; Rocco Pomodoro; Omar Abou El Kheir; Marco Bernasconi	2025	J. Chem. Phys. 163, 154501 (2025)	0	10.1063/5.0282855
Validation Workflow for Machine Learning Interatomic Potentials for Complex Ceramics	Kimia Ghaffari; Salil Bavdekar; Douglas E. Spearot; Ghatu Subhash	2024	arXiv preprint	0	10.1016/j.commatsci.2024.112983
Atomistic Insights into Cu/amorphous-Ta$_x$N Interfacial Adhesion via Machine Learning Interatomic Potentials: Effects of Stoichiometry and Interface Construction	Jeong Min Choi; Jaehoon Kim; Ji-Hwan Lee; Won-Joon Son; Seungwu Han	2025	arXiv preprint	0	(link)
Machine Learning Potential Powered Insights into the Mechanical Stability of Amorphous Li-Si Alloys	Zixiong Wei; Nongnuch Artrith	2024	arXiv preprint	0	(link)
Is the Future of Materials Amorphous? Challenges and Opportunities in Simulations of Amorphous Materials	Ata Madanchi; Emna Azek; Karim Zongo; Laurent K. Béland; Normand Mousseau; Lena Simine	2024	arXiv preprint	0	(link)
Accurate prediction of structural and mechanical properties on amorphous materials enabled through machine-learning potentials: a case study of silicon nitride	Ganesh Kumar Nayak; Prashanth Srinivasan; Juraj Todt; Rostislav Daniel; Paolo Nicolini; David Holec	2024	Comput. Mater. Sci. 249 (2025) 113629	0	10.1016/j.commatsci.2024.113629
Insights Into Radiation Damage in YBa$_2$Cu$_3$O$_{7-}$ From Machine-Learned Interatomic Potentials	Ashley Dickson; Niccolò Di Eugenio; Federico Ledda; Daniele Torsello; Francesco Laviano; Flyura Djurabekova; Jesper Byggmästar; Mark R. Gilbert; Duc Nguyen-Manh; Erik Gallo; Antonio Trotta; Davide Gambino; Samuel T. Murphy	2025	arXiv preprint	0	(link)
Machine-learning-driven modelling of amorphous and polycrystalline BaZrS$_{3}$	Laura-Bianca Paşca; Yuanbin Liu; Andy S. Anker; Ludmilla Steier; Volker L. Deringer	2025	arXiv preprint	0	(link)
Large-scale first-principle simulations of amorphous indium oxide	Matthew Bousquet; Francois Gygi; Giulia Galli	2026	arXiv preprint	0	(link)
Resolving Structural Avalanches in Amorphous Carbon with Arclength Continuation	Fraser Birks; Ibrahim Ghanem; Lars Pastewka; James Kermode; Maciej Buze	2026	arXiv preprint	0	10.1103/6n5m-rxc1
On phase separation and crystallization of Ge-rich GeSbTe alloys from atomistic simulations with a machine learning interatomic potential	Omar Abou El Kheir; Dario Baratella; Marco Bernasconi	2026	arXiv preprint	0	(link)
Thermal Conductivity and Temperature-Induced Band Gap Renormalization in Crystalline and Amorphous Ga$_2$O$_3$	Rustam Arabov; Jiaxuan Li; Xiaotong Chen; Nikita Rybin; Alexander Shapeev	2026	arXiv preprint	0	(link)
From oligomers to entangled polymers: How to train a transferable machine learning interatomic potential	Mirko Fischer; Andreas Heuer	2026	arXiv preprint	0	(link)

Title	Authors	Year	Venue	Cited	DOI
ATLAS: A Foundation Neural Sampler for Amorphous Materials	Mouyang Cheng; Denis Blessing; Botao Yu; Gerhard Neumann; Mingda Li; Carles Domingo-Enrich; Yuanqi Du	2026	arXiv preprint	0	(link)
Polymorphic crystallites model for monolayer amorphous materials	Le-Ye Zhu; Xi Zhang; Yun-Peng Wang; Jieheng Shi; Junwei Zhang; Shixuan Du; Yu-Yang Zhang	2026	arXiv preprint	0	(link)
A Unified Deep Neural Network Potential Capable of Predicting Thermal Conductivity of Silicon in Different Phases	Ruiyang Li; Eungkyu Lee; Tengfei Luo	2019	arXiv preprint	0	(link)
Training machine-learning potentials for crystal structure prediction using disordered structures	Changho Hong; Jeong Min Choi; Wonseok Jeong; Sungwoo Kang; Suyeon Ju; Kyeongpung Lee; Jisu Jung; Yong Youn; Seungwu Han	2020	Phys. Rev. B 102, 224104 (2020)	0	10.1103/PhysRevB.102.224104
Atomistic insights into hydrogen migration in IGZO from machine-learning interatomic potential: linking atomic diffusion to device performance	Hyunsung Cho; Minseok Moon; Jaehoon Kim; Eunkyung Koh; Hyeon-Deuk Kim; Rokyeon Kim; Gyehyun Park; Seungwu Han; Youngho Kang	2025	arXiv preprint	0	(link)
Thermal Conductivity Modeling using Machine Learning Potentials: Application to Crystalline and Amorphous Silicon	Xin Qian; Shenyou Peng; Xiaobo Li; Yujie Wei; Ronggui Yang	2019	arXiv preprint	0	(link)
Amorphous materials as a frontier challenge for universal interatomic potentials	Natascia L. Fragapane; Volker L. Deringer	2026	arXiv preprint	0	(link)
Efficient training of machine learning potentials for metallic glasses: CuZrAl validation	Antoni Wadowski; Anshul D. S. Parmar; Filip Kaśkosz; Jesper Byggmästar; Jan S. Wróbel; Mikko J. Alava; Silvia Bonfanti	2024	arXiv preprint	0	(link)
Constitutive relations for plasticity of amorphous carbon	Richard Jana; Julian von Lautz; S. Mostafa Khosrownejad; W. Beck Andrews; Michael Moseler; Lars Pastewka	2020	J. Phys. Mater. 3, 035005 (2020)	0	10.1088/2515-7639/ab953c
Crystal-like thermal transport in amorphous carbon	Jaeyun Moon; Zhiting Tian	2024	arXiv preprint	0	(link)
Structural and elastic properties of amorphous carbon from simulated quenching at low rates	Richard Jana; Daniele Savio; Volker L. Deringer; Lars Pastewka	2019	Modelling Simul. Mater. Sci. Eng. 27, 085009 (2019)	0	10.1088/1361-651X/ab45da
Exploration of amorphous V$_2$O$_5$ as cathode for magnesium batteries	Vijay Choyal; Debsundar Dey; Gopalakrishnan Sai Gautam	2025	arXiv preprint	0	10.1002/smll.202505851
A Gaussian Approximation Potential for Amorphous Si:H	Davis Unruh; Reza Vatan Meidanshahi; Stephen M. Goodnick; Gábor Csányi; Gergely T. Zimányi	2021	arXiv preprint	0	(link)
On the origin of in-gap states in amorphous Ge$_2$Sb$_2$Te$_5$	Omar Abou El Kheir; Marco Bernasconi	2026	arXiv preprint	0	(link)
A Systematic Approach to Generating Accurate Neural Network Potentials: the Case of Carbon	Yusuf Shaidu; Emine Kucukbenli; Ruggero Lot; Franco Pellegrini; Efthimios Kaxiras; Stefano de Gironcoli	2020	arXiv preprint	0	(link)
Modulation of structural short-range order due to chemical patterning in multi-component amorphous interfacial complexions	Esther C. Hessong; Zhengyu Zhang; Tianjiao Lei; Mingjie Xu; Toshihiro Aoki; Timothy J. Rupert	2025	arXiv preprint	0	(link)
Growth Mechanism and Origin of High sp^3 Content in Tetrahedral Amorphous Carbon	Miguel A. Caro; Volker L. Deringer; Jari Koskinen; Tomi Laurila; Gábor Csányi	2018	Phys. Rev. Lett. 120, 166101 (2018)	0	10.1103/PhysRevLett.120.166101
Medium-range structural order in amorphous arsenic	Yuanbin Liu; Yuxing Zhou; Richard Ademuwagun; Luc Walterbos; Janine George; Stephen R. Elliott; Volker L. Deringer	2025	arXiv preprint	0	(link)
From Bulk to Surface: Structure and Dynamics of Amorphous Alumina from Deep Potential Molecular Dynamics	Zheng Yu; Jiayan Xu; Abhirup Patra; Sharan Shetty; Detlef Hohl; Roberto Car	2026	arXiv preprint	0	(link)
Recent Advances in Metallic Glasses	Silvia Bonfanti; Ralf Busch; Jesper Byggmästar; Jeppe C. Dyre; Jürgen Eckert; Spencer Fajardo; Michael L. Falk; Isabella Gallino; Jamie J. Kruzic; Jiayin Lu; Giulio Monaco; Misaki Ozawa; Anshul D. S. Parmar; Chris H. Rycroft; Srikanth Sastry	2025	arXiv preprint	0	(link)
Bridging classical and quantum interpretation of chemical state analysis by XPS/HAXPES to resolve short-range order in amorphous alumina films	Simon Gramatte; Xing Wang; Michael Alejandro Hernández Bertrán; Claudia Cancellieri; Giovanni Pizzi; Deborah Prezzi; Iurii Timrov; Olivier Politano; Ivo Utke; Lars P. H. Jeurgens; Vladyslav Turlo	2024	arXiv preprint	0	(link)

Title	Authors	Year	Venue	Cited	DOI
Machine-Learning-Guided Insights into Solid-Electrolyte Interphase Conductivity: Are Amorphous Lithium Fluorophosphates the Key?	Peichen Zhong; Kristin A. Persson	2025	arXiv preprint	0	10.1021/acsenergylett.5c03526
Neutron and X-ray Diffraction Reveal the Limits of Long-Range Machine Learning Potentials for Medium-Range Order in Silica Glass	Sai Harshit Balantrapu; Atul C. Thakur; Chris Benmore; Ganesh Sivaraman	2026	arXiv preprint	0	(link)
Constructing first-principles phase diagrams of amorphous LixSi using machine-learning-assisted sampling with an evolutionary algorithm	Nongnuch Artrith; Alexander Urban; Gerbrand Ceder	2018	The Journal of Chemical Physics 148 (24), 241711 2018	0	10.1063/1.5017661
Atomic insight into Li$^+$ ion transport in amorphous electrolytes Li$_x$AlO$_y$Cl$_{3+x-2y}$ (0.5 $\leq$ x $\leq$ 1.5, 0.25 $\leq$ y $\leq$ 0.75)	Yang Qifan; Xu Jing; Fu Xiao; Lian Jingchen; Wang Liqi; Gong Xuhe; Xiao Ruijuan; Li Hong	2024	arXiv preprint	0	(link)
Turing Pattern Engineering Enables Kinetically Ultrastable yet Ductile Metallic Glasses	Huanrong Liu; Qingan Li; Shan Zhang; Rui Su; Yunjiang Wang; Pengfei Guan	2025	arXiv preprint	0	(link)
Realistic atomistic structure of amorphous silicon from machine-learning-driven molecular dynamics	Volker L. Deringer; Noam Bernstein; Albert P. Bartók; Matthew J. Cliffe; Rachel N. Kerber; Lauren E. Marbella; Clare P. Grey; Stephen R. Elliott; Gábor Csányi	2018	arXiv preprint	0	(link)
Density dependence of thermal conductivity in nanoporous and amorphous carbon with machine-learned molecular dynamics	Yanzhou Wang; Zheyong Fan; Ping Qian; Miguel A. Caro; Tapio Ala-Nissila	2024	arXiv preprint	0	(link)
Energy dissipation at the atomic scale explains how fracture energy depends on crack velocity in silica glass	Marthe Grønlie Guren; Sigbjørn Løland Bore; François Renard; Henrik Andersen Sveinsson	2026	arXiv preprint	0	(link)
The ab initio amorphous materials database: Empowering machine learning to decode diffusivity	Hui Zheng; Eric Sivonxay; Max Gallant; Ziyao Luo; Matthew McDermott; Patrick Huck; Kristin A. Persson	2024	arXiv preprint	0	(link)
Structural properties of amorphous Na$_3$OCl electrolyte by first-principles and machine learning molecular dynamics	T. -L. Pham; M. Guerboub; S. D. Wansi Wendj; A. Bouzid; C. Tugène; M. Boero; C. Massobrio; Y. -H. Shin; G. Ori	2024	arXiv preprint	0	(link)
On the Boroxol Ring Fraction in Melt-Quenched B$_2$O$_3$ Glass	Debendra Meher; Nikhil V. S. Avula; Sundaram Balasubramanian	2025	J. Chem. Phys. 164, 124504 (2026)	0	10.1063/5.0321441
Enhanced ionic conductivity through crystallization of glass-Li$_3$PS$_4$ by machine learning molecular dynamics simulations	Koji Shimizu; Parth Bahuguna; Shigeo Mori; Akitoshi Hayashi; Satoshi Watanabe	2023	arXiv preprint	0	(link)
Machine learning potential for predicting thermal conductivity of -phase and amorphous Tantalum Nitride	Zhicheng Zong; Yangjun Qin; Jiahong Zhan; Haisheng Fang; Nuo Yang	2025	arXiv preprint	0	(link)
Role of Characteristic Length Scale in Interface Graphitization-Induced Wear Resistance of Diamond and Amorphous Carbon	WenLiang Shi; YanYu Zhang; WenZheng Liao; Kai Xu; HongYu Wu; ZhiCheng Zhong; KeKe Chang	2026	arXiv preprint	0	(link)
Structure and mechanical properties of monolayer amorphous carbon and boron nitride	Xi Zhang; Yu-Tian Zhang; Yun-Peng Wang; Shiyu Li; Shixuan Du; Yu-Yang Zhang; Sokrates T. Pantelides	2023	arXiv preprint	0	10.1103/PhysRevB.109.174106
Optimization and Validation of a Deep Learning CuZr Atomistic Potential: Robust Applications for Crystalline and Amorphous Phases with near-DFT Accuracy	Christopher M. Andolina; Philip Williamson; Wissam A. Saidi	2020	J. Chem. Phys. 152, 154701 (2020)	0	10.1063/5.0005347
Mechanisms of alkali ionic transport in amorphous oxyhalides solid state conductors	Luca Binci; KyuJung Jun; Bowen Deng; Gerbrand Ceder	2026	arXiv preprint	0	(link)
Neural-Network Assisted Study of Nitrogen Atom Dynamics on Amorphous Solid Water – II. Diffusion	Viktor Zaverkin; Germán Molpeceres; Johannes Kästner	2021	arXiv preprint	0	10.1093/mnras/stab3631
Machine Learning Assisted Modeling of Amorphous TiO$_2$-Doped GeO$_2$ for Advanced LIGO Mirror Coatings	Jun Jiang; Rui Zhang; Kiran Prasai; Riccardo Bassiri; James N. Fry; Martin M. Fejer; Hai-Ping Cheng	2025	arXiv preprint	0	(link)
Fast Isotropic Li-Ion Diffusion in Zeolitic Imidazolate Framework Glass Electrolytes for Batteries	Yong Li; Tao Du; Timothée Jamin; Zhencai Li; Kasper Tolborg; Yuanzheng Yue; Morten M. Smedskjaer	2026	arXiv preprint	0	(link)

Title	Authors	Year	Venue	Cited	DOI
kALDo 2.0: Scalable Thermal Transport from First Principles and Machine Learning Potentials	Giuseppe Barbalinardo; Zekun Chen; Dylan Folkner; Bohan Li; Nicholas W. Lundgren; Nathaniel Troup; Alfredo Fiorentino; Davide Donadio	2026	arXiv preprint	0	(link)
Dynamic heterogeneity in sodium silicate melts via machine-learning potential	Kumpei Shiraishi; Rikuta Nozawa; Emi Minamitani	2026	arXiv preprint	0	(link)
Neural-Network Assisted Study of Nitrogen Atom Dynamics on Amorphous Solid Water. I. Adsorption & Desorption	Germán Molpeceres; Viktor Zaverkin; Johannes Kästner	2020	arXiv preprint	0	10.1093/mnras/staa2891
Simulations of High Temperature Decomposition of Metal-Organic Frameworks to form Amorphous Catalysts	Connor W. Edwards; Oliver M. Linder-Patton; Jack D. Evans	2026	arXiv preprint	0	(link)
Thermal conductivity of glasses above the plateau: first-principles theory and applications	Michele Simoncelli; Francesco Mauri; Nicola Marzari	2022	arXiv preprint	0	(link)
Simulating alternating bias assisted annealing of amorphous oxide tunnel junctions	Alexander C. Tyner; Alexander V. Balatsky	2025	arXiv preprint	0	(link)
Origin of negative thermal expansion and pressure induced amorphization in zirconium tungstate from machine-learning potential	Ri He; Hongyu Wu; Yi Lu; Zhicheng Zhong	2022	arXiv preprint	0	10.1103/PhysRevB.106.174101
Harnessing Structural Disorder: Unraveling Hydrogen Evolution in Monolayer Amorphous Carbon via First-Principles Simulations and Machine-Learned Potentials	Sreehari M S; Ashutosh Krishna Amaram; Raghavan Ranganathan	2026	arXiv preprint	0	(link)
Thermal transport of glasses via machine learning driven simulations	Paolo Pegolo; Federico Grasselli	2024	arXiv preprint	0	(link)
Machine learning Hamiltonian enables scalable and accurate defect calculations: The case of oxygen vacancies in amorphous SiO$_2$	Zhenxing Dai; Zhong Yang; Mingjue Ni; Menglin Huang; Hongjun Xiang; Xin-Gao Gong; Shiyou Chen	2026	arXiv preprint	0	(link)
Synthesizability, hardness, and stacking order in multicomponent transition metal carbides from machine-learned potentials	Xin Liu; Anirudh Raju Natarajan	2026	arXiv preprint	0	(link)
Efficient and accurate simulation of vitrification in multi-component metallic liquids with neural-network potentials	Rui Su; Jieyi Yu; Pengfei Guan; Weihua Wang	2022	arXiv preprint	0	(link)
Atomic cluster expansion potential for the Si-H system	Louise A. M. Rosset; Volker L. Deringer	2025	arXiv preprint	0	(link)
Study of vacancy ordering and the boson peak in metastable cubic Ge-Sb-Te using machine learning potentials	Young-Jae Choi; Minjae Ghim; Seung-Hoon Jhi	2023	Phys. Rev. Materials 8, 013606 (2024)	0	10.1103/PhysRevMaterials.8.013606
Mechanisms of temperature-dependent thermal transport in amorphous silica from machine-learning molecular dynamics	Ting Liang; Penghua Ying; Ke Xu; Zhenqiang Ye; Chao Ling; Zheyong Fan; Jianbin Xu	2023	Physical Review B 108, 184203 (2023)	0	10.1103/PhysRevB.108.184203
AENET-LAMMPS and AENET-TINKER: Interfaces for Accurate and Efficient Molecular Dynamics Simulations with Machine Learning Potentials	Michael S. Chen; Tobias Morawietz; Hideki Mori; Thomas E. Markland; Nongnuch Artrith	2021	J. Chem. Phys. 155, 074801 (2021)	0	10.1063/5.0063880
Atomic-scale factors that control the rate capability of nanostructured amorphous Si for high-energy-density batteries	Nongnuch Artrith; Alexander Urban; Yan Wang; Gerbrand Ceder	2019	arXiv preprint	0	(link)
CO Adsorption Sites on Interstellar Water Ices Explored with Machine Learning Potentials. Binding energy distributions and snowline	Giulia M. Bovolenta; Germán Molpeceres; Kenji Furuya; Johannes Kästner; Stefan Vogt-Geisse	2025	A&A 703, A172 (2025)	0	10.1051/0004-6361/202555836
Basic stability tests of machine learning potentials for molecular simulations in computational drug discovery	Kavindri Ranasinghe; Adam L. Baskerville; Geoffrey P. F. Wood; Gerhard Koenig	2025	J. Chem. Inf. Model. 2025, 65, 17, 8980-8999	0	10.1021/acs.jcim.5c01150
AI-Aided Mapping of the Structure-Composition-Conductivity Relationships of Glass-Ceramic Lithium Thiophosphate Electrolytes	Haoyue Guo; Qian Wang; Alexander Urban; Nongnuch Artrith	2022	arXiv preprint	0	(link)
Nine-element machine-learned interatomic potentials for multi-phase refractory alloys	Jesper Byggmästar; Tiago Lopes; Zheyong Fan; Tapio Ala-Nissila	2026	arXiv preprint	0	(link)

Title	Authors	Year	Venue	Cited	DOI
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Liquid anomalies and fragility of supercooled antimony	Flavio Giuliani; Francesco Guidarelli Mattioli; Yuhan Chen; Dario Baratella; Daniele Dragoni; Marco Bernasconi; John Russo; Lilia Boeri; Riccardo Mazzarello	2026	Proceedings of the National Academy of Sciences	1	10.1073/pnas.2531605123
Elucidating Lithium Transport Mechanisms in Disordered LiF from Machine-Learning Molecular Dynamics Simulations	; Nikhil Rampal; Nicole Adelstein; Liwen F. Wan	2026	ACS Materials Letters	0	10.1021/acsmaterialslett.6c00029
Machine Learning Guided Exploration of Amorphous Electrodes and Electrolytes	Sai Gautam Gopalakrishnan; Aqshat Seth; Debsundar Dey; Vijay Choyal	2025	ECS Meeting Abstracts	0	10.1149/ma2025-0271034mtgabs
Large-scale first-principle simulations of amorphous indium oxide	Matthew Bousquet; Francois Gygi; Giulia Galli	2026	arXiv (Cornell University)	0	(link)
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Amorphous boron nitride as an ultrathin copper diffusion barrier for advanced interconnects	Onurcan Kaya; Hyeongjoon Kim; Byeongkyu Kim; Thomas Galvani; Luigi Colombo; Mario Lanza; Hyeon-Jin Shin; Ivan Cole; Hyeon Suk Shin; Stephan Roche	2025	2D Materials	0	10.1088/2053-1583/ae2521
Large scale molecular modelling of basalt surfaces and fracturing in basaltic glass	Marthe Grønlie Guren; Henrik Andersen Sveinsson; Razvan Caracas; Anders Malthe-Sørenssen; François Renard	2024		0	10.5194/egusphere-egu24-11861
Structure and mechanical properties of monolayer amorphous carbon and boron nitride	Xi Zhang; Yutian Zhang; Yunpeng Wang; Yiran Li; Shixuan Du; Yuyang Zhang; Sokrates T. Pantelides	2023	arXiv (Cornell University)	0	10.48550/arxiv.2309.15352
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The ab initio amorphous materials database: Empowering machine learning to decode diffusivity	Hui Zheng; Patrick Huck	2024	OSTI OAI (U.S. Department of Energy Office of Scientific and Technical Information)	0	10.17188/mpcontribs/2281590
Thermal conductivity of Li$_3$PS$_4$ solid electrolytes with ab initio accuracy	Davide Tisi; Federico Grasselli; Lorenzo Gigli; Michele Ceriotti	2024	arXiv (Cornell University)	2	10.48550/arxiv.2401.12936
Energy dissipation at the atomic scale explains how fracture energy depends on crack velocity in silica glass	Marthe Grønlie Guren; Sigbjørn Løland Bore; François Renard; Henrik Andersen Sveinsson	2026	arXiv (Cornell University)	0	10.48550/arxiv.2605.03457
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Role of Characteristic Length Scale in Interface Graphitization-Induced Wear Resistance of Diamond and Amorphous Carbon	Wenliang Shi; YanYu Zhang; WenZheng Liao; Kai Xu; HongYu Wu; ZhiCheng Zhong; KeKe Chang	2026	arXiv (Cornell University)	0	(link)
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Visualizing a Li-depleted amorphous cathode–electrolyte interphase in sulfide solid-state batteries using in situ cryogenic electron microscopy	Yuki Nomura; Ryoma Sasaki; Huu Duc Luong; Misaki Hasegawa; Willy Shun Kai Bong; Koji Hiraoka; Kazuo Yamamoto; Yutaka Ito; Yoshiya Fujiwara; Yoshitaka Tateyama; Takuhiro Miyuki	2026	ChemRxiv	0	10.26434/chemrxiv.15003253/v1
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Phonon Transport in Disordered Two-Dimensional Graphene	Shoaib, Hassan	2025	eScholarship@McGill (McGill)	0	10.82308/27920
Atomic structure at the surface of warm basaltic glasses	Marthe Grønlie Guren; Henrik Anderson Sveinsson; Razvan Caracas; Anders Malthe-Sørenssen; François Renard	2025		0	10.5194/egusphere-egu25-12944
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Through rainbow-tinted glasses: machine learning-driven modeling of chromophores	Lindgren E	2026	Science and Technology Facilities Council	0	10.63959/chalmers.dt/5890
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Trapped charge driven degradation of perovskite solar cells	Namyoung Ahn; Kwisung Kwak; Min Seok Jang; Heetae Yoon; Byung Yang Lee; Jong-Kwon Lee; Peter V. Pikhitsa; Junseop Byun; Mansoo Choi	2016	arXiv preprint	0	10.1038/ncomms13422
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Growth of a One Dimensional Quasiperiodic Covering with Locally Determined Decorations	Hyeong-Chai Jeong	2008	arXiv preprint	0	(link)
Local 3D real space atomic structure of the simple icosahedral Ho11Mg15Zn74 quasicrystal from PDF data	S. Bruehne; E. Uhrig; C. Gross; W. Assmus	2003	Cryst. Res. Technol. 38(12) (2003) 1023-1036.	0	(link)
Mobility edge and intermediate phase in one-dimensional incommensurate lattice potentials	Xiao Li; S. Das Sarma	2018	Phys. Rev. B 101, 064203 (2020)	0	10.1103/PhysRevB.101.064203
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Local Rules for Computable Planar Tilings	Thomas Fernique; Mathieu Sablik	2012	EPTCS 90, 2012, pp. 133-141	0	10.4204/EPTCS.90.11
Local spectroscopy of moiré-induced electronic structure in gate-tunable twisted bilayer graphene	Dillon Wong; Yang Wang; Jeil Jung; Sergio Pezzini; Ashley M. DaSilva; Hsin-Zon Tsai; Han Sae Jung; Ramin Khajeh; Youngkyou Kim; Juwon Lee; Salman Kahn; Sajjad Tollabimazraehno; Haider Rasool; Kenji Watanabe; Takashi Taniguchi; Alex Zettl; Shaffique Adam; Allan H. MacDonald; Michael F. Crommie	2015	Phys. Rev. B 92, 155409 (2015)	0	10.1103/PhysRevB.92.155409
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Hidden dualities in 1D quasiperiodic lattice models	Miguel Gonçalves; Bruno Amorim; Eduardo V. Castro; Pedro Ribeiro	2021	SciPost Phys. 13, 046 (2022)	0	10.21468/SciPostPhys.13.3.046
An efficient saddle search method for ordered phase transitions involving translational invariance	Gang Cui; Kai Jiang; Tiejun Zhou	2023	arXiv preprint	0	(link)
Invariable mobility edge in a quasiperiodic lattice	Tong Liu; Yufei Zhu; Shujie Cheng; Feng Li; Hao Guo; Yong Pu	2021	arXiv preprint	0	10.1088/1674-1056/ac140e
Interplay of Quasiperiodic Criticality and the Non-Hermitian Skin Effect	Zhangyuan Chen; Xianqi Tong; Xiaosen Yang	2026	arXiv preprint	0	(link)
Computing with almost periodic functions	R. V. Moody; M. Nesterenko; J. Patera	2008	arXiv preprint	0	(link)
Quantum criticality and Kibble-Zurek scaling in the Aubry-André-Stark model	En-Wen Liang; Ling-Zhi Tang; Dan-Wei Zhang	2024	Phys. Rev. B 110, 024207 (2024)	0	10.1103/PhysRevB.110.024207
Single-particle and many-body analyses of a quasiperiodic integrable system after a quench	Kai He; Lea F. Santos; Tod M. Wright; Marcos Rigol	2013	Phys. Rev. A 87, 063637 (2013)	0	10.1103/PhysRevA.87.063637
Coexistence of topological Anderson insulator and multifractal critical phase in a non-Hermitian quasicrystal	Qi-Bo Zeng; Rong Lü	2026	Phys. Rev. B 113, 224203 (2026)	0	10.1103/n9rf-zk9t
Quasicrystalline Spin Liquid	Sunghoon Kim; Mohammad Saad; Dan Mao; Adhip Agarwala; Debanjan Chowdhury	2024	Phys. Rev. B 110, 214438, 2024	0	10.1103/PhysRevB.110.214438
Disordered, Quasicrystalline and Crystalline Phases of Densely Packed Tetrahedra	Amir Haji-Akbari; Michael Engel; Aaron S. Keys; Xiaoyu Zheng; Rolfe G. Petschek; Peter Palffy-Muhoray; Sharon C. Glotzer	2010	Nature 462, 773–777 (2009)	0	10.1038/nature08641
Photoinduced Phase Transition in Two-Band model on Penrose Tiling	Ken Inayoshi; Yuta Murakami; Akihisa Koga	2021	J. Phys.: Conf. Ser. 2164, 012050 (2022)	0	10.1088/1742-6596/2164/1/012050
Quasicrystalline electronic states in 30° rotated twisted bilayer graphene	Pilkyung Moon; Mikito Koshino; Young-Woo Son	2019	Phys. Rev. B 99, 165430 (2019)	0	10.1103/PhysRevB.99.165430
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Localized photonic bandedge modes and orbital angular momenta of light in a golden-angle spiral	Seng Fatt Liew; Heeso Noh; Jacob Trevino; Luca Dal Negro; Hui Cao	2011	Optics Express vol. 19, pp. 23631-23642 (2011)	0	10.1364/OE.19.023631

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Amplitude-dependent edge states and discrete breathers in nonlinear modulated phononic lattices	Matheus I. N. Rosa; Michael J. Leamy; Massimo Ruzzene	2022	arXiv preprint	0	(link)
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Chemical localization	V. F. Gantmakher	2002	UFN 172(11), 1283 (2002) [English translation in Physics-Uspekhi 45(11), (2002)]	0	10.1070/PU2002v045n11ABEH00124
Sub-diffusive electronic states in octagonal tiling	G. Trambly de Laissardière; C. Oguey; D. Mayou	2016	Journal of Physics: Conf. Series 809 (2017) 012020	0	10.1088/1742-6596/809/1/012020
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Higher-dimensional Hofstadter butterfly on Penrose lattice	Rasoul Ghadimi; Takanori Sugimoto; Takami Tohyama	2022	arXiv preprint	0	10.1103/PhysRevB.106.L201113
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Exploring Aperiodic Order in Photonic Time Crystals	Marino Coppolaro; Massimo Moccia; Giuseppe Castaldi; Vincenzo Galdi	2025	arXiv preprint	0	(link)
Ground state of a two dimensional quasiperiodic antiferromagnet	A. Jagannathan	2004	arXiv preprint	0	10.1103/PhysRevB.71.115101
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Complete Structural Determination of Mesostructural Dodecagonal Quasicrystalline Particles	Xueliang Zhang; Xi Wang; Nobuhisa Fujita; Osamu Terasaki; Lu Han	2026	arXiv preprint	0	(link)
Luminescence Properties of a Fibonacci Photonic Quasicrystal	Vasilios Passias; Nikesh Valappil; Zhou Shi; Lev Deych; Alexander Lisyansky; Vinod M. Menon	2008	arXiv preprint	0	10.1364/OE.17.006636
Antiferromagnetic order in the Hubbard Model on the Penrose Lattice	Akihisa Koga; Hirokazu Tsunetsugu	2017	Phys. Rev. B 96, 214402 (2017)	0	10.1103/PhysRevB.96.214402
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Cleaved surface of i-AlPdMn quasicrystals: Influence of the local temperature elevation at the crack tip on the fracture surface roughness	Laurent Ponson; Daniel Bonamy; Luc Barbier	2006	Physical Review B 74 (20/11/2006) 184205	0	10.1103/PhysRevB.74.184205
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Topological Pumping over a Photonic Fibonacci Quasicrystal	Verbin, Mor; Zilberberg, Oded; Lahini, Yoav; Kraus, Yaacov E.; Silberberg, Yaron	2014	Phys.Rev.B	0	10.1103/PhysRevB.91.064201
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Robust Anderson transition in non-Hermitian photonic quasicrystals	Longhi, Stefano	2024	Opt.Lett.	0	10.1364/OL.517182
Exact mobility edges and topological phase transition in two-dimensional non-Hermitian quasicrystals	Xu, Zhi-Hao; Xia, Xu; Chen, Shu	2021	Sci.China Phys.Mech.Astron.	0	10.1007/s11433-021-1802-4
Super-ballistic diffusion induced by nonlinear interactions in a one-dimensional quasiperiodic lattice	Dong, Ji-Long; Gu, Long-Hua; Zhang, Lu-Chen; Li, Zhi; Zhang, Dan-Wei	2024	Phys.Lett.A	0	10.1016/j.physleta.2024.129549
Support Vector Machine Kernels as Quantum Propagators	Kuo, Nan-Hong; Wong, Renata	2025	INSPIRE record	0	(link)
Anomalous localization in a kicked quasicrystal	Shimasaki, Toshihiko; Prichard, Max; Kondakci, H. Esat; Pagett, Jared E.; Bai, Yifei; Dotti, Peter; Cao, Alec; Dardia, Anna R.; Lu, Tsung-Cheng; Grover, Tarun; Weld, David M.	2022	Nature Phys.	0	10.1038/s41567-023-02329-4
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Non-Hermitian topological mobility edges and transport in photonic quantum walks	Longhi, Stefano	2022	Opt.Lett.	0	10.1364/OL.460484
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Topological Phase Transitions and Mobility Edges in Non-Hermitian Quasicrystals	Lin, Quan; Li, Tianyu; Xiao, Lei; Wang, Kunkun; Yi, Wei; Xue, Peng	2021	Phys.Rev.Lett.	0	10.1103/PhysRevLett.129.113601
Observing the two-dimensional Bose glass in an optical quasicrystal	Yu, Jr-Chiun; Bhave, Shaurya; Reeve, Lee; Song, Bo; Schneider, Ulrich	2023	Nature	0	10.1038/s41586-024-07875-2
Phasonic Spectroscopy of a Quantum Gas in a Quasicrystalline Lattice	Rajagopal, Shankari V.; Shimasaki, Toshihiko; Dotti, Peter; Račiūnas, Mantas; Senaratne, Ruwan; Anisimovas, Egidijus; Eckardt, André; Weld, David M.	2019	Phys.Rev.Lett.	0	10.1103/PhysRevLett.123.223201
Stability of quasicrystalline ultracold fermions to dipolar interactions	Molignini, Paolo	2024	Phys.Rev.Res.	0	10.1103/szdc-61nl
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Observing Localization in a 2D Quasicrystalline Optical Lattice	Matteo Sbroscia; Konrad Viebahn; Edward Carter; Jr-Chiun Yu; Alexander L. Gaunt; Ulrich Schneider	2020	Physical Review Letters	105	10.1103/physrevlett.125.200604
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Observation of Topological Corner State Arrays in Photonic Quasicrystals	Aoqian Shi; Yiwei Peng; Jiapei Jiang; Yuchen Peng; Peng Peng; Jianzhi Chen; Hongsheng Chen; Shuangchun Wen; Xiao Lin; Fei Gao; Jianjun Liu	2024	Laser & Photonics Review	35	10.1002/lpor.202300956
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Effects of graphene on light transmission spectra in Dodecanacci photonic quasicrystals	E.F. Silva; M.S. Vasconcelos; Carlos H. Costa; D.H.A.L. Anselmo; V.D. Mello	2019	Optical Materials	20	10.1016/j.optmat.2019.109450
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Basic Stability Tests of Machine Learning Potentials for Molecular Simulations in Computational Drug Discovery	Kavindri Ranasinghe (18990399); Adam L. Baskerville (4737792); Geoffrey P. F. Wood (7507880); Gerhard König (1338774)	2025	Figshare	0	10.1021/acs.jcim.5c01150.s001	openalex
Machine-Learning Molecular Dynamics Study on the Structure and Glass Transition of Calcium Aluminosilicate Glasses	Takeyuki Kato (277950); Ryuki Kayano (19787715); Takahiro Ohkubo (1273047)	2025	Figshare	0	10.1021/acs.jpcb.5c03404.s001	openalex
Site-Selective Initial Hydration of Boron at Multicomponent Borosilicate Glass–Water Interfaces Revealed Using Machine Learning Molecular Dynamics	Ryuki Kayano; Takahiro Ohkubo	2026	Zenodo (CERN European Organization for Nuclear Research)	0	10.5281/zenodo.19847865	openalex
Site-Selective Initial Hydration of Boron at Multicomponent Borosilicate Glass–Water Interfaces Revealed Using Machine Learning Molecular Dynamics	Ryuki Kayano; Takahiro Ohkubo	2026	Zenodo (CERN European Organization for Nuclear Research)	0	10.5281/zenodo.19847866	openalex
amorphous carbon ab-initio calculation dataset	Emi Minamitani	2023	Zenodo (CERN European Organization for Nuclear Research)	0	10.5281/zenodo.7905585	openalex
amorphous carbon ab-initio calculation dataset	Emi Minamitani	2023	Zenodo (CERN European Organization for Nuclear Research)	0	10.5281/zenodo.7905584	openalex

Title	Authors	Year	Venue	Cited	DOI	Source
Ion correlations explain kinetic selectivity in diffusion-limited solid state synthesis reactions: Transport coefficients in amorphous Ba-Ti-O interphases	Vir Karan; Max C. Gallant; Kristin Persson	2026	Figshare	0	10.6084/m9.figshare.28207292.v1	openalex
Ion correlations explain kinetic selectivity in diffusion-limited solid state synthesis reactions: Transport coefficients in amorphous Ba-Ti-O interphases	Vir Karan; Max C. Gallant; Kristin Persson	2026	Figshare	0	10.6084/m9.figshare.28207292.v3	openalex
Ion correlations explain kinetic selectivity in diffusion-limited solid state synthesis reactions: Transport coefficients in amorphous Ba-Ti-O interphases	Vir Karan; Max C. Gallant; Kristin Persson	2026	Figshare	0	10.6084/m9.figshare.28207292	openalex
Ion correlations explain kinetic selectivity in diffusion-limited solid state synthesis reactions: Transport coefficients in amorphous Ba-Ti-O interphases	Vir Karan; Max C. Gallant; Kristin Persson	2026	Figshare	0	10.6084/m9.figshare.28207292.v2	openalex
Amorphous Ni . Fe . O: melt-quench AIMD configurations, DFT+U/Bader oxidation-state analyses, ΔSCF O 1s core-level shifts, and a MACE interatomic potential	DAPENG ZHANG	2026	Figshare	0	10.6084/m9.figshare.33075551.v1	openalex
Amorphous Ni . Fe . O: melt-quench AIMD configurations, DFT+U/Bader oxidation-state analyses, ΔSCF O 1s core-level shifts, and a MACE interatomic potential	DAPENG ZHANG	2026	Figshare	0	10.6084/m9.figshare.33075551	openalex
Dataset: Atomic-Scale Origin of the Cation Field Strength Dependence of Mechanical Properties in Divalent-Cation Aluminosilicate Glasses	Takeyuki Kato; Ohkubo Takahiro	2026	Zenodo (CERN European Organization for Nuclear Research)	0	10.5281/zenodo.21700493	openalex
Dataset: Atomic-Scale Origin of the Cation Field Strength Dependence of Mechanical Properties in Divalent-Cation Aluminosilicate Glasses	Takeyuki Kato; Ohkubo Takahiro	2026	Zenodo (CERN European Organization for Nuclear Research)	0	10.5281/zenodo.21700494	openalex
Influence of Crystal Orientation and Polymorphism on the Shock Response of Boron Carbide	Kimia Ghaffari; Salil Bavdekar; Douglas E. Spearot; Ghatu Subhash	2025	SSRN Electronic Journal	0	10.2139/ssrn.5186595	openalex
MULTISCALE ATOMISTIC MODELING AND MACHINE LEARNING FOR NANOMATERIALS: STRUCTURAL EVOLUTION AND DEVICE APPLICATIONS	Akram Ibrahim	2025	Open MIND	0	10.13016/m2o06y-eqzn	openalex
A New Accurate, Efficient and Transferable Deep Learning Interatomic Potential for Carbon with Advanced Applications	Songyou Wang; Kai Chen; Riyi Yang; Zhefeng Wang; Wuyan Zhao; Youmin Xu; Huaijun Sun; Chao Zhang; Kai-Ming Ho; C. Z. Wang; Wan-Sheng Su	2024	SSRN Electronic Journal	0	10.2139/ssrn.4834446	openalex
Crystal Intentions: A Design-Integrated Roadmap for Engineering the Solid-State Properties of Active Pharmaceutical Ingredients	Ramarao Nadendla; Uppalapati Mohan Chandu; Ponnuru Venkata Suresh; Pallavi Vadlamudi; K N Rajini Kanth	2026	Preprints.org	0	10.20944/preprints202608.0453.v1	openalex
Atomistic Insights into Melt-Quench Rate on Structural Evolution and Thermal Transport of Silica: A Machine-Learning Study	Hui Li; Dongyuan Zhai; Jiwu Lu	2025	SSRN Electronic Journal	0	10.2139/ssrn.5202849	openalex
Origin of Doping-Induced Structural Amorphization and Improved Cycling Performance in Aluminum Anodes for Lithium-Ion Batteries	Jiwon Yu (6671279); Gyeong S. Hwang (1283838)	2025	Figshare	0	10.1021/acs.jpcc.5c06267.s001	openalex
Atomistic Insights into Stress-Driven Lithiation at Silicon Anode Crack Tips	Bowen Zhang (595078); Peiyao Zhang (1261464); Changguo Wang (10539339); Huifeng Tan (22386821); Liwei Dong (380035); Jia-Yan Liang (5277707); Yuanpeng Liu (10185905)	2025	Figshare	0	10.1021/acsami.5c11237.s001	openalex

Title	Authors	Year	Venue	Cited	DOI	Source
Competitive Hydrogen-Bond Partitioning in Deep Eutectic Solvents: From Cooperative Charge Spreading to Structure–Property Design Rules	Sergio de-la-Huerta-Sainz; Valentin Diez-Cabanes; Alberto Gutiérrez; Sara Santamaria; María A. Escobedo; Pedro A. Marcos; Alfredo Bol; Jose L. Trenzado; Mert Atilhan; Santiago Aparício	2026	Figshare	0	10.1021/acsomega.6c02376.s001	openalex
Competitive Hydrogen-Bond Partitioning in Deep Eutectic Solvents: From Cooperative Charge Spreading to Structure–Property Design Rules	Sergio de-la-Huerta-Sainz; Valentin Diez-Cabanes; Alberto Gutiérrez; Sara Santamaria; María A. Escobedo; Pedro A. Marcos; Alfredo Bol; Jose L. Trenzado; Mert Atilhan; Santiago Aparício	2026	Figshare	0	10.1021/acsomega.6c02376.s002	openalex
Machine-Learning-Accelerated Investigation of He, Kr, and Xe Gas Bubble Nucleation in U-10Zr Metallic Nuclear Fuel	Mangladeep Bhullar; Barbara Szpunar; Jerzy Szpunar	2026	SSRN Electronic Journal	0	10.2139/ssrn.7349007	openalex
Axionic Modulation of Band Spectra in Photonic Quasicrystals: Localization and Transmission Properties	C.G. Bezerra; Anny Araújo; Carlos H. Costa; S. Azevedo; G. M. Viswanathan; A. J. Chaves	2026	SSRN Electronic Journal	0	10.2139/ssrn.7073044	openalex
Exploring Diffusion and Localization Dynamics of Wave Packets in a One-Dimensional Quasiperiodic Lattice with Nonlinear Interactions	J. K. Dong; Long-Hua Gu; Luchen Zhang; Li Zhi; Dan-Wei Zhang	2024	SSRN Electronic Journal	0	10.2139/ssrn.4700417	openalex
Wave Particle Duality as a Spectral Transition in the Fibonacci Potential of Dark Geometry	Hugo Pascal Gilles Hertault	2026	Zenodo (CERN European Organization for Nuclear Research)	0	10.5281/zenodo.18972684	openalex
Wave Particle Duality as a Spectral Transition in the Fibonacci Potential of Dark Geometry	Hugo Pascal Gilles Hertault	2026	Zenodo (CERN European Organization for Nuclear Research)	0	10.5281/zenodo.18972683	openalex

Backend configuration

Backend	Endpoint	Auth	Paging
openalex	https://api.openalex.org/works	OPENALEX_API_KEY	cursor
arxiv	(driver: arxiv)	none	-
inspire	https://inspirehep.net/api/literature	none	page

prisma.json


```
{
  "_comment": "Manual screening stages of the PRISMA 2020 flow. Fill in the integers and re-run
report.py; null = not yet done. excluded_reasons maps a reason to a count. The other_* keys are the
right-hand column (records identified via other methods).",
  "records_screened": null,
  "records_excluded": null,
  "reports_sought": null,
  "reports_not_retrieved": null,
  "reports_assessed": null,
  "excluded_reasons": {},
  "other_sought": null,
  "other_not_retrieved": null,
  "other_assessed": null,
  "other_excluded_reasons": {},
  "studies_included": null,
  "reports_included": null,
  "citation_searching": "",
  "other_methods": "",
  "prior_work": "",
  "peer_review": ""
}
```


Run log

```
scitech-librarian run 20260828T160426
blocks:   PSC MLP NOV QC
backends: openalex arxiv inspire
mode:     full fetch (limit 300)

=== Block PSC: Perovskite solar cell degradation under humidity
    (grounding block -- expect thousands of hits; use it to sanity-check that every backend is reachable
and the counts are plausible)
```

```
    openalex           4,569 hits   (300 saved)   [6.3s]
    arxiv                175 hits   (175 saved)   [6.4s]
    inspire                0 hits   [0.7s]

=== Block MLP: Machine-learned interatomic potentials for amorphous materials
    (a narrower intersection -- expect hundreds; good for testing record fetch and RIS export)
    openalex             238 hits   (239 saved)   [4.4s]
                     (filtered 33 non-scholarly: Figshare, Open MIND, Preprints.org, SSRN Electronic
Journal,)
    arxiv                138 hits   (138 saved)   [6.1s]
    inspire                0 hits   [0.5s]

=== Block NOV: EXAMPLE NOVELTY CHECK: origami metamaterials as topological acoustic pumps
    (novelty-check pattern -- a SMALL number is the good outcome; read every hit by hand before claiming
a gap)
    openalex               0 hits   [0.6s]
    arxiv                  0 hits   [0.4s]
    inspire                0 hits   [0.5s]

=== Block QC: Quasicrystal photonics (arXiv-tuned example)
    (demonstrates arxiv_groups: arXiv chokes on deeply nested booleans, so name the two groups it should
get)
    openalex             154 hits   (154 saved)   [2.0s]
                     (filtered 4 non-scholarly: SSRN Electronic Journal, Zenodo)
    arxiv                414 hits   (300 saved)   [11.0s]
    inspire               17 hits   (17 saved)   [1.0s]


| Block | openalex | arxiv | inspire | Title |
|---|---|---|---|---|
| PSC | 4569 | 175 | 0 | Perovskite solar cell degradation under humidity |
| MLP | 238 | 138 | 0 | Machine-learned interatomic potentials for amorphous materials |
| NOV | 0 | 0 | 0 | EXAMPLE NOVELTY CHECK: origami metamaterials as topological acoustic pumps |
| QC | 154 | 414 | 17 | Quasicrystal photonics (arXiv-tuned example) |

1286 records fetched, 1226 unique after DOI/title dedup
```

Environment

Item	Value
Python	3.13.11
Platform	win32
Tool version	3.2
Report generated	2026-08-28 16:08

Suggestions

- Block PSC: openalex 4,569 hits – a generic term is probably driving this; tighten a group or add a more specific one before reading.
- Block PSC: counts differ >20x across backends (175 to 4,569); grammar and coverage diverge, so do not compare these numbers – discover here, quote one source.
- Block NOV: zero hits on every backend. Either the intersection is genuinely empty (a finding – check the synonyms first) or one group is too narrow; try dropping one group and rerunning.
- Block QC: counts differ >20x across backends (17 to 414); grammar and coverage diverge, so do not compare these numbers – discover here, quote one source.
- 2 block/backend pair(s) hit the `--limit` cap (300); raise it (largest total 4,569) if you need the complete record set rather than the most-cited slice.
- No citation-grade backend (Scopus, Web of Science, NASA ADS) was in this search; add ADS (free token) or Scopus before quoting counts.
- Open-access status was not looked up; rerun with `--pdfs` (optionally `--pdf-blocks`) to collect legal OA PDF links via Unpaywall.
- The PRISMA flow's manual stages are empty: fill in prisma.json (screened / excluded / assessed / included) and rerun report.py to complete the diagram.
- Block PSC: total hits changed from 14,620 (run 20260828T100120) to 4,744; count drift is expected as indexes grow, but a large jump usually means the query changed – diff queries.json between the runs.

- Block MLP: total hits changed from 838 (run 20260828T100120) to 376; count drift is expected as indexes grow, but a large jump usually means the query changed – diff queries.json between the runs.
- Block QC: total hits changed from 978 (run 20260828T100120) to 585; count drift is expected as indexes grow, but a large jump usually means the query changed – diff queries.json between the runs.